

IoT-Enabled Barangay Resident and Attendance Management System



IoT-Enabled Barangay Resident and Attendance Management System

Project Codebook

Table of Contents

Page 8	 __init__.py
Page 9	 apply_network_fix.sh
Page 11	 apply_permanent_fix.sh
Page 13	 appointments/models.py
Page 14	 appointments/urls.py
Page 15	 appointments/views.py
Page 17	 asgi.py
Page 18	 attendance/__init__.py
Page 19	 attendance/admin.py
Page 20	 attendance/apps.py
Page 21	 attendance/models.py
Page 25	 attendance/templatetags/__init__.py
Page 26	 attendance/templatetags/attendance_tags.py
Page 28	 attendance/tests.py
Page 29	 attendance/urls.py
Page 30	 attendance/views.py
Page 51	 barangay_project/__init__.py
Page 52	 barangay_project/asgi.py
Page 53	 barangay_project/settings.py
Page 57	 barangay_project/urls.py
Page 58	 barangay_project/wsgi.py
Page 59	 biometrics/__init__.py
Page 60	 biometrics/base.py
Page 61	 biometrics/stub.py
Page 62	 biometrics/utils.py
Page 63	 census/__init__.py
Page 64	 census/admin.py
Page 65	 census/apps.py
Page 66	 census/models.py
Page 67	 census/tests.py
Page 68	 census/urls.py
Page 69	 census/views.py
Page 72	 certifications/__init__.py
Page 73	 certifications/admin.py
Page 74	 certifications/apps.py
Page 75	 certifications/models.py
Page 82	 certifications/ra11261_views.py

IoT-Enabled Barangay Resident and Attendance Management System

Page 91	 certifications/templatetags/__init__.py
Page 92	 certifications/templatetags/ra11261_tags.py
Page 93	 certifications/tests.py
Page 94	 certifications/urls.py
Page 95	 certifications/utils.py
Page 105	 certifications/views.py
Page 119	 core/__init__.py
Page 120	 core/admin.py
Page 121	 core/apps.py
Page 122	 core/context_processors.py
Page 123	 core/decorators.py
Page 124	 core/management/__init__.py
Page 125	 core/management/commands/__init__.py
Page 126	 core/management/commands/seed_roles.py
Page 127	 core/middleware.py
Page 128	 core/models.py
Page 132	 core/test_usb_hardware.py
Page 133	 core/tests.py
Page 134	 core/urls.py
Page 135	 core/utils/backup.py
Page 139	 core/utils/biometric_discovery.py
Page 143	 core/views.py
Page 166	 documents/__init__.py
Page 167	 documents/admin.py
Page 168	 documents/apps.py
Page 169	 documents/models.py
Page 170	 documents/tests.py
Page 171	 documents/urls.py
Page 172	 documents/views.py
Page 174	 fix_hotspot.sh
Page 184	 generate_codebook.py
Page 186	 generate_pdf.py
Page 189	 install_internet.sh
Page 190	 manage.py
Page 191	 officials/__init__.py
Page 192	 officials/admin.py
Page 193	 officials/apps.py
Page 194	 officials/models.py
Page 199	 officials/tests.py
Page 200	 officials/urls.py

IoT-Enabled Barangay Resident and Attendance Management System

Page 201	 officials/views.py
Page 221	 ordinances/__init__.py
Page 222	 ordinances/admin.py
Page 223	 ordinances/apps.py
Page 224	 ordinances/models.py
Page 225	 ordinances/tests.py
Page 226	 ordinances/urls.py
Page 227	 ordinances/utils.py
Page 231	 ordinances/views.py
Page 235	 payments/apps.py
Page 236	 payments/models.py
Page 238	 payments/urls.py
Page 239	 payments/utils.py
Page 242	 payments/views.py
Page 246	 philsys/__init__.py
Page 247	 philsys/admin.py
Page 249	 philsys/apps.py
Page 250	 philsys/models.py
Page 251	 philsys/serializers.py
Page 252	 philsys/urls.py
Page 253	 philsys/views.py
Page 258	 refactor_auth.py
Page 259	 reports/__init__.py
Page 260	 reports/admin.py
Page 261	 reports/apps.py
Page 262	 reports/models.py
Page 263	 reports/tests.py
Page 264	 reports/urls.py
Page 265	 reports/views.py
Page 266	 residents/__init__.py
Page 267	 residents/admin.py
Page 268	 residents/apps.py
Page 269	 residents/models.py
Page 276	 residents/tests.py
Page 278	 residents/urls.py
Page 279	 residents/views.py
Page 295	 run_app.sh
Page 296	 seed_puroks.py
Page 297	 settings.py
Page 300	 setup_autostart.sh

IoT-Enabled Barangay Resident and Attendance Management System

Page 302	 setup_hostapd_hotspot.sh
Page 304	 setup_hotspot.sh
Page 306	 templates/appointments/book.html
Page 308	 templates/appointments/list.html
Page 309	 templates/appointments/manage.html
Page 311	 templates/attendance/biometric_attendance.html
Page 330	 templates/attendance/clock.html
Page 332	 templates/attendance/daily_report.html
Page 338	 templates/attendance/dashboard.html
Page 356	 templates/attendance/dtr.html
Page 359	 templates/attendance/dtr_detail.html
Page 363	 templates/attendance/dtr_print.html
Page 367	 templates/attendance/enroll.html
Page 370	 templates/attendance/history_calendar.html
Page 374	 templates/attendance/public_scan.html
Page 376	 templates/attendance/schedule.html
Page 382	 templates/base.html
Page 392	 templates/census/age_groups.html
Page 395	 templates/census/dashboard.html
Page 403	 templates/census/demographics.html
Page 406	 templates/certifications/fulfill_form.html
Page 412	 templates/certifications/issue.html
Page 420	 templates/certifications/list.html
Page 427	 templates/certifications/my_requests.html
Page 435	 templates/certifications/ra11261_admin_list.html
Page 441	 templates/certifications/ra11261_admin_roster.html
Page 446	 templates/certifications/ra11261_apply.html
Page 450	 templates/certifications/request_form.html
Page 458	 templates/certifications/request_list.html
Page 464	 templates/certifications/search.html
Page 467	 templates/certifications/view.html
Page 474	 templates/certifications/virtual_certificate.html
Page 483	 templates/components/back_button.html
Page 484	 templates/core/backup_choice.html
Page 489	 templates/core/barangay_map.html
Page 498	 templates/core/dashboard.html
Page 501	 templates/core/dashboard_captain.html
Page 505	 templates/core/dashboard_default.html
Page 507	 templates/core/dashboard_kagawad.html
Page 510	 templates/core/dashboard_lupon.html

IoT-Enabled Barangay Resident and Attendance Management System

Page 512		templates/core/dashboard_resident.html
Page 520		templates/core/dashboard_secretary.html
Page 523		templates/core/dashboard_sk.html
Page 526		templates/core/dashboard_treasurer.html
Page 529		templates/core/hardware_poller.html
Page 531		templates/core/login.html
Page 547		templates/core/notifications.html
Page 548		templates/core/registration_success.html
Page 552		templates/core/resident_signup.html
Page 569		templates/core/signup.html
Page 574		templates/documents/confirm_delete.html
Page 575		templates/documents/list.html
Page 577		templates/documents/pdf_export.html
Page 578		templates/documents/workspace.html
Page 596		templates/officials/biometric_register.html
Page 609		templates/officials/confirm_delete.html
Page 610		templates/officials/form.html
Page 616		templates/officials/list.html
Page 621		templates/officials/onboarding/approvals.html
Page 625		templates/officials/onboarding/invite_form.html
Page 627		templates/officials/onboarding/upload_documents.html
Page 632		templates/officials/onboarding/verify_otp.html
Page 636		templates/officials/org_chart.html
Page 644		templates/officials/view.html
Page 652		templates/ordinances/confirm_delete.html
Page 653		templates/ordinances/form.html
Page 656		templates/ordinances/list.html
Page 658		templates/ordinances/parse_review.html
Page 667		templates/ordinances/upload.html
Page 668		templates/ordinances/view.html
Page 672		templates/payments/dashboard.html
Page 677		templates/payments/select_payment.html
Page 691		templates/reports/dashboard.html
Page 697		templates/residents/form.html
Page 705		templates/residents/household_form.html
Page 712		templates/residents/household_list.html
Page 714		templates/residents/household_view.html
Page 728		templates/residents/list.html
Page 733		templates/residents/registration_detail.html
Page 739		templates/residents/registration_list.html

IoT-Enabled Barangay Resident and Attendance Management System

Page 743 templates/residents/view.html

Page 751 test_fpdf.py

Page 752 urls.py

Page 753 wsgi.py

IoT-Enabled Barangay Resident and Attendance Management System

File: __init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: apply_network_fix.sh

```
#!/bin/bash
# =====
# Barangay System - Persistent Network Fix
# Runs on every boot to ensure correct routing.
# =====

WIFI_IFACE="wlan0"

# Auto-detect Ethernet interface ? NEVER pick a wireless interface!
ETH_IFACE=""
for iface in end0 eth0 enp1s0 enp2s0 enp3s0; do
    if ip link show "$iface" >/dev/null 2>&1; then
        ETH_IFACE="$iface"
        break
    fi
done
if [ -z "$ETH_IFACE" ]; then
    ETH_IFACE="end0"
fi

HOTSPOT_IP="10.42.0.1"

log() { echo "[$(date '+%H:%M:%S')] $1" | tee -a /var/log/brgy_network.log; }

log "=== Barangay Network Fix Starting ==="

# 1. Ensure wlan0 is up
ip link set dev $WIFI_IFACE up
sleep 1

# 2. Remove any stray LAN IP from wlan0 (e.g. 192.168.1.x should NOT be on wlan0)
STRAY_IPS=$(ip addr show $WIFI_IFACE | grep 'inet ' | grep -v "$HOTSPOT_IP" | awk '{print $2}')
for IP in $STRAY_IPS; do
    log "Removing stray IP $IP from $WIFI_IFACE..."
    ip addr del "$IP" dev $WIFI_IFACE 2>/dev/null
done

# 3. Make sure wlan0 has ONLY the hotspot IP
if ! ip addr show $WIFI_IFACE | grep -q "$HOTSPOT_IP"; then
    log "Adding hotspot IP $HOTSPOT_IP to $WIFI_IFACE..."
    ip addr add "$HOTSPOT_IP/24" dev $WIFI_IFACE
fi

# 4. Remove any wrong default route via wlan0
WRONG_ROUTES=$(ip route show default dev $WIFI_IFACE)
if [ -n "$WRONG_ROUTES" ]; then
    log "Removing wrong default route via $WIFI_IFACE..."
    ip route del default dev $WIFI_IFACE 2>/dev/null
fi

# 5. Make sure default route goes through $ETH_IFACE (the real internet interface)
if [ -n "$ETH_IFACE" ]; then
    DEFAULT_ROUTE=$(ip route show default dev "$ETH_IFACE" 2>/dev/null | head -1)
    if [ -z "$DEFAULT_ROUTE" ]; then
        # lost its default route ? try to get it back via DHCP
        log "No default route via $ETH_IFACE ? requesting DHCP..."
    fi
fi
```

IoT-Enabled Barangay Resident and Attendance Management System

```
dhclient "$ETH_IFACE" 2>/dev/null &
sleep 3
fi
else
    log "Warning: No ethernet interface found."
fi

log "Current default routes:"
ip route show default | tee -a /var/log/brgy_network.log

# 6. Enable IP forwarding (for internet sharing to WiFi clients)
echo 1 > /proc/sys/net/ipv4/ip_forward

# 7. Set up iptables NAT (internet sharing from ETH_IFACE to wlan0)
if [ -n "$ETH_IFACE" ]; then
    iptables -t nat -F
    iptables -F FORWARD
    iptables -t nat -A POSTROUTING -o $ETH_IFACE -j MASQUERADE
    iptables -A FORWARD -i $ETH_IFACE -o $WIFI_IFACE -m state --state RELATED,ESTABLISHED
-j ACCEPT
    iptables -A FORWARD -i $WIFI_IFACE -o $ETH_IFACE -j ACCEPT
    log "iptables NAT rules applied using $ETH_IFACE."
fi

# 8. Force-release wlan0 from NetworkManager, then restart hostapd and dnsmasq
log "Releasing wlan0 from NetworkManager..."
nmcli dev set $WIFI_IFACE managed no 2>/dev/null || true
sleep 2

systemctl restart dnsmasq
sleep 1
# Give hostapd a clean start
systemctl stop hostapd 2>/dev/null
sleep 2
systemctl start hostapd
sleep 3

HOSTAPD_STATUS=$(systemctl is-active hostapd)
log "hostapd status: $HOSTAPD_STATUS"

if [ "$HOSTAPD_STATUS" = "active" ]; then
    log "=== SUCCESS: Barangay_System_WiFi is broadcasting ==="
else
    log "=== ERROR: hostapd failed to start. Check: journalctl -u hostapd ==="
fi
```

IoT-Enabled Barangay Resident and Attendance Management System

File: apply_permanent_fix.sh

```
#!/bin/bash
echo "=====
echo "   INSTALLING PERMANENT HOTSPOT & INTERNET FIX   "
echo "=====

WIFI_IFACE="wlan0"
ETH_IFACE="end0"

echo "[1/6] Stopping interfering services (wpa_supplicant)..."
sudo systemctl stop wpa_supplicant 2>/dev/null
sudo systemctl disable wpa_supplicant 2>/dev/null
sudo systemctl mask wpa_supplicant 2>/dev/null

echo "[2/6] Telling NetworkManager to completely ignore WiFi..."
cat << EOF | sudo tee /etc/NetworkManager/conf.d/99-unmanaged-devices.conf
[keyfile]
unmanaged-devices=interface-name:$WIFI_IFACE
EOF
sudo systemctl restart NetworkManager
sleep 3

echo "[3/6] Forcing DNS resolution to Google..."
sudo rm -f /etc/resolv.conf
echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
echo "nameserver 8.8.4.4" | sudo tee -a /etc/resolv.conf

echo "[4/6] Creating the permanent boot service..."
cat << EOF | sudo tee /usr/local/bin/brgy-boot.sh
#!/bin/bash
# 1. Clear any stray IPs from WiFi just in case
ip addr flush dev $WIFI_IFACE
ip route del default dev $WIFI_IFACE 2>/dev/null

# 2. Set the static IP for the hotspot
ip addr add 10.42.0.1/24 dev $WIFI_IFACE

# 3. Enable Internet Sharing (NAT)
sysctl -w net.ipv4.ip_forward=1
iptables -t nat -A POSTROUTING -o $ETH_IFACE -j MASQUERADE
iptables -A FORWARD -i $ETH_IFACE -o $WIFI_IFACE -m state --state RELATED,ESTABLISHED -j
ACCEPT
iptables -A FORWARD -i $WIFI_IFACE -o $ETH_IFACE -j ACCEPT

# 4. Force DNS
rm -f /etc/resolv.conf
echo "nameserver 8.8.8.8" > /etc/resolv.conf
echo "nameserver 8.8.4.4" >> /etc/resolv.conf

# 5. Restart services
systemctl restart hostapd
systemctl restart dnsmasq
exit 0
EOF

sudo chmod +x /usr/local/bin/brgy-boot.sh

cat << EOF | sudo tee /etc/systemd/system/brgy-boot.service
```

IoT-Enabled Barangay Resident and Attendance Management System

```
[Unit]
Description=Barangay System Permanent Network Fix
After=network-online.target NetworkManager.service
Wants=network-online.target

[Service]
Type=oneshot
ExecStart=/usr/local/bin/brgy-boot.sh
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
EOF

echo "[5/6] Enabling the boot service so it runs on every restart..."
sudo systemctl daemon-reload
sudo systemctl enable brgy-boot.service
sudo systemctl start brgy-boot.service

echo "[6/6] Cleaning up old broken scripts..."
sudo systemctl disable brgy-network.service 2>/dev/null
sudo rm -f /etc/rc.local

echo "======"
echo " SUCCESS! The Orange Pi is now permanently fixed."
echo " The hotspot will be 100% stable and the internet"
echo " routing will survive every reboot flawlessly!"
echo "======"
```

IoT-Enabled Barangay Resident and Attendance Management System

File: appointments/models.py

```
from django.db import models
from django.conf import settings
from residents.models import Resident

class Appointment(models.Model):
    PURPOSE_CHOICES = [
        ('certification', 'Request Certification'),
        ('clearance', 'Barangay Clearance'),
        ('indigency', 'Certificate of Indigency'),
        ('meeting', 'Meeting with Officials'),
        ('complaint', 'File a Complaint/Blotter'),
        ('business', 'Business Permit'),
        ('other', 'Other Purpose'),
    ]

    STATUS_CHOICES = [
        ('pending', 'Pending Approval'),
        ('approved', 'Approved'),
        ('completed', 'Completed'),
        ('cancelled', 'Cancelled'),
        ('no_show', 'No Show'),
    ]

    resident = models.ForeignKey(Resident, on_delete=models.CASCADE,
related_name='appointments')
    # Which official handles this appointment (optional)
    official = models.ForeignKey(
        'officials.Official',
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name='appointments',
    )
    purpose = models.CharField(max_length=50, choices=PURPOSE_CHOICES)
    specification = models.TextField(blank=True, help_text="Specific details about the
request")
    appointment_date = models.DateField()
    appointment_time = models.TimeField(null=True, blank=True)
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='pending')
    remarks = models.TextField(blank=True)
    notes = models.TextField(blank=True, help_text="Internal notes from the handling
official")
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    class Meta:
        ordering = ['appointment_date', 'appointment_time']
        verbose_name = "Appointment"
        verbose_name_plural = "Appointments"

    def __str__(self):
        return f"{self.resident.full_name} - {self.get_purpose_display()} on
{self.appointment_date}"
```

IoT-Enabled Barangay Resident and Attendance Management System

File: appointments/urls.py

```
from django.urls import path
from . import views

urlpatterns = [
    path('book/', views.book_appointment, name='book_appointment'),
    path('list/', views.appointment_list, name='appointment_list'),
    path('update/<int:pk>/<str:status>/', views.appointment_status_update,
name='appointment_status_update'),
]
```

IoT-Enabled Barangay Resident and Attendance Management System

File: appointments/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.contrib import messages
from django import forms
from .models import Appointment
from residents.models import Resident

class AppointmentForm(forms.ModelForm):
    class Meta:
        model = Appointment
        fields = ['purpose', 'specification', 'appointment_date', 'appointment_time']
        widgets = {
            'appointment_date': forms.DateInput(attrs={'type': 'date'}),
            'appointment_time': forms.TimeInput(attrs={'type': 'time'}),
            'specification': forms.Textarea(attrs={'rows': 3, 'placeholder': 'Optional:
Add more details...'}),
        }

@login_required
def book_appointment(request):
    try:
        resident = request.user.profile.resident
    except AttributeError:
        messages.error(request, 'Your account is not linked to a resident record. Please
contact the secretary.')
        return redirect('dashboard')

    if not resident:
        messages.error(request, 'Your account is not linked to a resident record. Please
contact the secretary.')
        return redirect('dashboard')

    if request.method == 'POST':
        form = AppointmentForm(request.POST)
        if form.is_valid():
            appointment = form.save(commit=False)
            appointment.resident = resident
            appointment.save()
            messages.success(request, 'Appointment booked successfully! Please wait for
approval.')
            return redirect('appointment_list')
        else:
            form = AppointmentForm()

    return render(request, 'appointments/book.html', {'form': form})

@login_required
def appointment_list(request):
    try:
        role = request.user.profile.role
    except AttributeError:
        role = 'resident'

    if role in ('captain', 'secretary', 'admin', 'staff'):
        appointments = Appointment.objects.all().order_by('-created_at')
        template = 'appointments/manage.html'
    else:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    resident = request.user.profile.resident
    appointments =
Appointment.objects.filter(resident=resident).order_by('-created_at') if resident else []
    template = 'appointments/list.html'

    return render(request, template, {'appointments': appointments})

@login_required
def appointment_status_update(request, pk, status):
    if request.user.profile.role not in ('captain', 'secretary', 'admin'):
        messages.error(request, 'Unauthorized.')
        return redirect('dashboard')

    appointment = get_object_or_404(Appointment, pk=pk)
    appointment.status = status
    appointment.save()
    messages.success(request, f'Appointment marked as {status}.')
    return redirect('appointment_list')
```


IoT-Enabled Barangay Resident and Attendance Management System

File: asgi.py

```
"""
ASGI config for barangay_system project.

It exposes the ASGI callable as a module-level variable named ``application``.

For more information on this file, see
https://docs.djangoproject.com/en/6.0/howto/deployment/asgi/
"""

import os

from django.core.asgi import get_asgi_application

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'barangay_system.settings')

application = get_asgi_application()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/admin.py

```
from django.contrib import admin
from .models import AttendanceLog, FaceEncoding

@admin.register(FaceEncoding)
class FaceEncodingAdmin(admin.ModelAdmin):
    list_display = ('resident', 'enrolled_at')

@admin.register(AttendanceLog)
class AttendanceLogAdmin(admin.ModelAdmin):
    list_display = ('official', 'date', 'time_in', 'time_out', 'status', 'method')
    list_filter = ('date', 'status', 'method')
    search_fields = ('official__resident__last_name', 'official__resident__first_name')
```

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/apps.py

```
from django.apps import AppConfig

class AttendanceConfig(AppConfig):
    name = 'attendance'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/models.py

```
from django.db import models
import json

class Biometric(models.Model):
    """
    Consolidated biometric record for a resident.
    Replaces the duplicated fingerprint_template on RESIDENT/OFFICIAL
    and the formerly separate FaceEncoding table.
    """
    resident = models.OneToOneField(
        'residents.Resident',
        on_delete=models.CASCADE,
        related_name='biometric',
    )
    # Face recognition
    face_encoding_data = models.TextField(
        blank=True, null=True,
        help_text='JSON-encoded face encoding array',
    )
    face_photo = models.ImageField(upload_to='biometrics/face/', blank=True, null=True)
    # Fingerprint
    fingerprint_template = models.TextField(
        blank=True, null=True,
        help_text='Base64-encoded fingerprint template',
    )
    enrolled_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def set_face_encoding(self, encoding_array):
        self.face_encoding_data = json.dumps(
            encoding_array.tolist() if hasattr(encoding_array, 'tolist') else
            list(encoding_array)
        )

    def get_face_encoding(self):
        import numpy as np
        return np.array(json.loads(self.face_encoding_data))

    def __str__(self):
        return f"Biometric record ? {self.resident}"

    class Meta:
        verbose_name = 'Biometric'
        verbose_name_plural = 'Biometrics'

# -----
# Legacy alias so existing imports of FaceEncoding don't hard-break immediately
# TODO: remove after all call sites updated
FaceEncoding = Biometric

class ShiftConfiguration(models.Model):
    DAY_CHOICES = [
        ('mon', 'Monday'),
        ('tue', 'Tuesday'),
```

IoT-Enabled Barangay Resident and Attendance Management System

```
( 'wed', 'Wednesday'),
( 'thu', 'Thursday'),
( 'fri', 'Friday'),
( 'sat', 'Saturday'),
( 'sun', 'Sunday'),
]
day = models.CharField(max_length=3, choices=DAY_CHOICES, unique=True)
am_in = models.TimeField(default='08:00')
am_out = models.TimeField(default='12:00')
pm_in = models.TimeField(default='13:00')
pm_out = models.TimeField(default='17:00')
is_day_off = models.BooleanField(default=False)
grace_period = models.IntegerField(default=15)
updated_at = models.DateTimeField(auto_now=True)

def __str__(self):
    return f"{self.get_day_display()} Configuration"

class Meta:
    verbose_name = 'Shift Configuration'
    verbose_name_plural = 'Shift Configurations'

class AttendanceLog(models.Model):
    """Daily attendance record for officials/staff."""
    METHOD_CHOICES = [
        ('face', 'Face Recognition'),
        ('manual', 'Manual Entry'),
        ('biometric', 'Biometric/Fingerprint'),
    ]
    STATUS_CHOICES = [
        ('present', 'Present'),
        ('late', 'Late'),
        ('absent', 'Absent'),
        ('half_day', 'Half Day'),
        ('on_leave', 'On Leave'),
    ]

    official = models.ForeignKey('officials.Official', on_delete=models.CASCADE,
related_name='attendance_logs')
    date = models.DateField()

    # Legacy fields (kept for backward compatibility during transition)
    time_in = models.TimeField(null=True, blank=True)
    time_out = models.TimeField(null=True, blank=True)

    # New Morning/Afternoon fields
    am_in = models.TimeField(null=True, blank=True)
    am_out = models.TimeField(null=True, blank=True)
    pm_in = models.TimeField(null=True, blank=True)
    pm_out = models.TimeField(null=True, blank=True)

    method = models.CharField(max_length=20, choices=METHOD_CHOICES, default='manual')
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='present')
    remarks = models.TextField(blank=True)
    photo_in = models.ImageField(upload_to='attendance/photos/', blank=True, null=True)
    photo_out = models.ImageField(upload_to='attendance/photos/', blank=True, null=True)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
def __str__(self):
    return f"{self.official} - {self.date} ({self.get_status_display()})"

@property
def hours_worked(self):
    total_seconds = 0
    from datetime import datetime

    # AM Session
    if self.am_in and self.am_out:
        dt_in = datetime.combine(self.date, self.am_in)
        dt_out = datetime.combine(self.date, self.am_out)
        total_seconds += (dt_out - dt_in).total_seconds()

    # PM Session
    if self.pm_in and self.pm_out:
        dt_in = datetime.combine(self.date, self.pm_in)
        dt_out = datetime.combine(self.date, self.pm_out)
        total_seconds += (dt_out - dt_in).total_seconds()

    if total_seconds > 0:
        return round(total_seconds / 3600, 2)

    # Fallback to legacy fields if new fields are empty
    if self.time_in and self.time_out:
        dt_in = datetime.combine(self.date, self.time_in)
        dt_out = datetime.combine(self.date, self.time_out)
        return round((dt_out - dt_in).total_seconds() / 3600, 2)

    return 0

class Meta:
    ordering = ['-date', 'official']
    unique_together = ['official', 'date']
    verbose_name = 'Attendance Log'
    verbose_name_plural = 'Attendance Logs'

class SpecialDate(models.Model):
    """Custom holidays or specific non-working dates."""
    date = models.DateField(unique=True)
    is_day_off = models.BooleanField(default=True)
    description = models.CharField(max_length=100, blank=True)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return f"{self.date} - {'Day Off' if self.is_day_off else 'Working Day'}"

    class Meta:
        ordering = ['-date']
        verbose_name = 'Special Date'
        verbose_name_plural = 'Special Dates'

class WorkSchedule(models.Model):
    """Customizable work schedule per official per date."""
    SHIFT_CHOICES = [
        ('regular', 'Regular (8 AM - 5 PM)'),
        ('morning', 'Morning Shift (8 AM - 12 PM)'),
        ('afternoon', 'Afternoon Shift (1 PM - 5 PM)'),
        ('night', 'Night Shift (10 PM - 6 AM)'),
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        ('day_off', 'Day Off / Rest Day'),
        ('leave', 'On Leave'),
    ]

    official = models.ForeignKey('officials.Official', on_delete=models.CASCADE,
related_name='work_schedules')
    date = models.DateField()
    shift_type = models.CharField(max_length=20, choices=SHIFT_CHOICES, default='regular')

    # Optional override times (if a shift needs custom hours)
    start_time = models.TimeField(null=True, blank=True)
    end_time = models.TimeField(null=True, blank=True)

    notes = models.CharField(max_length=255, blank=True, help_text="Reason for leave,
extra info, etc.")

    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def __str__(self):
        return f"{self.official} - {self.date} ({self.get_shift_type_display()})"

    def save(self, *args, **kwargs):
        # Auto-set default times based on shift type if not provided
        from datetime import time
        if not self.start_time or not self.end_time:
            if self.shift_type == 'regular':
                self.start_time = time(8, 0)
                self.end_time = time(17, 0)
            elif self.shift_type == 'morning':
                self.start_time = time(8, 0)
                self.end_time = time(12, 0)
            elif self.shift_type == 'afternoon':
                self.start_time = time(13, 0)
                self.end_time = time(17, 0)
            elif self.shift_type == 'night':
                self.start_time = time(22, 0)
                self.end_time = time(6, 0)
            elif self.shift_type in ['day_off', 'leave']:
                self.start_time = None
                self.end_time = None
        super().save(*args, **kwargs)

    class Meta:
        ordering = ['date', 'official']
        unique_together = ['official', 'date']
        verbose_name = 'Work Schedule'
        verbose_name_plural = 'Work Schedules'
```


IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/templatetags/__init__.py

```
# Attendance template tags
```

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/templatetags/attendance_tags.py

```
from django import template

register = template.Library()

@register.filter
def get_item(dictionary, key):
    return dictionary.get(key)

@register.filter
def add_str(value, arg):
    return f"{value}{arg}"

@register.filter
def range_to(start, end):
    return range(start, end)

@register.filter
def month_name_short(month_num):
    import calendar
    return calendar.month_name[month_num][:3]

@register.filter
def late_duration(actual_time, shift_time):
    if not actual_time or not shift_time:
        return ""
    from datetime import datetime, date
    # Use a dummy date to ensure we only compare times
    dummy = date(2000, 1, 1)
    actual_dt = datetime.combine(dummy, actual_time)
    shift_dt = datetime.combine(dummy, shift_time)
    diff_sec = (actual_dt - shift_dt).total_seconds()

    if diff_sec <= 0:
        return ""

    if diff_sec < 60:
        return f"{int(diff_sec)} sec"
    elif diff_sec < 3600:
        return f"{int(diff_sec // 60)} mins"
    else:
        hrs = int(diff_sec // 3600)
        mins = int((diff_sec % 3600) // 60)
        if mins > 0:
            return f"{hrs} hrs {mins} mins"
        return f"{hrs} hrs"

@register.simple_tag
def get_lateness(actual_time, shift_time, grace_period=15):
    """
    Returns the lateness duration string ONLY if it exceeds the grace period.
    Otherwise returns an empty string.
    """
    if not actual_time or not shift_time:
        return ""

    from datetime import datetime, date
    dummy = date(2000, 1, 1)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
actual_dt = datetime.combine(dummy, actual_time)
shift_dt = datetime.combine(dummy, shift_time)

diff_sec = (actual_dt - shift_dt).total_seconds()

# If not late at all or within grace period, return empty
if diff_sec <= (grace_period * 60):
    return ""

# Return the duration string using the existing filter logic
return late_duration(actual_time, shift_time)

@register.filter
def is_absent_visible(log_date, shift_out_time):
    """
    Returns True if the 'Absent' indicator should be shown.
    For past dates, it's always shown.
    For today, it's only shown after the shift end time (shift_out_time).
    """
    from datetime import datetime, date
    if not log_date: return True

    today = date.today()
    if log_date < today:
        return True
    if log_date > today:
        return False

    # It's today - check if current time is past shift end
    if not shift_out_time: return True
    return datetime.now().time() > shift_out_time
```

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/tests.py

```
from django.test import TestCase
```

```
# Create your tests here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/urls.py

```
from django.urls import path
from . import views

app_name = 'attendance'

urlpatterns = [
    path('', views.attendance_dashboard, name='dashboard'),
    path('clock/', views.clock_in_out, name='clock'),
    path('enroll/', views.face_enroll, name='enroll'),
    path('dtr/', views.dtr_list, name='dtr'),
    path('dtr/<int:official_id>', views.dtr_detail, name='dtr_detail'),
    path('dtr/<int:official_id>/print/', views.dtr_print, name='dtr_print'),
    path('api/recognize/', views.api_face_recognize, name='api_recognize'),
    path('public-scan/', views.public_dtr_scan, name='public_scan'),
    path('api/biometric-verify/', views.api_biometric_verify,
name='api_biometric_verify'),
    path('biometric-attendance/', views.biometric_attendance,
name='biometric_attendance'),
    path('api/event-attendance-list/', views.api_event_attendance_list,
name='api_event_attendance_list'),
    path('api/event-attendance-pdf/', views.event_attendance_pdf,
name='event_attendance_pdf'),
    path('history/', views.attendance_history_calendar, name='history_calendar'),
    path('history/<str:date_str>', views.daily_attendance_report, name='daily_report'),
    path('api/shift-settings/', views.api_get_shift_settings,
name='api_get_shift_settings'),
    path('api/shift-settings/save/', views.api_save_shift_settings,
name='api_save_shift_settings'),
    path('api/toggle-special-date/', views.api_toggle_special_date,
name='api_toggle_special_date'),

    # Work Scheduling
    path('schedule/', views.work_schedule_view, name='work_schedule'),
    path('api/schedule/generate/', views.api_generate_schedule,
name='api_generate_schedule'),
    path('api/schedule/update/', views.api_update_schedule, name='api_update_schedule'),
    path('api/schedule/replacements/', views.api_get_replacements,
name='api_get_replacements'),
]
```

IoT-Enabled Barangay Resident and Attendance Management System

File: attendance/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.views.decorators.http import require_POST
from django.contrib import messages
from django.http import JsonResponse
from django.views.decorators.csrf import csrf_exempt
from django.core.paginator import Paginator
from django.core.cache import cache
from datetime import date, datetime, time, timedelta
from django.utils import timezone
from .models import AttendanceLog, FaceEncoding, ShiftConfiguration, SpecialDate,
WorkSchedule
from officials.models import Official
import json
from biometrics.utils import get_biometric_provider
from django.http import HttpResponse
from reportlab.pdfgen import canvas
from reportlab.lib.pagesizes import letter
from reportlab.lib import colors
from reportlab.platypus import SimpleDocTemplate, Table, TableStyle, Paragraph, Spacer
from reportlab.lib.styles import getSampleStyleSheet

@login_required
def attendance_dashboard(request):
    """Attendance overview."""
    today = timezone.localdate()
    today_logs =
AttendanceLog.objects.filter(date=today).select_related('official__resident')
    officials = Official.objects.filter(status='active').select_related('resident')

    # Get custom schedules for today
    today_schedules = WorkSchedule.objects.filter(date=today)
    sched_dict = {s.official_id: s for s in today_schedules}

    # Get shift config for threshold (fallback)
    day_names = {'mon': 0, 1: 'tue', 2: 'wed', 3: 'thu', 4: 'fri', 5: 'sat', 6: 'sun'}
    day_code = day_names[today.weekday()]
    shift_config = ShiftConfiguration.objects.filter(day=day_code).first()

    log_dict = {log.official_id: log for log in today_logs}
    combined_logs = []

    expected_count = 0
    absent_count = 0

    for official in officials:
        log = log_dict.get(official.id)
        sched = sched_dict.get(official.id)

        # Determine status based on schedule and log
        if log:
            status = log.status
            display_status = log.get_status_display()
        else:
            if sched:
                if sched.shift_type == 'leave':
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        status = 'leave'
        display_status = 'On Leave'
    elif sched.shift_type == 'day_off':
        status = 'day_off'
        display_status = 'Rest Day'
    else:
        status = 'absent'
        display_status = 'Absent'
        expected_count += 1
        absent_count += 1
    else:
        # No schedule = Unscheduled/Rest Day
        status = 'unscheduled'
        display_status = 'Not Scheduled'

    if log:
        expected_count += 1

    combined_logs.append({
        'official': official,
        'date': today,
        'am_in': log.am_in if log else None,
        'am_out': log.am_out if log else None,
        'pm_in': log.pm_in if log else None,
        'pm_out': log.pm_out if log else None,
        'status': status,
        'display_status': display_status,
        'has_log': bool(log),
        'shift_label': sched.get_shift_type_display() if sched else 'No Schedule'
    })

    # Sort combined logs: Present/Late first, then Absent, then Leave/Rest Day
    status_order = {'present': 0, 'late': 1, 'absent': 2, 'leave': 3, 'day_off': 4,
                    'unscheduled': 5}
    combined_logs.sort(key=lambda x: (status_order.get(x['status'], 6),
                                       x['official'].resident.last_name))

    context = {
        'today_logs': combined_logs,
        'total_officials': officials.count(),
        'present_count': today_logs.filter(status__in=['present', 'late']).count(),
        'late_count': today_logs.filter(status='late').count(),
        'absent_count': absent_count,
        'shift_config': shift_config
    }
    return render(request, 'attendance/dashboard.html', context)

@login_required
def clock_in_out(request):
    """Manual clock in/out page."""
    today = timezone.localtime()

    if request.method == 'POST':
        official_id = request.POST.get('official')
        action = request.POST.get('action') # 'in' or 'out'
        official = get_object_or_404(Official, pk=official_id)

        now = datetime.now().time()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# Get custom schedule
sched = WorkSchedule.objects.filter(official=official, date=today).first()

# Get shift config for threshold (fallback)
day_names = {0:'mon', 1:'tue', 2:'wed', 3:'thu', 4:'fri', 5:'sat', 6:'sun'}
day_code = day_names[today.weekday()]
shift = ShiftConfiguration.objects.filter(day=day_code).first()

# Block if no schedule assigned
if not sched:
    messages.warning(request, f'{official.resident.full_name} is not scheduled
today. No attendance recorded.')
    return redirect('attendance:clock')

# Block day_off or leave
if sched.shift_type in ['day_off', 'leave']:
    messages.info(request, f'{official.resident.full_name} is on
{sched.get_shift_type_display()} today. No attendance recorded.')
    return redirect('attendance:clock')

log, created = AttendanceLog.objects.get_or_create(
    official=official,
    date=today,
    defaults={'method': 'manual', 'status': 'present'}
)

if action == 'in':
    if not log.am_in:
        log.am_in = now
        action_label = "Morning Clock IN"
        # Check lateness
        if sched and sched.start_time:
            threshold = sched.start_time
        else:
            threshold = shift.am_in if shift else time(8, 0)

        grace = shift.grace_period if shift else 15
        threshold_dt = datetime.combine(today, threshold) +
timedelta(minutes=grace)

        if datetime.now() > threshold_dt:
            log.status = 'late'
        else:
            log.status = 'present'
    elif not log.pm_in:
        log.pm_in = now
        action_label = "Afternoon Clock IN"
    else:
        messages.warning(request, f'{official.resident.full_name} has already
clocked IN for both shifts.')
        return redirect('attendance:clock')

    log.save()
    messages.success(request, f'{official.resident.full_name} {action_label} at
{now.strftime("%I:%M %p")}')
elif action == 'out':
    if not log.am_out:
        log.am_out = now
        action_label = "Morning Clock OUT"
    elif not log.pm_out:
```


IoT-Enabled Barangay Resident and Attendance Management System

```
        log.pm_out = now
        action_label = "Afternoon Clock OUT"
    else:
        # Allow updating PM OUT
        log.pm_out = now
        action_label = "Afternoon Clock OUT (Updated)"

    log.save()
    messages.success(request, f'{official.resident.full_name} {action_label} at
{now.strftime("%I:%M %p")}')

    return redirect('attendance:clock')

officials = Official.objects.filter(status='active').select_related('resident')
today_logs = {log.official_id: log for log in
AttendanceLog.objects.filter(date=today)}

context = {
    'officials': officials,
    'today_logs': today_logs,
    'current_time': datetime.now(),
}
return render(request, 'attendance/clock.html', context)

@login_required
def face_enroll(request):
    """Enroll face for an official."""
    if request.method == 'POST':
        official_id = request.POST.get('official')
        official = get_object_or_404(Official, pk=official_id)

        if request.FILES.get('photo'):
            face_enc, created = FaceEncoding.objects.get_or_create(official=official)
            face_enc.photo = request.FILES['photo']
            # In production, face encoding would be computed here
            face_enc.encoding_data = '[]'
            face_enc.save()
            messages.success(request, f'Face enrolled for {official.resident.full_name}')
        else:
            messages.error(request, 'Please upload a photo.')

    return redirect('attendance:enroll')

officials = Official.objects.filter(status='active').select_related('resident')
enrolled = set(FaceEncoding.objects.values_list('official_id', flat=True))

context = {
    'officials': officials,
    'enrolled': enrolled,
}
return render(request, 'attendance/enroll.html', context)

@login_required
def dtr_list(request):
    """DTR summary for all officials."""
    month = request.GET.get('month', date.today().month)
    year = request.GET.get('year', date.today().year)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
officials = Official.objects.filter(status='active').select_related('resident')

dtr_summary = []
current_date = date.today()

for official in officials:
    logs = AttendanceLog.objects.filter(
        official=official,
        date__month=month,
        date__year=year,
    )

    # Calculate true absences: scheduled working days up to today with no attendance
log    scheduled_dates = WorkSchedule.objects.filter(
        official=official,
        date__month=month,
        date__year=year,
        date__lte=current_date
    ).exclude(shift_type__in=['day_off', 'leave']).values_list('date', flat=True)

    log_dates = set(logs.values_list('date', flat=True))
    absent_count = sum(1 for d in scheduled_dates if d not in log_dates)

    total_hours = sum(log.hours_worked for log in logs)
    dtr_summary.append({
        'official': official,
        'present': logs.filter(status='present').count(),
        'late': logs.filter(status='late').count(),
        'absent': absent_count,
        'total_hours': round(total_hours, 2),
        'total_days': logs.count(),
    })

context = {
    'dtr_summary': dtr_summary,
    'month': int(month),
    'year': int(year),
}
return render(request, 'attendance/dtr.html', context)

@login_required
def dtr_detail(request, official_id):
    """Detailed DTR for a specific official."""
    official = get_object_or_404(Official, pk=official_id)
    month = request.GET.get('month', date.today().month)
    year = request.GET.get('year', date.today().year)

    logs = AttendanceLog.objects.filter(
        official=official,
        date__month=month,
        date__year=year,
    ).order_by('-date')

    total_hours = sum(log.hours_worked for log in logs)

    # Get all shift configs for lookup
    shifts = ShiftConfiguration.objects.all()
    shift_map = {s.day: s for s in shifts}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
day_names = {0:'mon', 1:'tue', 2:'wed', 3:'thu', 4:'fri', 5:'sat', 6:'sun'}

# Enrich logs with their specific shift times for the template
for log in logs:
    d_code = day_names[log.date.weekday()]
    s_cfg = shift_map.get(d_code)
    if s_cfg:
        log.shift_am_in = s_cfg.am_in
        log.shift_am_out = s_cfg.am_out
        log.shift_pm_in = s_cfg.pm_in
        log.shift_pm_out = s_cfg.pm_out
        log.grace_period = s_cfg.grace_period

import calendar
context = {
    'official': official,
    'logs': logs,
    'month': int(month),
    'month_name': calendar.month_name[int(month)],
    'year': int(year),
    'total_hours': round(total_hours, 2),
}
return render(request, 'attendance/dtr_detail.html', context)

@login_required
def dtr_print(request, official_id):
    """Print-friendly DTR."""
    official = get_object_or_404(Official, pk=official_id)
    month = request.GET.get('month', date.today().month)
    year = request.GET.get('year', date.today().year)

    logs = AttendanceLog.objects.filter(
        official=official,
        date__month=month,
        date__year=year,
    ).order_by('date')

    total_hours = sum(log.hours_worked for log in logs)

    import calendar
    context = {
        'official': official,
        'logs': logs,
        'month': int(month),
        'month_name': calendar.month_name[int(month)],
        'year': int(year),
        'total_hours': round(total_hours, 2),
    }
    return render(request, 'attendance/dtr_print.html', context)

def public_dtr_scan(request):
    """Public DTR page for functionaries to scan fingerprints without logging in."""
    return render(request, 'attendance/public_scan.html')

import requests

def match_1_to_n(target_template, target_type='official'):
```

IoT-Enabled Barangay Resident and Attendance Management System

```
"""
Perform 1:N biometric matching using the configured provider.
"""
if not target_template:
    return None

provider = get_biometric_provider()

if target_type == 'official':
    candidates = Official.objects.exclude(fingerprint_template__isnull=True).exclude(fingerprint_template='')
    for cand in candidates:
        result = provider.verify(user_id=cand.id, scan_data={'template': target_template, 'stored_template': cand.fingerprint_template})
        if result.get('status') == 'success':
            return cand
    return None

@csrf_exempt
def api_biometric_verify(request):
    """
    API endpoint for biometric verification (Clock In/Out).
    """
    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'POST required'}, status=405)

    try:
        data = json.loads(request.body)
        template = data.get('template')
        official_id = data.get('official_id')
        is_auto = data.get('auto', False)

        official = None
        if is_auto:
            att_rid = request.session.get('biometric_attendance_request_id')
            if att_rid:
                att_state = cache.get(f"biometric_attendance:{att_rid}")
                if att_state and att_state.get('status') == 'authenticated':
                    oid = att_state.get('official_id')
                    if oid:
                        official = Official.objects.filter(pk=oid, status='active').first()

            if not official:
                request_id = request.session.get('biometric_request_id')
                if request_id:
                    state = cache.get(f"biometric:{request_id}")
                    if state and state.get('status') == 'authenticated':
                        matched_user_id = state.get('user_id')
                        if matched_user_id:
                            from django.contrib.auth import get_user_model
                            User = get_user_model()
                            try:
                                user = User.objects.get(id=matched_user_id)
                                official = getattr(user, 'official_profile', None)
                            except User.DoesNotExist:
                                pass

            if not official and request.user.is_authenticated:
                official = getattr(request.user, 'official_profile', None)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
elif official_id:
    official = Official.objects.filter(pk=official_id).first()
elif template:
    official = match_1_to_n(template)

if not official:
    return JsonResponse({'status': 'failed', 'message': 'Fingerprint not
recognized or session expired'})

device_time_str = data.get('device_time')
device_date_str = data.get('device_date')

is_auto = data.get('auto', False)
if is_auto and (not device_time_str or not device_date_str):
    att_rid = request.session.get('biometric_attendance_request_id') or
'global_hardware_scan'
    if att_rid:
        att_state = cache.get(f"biometric_attendance:{att_rid}")
        if att_state:
            device_time_str = device_time_str or att_state.get('device_time')
            device_date_str = device_date_str or att_state.get('device_date')

today = timezone.localdate()
now = timezone.localtime().time()

if device_time_str and device_date_str:
    try:
        parsed_dt = datetime.strptime(f"{device_date_str} {device_time_str}",
"%Y-%m-%d %H:%M:%S")
        if parsed_dt.year >= 2026:
            today = parsed_dt.date()
            now = parsed_dt.time()
    except Exception as e:
        print(f"[Biometric] Failed to parse device date/time: {e}")

requested_action = data.get('action')
if requested_action == 'attendance' or not requested_action:
    requested_action = None

log = AttendanceLog.objects.filter(official=official, date=today).first()

# Get custom schedule
sched = WorkSchedule.objects.filter(official=official, date=today).first()

# Get shift configuration for today (fallback)
day_names = {0:'mon', 1:'tue', 2:'wed', 3:'thu', 4:'fri', 5:'sat', 6:'sun'}
day_code = day_names[today.weekday()]
shift = ShiftConfiguration.objects.filter(day=day_code).first()

if sched:
    if sched.shift_type in ['day_off', 'leave']:
        return JsonResponse({'status': 'info', 'message': f'Today is a
{sched.get_shift_type_display()} for {official.resident.full_name}.'})
    elif shift and shift.is_day_off:
        return JsonResponse({'status': 'info', 'message': f'Today is a Day Off for
{official.resident.full_name}.'})

# ?? NO SCHEDULE CHECK ?? If there is no WorkSchedule at all, block the scan
if not sched:
    try:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        requests.post(f"{esp32_base_url}/error-feedback",
                      data={'reason': 'Not Scheduled'},
                      timeout=2,
                      proxies={'http': None, 'https': None})
    except Exception:
        pass
    return JsonResponse({
        'status': 'not_scheduled',
        'message': f'{official.resident.full_name} ? Not Scheduled today.'
    })
midpoint = time(12, 30) # Default midpoint
if shift:
    # mid-point between AM OUT and PM IN
    h1, m1 = shift.am_out.hour, shift.am_out.minute
    h2, m2 = shift.pm_in.hour, shift.pm_in.minute
    mid_total_mins = ((h1 * 60 + m1) + (h2 * 60 + m2)) // 2
    midpoint = time(mid_total_mins // 60, mid_total_mins % 60)

is_morning = now < midpoint
action_label = ""

from core.utils.biometric_discovery import get_esp32_base_url
esp32_base_url = get_esp32_base_url()

if not log:
    log = AttendanceLog.objects.create(official=official, date=today,
method='biometric')

target_field = None
error_msg = ""

# Determine target action and field based on time and requested action
if requested_action == 'in':
    if is_morning:
        target_field = 'am_in'
        action_label = "Morning Clock IN"
        error_msg = "Morning IN already recorded."
    else:
        target_field = 'pm_in'
        action_label = "Afternoon Clock IN"
        error_msg = "Afternoon IN already recorded."
elif requested_action == 'out':
    if is_morning:
        target_field = 'am_out'
        action_label = "Morning Clock OUT"
        error_msg = "Morning OUT already recorded."
    else:
        target_field = 'pm_out'
        action_label = "Afternoon Clock OUT"
        error_msg = "Afternoon OUT already recorded."
else:
    # Auto-detect (requested_action is None or 'attendance')
    if is_morning:
        if not log.am_in:
            target_field = 'am_in'
            action_label = "Morning Clock IN"
        elif not log.am_out:
            target_field = 'am_out'
            action_label = "Morning Clock OUT"
        else:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# Clear state
if data.get('auto'):
    att_rid = request.session.get('biometric_attendance_request_id')
    if att_rid: cache.delete(f"biometric_attendance:{att_rid}")
    try:
        requests.post(f"{esp32_base_url}/error-feedback", data={'reason':
'Already Timed Out'}, timeout=2, proxies={'http': None, 'https': None})
    except Exception:
        pass
    return JsonResponse({'status': 'failed', 'message': 'Morning shift
already completed.'})
else:
    # Afternoon
    if not log.pm_in:
        target_field = 'pm_in'
        action_label = "Afternoon Clock IN"
    elif not log.pm_out:
        target_field = 'pm_out'
        action_label = "Afternoon Clock OUT"
    else:
        # Clear state
        if data.get('auto'):
            att_rid = request.session.get('biometric_attendance_request_id')
            if att_rid: cache.delete(f"biometric_attendance:{att_rid}")
            try:
                requests.post(f"{esp32_base_url}/error-feedback", data={'reason':
'Already Timed Out'}, timeout=2, proxies={'http': None, 'https': None})
            except Exception:
                pass
            return JsonResponse({'status': 'failed', 'message': 'Afternoon shift
already completed.'})

# Check if already recorded
if getattr(log, target_field) is not None:
    if data.get('auto'):
        att_rid = request.session.get('biometric_attendance_request_id')
        if att_rid: cache.delete(f"biometric_attendance:{att_rid}")
        try:
            reason_err = 'Already Timed In' if 'IN' in action_label else 'Already
Timed Out'
            requests.post(f"{esp32_base_url}/error-feedback", data={'reason':
reason_err}, timeout=2, proxies={'http': None, 'https': None})
        except Exception:
            pass
        return JsonResponse({'status': 'failed', 'message': error_msg or
f"{action_label} already recorded."})

# Save the time
setattr(log, target_field, now)

# Handle lateness / status calculation for clock in
local_now = timezone.localtime().replace(tzinfo=None)

# Determine fallback shift config
shift_am_in = shift.am_in if shift else time(8, 0)
shift_pm_in = shift.pm_in if shift else time(13, 0)
grace = shift.grace_period if shift else 15

if target_field == 'am_in':
    threshold = sched.start_time if sched and sched.start_time else shift_am_in
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        threshold_dt = datetime.combine(today, threshold) + timedelta(minutes=grace)
        log.status = 'late' if local_now > threshold_dt else 'present'
    elif target_field == 'pm_in':
        threshold = sched.start_time if sched and sched.start_time and
        sched.shift_type in ['afternoon', 'night'] else shift_pm_in
        threshold_dt = datetime.combine(today, threshold) + timedelta(minutes=grace)
        if local_now > threshold_dt and log.status != 'late':
            log.status = 'late'

    log.save()

    # Clear active scan state
    is_auto = data.get('auto', False)
    if is_auto:
        att_rid = request.session.get('biometric_attendance_request_id')
        if att_rid:
            cache.delete(f"biometric_attendance:{att_rid}")

    return JsonResponse({
        'status': 'success',
        'message': f'{action_label} successful',
        'name': official.resident.full_name,
        'time': now.strftime("%I:%M %p"),
        'action': action_label,
        'attendance_status': log.status if 'IN' in action_label else None
    })
except Exception as e:
    return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

@csrf_exempt
def api_face_recognize(request):
    if request.method != 'POST':
        return JsonResponse({'error': 'POST required'}, status=405)
    return JsonResponse({'status': 'success', 'message': 'Face recognition placeholder',
        'recognized': False})

@login_required
def biometric_attendance(request):
    return render(request, 'attendance/biometric_attendance.html')

@login_required
def api_event_attendance_list(request):
    date_str = request.GET.get('date')
    if not date_str:
        return JsonResponse({'status': 'error', 'message': 'Date required'}, status=400)
    try:
        query_date = datetime.strptime(date_str, '%Y-%m-%d').date()
    except ValueError:
        return JsonResponse({'status': 'error', 'message': 'Invalid date format'},
            status=400)
    logs =
AttendanceLog.objects.filter(date=query_date).select_related('official__resident')
    data = [{'name': l.official.resident.full_name, 'position':
l.official.get_position_display(), 'time_in': l.time_in.strftime('%I:%M %p') if l.time_in
else '-', 'status': l.get_status_display()} for l in logs]
    return JsonResponse({'status': 'success', 'data': data})
```


IoT-Enabled Barangay Resident and Attendance Management System

```
@login_required
def event_attendance_pdf(request):
    date_str = request.GET.get('date')
    event_name = request.GET.get('event_name', 'Event Attendance')
    if not date_str: return HttpResponse('Date required', status=400)
    try: query_date = datetime.strptime(date_str, '%Y-%m-%d').date()
    except ValueError: return HttpResponse('Invalid date format', status=400)
    logs = AttendanceLog.objects.filter(date=query_date).select_related('official__resident').order_by('time_in')
    response = HttpResponse(content_type='application/pdf')
    response['Content-Disposition'] = f'attachment; filename="attendance_{date_str}.pdf"'
    doc = SimpleDocTemplate(response, pagesize=letter)
    elements = []
    styles = getSampleStyleSheet()
    elements.append(Paragraph(f"<b>BARANGAY SICO-SICO MANAGEMENT SYSTEM</b>",
    styles['Title']))
    elements.append(Paragraph(f"Attendance Report: {event_name}", styles['Heading2']))
    elements.append(Paragraph(f"Date: {query_date.strftime('%B %d, %Y')}",
    styles['Normal']))
    elements.append(Spacer(1, 12))
    data = [['Official Name', 'Position', 'Time In', 'Status']]
    for log in logs: data.append([log.official.resident.full_name,
    log.official.get_position_display(), log.time_in.strftime('%I:%M %p') if log.time_in else
    '-', log.get_status_display()])
    if len(data) == 1: data.append(['No records found', '', '', ''])
    t = Table(data, colWidths=[200, 150, 80, 80])
    t.setStyle(TableStyle([('BACKGROUND', (0, 0), (-1, 0), colors.HexColor('#10b981')),
    ('TEXTCOLOR', (0, 0), (-1, 0), colors.whitesmoke), ('ALIGN', (0, 0), (-1, -1), 'CENTER'),
    ('FONTNAME', (0, 0), (-1, 0), 'Helvetica-Bold'), ('FONTSIZE', (0, 0), (-1, 0), 12),
    ('BOTTOMPADDING', (0, 0), (-1, 0), 12), ('BACKGROUND', (0, 1), (-1, -1), colors.white),
    ('GRID', (0, 0), (-1, -1), 1, colors.grey)]))
    elements.append(t)
    elements.append(Spacer(1, 24))
    elements.append(Paragraph(f"Generated on: {datetime.now().strftime('%Y-%m-%d
%H:%M:%S')}", styles['Italic']))
    doc.build(elements)
    return response

@login_required
def attendance_history_calendar(request):
    import calendar
    from datetime import date, timedelta
    today = date.today()
    year = int(request.GET.get('year', today.year))
    month = int(request.GET.get('month', today.month))
    logs = AttendanceLog.objects.filter(date__year=year, date__month=month).values('date',
    'status')
    month_data = {}
    for d in range(1, 32):
        try:
            curr_date = date(year, month, d)
            month_data[d] = {'present': 0, 'late': 0, 'absent': 0, 'date_str':
            curr_date.strftime('%Y-%m-%d')}
        except ValueError: break
    active_officials_count = Official.objects.filter(status='active').count()

    # Build a per-day count of officials with an actual working schedule (not
    day_off/leave)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# This prevents unscheduled officials from being counted as absent
from django.db.models import Count as _Count
scheduled_counts_qs = WorkSchedule.objects.filter(
    date__year=year, date__month=month,
).exclude(shift_type__in=['day_off',
'leave']).values('date').annotate(cnt=_Count('id'))
scheduled_counts = {s['date'].day: s['cnt'] for s in scheduled_counts_qs}

for log in logs:
    d = log['date'].day
    status = log['status']
    if d in month_data:
        if status == 'late': month_data[d]['late'] += 1
        elif status == 'present': month_data[d]['present'] += 1

# Get day off settings
day_offs = ShiftConfiguration.objects.filter(is_day_off=True).values_list('day',
flat=True)
day_names_inv = {'mon':0, 'tue':1, 'wed':2, 'thu':3, 'fri':4, 'sat':5, 'sun':6}
day_off_weekdays = [day_names_inv[d] for d in day_offs if d in day_names_inv]

# Get date-specific overrides (SpecialDate)
special_dates = {s.date: s.is_day_off for s in
SpecialDate.objects.filter(date__year=year, date__month=month)}

# Calculate absent for each day that has passed or is today
for d, data in month_data.items():
    curr_date = date(year, month, d)
    data['is_future'] = curr_date > today

# Only SpecialDate overrides mark a day off
data['is_day_off'] = special_dates.get(curr_date, False)

if not data['is_future']:
    if data['is_day_off']:
        data['absent'] = 0
    else:
        # Use scheduled count for this day; fall back to all active officials
        # if no WorkSchedule data exists (e.g. dates before scheduling was
adopted)
        scheduled = scheduled_counts.get(d)
        base = scheduled if scheduled is not None else active_officials_count
        data['absent'] = max(0, base - (data['present'] + data['late']))

cal = calendar.Calendar(firstweekday=6)
month_days = cal.monthdayscalendar(year, month)
prev_month_date = date(year, month, 1) - timedelta(days=1)
next_month_date = date(year, month, 28) + timedelta(days=5)
next_month_date = next_month_date.replace(day=1)

# Dynamic year range: 5 years back and 5 years forward
year_range = range(today.year - 5, today.year + 6)

context = {
    'month_days': month_days,
    'month_name': calendar.month_name[month],
    'year': year,
    'month': month,
    'month_data': month_data,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'prev_month': prev_month_date,
'next_month': next_month_date,
'today': today,
'year_range': year_range
}
return render(request, 'attendance/history_calendar.html', context)

def get_current_week_dates():
    today = date.today()
    start_of_week = today - timedelta(days=today.weekday())
    return {
        'mon': start_of_week,
        'tue': start_of_week + timedelta(days=1),
        'wed': start_of_week + timedelta(days=2),
        'thu': start_of_week + timedelta(days=3),
        'fri': start_of_week + timedelta(days=4),
        'sat': start_of_week + timedelta(days=5),
        'sun': start_of_week + timedelta(days=6),
    }

@login_required
def api_get_shift_settings(request):
    """Fetch all shift configurations from the database."""
    configs = ShiftConfiguration.objects.all()
    week_dates = get_current_week_dates()
    data = {}
    for cfg in configs:
        target_date = week_dates[cfg.day]
        special = SpecialDate.objects.filter(date=target_date).first()
        is_day_off = special.is_day_off if special else cfg.is_day_off

        data[cfg.day] = {
            'time_in': cfg.am_in.strftime('%H:%M'),
            'am_out': cfg.am_out.strftime('%H:%M'),
            'pm_in': cfg.pm_in.strftime('%H:%M'),
            'time_out': cfg.pm_out.strftime('%H:%M'),
            'is_day_off': is_day_off,
            'grace_period': cfg.grace_period
        }
    return JsonResponse({'status': 'success', 'settings': data})

@login_required
@csrf_exempt
def api_save_shift_settings(request):
    """Save shift configurations to the database."""
    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'POST required'}, status=405)

    try:
        data = json.loads(request.body)
        settings = data.get('settings', {})
        bulk_apply = data.get('bulk_apply', False)
        week_dates = get_current_week_dates()

        if bulk_apply:
            # Apply one day's settings to all
            source_day = data.get('source_day', 'mon')
            source_cfg = settings.get(source_day)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if source_cfg:
    days = ['mon', 'tue', 'wed', 'thu', 'fri', 'sat', 'sun']
    for d in days:
        is_off = source_cfg.get('is_day_off', False) if d == source_day else
False

        ShiftConfiguration.objects.update_or_create(
            day=d,
            defaults={
                'am_in': source_cfg.get('time_in', '08:00'),
                'am_out': source_cfg.get('am_out', '12:00'),
                'pm_in': source_cfg.get('pm_in', '13:00'),
                'pm_out': source_cfg.get('time_out', '17:00'),
                'grace_period': source_cfg.get('grace_period', 15)
            }
        )
        # Apply day off ONLY to current week
        SpecialDate.objects.update_or_create(
            date=week_dates[d],
            defaults={'is_day_off': is_off, 'description': 'Modal Override'}
        )
    else:
        # Save individual day settings
        for day, cfg in settings.items():
            ShiftConfiguration.objects.update_or_create(
                day=day,
                defaults={
                    'am_in': cfg.get('time_in', '08:00'),
                    'am_out': cfg.get('am_out', '12:00'),
                    'pm_in': cfg.get('pm_in', '13:00'),
                    'pm_out': cfg.get('time_out', '17:00'),
                    'grace_period': cfg.get('grace_period', 15)
                }
            )
            # Apply day off ONLY to current week
            SpecialDate.objects.update_or_create(
                date=week_dates[day],
                defaults={'is_day_off': cfg.get('is_day_off', False), 'description':
'Modal Override'}
            )

        return JsonResponse({'status': 'success', 'message': 'Shift settings saved
successfully for the current week'})
    except Exception as e:
        return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

@login_required
def daily_attendance_report(request, date_str):
    try: query_date = datetime.strptime(date_str, '%Y-%m-%d').date()
    except ValueError: return redirect('attendance:history_calendar')
    logs = AttendanceLog.objects.filter(date=query_date).select_related('official__residen
t').order_by('official__resident__last_name')
    officials = Official.objects.filter(status='active')
    log_dict = {log.official_id: log for log in logs}
    today = datetime.now().date()
    is_future = query_date > today

    # Get shift config for threshold
    day_names = {0:'mon', 1:'tue', 2:'wed', 3:'thu', 4:'fri', 5:'sat', 6:'sun'}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
day_code = day_names[query_date.weekday()]
shift_config = ShiftConfiguration.objects.filter(day=day_code).first()

# Check for date-specific override (SpecialDate)
special_date = SpecialDate.objects.filter(date=query_date).first()
if special_date:
    is_day_off = special_date.is_day_off
else:
    is_day_off = shift_config.is_day_off if shift_config else False

# Get WorkSchedule for each official on that date
work_schedules = WorkSchedule.objects.filter(date=query_date)
sched_dict = {s.official_id: s for s in work_schedules}

full_report = []
stats = {'present': 0, 'late': 0, 'absent': 0}

if not is_future:
    for official in officials:
        log = log_dict.get(official.id)
        sched = sched_dict.get(official.id)

        if log:
            status = log.status
        elif sched:
            if sched.shift_type == 'leave':
                status = 'leave'
            elif sched.shift_type == 'day_off':
                status = 'day_off'
            else:
                status = 'absent'
        else:
            # No schedule = Not scheduled, do NOT count as absent
            status = 'unscheduled'

        full_report.append({
            'official': official,
            'log': log,
            'status': status,
            'shift_label': sched.get_shift_type_display() if sched else None,
        })

    # Sort: present/late first, then absent, then leave/day_off/unscheduled
    status_order = {'present': 0, 'late': 1, 'absent': 2, 'leave': 3, 'day_off': 4,
                    'unscheduled': 5}
    full_report.sort(key=lambda x: (status_order.get(x['status'], 6),
                                    x['official'].resident.last_name))

    stats = {
        'present': sum(1 for r in full_report if r['status'] in ['present', 'late']),
        'late': sum(1 for r in full_report if r['status'] == 'late'),
        'absent': sum(1 for r in full_report if r['status'] == 'absent'),
    }

context = {
    'date': query_date,
    'report': full_report,
    'stats': stats,
    'today': today,
    'is_future': is_future,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'shift_config': shift_config,
'is_day_off': is_day_off
}
return render(request, 'attendance/daily_report.html', context)

@login_required
@require_POST
def api_toggle_special_date(request):
    """API to toggle a date as a special day off."""
    date_str = request.POST.get('date')
    if not date_str:
        return JsonResponse({'status': 'error', 'message': 'Date required'}, status=400)

    try:
        target_date = datetime.strptime(date_str, '%Y-%m-%d').date()
        special_date, created = SpecialDate.objects.get_or_create(date=target_date)

        # If it was already there, toggle it. If new, default is Day Off (True)
        if not created:
            special_date.is_day_off = not special_date.is_day_off
            special_date.save()

        return JsonResponse({
            'status': 'success',
            'is_day_off': special_date.is_day_off,
            'message': f"Date marked as {'Day Off' if special_date.is_day_off else
'Working Day'}"
        })
    except Exception as e:
        return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

# =====
# WORK SCHEDULING UI
# =====

from .models import WorkSchedule

@login_required
def work_schedule_view(request):
    """View and manage weekly work schedules."""
    import calendar
    today = timezone.localdate()

    # Check if user requested a specific week
    date_str = request.GET.get('date')
    if date_str:
        try:
            target_date = datetime.strptime(date_str, '%Y-%m-%d').date()
        except ValueError:
            target_date = today
    else:
        target_date = today

    # Get the start of the week (Monday)
    start_of_week = target_date - timedelta(days=target_date.weekday())

    week_dates = []
```

IoT-Enabled Barangay Resident and Attendance Management System

```
for i in range(7):
    current_date = start_of_week + timedelta(days=i)
    week_dates.append({
        'date': current_date,
        'day_name': calendar.day_name[current_date.weekday()],
        'date_str': current_date.strftime('%Y-%m-%d'),
        'is_today': current_date == today
    })

officials =
Official.objects.filter(status='active').select_related('resident').order_by('position',
'resident__last_name')

# Get schedules for this week
end_of_week = start_of_week + timedelta(days=6)
schedules = WorkSchedule.objects.filter(
    date__range=[start_of_week, end_of_week]
)

schedule_dict = {}
for sched in schedules:
    if sched.official_id not in schedule_dict:
        schedule_dict[sched.official_id] = {}
    schedule_dict[sched.official_id][sched.date.strftime('%Y-%m-%d')] = {
        'shift_type': sched.shift_type,
        'shift_label': sched.get_shift_type_display(),
        'notes': sched.notes
    }

# Group officials by category for better UI
categories = {
    'Council & Exec': ['captain', 'kagawad', 'sk_chairman', 'sk_kagawad'],
    'Admin & Finance': ['secretary', 'treasurer', 'clerk', 'staff'],
    'Health & Safety': ['tanod', 'bhw', 'nutrition_scholar', 'day_care_worker',
'lupon']
}

grouped_officials = {cat: [] for cat in categories}
grouped_officials['Others'] = []

for official in officials:
    found = False
    for cat, pos_list in categories.items():
        if official.position in pos_list:
            grouped_officials[cat].append(official)
            found = True
            break
    if not found:
        grouped_officials['Others'].append(official)

# Remove empty groups
grouped_officials = {k: v for k, v in grouped_officials.items() if v}

context = {
    'week_dates': week_dates,
    'grouped_officials': grouped_officials,
    'schedule_dict': schedule_dict,
    'start_of_week': start_of_week,
    'end_of_week': end_of_week,
    'prev_week': (start_of_week - timedelta(days=7)).strftime('%Y-%m-%d'),
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'next_week': (start_of_week + timedelta(days=7)).strftime('%Y-%m-%d'),
'shift_choices': WorkSchedule.SHIFT_CHOICES
}
return render(request, 'attendance/schedule.html', context)

@login_required
@require_POST
def api_generate_schedule(request):
    """Auto-generate default schedules for the specified week."""
    import json
    data = json.loads(request.body)
    start_date_str = data.get('start_date')

    if not start_date_str:
        return JsonResponse({'status': 'error', 'message': 'Start date required'},
            status=400)

    start_date = datetime.strptime(start_date_str, '%Y-%m-%d').date()
    officials = Official.objects.filter(status='active')

    generated_count = 0
    # Monday to Friday (5 days)
    for i in range(5):
        current_date = start_date + timedelta(days=i)
        for official in officials:
            # Create if doesn't exist
            _, created = WorkSchedule.objects.get_or_create(
                official=official,
                date=current_date,
                defaults={'shift_type': 'regular'})
            if created:
                generated_count += 1

    # Sat/Sun as day off
    for i in [5, 6]:
        current_date = start_date + timedelta(days=i)
        for official in officials:
            _, created = WorkSchedule.objects.get_or_create(
                official=official,
                date=current_date,
                defaults={'shift_type': 'day_off'})
            if created:
                generated_count += 1

    return JsonResponse({
        'status': 'success',
        'message': f'Auto-generated {generated_count} new schedule entries.'
    })

@login_required
@require_POST
def api_update_schedule(request):
    """Update a specific shift assignment."""
    import json
    data = json.loads(request.body)
    official_id = data.get('official_id')
    date_str = data.get('date')
    shift_type = data.get('shift_type')
```


IoT-Enabled Barangay Resident and Attendance Management System

```
notes = data.get('notes', '')

if not all([official_id, date_str, shift_type]):
    return JsonResponse({'status': 'error', 'message': 'Missing data'}, status=400)

try:
    official = Official.objects.get(pk=official_id)
    target_date = datetime.strptime(date_str, '%Y-%m-%d').date()

    schedule, created = WorkSchedule.objects.update_or_create(
        official=official,
        date=target_date,
        defaults={
            'shift_type': shift_type,
            'notes': notes
        }
    )

    return JsonResponse({'status': 'success', 'message': 'Schedule updated successfully'})
except Exception as e:
    return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

@login_required
def api_get_replacements(request):
    """Get suggested replacements for a given official on a given date."""
    official_id = request.GET.get('official_id')
    date_str = request.GET.get('date')

    if not all([official_id, date_str]):
        return JsonResponse({'status': 'error', 'message': 'Missing parameters'},
            status=400)

    try:
        official = Official.objects.get(pk=official_id)
        target_date = datetime.strptime(date_str, '%Y-%m-%d').date()

        # Find active officials with the SAME position
        peers = Official.objects.filter(
            position=official.position,
            status='active'
        ).exclude(pk=official_id)

        # See what these peers are doing on that date
        peer_schedules = WorkSchedule.objects.filter(
            official__in=peers,
            date=target_date
        )
        peer_schedule_dict = {s.official_id: s.shift_type for s in peer_schedules}

        working_shifts = ['regular', 'morning', 'afternoon', 'night']

        replacements = []
        for peer in peers:
            shift = peer_schedule_dict.get(peer.id)
            if shift not in working_shifts:
                # They are either unassigned, on leave, or have a day off
                if shift == 'leave':
                    status_desc = 'Also on Leave'
                elif shift == 'day_off':
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        status_desc = 'Rest Day'
    else:
        status_desc = 'No Schedule'

    replacements.append({
        'id': peer.id,
        'name': peer.resident.full_name,
        'status': status_desc
    })

    return JsonResponse({'status': 'success', 'replacements': replacements})
except Exception as e:
    return JsonResponse({'status': 'error', 'message': str(e)}, status=500)
```

IoT-Enabled Barangay Resident and Attendance Management System

File: barangay_project/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: barangay_project/asgi.py

```
"""
ASGI config for barangay_project project.

It exposes the ASGI callable as a module-level variable named ``application``.

For more information on this file, see
https://docs.djangoproject.com/en/6.0/howto/deployment/asgi/
"""

import os

from django.core.asgi import get_asgi_application

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'barangay_project.settings')

application = get_asgi_application()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: barangay_project/settings.py

```
import os
from pathlib import Path

BASE_DIR = Path(__file__).resolve().parent.parent

SECRET_KEY = "***REDACTED***"

DEBUG = True

ALLOWED_HOSTS = ["*", "localhost", "127.0.0.1", "barangay.local", "brgysicosico.local"]

# Local ESP32 fingerprint module proxy target (Replace with your ESP32's actual IP if
# .local fails)
ESP32_BASE_URL = os.environ.get("ESP32_BASE_URL", "http://192.168.1.50")
# Ceiling for slot IDs; effective slots = min(this, sensor-reported capacity from ESP32
# /status).
# If the scanner is unreachable, enrollment uses min(this, 300) for safety on common R307
# modules.
FINGERPRINT_SENSOR_MAX_SLOTS = int(os.environ.get("FINGERPRINT_SENSOR_MAX_SLOTS", "1000"))

CSRF_TRUSTED_ORIGINS = [
    "http://127.0.0.1:8001",
    "http://localhost:8001",
    "http://barangay.local",
    "http://brgysicosico.local",
    "http://brgysicosico.local:8001",
    "http://brgysicosico.local:8001",
    "http://brgysicosico:8001",
    "https://*.trycloudflare.com",
    "https://*.ts.net",
]

INSTALLED_APPS = [
    "django.contrib.admin",
    "django.contrib.auth",
    "django.contrib.contenttypes",
    "django.contrib.sessions",
    "django.contrib.messages",
    "django.contrib.staticfiles",
    "axes",
    # Barangay apps
    "core",
    "residents",
    "certifications",
    "attendance",
    "census",
    "ordinances",
    "officials",
    "reports",
    "philsys",
    "appointments",
    "payments",
    "biometrics",
    "documents",
]

MIDDLEWARE = [
```

IoT-Enabled Barangay Resident and Attendance Management System

```
"django.middleware.security.SecurityMiddleware",
"django.contrib.sessions.middleware.SessionMiddleware",
"django.middleware.common.CommonMiddleware",
"django.middleware.csrf.CsrfViewMiddleware",
"django.contrib.auth.middleware.AuthenticationMiddleware",
"django.contrib.messages.middleware.MessageMiddleware",
"django.middleware.clickjacking.XFrameOptionsMiddleware",
"axes.middleware.AxesMiddleware",
"core.middleware.AutoLogoutMiddleware",
]

X_FRAME_OPTIONS = 'SAMEORIGIN'

ROOT_URLCONF = "barangay_project.urls"

TEMPLATES = [
    {
        "BACKEND": "django.template.backends.django.DjangoTemplates",
        "DIRS": [BASE_DIR / "templates"],
        "APP_DIRS": True,
        "OPTIONS": {
            "context_processors": [
                "django.template.context_processors.debug",
                "django.template.context_processors.request",
                "django.contrib.auth.context_processors.auth",
                "django.contrib.messages.context_processors.messages",
                "core.context_processors.user_role_context",
            ],
        },
    },
]

WSGI_APPLICATION = "barangay_project.wsgi.application"

# Use PostgreSQL if DB_TYPE is set to 'postgres' or if DB_HOST is provided, otherwise
# fallback to SQLite
# Using Orange Pi Zero 3 (Hostname: brgysicosico.local) as the Database Server
DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.postgresql",
        "NAME": "brgy_db",
        "USER": "brgy_user",
        "PASSWORD": "****REDACTED****",
        "HOST": "127.0.0.1",
        "PORT": "5432",
    }
}

AUTH_PASSWORD_VALIDATORS = [
    {
        "NAME": "django.contrib.auth.password_validation.UserAttributeSimilarityValidator"
    },
    {"NAME": "django.contrib.auth.password_validation.MinimumLengthValidator"},
    {"NAME": "django.contrib.auth.password_validation.CommonPasswordValidator"},
    {"NAME": "django.contrib.auth.password_validation.NumericPasswordValidator"},
]

LANGUAGE_CODE = "en-us"
TIME_ZONE = "Asia/Manila"
USE_I18N = True
```

IoT-Enabled Barangay Resident and Attendance Management System

```
USE_TZ = True

STATIC_URL = "/static/"
STATICFILES_DIRS = [BASE_DIR / "static"]
STATIC_ROOT = BASE_DIR / "staticfiles"

MEDIA_URL = "/media/"
MEDIA_ROOT = BASE_DIR / "media"

DEFAULT_AUTO_FIELD = "django.db.models.BigAutoField"

LOGIN_URL = "/login/"
LOGIN_REDIRECT_URL = "/"
LOGOUT_REDIRECT_URL = "/login/"

AUTH_USER_MODEL = 'core.User'

AUTHENTICATION_BACKENDS = [
    'axes.backends.AxesBackend',
    'django.contrib.auth.backends.ModelBackend',
]

# Security configuration
SESSION_COOKIE_AGE = 1800 # 30 minutes
SESSION_COOKIE_HTTPONLY = True
SESSION_COOKIE_SECURE = False # Set to True in production over HTTPS
CSRF_COOKIE_SECURE = False # Set to True in production over HTTPS
SESSION_EXPIRE_AT_BROWSER_CLOSE = True

# Axes Configuration
AXES_FAILURE_LIMIT = 5
AXES_COOLOFF_TIME = 1
AXES_LOCK_OUT_AT_FAILURE = True
AXES_RESET_ON_SUCCESS = True

# Barangay Settings
BARANGAY_NAME = "Barangay Sico-Sico"
BARANGAY_MUNICIPALITY = "Gigaquit"
BARANGAY_PROVINCE = "Surigao del Norte"
BARANGAY_REGION = "XIII (Caraga)"
BARANGAY_CAPTAIN = "HON. MARITES R. MANONGAS"

# Biometric Settings
BIOMETRIC_PROVIDER = 'biometrics.stub.StubBiometricProvider'
ATTENDANCE_START_HOUR = 7
ATTENDANCE_END_HOUR = 18
LATE_THRESHOLD_MINUTES = 30

# Pagination
ITEMS_PER_PAGE = 25

# Use SQLite for testing to bypass Postgres DB creation permission errors
import sys
if 'test' in sys.argv or 'test_coverage' in sys.argv:
    DATABASES = {
        "default": {
            "ENGINE": "django.db.backends.sqlite3",
            "NAME": ":memory:",
        }
    }
AUTHENTICATION_BACKENDS = [
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'django.contrib.auth.backends.ModelBackend',  
]
```


IoT-Enabled Barangay Resident and Attendance Management System

File: barangay_project/urls.py

```
from django.contrib import admin
from django.contrib import admin
from django.urls import path, include
from django.conf import settings
from django.conf.urls.static import static
from django.views.decorators.csrf import csrf_exempt
from core import views as core_views

urlpatterns = [
    path('admin/', admin.site.urls),
    path('login/', core_views.login_view, name='login'),
    path('logout/', core_views.logout_view, name='logout'),
    path('signup/', core_views.signup_view, name='signup'),
    path('resident-signup/', core_views.resident_signup_view, name='resident_signup'),
    path('registration-success/<str:ref>/', core_views.registration_success_view,
name='registration_success'),
    path('dashboard/', core_views.dashboard, name='dashboard'),
    path('biometric-login-start/', csrf_exempt(core_views.biometric_login_start),
name='biometric_login_start'),
    path('biometric-status-check/', csrf_exempt(core_views.biometric_status_check),
name='biometric_status_check'),

    path('', include('core.urls')),
    path('residents/', include('residents.urls')),
    path('certifications/', include('certifications.urls')),
    path('attendance/', include('attendance.urls')),
    path('census/', include('census.urls')),
    path('ordinances/', include('ordinances.urls')),
    path('officials/', include('officials.urls')),
    path('reports/', include('reports.urls')),
    path('appointments/', include('appointments.urls')),
    path('payments/', include('payments.urls')),
    path('documents/', include('documents.urls')),
]

if settings.DEBUG:
    urlpatterns += static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT)
```

IoT-Enabled Barangay Resident and Attendance Management System

File: barangay_project/wsgi.py

```
"""
WSGI config for barangay_project project.

It exposes the WSGI callable as a module-level variable named ``application``.

For more information on this file, see
https://docs.djangoproject.com/en/6.0/howto/deployment/wsgi/
"""

import os

from django.core.wsgi import get_wsgi_application

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'barangay_project.settings')

application = get_wsgi_application()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: biometrics/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: biometrics/base.py

```
from abc import ABC, abstractmethod

class BiometricProvider(ABC):
    """Abstract base class for biometric providers."""

    @abstractmethod
    def enroll(self, user_id, template_data):
        """Enroll a new biometric template for a user."""
        pass

    @abstractmethod
    def verify(self, user_id, scan_data):
        """Verify a user against a provided scan."""
        pass

    @abstractmethod
    def delete_template(self, user_id):
        """Delete a biometric template for a user."""
        pass
```

IoT-Enabled Barangay Resident and Attendance Management System

File: biometrics/stub.py

```
from .base import BiometricProvider

class StubBiometricProvider(BiometricProvider):
    """Stub implementation of a biometric provider for no-op responses."""

    def enroll(self, user_id, template_data):
        return {"status": "success", "message": "Stub: Enrolled template for user {}".format(user_id)}

    def verify(self, user_id, scan_data):
        return {"status": "success", "message": "Stub: Verified user {}".format(user_id)}

    def delete_template(self, user_id):
        return {"status": "success", "message": "Stub: Deleted template for user {}".format(user_id)}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: biometrics/utils.py

```
from django.conf import settings
from django.utils.module_loading import import_string

def get_biometric_provider():
    """Factory function to load the configured biometric provider."""
    provider_path = getattr(settings, 'BIOMETRIC_PROVIDER',
'biometrics.stub.StubBiometricProvider')
    provider_class = import_string(provider_path)
    return provider_class()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: census/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: census/admin.py

```
from django.contrib import admin
```

```
# Register your models here.
```


IoT-Enabled Barangay Resident and Attendance Management System

File: census/apps.py

```
from django.apps import AppConfig
```

```
class CensusConfig(AppConfig):  
    name = 'census'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: census/models.py

```
from django.db import models
```

```
# Create your models here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: census/tests.py

```
from django.test import TestCase
```

```
# Create your tests here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: census/urls.py

```
from django.urls import path
from . import views

app_name = 'census'

urlpatterns = [
    path('', views.census_dashboard, name='dashboard'),
    path('age-groups/', views.age_groups, name='age_groups'),
    path('demographics/', views.demographics, name='demographics'),
]
```

IoT-Enabled Barangay Resident and Attendance Management System

File: census/views.py

```
from django.shortcuts import render
from django.contrib.auth.decorators import login_required
from django.db.models import Count, Q
from residents.models import Resident, Household
from datetime import date

@login_required
def census_dashboard(request):
    """Census overview with population stats."""
    residents = Resident.objects.filter(is_active=True)
    total = residents.count()

    total_households = Household.objects.count()
    avg_household_size = round(total / total_households, 1) if total_households > 0 else 0

    # Age groups
    today = date.today()
    age_groups = {
        'Infant (0-1)': 0,
        'Early Childhood (2-5)': 0,
        'Child (6-12)': 0,
        'Teenager (13-17)': 0,
        'Young Adult (18-24)': 0,
        'Adult (25-59)': 0,
        'Senior Citizen (60+)': 0,
    }

    for r in residents:
        ag = r.age_group
        if ag == 'Infant':
            age_groups['Infant (0-1)'] += 1
        elif ag == 'Early Childhood':
            age_groups['Early Childhood (2-5)'] += 1
        elif ag == 'Child':
            age_groups['Child (6-12)'] += 1
        elif ag == 'Teenager':
            age_groups['Teenager (13-17)'] += 1
        elif ag == 'Young Adult':
            age_groups['Young Adult (18-24)'] += 1
        elif ag == 'Adult':
            age_groups['Adult (25-59)'] += 1
        elif ag == 'Senior Citizen':
            age_groups['Senior Citizen (60+)'] += 1

    # Gender
    male_count = residents.filter(gender='M').count()
    female_count = residents.filter(gender='F').count()

    # Civil status
    civil_stats = residents.values('civil_status').annotate(count=Count('id'))

    # Purok
    purok_stats = residents.values('purok').annotate(count=Count('id')).order_by('purok')

    # Voter
    voters = residents.filter(is_registered_voter=True).count()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
pwd = residents.filter(is_pwd=True).count()
senior = residents.filter(is_senior_citizen=True).count()
fourps = residents.filter(is_4ps_member=True).count()

context = {
    'total_population': total,
    'total_households': total_households,
    'avg_household_size': avg_household_size,
    'age_groups': age_groups,
    'male_count': male_count,
    'female_count': female_count,
    'civil_stats': {cs['civil_status']: cs['count'] for cs in civil_stats},
    'purok_stats': list(purok_stats),
    'voters': voters,
    'pwd': pwd,
    'senior': senior,
    'fourps': fourps,
    'age_group_labels': list(age_groups.keys()),
    'age_group_values': list(age_groups.values()),
}
return render(request, 'census/dashboard.html', context)

@login_required
def age_groups(request):
    """Detailed age group breakdown."""
    residents = Resident.objects.filter(is_active=True)

    groups = {}
    for r in residents:
        ag = r.age_group
        if ag not in groups:
            groups[ag] = {'male': 0, 'female': 0, 'total': 0, 'residents': []}
        groups[ag]['total'] += 1
        if r.gender == 'M':
            groups[ag]['male'] += 1
        else:
            groups[ag]['female'] += 1
        groups[ag]['residents'].append(r)

    ordered_groups = ['Infant', 'Early Childhood', 'Child', 'Teenager', 'Young Adult',
'Adult', 'Senior Citizen']
    ordered_data = []
    for g in ordered_groups:
        if g in groups:
            ordered_data.append({'name': g, **groups[g]})

    return render(request, 'census/age_groups.html', {'age_groups': ordered_data})

@login_required
def demographics(request):
    """Demographics data with charts."""
    residents = Resident.objects.filter(is_active=True)

    # By purok
    purok_data = residents.values('purok').annotate(
        total=Count('id'),
        male=Count('id', filter=Q(gender='M')),
        female=Count('id', filter=Q(gender='F')),
    )
```

IoT-Enabled Barangay Resident and Attendance Management System

```
).order_by('purok')

# By nationality
nationality_data = residents.values('nationality').annotate(count=Count('id'))

# By religion
religion_data =
residents.values('religion').annotate(count=Count('id')).order_by('-count')[:10]

# By occupation
occupation_data =
residents.values('occupation').annotate(count=Count('id')).order_by('-count')[:10]

context = {
    'purok_data': list(purok_data),
    'nationality_data': list(nationality_data),
    'religion_data': list(religion_data),
    'occupation_data': list(occupation_data),
}
return render(request, 'census/demographics.html', context)
```

IoT-Enabled Barangay Resident and Attendance Management System

File: certifications/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: certifications/admin.py

```
from django.contrib import admin
from .models import Certificate

@admin.register(Certificate)
class CertificateAdmin(admin.ModelAdmin):
    list_display = ('control_number', 'resident', 'get_cert_type', 'date_issued',
'amount_paid', 'status')
    list_filter = ('status', 'date_issued')
    search_fields = ('control_number', 'resident__last_name', 'resident__first_name')
    readonly_fields = ('date_issued', 'created_at')

@admin.display(description='Certificate Type')
def get_cert_type(self, obj):
    return obj.get_cert_type_display() or '?'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: certifications/apps.py

```
from django.apps import AppConfig

class CertificationsConfig(AppConfig):
    name = 'certifications'
```

File: certifications/models.py

```
from django.db import models
from django.conf import settings
from decimal import Decimal

def generate_tracking_code():
    """Generate a random 9-character alphanumeric tracking code."""
    import string
    import random
    chars = string.ascii_uppercase + string.digits
    return "".join(random.choices(chars, k=9))

class Certificate(models.Model):
    """
    Certificate / clearance issued by the barangay.

    cert_type, purpose, and status are no longer stored here directly ?
    they are read from the linked CertificateRequest to avoid duplication.
    Access via: certificate.request.cert_type / .purpose / .status
    """
    TYPE_CHOICES = [
        ('clearance', 'Barangay Clearance'),
        ('residency', 'Certificate of Residency'),
        ('indigency', 'Certificate of Indigency'),
        ('good_moral', 'Good Moral Certificate'),
        ('business_permit', 'Business Permit/Clearance'),
        ('comelec', 'COMELEC Certificate'),
        ('cedula', 'Community Tax Certificate'),
        ('late_registration', 'Certificate of Late Registration'),
    ]

    STATUS_CHOICES = [
        ('pending', 'Pending'),
        ('issued', 'Issued'),
        ('cancelled', 'Cancelled'),
    ]

    # Links
    resident = models.ForeignKey(
        'residents.Resident',
        on_delete=models.CASCADE,
        related_name='certificates',
    )
    # FK back to the request that originated this certificate
    certificate_request = models.OneToOneField(
        'CertificateRequest',
        on_delete=models.SET_NULL,
        null=True, blank=True,
        related_name='issued_certificate',
    )

    control_number = models.CharField(max_length=50, unique=True)
    date_issued = models.DateField(auto_now_add=True)

    # For business permit
    business_name = models.CharField(max_length=200, blank=True)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
business_address = models.TextField(blank=True)
business_type = models.CharField(max_length=200, blank=True)

# For late registration of birth
child_name = models.CharField(max_length=200, blank=True)
child_birth_date = models.DateField(null=True, blank=True)
child_birth_place = models.CharField(max_length=200, blank=True)
father_name = models.CharField(max_length=200, blank=True)
mother_name = models.CharField(max_length=200, blank=True)

# Status (kept here so a certificate can be cancelled independently of the request)
status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='issued')
issued_by = models.ForeignKey(
    settings.AUTH_USER_MODEL,
    on_delete=models.SET_NULL,
    null=True,
    related_name='certificates_issued',
)

# Document
pdf_file = models.FileField(upload_to='certificates/', blank=True, null=True)

created_at = models.DateTimeField(auto_now_add=True)
updated_at = models.DateTimeField(auto_now=True)

# -----
# Delegated properties ? read from linked request to avoid duplication
# -----
@property
def cert_type(self):
    return self.certificate_request.cert_type if self.certificate_request else None

@property
def purpose(self):
    return self.certificate_request.purpose if self.certificate_request else ''

@property
def or_number(self):
    """
    OR number lives exclusively in OfficialReceipt.
    Traverse: certificate ? request ? payment ? official_receipt.
    """
    try:
        return self.certificate_request.payment.official_receipt.or_number
    except Exception:
        return None

@property
def amount_paid(self):
    """Payment amount via the linked payment record."""
    try:
        return self.certificate_request.payment.amount
    except Exception:
        return 0

def get_cert_type_display(self):
    if self.cert_type:
        mapping = dict(self.TYPE_CHOICES)
        return mapping.get(self.cert_type, self.cert_type)
    return ''
```

IoT-Enabled Barangay Resident and Attendance Management System

```
def __str__(self):
    return f"{self.get_cert_type_display()} - {self.control_number}"

class Meta:
    ordering = ['-date_issued', '-id']
    verbose_name = 'Certificate'
    verbose_name_plural = 'Certificates'

class Cedula(models.Model):
    """Specific details for Community Tax Certificate (Cedula)."""
    TAXPAYER_TYPES = [
        ('individual', 'Individual'),
        ('corporation', 'Corporation'),
    ]

    certificate = models.OneToOneField(Certificate, on_delete=models.CASCADE,
related_name='cedula_details')
    ctc_number = models.CharField(max_length=50, unique=True, verbose_name='CTC Number')
    taxpayer_type = models.CharField(max_length=20, choices=TAXPAYER_TYPES,
default='individual')

    # Personal Info (matching official form)
    place_of_issue = models.CharField(max_length=200)
    height = models.CharField(max_length=20, blank=True, help_text="e.g. 170 cm")
    weight = models.CharField(max_length=20, blank=True, help_text="e.g. 65 kg")

    # Financial Info - Raw Taxable Amounts
    raw_taxable_property = models.DecimalField(max_digits=15, decimal_places=2,
default=0.00, verbose_name='Taxable Property Value')
    raw_taxable_business = models.DecimalField(max_digits=15, decimal_places=2,
default=0.00, verbose_name='Taxable Gross Receipts')
    raw_taxable_income = models.DecimalField(max_digits=15, decimal_places=2,
default=0.00, verbose_name='Taxable Annual Income')

    # Calculated Tax Amounts
    basic_tax = models.DecimalField(max_digits=10, decimal_places=2, default=5.00)
    additional_tax_property = models.DecimalField(max_digits=10, decimal_places=2,
default=0.00, verbose_name='Additional Tax (Property)')
    additional_tax_business = models.DecimalField(max_digits=10, decimal_places=2,
default=0.00, verbose_name='Additional Tax (Business)')
    additional_tax_income = models.DecimalField(max_digits=10, decimal_places=2,
default=0.00, verbose_name='Additional Tax (Income)')

    interest = models.DecimalField(max_digits=10, decimal_places=2, default=0.00)

    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    @property
    def total_additional_tax(self):
        def to_d(val):
            if val is None: return Decimal('0.00')
            try:
                return Decimal(str(val))
            except:
                return Decimal('0.00')

        prop = to_d(self.additional_tax_property)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
bus = to_d(self.additional_tax_business)
inc = to_d(self.additional_tax_income)
return prop + bus + inc

@property
def total_amount(self):
    def to_d(val):
        if val is None: return Decimal('0.00')
        try:
            return Decimal(str(val))
        except:
            return Decimal('0.00')

    basic = to_d(self.basic_tax)
    interest = to_d(self.interest)
    add_tax = self.total_additional_tax

    if self.taxpayer_type == 'individual':
        cap = Decimal('5000.00')
    else:
        cap = Decimal('10000.00')

    capped_additional_tax = add_tax
    if add_tax > cap:
        capped_additional_tax = cap

    return basic + capped_additional_tax + interest

def save(self, *args, **kwargs):
    if not self.basic_tax or self.basic_tax == 0:
        self.basic_tax = Decimal('5.00') if self.taxpayer_type == 'individual' else
Decimal('500.00')

    if self.taxpayer_type == 'individual':
        if not self.additional_tax_property and self.raw_taxable_property:
            self.additional_tax_property = (self.raw_taxable_property /
1000).quantize(Decimal('0.01'))
        if not self.additional_tax_business and self.raw_taxable_business:
            self.additional_tax_business = (self.raw_taxable_business /
1000).quantize(Decimal('0.01'))
        if not self.additional_tax_income and self.raw_taxable_income:
            self.additional_tax_income = (self.raw_taxable_income /
1000).quantize(Decimal('0.01'))
    else:
        if not self.additional_tax_property and self.raw_taxable_property:
            self.additional_tax_property = ((self.raw_taxable_property / 5000) *
2).quantize(Decimal('0.01'))
        if not self.additional_tax_business and self.raw_taxable_business:
            self.additional_tax_business = ((self.raw_taxable_business / 5000) *
2).quantize(Decimal('0.01'))

    # Synchronize payment amount ? traverse via the request's payment
    try:
        payment = self.certificate.certificate_request.payment
        if payment:
            payment.amount = self.total_amount
            payment.save(update_fields=['amount'])
    except Exception:
        pass
    super().save(*args, **kwargs)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
def __str__(self):
    return f"Cedula ({self.get_taxpayer_type_display()}) {self.ctc_number} - {self.certificate.resident}"

class Meta:
    verbose_name = 'Cedula'
    verbose_name_plural = 'Cedulas'

class CertificateRequest(models.Model):
    """A resident's request for a barangay certificate."""

    STATUS_CHOICES = [
        ('pending', 'Pending'),
        ('issued', 'Issued'),
        ('rejected', 'Rejected'),
    ]

    resident = models.ForeignKey(
        'residents.Resident',
        on_delete=models.CASCADE,
        related_name='certificate_requests',
    )
    tracking_code = models.CharField(max_length=20, unique=True,
default=generate_tracking_code)
    cert_type = models.CharField(max_length=30, choices=Certificate.TYPE_CHOICES)
    purpose = models.TextField()
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='pending')
    is_first_time_jobseeker = models.BooleanField(default=False)

    # Payment integration
    payment = models.OneToOneField(
        'payments.Payment',
        on_delete=models.SET_NULL,
        null=True, blank=True,
        related_name='certificate_request'
    )

    # Specific details for the request
    # Business
    business_name = models.CharField(max_length=200, blank=True)
    business_address = models.TextField(blank=True)
    business_type = models.CharField(max_length=200, blank=True)

    # Cedula
    taxpayer_type = models.CharField(max_length=20, blank=True)
    place_of_issue = models.CharField(max_length=200, blank=True)
    height = models.CharField(max_length=20, blank=True)
    weight = models.CharField(max_length=20, blank=True)
    raw_taxable_property = models.DecimalField(max_digits=15, decimal_places=2,
default=0.00)
    raw_taxable_business = models.DecimalField(max_digits=15, decimal_places=2,
default=0.00)
    raw_taxable_income = models.DecimalField(max_digits=15, decimal_places=2,
default=0.00)

    # Late Registration
    child_name = models.CharField(max_length=200, blank=True)
    child_birth_date = models.DateField(null=True, blank=True)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
child_birth_place = models.CharField(max_length=200, blank=True)
father_name = models.CharField(max_length=200, blank=True)
mother_name = models.CharField(max_length=200, blank=True)

# Admin processing
notes = models.TextField(blank=True, help_text='Admin notes (shown to resident on rejection)')
processed_by = models.ForeignKey(
    settings.AUTH_USER_MODEL,
    on_delete=models.SET_NULL,
    null=True, blank=True,
    related_name='processed_cert_requests',
)
processed_at = models.DateTimeField(null=True, blank=True)
created_at = models.DateTimeField(auto_now_add=True)
updated_at = models.DateTimeField(auto_now=True)

def __str__(self):
    return f"Request #{self.pk} ? {self.get_cert_type_display()} ({self.get_status_display()})"

class Meta:
    ordering = ['-created_at']
    verbose_name = 'Certificate Request'
    verbose_name_plural = 'Certificate Requests'

class FirstTimeJobseekerRoster(models.Model):
    """Registry of residents who have availed the RA 11261 benefit (First Time Jobseeker)."""
    resident = models.OneToOneField('residents.Resident', on_delete=models.CASCADE, related_name='ra11261_roster')
    certificate_request = models.ForeignKey(CertificateRequest, on_delete=models.CASCADE, related_name='ra11261_availments')
    certification_date = models.DateField(auto_now_add=True)
    expiry_date = models.DateField()
    availed = models.BooleanField(default=True)
    added_by = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.SET_NULL, null=True)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def __str__(self):
        return f"{self.resident.full_name} (Expires: {self.expiry_date})"

    class Meta:
        ordering = ['-certification_date']

class RA11261Application(models.Model):
    """Application submitted by resident to avail the RA 11261 benefit."""
    STATUS_CHOICES = [
        ('PENDING', 'Pending'),
        ('FOR_REVIEW', 'For Review'),
        ('APPROVED', 'Approved'),
        ('REJECTED', 'Rejected'),
    ]

    resident = models.ForeignKey('residents.Resident', on_delete=models.CASCADE, related_name='ra11261_applications')
```


IoT-Enabled Barangay Resident and Attendance Management System

```
status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='PENDING')
declaration_confirmed = models.BooleanField()
uploaded_docs = models.FileField(upload_to='ra11261_docs/')
rejection_reason = models.TextField(blank=True, null=True)
created_at = models.DateTimeField(auto_now_add=True)
updated_at = models.DateTimeField(auto_now=True)

def __str__(self):
    return f"Application by {self.resident.full_name} - {self.status}"

class Meta:
    ordering = ['-created_at']
```

IoT-Enabled Barangay Resident and Attendance Management System

File: certifications/ra11261_views.py

```
import csv
from datetime import date, timedelta
from django.utils import timezone
from django.http import HttpResponse, JsonResponse
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.contrib import messages
import io

from residents.models import Resident
from payments.models import Payment, OfficialReceipt
from certifications.models import CertificateRequest, FirstTimeJobseekerRoster,
RA11261Application, Certificate

def ra11261_apply(request):
    """Resident submits application via a Django form."""
    if request.method == 'POST':
        resident_id = request.POST.get('resident')
        declaration = request.POST.get('declaration_confirmed')
        docs = request.FILES.get('uploaded_docs')

        if not resident_id or not declaration or not docs:
            messages.error(request, "Please fill in all required fields and upload the
document.")
            return redirect('certifications:ra11261_apply')

        try:
            resident = Resident.objects.get(id=resident_id)
        except Resident.DoesNotExist:
            messages.error(request, "Resident not found.")
            return redirect('certifications:ra11261_apply')

        application = RA11261Application(
            resident=resident,
            declaration_confirmed=(declaration == 'on' or declaration == 'true'),
            uploaded_docs=docs
        )

        # Auto-validation Step 1: Check existing availment
        if FirstTimeJobseekerRoster.objects.filter(resident=resident).exists():
            application.status = 'REJECTED'
            application.rejection_reason = "Already availed RA 11261 benefit"
            application.save()
            messages.error(request, f"Application Auto-Rejected:
{application.rejection_reason}")
            return redirect('certifications:ra11261_apply')

        # Auto-validation Step 2: Check residency requirement (180 days)
        if resident.residency_start_date:
            days_resident = (date.today() - resident.residency_start_date).days
            if days_resident < 180:
                application.status = 'REJECTED'
                application.rejection_reason = "Residency requirement not met (minimum 6
months)"
                application.save()
                messages.error(request, f"Application Auto-Rejected:
{application.rejection_reason}")
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        return redirect('certifications:ra11261_apply')

    # Passed auto-checks
    application.status = 'FOR_REVIEW'
    application.save()
    messages.success(request, "Application submitted successfully. It is now FOR
REVIEW.")
    return redirect('certifications:ra11261_apply')

    return render(request, 'certifications/ra11261_apply.html')

@login_required
def ra11261_admin_list(request):
    """Admin views list of applications."""
    status_filter = request.GET.get('status', 'ALL')
    search = request.GET.get('search', '')

    applications = RA11261Application.objects.select_related('resident').all()

    if status_filter != 'ALL':
        applications = applications.filter(status=status_filter)

    if search:
        applications = applications.filter(resident__first_name__icontains=search) | \
            applications.filter(resident__last_name__icontains=search)

    context = {
        'applications': applications,
        'status_filter': status_filter,
        'search': search,
    }
    return render(request, 'certifications/ra11261_admin_list.html', context)

@login_required
def ra11261_admin_review(request, pk):
    """Admin approves or rejects the application via POST."""
    if request.method == 'POST':
        application = get_object_or_404(RA11261Application, pk=pk)
        new_status = request.POST.get('status')
        rejection_reason = request.POST.get('rejection_reason', '')

        if new_status not in ['APPROVED', 'REJECTED']:
            messages.error(request, "Invalid status.")
            return redirect('certifications:ra11261_admin_list')

        if new_status == 'APPROVED':
            if
FirstTimeJobseekerRoster.objects.filter(resident=application.resident).exists():
                messages.error(request, "Resident already has an active or expired roster
record.")
                return redirect('certifications:ra11261_admin_list')

            # Issue Certificate Request & waive fee
            cert_req = CertificateRequest.objects.filter(resident=application.resident,
is_first_time_jobseeker=True, status='pending').first()
            if not cert_req:
                cert_req = CertificateRequest.objects.create(
                    resident=application.resident,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        cert_type='clearance',
        purpose='First Time Jobseeker (RA 11261)',
        is_first_time_jobseeker=True,
        status='issued',
        processed_by=request.user,
        processed_at=timezone.now()
    )
else:
    cert_req.status = 'issued'
    cert_req.processed_by = request.user
    cert_req.processed_at = timezone.now()
    cert_req.save()

if not cert_req.payment:
    payment = Payment.objects.create(amount=0, status='paid', method='waived',
verified_by=request.user, waive_reason='RA 11261 Benefit Waived')
    cert_req.payment = payment
    cert_req.save()
else:
    payment = cert_req.payment
    payment.amount = 0
    payment.status = 'paid'
    payment.method = 'waived'
    payment.waive_reason = 'RA 11261 Benefit Waived'
    payment.save()

if not payment.official_receipt:
    receipt = OfficialReceipt.objects.create(
        or_number=f"EXEMPT-RA11261-{cert_req.id}",
        resident=application.resident,
        amount=0,
        particulars="RA 11261 Benefit Waived",
        issued_by=request.user
    )
    payment.official_receipt = receipt
    payment.save()
else:
    receipt = payment.official_receipt
    receipt.or_number = f"EXEMPT-RA11261-{cert_req.id}"
    receipt.amount = 0
    receipt.particulars = "RA 11261 Benefit Waived"
    receipt.save()

cert_date = date.today()
expiry_date = cert_date + timedelta(days=365)

FirstTimeJobseekerRoster.objects.create(
    resident=application.resident,
    certificate_request=cert_req,
    certification_date=cert_date,
    expiry_date=expiry_date,
    added_by=request.user
)

from certifications.views import generate_control_number
Certificate.objects.get_or_create(
    resident=application.resident,
    certificate_request=cert_req,
    defaults={
        'issued_by': request.user,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        'status': 'issued',
        'control_number': generate_control_number(cert_req.cert_type)
    }
)
messages.success(request, "Application APPROVED and Roster updated.")

elif new_status == 'REJECTED':
    if not rejection_reason:
        messages.error(request, "Rejection reason is required.")
        return redirect('certifications:ra11261_admin_list')
    messages.warning(request, "Application REJECTED.")

application.status = new_status
application.rejection_reason = rejection_reason if new_status == 'REJECTED' else
None
application.save()

return redirect('certifications:ra11261_admin_list')

@login_required
def ra11261_admin_roster(request):
    """Admin views full roster."""
    search = request.GET.get('search', '')
    roster = FirstTimeJobseekerRoster.objects.select_related('resident').all()

    if search:
        roster = roster.filter(resident__first_name__icontains=search) | \
            roster.filter(resident__last_name__icontains=search)

    active_count = 0
    expired_count = 0
    today = date.today()
    for r in roster:
        if r.expiry_date >= today:
            active_count += 1
        else:
            expired_count += 1

    # Fetch all residents who are NOT already in the roster, so admin can select them.
    roster_resident_ids = roster.values_list('resident_id', flat=True)
    available_residents =
Resident.objects.exclude(id__in=roster_resident_ids).order_by('last_name')

    context = {
        'roster': roster,
        'search': search,
        'total_availed': roster.count(),
        'active_count': active_count,
        'expired_count': expired_count,
        'available_residents': available_residents,
    }
    return render(request, 'certifications/ra11261_admin_roster.html', context)

@login_required
def ra11261_admin_roster_add(request):
    """Admin manually adds a walk-in resident to the Roster."""
    if request.method == 'POST':
        resident_id = request.POST.get('resident_id')
        if not resident_id:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    messages.error(request, "Please select a resident.")
    return redirect('certifications:ra11261_admin_roster')

resident = get_object_or_404(Resident, pk=resident_id)

if FirstTimeJobseekerRoster.objects.filter(resident=resident).exists():
    messages.error(request, "Resident already exists in the RA 11261 roster.")
    return redirect('certifications:ra11261_admin_roster')

# To keep data consistent, we create an approved RA11261Application representing
the walk-in
application = RA11261Application.objects.create(
    resident=resident,
    declaration_confirmed=True, # Assumed they signed it physically
    status='APPROVED'
)

# Issue Certificate Request & waive fee
cert_req = CertificateRequest.objects.create(
    resident=resident,
    cert_type='clearance',
    purpose='First Time Jobseeker (RA 11261) - Walk In',
    is_first_time_jobseeker=True,
    status='issued',
    processed_by=request.user,
    processed_at=timezone.now()
)

payment = Payment.objects.create(amount=0, status='paid', method='waived',
verified_by=request.user, waive_reason='RA 11261 Benefit Waived (Walk-In)')
cert_req.payment = payment
cert_req.save()

receipt = OfficialReceipt.objects.create(
    or_number=f"EXEMPT-RA11261-{cert_req.id}",
    resident=resident,
    amount=0,
    particulars="RA 11261 Benefit Waived (Walk-In)",
    issued_by=request.user
)
payment.official_receipt = receipt
payment.save()

cert_date = date.today()
expiry_date = cert_date + timedelta(days=365)

FirstTimeJobseekerRoster.objects.create(
    resident=resident,
    certificate_request=cert_req,
    certification_date=cert_date,
    expiry_date=expiry_date,
    added_by=request.user
)

from certifications.views import generate_control_number
Certificate.objects.create(
    resident=resident,
    certificate_request=cert_req,
    issued_by=request.user,
    status='issued',
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        control_number=generate_control_number(cert_req.cert_type)
    )
    messages.success(request, f"Successfully added {resident.full_name} to the RA
11261 Roster.")

    return redirect('certifications:ra11261_admin_roster')

@login_required
def ra11261_export_csv(request):
    """Export roster to CSV."""
    response = HttpResponse(content_type='text/csv')
    response['Content-Disposition'] = 'attachment; filename="ra11261_roster.csv"'

    writer = csv.writer(response)
    writer.writerow(['Resident Name', 'Certification Date', 'Expiry Date', 'Status',
'Added By'])

    roster = FirstTimeJobseekerRoster.objects.select_related('resident', 'added_by').all()
    today = date.today()

    for r in roster:
        status_text = 'Active' if r.expiry_date >= today else 'Expired'
        added_by = r.added_by.username if r.added_by else 'System'
        writer.writerow([
            r.resident.full_name,
            r.certification_date,
            r.expiry_date,
            status_text,
            added_by
        ])

    return response

def ra11261_certification_pdf(request, pk):
    """Generate PDF for the RA 11261 Certification ? matching the standard certificate
layout."""
    import os
    from functools import partial
    from reportlab.lib.pagesizes import letter
    from reportlab.lib.units import inch
    from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
    from reportlab.lib.enums import TA_CENTER, TA_JUSTIFY, TA_LEFT
    from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table,
TableStyle, Image
    from reportlab.lib import colors
    from django.conf import settings as django_settings
    from certifications.utils import draw_watermark
    from officials.models import Official

    try:
        application = RA11261Application.objects.get(pk=pk)
        roster = FirstTimeJobseekerRoster.objects.get(resident=application.resident)
    except (RA11261Application.DoesNotExist, FirstTimeJobseekerRoster.DoesNotExist):
        return HttpResponse("Certification not found or not approved yet.", status=404)

    buffer = io.BytesIO()
    doc = SimpleDocTemplate(buffer, pagesize=letter,
        topMargin=0.3 * inch, bottomMargin=0.3 * inch,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
leftMargin=0.5 * inch, rightMargin=0.5 * inch)

styles = getSampleStyleSheet()

# Custom Styles (same as generate_certificate_pdf)
styles.add(ParagraphStyle(name='CertHeaderLabel', fontSize=10, alignment=TA_CENTER,
                           fontName='Helvetica', leading=12))
styles.add(ParagraphStyle(name='CertHeaderBrgy', fontSize=12, alignment=TA_CENTER,
                           fontName='Helvetica-Bold', leading=14))
styles.add(ParagraphStyle(name='CertOffice', fontSize=16, alignment=TA_CENTER,
                           spaceBefore=20, spaceAfter=10, fontName='Times-Bold'))
styles.add(ParagraphStyle(name='CertTitleLarge', fontSize=22, alignment=TA_CENTER,
                           spaceAfter=25, fontName='Helvetica-Bold', leading=26))
styles.add(ParagraphStyle(name='CertSubTitle', fontSize=13, alignment=TA_CENTER,
                           spaceAfter=20, fontName='Helvetica-Oblique', leading=16))
styles.add(ParagraphStyle(name='CertToWhom', fontSize=14, alignment=TA_LEFT,
                           spaceAfter=15, fontName='Helvetica-Bold'))
styles.add(ParagraphStyle(name='CertBody', fontSize=12, alignment=TA_JUSTIFY,
                           spaceAfter=12, fontName='Helvetica', leading=18))
styles.add(ParagraphStyle(name='CertSignName', fontSize=14, alignment=TA_CENTER,
                           fontName='Helvetica-Bold'))
styles.add(ParagraphStyle(name='CertSignPos', fontSize=11, alignment=TA_CENTER,
                           fontName='Helvetica'))
styles.add(ParagraphStyle(name='CertWarning', fontSize=10, alignment=TA_CENTER,
                           textColor=colors.red, fontName='Helvetica-BoldOblique'))

elements = []

# ?? HEADER (same as other certificates) ?????????????????????????????
sico_logo_path = os.path.join(django_settings.BASE_DIR,
                              'static/images/sico_sico_logo.jpg')
bagong_pilipinas_path = os.path.join(django_settings.BASE_DIR,
                                      'static/images/bagong_pilipinas.png')
gigaquit_logo_path = os.path.join(django_settings.BASE_DIR,
                                   'static/images/gigaquit_logo.png')

logo_w = 0.8 * inch
logo_h = 0.8 * inch

img_sico = Image(sico_logo_path, width=logo_w, height=logo_h, kind='bound') if
os.path.exists(sico_logo_path) else Spacer(logo_w, logo_h)
img_bagong = Image(bagong_pilipinas_path, width=1.0*inch, height=0.7*inch,
kind='bound') if os.path.exists(bagong_pilipinas_path) else Spacer(1.0*inch, logo_h)
img_gigaquit = Image(gigaquit_logo_path, width=logo_w, height=logo_h, kind='bound') if
os.path.exists(gigaquit_logo_path) else Spacer(logo_w, logo_h)

brgy_name = getattr(django_settings, 'BARANGAY_NAME', 'Sico-Sico')
municipality = getattr(django_settings, 'BARANGAY_MUNICIPALITY', 'Gigaquit')
province = getattr(django_settings, 'BARANGAY_PROVINCE', 'Surigao del Norte')

header_brgy_title = brgy_name.upper() if "BARANGAY" in brgy_name.upper() else
f"BARANGAY {brgy_name.upper()}"

center_text = [
    Paragraph('REPUBLIC OF THE PHILIPPINES', styles['CertHeaderLabel']),
    Paragraph(f'Province of {province}', styles['CertHeaderLabel']),
    Paragraph(f'Municipality of {municipality}', styles['CertHeaderLabel']),
    Paragraph(header_brgy_title, styles['CertHeaderBrgy']),
]
```


IoT-Enabled Barangay Resident and Attendance Management System

```
col_widths = [1.1*inch, 1.2*inch, 3.0*inch, 1.1*inch, 1.1*inch]
header_table_data = [[img_sico, img_bagong, center_text, img_gigaquit, '']]

header_table = Table(header_table_data, colWidths=col_widths)
header_table.setStyle(TableStyle([
    ('VALIGN', (0,0), (-1,-1), 'MIDDLE'),
    ('ALIGN', (0,0), (1,0), 'CENTER'),
    ('ALIGN', (2,0), (2,0), 'CENTER'),
    ('ALIGN', (3,0), (3,0), 'CENTER'),
    ('LEFTPADDING', (0,0), (-1,-1), 2),
    ('RIGHTPADDING', (0,0), (-1,-1), 2),
    ('TOPPADDING', (0,0), (-1,-1), 2),
    ('BOTTOMPADDING', (0,0), (-1,-1), 2),
]))

elements.append(header_table)
elements.append(Paragraph('OFFICE OF THE SANGGUNIANG BARANGAY', styles['CertOffice']))
elements.append(Paragraph('BARANGAY CERTIFICATION', styles['CertTitleLarge']))
elements.append(Paragraph('First Time Jobseekers Assistance Act (RA 11261)',
styles['CertSubTitle']))

# ?? BODY ?????????????????????????????????????????????????????????????
resident = application.resident
display_brgy = brgy_name if "BARANGAY" in brgy_name.upper() else f"Barangay
{brgy_name}"

elements.append(Paragraph('TO WHOM IT MAY CONCERN;', styles['CertToWhom']))

body_text = (
    f"This is to certify that <b>{resident.full_name.upper()}</b>, "
    f"{resident.age} years old, {resident.get_civil_status_display()}, Filipino, "
    f"a resident of {display_brgy}, {municipality}, {province}, "
    f"is a qualified <b>First Time Jobseeker</b> under Republic Act No. 11261, "
    f"otherwise known as the \"First Time Jobseekers Assistance Act\"."
)
elements.append(Paragraph(body_text, styles['CertBody']))

benefit_text = (
    "The above-named person has not previously availed of the benefits under RA 11261
"
    "and is hereby granted exemption from the payment of fees and charges for the "
    "issuance of this Barangay Certification."
)
elements.append(Paragraph(benefit_text, styles['CertBody']))

conclusion_text = (
    "This certification is issued upon the request of the above-named person "
    "in compliance with RA 11261 for employment purposes."
)
elements.append(Paragraph(conclusion_text, styles['CertBody']))

# ?? ISSUED DATE ?????????????????????????????????????????????????????????
def get_ordinal(n):
    if 11 <= (n % 100) <= 13:
        return f"{n}th"
    return f"{n}{'th', 'st', 'nd', 'rd', 'th'}[min(n % 10, 4)]"

cert_date = roster.certification_date
issued_date_text = (
    f"Given this <b>{get_ordinal(cert_date.day)}</b> day of "
```

IoT-Enabled Barangay Resident and Attendance Management System

```
f"<b>{cert_date.strftime('%B')}, {cert_date.year}</b>, "  
f"at the Office of Punong Barangay of {display_brgy}, {municipality}, Surigao del  
Norte."  
)  
elements.append(Spacer(1, 10))  
elements.append(Paragraph(issued_date_text, styles['CertBody']))  
  
validity_text = f"<b>Valid until: {roster.expiry_date.strftime('%B %d, %Y')}</b>"  
elements.append(Paragraph(validity_text, styles['CertBody']))  
  
elements.append(Spacer(1, 40))  
  
# ?? CAPTAIN SIGNATURE ?????????????????????????????????????????????  
captain = Official.objects.filter(position='captain', status='active').first()  
captain_name = f"HON. {captain.resident.full_name.upper()}" if captain else  
getattr(django_settings, 'BARANGAY_CAPTAIN', 'NO ACTIVE CAPTAIN')  
sig_content = [  
    Paragraph(captain_name, styles['CertSignName']),  
    Paragraph('Punong Barangay', styles['CertSignPos'])  
]  
sig_table = Table(['', sig_content], colWidths=[3.5*inch, 4*inch])  
sig_table.setStyle(TableStyle([  
    ('VALIGN', (0,0), (-1,-1), 'TOP'),  
    ('ALIGN', (1,0), (1,0), 'CENTER'),  
]))  
elements.append(sig_table)  
  
# ?? WARNING ??????????????????????????????????????????????????????  
elements.append(Spacer(1, 20))  
elements.append(Paragraph('\u201cNOT VALID WITHOUT SEAL\u201d',  
styles['CertWarning']))  
  
# ?? BUILD PDF ??????????????????????????????????????????????????????  
watermark_func = partial(draw_watermark, is_resident=False)  
doc.build(elements, onFirstPage=watermark_func, onLaterPages=watermark_func)  
buffer.seek(0)  
  
response = HttpResponse(buffer, content_type='application/pdf')  
response['Content-Disposition'] = f'inline;  
filename="RA11261_{resident.full_name}.pdf"'  
return response
```

IoT-Enabled Barangay Resident and Attendance Management System

File: certifications/templatetags/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: `certifications/templatetags/ra11261_tags.py`

```
from django import template
from datetime import date

register = template.Library()

@register.filter
def is_expired(roster_obj):
    return roster_obj.expiry_date < date.today()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: `certifications/tests.py`

```
from django.test import TestCase
```

```
# Create your tests here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: certifications/urls.py

```
from django.urls import path
from . import views
from . import ra11261_views

app_name = 'certifications'

urlpatterns = [
    path('', views.certificate_list, name='list'),
    path('issue/', views.certificate_issue, name='issue'),
    path('<int:pk>', views.certificate_view, name='view'),
    path('<int:pk>/pdf/', views.certificate_pdf, name='pdf'),
    path('<int:pk>/receipt/', views.certificate_receipt_pdf, name='receipt'),
    path('search/', views.certificate_search, name='search'),
    path('check-uniqueness/', views.check_uniqueness, name='check_uniqueness'),
    path('get-next-numbers/', views.get_next_numbers, name='get_next_numbers'),
    path('<int:pk>/delete/', views.certificate_delete, name='delete'),
    # Certificate Requests
    path('requests/', views.request_list, name='request_list'),
    path('requests/new/', views.request_certificate, name='request_certificate'),
    path('my-requests/', views.my_requests, name='my_requests'),
    path('requests/<int:pk>/fulfill/', views.fulfill_request, name='fulfill_request'),
    path('requests/<int:pk>/reject/', views.reject_request, name='reject_request'),
    path('<int:pk>/virtual/', views.virtual_certificate, name='virtual_certificate'),

    # RA 11261 URLs
    path('ra11261/apply/', ra11261_views.ra11261_apply, name='ra11261_apply'),
    path('ra11261/admin/applications/', ra11261_views.ra11261_admin_list,
name='ra11261_admin_list'),
    path('ra11261/admin/applications/<int:pk>/review/',
ra11261_views.ra11261_admin_review, name='ra11261_admin_review'),
    path('ra11261/admin/roster/', ra11261_views.ra11261_admin_roster,
name='ra11261_admin_roster'),
    path('ra11261/admin/roster/add/', ra11261_views.ra11261_admin_roster_add,
name='ra11261_admin_roster_add'),
    path('ra11261/admin/roster/export/', ra11261_views.ra11261_export_csv,
name='ra11261_export_csv'),
    path('ra11261/certification/<int:pk>/pdf/', ra11261_views.ra11261_certification_pdf,
name='ra11261_pdf'),
]
```

IoT-Enabled Barangay Resident and Attendance Management System

File: certifications/utils.py

```
import io
import os
from datetime import date
from decimal import Decimal
from reportlab.lib.pagesizes import letter
from reportlab.lib.units import inch
from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
from reportlab.lib.enums import TA_CENTER, TA_JUSTIFY, TA_LEFT, TA_RIGHT
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle,
Image, Frame, PageTemplate
from reportlab.lib import colors
from django.conf import settings
from django.contrib.staticfiles import finders

def draw_watermark(canvas, doc, is_resident=False):
    """Draw a faded logo as a watermark in the center of the page."""
    canvas.saveState()
    # Path to the barangay logo
    logo_path = os.path.join(settings.BASE_DIR, 'static/images/sico_sico_logo.jpg')
    if not os.path.exists(logo_path):
        # Fallback if specific name doesn't exist
        logo_path = os.path.join(settings.BASE_DIR, 'static/images/logo.png')

    if os.path.exists(logo_path):
        canvas.setFillAlpha(0.1) # Set opacity to 10%
        # Calculate center
        img_width = 5 * inch
        img_height = 5 * inch
        x = (letter[0] - img_width) / 2
        y = (letter[1] - img_height) / 2
        canvas.drawImage(logo_path, x, y, width=img_width, height=img_height, mask='auto')

    # Add text watermark if resident
    if is_resident:
        canvas.setFillAlpha(0.15)
        canvas.setFont('Helvetica-Bold', 60)
        canvas.setFillColor(colors.red)
        canvas.translate(letter[0]/2, letter[1]/2)
        canvas.rotate(45)
        # Watermark text removed per user request

    canvas.restoreState()

def generate_certificate_pdf(certificate, is_resident=False):
    """Generate a styled PDF for any certificate type."""
    if certificate.cert_type == 'cedula':
        return generate_cedula_pdf(certificate)

    buffer = io.BytesIO()
    doc = SimpleDocTemplate(buffer, pagesize=letter,
                           topMargin=0.3 * inch, bottomMargin=0.3 * inch,
                           leftMargin=0.5 * inch, rightMargin=0.5 * inch)

    styles = getSampleStyleSheet()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# Custom Styles
styles.add(ParagraphStyle(name='CertHeaderLabel', fontSize=10, alignment=TA_CENTER,
                           fontName='Helvetica', leading=12))
styles.add(ParagraphStyle(name='CertHeaderBrgy', fontSize=12, alignment=TA_CENTER,
                           fontName='Helvetica-Bold', leading=14))
styles.add(ParagraphStyle(name='CertOffice', fontSize=16, alignment=TA_CENTER,
                           spaceBefore=20, spaceAfter=10, fontName='Times-Bold'))
styles.add(ParagraphStyle(name='CertTitleLarge', fontSize=22, alignment=TA_CENTER,
                           spaceAfter=25, fontName='Helvetica-Bold', leading=26))
styles.add(ParagraphStyle(name='CertToWhom', fontSize=14, alignment=TA_LEFT,
                           spaceAfter=15, fontName='Helvetica-Bold'))
styles.add(ParagraphStyle(name='CertBody', fontSize=12, alignment=TA_JUSTIFY,
                           spaceAfter=12, fontName='Helvetica', leading=18))
styles.add(ParagraphStyle(name='CertSignName', fontSize=14, alignment=TA_CENTER,
                           fontName='Helvetica-Bold'))
styles.add(ParagraphStyle(name='CertSignPos', fontSize=11, alignment=TA_CENTER,
                           fontName='Helvetica'))
styles.add(ParagraphStyle(name='CertWarning', fontSize=10, alignment=TA_CENTER,
                           textColor=colors.red, fontName='Helvetica-BoldOblique'))

elements = []

# Paths to logos
sico_logo_path = os.path.join(settings.BASE_DIR, 'static/images/sico_sico_logo.jpg')
bagong_pilipinas_path = os.path.join(settings.BASE_DIR,
'static/images/bagong_pilipinas.png')
gigaquit_logo_path = os.path.join(settings.BASE_DIR,
'static/images/gigaquit_logo.png')

# Standardized logo size
logo_w = 0.8 * inch
logo_h = 0.8 * inch

# Header Components
img_sico = Image(sico_logo_path, width=logo_w, height=logo_h, kind='bound') if
os.path.exists(sico_logo_path) else Spacer(logo_w, logo_h)
img_bagong = Image(bagong_pilipinas_path, width=1.0*inch, height=0.7*inch,
kind='bound') if os.path.exists(bagong_pilipinas_path) else Spacer(1.0*inch, logo_h)
img_gigaquit = Image(gigaquit_logo_path, width=logo_w, height=logo_h, kind='bound') if
os.path.exists(gigaquit_logo_path) else Spacer(logo_w, logo_h)

brgy_name = getattr(settings, 'BARANGAY_NAME', 'Sico-Sico')
municipality = getattr(settings, 'BARANGAY_MUNICIPALITY', 'Gigaquit')
province = getattr(settings, 'BARANGAY_PROVINCE', 'Surigao del Norte')

# Fix redundant "Barangay" in title
header_brgy_title = brgy_name.upper() if "BARANGAY" in brgy_name.upper() else
f"BARANGAY {brgy_name.upper()}"

center_text = [
    Paragraph('REPUBLIC OF THE PHILIPPINES', styles['CertHeaderLabel']),
    Paragraph(f'Province of {province}', styles['CertHeaderLabel']),
    Paragraph(f'Municipality of {municipality}', styles['CertHeaderLabel']),
    Paragraph(header_brgy_title, styles['CertHeaderBrgy']),
]

# Create 5-column table to balance the logos and keep text perfectly centered
# [Logo Sico] [Logo Bagong] [Center Text] [Logo Gigaquit] [Empty Balance]
col_widths = [1.1*inch, 1.2*inch, 3.0*inch, 1.1*inch, 1.1*inch]
header_table_data = [[img_sico, img_bagong, center_text, img_gigaquit, '']]
```


IoT-Enabled Barangay Resident and Attendance Management System

```
header_table = Table(header_table_data, colWidths=col_widths)
header_table.setStyle(TableStyle([
    ('VALIGN', (0,0), (-1,-1), 'MIDDLE'),
    ('ALIGN', (0,0), (1,0), 'CENTER'),
    ('ALIGN', (2,0), (2,0), 'CENTER'),
    ('ALIGN', (3,0), (3,0), 'CENTER'),
    ('LEFTPADDING', (0,0), (-1,-1), 2),
    ('RIGHTPADDING', (0,0), (-1,-1), 2),
    ('TOPPADDING', (0,0), (-1,-1), 2),
    ('BOTTOMPADDING', (0,0), (-1,-1), 2),
]))

# Certificate Type Title Mapping
cert_titles_map = {
    'clearance': 'BARANGAY CLEARANCE',
    'residency': 'CERTIFICATE OF RESIDENCY',
    'indigency': 'CERTIFICATE OF INDIGENCY',
    'good_moral': 'CERTIFICATE OF GOOD MORAL CHARACTER',
    'business_permit': 'BARANGAY BUSINESS CLEARANCE',
    'comelec': 'COMELEC CERTIFICATE',
    'cedula': 'COMMUNITY TAX CERTIFICATE',
    'late_registration': 'CERTIFICATE OF LATE REGISTRATION',
}
cert_display_title = cert_titles_map.get(certificate.cert_type, 'CERTIFICATE')

elements.append(header_table)
elements.append(Paragraph('OFFICE OF THE SANGGUNIANG BARANGAY', styles['CertOffice']))
elements.append(Paragraph(cert_display_title, styles['CertTitleLarge']))

resident = certificate.resident
day = date.today().day
month = date.today().strftime('%B')
year = date.today().year

display_brgy = brgy_name if "BARANGAY" in brgy_name.upper() else f"Barangay
{brgy_name}"

# Body text based on certificate type
if certificate.cert_type == 'clearance':
    body_text = f"This Clearance is hereby granted to
<b>{resident.full_name.upper()}</b>, with residence at {display_brgy}, {municipality},
Surigao del Norte, in connection with this application for <b>{certificate.purpose}</b>."
    conclusion_text = f"It is understood that the issuance of this clearance shall not
exempt the applicant/s from other requirements prescribed under the existing barangay
ordinance of {display_brgy}, {municipality}, Surigao del Norte."
elif certificate.cert_type == 'residency':
    body_text = f"This is to certify that <b>{resident.full_name.upper()}</b>,
{resident.age} years old, {resident.get_civil_status_display()}, Filipino, is a bonafide
resident of {display_brgy}, {municipality}, {province}."
    conclusion_text = f"This certification is issued upon the request of the above-
named person for <b>{certificate.purpose}</b>."
elif certificate.cert_type == 'indigency':
    body_text = f"This is to certify that <b>{resident.full_name.upper()}</b>,
{resident.age} years old, {resident.get_civil_status_display()}, Filipino, a resident of
{display_brgy}, {municipality}, {province}, belongs to an indigent family in this
barangay."
    conclusion_text = f"This certification is issued upon the request of the above-
named person for <b>{certificate.purpose}</b>."
elif certificate.cert_type == 'good_moral':
```

IoT-Enabled Barangay Resident and Attendance Management System

```
body_text = f"This is to certify that <b>{resident.full_name.upper()}</b>,\n{resident.age} years old, {resident.get_civil_status_display()}, Filipino, a resident of\n{display_brgy}, {municipality}, {province}, is known to me to be a person of good moral\ncharacter and has no derogatory or criminal record in this barangay."
conclusion_text = f"This certification is issued upon the request of the above-\nnamed person for <b>{certificate.purpose}</b>."
elif certificate.cert_type == 'business_permit':
    body_text = f"This is to certify that <b>{certificate.business_name.upper()}</b>\nowned/managed by <b>{resident.full_name.upper()}</b>, located at\n{certificate.business_address or resident.address}, {display_brgy}, {municipality},\n{province}, is hereby granted clearance to operate within the jurisdiction of this\nbarangay.<br/><br/>Type of Business: <b>{certificate.business_type}</b>"
    conclusion_text = f"This certification is issued for\n<b>{certificate.purpose}</b>."
else:
    body_text = f"This is to certify that <b>{resident.full_name.upper()}</b>,\n{resident.age} years old, {resident.get_civil_status_display()}, Filipino, a resident of\n{display_brgy}, {municipality}, {province}."
    conclusion_text = f"This certification is issued for\n<b>{certificate.purpose}</b>."

def get_ordinal(n):
    if 11 <= (n % 100) <= 13:\n        return f"{n}th"\n    return f"{n}{'th', 'st', 'nd', 'rd', 'th'}[min(n % 10, 4)]"

issued_date_text = f"Given this <b>{get_ordinal(day)}</b> day of <b>{month},\n{year}</b>, at the Office of Punong Barangay of {display_brgy}, {municipality}, Surigao\ndel Norte."

elements.append(Paragraph('TO WHOM IT MAY CONCERN;', styles['CertToWhom']))
elements.append(Paragraph(body_text, styles['CertBody']))
elements.append(Paragraph(conclusion_text, styles['CertBody']))
elements.append(Paragraph(f'Certification is issued upon the request of the above\nnamed for whatever legal purpose it may serve him/her best.', styles['CertBody']))
elements.append(Spacer(1, 15))
elements.append(Paragraph(issued_date_text, styles['CertBody']))
elements.append(Spacer(1, 40))

# Captain Signature - Optimized for full width to prevent clipping
from officials.models import Official
captain = Official.objects.filter(position='captain', status='active').first()
captain_name = f"HON. {captain.resident.full_name.upper()}" if captain else
getattr(settings, 'BARANGAY_CAPTAIN', 'NO ACTIVE CAPTAIN')
sig_content = [\n    Paragraph(captain_name, styles['CertSignName']),\n    Paragraph('Punong Barangay', styles['CertSignPos'])\n]\nsig_table = Table(['', sig_content], colwidths=[3.5*inch, 4*inch])
sig_table.setStyle(TableStyle([\n    ('VALIGN', (0,0), (-1,-1), 'TOP'),\n    ('ALIGN', (1,0), (1,0), 'CENTER'),\n]))
elements.append(sig_table)

# Final warning
elements.append(Spacer(1, 20))
elements.append(Paragraph('?NOT VALID WITHOUT SEAL?', styles['CertWarning']))

# Build PDF
```

IoT-Enabled Barangay Resident and Attendance Management System

```
from functools import partial
watermark_func = partial(draw_watermark, is_resident=is_resident)
doc.build(elements, onFirstPage=watermark_func, onLaterPages=watermark_func)
buffer.seek(0)
return buffer

def generate_cedula_pdf(certificate, is_resident=False):
    """Generate an authentic-looking Cedula (Community Tax Certificate)."""
    buffer = io.BytesIO()
    doc = SimpleDocTemplate(buffer, pagesize=letter,
                           topMargin=0.5 * inch, bottomMargin=0.5 * inch,
                           leftMargin=0.5 * inch, rightMargin=0.5 * inch)

    styles = getSampleStyleSheet()
    styles.add(ParagraphStyle(name='CedulaHeader', fontSize=14, alignment=TA_CENTER,
                              fontName='Helvetica-Bold'))
    styles.add(ParagraphStyle(name='CedulaSubHeader', fontSize=10, alignment=TA_CENTER,
                              fontName='Helvetica'))
    styles.add(ParagraphStyle(name='CedulaLabel', fontSize=8, fontName='Helvetica-Bold'))
    styles.add(ParagraphStyle(name='CedulaValue', fontSize=10, fontName='Helvetica'))
    styles.add(ParagraphStyle(name='CedulaMoney', fontSize=10, fontName='Helvetica',
                              alignment=TA_RIGHT))

    elements = []

    cedula = certificate.cedula_details
    resident = certificate.resident
    is_corp = cedula.taxpayer_type == 'corporation'

    # Header
    header_title = 'COMMUNITY TAX CERTIFICATE'
    if is_corp:
        header_title += ' - CORPORATION'
    elements.append(Paragraph(header_title, styles['CedulaHeader']))
    elements.append(Paragraph(cedula.get_taxpayer_type_display().upper(),
                              styles['CedulaSubHeader']))
    elements.append(Spacer(1, 10))

    # Place and Date of Issue
    header_data = [
        [Paragraph('YEAR', styles['CedulaLabel']), Paragraph('PLACE OF ISSUE (City /
Municipality / Province)', styles['CedulaLabel']), Paragraph('DATE ISSUED',
styles['CedulaLabel'])],
        [Paragraph(str(date.today().year), styles['CedulaValue']),
Paragraph(cedula.place_of_issue, styles['CedulaValue']),
Paragraph(date.today().strftime('%m/%d/%Y'), styles['CedulaValue'])]
    ]
    header_table = Table(header_data, colWidths=[1*inch, 4.5*inch, 1.5*inch])
    header_table.setStyle(TableStyle([
        ('BOX', (0,0), (-1,-1), 1, colors.black),
        ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
        ('VALIGN', (0,0), (-1,-1), 'TOP'),
    ]))
    elements.append(header_table)

    # CTC Number
    ctc_data = [[Paragraph('CTC NUMBER', styles['CedulaLabel']),
Paragraph(cedula.ctc_number, styles['CedulaValue'])]]
    ctc_table = Table(ctc_data, colWidths=[1.5*inch, 5.5*inch])
```

IoT-Enabled Barangay Resident and Attendance Management System

```
ctc_table.setStyle(TableStyle([
    ('BOX', (0,0), (-1,-1), 1, colors.black),
    ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
]))
elements.append(ctc_table)

# Name/Company
name_label = 'NAME (Surname, First Name, Middle Name)' if not is_corp else 'COMPANY
NAME'
name_data = [
    [Paragraph(name_label, styles['CedulaLabel']), Paragraph('TIN (If any)',
styles['CedulaLabel'])],
    [Paragraph(resident.full_name.upper(), styles['CedulaValue']), Paragraph('',
styles['CedulaValue'])]
]
name_table = Table(name_data, colWidths=[5.5*inch, 1.5*inch])
name_table.setStyle(TableStyle([
    ('BOX', (0,0), (-1,-1), 1, colors.black),
    ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
]))
elements.append(name_table)

addr_data = [[Paragraph('ADDRESS', styles['CedulaLabel']), Paragraph(resident.address,
styles['CedulaValue'])]]
addr_table = Table(addr_data, colWidths=[1*inch, 6*inch])
addr_table.setStyle(TableStyle([
    ('BOX', (0,0), (-1,-1), 1, colors.black),
    ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
]))
elements.append(addr_table)

if not is_corp:
    info_row1 = [
        [Paragraph('CITIZENSHIP', styles['CedulaLabel']), Paragraph('CIVIL STATUS',
styles['CedulaLabel']), Paragraph('GENDER', styles['CedulaLabel']), Paragraph('BIRTH
DATE', styles['CedulaLabel'])],
        [Paragraph(resident.nationality, styles['CedulaValue']),
Paragraph(resident.get_civil_status_display(), styles['CedulaValue']),
Paragraph(resident.get_gender_display(), styles['CedulaValue']),
Paragraph(resident.birthdate.strftime('%m/%d/%Y'), styles['CedulaValue'])]
    ]
    info_table1 = Table(info_row1, colWidths=[1.75*inch, 1.75*inch, 1.75*inch,
1.75*inch])
    info_table1.setStyle(TableStyle([
        ('BOX', (0,0), (-1,-1), 1, colors.black),
        ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
    ]))
    elements.append(info_table1)

    info_row2 = [
        [Paragraph('PLACE OF BIRTH', styles['CedulaLabel']), Paragraph('HEIGHT',
styles['CedulaLabel']), Paragraph('WEIGHT', styles['CedulaLabel']), Paragraph('PROFESSION
/ OCCUPATION', styles['CedulaLabel'])],
        [Paragraph(resident.birthplace, styles['CedulaValue']),
Paragraph(cedula.height, styles['CedulaValue']), Paragraph(cedula.weight,
styles['CedulaValue']), Paragraph(resident.occupation, styles['CedulaValue'])]
    ]
    info_table2 = Table(info_row2, colWidths=[1.75*inch, 1.1*inch, 1.1*inch,
3.05*inch])
    info_table2.setStyle(TableStyle([
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        ('BOX', (0,0), (-1,-1), 1, colors.black),
        ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
    )))
    elements.append(info_table2)
else:
    corp_info = [[Paragraph('DATE OF INCORPORATION / REGISTRATION',
styles['CedulaLabel']], Paragraph('NATURE OF BUSINESS', styles['CedulaLabel']))]
    corp_table = Table(corp_info, colWidths=[3.5*inch, 3.5*inch])
    corp_table.setStyle(TableStyle([
        ('BOX', (0,0), (-1,-1), 1, colors.black),
        ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
    ]))
    elements.append(corp_table)

elements.append(Spacer(1, 10))

# Financial Section
basic_label = 'BASIC COMMUNITY TAX (?5.00) Voluntary or Exempted (?1.00)' if not
is_corp else 'BASIC COMMUNITY TAX (?500.00)'
    fin_header = [[Paragraph('COMMUNITY TAX DUE', styles['CedulaLabel']),
Paragraph(basic_label, styles['CedulaLabel']), Paragraph('AMOUNT',
styles['CedulaLabel'])]]
    fin_table_h = Table(fin_header, colWidths=[3.5*inch, 2.5*inch, 1.5*inch])
    fin_table_h.setStyle(TableStyle([
        ('BOX', (0,0), (-1,-1), 1, colors.black),
        ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
        ('BACKGROUND', (0,0), (-1,0), colors.lightgrey),
    ]))
    elements.append(fin_table_h)

# Calculate capped additional tax for display
raw_additional = cedula.total_additional_tax
addl_cap = Decimal('5000.00') if not is_corp else Decimal('10000.00')
capped_additional = min(raw_additional, addl_cap)

if not is_corp:
    fin_data = [
        [Paragraph('1. BASIC COMMUNITY TAX', styles['CedulaValue']), '',
Paragraph(f"{cedula.basic_tax:,.2f}", styles['CedulaMoney'])],
        [Paragraph('2. ADDITIONAL COMMUNITY TAX (Taxable amount not to exceed
?5,000.00)', styles['CedulaValue']), '', ''],
        [Paragraph('    (a) GROSS ANNUAL SALARY OR EARNINGS DERIVED FROM EXERCISE OF
PROFESSION (?1.00 for every ?1,000)', styles['CedulaValue']), '',
Paragraph(f"{cedula.additional_tax_income:,.2f}", styles['CedulaMoney'])],
        [Paragraph('    (b) GROSS RECEIPTS OR EARNINGS DERIVED FROM BUSINESS DURING THE
PRECEDING YEAR (?1.00 for every ?1,000)', styles['CedulaValue']), '',
Paragraph(f"{cedula.additional_tax_business:,.2f}", styles['CedulaMoney'])],
        [Paragraph('    (c) INCOME FROM REAL PROPERTY (?1.00 for every ?1,000)',
styles['CedulaValue']), '', Paragraph(f"{cedula.additional_tax_property:,.2f}",
styles['CedulaMoney'])],
    ]
else:
    fin_data = [
        [Paragraph('1. BASIC COMMUNITY TAX', styles['CedulaValue']), '',
Paragraph(f"{cedula.basic_tax:,.2f}", styles['CedulaMoney'])],
        [Paragraph('2. ADDITIONAL COMMUNITY TAX (Taxable amount not to exceed
?10,000.00)', styles['CedulaValue']), '', ''],
        [Paragraph('    (a) ASSESSED VALUE OF REAL PROPERTY OWNED (?2.00 for every
?5,000)', styles['CedulaValue']), '', Paragraph(f"{cedula.additional_tax_property:,.2f}",
styles['CedulaMoney'])],
    ]

```

IoT-Enabled Barangay Resident and Attendance Management System

```
[Paragraph(' (b) GROSS RECEIPTS OR EARNINGS FROM BUSINESS (?2.00 for every
?5,000)', styles['CedulaValue']), '', Paragraph(f"{cedula.additional_tax_business:,.2f}",
styles['CedulaMoney'])],
]

fin_data.extend([
    Paragraph('TOTAL (Basic + Capped Additional)', styles['CedulaValue']), '',
    Paragraph(f"{(cedula.basic_tax + capped_additional):,.2f}", styles['CedulaMoney']),
    Paragraph('INTEREST', styles['CedulaValue']), '',
    Paragraph(f"{cedula.interest:,.2f}", styles['CedulaMoney']),
    Paragraph('TOTAL AMOUNT PAID', styles['CedulaHeader']), '',
    Paragraph(f"?{cedula.total_amount:,.2f}", styles['CedulaHeader']),
])

fin_table = Table(fin_data, colWidths=[3.5*inch, 2.5*inch, 1.5*inch])
fin_table.setStyle(TableStyle([
    ('BOX', (0,0), (-1,-1), 1, colors.black),
    ('INNERGRID', (0,0), (-1,-1), 0.5, colors.black),
    ('ALIGN', (2,0), (2,-1), 'RIGHT'),
    ('VALIGN', (0,0), (-1,-1), 'MIDDLE'),
]))
elements.append(fin_table)

elements.append(Spacer(1, 40))

# Signatures
label1 = "Taxpayer's Signature" if not is_corp else "Authorized Representative"
label2 = "Municipal/City Treasurer"

sig_data = [
    Paragraph('_____', styles['CedulaValue']),
    Paragraph('_____', styles['CedulaValue']),
    Paragraph(label1, styles['CedulaSubHeader']), Paragraph(label2,
styles['CedulaSubHeader'])]
]
sig_table = Table(sig_data, colWidths=[3.5*inch, 3.5*inch])
sig_table.setStyle(TableStyle([
    ('ALIGN', (0,0), (-1,-1), 'CENTER'),
]))
elements.append(sig_table)

from functools import partial
watermark_func = partial(draw_watermark, is_resident=is_resident)
doc.build(elements, onFirstPage=watermark_func)
buffer.seek(0)
return buffer

def generate_receipt_pdf(certificate):
    """
    Generate a 2-copy printable receipt (Resident & Barangay copies) on a single A4/Letter
    page.
    """
    buffer = io.BytesIO()
    # Using letter size which is 8.5 x 11 inches
    doc = SimpleDocTemplate(buffer, pagesize=letter,
                           topMargin=0.5 * inch, bottomMargin=0.5 * inch,
                           leftMargin=0.5 * inch, rightMargin=0.5 * inch)

    styles = getSampleStyleSheet()
```

IoT-Enabled Barangay Resident and Attendance Management System

```

# Custom styles for receipt
styles.add(ParagraphStyle(name='ReceiptHeader', fontSize=10, alignment=TA_CENTER,
fontName='Helvetica-Bold'))
styles.add(ParagraphStyle(name='CertHeaderLabel', fontSize=10, alignment=TA_CENTER,
fontName='Helvetica', leading=12))
styles.add(ParagraphStyle(name='ReceiptTitle', fontSize=14, alignment=TA_CENTER,
fontName='Helvetica-Bold', spaceBefore=10, spaceAfter=10))
styles.add(ParagraphStyle(name='ReceiptLabel', fontSize=11, fontName='Helvetica-
Bold'))
styles.add(ParagraphStyle(name='ReceiptValue', fontSize=11, fontName='Helvetica'))
styles.add(ParagraphStyle(name='CopyLabel', fontSize=9, alignment=TA_RIGHT,
fontName='Helvetica-Oblique', textColor=colors.grey))

def create_receipt_elements(copy_type):
    elements = []

    # Copy indicator
    elements.append(Paragraph(f"{copy_type} COPY", styles['CopyLabel']))

    # Header
    brgy_name = getattr(settings, 'BARANGAY_NAME', 'Sico-Sico')
    municipality = getattr(settings, 'BARANGAY_MUNICIPALITY', 'Gigaquit')
    province = getattr(settings, 'BARANGAY_PROVINCE', 'Surigao del Norte')

    elements.append(Paragraph('REPUBLIC OF THE PHILIPPINES', styles['ReceiptHeader']))
    elements.append(Paragraph(f'Barangay {brgy_name.upper()}, {municipality},
{province}', styles['ReceiptHeader']))
    elements.append(Paragraph('OFFICIAL RECEIPT', styles['ReceiptTitle']))

    # Receipt Data Table
    data = [
        [Paragraph('Control Number:', styles['ReceiptLabel']),
Paragraph(certificate.control_number, styles['ReceiptValue'])],
        [Paragraph('OR Number:', styles['ReceiptLabel']),
Paragraph(certificate.or_number or "N/A", styles['ReceiptValue'])],
        [Paragraph('Date Issued:', styles['ReceiptLabel']),
Paragraph(certificate.created_at.strftime('%B %d, %Y'), styles['ReceiptValue'])],
        [Paragraph('Received From:', styles['ReceiptLabel']),
Paragraph(certificate.resident.full_name, styles['ReceiptValue'])],
        [Paragraph('Nature of Payment:', styles['ReceiptLabel']),
Paragraph(certificate.get_cert_type_display(), styles['ReceiptValue'])],
        [Paragraph('Purpose:', styles['ReceiptLabel']), Paragraph(certificate.purpose,
styles['ReceiptValue'])],
        [Paragraph('Amount Paid:', styles['ReceiptLabel']), Paragraph(f"<b>PHP
{certificate.amount_paid:,.2f}</b>", styles['ReceiptValue'])],
    ]

    t = Table(data, colWidths=[1.8 * inch, 5.2 * inch])
    t.setStyle(TableStyle([
        ('GRID', (0, 0), (-1, -1), 0.5, colors.grey),
        ('PADDING', (0, 0), (-1, -1), 6),
        ('VALIGN', (0, 0), (-1, -1), 'MIDDLE'),
    ]))
    elements.append(t)
    elements.append(Spacer(1, 0.4 * inch))

    # Signature block
    sig_data = [
        [Paragraph('_____', styles['ReceiptHeader']), Spacer(1,

```

IoT-Enabled Barangay Resident and Attendance Management System

```
1), Paragraph('_____', styles['ReceiptHeader'])),
    [Paragraph('Resident Signature', styles['CertHeaderLabel']), Spacer(1, 1),
Paragraph('Barangay Collector/Treasurer', styles['CertHeaderLabel'])]
    ]
    sig_table = Table(sig_data, colWidths=[3 * inch, 1 * inch, 3 * inch])
    elements.append(sig_table)

    return elements

# Combine Resident and Barangay copies
all_elements = []

# Resident Copy
all_elements.extend(create_receipt_elements("RESIDENT"))

# Separation Dash Line
all_elements.append(Spacer(1, 0.5 * inch))
all_elements.append(Paragraph('-', ' ' * 140, styles['ReceiptHeader']))
all_elements.append(Spacer(1, 0.5 * inch))

# Barangay Copy
all_elements.extend(create_receipt_elements("BARANGAY"))

doc.build(all_elements)
buffer.seek(0)
return buffer
```


File: certifications/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.contrib import messages
from django.http import HttpResponse, FileResponse, JsonResponse
from django.core.paginator import Paginator
from django.db.models import Q
from .models import Certificate, Cedula, CertificateRequest
from officials.models import Official
from residents.models import Resident
from datetime import date
from decimal import Decimal
import uuid
from core.models import SystemSettings

def generate_control_number(cert_type):
    """Generate unique control number for certificate following Prefix-Sequence-Year."""
    prefix_map = {
        'clearance': 'BC',
        'residency': 'CR',
        'indigency': 'CI',
        'good_moral': 'GM',
        'business_permit': 'BP',
        'comelec': 'CC',
        'cedula': 'CTC',
        'late_registration': 'LR',
    }
    prefix = prefix_map.get(cert_type, 'CERT')
    year = date.today().year

    # Count all certificates of this type in the current year
    count = Certificate.objects.filter(
        certificate_request__cert_type=cert_type,
        created_at__year=year
    ).count() + 1

    return f"{prefix}-{count:03d}-{year}"

def generate_or_number():
    """Generate the next OR number by incrementing the last numeric one from
    OfficialReceipt."""
    from payments.models import OfficialReceipt
    import re
    receipts = OfficialReceipt.objects.order_by('-id')[:20]
    base_or = None
    for r in receipts:
        if re.search(r'\d+$', r.or_number):
            base_or = r.or_number
            break
    if not base_or:
        return OfficialReceipt.generate_next_or_number()
    try:
        match = re.search(r'(\d+)$', base_or)
        number_str = match.group(1)
        number = int(number_str)
        prefix = base_or[:match.start()]
    
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        return f"{prefix}{number + 1:0{len(number_str)}d}"
    except Exception:
        return OfficialReceipt.generate_next_or_number()

def generate_ctc_number():
    """Generate the next CTC (Cedula) number."""
    from certifications.models import Cedula
    last_cedula = Cedula.objects.order_by('-id').first()
    year = date.today().year

    if not last_cedula or not last_cedula.ctc_number:
        return f"CCIC{year}001"

    import re
    match = re.search(r'(\d+)$', last_cedula.ctc_number)
    if match:
        number_str = match.group(1)
        # Handle cases where the year might be part of the number string
        # e.g., 2026001. If it's too long, we might just want the last 3 digits.
        # But let's assume the prefix is constant for the year
        prefix = last_cedula.ctc_number[:match.start()]
        number = int(number_str)
        return f"{prefix}{number + 1:0{len(number_str)}d}"

    return f"CCIC{year}001"

def get_certificate_rate(cert_type):
    """Return the standard rate for a certificate type."""
    rates = {
        'clearance': Decimal('100.00'),
        'residency': Decimal('50.00'),
        'indigency': Decimal('0.00'),
        'good_moral': Decimal('50.00'),
        'business_permit': Decimal('500.00'),
        'comelec': Decimal('50.00'),
        'cedula': Decimal('0.00'), # Calculated dynamically
        'late_registration': Decimal('50.00'),
    }
    return rates.get(cert_type, Decimal('50.00'))

@login_required
def get_next_numbers(request):
    """AJAX view to suggest next Control and OR numbers."""
    cert_type = request.GET.get('type')
    if not cert_type:
        return JsonResponse({})

    next_control = generate_control_number(cert_type)
    next_or = generate_or_number()
    next_ctc = generate_ctc_number() if cert_type == 'cedula' else ""
    rate = get_certificate_rate(cert_type)

    return JsonResponse({
        'control_number': next_control,
        'or_number': next_or,
        'ctc_number': next_ctc,
        'rate': format(rate, '.2f')
    })
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}}
```

```
@login_required
def certificate_list(request):
    """List all certificates."""
    query = request.GET.get('q', '')
    cert_type = request.GET.get('type', '')
    date_from = request.GET.get('date_from', '')
    date_to = request.GET.get('date_to', '')

    certs = Certificate.objects.select_related('resident', 'issued_by')

    if query:
        certs = certs.filter(
            Q(control_number__icontains=query) |
            Q(resident__first_name__icontains=query) |
            Q(resident__last_name__icontains=query)
        )
    if cert_type:
        certs = certs.filter(certificate_request__cert_type=cert_type)
    if date_from:
        certs = certs.filter(date_issued__gte=date_from)
    if date_to:
        certs = certs.filter(date_issued__lte=date_to)

    # Residents can only see their own certificates
    role = 'staff'
    try:
        role = request.user.profile.role
    except Exception:
        pass

    if role == 'resident':
        try:
            own_resident = request.user.profile.resident
            if own_resident:
                certs = certs.filter(resident=own_resident)
            else:
                certs = Certificate.objects.none()
        except Exception:
            certs = Certificate.objects.none()

    paginator = Paginator(certs, 25)
    page = request.GET.get('page')
    certs = paginator.get_page(page)
    return render(request, 'certifications/list.html', {
        'certificates': certs,
        'query': query,
        'selected_type': cert_type,
        'cert_types': Certificate.TYPE_CHOICES,
        'is_resident_view': True,
    })

paginator = Paginator(certs, 25)
page = request.GET.get('page')
certs = paginator.get_page(page)

context = {
    'certificates': certs,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'query': query,
'selected_type': cert_type,
'cert_types': Certificate.TYPE_CHOICES,
}
return render(request, 'certifications/list.html', context)

@login_required
def certificate_issue(request):
    """Issue a new certificate ? admin/staff only."""
    # Residents cannot issue certificates ? refer to barangay office
    role = 'staff'
    try:
        role = request.user.profile.role
    except Exception:
        pass

    if role == 'resident':
        messages.error(request, "Certificate issuance must be done at the barangay office.
Please visit us in person.")
        return redirect('core:dashboard')

    if request.method == 'POST':
        cert_type = request.POST.get('cert_type')
        resident_id = request.POST.get('resident')
        resident = get_object_or_404(Resident, pk=resident_id)

        def to_decimal(val):
            if not val: return Decimal('0.00')
            try:
                clean_val = str(val).replace(',', '')
                return Decimal(clean_val)
            except:
                return Decimal('0.00')

        control_number = request.POST.get('control_number') or
generate_control_number(cert_type)

        # Build or find a backing CertificateRequest so Certificate can reference it
        cert_req, _ = CertificateRequest.objects.get_or_create(
            resident=resident,
            cert_type=cert_type,
            status='pending',
            defaults={
                'purpose': request.POST.get('purpose', ''),
                'business_name': request.POST.get('business_name', ''),
                'business_address': request.POST.get('business_address', ''),
                'business_type': request.POST.get('business_type', ''),
                'child_name': request.POST.get('child_name', ''),
                'child_birth_date': request.POST.get('child_birth_date') or None,
                'child_birth_place': request.POST.get('child_birth_place', ''),
                'father_name': request.POST.get('father_name', ''),
                'mother_name': request.POST.get('mother_name', ''),
            }
        )

        cert = Certificate(
            control_number=control_number,
            resident=resident,
            certificate_request=cert_req,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        business_name=request.POST.get('business_name', ''),
        business_address=request.POST.get('business_address', ''),
        business_type=request.POST.get('business_type', ''),
        issued_by=request.user,
        status='issued',
    )

    if cert_type == 'late_registration':
        cert.child_name = request.POST.get('child_name', '')
        cert.child_birth_date = request.POST.get('child_birth_date') or None
        cert.child_birth_place = request.POST.get('child_birth_place', '')
        cert.father_name = request.POST.get('father_name', '')
        cert.mother_name = request.POST.get('mother_name', '')

    cert.save()

    # Capture OR Number and Amount Paid for manual issuance
    or_number = request.POST.get('or_number')
    amount_paid = to_decimal(request.POST.get('amount_paid', 0))

    if or_number:
        from payments.models import Payment, OfficialReceipt

        # Create a Payment record linked to the request (which is linked to the
certificate)
        payment = Payment.objects.create(
            amount=amount_paid,
            status='paid',
            method='cash'
        )
        cert_req.payment = payment
        cert_req.save()

        # Create the OfficialReceipt
        existing_receipt = OfficialReceipt.objects.filter(or_number=or_number).first()
        if existing_receipt:
            payment.official_receipt = existing_receipt
            payment.save()
        else:
            receipt = OfficialReceipt.objects.create(
                or_number=or_number,
                resident=resident,
                amount=amount_paid,
                particulars=f"{cert.get_cert_type_display()} - {cert_req.purpose}",
                issued_by=request.user
            )
            payment.official_receipt = receipt
            payment.save()

    # Mark request as issued
    cert_req.status = 'issued'
    cert_req.processed_by = request.user
    from django.utils import timezone as tz
    cert_req.processed_at = tz.now()
    cert_req.save()

    if cert_type == 'cedula':
        from certifications.models import Cedula
        Cedula.objects.create(
            certificate=cert,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        ctc_number=request.POST.get('ctc_number'),
        taxpayer_type=request.POST.get('taxpayer_type', 'individual'),
        place_of_issue=request.POST.get('place_of_issue', 'Sico-Sico, Gigaquit'),
        height=request.POST.get('height', ''),
        weight=request.POST.get('weight', ''),
        raw_taxable_property=to_decimal(request.POST.get('raw_taxable_property')),
        raw_taxable_business=to_decimal(request.POST.get('raw_taxable_business')),
        raw_taxable_income=to_decimal(request.POST.get('raw_taxable_income')),
        basic_tax=to_decimal(request.POST.get('basic_tax')),

    additional_tax_property=to_decimal(request.POST.get('additional_tax_property')),

    additional_tax_business=to_decimal(request.POST.get('additional_tax_business')),

    additional_tax_income=to_decimal(request.POST.get('additional_tax_income')),
    interest=to_decimal(request.POST.get('interest'))
)

    messages.success(request, f'{cert.get_cert_type_display()} issued for
{resident.full_name}. Control #: {cert.control_number}')
    return redirect('certifications:view', pk=cert.pk)

    residents = Resident.objects.filter(is_active=True)
    query = request.GET.get('q', '')
    if query:
        residents = residents.filter(
            Q(first_name__icontains=query) |
            Q(last_name__icontains=query)
        )
    context = {
        'residents': residents[:50],
        'cert_types': Certificate.TYPE_CHOICES,
        'query': query,
    }
    return render(request, 'certifications/issue.html', context)

@login_required
def certificate_view(request, pk):
    """View certificate details."""
    cert = get_object_or_404(Certificate.objects.select_related('resident', 'issued_by'),
pk=pk)
    return render(request, 'certifications/view.html', {'certificate': cert})

@login_required
def certificate_pdf(request, pk):
    """Generate and download certificate PDF."""
    from .utils import generate_certificate_pdf
    cert = get_object_or_404(Certificate.objects.select_related('resident'), pk=pk)
    role = 'resident'
    try:
        role = request.user.profile.role
    except Exception:
        pass

    pdf_buffer = generate_certificate_pdf(cert, is_resident=(role == 'resident'))
    response = HttpResponse(pdf_buffer, content_type='application/pdf')
    response['Content-Disposition'] = f'inline; filename="{cert.control_number}.pdf"'
    return response
```

IoT-Enabled Barangay Resident and Attendance Management System

```
@login_required
def certificate_search(request):
    """Search certificates by control number."""
    control = request.GET.get('control', '')
    cert = None
    if control:
        cert =
Certificate.objects.filter(control_number=control).select_related('resident').first()
    return render(request, 'certifications/search.html', {
        'certificate': cert,
        'control': control,
    })

@login_required
def check_uniqueness(request):
    """AJAX view to check if an OR Number or CTC Number is already taken."""
    from django.http import JsonResponse
    from .models import Certificate, Cedula

    field = request.GET.get('field')
    value = request.GET.get('value')

    if not field or not value:
        return JsonResponse({'available': True})

    if field == 'or_number':
        from payments.models import OfficialReceipt
        exists = OfficialReceipt.objects.filter(or_number=value).exists()
    elif field == 'ctc_number':
        exists = Cedula.objects.filter(ctc_number=value).exists()
    else:
        exists = False

    return JsonResponse({'available': not exists})

@login_required
def certificate_receipt_pdf(request, pk):
    """Generate and download transaction receipt PDF."""
    from .utils import generate_receipt_pdf
    cert = get_object_or_404(Certificate.objects.select_related('resident'), pk=pk)
    pdf_buffer = generate_receipt_pdf(cert)
    response = HttpResponse(pdf_buffer, content_type='application/pdf')
    response['Content-Disposition'] = f'inline;
filename="RECEIPT-{cert.control_number}.pdf"'
    return response

@login_required
def certificate_delete(request, pk):
    """Delete a certificate (POST only) ? admin/staff only."""
    role = 'staff'
    try:
        role = request.user.profile.role
    except Exception:
        pass
    if role == 'resident':
        messages.error(request, "You do not have permission to perform this action.")
        return redirect('certifications:list')
```

IoT-Enabled Barangay Resident and Attendance Management System

```

cert = get_object_or_404(Certificate, pk=pk)
if request.method == 'POST':
    control_number = cert.control_number
    cert.delete()
    messages.success(request, f'Certificate {control_number} has been deleted.')
    return redirect('certifications:list')

# ??? Certificate Requests ?????????????????????????????????????????????????????????????

@login_required
def request_certificate(request):
    """Resident submits a certificate request."""
    role = 'staff'
    own_resident = None
    try:
        role = request.user.profile.role
        own_resident = request.user.profile.resident
    except Exception:
        pass

    if role != 'resident':
        messages.error(request, "This page is for residents only.")
        return redirect('core:dashboard')

    if own_resident is None:
        messages.error(request, "Your account is not linked to a resident record. Please contact the barangay office.")
        return redirect('core:dashboard')

    if request.method == 'POST':
        cert_type = request.POST.get('cert_type')
        purpose = request.POST.get('purpose', '').strip()

        if not cert_type or not purpose:
            messages.error(request, "Please fill in all required fields.")
        elif CertificateRequest.objects.filter(resident=own_resident, cert_type=cert_type, status='pending').exists():
            messages.warning(request, "You already have a pending request for this certificate type.")
        else:
            cert_req = CertificateRequest.objects.create(
                resident=own_resident,
                cert_type=cert_type,
                purpose=purpose,

                # Business fields
                business_name=request.POST.get('business_name', ''),
                business_address=request.POST.get('business_address', ''),
                business_type=request.POST.get('business_type', ''),

                # Cedula fields
                taxpayer_type=request.POST.get('taxpayer_type', ''),
                place_of_issue=request.POST.get('place_of_issue', ''),
                height=request.POST.get('height', ''),
                weight=request.POST.get('weight', ''),
                raw_taxable_property=Decimal(request.POST.get('raw_taxable_property') or '0.00'),
                raw_taxable_business=Decimal(request.POST.get('raw_taxable_business') or '0.00'),
            )

```


IoT-Enabled Barangay Resident and Attendance Management System

```
raw_taxable_income=Decimal(request.POST.get('raw_taxable_income') or
'0.00'),

    # Late Registration details
    child_name=request.POST.get('child_name', ''),
    child_birth_date=request.POST.get('child_birth_date') or None,
    child_birth_place=request.POST.get('child_birth_place', ''),
    father_name=request.POST.get('father_name', ''),
    mother_name=request.POST.get('mother_name', ''),

    # Waiver flags
    is_first_time_jobseeker=(request.POST.get('is_first_time_jobseeker') ==
'on'),
    )
    messages.success(request, f"Your request for a {cert_type} has been submitted!
Tracking Code: {cert_req.tracking_code}")
    return redirect('payments:choose_payment', request_id=cert_req.id)

    return render(request, 'certifications/request_form.html', {
        'cert_types': Certificate.TYPE_CHOICES,
    })

@login_required
def my_requests(request):
    """Resident's own certificate request history."""
    role = 'staff'
    own_resident = None
    try:
        role = request.user.profile.role
        own_resident = request.user.profile.resident
    except Exception:
        pass

    if role != 'resident':
        return redirect('certifications:request_list')

    reqs =
CertificateRequest.objects.filter(resident=own_resident).order_by('-created_at')
    return render(request, 'certifications/my_requests.html', {'requests': reqs})

@login_required
def request_list(request):
    """Admin view: list all certificate requests with status tabs."""
    role = 'staff'
    try:
        role = request.user.profile.role
    except Exception:
        pass

    if role == 'resident':
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    status_filter = request.GET.get('status', 'pending')
    reqs = CertificateRequest.objects.select_related('resident', 'processed_by').all()
    if status_filter:
        reqs = reqs.filter(status=status_filter)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
paginator = Paginator(reqs, 20)
reqs = paginator.get_page(request.GET.get('page'))

pending_count = CertificateRequest.objects.filter(status='pending').count()

return render(request, 'certifications/request_list.html', {
    'requests': reqs,
    'status_filter': status_filter,
    'pending_count': pending_count,
})

@login_required
def fulfill_request(request, pk):
    """Admin: fulfill a pending request by issuing a certificate."""
    from django.utils import timezone
    from core.models import Notification

    role = 'staff'
    try:
        role = request.user.profile.role
    except Exception:
        pass

    if role == 'resident':
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    cert_request = get_object_or_404(CertificateRequest, pk=pk)

    if cert_request.status != 'pending':
        messages.error(request, "This request has already been processed.")
        return redirect('certifications:request_list')

    # Bypassed payment status blocker to allow admin to open fulfillment form
    if not hasattr(cert_request, 'payment') or not cert_request.payment:
        # If for some reason no payment object exists, we should probably create one or
        handle it
        pass

    if request.method == 'POST':
        def to_decimal(val):
            if not val:
                return Decimal('0.00')
            try:
                return Decimal(str(val).replace(',', ' '))
            except Exception:
                return Decimal('0.00')

        control_number = request.POST.get('control_number') or
        generate_control_number(cert_request.cert_type)

        # Update cert_request with potentially modified details from sender
        cert_request.purpose = request.POST.get('purpose', cert_request.purpose)
        cert_request.business_name = request.POST.get('business_name',
        cert_request.business_name)
        cert_request.business_address = request.POST.get('business_address',
        cert_request.business_address)
        cert_request.business_type = request.POST.get('business_type',
        cert_request.business_type)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        if cert_request.cert_type == 'late_registration':
            cert_request.child_name = request.POST.get('child_name',
cert_request.child_name)
            cert_request.child_birth_date = request.POST.get('child_birth_date') or
cert_request.child_birth_date
            cert_request.child_birth_place = request.POST.get('child_birth_place',
cert_request.child_birth_place)
            cert_request.father_name = request.POST.get('father_name',
cert_request.father_name)
            cert_request.mother_name = request.POST.get('mother_name',
cert_request.mother_name)

        cert_request.save()

    cert = Certificate.objects.create(
        control_number=control_number,
        resident=cert_request.resident,
        certificate_request=cert_request,
        issued_by=request.user,
        status='issued',
        business_name=cert_request.business_name,
        business_address=cert_request.business_address,
        business_type=cert_request.business_type,
        child_name=cert_request.child_name,
        child_birth_date=cert_request.child_birth_date,
        child_birth_place=cert_request.child_birth_place,
        father_name=cert_request.father_name,
        mother_name=cert_request.mother_name,
    )

    if cert.cert_type == 'cedula':
        from certifications.models import Cedula
        ctc_number = request.POST.get('ctc_number', '')
        Cedula.objects.create(
            certificate=cert,
            ctc_number=ctc_number,
            taxpayer_type=request.POST.get('taxpayer_type', 'individual'),
            place_of_issue=request.POST.get('place_of_issue', 'Sico-Sico, Gigaquit'),
            height=request.POST.get('height', ''),
            weight=request.POST.get('weight', ''),
            raw_taxable_property=to_decimal(request.POST.get('raw_taxable_property',
0)),
            raw_taxable_business=to_decimal(request.POST.get('raw_taxable_business',
0)),
            raw_taxable_income=to_decimal(request.POST.get('raw_taxable_income', 0)),
        )

    # Capture OR Number and Amount Paid
    or_number = request.POST.get('or_number')
    amount_paid = to_decimal(request.POST.get('amount_paid', 0))
    is_exempt = request.POST.get('is_exempt') == 'on'

    # Determine status and method based on exemption
    payment_status = 'waived' if is_exempt else 'paid'
    payment_method = 'waived' if is_exempt else 'cash'

    if or_number:
        from payments.models import Payment, OfficialReceipt
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# 1. Ensure Payment exists
if not hasattr(cert_request, 'payment') or not cert_request.payment:
    payment = Payment.objects.create(
        amount=amount_paid,
        status=payment_status,
        method=payment_method
    )
    cert_request.payment = payment
    cert_request.save()
else:
    payment = cert_request.payment
    payment.amount = amount_paid
    payment.status = payment_status
    payment.method = payment_method
    payment.save()

# 2. Ensure OfficialReceipt exists
if not payment.official_receipt:
    # Check if OR number already exists to avoid unique constraint error
    existing_receipt =
OfficialReceipt.objects.filter(or_number=or_number).first()
    if existing_receipt:
        payment.official_receipt = existing_receipt
        payment.save()
    else:
        receipt = OfficialReceipt.objects.create(
            or_number=or_number,
            resident=cert_request.resident,
            amount=amount_paid,
            particulars=f"{cert_request.get_cert_type_display()} -
{cert_request.purpose}" if is_exempt else ''',
            issued_by=request.user
        )
        payment.official_receipt = receipt
        payment.save()
    else:
        receipt = payment.official_receipt
        receipt.or_number = or_number
        receipt.amount = amount_paid
        receipt.save()

cert_request.status = 'issued'
cert_request.processed_by = request.user
cert_request.processed_at = timezone.now()
cert_request.save()

# Notify resident
try:
    resident_user = cert_request.resident.user_profile.user
    Notification.objects.create(
        user=resident_user,
        message=f'Your request for a {cert_request.get_cert_type_display()} has
been issued! Control #: {cert.control_number}.',
        link=f'/certifications/{cert.pk}/',
    )
except Exception:
    pass

messages.success(request, f'Certificate {cert.control_number} issued and resident
notified.')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        return redirect('certifications:request_list')

# GET: show the fulfillment form
return render(request, 'certifications/fulfill_form.html', {
    'cert_request': cert_request,
    'suggested_control': generate_control_number(cert_request.cert_type),
    'suggested_or': generate_or_number(),
    'suggested_ctc': generate_ctc_number() if cert_request.cert_type == 'cedula' else
    "",
    'base_amount': get_certificate_rate(cert_request.cert_type),
})

@login_required
def reject_request(request, pk):
    """Admin: reject a pending certificate request."""
    from django.utils import timezone
    from core.models import Notification

    role = 'staff'
    try:
        role = request.user.profile.role
    except Exception:
        pass

    if role == 'resident':
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    cert_request = get_object_or_404(CertificateRequest, pk=pk)

    if cert_request.status != 'pending':
        messages.error(request, "This request has already been processed.")
        return redirect('certifications:request_list')

    if request.method == 'POST':
        notes = request.POST.get('notes', '').strip()
        cert_request.status = 'rejected'
        cert_request.notes = notes
        cert_request.processed_by = request.user
        cert_request.processed_at = timezone.now()
        cert_request.save()

        # Notify resident
        try:
            resident_user = cert_request.resident.user_profile.user
            note_text = f' Reason: {notes}' if notes else ''
            Notification.objects.create(
                user=resident_user,
                message=f'Your request for a {cert_request.get_cert_type_display()} was
not approved.{note_text}',
                link='/certifications/my-requests/',
            )
        except Exception:
            pass

        messages.warning(request, f'Request #{pk} has been rejected.')
        return redirect('certifications:request_list')

    return redirect('certifications:request_list')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
@login_required
def virtual_certificate(request, pk):
    """View certificate as a 'virtual' document (card/styled)."""
    cert = get_object_or_404(Certificate.objects.select_related('resident', 'issued_by'),
                             pk=pk)

    # Check if resident owns this certificate
    try:
        role = request.user.profile.role
        if role == 'resident':
            if cert.resident != request.user.profile.resident:
                messages.error(request, "Permission denied.")
                return redirect('core:dashboard')
    except Exception:
        pass

    settings = SystemSettings.objects.first()

    # Logic for current captain
    captain = Official.objects.filter(position='captain', status='active').first()
    captain_name = captain.resident.full_name if captain else "OFFICIAL CAPTAIN NAME"

    role = 'resident'
    try:
        role = request.user.profile.role
    except Exception:
        pass

    return render(request, 'certifications/virtual_certificate.html', {
        'certificate': cert,
        'settings': settings,
        'captain_name': captain_name,
        'user_role': role
    })
```

File: core/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: core/admin.py

```
from django.contrib import admin
from django.contrib.auth.admin import UserAdmin as BaseUserAdmin
from django.contrib.auth import get_user_model
from .models import Role, UserProfile, SystemSettings
User = get_user_model()

# Unregister default User admin if already registered
try: admin.site.unregister(User)
except admin.sites.NotRegistered: pass

@admin.register(User)
class UserAdmin(BaseUserAdmin):
    list_display = ('username', 'email', 'first_name', 'last_name', 'get_role',
                    'is_staff')
    list_filter = ('role', 'is_staff', 'is_superuser', 'is_active')
    search_fields = ('username', 'first_name', 'last_name', 'email')

    def get_role(self, obj):
        return obj.get_role_display()
    get_role.short_description = 'Role'

@admin.register(Role)
class RoleAdmin(admin.ModelAdmin):
    list_display = ('display_name', 'name', 'permission_level')
    search_fields = ('name', 'display_name')
    list_editable = ('permission_level',)

@admin.register(SystemSettings)
class SystemSettingsAdmin(admin.ModelAdmin):
    list_display = ('barangay_name', 'municipality', 'province')
```


IoT-Enabled Barangay Resident and Attendance Management System

File: core/apps.py

```
from django.apps import AppConfig
```

```
class CoreConfig(AppConfig):  
    name = 'core'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/context_processors.py

```
from .models import UserProfile, Notification

def user_role_context(request):
    """Inject user_role, user_profile, and notification data into every template
    context."""
    if request.user.is_authenticated:
        try:
            profile = request.user.profile
            role = profile.role
        except UserProfile.DoesNotExist:
            profile = None
            role = 'admin' if request.user.is_superuser else 'staff'
    else:
        profile = None
        role = None

    pending_count = 0
    if role in ['captain', 'admin', 'secretary', 'treasurer']:
        from residents.models import ResidentRegistration
        pending_count = ResidentRegistration.objects.filter(status='pending').count()

    # Pending certificate requests count (for admin nav badge)
    pending_cert_requests = 0
    if role and role != 'resident':
        from certifications.models import CertificateRequest
        pending_cert_requests = CertificateRequest.objects.filter(status='pending').count()

    # Notifications for the current user
    unread_notifications_count = 0
    recent_notifications = []
    if request.user.is_authenticated:
        unread_notifications_count = Notification.objects.filter(
            user=request.user, is_read=False
        ).count()
        recent_notifications = list(
            Notification.objects.filter(user=request.user).order_by('-created_at')[:5]
        )

    return {
        'user_role': role,
        'user_profile': profile,
        'pending_registrations_count': pending_count,
        'pending_cert_requests_count': pending_cert_requests,
        'unread_notifications_count': unread_notifications_count,
        'recent_notifications': recent_notifications,
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/decorators.py

```
from django.http import HttpResponseRedirect
from functools import wraps

def role_required(*allowed_groups):
    """
    Decorator for views that checks whether a user is in the required groups,
    based on the 3 permission levels (admin, staff, resident).
    """
    def decorator(view_func):
        @wraps(view_func)
        def _wrapped_view(request, *args, **kwargs):
            if not request.user.is_authenticated:
                from django.contrib.auth.views import redirect_to_login
                return redirect_to_login(request.get_full_path())

            # Allow superusers by default
            if request.user.is_superuser:
                return view_func(request, *args, **kwargs)

            # Check if user has any of the allowed groups
            user_groups = request.user.groups.values_list('name', flat=True)
            user_groups_set = set(g.lower() for g in user_groups)
            allowed_set = set(a.lower() for a in allowed_groups)

            if user_groups_set.intersection(allowed_set):
                return view_func(request, *args, **kwargs)

            return HttpResponseRedirect("You do not have permission to access this
page.")
        return _wrapped_view
    return decorator
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/management/__init__.py

```
# init
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/management/commands/__init__.py

```
# init
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/management/commands/seed_roles.py

```
from django.core.management.base import BaseCommand
from core.models import Role
from officials.models import Official

class Command(BaseCommand):
    help = 'Seed roles based on official positions'

    def handle(self, *args, **options):
        # Format: (name, display_name, permission_level)
        roles_to_create = [
            ('admin', 'Administrator', 1),
            ('captain', 'Punong Barangay (Captain)', 1),
            ('secretary', 'Barangay Secretary', 2),
            ('treasurer', 'Barangay Treasurer', 2),
            ('kagawad', 'Sangguniang Barangay Member', 2),
            ('sk_chairman', 'SK Chairperson', 2),
            ('sk_kagawad', 'SK Kagawad', 2),
            ('tanod', 'Barangay Tanod', 2),
            ('bhw', 'Barangay Health Worker', 2),
            ('nutrition_scholar', 'Barangay Nutrition Scholar', 2),
            ('day_care_worker', 'Day Care Worker', 2),
            ('lupon', 'Lupon Member', 2),
            ('clerk', 'Barangay Clerk', 2),
            ('staff', 'Barangay Staff', 2),
            ('resident', 'Resident', 3),
        ]

        for name, display_name, level in roles_to_create:
            role, created = Role.objects.update_or_create(
                name=name,
                defaults={
                    'display_name': display_name,
                    'permission_level': level
                }
            )
            if created:
                self.stdout.write(self.style.SUCCESS(f'Created role: {name}'))
            else:
                self.stdout.write(f'Updated role: {name}')

        self.stdout.write(self.style.SUCCESS('Successfully seeded roles'))
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/middleware.py

```
import time
from django.conf import settings
from django.contrib.auth import logout
from django.shortcuts import redirect
from django.contrib import messages

class AutoLogoutMiddleware:
    """
    Auto-logout middleware after 30 minutes of inactivity.
    """
    def __init__(self, get_response):
        self.get_response = get_response

    def __call__(self, request):
        if request.user.is_authenticated:
            current_time = time.time()
            last_activity = request.session.get('last_activity', None)

            if last_activity:
                # Logout if inactive for longer than SESSION_COOKIE_AGE settings
                if (current_time - last_activity) > settings.SESSION_COOKIE_AGE:
                    logout(request)
                    messages.info(request, "Your session has expired due to 30 minutes of inactivity.")
                    return redirect(settings.LOGIN_URL)

            # Update the last recorded activity time
            request.session['last_activity'] = current_time

        return self.get_response(request)
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/models.py

```
from django.db import models
from django.contrib.auth.models import AbstractUser, Group
from django.db.models.signals import post_save
from django.dispatch import receiver
from django.conf import settings

class Role(models.Model):
    """Lookup table for user roles."""
    name = models.CharField(max_length=30, unique=True)
    display_name = models.CharField(max_length=100)
    permission_level = models.IntegerField(
        default=3,
        help_text="1=Admin, 2=Staff, 3=Resident",
    )

    def __str__(self):
        return self.display_name

    class Meta:
        ordering = ['permission_level', 'name']
        verbose_name = 'Role'
        verbose_name_plural = 'Roles'

class User(AbstractUser):
    role = models.ForeignKey(
        Role,
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name='users',
    )
    avatar = models.ImageField(upload_to="avatars/", blank=True, null=True)

    def get_permission_level(self):
        if self.is_superuser:
            return 1
        if self.role:
            return self.role.permission_level
        return 3

    def get_role_display(self):
        if self.is_superuser:
            return "Administrator"
        if self.role:
            return self.role.display_name
        return "Resident"

class UserProfile(models.Model):
    """Extended user profile ? bridges User ? Resident."""

    user = models.OneToOneField(settings.AUTH_USER_MODEL, on_delete=models.CASCADE,
        related_name="profile")
    resident = models.OneToOneField("residents.Resident", on_delete=models.SET_NULL,
        null=True, blank=True, related_name="user_profile")
```


IoT-Enabled Barangay Resident and Attendance Management System

```
created_at = models.DateTimeField(auto_now_add=True)

@property
def role(self):
    return self.user.role.name if self.user.role else None

@role.setter
def role(self, value):
    try:
        role_obj = Role.objects.get(name=value)
        self.user.role = role_obj
        self.user.save()
    except Role.DoesNotExist:
        pass

@property
def avatar(self):
    return self.user.avatar

@property
def middle_name(self):
    return self.resident.middle_name if self.resident else ""

@middle_name.setter
def middle_name(self, value):
    if self.resident:
        self.resident.middle_name = value
        self.resident.save()

@property
def phone(self):
    return self.resident.contact_number if self.resident else ""

@phone.setter
def phone(self, value):
    if self.resident:
        self.resident.contact_number = value
        self.resident.save()

@property
def philSys_id(self):
    return self.resident.philSys_number if self.resident else None

@philSys_id.setter
def philSys_id(self, value):
    if self.resident:
        self.resident.philSys_number = value
        self.resident.save()

@property
def is_philsys_verified(self):
    return self.resident.is_philsys_verified if self.resident else False

@is_philsys_verified.setter
def is_philsys_verified(self, value):
    if self.resident:
        self.resident.is_philsys_verified = value
        self.resident.save()

@property
```

IoT-Enabled Barangay Resident and Attendance Management System

```
def philsys_verified_at(self):
    return self.resident.philsys_verified_at if self.resident else None

@philsys_verified_at.setter
def philsys_verified_at(self, value):
    if self.resident:
        self.resident.philsys_verified_at = value
        self.resident.save()

@property
def fingerprint_template(self):
    return self.resident.fingerprint_template if self.resident else None

@fingerprint_template.setter
def fingerprint_template(self, value):
    if self.resident:
        self.resident.fingerprint_template = value
        self.resident.save()

def __str__(self):
    return f"{self.user.get_full_name()} - {self.user.get_role_display()}"

class Meta:
    verbose_name = "User Profile"
    verbose_name_plural = "User Profiles"

class SystemSettings(models.Model):
    """Barangay system settings."""

    barangay_name = models.CharField(max_length=200, default="Barangay Sample")
    municipality = models.CharField(max_length=200, default="Municipality of Sample")
    province = models.CharField(max_length=200, default="Province of Sample")
    region = models.CharField(max_length=200, default="Region Sample")
    barangay_logo = models.ImageField(upload_to="settings/", blank=True, null=True)

    def __str__(self):
        return self.barangay_name

    class Meta:
        verbose_name = "System Settings"
        verbose_name_plural = "System Settings"

class Notification(models.Model):
    """In-app notification for a specific user."""
    user = models.ForeignKey(
        settings.AUTH_USER_MODEL, on_delete=models.CASCADE, related_name='notifications'
    )
    message = models.TextField()
    link = models.CharField(max_length=300, blank=True, help_text='Optional URL path for the notification')
    is_read = models.BooleanField(default=False)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return f"Notification for {self.user.username}: {self.message[:50]}"

    class Meta:
        ordering = ['-created_at']
```

IoT-Enabled Barangay Resident and Attendance Management System

```
verbose_name = 'Notification'
verbose_name_plural = 'Notifications'

@receiver(post_save, sender=settings.AUTH_USER_MODEL)
def assign_superuser_role(sender, instance, created, **kwargs):
    """Automatically assign Admin role to superusers."""
    if created and instance.is_superuser and not instance.role:
        try:
            admin_role = Role.objects.filter(models.Q(name='admin') |
models.Q(name='captain')).first()
            if admin_role:
                instance.role = admin_role
                instance.save()
        except Exception:
            pass

@receiver(post_save, sender=settings.AUTH_USER_MODEL)
def assign_user_group(sender, instance, created, **kwargs):
    """Assign users to proper groups based on permission level."""
    level = instance.get_permission_level()
    group_name = None
    if level == 1:
        group_name = "Admin"
    elif level == 2:
        group_name = "Staff"
    elif level == 3:
        group_name = "Resident"

    if group_name:
        group, _ = Group.objects.get_or_create(name=group_name)
        # Clear existing groups to strictly enforce current role level
        instance.groups.clear()
        instance.groups.add(group)

from django.db.models.signals import pre_delete

@receiver(pre_delete, sender=settings.AUTH_USER_MODEL)
def cleanup_resident_on_user_delete(sender, instance, **kwargs):
    """Ensure Resident record is deleted when User is deleted."""
    try:
        # Check if there was an associated resident profile
        # Use hasattr or try/except because profile might have been deleted already by
CASCADE
        if hasattr(instance, 'profile') and instance.profile.resident:
            instance.profile.resident.delete()
    except Exception:
        pass
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/test_usb_hardware.py

```
import usb.core
import usb.util
import sys

def test_hardware():
    VID = 0x1b55
    PID = 0x0124

    print(f"Searching for registered fingerprint USB hardware (ID
{VID:04x}:{PID:04x})...")

    # Find device
    dev = usb.core.find(idVendor=VID, idProduct=PID)

    if dev is None:
        print("Error: fingerprint hardware NOT found on USB bus.")
        print("Please ensure it is plugged in and check 'lsusb' output.")
        return

    print("Success: fingerprint hardware DETECTED!")

    try:
        # Check if we can read the device descriptor
        print(f"Device Manufacturer: {usb.util.get_string(dev, dev.iManufacturer)}")
        print(f"Device Product: {usb.util.get_string(dev, dev.iProduct)}")

        # Try to set configuration (requires udev rule permissions)
        try:
            dev.set_configuration()
            print("Success: Device configuration SET (permissions OK).")
        except usb.core.USBError as e:
            if e.errno == 13:
                print("Error: Permission denied. Please ensure the udev rules are
installed and correctly triggered.")
            else:
                print(f"Error setting configuration: {e}")

    except Exception as e:
        print(f"Successfully detected, but could not read details: {e}")
        print("This is usually a permission issue. Please ensure udev rules are set
correctly.")

if __name__ == "__main__":
    test_hardware()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/tests.py

```
from django.test import TestCase
```

```
# Create your tests here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/urls.py

```
from django.urls import path
from . import views

app_name = 'core'

urlpatterns = [
    path('', views.dashboard, name='dashboard'),
    path('login/', views.login_view, name='login'),
    path('signup/', views.signup_view, name='signup'),
    path('logout/', views.logout_view, name='logout'),
    path('map/', views.barangay_map, name='barangay_map'),

    # Biometric APIs
    path('biometric-templates/', views.biometric_templates, name='biometric_templates'),
    path('biometric-verify-login/', views.biometric_verify_login,
name='biometric_verify_login'),
    path('biometric-verify-login-start/', views.biometric_verify_login_start,
name='biometric_verify_login_start'),
    path('biometric-login-start/', views.biometric_login_start,
name='biometric_login_start'),
    path('biometric-status-check/', views.biometric_status_check,
name='biometric_status_check'),
    path('biometric-attendance-start/', views.biometric_attendance_start,
name='biometric_attendance_start'),
    path(
        'biometric-attendance-status-check/',
        views.biometric_attendance_status_check,
        name='biometric_attendance_status_check',
    ),
    path('biometric-login-complete/', views.biometric_login_complete,
name='biometric_login_complete'),
    path('biometric-reset-all/', views.biometric_reset_all, name='biometric_reset_all'),

    # ESP32 self-registration ? called by the device on every boot to announce its current
    IP
    path('api/esp32-heartbeat/', views.esp32_heartbeat, name='esp32_heartbeat'),

    path('backup/choose/', views.backup_setup, name='backup_setup'),
    path('backup/download/', views.backup_download, name='backup_download'),
    path('backup/execute/', views.backup_execute, name='backup_execute'),
    path('backup/mount/', views.backup_mount_drive, name='backup_mount_drive'),
    path('backup/check-dests/', views.backup_check_dests, name='backup_check_dests'),
    # Notifications
    path('notifications/', views.notifications_view, name='notifications'),
    path('notifications/<int:pk>/read/', views.mark_notification_read,
name='mark_notification_read'),
]
```

File: core/utils/backup.py

```
import os
import shutil
import tarfile
import json
import subprocess
from django.conf import settings
from django.utils.timezone import now

def get_backup_destinations():
    """Returns a list of potential backup destinations using robust detection."""
    destinations = []

    # 1. Local backup folder (always available)
    local_backup_dir = os.path.join(settings.BASE_DIR, 'backups')
    os.makedirs(local_backup_dir, exist_ok=True)

    # 2. Use lsblk -J to find removable drives
    try:
        output = subprocess.check_output(['lsblk', '-J', '-o',
'NAME,MOUNTPOINT,RM,TYPE,FSTYPE'], text=True)
        data = json.loads(output)

        def process_devices(devices):
            for dev in devices:
                # Check for removable drives (RM="1") that are disks or partitions
                is_removable = str(dev.get('rm', '0')) == '1' or str(dev.get('rm', '0'))
                is_valid_type = dev.get('type') in ('part', 'disk')
                mountpoint = dev.get('mountpoint')
                name = dev.get('name')

                if is_removable and is_valid_type:
                    dest = {
                        'path': mountpoint if mountpoint and mountpoint != 'None' else
                        'name': name,
                        'device': f"/dev/{name}",
                        'is_mounted': bool(mountpoint and mountpoint != 'None'),
                        'is_external': True
                    }
                    # If mounted, verify write access
                    if dest['is_mounted'] and os.path.isdir(mountpoint) and
os.access(mountpoint, os.W_OK):
                        # Avoid duplicates
                        if not any(d.get('path') == dest['path'] for d in destinations if
isinstance(d, dict)):
                            destinations.append(dest)
                    elif not dest['is_mounted']:
                        # Add unmounted drives so the user can see they are detected
                        destinations.append(dest)

                # Recurse children if any
                if 'children' in dev:
                    process_devices(dev['children'])

        if 'blockdevices' in data:
            process_devices(data['blockdevices'])
```

IoT-Enabled Barangay Resident and Attendance Management System

```
except Exception:
    pass

# 3. Fallback: Common mount points scan
mount_bases = ['/media', '/mnt', '/run/media']
extended_bases = set(mount_bases)
for base in mount_bases:
    if os.path.exists(base):
        try:
            for entry in os.listdir(base):
                full_path = os.path.join(base, entry)
                if os.path.isdir(full_path):
                    extended_bases.add(full_path)
        except (PermissionError, OSError):
            continue

for base in extended_bases:
    if os.path.exists(base):
        try:
            for entry in os.listdir(base):
                full_path = os.path.join(base, entry)
                if os.path.isdir(full_path) and os.access(full_path, os.W_OK):
                    dest = {
                        'path': os.path.abspath(full_path),
                        'name': entry,
                        'device': None,
                        'is_mounted': True,
                        'is_external': True
                    }
                    # Avoid duplicates
                    if not any(d.get('path') == dest['path'] for d in destinations if
isinstance(d, dict)):
                        destinations.append(dest)
        except (PermissionError, OSError):
            continue

# Final cleanup and local fallback addition
final_destinations = []

# Add local backup first
final_destinations.append({
    'path': os.path.abspath(local_backup_dir),
    'name': 'backups',
    'device': None,
    'is_mounted': True,
    'is_external': False
})

for d in destinations:
    if not isinstance(d, dict): continue

    if d.get('path'):
        abs_path = os.path.abspath(d['path'])
        # Don't add if it's already the local backup folder or already in final
        if abs_path == os.path.abspath(local_backup_dir): continue
        if not any(fd.get('path') == abs_path for fd in final_destinations):
            d['path'] = abs_path
            final_destinations.append(d)
    elif not d.get('is_mounted'):
        # Keep unmounted ones, avoiding device duplicates
```


IoT-Enabled Barangay Resident and Attendance Management System

```
        if not any(fd.get('device') == d.get('device') for fd in final_destinations):
            final_destinations.append(d)

    return final_destinations

def perform_backup(target_base=None):
    """Performs the backup of the database and media files."""
    timestamp = now().strftime('%Y%m%d_%H%M%S')
    backup_folder_name = f"backup_{timestamp}"

    if not target_base:
        all_dests = get_backup_destinations()
        mounted_dests = [d for d in all_dests if d.get('is_mounted')]
        if not mounted_dests:
            return False, "No writeable backup destinations found."

        # Prefer external drives
        target_base = mounted_dests[0]['path']
        for d in mounted_dests:
            if d.get('is_external'):
                target_base = d['path']
                break

    target_dir = os.path.join(target_base, 'BrgySystemBackups', backup_folder_name)
    os.makedirs(target_dir, exist_ok=True)

    try:
        # 1. Backup SQLite database
        db_path = settings.DATABASES['default']['NAME']
        if os.path.exists(db_path):
            shutil.copy2(db_path, os.path.join(target_dir, 'db.sqlite3'))

        # 2. Backup Media files
        media_root = settings.MEDIA_ROOT
        if os.path.exists(media_root) and os.listdir(media_root):
            media_tar = os.path.join(target_dir, 'media.tar.gz')
            with tarfile.open(media_tar, "w:gz") as tar:
                tar.add(media_root, arcname=os.path.basename(media_root))

        # 3. Create a metadata file
        with open(os.path.join(target_dir, 'metadata.txt'), 'w') as f:
            f.write(f"Backup created on: {now().isoformat()}\n")
            f.write(f"Source: {settings.BASE_DIR}\n")

        return True, f"Backup successfully created at: {target_dir}"
    except Exception as e:
        if os.path.exists(target_dir):
            shutil.rmtree(target_dir)
        return False, f"Backup failed: {str(e)}"

def create_backup_archive():
    """Creates a temporary ZIP archive containing the database and media files."""
    import tempfile
    import zipfile

    timestamp = now().strftime('%Y%m%d_%H%M%S')
    filename = f"BrgySystem_Backup_{timestamp}.zip"
    temp_dir = tempfile.mkdtemp()
    zip_path = os.path.join(temp_dir, filename)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
try:
    with zipfile.ZipFile(zip_path, 'w', zipfile.ZIP_DEFLATED) as zipf:
        # 1. Add database
        db_path = settings.DATABASES['default']['NAME']
        if os.path.exists(db_path):
            zipf.write(db_path, 'db.sqlite3')

        # 2. Add media files
        media_root = settings.MEDIA_ROOT
        if os.path.exists(media_root):
            for root, dirs, files in os.walk(media_root):
                for file in files:
                    file_path = os.path.join(root, file)
                    arcname = os.path.join('media', os.path.relpath(file_path,
media_root))
                    zipf.write(file_path, arcname)

        # 3. Add metadata
        metadata_content = f"Backup created on: {now().isoformat()}\nSource:
{settings.BASE_DIR}\n"
        zipf.writestr('metadata.txt', metadata_content)

    return zip_path
except Exception as e:
    if os.path.exists(temp_dir):
        shutil.rmtree(temp_dir)
    raise e
```

File: core/utils/biometric_discovery.py

```
import os
import socket
import requests
from concurrent.futures import ThreadPoolExecutor
from django.conf import settings

# =====
# ESP32 DISCOVERY STRATEGY (in priority order):
#
# DEPLOYMENT MODE: Orange Pi runs as a WiFi hotspot (10.42.0.1)
# The ESP32 ALWAYS connects to "Barangay_System_WiFi" and gets
# a DHCP address in the 10.42.0.10 ? 10.42.0.100 range.
#
# Priority 1: Heartbeat file      ? IP saved by ESP32's boot POST
# Priority 2: Cached .esp32_ip    ? last known working IP
# Priority 3: mDNS hostname       ? http://esp32-fingerprint.local
# Priority 4: Hotspot scan        ? brute-force scan of 10.42.0.10?100
# Priority 5: settings fallback ? ESP32_BASE_URL (last resort)
#
# Timeouts are tight since the ESP32 is on the local hotspot ?
# no internet routing delay, sub-millisecond LAN latency.
# =====

PROBE_TIMEOUT = 0.8 # seconds ? fast on local hotspot LAN
CONNECT_TIMEOUT = 1.0 # seconds ? TCP connect on local LAN
READ_TIMEOUT = 3.0 # seconds ? HTTP read (sensor may be busy)

_MDNS_HOSTNAME = "esp32-fingerprint.local"
_MDNS_URL = f"http://{_MDNS_HOSTNAME}"

# The Orange Pi's private hotspot subnet (fixed ? never changes)
_HOTSPOT_SUBNET = "10.42.0."
_HOTSPOT_DHCP_START = 10
_HOTSPOT_DHCP_END = 100

# -----
# File helpers
# -----

def get_local_ip():
    """Return this machine's LAN IP (not 127.0.0.1).
    On the Orange Pi, this will be 10.42.0.1 when routing to the hotspot subnet.
    """
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    try:
        # Try routing to the ESP32's subnet ? will pick the wlan0 (hotspot) interface
        s.connect(('10.42.0.10', 1))
        IP = s.getsockname()[0]
    except Exception:
        try:
            s.connect(('8.8.8.8', 1))
            IP = s.getsockname()[0]
        except Exception:
            try:
                IP = socket.gethostbyname(socket.gethostname())
                if IP.startswith('127.'):

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        IP = socket.gethostbyname(socket.getfqdn())
    except Exception:
        IP = '127.0.0.1'
finally:
    s.close()
return IP

def get_esp32_ip_file_path():
    return os.path.join(settings.BASE_DIR, '.esp32_ip')

def get_heartbeat_file_path():
    """Separate file written when ESP32 POSTs its own IP on boot."""
    return os.path.join(settings.BASE_DIR, '.esp32_heartbeat_ip')

def get_saved_esp32_url():
    """Read the last-scanned/cached URL from .esp32_ip."""
    ip_file = get_esp32_ip_file_path()
    if os.path.exists(ip_file):
        try:
            with open(ip_file, 'r') as f:
                url = f.read().strip()
                if url:
                    return url
        except Exception:
            pass
    return None

def get_heartbeat_url():
    """Read the URL that the ESP32 registered itself with on last boot."""
    hb_file = get_heartbeat_file_path()
    if os.path.exists(hb_file):
        try:
            with open(hb_file, 'r') as f:
                url = f.read().strip()
                if url:
                    return url
        except Exception:
            pass
    return None

def save_esp32_url(url):
    """Persist the working URL to .esp32_ip so next boot is instant."""
    ip_file = get_esp32_ip_file_path()
    try:
        with open(ip_file, 'w') as f:
            f.write(url)
    except Exception:
        pass

def save_heartbeat_url(url):
    """Called by the heartbeat view when ESP32 registers itself."""
    hb_file = get_heartbeat_file_path()
    try:
        with open(hb_file, 'w') as f:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
f.write(url)
# Also update the main cache so every path benefits
save_esp32_url(url)
except Exception:
    pass

# -----
# Probe helpers
# -----

def _is_esp32(url: str, timeout: float = PROBE_TIMEOUT) -> bool:
    """
    Return True if `url` hosts a valid ESP32 fingerprint module.
    Checks /status for the 'fingerprint_initialized' key.
    """
    try:
        resp = requests.get(
            f"{url}/status",
            timeout=(CONNECT_TIMEOUT, timeout),
            proxies={'http': None, 'https': None},
        )
        if resp.status_code == 200:
            data = resp.json()
            if "fingerprint_initialized" in data:
                return True
    except Exception:
        pass
    return False

def _test_ip_for_esp32(ip: str):
    """Used by the hotspot subnet scanner ThreadPoolExecutor."""
    url = f"http://{ip}"
    if _is_esp32(url, timeout=PROBE_TIMEOUT):
        return url
    return None

# -----
# Hotspot subnet scanner (fast ? scans only the dnsmasq DHCP range)
# -----

def scan_hotspot_subnet_for_esp32():
    """
    Scan only the hotspot DHCP range (10.42.0.10 ? 10.42.0.100).
    Much faster than a full /24 scan since we know exactly where the ESP32 will be.
    Runs in parallel with 30 workers (91 hosts max ? completes in < 2 seconds).
    """
    ips = [f"{_HOTSPOT_SUBNET}{i}" for i in range(_HOTSPOT_DHCP_START, _HOTSPOT_DHCP_END +
1)]
    print(f"[Biometric] Scanning hotspot subnet {_HOTSPOT_SUBNET}{_HOTSPOT_DHCP_START}"
        f"?{_HOTSPOT_DHCP_END} for ESP32...")
    with ThreadPoolExecutor(max_workers=30) as executor:
        results = executor.map(_test_ip_for_esp32, ips)
        for r in results:
            if r:
                return r
    return None
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# -----
# Main entry point
# -----

def get_esp32_base_url(force_scan: bool = False) -> str:
    """
    Return the base URL of the ESP32 fingerprint module.

    Discovery order (skipped when force_scan=True):
    1. Heartbeat-registered IP ? ESP32 POSTs its IP on every boot
    2. Cached .esp32_ip file ? last known working IP
    3. mDNS hostname ? http://esp32-fingerprint.local

    Then falls through to active hotspot subnet scan (fast, < 2 seconds).
    """
    default_url = getattr(settings, 'ESP32_BASE_URL', 'http://10.42.0.10').rstrip('/')

    if not force_scan:
        # ---- 1. Heartbeat-registered IP (highest priority ? set on ESP32 boot) ----
        hb_url = get_heartbeat_url()
        if hb_url and _is_esp32(hb_url):
            print(f"[Biometric] ESP32 found via heartbeat cache: {hb_url}")
            save_esp32_url(hb_url)
            return hb_url

        # ---- 2. Last cached IP from .esp32_ip ----
        cached_url = get_saved_esp32_url()
        if cached_url and cached_url != hb_url and _is_esp32(cached_url):
            print(f"[Biometric] ESP32 found via IP cache: {cached_url}")
            return cached_url

        # ---- 3. mDNS hostname (works if avahi / mDNS is available) ----
        try:
            resp = requests.get(
                f"{_MDNS_URL}/status",
                timeout=(CONNECT_TIMEOUT, READ_TIMEOUT),
                proxies={'http': None, 'https': None},
            )
            if resp.ok and "fingerprint_initialized" in resp.json():
                print(f"[Biometric] ESP32 found via mDNS: {_MDNS_URL}")
                save_esp32_url(_MDNS_URL)
                return _MDNS_URL
        except Exception:
            pass

        # ---- 4. Hotspot subnet scan (fast ? only scans DHCP range 10.42.0.10?100) ----
        discovered = scan_hotspot_subnet_for_esp32()
        if discovered:
            print(f"[Biometric] ESP32 found via hotspot scan: {discovered}")
            save_esp32_url(discovered)
            return discovered

        # ---- 5. Settings default (last resort ? avoids crash if nothing works) ----
        print(f"[Biometric] ESP32 not found. Falling back to default: {default_url}")
        return default_url
```

IoT-Enabled Barangay Resident and Attendance Management System

File: core/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth import authenticate, login, logout
from django.contrib.auth.decorators import login_required
from django.contrib import messages
from django.contrib.auth import get_user_model
User = get_user_model()
from django.http import JsonResponse, FileResponse
from django import forms
from django.utils.text import slugify
from django.utils.timezone import now
from django.core.cache import cache
import secrets
import os
import sys
import subprocess
from .models import UserProfile
from residents.models import Resident, Household, ResidentRegistration
from certifications.models import Certificate
from officials.models import Official
from attendance.models import AttendanceLog
from ordinances.models import Ordinance
from datetime import date
from .utils.backup import perform_backup, get_backup_destinations, create_backup_archive
from biometrics.utils import get_biometric_provider
from django.conf import settings
from django.db.models import Max
from core.utils.biometric_discovery import get_esp32_base_url, save_heartbeat_url
import requests

OFFICIAL_ROLE_CHOICES = [
    ('captain', 'Barangay Captain'),
    ('secretary', 'Secretary'),
    ('treasurer', 'Treasurer'),
    ('bhw', 'Barangay BHW'),
    ('resident', 'Resident'),
]

OFFICIAL_USERNAME_BY_ROLE = {
    'captain': 'captain',
    'secretary': 'secretary',
    'treasurer': 'treasurer',
    'bhw': 'bhw',
    'admin': 'admin',
}

MIN_FINGERPRINT_TEMPLATE_LEN = 10

# ESP32 passwordless login: only these positions may complete Django login via
# fingerprint.
BIOMETRIC_LOGIN_ALLOWED_POSITIONS = ('captain', 'secretary', 'treasurer')

ROLE_TEMPLATE_MAP = {
    'captain': 'core/dashboard_captain.html',
    'admin': 'core/dashboard_captain.html',
    'secretary': 'core/dashboard_secretary.html',
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'treasurer': 'core/dashboard_treasurer.html',
'kagawad': 'core/dashboard_kagawad.html',
'sk_chairman': 'core/dashboard_sk.html',
'sk_chairperson': 'core/dashboard_sk.html', # alias
'lupon': 'core/dashboard_lupon.html',
'lupong_member': 'core/dashboard_lupon.html', # alias
'bhw': 'core/dashboard_default.html',
'tanod': 'core/dashboard_default.html',
'staff': 'core/dashboard_default.html',
'clerk': 'core/dashboard_default.html',
'resident': 'core/dashboard_resident.html',
}

@login_required
def dashboard(request):
    """Role-based dashboard dispatcher."""
    # Consume biometric flags set by biometric_login_complete
    biometrically_verified = request.session.pop('biometrically_verified', False)
    biometric_position = request.session.pop('biometric_position', None)

    today = date.today()

    # Determine role:
    # Priority 1: role explicitly set during biometric login
    # Priority 2: active Official record linked to this user
    # Priority 3: UserProfile / User.role fallback
    role = 'resident'

    if biometric_position:
        # Trust the position recorded at biometric login time
        role = biometric_position
    else:
        # 1. Check if user has an active official position
        try:
            official = request.user.official_profile
            if official and official.status == 'active':
                role = official.position
        except Exception:
            pass

        if role == 'resident':
            # 2. Fallback to UserProfile / User.role
            try:
                profile_role = request.user.profile.role
                if profile_role:
                    role = profile_role
            except Exception:
                pass

    # Superuser always gets admin/captain view
    if request.user.is_superuser:
        role = 'admin'

    # Shared base context
    context = {
        'today': today,
        'total_residents': Resident.objects.filter(is_active=True).count(),
        'total_households': Household.objects.count(),
        'total_officials': Official.objects.filter(status='active').count(),
    }
```


IoT-Enabled Barangay Resident and Attendance Management System

```
'total_ordinances': Ordinance.objects.filter(status='active').count(),
'certificates_today': Certificate.objects.filter(date_issued=today).count(),
'certificates_total': Certificate.objects.count(),
'attendance_today': AttendanceLog.objects.filter(date=today).count(),
'pending_registrations_count':
ResidentRegistration.objects.filter(status='pending').count(),
'pending_registrations_today':
ResidentRegistration.objects.filter(status='pending', created_at__date=today).count(),
'recent_certificates':
Certificate.objects.select_related('resident').order_by('-date_issued')[:5],
'recent_residents': Resident.objects.order_by('-created_at')[:5],
'male_count': Resident.objects.filter(gender='M', is_active=True).count(),
'female_count': Resident.objects.filter(gender='F', is_active=True).count(),
}

# Role-specific extra context
if role in ('captain', 'admin'):
    context['recent_officials'] =
Official.objects.select_related('resident').order_by('-created_at')[:5]
    context['recent_ordinances'] = Ordinance.objects.order_by('-date_enacted')[:5]

elif role == 'secretary':
    context['pending_certs'] = Certificate.objects.filter(date_issued=today).count()
    context['recent_ordinances'] = Ordinance.objects.order_by('-date_enacted')[:5]

elif role == 'treasurer':
    context['attendance_logs'] =
AttendanceLog.objects.select_related('official__resident').filter(date=today)[:10]

elif role == 'kagawad':
    context['recent_ordinances'] = Ordinance.objects.order_by('-date_enacted')[:5]

elif role == 'sk_chairperson':
    from django.utils.timezone import now
    from datetime import timedelta
    cutoff = date.today().replace(year=date.today().year - 30)
    context['youth_count'] = Resident.objects.filter(is_active=True,
birthdate__gte=cutoff).count()

elif role == 'lupong_member':
    context['recent_ordinances'] = Ordinance.objects.order_by('-date_enacted')[:5]

elif role == 'resident':
    # Personalized resident portal context
    from certifications.models import Certificate as Cert
    try:
        resident_record = request.user.profile.resident
        context['resident_record'] = resident_record
        if resident_record:
            context['my_certificates'] = Cert.objects.filter(
                resident=resident_record
            ).order_by('-date_issued')[:5]
        else:
            context['my_certificates'] = []
    except Exception:
        context['resident_record'] = None
        context['my_certificates'] = []
    context['cert_types'] = Cert.TYPE_CHOICES

template = ROLE_TEMPLATE_MAP.get(role, 'core/dashboard_default.html')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
return render(request, template, context)

@login_required
def barangay_map(request):
    """Render the MapLibre GL JS Barangay Map for Puroks."""
    return render(request, 'core/barangay_map.html')

from django.views.decorators.csrf import csrf_exempt
from django.views.decorators.http import require_http_methods
from django.db.models import Q
import json

# =====
# ESP32 HEARTBEAT ? Called by ESP32 on every boot via HTTP POST
# Registers its current DHCP IP so Django never needs to scan.
# No authentication required (device has no credentials).
# =====
@csrf_exempt
@require_http_methods(["POST", "GET"])
def esp32_heartbeat(request):
    """
    The ESP32 POSTs its own local IP here immediately after connecting to Wi-Fi.
    Django saves it so it can be used as the highest-priority discovery address.
    Accepts both POST (from ESP32) and GET (for health-check / browser testing).
    """
    if request.method == 'GET':
        # Simple health-check ? ESP32 can confirm the endpoint is alive
        return JsonResponse({'status': 'ok', 'message': 'Heartbeat endpoint ready'})

    try:
        # Prefer X-ESP32-IP header; fall back to request body JSON
        ip = request.META.get('HTTP_X_ESP32_IP', '').strip()
        if not ip:
            try:
                body = json.loads(request.body)
                ip = body.get('ip', '').strip()
            except Exception:
                ip = ''

        # Last resort: use the remote IP of the incoming connection
        if not ip:
            ip = request.META.get('REMOTE_ADDR', '')

        if ip and ip != '127.0.0.1':
            url = f"http://{ip}"
            save_heartbeat_url(url)
            print(f"[Biometric] ESP32 heartbeat received ? registered at {url}")
            return JsonResponse({'status': 'ok', 'registered_ip': ip})

        return JsonResponse({'status': 'error', 'message': 'Could not determine ESP32 IP'}, status=400)
    except Exception as e:
        return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

@csrf_exempt
def biometric_templates(request):
```

IoT-Enabled Barangay Resident and Attendance Management System

```
"""API for the 32-bit service to fetch all registered templates."""
role = (request.GET.get('role') or '').strip()
username_param = (request.GET.get('username') or '').strip()
profiles = UserProfile.objects.exclude(fingerprint_template__isnull=True).exclude(fingerprint_template='')

if username_param:
    profiles = profiles.filter(user__username=username_param)
elif role:
    username = OFFICIAL_USERNAME_BY_ROLE.get(role)
    if username:
        profiles = profiles.filter(user__username=username)
    else:
        profiles = profiles.filter(role=role)
templates = []
for p in profiles:
    if p.fingerprint_template and len(p.fingerprint_template) >=
MIN_FINGERPRINT_TEMPLATE_LEN:
        templates.append({
            'id': p.user.id,
            'template': p.fingerprint_template
        })
return JsonResponse({'templates': templates})

@csrf_exempt
@require_http_methods(["POST"])
def biometric_verify_login(request):
    """API for the 32-bit service to notify successful verification."""
    try:
        data = json.loads(request.body)
        user_id = data.get('user_id')
        request_id = data.get('request_id')
        if not request_id:
            return JsonResponse({'status': 'error', 'message': 'Missing request_id'},
status=400)

        expected = cache.get(f"biometric:{request_id}") or {}
        expected_role = (expected.get('role') or '').strip()

        user = User.objects.get(id=user_id)

        # Biometric login: allow classic long template OR R307 slot on linked resident
        profile = UserProfile.objects.filter(user=user).first()
        resident = profile.resident if profile else None
        tpl_ok = (
            profile
            and profile.fingerprint_template
            and len(profile.fingerprint_template) >= MIN_FINGERPRINT_TEMPLATE_LEN
        )
        sensor_ok = resident is not None and resident.fingerprint_id is not None
        if not profile or (not tpl_ok and not sensor_ok):
            if request_id:
                cache.set(
                    f"biometric:{request_id}",
                    {"status": "failed", "reason": "Fingerprint not registered"},
                    timeout=120,
                )
            return JsonResponse({'status': 'error', 'message': 'Fingerprint not
registered'}, status=403)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if expected_role:
    expected_username = OFFICIAL_USERNAME_BY_ROLE.get(expected_role)
    if expected_username and (user.username or '').strip() != expected_username:
        cache.set(
            f"biometric:{request_id}",
            {"status": "failed", "reason": "Role mismatch"},
            timeout=120,
        )
        return JsonResponse({'status': 'error', 'message': 'Role mismatch'},
status=403)

    if (profile.role or '').strip() and (profile.role or '').strip() !=
expected_role:
        cache.set(
            f"biometric:{request_id}",
            {"status": "failed", "reason": "Role mismatch"},
            timeout=120,
        )
        return JsonResponse({'status': 'error', 'message': 'Role mismatch'},
status=403)

    if expected_role and not OFFICIAL_USERNAME_BY_ROLE.get(expected_role) and
(profile.role or '').strip() != expected_role:
        cache.set(
            f"biometric:{request_id}",
            {"status": "failed", "reason": "Role mismatch"},
            timeout=120,
        )
        return JsonResponse({'status': 'error', 'message': 'Role mismatch'},
status=403)

    cache.set(
        f"biometric:{request_id}",
        {"status": "authenticated", "user_id": user.id},
        timeout=120,
    )
    return JsonResponse({'status': 'success', 'message': f'Verified as
{user.username}'})
except Exception as e:
    return JsonResponse({'status': 'error', 'message': str(e)}, status=400)

@login_required
def biometric_verify_login_start(request):
    """View to initiate biometric verification."""
    try:
        request_id = secrets.token_urlsafe(16)
        request.session['biometric_request_id'] = request_id
        cache.set(f"biometric:{request_id}", {"status": "pending"}, timeout=120)

        # Use BiometricProvider interface
        provider = get_biometric_provider()
        result = provider.verify(user_id=request.user.id, scan_data={'request_id':
request_id})

        return JsonResponse({
            'status': 'success',
            'message': result.get('message', 'Verification service initiated'),
            'request_id': request_id
        })
    except Exception as e:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

@csrf_exempt
def biometric_login_start(request):
    """Public view to initiate biometric verification for login."""
    try:
        role = ''
        if request.method == 'POST':
            try:
                payload = json.loads(request.body) if request.body else {}
            except json.JSONDecodeError:
                payload = {}
            role = (payload.get('role') or '').strip()
            username = (payload.get('username') or '').strip()
            role = role or (request.GET.get('role') or '').strip()
            username = username or (request.GET.get('username') or '').strip()

            if role and role != 'auto' and role not in dict(OFFICIAL_ROLE_CHOICES):
                return JsonResponse({'status': 'error', 'message': 'Invalid role'},
                                    status=400)

            request_id = secrets.token_urlsafe(16)
            request.session['biometric_request_id'] = request_id

            _esp32_trigger_start_verification()

            cache.set(f"biometric:{request_id}", {"status": "pending", "role": role,
                                                    "username": username}, timeout=120)

            return JsonResponse({
                'status': 'success',
                'message': 'Biometric verification initiated. Please scan your finger.',
                'request_id': request_id
            })
        except Exception as e:
            return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

@csrf_exempt
def biometric_status_check(request):
    """Check if the biometric verification was successful by polling ESP32 status."""
    request_id = request.session.get('biometric_request_id')
    if not request_id:
        return JsonResponse({'status': 'none'})

    state = cache.get(f"biometric:{request_id}") or {"status": "pending"}

    if state.get('status') == 'failed':
        return JsonResponse({'status': 'failed', 'reason': state.get('reason')})

    if state.get('status') == 'authenticated':
        # Already authenticated in a previous poll
        return JsonResponse({'status': 'authenticated'})

    # Poll ESP32 for detection
    esp32_base_url = get_esp32_base_url()
    try:
        resp = requests.get(
            f"{esp32_base_url}/status",
            timeout=(3, 5),
            proxies={'http': None, 'https': None},
```

IoT-Enabled Barangay Resident and Attendance Management System

```

    )
except Exception:
    esp32_base_url = get_esp32_base_url(force_scan=True)
    try:
        resp = requests.get(
            f"{esp32_base_url}/status",
            timeout=(3, 5),
            proxies={'http': None, 'https': None},
        )
    except Exception as e:
        print(f"[Biometric] Status poll failed after scan: {str(e)}")
        return JsonResponse({'status': 'pending'})
    try:
        if resp.ok:
            data = resp.json()

            # Only fail on fingerprint not found errors during verification mode
            esp32_state_raw = data.get('state', '')
            if esp32_state_raw == 'auth_failed':
                reason = data.get('last_error', 'Fingerprint not recognized.')
                if reason.startswith("Auth Failed: "):
                    reason = reason[len("Auth Failed: "):]
                cache.set(f"biometric:{request_id}", {"status": "failed", "reason":
reason}, timeout=120)
                return JsonResponse({'status': 'failed', 'reason': reason})

            # Fingerprint matched only when state is 'detected' (verify success pulse) or
            standby after.
            # fingerprint_id is 0-based (page 0 is valid); ESP omits the key when no match
            pending.
            esp32_state = data.get('state', '')
            has_fp = 'fingerprint_id' in data
            matched_id = data.get('fingerprint_id') if has_fp else None

            # Fingerprint matched ? only Captain, Secretary, Treasurer may use biometric
*login*
            if esp32_state in ('detected', 'standby') and has_fp and matched_id is not
None:
                attendance_mode = data.get('attendance_mode') or 'none'

                # Cross-route if this is an attendance scan (mode is 'in', 'out', or
'attendance')
                if attendance_mode in ('in', 'out', 'attendance'):
                    from officials.models import Official as OfficialModel
                    official_rec = OfficialModel.objects.select_related('resident',
'user').filter(
                        resident__fingerprint_id=matched_id,
                        resident__is_active=True,
                        status='active',
                    ).first()
                    if official_rec:
                        att_rid = request.session.get('biometric_attendance_request_id')
or 'global_hardware_scan'
                        cache.set(
                            f"biometric_attendance:{att_rid}",
                            {
                                'status': 'authenticated',
                                'official_id': official_rec.id,
                                'user_id': official_rec.user_id,
                                'attendance_mode': attendance_mode,

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        'device_time': data.get('device_time') or '',
        'device_date': data.get('device_date') or '',
    },
    timeout=60,
)
return JsonResponse({'status': 'pending'})

from officials.models import Official as OfficialModel
user = None

# Only allow active officials with an authorised position
try:
    official_rec = OfficialModel.objects.select_related('resident',
'user').filter(
        resident__fingerprint_id=matched_id,
        resident__is_active=True,
        status='active',
        position__in=BIOMETRIC_LOGIN_ALLOWED_POSITIONS,
        user__isnull=False
    ).first()
    if official_rec:
        print(f"[Biometric] Matched ID {matched_id}: "
            f"{official_rec.resident.full_name}
({official_rec.position})")
        user = official_rec.user
    except Exception as e:
        print(f"[Biometric] Official lookup failed: {e}")

# If no authorised official was found, reject immediately
if not user:
    # Check if any resident/other official owns this fingerprint_id
    # so we can give a meaningful error
    unauthorized = Resident.objects.filter(
        fingerprint_id=matched_id, is_active=True
    ).exists()
    reason = (
        "Fingerprint is not authorised for biometric login. "
        "Only the Barangay Captain, Secretary, and Treasurer may sign in
with fingerprint."
        if unauthorized
        else "Fingerprint not registered for biometric login."
    )
    try:
        requests.post(f"{esp32_base_url}/error-feedback", data={'reason':
'Not Registered'}, timeout=2,
                    proxies={'http': None, 'https': None})
    except Exception:
        pass
    cache.set(f"biometric:{request_id}", {"status": "failed", "reason":
reason}, timeout=30)
    return JsonResponse({'status': 'failed', 'reason': reason})

# Store user_id in cache ? biometric_login_complete will do the actual
login()
cache.set(f"biometric:{request_id}", {"status": "authenticated",
"user_id": user.id}, timeout=60)

# Stop ESP32 verification
try:
    requests.post(f"{esp32_base_url}/stop-enrollment", timeout=1,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        proxies={'http': None, 'https': None})
    except Exception:
        pass

    return JsonResponse({'status': 'authenticated'})
    # Wrong finger / bad scan during verify ? ESP stays in session (soft retry).
    # Surface a hint to the login page without ending the biometric poll.
    esp_st = (data.get('state') or '').strip()
    last_err = (data.get('last_error') or '').strip()
    low = last_err.lower()
    if esp_st in ('wait_remove', 'verifying') and (
        'not recognized' in low or 'poor scan' in low or 'remove finger' in low
    ):
        return JsonResponse({
            'status': 'pending',
            'hint': 'Fingerprint not recognized. Remove your finger from the
sensor, then try again.',
        })
    except Exception as e:
        print(f"[Biometric] Status poll failed: {str(e)}")

    return JsonResponse({'status': 'pending'})

def _esp32_trigger_start_verification(mode=None):
    """POST /start-verification on the ESP32. Returns True if ESP32 acknowledged, False
otherwise."""
    esp32_base_url = get_esp32_base_url()
    max_slot = (
        Resident.objects.filter(is_active=True, fingerprint_id__isnull=False)
        .aggregate(m=Max('fingerprint_id'))
        .get('m')
    )

    payload = {}
    if max_slot is not None:
        payload['max_page'] = str(int(max_slot))
    if mode:
        payload['mode'] = mode

    # NEW: Pass our own server IP so ESP32 knows who to send heartbeat/callbacks to
    immediately
    from django.conf import settings
    # Try to grab the IP from the network interfaces, or fallback to the one in
    esp32_base_url
    server_ip = None
    if esp32_base_url:
        import socket
        try:
            # Create a dummy socket to figure out which local IP routes to the ESP32
            esp32_ip = esp32_base_url.split('/')[1].split(':')[0]
            s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
            s.connect((esp32_ip, 80))
            server_ip = s.getsockname()[0]
            s.close()
        except Exception:
            pass

    if not server_ip:
        server_ip = "10.42.0.1"
```


IoT-Enabled Barangay Resident and Attendance Management System

```
if server_ip:
    payload['server_ip'] = server_ip

try:
    resp = requests.post(
        f"{esp32_base_url}/start-verification",
        data=payload if payload else None,
        timeout=3,
        proxies={'http': None, 'https': None},
    )
    print(f"[Biometric] ESP32 /start-verification response: {resp.status_code}
{resp.text[:100]}")
    return resp.ok
except Exception as e:
    print(f"[Biometric] Failed to trigger ESP32: {str(e)}. Scanning...")
    esp32_base_url = get_esp32_base_url(force_scan=True)
    try:
        resp = requests.post(
            f"{esp32_base_url}/start-verification",
            data=payload if payload else None,
            timeout=3,
            proxies={'http': None, 'https': None},
        )
        return resp.ok
    except Exception as retry_err:
        print(f"[Biometric] Failed to trigger ESP32 after scan: {str(retry_err)}")
        return False

@csrf_exempt
@login_required
def biometric_attendance_start(request):
    """
    Start ESP32 verification for clock in/out.
    """
    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'POST required'}, status=405)

    mode = 'attendance'
    try:
        data = json.loads(request.body)
        if 'mode' in data and data['mode']:
            mode = data['mode']
    except Exception:
        pass

    request_id = secrets.token_urlsafe(16)
    request.session['biometric_attendance_request_id'] = request_id
    request.session.save()
    esp32_ok = _esp32_trigger_start_verification(mode=mode)
    cache.set(f"biometric_attendance:{request_id}", {'status': 'pending'}, timeout=120)
    print(f"[Biometric] attendance_start: request_id={request_id}, esp32_ok={esp32_ok}")
    return JsonResponse({
        'status': 'success',
        'message': 'Attendance fingerprint scan started.',
        'request_id': request_id,
        'esp32_connected': esp32_ok,
    })
})
```

IoT-Enabled Barangay Resident and Attendance Management System

```
@csrf_exempt
def biometric_attendance_status_check(request):
    """Poll ESP32 after biometric_attendance_start; accept any active official with a
    matching slot.
    Also handles global hardware button polls without a request_id.
    """
    request_id = request.session.get('biometric_attendance_request_id')

    if request_id:
        state = cache.get(f"biometric_attendance:{request_id}") or {'status': 'pending'}
        if state.get('status') == 'failed':
            return JsonResponse({'status': 'failed', 'reason': state.get('reason')})
        if state.get('status') == 'authenticated':
            return JsonResponse({'status': 'authenticated', 'attendance_mode':
state.get('attendance_mode', 'none')})

    esp32_base_url = get_esp32_base_url()
    try:
        resp = requests.get(
            f"{esp32_base_url}/status",
            timeout=(3, 5),
            proxies={'http': None, 'https': None},
        )
    except Exception:
        esp32_base_url = get_esp32_base_url(force_scan=True)
        try:
            resp = requests.get(
                f"{esp32_base_url}/status",
                timeout=(3, 5),
                proxies={'http': None, 'https': None},
            )
        except Exception as e:
            print(f"[Biometric] Attendance status poll failed after scan: {str(e)}")
            return JsonResponse({'status': 'pending'})

    try:
        if resp.ok:
            data = resp.json()
            esp32_state_raw = data.get('state', '')
            last_error = (data.get('last_error') or '').lower()
            if esp32_state_raw == 'auth_failed':
                reason = data.get('last_error', 'Fingerprint not recognized.')
                if reason.startswith("Auth Failed: "):
                    reason = reason[len("Auth Failed: "):]
                cache.set(
                    f"biometric_attendance:{request_id}",
                    {'status': 'failed', 'reason': reason},
                    timeout=120,
                )
                return JsonResponse({'status': 'failed', 'reason': reason})

            esp32_state = data.get('state', '')
            has_fp = 'fingerprint_id' in data
            matched_id = data.get('fingerprint_id') if has_fp else None

            if esp32_state in ('detected', 'standby') and has_fp and matched_id is not
None:
                attendance_mode = data.get('attendance_mode') or 'none'

                # Cross-route if this is a login scan (mode is 'none')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if attendance_mode == 'none':
    from officials.models import Official as OfficialModel
    official_rec = OfficialModel.objects.select_related('resident',
'user').filter(
        resident__fingerprint_id=matched_id,
        resident__is_active=True,
        status='active',
    ).first()
    if official_rec and official_rec.user_id:
        login_req_id = request.session.get('biometric_request_id')
        if login_req_id:
            cache.set(
                f"biometric:{login_req_id}",
                {
                    'status': 'authenticated',
                    'user_id': official_rec.user_id,
                },
                timeout=60,
            )
            return JsonResponse({'status': 'pending'})

    from officials.models import Official as OfficialModel
    official_rec = OfficialModel.objects.select_related('resident',
'user').filter(
        resident__fingerprint_id=matched_id,
        resident__is_active=True,
        status='active',
    ).first()
    if not official_rec:
        if Resident.objects.filter(fingerprint_id=matched_id,
is_active=True).exists():
            reason = (
                'This fingerprint is not linked to an active barangay
functionary. '
                'Personnel must be enrolled as an official before using
attendance scan.'
            )
        else:
            reason = 'Fingerprint not registered in this system.'
        try:
            requests.post(
                f"{esp32_base_url}/error-feedback",
                data={'reason': 'Not Registered'},
                timeout=2,
                proxies={'http': None, 'https': None},
            )
        except Exception:
            pass
        cache.set(
            f"biometric_attendance:{request_id}",
            {'status': 'failed', 'reason': reason},
            timeout=30,
        )
        return JsonResponse({'status': 'failed', 'reason': reason})

device_time = data.get('device_time') or ''
device_date = data.get('device_date') or ''

# ?? NOT SCHEDULED CHECK ?? Block if official has no WorkSchedule today
from attendance.models import WorkSchedule as WS
```

IoT-Enabled Barangay Resident and Attendance Management System

```

from django.utils import timezone as tz
scan_date = tz.localdate()
if device_date:
    try:
        from datetime import datetime as _dt
        _parsed = _dt.strptime(device_date, '%Y-%m-%d').date()
        if _parsed.year >= 2026:
            scan_date = _parsed
    except Exception:
        pass
official_sched = WS.objects.filter(official=official_rec,
date=scan_date).first()
if not official_sched or official_sched.shift_type in ['day_off',
'leave']:
    block_reason = 'Not Scheduled' if not official_sched else
official_sched.get_shift_type_display()
    try:
        requests.post(
            f"{esp32_base_url}/error-feedback",
            data={'reason': 'Not Scheduled'},
            timeout=2,
            proxies={'http': None, 'https': None},
        )
    except Exception:
        pass
    target_req_id = request_id or 'global_hardware_scan'
    cache.set(
        f"biometric_attendance:{target_req_id}",
        {'status': 'failed', 'reason': f'{official_rec.resident.full_name}
? {block_reason} today.'},
        timeout=30,
    )
    return JsonResponse({
        'status': 'not_scheduled',
        'reason': f'{official_rec.resident.full_name} ? {block_reason}
today.'
    })

# Hardware validation: Reject TIME OUT if no TIME IN is recorded for today
if attendance_mode == 'out':
    from datetime import date, datetime
    from django.utils import timezone
    check_date = timezone.localdate()
    if device_date:
        try:
            parsed_date = datetime.strptime(device_date,
"%Y-%m-%d").date()
            if parsed_date.year >= 2026:
                check_date = parsed_date
        except Exception:
            pass

    log = AttendanceLog.objects.filter(official=official_rec,
date=check_date).first()
    if not log or (not log.am_in and not log.pm_in):
        reason = 'No Clock-In'
        try:
            requests.post(
                f"{esp32_base_url}/error-feedback",
                data={'reason': reason},

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        timeout=2,
        proxies={'http': None, 'https': None},
    )
except Exception:
    pass

    target_req_id = request_id or 'global_hardware_scan'
    cache.set(
        f"biometric_attendance:{target_req_id}",
        {
            'status': 'failed',
            'reason': f"Clock-Out Rejected: No Clock-In recorded today
for {official_rec.resident.full_name}."
        },
        timeout=30,
    )
    return JsonResponse({
        'status': 'failed',
        'reason': f"Clock-Out Rejected: No Clock-In recorded today for
{official_rec.resident.full_name}."
    })

if request_id:
    cache.set(
        f"biometric_attendance:{request_id}",
        {
            'status': 'authenticated',
            'official_id': official_rec.id,
            'user_id': official_rec.user_id,
            'attendance_mode': attendance_mode,
            'device_time': device_time,
            'device_date': device_date,
        },
        timeout=60,
    )
else:
    # Global poll hit
    temp_request_id = 'global_hardware_scan'
    request.session['biometric_attendance_request_id'] = temp_request_id
    cache.set(
        f"biometric_attendance:{temp_request_id}",
        {
            'status': 'authenticated',
            'official_id': official_rec.id,
            'user_id': official_rec.user_id,
            'attendance_mode': attendance_mode,
            'device_time': device_time,
            'device_date': device_date,
        },
        timeout=60,
    )
try:
    requests.post(
        f"{esp32_base_url}/stop-enrollment",
        timeout=1,
        proxies={'http': None, 'https': None},
    )
except Exception:
    pass
return JsonResponse({
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        'status': 'authenticated',
        'attendance_mode': data.get('attendance_mode') or 'none',
        'device_time': device_time,
        'device_date': device_date,
    })

    esp_st = (data.get('state') or '').strip()
    last_err = (data.get('last_error') or '').strip()
    low = last_err.lower()
    if esp_st in ('wait_remove', 'verifying') and (
        'not recognized' in low or 'poor scan' in low or 'remove finger' in low
    ):
        return JsonResponse({
            'status': 'pending',
            'esp32_state': esp_st,
            'hint': 'Fingerprint not recognized. Remove your finger from the
sensor, then try again.',
        })

    return JsonResponse({'status': 'pending', 'esp32_state': esp_st})
except Exception as e:
    print(f"[Biometric] Attendance status poll failed: {str(e)}")

return JsonResponse({'status': 'pending'})

@csrf_exempt
def biometric_login_complete(request):
    """
    Final redirect step for biometric login.
    Called by the frontend after receiving 'authenticated' from the polling endpoint.
    Performs the actual Django login() here (full HTTP request, not AJAX) so the
    session cookie is reliably set before the browser follows the redirect.
    Biometric login is restricted to Captain, Secretary, and Treasurer only.
    """
    request_id = request.session.get('biometric_request_id')
    if not request_id:
        messages.error(request, 'Biometric session expired. Please scan again.')
        return redirect('core:login')

    state = cache.get(f"biometric:{request_id}") or {}
    if state.get('status') != 'authenticated':
        messages.error(request, 'Biometric authentication not completed. Please try
again.')
        return redirect('core:login')

    user_id = state.get('user_id')
    if not user_id:
        messages.error(request, 'No user matched. Please try again.')
        return redirect('core:login')

    try:
        user = User.objects.get(id=user_id)
    except User.DoesNotExist:
        messages.error(request, 'User account not found.')
        return redirect('core:login')

    # Final role guard: confirm the matched user is an authorised official
    from officials.models import Official as OfficialModel
    official_rec = OfficialModel.objects.filter(
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        user=user,
        status='active',
        position__in=BIOMETRIC_LOGIN_ALLOWED_POSITIONS,
    ).first()
if not official_rec:
    cache.delete(f"biometric:{request_id}")
    messages.error(
        request,
        'Access denied. Biometric login is only available to the Barangay Captain, '
        'Secretary, and Treasurer.'
    )
    return redirect('core:login')

# Perform the real Django login here (full HTTP request)
if not hasattr(user, 'backend'):
    user.backend = 'django.contrib.auth.backends.ModelBackend'
login(request, user)
request.session['biometrically_verified'] = True

# Clean up the cache entry
cache.delete(f"biometric:{request_id}")

# Redirect directly to the role-specific dashboard
position = official_rec.position # 'captain', 'secretary', or 'treasurer'
role_dashboard_map = {
    'captain': 'core:dashboard',
    'secretary': 'core:dashboard',
    'treasurer': 'core:dashboard',
}
# Set session flag so dashboard() picks the right role even before profile refresh
request.session['biometric_position'] = position
return redirect('core:dashboard')

@login_required
def biometric_reset_all(request):
    """Clear all fingerprint data from the database and sync with ESP32."""
    role = getattr(request.user.profile, 'role', None) if hasattr(request.user, 'profile')
    else None
    if not (request.user.is_superuser or role in ['captain', 'secretary', 'treasurer',
'admin']):
        return JsonResponse({'status': 'error', 'message': 'Permission denied'},
status=403)

    # 1. Clear Django Database
    Resident.objects.all().update(fingerprint_id=None, fingerprint_template=None)

    # 2. Erase all templates on the R307 module (flash can take several seconds)
    esp32_base_url = get_esp32_base_url()
    esp32_status = "offline"
    resp = None
    try:
        resp = requests.post(
            f"{esp32_base_url}/empty-library",
            timeout=10,
            proxies={'http': None, 'https': None},
        )
    except Exception:
        esp32_base_url = get_esp32_base_url(force_scan=True)
        try:
            resp = requests.post(
```

IoT-Enabled Barangay Resident and Attendance Management System

```
f"{esp32_base_url}/empty-library",
    timeout=10,
    proxies={'http': None, 'https': None},
)
except Exception:
    pass

if resp and resp.ok:
    try:
        payload = resp.json()
        if payload.get('status') == 'success':
            esp32_status = "sensor cleared"
        else:
            esp32_status = f"sensor error: {payload.get('message', resp.text)[:120]}"
    except ValueError:
        esp32_status = "sensor cleared" if resp.status_code == 200 else f"HTTP
{resp.status_code}"
    elif resp is not None:
        esp32_status = f"HTTP {resp.status_code}"

    messages.success(
        request,
        f"Database biometric fields cleared. Fingerprint module: {esp32_status}. "
        "You can register fingerprints again from the enrollment page.",
    )
    return redirect('officials:biometric_register')

def login_view(request):
    """User login."""
    if request.user.is_authenticated:
        return redirect('dashboard')

    if request.method == 'POST':
        user_type = request.POST.get('user_type', 'official')
        role = request.POST.get('role') if user_type == 'official' else 'resident'
        password = request.POST.get('password')

        if user_type == 'resident':
            username = request.POST.get('username')
        else:
            username = request.POST.get('official_username')
            role = None # Role is no longer used for identification

        if not username:
            messages.error(request, 'Username is required.')
            return render(request, 'core/login.html')

        user = authenticate(request, username=username, password=password)

        if user is not None:
            profile = getattr(user, 'profile', None)
            user_role_name = profile.role if profile else (user.role.name if user.role
else 'resident')
            # Use official record as fallback check for role permission
            from officials.models import Official
            has_active_official_record = Official.objects.filter(user=user,
status='active').exists()
            is_official_user = user.is_superuser or (user.role and
user.role.permission_level < 3) or has_active_official_record
```


IoT-Enabled Barangay Resident and Attendance Management System

```
# Enforce separation as requested by the user
if user_type == 'resident' and is_official_user:
    messages.error(request, f'Account "{username}" is assigned as an Official
({user_role_name}). Please log in using the "Official" tab with your role.')
    return render(request, 'core/login.html')

if user_type == 'official':
    if not is_official_user:
        messages.error(request, f'Account "{username}" is a Resident account.
Please log in using the "Resident" tab.')
        return render(request, 'core/login.html')

    login(request, user)
    messages.success(request, f'Welcome back, {user.get_full_name() or
user.username}!')
    return redirect('dashboard')
else:
    messages.error(request, f'Invalid credentials for "{username}". Please check
your username and password.')

    return render(request, 'core/login.html')

def signup_view(request):
    messages.info(request, 'Public registration for officials is disabled. Please contact
the Captain for an invite.')
    return redirect('core:login')

def resident_signup_view(request):
    if request.user.is_authenticated:
        return redirect('dashboard')

    if request.method == 'POST':
        # Collect all fields from the multi-step form
        try:
            from django.contrib.auth.hashers import make_password

            # Manual extraction from POST to support dynamic steps without a complex Form
class
    reg = ResidentRegistration(
        # Step 1
        first_name=request.POST.get('first_name', '').strip(),
        middle_name=request.POST.get('middle_name', '').strip(),
        last_name=request.POST.get('last_name', '').strip(),
        suffix=request.POST.get('suffix', '').strip(),
        birthdate=request.POST.get('birthdate'),
        birthplace=request.POST.get('birthplace', '').strip(),
        gender=request.POST.get('gender'),
        civil_status=request.POST.get('civil_status'),
        nationality=request.POST.get('citizenship', 'Filipino').strip(),
        religion=request.POST.get('religion', '').strip(),
        highest_education=request.POST.get('highest_education'),
        occupation=request.POST.get('occupation', '').strip(),

        # Step 2
        mobile_number=request.POST.get('mobile_number', '').strip(),
        email=request.POST.get('email', '').strip(),
        house_number=request.POST.get('house_number', '').strip(),
        street=request.POST.get('street', '').strip(),
```

IoT-Enabled Barangay Resident and Attendance Management System

```
purok=request.POST.get('purok', '').strip(),
barangay=request.POST.get('barangay', '').strip(),
municipality=request.POST.get('municipality', '').strip(),
city=request.POST.get('city', '').strip(),
zip_code=request.POST.get('zip_code', '').strip(),
years_of_residency=int(request.POST.get('years_of_residency', 0)),
philSys_number=request.POST.get('philSys_number', '').strip() or None,

# Step 3
is_joining_household=(request.POST.get('household_action') == 'join'),
household_number=request.POST.get('household_number', '').strip(),
is_pwd=(request.POST.get('is_pwd') == 'on'),
is_senior_citizen=(request.POST.get('is_senior_citizen') == 'on'),
is_4ps_member=(request.POST.get('is_4ps_member') == 'on'),
is_sole_parent=(request.POST.get('is_sole_parent') == 'on'),
is_registered_voter=(request.POST.get('is_registered_voter') == 'on'),

guardian_name=request.POST.get('guardian_name', '').strip(),
guardian_relationship=request.POST.get('guardian_relationship',
').strip(),

guardian_contact=request.POST.get('guardian_contact', '').strip(),
guardian_id_number=request.POST.get('guardian_id_number', '').strip(),

# Step 4
username=request.POST.get('username', '').strip(),
password=make_password(request.POST.get('password1')),
data_privacy_consent=(request.POST.get('data_privacy_consent') == 'on'),
)

# File Uploads
if request.FILES.get('photo'):
    reg.photo = request.FILES['photo']
if request.FILES.get('id_card'):
    reg.id_card = request.FILES['id_card']
if request.FILES.get('birth_certificate'):
    reg.birth_certificate = request.FILES['birth_certificate']
if request.FILES.get('proof_of_residency'):
    reg.proof_of_residency = request.FILES['proof_of_residency']

# Check for duplicate username
if User.objects.filter(username=reg.username).exists() or
ResidentRegistration.objects.filter(username=reg.username, status='pending').exists():
    heads = Resident.objects.filter(is_household_head=True,
is_active=True).select_related('household')
    messages.error(request, 'Username is already taken or pending approval.')
    return render(request, 'core/resident_signup.html',
{'heads_of_households': heads})

# Delete any existing rejected/approved registrations to avoid UNIQUE
constraint violation
# (Approved registrations have already been converted to User/Resident
records)
ResidentRegistration.objects.filter(username=reg.username).exclude(status='pen
ding').delete()

reg.save()
return redirect('registration_success', ref=reg.reference_number)

except Exception as e:
    heads = Resident.objects.filter(is_household_head=True,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
is_active=True).select_related('household')
    messages.error(request, f'Registration failed: {str(e)}')
    return render(request, 'core/resident_signup.html', {'heads_of_households':
heads}})

    heads = Resident.objects.filter(is_household_head=True,
is_active=True).select_related('household')
    return render(request, 'core/resident_signup.html', {'heads_of_households': heads})


def registration_success_view(request, ref):
    registration = get_object_or_404(ResidentRegistration, reference_number=ref)
    return render(request, 'core/registration_success.html', {'registration':
registration})


@login_required
def logout_view(request):
    """User logout."""
    logout(request)
    messages.info(request, 'You have been logged out.')
    return redirect('login')


@login_required
def backup_download(request):
    """Trigger data backup as a browser download."""
    role = 'staff'
    try: role = request.user.profile.role
    except Exception: pass

    if not (request.user.is_superuser or role in ('admin', 'captain', 'secretary',
'treasurer')):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    try:
        zip_path = create_backup_archive()
        response = FileResponse(open(zip_path, 'rb'), as_attachment=True)
        response['Content-Disposition'] = f'attachment;
filename="{os.path.basename(zip_path)}"'
        return response
    except Exception as e:
        messages.error(request, f"Download failed: {str(e)}")
        return redirect('core:backup_setup')


@login_required
def backup_execute(request):
    """Trigger data backup to a server-side destination."""
    role = 'staff'
    try: role = request.user.profile.role
    except Exception: pass

    if not (request.user.is_superuser or role in ('admin', 'captain', 'secretary',
'treasurer')):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if request.method == 'POST':
    target_path = request.POST.get('target_path')
    success, message = perform_backup(target_base=target_path)
    if success:
        messages.success(request, f"Done! {message}")
    else:
        messages.error(request, message)

return redirect('core:backup_setup')

@login_required
def backup_mount_drive(request):
    """Attempt to mount a removable drive via udiskctl."""
    role = 'staff'
    try: role = request.user.profile.role
    except Exception: pass

    if not (request.user.is_superuser or role in ('admin', 'captain', 'secretary',
'treasurer')):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    device = request.POST.get('device')
    do_backup = request.POST.get('perform_backup') == 'true'

    if device:
        import subprocess
        try:
            # Attempt to mount
            subprocess.run(['udiskctl', 'mount', '-b', device], check=True, timeout=10)

            if do_backup:
                import time
                from core.utils.backup import get_backup_destinations, perform_backup

                new_mount = None
                # OS might take a second to settle the mount, so we retry a few times
                for attempt in range(3):
                    time.sleep(1.5) # Wait for OS
                    all_dests = get_backup_destinations()
                    new_mount = next((d['path'] for d in all_dests if d.get('device') ==
device and d.get('is_mounted')), None)
                    if new_mount: break

                if new_mount:
                    success, message = perform_backup(target_base=new_mount)
                    if success:
                        messages.success(request, f"Drive activated and {message}")
                    else:
                        messages.warning(request, f"Drive activated, but backup failed:
{message}")
                else:
                    messages.success(request, f"Drive {device} activated successfully.
Please click 'Save' below.")
            else:
                messages.success(request, f"Drive {device} activated successfully.")

        except subprocess.TimeoutExpired:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        messages.warning(request, "Mount command timed out. Your drive might be ready,
please refresh.")
    except Exception as e:
        messages.error(request, f"Failed to activate drive: {str(e)}")

    return redirect('core:backup_setup')

from django.http import JsonResponse
from core.utils.backup import get_backup_destinations

@login_required
def backup_check_dests(request):
    """JSON endpoint for polling backup destinations."""
    return JsonResponse({'destinations': get_backup_destinations()})

@login_required
def backup_setup(request):
    """Show available backup destinations (GUI for choosing location)."""
    role = 'staff'
    try: role = request.user.profile.role
    except Exception: pass

    if not (request.user.is_superuser or role in ('admin', 'captain', 'secretary',
'treasurer')):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    destinations = get_backup_destinations()
    return render(request, 'core/backup_choice.html', {'destinations': destinations})

# ??? Notifications ?????????????????????????????????????????????????????????

@login_required
def notifications_view(request):
    """Full notifications list for the current user."""
    from .models import Notification
    notifs = Notification.objects.filter(user=request.user).order_by('-created_at')
    # Mark all as read on page open
    Notification.objects.filter(user=request.user, is_read=False).update(is_read=True)
    return render(request, 'core/notifications.html', {'notifications': notifs})

@login_required
def mark_notification_read(request, pk):
    """Mark a single notification as read (AJAX or redirect)."""
    from .models import Notification
    notif = get_object_or_404(Notification, pk=pk, user=request.user)
    notif.is_read = True
    notif.save()
    next_url = request.GET.get('next') or notif.link or '/'
    return redirect(next_url)
```

IoT-Enabled Barangay Resident and Attendance Management System

File: documents/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: documents/admin.py

```
from django.contrib import admin
```

```
# Register your models here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: documents/apps.py

```
from django.apps import AppConfig

class DocumentsConfig(AppConfig):
    default_auto_field = 'django.db.models.BigAutoField'
    name = 'documents'
```


IoT-Enabled Barangay Resident and Attendance Management System

File: documents/models.py

```
from django.db import models
from django.conf import settings

class Document(models.Model):
    DOCUMENT_TYPES = [
        ('clearance', 'Barangay Clearance'),
        ('residency', 'Certificate of Residency'),
        ('indigency', 'Certificate of Indigency'),
        ('business', 'Business Permit'),
        ('ordinance', 'Ordinance'),
        ('other', 'Other'),
    ]

    title = models.CharField(max_length=255)
    document_type = models.CharField(max_length=20, choices=DOCUMENT_TYPES,
default='other')
    content = models.TextField(blank=True, null=True)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)
    created_by = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
null=True, blank=True, related_name='created_documents')

    def __str__(self):
        return f"{self.title} ({self.get_document_type_display()})"

class Meta:
    ordering = ['-updated_at']
```

IoT-Enabled Barangay Resident and Attendance Management System

File: documents/tests.py

```
from django.test import TestCase
```

```
# Create your tests here.
```

File: documents/urls.py

```
from django.urls import path
from . import views

app_name = 'documents'

urlpatterns = [
    path('', views.DocumentListView.as_view(), name='list'),
    path('workspace/', views.DocumentCreateView.as_view(), name='workspace'),
    path('<int:pk>/edit/', views.DocumentUpdateView.as_view(), name='edit'),
    path('<int:pk>/delete/', views.DocumentDeleteView.as_view(), name='delete'),
    path('<int:pk>/pdf/', views.DocumentPDFView.as_view(), name='pdf'),
]
```

IoT-Enabled Barangay Resident and Attendance Management System

File: documents/views.py

```
from django.shortcuts import render
from django.urls import reverse_lazy
from django.views.generic import ListView, CreateView, UpdateView, DeleteView
from django.contrib.auth.mixins import LoginRequiredMixin
from .models import Document

class DocumentListView(LoginRequiredMixin, ListView):
    model = Document
    template_name = 'documents/list.html'
    context_object_name = 'documents'

    def get_queryset(self):
        # Allow all users to see documents for now. Can restrict based on roles if needed.
        return Document.objects.all().order_by('-updated_at')

class DocumentCreateView(LoginRequiredMixin, CreateView):
    model = Document
    fields = ['title', 'document_type', 'content']
    template_name = 'documents/workspace.html'
    success_url = reverse_lazy('documents:list')

    def form_valid(self, form):
        form.instance.created_by = self.request.user
        return super().form_valid(form)

    def get_context_data(self, **kwargs):
        context = super().get_context_data(**kwargs)
        context['title'] = 'New Document'
        context['document_types'] = Document.DOCUMENT_TYPES
        return context

class DocumentUpdateView(LoginRequiredMixin, UpdateView):
    model = Document
    fields = ['title', 'document_type', 'content']
    template_name = 'documents/workspace.html'
    success_url = reverse_lazy('documents:list')

    def get_context_data(self, **kwargs):
        context = super().get_context_data(**kwargs)
        context['title'] = 'Edit Document'
        context['document_types'] = Document.DOCUMENT_TYPES
        return context

class DocumentDeleteView(LoginRequiredMixin, DeleteView):
    model = Document
    template_name = 'documents/confirm_delete.html'
    success_url = reverse_lazy('documents:list')

from django.http import HttpResponse
from django.shortcuts import get_object_or_404
from django.template.loader import render_to_string
from django.views import View

class DocumentPDFView(LoginRequiredMixin, View):
    def get(self, request, pk, *args, **kwargs):
        document = get_object_or_404(Document, pk=pk)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# Render the HTML template with the document content
html_string = render_to_string('documents/pdf_export.html', {'document':
document})

try:
    import weasyprint
    # Generate PDF using WeasyPrint
    pdf_file = weasyprint.HTML(string=html_string,
base_url=request.build_absolute_uri()).write_pdf()

    # Return as HTTP response with PDF content type
    response = HttpResponse(pdf_file, content_type='application/pdf')
    filename = f"{document.title or 'document'}.pdf"
    response['Content-Disposition'] = f'attachment; filename="{filename}"'
    return response
except OSError as e:
    # Catch the Windows GTK3 missing error (libgobject-2.0-0)
    return HttpResponse(
        "<h2>PDF Export Failed</h2>"
        "<p>WeasyPrint requires GTK3 binaries to be installed on Windows.</p>"
        f"<p>Error details: {str(e)}</p>"
        "<p>To fix this, please install the GTK3 runtime for Windows and add it to
your system PATH.</p>",
        status=500
    )
```

IoT-Enabled Barangay Resident and Attendance Management System

File: fix_hotspot.sh

```
#!/bin/bash
# =====
# Barangay System - Hotspot Diagnostic & Fix Script (PERMANENT FIX v2)
# =====
# Run this on the Orange Pi to:
# 1. Permanently ban NetworkManager from touching wlan0
# 2. Delete all rogue Wi-Fi profiles that steal the interface
# 3. Set up hostapd + dnsmasq for the hotspot
# 4. Enable NAT internet sharing from Ethernet to Wi-Fi
# 5. Persist ALL settings across reboots
#
# Usage: sudo ./fix_hotspot.sh
# =====

RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
CYAN='\033[0;36m'
NC='\033[0m' # No Color

if [ "$EUID" -ne 0 ]; then
    echo -e "${RED}[ERROR] Please run with sudo: sudo ./fix_hotspot.sh${NC}"
    exit 1
fi

echo -e "${CYAN}======"
echo -e "  Barangay System - Hotspot Diagnostic & Fix (v2)"
echo -e "===== ${NC}"

# =====
# STEP 0: PERMANENTLY BAN NETWORKMANAGER FROM TOUCHING WLAN0
# =====
# THIS IS THE ROOT CAUSE OF ALL INTERNET DROPS.
# NetworkManager keeps creating Wi-Fi connections on wlan0, which:
# - Steals the interface from hostapd
# - Adds a bad default route via wlan0 (instead of end0)
# - Assigns 192.168.x.x to wlan0 (conflicts with the hotspot 10.42.0.1)
# =====
echo -e "\n${YELLOW}[STEP 0] Permanently banning NetworkManager from wlan0...${NC}"

# 0a. Create the permanent ignore rule
mkdir -p /etc/NetworkManager/conf.d/
cat > /etc/NetworkManager/conf.d/10-ignore-wifi.conf <<EOF
[keyfile]
unmanaged-devices=interface-name:wlan0
EOF
echo -e "${GREEN}[OK] Created /etc/NetworkManager/conf.d/10-ignore-wifi.conf${NC}"

# 0b. Also add it to the main NetworkManager config as a backup
if [ -f /etc/NetworkManager/NetworkManager.conf ]; then
    if ! grep -q "unmanaged-devices=interface-name:wlan0"
/etc/NetworkManager/NetworkManager.conf; then
        # Add under [keyfile] section if it exists, otherwise append
        if grep -q "[keyfile]" /etc/NetworkManager/NetworkManager.conf; then
            sed -i '/[keyfile]/a unmanaged-devices=interface-name:wlan0'
/etc/NetworkManager/NetworkManager.conf
        else
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        echo -e "\n[keyfile]\nunmanaged-devices=interface-name:wlan0" >>
/etc/NetworkManager/NetworkManager.conf
    fi
    echo -e "${GREEN}[OK] Also patched NetworkManager.conf${NC}"
fi
fi

# 0c. Delete ALL saved Wi-Fi connections that could hijack wlan0
echo -e "${YELLOW}    Deleting rogue Wi-Fi profiles from NetworkManager...${NC}"
# Get all wifi-type connections and delete them
nmcli -t -f NAME,TYPE connection show 2>/dev/null | grep ".*wifi" | cut -d: -f1 | while
read -r conn; do
    echo -e "    Deleting: $conn"
    nmcli connection delete "$conn" 2>/dev/null || true
done

# Also delete known problematic connection names (in case they have wrong type)
for conn in "Hotspot" "Hotspot-1" "Hotspot-2" "netplan-wlan0" "Wi-Fi connection 1" "Wi-Fi
connection 2"; do
    nmcli connection delete "$conn" 2>/dev/null || true
done
echo -e "${GREEN}[OK] All rogue Wi-Fi profiles deleted.${NC}"

# 0d. Tell NetworkManager to unmanage wlan0 right now
nmcli device set wlan0 managed no 2>/dev/null || true

# 0e. Restart NetworkManager to apply the ignore rule
systemctl restart NetworkManager
sleep 2
echo -e "${GREEN}[OK] NetworkManager restarted. wlan0 is now permanently unmanaged.${NC}"

# 0f. Remove any bad IP addresses and routes that NetworkManager left behind
echo -e "${YELLOW}    Cleaning up leftover routes and IPs on wlan0...${NC}"
# Delete any default route pointing to wlan0
ip route del default dev wlan0 2>/dev/null || true
# Delete any 192.168.x.x address on wlan0 (these are from the router, NOT the hotspot)
ip addr show wlan0 2>/dev/null | grep "inet 192\." | awk '{print $2}' | while read -r
addr; do
    echo -e "    Removing bad IP: $addr from wlan0"
    ip addr del "$addr" dev wlan0 2>/dev/null || true
done
echo -e "${GREEN}[OK] Cleanup complete.${NC}"

# =====
# STEP 1: DETECT THE ACTUAL WIFI INTERFACE NAME
# =====
echo -e "\n${YELLOW}[STEP 1] Detecting WiFi interface name...${NC}"

# Show all network interfaces
echo "--- All interfaces found: ---"
ip link show | grep -E "^[0-9]+:" | awk '{print $2}' | sed 's:///'

# Try to find the WiFi interface
WIFI_IFACE=$(iw dev 2>/dev/null | awk ' $1=="Interface"{print $2}' | head -n 1)

if [ -z "$WIFI_IFACE" ]; then
    # Fallback: try common names
    for iface in wlan0 wlan1 wlx0 wlp2s0; do
        if ip link show "$iface" &>/dev/null; then
            WIFI_IFACE="$iface"
        fi
    done
fi
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        break
    fi
done
fi

if [ -z "$WIFI_IFACE" ]; then
    echo -e "${RED}[FATAL] No WiFi interface detected!"
    echo -e "        Your Orange Pi may not have a WiFi chip or it is disabled.${NC}"
    echo ""
    echo "Try running: lsusb (to check for USB WiFi adapter)"
    echo "Try running: lspci (to check for PCIe WiFi adapter)"
    exit 1
fi

echo -e "${GREEN}[OK] WiFi interface found: $WIFI_IFACE${NC}"

# =====
# STEP 2: CHECK IF INTERFACE SUPPORTS AP MODE
# =====
echo -e "\n${YELLOW}[STEP 2] Checking if $WIFI_IFACE supports Access Point (AP)
mode...${NC}"

AP_SUPPORT=$(iw phy 2>/dev/null | grep -A 10 "Supported interface modes" | grep "AP")
if [ -z "$AP_SUPPORT" ]; then
    echo -e "${RED}[FATAL] Your WiFi chip does NOT support AP mode!"
    echo -e "        hostapd will NEVER work with this hardware.${NC}"
    echo ""
    echo "SOLUTION: Buy a USB WiFi adapter that supports AP mode."
    echo "        Recommended: TP-Link TL-WN823N or any Realtek RTL8188/RTL8192"
    exit 1
fi
echo -e "${GREEN}[OK] AP mode is supported.${NC}"

# =====
# STEP 3: CHECK RFKILL (HARDWARE/SOFTWARE BLOCK)
# =====
echo -e "\n${YELLOW}[STEP 3] Checking rfkill (radio blocks)...${NC}"
rfkill list all
SOFT_BLOCKED=$(rfkill list all | grep -i "Soft blocked: yes")
HARD_BLOCKED=$(rfkill list all | grep -i "Hard blocked: yes")

if [ -n "$HARD_BLOCKED" ]; then
    echo -e "${RED}[FATAL] WiFi is HARDWARE BLOCKED. Check physical WiFi switch on
device.${NC}"
    exit 1
fi

if [ -n "$SOFT_BLOCKED" ]; then
    echo -e "${YELLOW}[FIX] WiFi is soft-blocked. Unblocking...${NC}"
    rfkill unblock wifi
    rfkill unblock all
    echo -e "${GREEN}[OK] Unblocked.${NC}"
else
    echo -e "${GREEN}[OK] No rfkill blocks.${NC}"
fi

# =====
# STEP 4: CHECK HOSTAPD SERVICE STATUS
# =====
echo -e "\n${YELLOW}[STEP 4] Checking hostapd status...${NC}"
```


IoT-Enabled Barangay Resident and Attendance Management System

```
if ! command -v hostapd &>/dev/null; then
    echo -e "${RED}[ERROR] hostapd is NOT installed!${NC}"
    echo -e "${YELLOW}[FIX] Installing hostapd and dnsmasq...${NC}"
    apt-get update -qq
    apt-get install -y hostapd dnsmasq iptables-persistent
    echo -e "${GREEN}[OK] Installed.${NC}"
else
    echo -e "${GREEN}[OK] hostapd is installed.${NC}"
fi

# Check if masked
MASKED=$(systemctl status hostapd 2>&1 | grep -i "masked")
if [ -n "$MASKED" ]; then
    echo -e "${YELLOW}[FIX] hostapd is MASKED. Unmasking...${NC}"
    systemctl unmask hostapd
    echo -e "${GREEN}[OK] Unmasked.${NC}"
fi

# Show hostapd status
echo "--- hostapd service status ---"
systemctl status hostapd --no-pager -l 2>&1 | tail -20

# =====
# STEP 5: DISABLE CONFLICTING NETWORK SERVICES
# =====
echo -e "\n${YELLOW}[STEP 5] Disabling conflicting network services...${NC}"

# Disable dhcpd if running (fights with NetworkManager)
if systemctl is-active dhcpd &>/dev/null; then
    echo -e "${YELLOW}[FIX] dhcpd is running ? disabling it...${NC}"
    systemctl stop dhcpd 2>/dev/null
    systemctl disable dhcpd 2>/dev/null
    echo -e "${GREEN}[OK] dhcpd disabled.${NC}"
fi

# MASK systemd-networkd (it aggressively re-assigns IPs via socket activation)
echo -e "${YELLOW}[FIX] Nuking systemd-networkd and its sockets to prevent
interference...${NC}"
systemctl stop systemd-networkd.socket 2>/dev/null || true
systemctl disable systemd-networkd.socket 2>/dev/null || true
systemctl stop systemd-networkd.service 2>/dev/null || true
systemctl disable systemd-networkd.service 2>/dev/null || true
systemctl mask systemd-networkd.service 2>/dev/null || true
echo -e "${GREEN}[OK] systemd-networkd completely masked.${NC}"

# Delete any rogue systemd-networkd configuration files specifically targeting wlan0
if ls /etc/systemd/network/*wlan0* 1> /dev/null 2>&1; then
    echo -e "${YELLOW}[FIX] Found rogue systemd-networkd wlan0 configs. Deleting
them...${NC}"
    rm -f /etc/systemd/network/*wlan0*
    echo -e "${GREEN}[OK] Rogue configs deleted.${NC}"
fi

# Verify wlan0 is truly unmanaged by NetworkManager
NM_STATUS=$(nmcli dev status 2>/dev/null | grep "$WIFI_IFACE" | awk '{print $3}')
if [ "$NM_STATUS" != "unmanaged" ] && [ -n "$NM_STATUS" ]; then
    echo -e "${YELLOW}[FIX] wlan0 is STILL managed by NetworkManager ($NM_STATUS). Forcing
unmanaged...${NC}"
    nmcli device set "$WIFI_IFACE" managed no 2>/dev/null || true
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    sleep 1
fi
echo -e "${GREEN}[OK] No conflicting services.${NC}"

# =====
# STEP 6: DETECT ETHERNET INTERFACE (NEVER pick wlan0!)
# =====
echo -e "\n${YELLOW}[STEP 6] Detecting Ethernet interface for internet sharing...${NC}"

# CRITICAL: We must NEVER use wlan0/wlan1 as the ethernet interface.
# Always detect by checking for physical ethernet ports first.
ETH_IFACE=""
for iface in end0 eth0 enp1s0 enp2s0 enp3s0; do
    if ip link show "$iface" &>/dev/null; then
        ETH_IFACE="$iface"
        break
    fi
done

if [ -z "$ETH_IFACE" ]; then
    echo -e "${RED}[ERROR] No physical ethernet interface found (checked end0, eth0,
enp1s0, enp2s0).${NC}"
    echo -e "${YELLOW}          Defaulting to end0. Make sure LAN cable is plugged in.${NC}"
    ETH_IFACE="end0"
fi

echo -e "${GREEN}[OK] Ethernet interface: $ETH_IFACE${NC}"

# STEP 6b: Make sure the ethernet interface has an IP via DHCP
echo -e "${YELLOW}[STEP 6b] Ensuring $ETH_IFACE has internet (DHCP)...${NC}"
ip link set dev $ETH_IFACE up

# Check if NetworkManager has a connection for end0, if not create one
if ! nmcli -t -f NAME con show --active 2>/dev/null | grep -qi "$ETH_IFACE"; then
    echo -e "${YELLOW}[FIX] $ETH_IFACE is not active. Activating via
NetworkManager...${NC}"
    # Try to bring up any existing connection for this device
    nmcli dev connect $ETH_IFACE 2>/dev/null || true
    sleep 3
fi

# Verify it got an IP
ETH_IP=$(ip addr show $ETH_IFACE 2>/dev/null | grep 'inet ' | grep -v '127.0.0' | awk
'{print $2}' | head -1)
if [ -n "$ETH_IP" ]; then
    echo -e "${GREEN}[OK] $ETH_IFACE has IP: $ETH_IP${NC}"
else
    echo -e "${YELLOW}[WARN] $ETH_IFACE has no IP yet. Trying nmcli...${NC}"
    nmcli dev connect $ETH_IFACE 2>/dev/null || true
    sleep 5
fi

# =====
# STEP 7: WRITE CORRECT HOSTAPD CONFIG
# =====
echo -e "\n${YELLOW}[STEP 7] Writing correct hostapd configuration...${NC}"

cat > /etc/hostapd/hostapd.conf <<EOF
interface=$WIFI_IFACE
driver=nl80211
```

IoT-Enabled Barangay Resident and Attendance Management System

```
ssid=Barangay_System_WiFi
hw_mode=g
channel=6
ieee80211n=1
wmm_enabled=1
macaddr_acl=0
auth_algs=1
ignore_broadcast_ssid=0
wpa=2
wpa_passphrase=barangay123
wpa_key_mgmt=WPA-PSK
wpa_pairwise=CCMP
rsn_pairwise=CCMP
EOF

echo -e "${GREEN}[OK] hostapd.conf written with interface=$WIFI_IFACE${NC}"

# Make sure /etc/default/hostapd points to our config
if [ -f /etc/default/hostapd ]; then
    # Remove old DAEMON_CONF lines and add correct one
    sed -i '/^DAEMON_CONF/d' /etc/default/hostapd
    sed -i '/^#DAEMON_CONF/d' /etc/default/hostapd
    echo 'DAEMON_CONF="/etc/hostapd/hostapd.conf"' >> /etc/default/hostapd
    echo -e "${GREEN}[OK] /etc/default/hostapd updated.${NC}"
fi

# =====
# STEP 8: SET STATIC IP ON WIFI INTERFACE
# =====
echo -e "\n${YELLOW}[STEP 8] Setting static IP 10.42.0.1 on $WIFI_IFACE...${NC}"

ip link set dev $WIFI_IFACE up
sleep 1
ip addr flush dev $WIFI_IFACE
ip addr add 10.42.0.1/24 dev $WIFI_IFACE
echo -e "${GREEN}[OK] IP set.${NC}"

# =====
# STEP 9: CONFIGURE DNSMASQ (DHCP SERVER)
# =====
echo -e "\n${YELLOW}[STEP 9] Configuring dnsmasq DHCP...${NC}"

# Stop dnsmasq if it is fighting with systemd-resolved on port 53
systemctl stop dnsmasq 2>/dev/null
systemctl disable systemd-resolved 2>/dev/null
systemctl stop systemd-resolved 2>/dev/null

# Check if port 53 is already occupied
PORT53=$(ss -tlnp | grep ':53')
if [ -n "$PORT53" ]; then
    echo -e "${YELLOW}[WARN] Port 53 is in use. Killing the occupant...${NC}"
    fuser -k 53/tcp 2>/dev/null
    fuser -k 53/udp 2>/dev/null
fi

# Back up original
[ -f /etc/dnsmasq.conf ] && cp /etc/dnsmasq.conf /etc/dnsmasq.conf.bak

cat > /etc/dnsmasq.conf <<EOF
# Barangay System DHCP Config
```

IoT-Enabled Barangay Resident and Attendance Management System

```
interface=$WIFI_IFACE
bind-interfaces
except-interface=lo
dhcp-range=10.42.0.10,10.42.0.100,255.255.255.0,24h
domain=local
address=/brgysicosico.local/10.42.0.1
# Prevent dnsmasq from touching /etc/resolv.conf
no-resolv
server=8.8.8.8
server=8.8.4.4
EOF

echo -e "${GREEN}[OK] dnsmasq configured.${NC}"

# =====
# STEP 10: FIX DNS RESOLUTION ON THE ORANGE PI ITSELF
# =====
echo -e "\n${YELLOW}[STEP 10] Fixing DNS resolution on the Pi...${NC}"

# Remove symlink if it exists and write a proper resolv.conf
rm -f /etc/resolv.conf
cat > /etc/resolv.conf <<EOF
nameserver 8.8.8.8
nameserver 8.8.4.4
EOF

# Prevent anything from overwriting our resolv.conf
chattr +i /etc/resolv.conf 2>/dev/null || true
echo -e "${GREEN}[OK] DNS resolvers set to 8.8.8.8 and 8.8.4.4 (locked).${NC}"

# =====
# STEP 11: ENABLE IP FORWARDING AND NAT (INTERNET SHARING) ? PERMANENTLY
# =====
echo -e "\n${YELLOW}[STEP 11] Enabling Internet sharing (NAT) permanently...${NC}"

# Enable immediately
echo 1 > /proc/sys/net/ipv4/ip_forward

# Make persistent in sysctl.conf
grep -q "net.ipv4.ip_forward=1" /etc/sysctl.conf || echo "net.ipv4.ip_forward=1" >>
/etc/sysctl.conf
sed -i 's/^#*net.ipv4.ip_forward.*net.ipv4.ip_forward=1/' /etc/sysctl.conf
sysctl -p

# SAFETY CHECK: Refuse to set up NAT if ETH_IFACE is a wireless interface
if echo "$ETH_IFACE" | grep -qE '^(wlan|wlp|wlx)'; then
    echo -e "${RED}[FATAL] ETH_IFACE='$ETH_IFACE' is a WiFi interface! Refusing to set up
    NAT.${NC}"
    echo -e "${RED}          This would kill your internet. Skipping NAT setup.${NC}"
else
    # FLUSH old rules first to prevent duplicates
    iptables -t nat -F
    iptables -F FORWARD

    # iptables NAT rules
    iptables -t nat -A POSTROUTING -o $ETH_IFACE -j MASQUERADE
    iptables -A FORWARD -i $ETH_IFACE -o $WIFI_IFACE -m state --state RELATED,ESTABLISHED
    -j ACCEPT
    iptables -A FORWARD -i $WIFI_IFACE -o $ETH_IFACE -j ACCEPT
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# Save iptables rules PERMANENTLY
if command -v netfilter-persistent &>/dev/null; then
    netfilter-persistent save
elif command -v iptables-save &>/dev/null; then
    mkdir -p /etc/iptables
    iptables-save > /etc/iptables/rules.v4 2>/dev/null || true
fi

echo -e "${GREEN}[OK] NAT routing enabled: $WIFI_IFACE <-> $ETH_IFACE${NC}"
fi

# =====
# STEP 12: DOUBLE-CHECK ? REMOVE ANY BAD ROUTES ON wlan0
# =====
echo -e "\n${YELLOW}[STEP 12] Final route cleanup (removing any bad wlan0 routes)...${NC}"

# Delete any default route that goes through wlan0 (the #1 cause of all problems)
ip route del default dev $WIFI_IFACE 2>/dev/null || true

# Delete any 192.168.x.x routes on wlan0 (these belong to end0, not the hotspot)
ip route | grep "$WIFI_IFACE" | grep "192.168" | while read -r route; do
    echo -e "    Deleting bad route: $route"
    ip route del $route 2>/dev/null || true
done

# Make sure the ONLY default route goes through the Ethernet interface
DEFAULT_COUNT=$(ip route | grep "^default" | wc -l)
if [ "$DEFAULT_COUNT" -eq 0 ]; then
    echo -e "${YELLOW}[FIX] No default route found! Adding one via $ETH_IFACE...${NC}"
    # Get the gateway from DHCP
    GW=$(nmcli -t -f IP4.GATEWAY dev show $ETH_IFACE 2>/dev/null | cut -d: -f2)
    if [ -n "$GW" ]; then
        ip route add default via $GW dev $ETH_IFACE
        echo -e "${GREEN}[OK] Default route added via $GW${NC}"
    else
        echo -e "${RED}[WARN] Could not detect gateway. Internet may not work.${NC}"
    fi
fi

echo -e "${GREEN}[OK] Route table is clean.${NC}"

# =====
# STEP 13: START SERVICES
# =====
echo -e "\n${YELLOW}[STEP 13] Starting services...${NC}"

systemctl enable hostapd
systemctl enable dnsmasq

systemctl restart dnsmasq
sleep 1
echo "--- dnsmasq status ---"
systemctl status dnsmasq --no-pager -l | tail -10

systemctl restart hostapd
sleep 3
echo "--- hostapd status ---"
systemctl status hostapd --no-pager -l | tail -20

# =====
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# STEP 14: INSTALL SYSTEMD SERVICE FOR BOOT PERSISTENCE
# =====
echo -e "\n${YELLOW}[STEP 14] Installing boot service for persistence...${NC}"

SCRIPT_PATH="$(cd "$(dirname "$0")" && pwd)/$(basename "$0")"

cat > /etc/systemd/system/brgy-network.service <<EOF
[Unit]
Description=Barangay System Network Fix (Hotspot + Internet Routing)
After=network-online.target NetworkManager.service
Wants=network-online.target

[Service]
Type=oneshot
ExecStartPre=/bin/sleep 10
ExecStart=/bin/bash $SCRIPT_PATH
RemainAfterExit=yes
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
EOF

systemctl daemon-reload
systemctl enable brgy-network.service
echo -e "${GREEN}[OK] brgy-network.service installed and enabled.${NC}"
echo -e "${GREEN}      This script will run automatically on every boot.${NC}"

# =====
# STEP 15: VERIFY
# =====
echo -e "\n${YELLOW}[STEP 15] Final verification...${NC}"
echo "--- $WIFI_IFACE IP address ---"
ip addr show $WIFI_IFACE

echo ""
echo "--- Routing table ---"
ip route

echo ""
echo "--- Testing internet connectivity ---"
if ping -c 2 -W 3 8.8.8.8 &>/dev/null; then
    INET_STATUS="${GREEN}WORKING${NC}"
else
    INET_STATUS="${RED}FAILED${NC}"
fi

HOSTAPD_RUNNING=$(systemctl is-active hostapd)
DNSMASQ_RUNNING=$(systemctl is-active dnsmasq)

echo ""
echo -e "${CYAN}======"
echo -e "  RESULT SUMMARY"
echo -e "===== ${NC}"
echo -e "  WiFi Interface : ${GREEN}$WIFI_IFACE${NC}"
echo -e "  Ethernet       : ${GREEN}$ETH_IFACE${NC}"
echo -e "  hostapd        : $([ "$HOSTAPD_RUNNING" = "active" ] && echo -e  
"${GREEN}RUNNING${NC}" || echo -e "${RED}FAILED${NC}")"
echo -e "  dnsmasq        : $([ "$DNSMASQ_RUNNING" = "active" ] && echo -e
```

IoT-Enabled Barangay Resident and Attendance Management System

```
"${GREEN}RUNNING${NC}" || echo -e "${RED}FAILED${NC}")"
echo -e " Internet      : $INET_STATUS"
echo -e " NM ignores wlan: ${GREEN}YES (permanent)${NC}"
echo -e " Boot service   : ${GREEN}ENABLED${NC}"
echo ""

if [ "$HOSTAPD_RUNNING" != "active" ]; then
    echo -e "${RED}[HOSTAPD FAILED] Full error log:${NC}"
    journalctl -u hostapd -n 30 --no-pager
    echo ""
    echo -e "${YELLOW}MANUAL DEBUG: Run this to test hostapd directly:${NC}"
    echo " sudo hostapd -d /etc/hostapd/hostapd.conf"
else
    echo -e "${GREEN}=== SUCCESS! WiFi Hotspot is broadcasting ==="
    echo -e " SSID      : Barangay_System_WiFi"
    echo -e " Password : barangay123"
    echo -e " IP        : 10.42.0.1${NC}"
fi

echo ""
echo -e "${CYAN}This fix is now PERMANENT. It will survive reboots.${NC}"
echo -e "${CYAN}If you ever need to re-run manually: sudo bash $(basename $0)${NC}"
```

File: generate_codebook.py

```
import os
import re
import urllib.parse

# Configuration
ROOT_DIR = "."
OUTPUT_MD = "codebook.md"
INCLUDE_EXTENSIONS = {'.py', '.html', '.css', '.js', '.sh'}
EXCLUDE_DIRS = {
    '.git', 'node_modules', '__pycache__', '.venv', 'venv', 'env',
    '.agents', '.qodo', 'migrations', 'media', 'static', 'staticfiles',
    'scratch', '.gemini'
}

def sanitize_line(line):
    # Redact Django SECRET_KEY
    line = re.sub(r'(SECRET_KEY\s*=\s*)([\'"]).*?\2', r'\1\2***REDACTED***\2', line)
    # Redact database passwords
    line = re.sub(r'([\'"]PASSWORD[\'"]\s*=\s*)([\'"]).*?\2', r'\1\2***REDACTED***\2',
line)
    # Redact other common password patterns
    line = re.sub(r'([Pp]assword\s*=\s*)([\'"]).*?\2', r'\1\2***REDACTED***\2', line)
    return line

def generate_codebook():
    files_to_process = []

    # Traverse directory
    for root, dirs, files in os.walk(ROOT_DIR):
        # Exclude directories
        dirs[:] = [d for d in dirs if d not in EXCLUDE_DIRS and not d.startswith('.')]

        for file in files:
            ext = os.path.splitext(file)[1]
            if ext in INCLUDE_EXTENSIONS:
                files_to_process.append(os.path.join(root, file))

    # Sort files for consistency
    files_to_process.sort()

    with open(OUTPUT_MD, 'w', encoding='utf-8') as md_file:
        md_file.write("# Project Codebook\n\n")
        md_file.write("This document compiles the source code of the project. Sensitive
credentials have been redacted.\n\n")

        md_file.write("## Table of Contents\n\n")
        for filepath in files_to_process:
            # Normalize path for Markdown links
            rel_path = os.path.relpath(filepath, ROOT_DIR).replace('\\', '/')
            anchor = rel_path.replace('/', '').replace('.', '')
            md_file.write(f"- [{rel_path}]({anchor})\n")

        md_file.write("\n--\n\n")

        for filepath in files_to_process:
            rel_path = os.path.relpath(filepath, ROOT_DIR).replace('\\', '/')
            anchor = rel_path.replace('/', '').replace('.', '')
```


IoT-Enabled Barangay Resident and Attendance Management System

```
ext = os.path.splitext(file)[1][1:] # e.g. 'py', 'html'
if ext == 'sh':
    ext = 'bash'

md_file.write(f"## <a name=\"{anchor}\"></a> {rel_path}\n\n")
md_file.write(f"`{ext}\n")

try:
    with open(filepath, 'r', encoding='utf-8') as f:
        for line in f:
            md_file.write(sanitize_line(line))
except UnicodeDecodeError:
    md_file.write("Could not decode file content.\n")
except Exception as e:
    md_file.write(f"Error reading file: {e}\n")

md_file.write(f"`\n\n")

if __name__ == "__main__":
    print("Generating markdown codebook...")
    generate_codebook()
    print(f"Successfully generated {OUTPUT_MD}")
```

IoT-Enabled Barangay Resident and Attendance Management System

File: generate_pdf.py

```
from fpdf import FPDF
import os
import re
import textwrap

ROOT_DIR = "."
OUTPUT_PDF = "codebook.pdf"
INCLUDE_EXTENSIONS = {'.py', '.html', '.css', '.js', '.sh'}
EXCLUDE_DIRS = {
    '.git', 'node_modules', '__pycache__', '.venv', 'venv', 'env',
    '.agents', '.qodo', 'migrations', 'media', 'static', 'staticfiles',
    'scratch', '.gemini'
}

def sanitize_line(line):
    line = re.sub(r'(SECRET_KEY\s*=\s*)([\'"]).*?\2', r'\1\2***REDACTED***\2', line)
    line = re.sub(r'([\'"]PASSWORD[\'"]\s*=\s*)([\'"]).*?\2', r'\1\2***REDACTED***\2',
line)
    line = re.sub(r'([Pp]assword\s*=\s*)([\'"]).*?\2', r'\1\2***REDACTED***\2', line)
    return line

class PDF(FPDF):
    def header(self):
        self.set_font("Courier", "B", 10)
        self.cell(0, 10, "IoT-Enabled Barangay Resident and Attendance Management System",
border=False, align="C", new_x="LMARGIN", new_y="NEXT")
        self.ln(5)

    def footer(self):
        self.set_y(-15)
        self.set_font("Courier", "I", 8)
        self.cell(0, 10, f"Page {self.page_no()}", align="C")

def render_title_page(pdf):
    pdf.add_page()
    logo_path = os.path.join(ROOT_DIR, "static", "images", "ibrams_logo.png")
    if os.path.exists(logo_path):
        pdf.image(logo_path, x=65, y=50, w=80)

    pdf.ln(130)
    pdf.set_font("Courier", "B", 16)
    pdf.multi_cell(0, 10, "IoT-Enabled Barangay Resident and Attendance Management
System", align="C")
    pdf.ln(10)
    pdf.set_font("Courier", "", 14)
    pdf.cell(0, 10, "Project Codebook", new_x="LMARGIN", new_y="NEXT", align="C")

def render_file_content(pdf, filepath, rel_path):
    pdf.add_page()
    pdf.set_font("Courier", "B", 12)
    pdf.cell(0, 8, f"File: {rel_path}", new_x="LMARGIN", new_y="NEXT")
    pdf.ln(4)

    pdf.set_font("Courier", "", 8)

    try:
        with open(filepath, 'r', encoding='utf-8') as f:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
for line in f:
    sanitized = sanitize_line(line)
    sanitized = sanitized.replace('\t', ' ').replace('\n', '')
    sanitized = sanitized.encode('latin-1', 'replace').decode('latin-1')

    if not sanitized.strip():
        pdf.cell(0, 4, "", new_x="LMARGIN", new_y="NEXT")
        continue

    wrapped_lines = textwrap.wrap(sanitized, width=90,
replace_whitespace=False, drop_whitespace=False)
    for w_line in wrapped_lines:
        pdf.cell(0, 4, w_line, new_x="LMARGIN", new_y="NEXT")
except Exception as e:
    pdf.cell(0, 4, f"Error reading file: {e}", new_x="LMARGIN", new_y="NEXT")

def generate_pdf():
    print("Generating codebook.pdf...")

    files_to_process = []
    for root, dirs, files in os.walk(ROOT_DIR):
        dirs[:] = [d for d in dirs if d not in EXCLUDE_DIRS and not d.startswith('.')]
        for file in files:
            ext = os.path.splitext(file)[1]
            if ext in INCLUDE_EXTENSIONS:
                files_to_process.append(os.path.join(root, file))

    files_to_process.sort()

    # Pass 1: Generate body to find exact page numbers relative to body start
    print("Pass 1: Calculating page numbers...")
    pdf1 = PDF()
    pdf1.set_margins(left=20, top=20, right=20)
    pdf1.set_auto_page_break(auto=True, margin=20)

    file_pages = []
    for filepath in files_to_process:
        rel_path = os.path.relpath(filepath, ROOT_DIR).replace('\\', '/')
        file_pages.append((rel_path, pdf1.page_no() + 1)) # Starts at page 1 of body
        render_file_content(pdf1, filepath, rel_path)

    # Pass 2: Calculate TOC length
    print("Pass 2: Calculating TOC pages...")
    pdf2 = PDF()
    pdf2.set_margins(left=20, top=20, right=20)
    pdf2.set_auto_page_break(auto=True, margin=20)
    render_title_page(pdf2)

    toc_start_page = pdf2.page_no() + 1
    pdf2.add_page()
    pdf2.set_font("Courier", "B", 14)
    pdf2.cell(0, 10, "Table of Contents", new_x="LMARGIN", new_y="NEXT", align="C")
    pdf2.ln(5)
    pdf2.set_font("Courier", "", 10)

    # We use a large dummy offset first to get the max possible size string
    for rel_path, _ in file_pages:
        pdf2.cell(0, 6, f"Page 999 .... {rel_path}", new_x="LMARGIN", new_y="NEXT")

    toc_end_page = pdf2.page_no()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
toc_pages_count = toc_end_page - toc_start_page + 1

body_start_page = toc_end_page + 1

# Pass 3: Final PDF Generation
print("Pass 3: Generating final PDF...")
pdf_final = PDF()
pdf_final.set_margins(left=20, top=20, right=20)
pdf_final.set_auto_page_break(auto=True, margin=20)

render_title_page(pdf_final)

pdf_final.add_page()
pdf_final.set_font("Courier", "B", 14)
pdf_final.cell(0, 10, "Table of Contents", new_x="LMARGIN", new_y="NEXT", align="C")
pdf_final.ln(5)
pdf_final.set_font("Courier", "", 10)

for filepath, rel_path in zip(files_to_process, [x[0] for x in file_pages]):
    # Calculate original body page relative to 1, add the offset (body_start_page - 1)
    original_body_page = file_pages[files_to_process.index(filepath)][1]
    final_page = original_body_page + body_start_page - 1
    pdf_final.cell(0, 6, f"Page {final_page:<4} .... {rel_path}", new_x="LMARGIN",
new_y="NEXT")

for filepath in files_to_process:
    rel_path = os.path.relpath(filepath, ROOT_DIR).replace('\\', '/')
    render_file_content(pdf_final, filepath, rel_path)

pdf_final.output(OUTPUT_PDF)
print("PDF generation complete!")

if __name__ == "__main__":
    generate_pdf()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: install_internet.sh

```
#!/bin/bash
cat << 'EOF' > /etc/rc.local
#!/bin/bash
sysctl -w net.ipv4.ip_forward=1
iptables -t nat -A POSTROUTING -o end0 -j MASQUERADE
iptables -A FORWARD -i end0 -o wlan0 -m state --state RELATED,ESTABLISHED -j ACCEPT
iptables -A FORWARD -i wlan0 -o end0 -j ACCEPT
rm -f /etc/resolv.conf
echo "nameserver 8.8.8.8" > /etc/resolv.conf
echo "nameserver 8.8.4.4" >> /etc/resolv.conf
systemctl restart dnsmasq
exit 0
EOF
chmod +x /etc/rc.local
/etc/rc.local
echo "Internet routing installed permanently and started!"
```

IoT-Enabled Barangay Resident and Attendance Management System

File: manage.py

```
#!/usr/bin/env python
"""Django's command-line utility for administrative tasks."""
import os
import sys

def main():
    """Run administrative tasks."""
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'barangay_project.settings')
    if len(sys.argv) == 2 and sys.argv[1] == 'runserver':
        sys.argv.append('127.0.0.1:8001')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
    execute_from_command_line(sys.argv)

if __name__ == '__main__':
    main()
```

IoT-Enabled Barangay Resident and Attendance Management System

File: officials/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: officials/admin.py

```
from django.contrib import admin
from .models import Official

@admin.register(Official)
class OfficialAdmin(admin.ModelAdmin):
    list_display = ('resident', 'position', 'status', 'term_start', 'term_end', 'user')
    list_filter = ('status', 'position', 'term_start')
    search_fields = ('resident__last_name', 'resident__first_name', 'employee_id',
'user__username')
    list_per_page = 20
```


IoT-Enabled Barangay Resident and Attendance Management System

File: officials/apps.py

```
from django.apps import AppConfig
```

```
class OfficialsConfig(AppConfig):  
    name = 'officials'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: officials/models.py

```
from django.db import models
from django.conf import settings

class Official(models.Model):
    """Barangay official/staff member."""
    POSITION_CHOICES = [
        ('captain', 'Punong Barangay (Captain)'),
        ('kagawad', 'Sangguniang Barangay Member (Kagawad)'),
        ('secretary', 'Barangay Secretary'),
        ('treasurer', 'Barangay Treasurer'),
        ('sk_chairman', 'SK Chairperson'),
        ('sk_kagawad', 'SK Kagawad'),
        ('tanod', 'Barangay Tanod'),
        ('bhw', 'Barangay Health Worker (BHW)'),
        ('nutrition_scholar', 'Barangay Nutrition Scholar (BNS)'),
        ('day_care_worker', 'Day Care Worker'),
        ('lupon', 'Lupon Member'),
        ('clerk', 'Barangay Clerk'),
        ('staff', 'Barangay Staff'),
    ]
    STATUS_CHOICES = [
        ('active', 'Active'),
        ('inactive', 'Inactive'),
        ('on_leave', 'On Leave'),
    ]

    resident = models.OneToOneField('residents.Resident', on_delete=models.CASCADE,
related_name='official_record')
    user = models.OneToOneField(settings.AUTH_USER_MODEL, on_delete=models.CASCADE,
null=True, blank=True, related_name='official_profile')
    position = models.CharField(max_length=30, choices=POSITION_CHOICES)
    committee = models.CharField(max_length=200, blank=True, help_text='Committee
assignment')
    term_start = models.DateField()
    term_end = models.DateField(null=True, blank=True)
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='active')
    salary = models.DecimalField(max_digits=10, decimal_places=2, default=0)
    employee_id = models.CharField(max_length=50, blank=True, null=True, unique=True)
    is_active = models.BooleanField(default=True, help_text='Uncheck to soft-delete
without losing records')

    # NOTE: fingerprint_template removed ? use official.resident.fingerprint_template
instead

    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def clean(self):
        """Enforce limits on active official positions."""
        from django.core.exceptions import ValidationError

        if self.status == 'active':
            # Count other active officials for the same position
            active_count = Official.objects.filter(position=self.position,
status='active')
            if self.pk:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        active_count = active_count.exclude(pk=self.pk)
    active_count = active_count.count()

    limits = {
        'captain': 1,
        'treasurer': 1,
        'secretary': 1,
        'kagawad': 7,
    }

    if self.position in limits:
        max_allowed = limits[self.position]
        if active_count >= max_allowed:
            raise ValidationError({
                'status': f"There can only be {max_allowed} active
{self.get_position_display()} at a time."
            })

def save(self, *args, **kwargs):
    is_new = self.pk is None
    old_status = None
    if not is_new:
        try:
            old_instance = Official.objects.get(pk=self.pk)
            old_status = old_instance.status
        except Official.DoesNotExist:
            pass

    self.full_clean()
    super().save(*args, **kwargs)

    # Sync resident.is_official
    if self.resident:
        # If status is active, resident.is_official MUST be True
        # If status is inactive/on_leave, resident.is_official should be False
        new_is_official = (self.status == 'active')
        if self.resident.is_official != new_is_official:
            self.resident.is_official = new_is_official
            # Only save if we are NOT already in the middle of a biometric sync
            # (which does its own resident.save)
            self.resident.save(update_fields=['is_official'])

    # Handle role transition
    if self.user:
        from core.models import Role
        if self.status == 'active':
            # Sync role to position
            role_obj = Role.objects.filter(name=self.position).first()
            if not role_obj:
                # Auto-create the role if it's missing to prevent login issues
                display_name = dict(self.POSITION_CHOICES).get(self.position,
self.position.replace('_', ' ').title())
                level = 1 if self.position in ['captain', 'admin'] else 2
                role_obj, _ = Role.objects.get_or_create(
                    name=self.position,
                    defaults={'display_name': display_name, 'permission_level': level}
                )

            if self.user.role != role_obj:
                self.user.role = role_obj
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        self.user.save(update_fields=['role'])
    elif self.status == 'inactive' and old_status == 'active':
        # Reset to resident
        resident_role = Role.objects.filter(name='resident').first()
        if resident_role:
            self.user.role = resident_role
            self.user.save(update_fields=['role'])

@property
def fingerprint_template(self):
    """Proxy to resident's fingerprint for backward compatibility."""
    return self.resident.fingerprint_template if self.resident else None

@fingerprint_template.setter
def fingerprint_template(self, value):
    """Proxy setter to resident's fingerprint."""
    if self.resident:
        self.resident.fingerprint_template = value
        self.resident.save()

@property
def full_name(self):
    return self.resident.full_name if self.resident else ""

@property
def display_photo(self):
    return self.resident.photo if self.resident and self.resident.photo else None

def __str__(self):
    return f"{self.full_name} - {self.get_position_display()}"

class Meta:
    ordering = ['position']
    verbose_name = 'Official'
    verbose_name_plural = 'Officials'

import uuid
from django.utils import timezone

class OfficialInvite(models.Model):
    """Model to track official invitations and onboarding."""
    STATUS_CHOICES = [
        ('pending_documents', 'Pending Documents'),
        ('pending_approval', 'Pending Approval'),
        ('pending_otp', 'Pending OTP'),
        ('activated', 'Activated'),
        ('rejected', 'Rejected'),
    ]

    token = models.UUIDField(default=uuid.uuid4, editable=False, unique=True)
    invited_by = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
    null=True, related_name='sent_official_invites')

    first_name = models.CharField(max_length=150)
    last_name = models.CharField(max_length=150)
    position = models.CharField(max_length=30, choices=Official.POSITION_CHOICES)
    phone_number = models.CharField(max_length=20)

    # Document Uploads
```

IoT-Enabled Barangay Resident and Attendance Management System

```
appointment_letter = models.FileField(upload_to='official_docs/appointment/',
null=True, blank=True)
valid_id = models.FileField(upload_to='official_docs/id/', null=True, blank=True)

# Approvals
captain_approved = models.BooleanField(default=False)
secretary_approved = models.BooleanField(default=False)

# OTP Tracking
otp_code = models.CharField(max_length=6, null=True, blank=True)
otp_expires_at = models.DateTimeField(null=True, blank=True)

status = models.CharField(max_length=20, choices=STATUS_CHOICES,
default='pending_documents')

created_at = models.DateTimeField(auto_now_add=True)
updated_at = models.DateTimeField(auto_now=True)

def is_otp_valid(self, code):
    if not self.otp_code or not self.otp_expires_at:
        return False
    if timezone.now() > self.otp_expires_at:
        return False
    return self.otp_code == code.strip()

def __str__(self):
    return f"{self.first_name} {self.last_name} - {self.get_position_display()}
({self.status})"

class OnboardingAuditLog(models.Model):
    """To track the exact steps of an official invite activation."""
    invite = models.ForeignKey(OfficialInvite, on_delete=models.CASCADE,
related_name='logs')
    actor = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
null=True, blank=True)
    action = models.CharField(max_length=255)
    ip_address = models.GenericIPAddressField(null=True, blank=True)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return f"{self.invite.first_name} - {self.action} at {self.created_at}"

from django.db.models.signals import pre_delete
from django.dispatch import receiver

@receiver(pre_delete, sender=Official)
def handle_official_deletion(sender, instance, **kwargs):
    """Ensure resident.is_official is False and user role is reset when official record is
deleted."""
    try:
        if instance.resident:
            instance.resident.is_official = False
            instance.resident.save(update_fields=['is_official'])

        if instance.user:
            from core.models import Role
            resident_role = Role.objects.filter(name='resident').first()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        if resident_role:
            instance.user.role = resident_role
            instance.user.save(update_fields=['role'])
except Exception:
    pass
```

IoT-Enabled Barangay Resident and Attendance Management System

File: officials/tests.py

```
from django.test import TestCase
```

```
# Create your tests here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: officials/urls.py

```
from django.urls import path
from . import views

app_name = 'officials'

urlpatterns = [
    path('', views.official_list, name='list'),
    path('add/', views.official_add, name='add'),
    path('org-chart/', views.org_chart, name='org_chart'),
    path('api/by-category/', views.get_officials_by_category, name='get_by_category'),
    path('biometric/register/', views.biometric_register, name='biometric_register'),
    path('biometric/status/', views.biometric_status, name='biometric_status'),
    path('esp32/status/', views.esp32_status_proxy, name='esp32_status_proxy'),
    path('esp32/start-enrollment/', views.esp32_start_enrollment_proxy,
name='esp32_start_enrollment_proxy'),
    path('esp32/stop-enrollment/', views.esp32_stop_enrollment_proxy,
name='esp32_stop_enrollment_proxy'),
    path('<int:pk>/', views.official_view, name='official_view'),
    path('<int:pk>/capture/', views.official_capture_fingerprint,
name='official_capture_fingerprint'),
    path('<int:pk>/enrollment/start/', views.start_multi_scan_enrollment,
name='enrollment_start'),
    path('<int:pk>/enrollment/scan-status/', views.get_scan_status, name='scan_status'),
    path('<int:pk>/enrollment/sync-progress/', views.sync_scan_progress,
name='sync_scan_progress'),
    path('<int:pk>/enrollment/register/', views.register_fingerprint_after_scans,
name='register_fingerprint'),
    path('<int:pk>/enrollment/remove/', views.remove_fingerprint,
name='remove_fingerprint'),
    path('<int:pk>/update-fingerprint/', views.official_update_fingerprint,
name='official_update_fingerprint'),
    path('profile/<int:pk>/update-fingerprint/', views.profile_update_fingerprint,
name='profile_update_fingerprint'),
    path('<int:pk>/edit/', views.official_edit, name='edit'),
    path('<int:pk>/delete/', views.official_delete, name='delete'),

    # Onboarding Flows
    path('onboarding/invite/', views.invite_official, name='invite_official'),
    path('onboarding/approvals/', views.onboard_approvals_list,
name='onboard_approvals_list'),
    path('onboarding/<uuid:token>/upload/', views.onboard_upload_docs,
name='onboard_upload_docs'),
    path('onboarding/<uuid:token>/approve/', views.onboard_approve,
name='onboard_approve'),
    path('onboarding/<uuid:token>/activate/', views.onboard_verify_otp,
name='onboard_verify_otp'),
]
```


IoT-Enabled Barangay Resident and Attendance Management System

File: officials/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.contrib import messages
from django.http import JsonResponse
from django.views.decorators.csrf import csrf_exempt
from django.conf import settings
import json
import time
import requests
from django.contrib.auth import get_user_model
User = get_user_model()
from .models import Official
from core.models import UserProfile
from residents.models import Resident
from core.utils.biometric_discovery import get_esp32_base_url

OFFICIAL_USERNAME_BY_ROLE = {
    'captain': 'captain',
    'secretary': 'secretary',
    'treasurer': 'treasurer',
}

from django.contrib.auth.decorators import user_passes_test

def is_barangay_admin(user):
    """Check if user has administrative access (Captain, Secretary, Treasurer, or
Admin)."""
    return hasattr(user, 'profile') and user.profile.role in ['captain', 'secretary',
'treasurer', 'admin']

@login_required
@user_passes_test(is_barangay_admin)
def get_officials_by_category(request):
    category = request.GET.get('category', 'all')

    officials = Official.objects.filter(status='active').select_related('resident')

    if category == 'officials':
        # Punong Barangay, Kagawad, SK Chairman, Secretary, Treasurer
        officials = officials.filter(position__in=['captain', 'kagawad', 'secretary',
'treasurer', 'sk_chairman', 'sk_kagawad'])
    elif category == 'bhw':
        officials = officials.filter(position='bhw')
    elif category == 'tanod':
        officials = officials.filter(position='tanod')
    elif category == 'staff':
        # Clerk, BNS, Day Care, Lupon, Staff, etc.
        staff_positions = ['nutrition_scholar', 'day_care_worker', 'lupon', 'clerk',
'staff']
        officials = officials.filter(position__in=staff_positions)
    # If category is 'all' or unknown, officials remains the full list of active
officials/staff

    data = [
        {
            'id': official.id,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        'resident_id': official.resident.id,
        'full_name': official.resident.full_name,
        'position': official.get_position_display(),
        'photo': official.resident.photo.url if official.resident.photo else None,
        'has_fingerprint': bool(official.fingerprint_template and
len(official.fingerprint_template) > 10)
    }
    for official in officials
]

return JsonResponse({'status': 'success', 'officials': data})

@login_required
def official_capture_fingerprint(request, pk):
    """Start biometric enrollment using ESP32 R307 fingerprint module."""
    official_id = request.GET.get('official_id') or pk

    # Enrollment is handled by the external ESP32 R307 fingerprint module.
    return JsonResponse({
        'status': 'success',
        'message': 'Biometric enrollment initiated using the connected ESP32 R307
fingerprint module.',
        'official_id': official_id
    })

@login_required
def biometric_status(request):
    official_id = request.GET.get('official_id')
    profile_id = request.GET.get('profile_id')

    has_template = False

    if official_id:
        official = Official.objects.filter(pk=official_id).first()
        has_template = bool(official and official.fingerprint_template and
len(official.fingerprint_template) > 10)
    elif profile_id:
        profile = UserProfile.objects.filter(pk=profile_id).first()
        has_template = bool(profile and profile.fingerprint_template and
len(profile.fingerprint_template) > 10)
    else:
        profile = getattr(request.user, 'profile', None)
        has_template = bool(profile and profile.fingerprint_template and
len(profile.fingerprint_template) > 10)

    return JsonResponse({'has_template': has_template})

@csrf_exempt
def esp32_status_proxy(request):
    if not request.user.is_authenticated:
        return JsonResponse({'status': 'error', 'message': 'Authentication required'},
status=401)

    esp32_base_url = get_esp32_base_url()
    try:
        # Bypassing proxies for local network reliability
        response = requests.get(f"{esp32_base_url}/status", timeout=4, proxies={'http':
None, 'https': None})
        response.raise_for_status()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
try:
    payload = response.json()
except ValueError:
    payload = {'status': 'success', 'message': response.text}
    return JsonResponse(payload, status=response.status_code)
except Exception:
    # Retry with force scan
    esp32_base_url = get_esp32_base_url(force_scan=True)
    try:
        response = requests.get(f"{esp32_base_url}/status", timeout=4,
proxies={'http': None, 'https': None})
        response.raise_for_status()
        try:
            payload = response.json()
        except ValueError:
            payload = {'status': 'success', 'message': response.text}
            return JsonResponse(payload, status=response.status_code)
        except requests.RequestException as exc:
            if exc.response is not None:
                try:
                    return JsonResponse(exc.response.json(),
status=exc.response.status_code)
                except Exception:
                    return JsonResponse({'status': 'error', 'message': f'ESP32 Error:
{exc.response.text}'}, status=exc.response.status_code)
            return JsonResponse({
                'status': 'error',
                'message': f'ESP32 disconnected or IP changed. Tried: {esp32_base_url}.
Error: {str(exc)}'
            }, status=503)
    except Exception as exc:
        return JsonResponse({
            'status': 'error',
            'message': f'ESP32 status proxy error: {str(exc)}'
        }, status=500)

@csrf_exempt
def esp32_start_enrollment_proxy(request):
    if not request.user.is_authenticated:
        return JsonResponse({'status': 'error', 'message': 'Authentication required'},
status=401)

    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'POST required'}, status=405)

    esp32_base_url = get_esp32_base_url()
    slot_id = request.GET.get('id') or request.POST.get('id')

    url = f"{esp32_base_url}/start-enrollment"
    if slot_id:
        url += f"?id={slot_id}"

    try:
        # Bypassing proxies for local network reliability
        response = requests.post(url, timeout=25, proxies={'http': None, 'https': None})
        try:
            payload = response.json()
        except ValueError:
            payload = {
                'status': 'error',
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        'message': response.text or 'Invalid response from ESP32',
    }
    if response.status_code >= 400 and payload.get('status') != 'error':
        payload['status'] = 'error'
    return JsonResponse(payload, status=response.status_code)
except Exception:
    esp32_base_url = get_esp32_base_url(force_scan=True)
    url = f"{esp32_base_url}/start-enrollment"
    if slot_id:
        url += f"?id={slot_id}"
    try:
        response = requests.post(url, timeout=25, proxies={'http': None, 'https':
None})
    try:
        payload = response.json()
    except ValueError:
        payload = {
            'status': 'error',
            'message': response.text or 'Invalid response from ESP32',
        }
    if response.status_code >= 400 and payload.get('status') != 'error':
        payload['status'] = 'error'
    return JsonResponse(payload, status=response.status_code)
except requests.RequestException as exc:
    if exc.response is not None:
        try:
            return JsonResponse(exc.response.json(),
status=exc.response.status_code)
        except Exception:
            return JsonResponse({'status': 'error', 'message': f'ESP32 Error:
{exc.response.text}'}, status=exc.response.status_code)
    return JsonResponse({
        'status': 'error',
        'message': f'ESP32 unreachable at {esp32_base_url}. Error: {str(exc)}'
    }, status=503)
except Exception as exc:
    return JsonResponse({
        'status': 'error',
        'message': f'ESP32 start-enrollment proxy error: {str(exc)}'
    }, status=500)

@csrf_exempt
def esp32_stop_enrollment_proxy(request):
    if not request.user.is_authenticated:
        return JsonResponse({'status': 'error', 'message': 'Authentication required'},
status=401)

    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'POST required'}, status=405)

    esp32_base_url = get_esp32_base_url()
    try:
        response = requests.post(f"{esp32_base_url}/stop-enrollment", timeout=4,
proxies={'http': None, 'https': None})
        response.raise_for_status()
    try:
        payload = response.json()
    except ValueError:
        payload = {'status': 'success', 'message': response.text}
    return JsonResponse(payload, status=response.status_code)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
except requests.RequestException as exc:
    return JsonResponse({
        'status': 'error',
        'message': f'ESP32 stop-enrollment proxy failed: {str(exc)}'
    }, status=503)
except Exception as exc:
    return JsonResponse({
        'status': 'error',
        'message': f'ESP32 stop-enrollment proxy error: {str(exc)}'
    }, status=500)

@csrf_exempt
def official_update_fingerprint(request, pk):
    """Update official fingerprint template."""
    remote_addr = request.META.get('REMOTE_ADDR')
    if remote_addr not in ('127.0.0.1', '::1'):
        return JsonResponse({'status': 'error', 'message': 'Forbidden'}, status=403)
    official = get_object_or_404(Official, pk=pk)
    if request.method == 'GET':
        return JsonResponse({
            'status': 'ok',
            'message': 'POST fingerprint template to this endpoint.',
            'expected': {'method': 'POST', 'body': {'template': '<base64>'}},
        })

    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'Method not allowed'},
            status=405)

    try:
        template = None

        # Prefer JSON body when available
        if request.body:
            try:
                data = json.loads(request.body)
                template = data.get('template')
            except json.JSONDecodeError:
                template = None

        # Fallback to form POST (application/x-www-form-urlencoded or multipart)
        if not template:
            template = request.POST.get('template')

        if not template:
            return JsonResponse({'status': 'error', 'message': 'Missing template'},
                status=400)

        print(f"[Biometric] official_update_fingerprint pk={pk} remote={remote_addr}")
        template_len=len(template)
        official.fingerprint_template = template
        official.save()
        return JsonResponse({'status': 'success'})
    except Exception as e:
        return JsonResponse({'status': 'error', 'message': str(e)}, status=400)

@csrf_exempt
def profile_update_fingerprint(request, pk):
    """Update UserProfile fingerprint template."""
    remote_addr = request.META.get('REMOTE_ADDR')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if remote_addr not in ('127.0.0.1', '::1'):
    return JsonResponse({'status': 'error', 'message': 'Forbidden'}, status=403)
profile = get_object_or_404(UserProfile, pk=pk)
if request.method == 'GET':
    return JsonResponse({
        'status': 'ok',
        'message': 'POST fingerprint template to this endpoint.',
        'expected': {'method': 'POST', 'body': {'template': '<base64>'}},
    })

if request.method != 'POST':
    return JsonResponse({'status': 'error', 'message': 'Method not allowed'},
status=405)

try:
    template = None

    # Prefer JSON body when available
    if request.body:
        try:
            data = json.loads(request.body)
            template = data.get('template')
        except json.JSONDecodeError:
            template = None

    # Fallback to form POST (application/x-www-form-urlencoded or multipart)
    if not template:
        template = request.POST.get('template')

    if not template:
        return JsonResponse({'status': 'error', 'message': 'Missing template'},
status=400)

    print(f"[Biometric] profile_update_fingerprint pk={pk} remote={remote_addr}
template_len={len(template)}")
    profile.fingerprint_template = template
    profile.save()
    return JsonResponse({'status': 'success'})
except Exception as e:
    return JsonResponse({'status': 'error', 'message': str(e)}, status=400)

@login_required
def official_list(request):
    """List all officials categorized by position."""
    status_filter = request.GET.get('status', 'active')
    pos_filter = request.GET.get('pos', 'all')

    officials = Official.objects.select_related('resident').all()

    if status_filter:
        officials = officials.filter(status=status_filter)

    if pos_filter != 'all':
        officials = officials.filter(position=pos_filter)

    # Categories definition
    categories = {
        'council': {
            'title': 'Barangay Council',
            'positions': ['captain', 'kagawad', 'secretary', 'treasurer'],
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        'officials': []
    },
    'sk': {
        'title': 'Sangguniang Kabataan (SK)',
        'positions': ['sk_chairman', 'sk_kagawad'],
        'officials': []
    },
    'tanod': {
        'title': 'Barangay Tanod',
        'positions': ['tanod'],
        'officials': []
    },
    'health': {
        'title': 'Health & Nutrition',
        'positions': ['health_worker', 'nutrition_scholar'],
        'officials': []
    },
    'staff': {
        'title': 'Administrative & Support Staff',
        'positions': ['clerk', 'staff', 'day_care_worker', 'lupon'],
        'officials': []
    }
}

# Group officials into categories
for official in officials:
    found = False
    for cat_key, cat_data in categories.items():
        if official.position in cat_data['positions']:
            cat_data['officials'].append(official)
            found = True
            break
    if not found:
        if 'other' not in categories:
            categories['other'] = {'title': 'Other Functionaries', 'positions': [],
'officials': []}
            categories['other']['officials'].append(official)

# Remove empty categories
active_categories = {k: v for k, v in categories.items() if v['officials']]

context = {
    'categories': active_categories,
    'selected_status': status_filter,
    'selected_pos': pos_filter,
    'all_positions': Official.POSITION_CHOICES,
}
return render(request, 'officials/list.html', context)

@login_required
def org_chart(request):
    """Display the organizational chart with dynamic data."""
    officials = Official.objects.filter(status='active').select_related('resident',
'user')

    # Organize by position
    captain = officials.filter(position='captain').first()
    secretary = officials.filter(position='secretary').first()
    treasurer = officials.filter(position='treasurer').first()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
kagawads = officials.filter(position='kagawad')
sk_chairman = officials.filter(position='sk_chairman').first()

def format_node(official, type_name):
    if not official: return None
    # Extract initials
    names = official.resident.full_name.split()
    initials = "".join([n[0] for n in names if n[:2]].upper())

    return {
        'id': str(official.id),
        'resident_id': str(official.resident.id),
        'name': official.resident.full_name,
        'role': official.get_position_display(),
        'initials': initials,
        'photo': official.resident.photo.url if official.resident.photo else None,
        'email': official.user.email if (official.user and official.user.email) else
'N/A',
        'phone': official.resident.contact_number or 'N/A',
        'type': type_name,
        'children': []
    }

root = format_node(captain, 'executive')
if not root:
    root = {
        'id': 'root',
        'name': 'Barangay Office',
        'role': 'Punong Barangay (Vacant)',
        'initials': 'BR',
        'type': 'executive',
        'children': []
    }

# Admin Staff Group
admin_group = []
sec_node = format_node(secretary, 'admin')
if sec_node:
    staff = officials.filter(position__in=['clerk', 'staff', 'lupon',
'day_care_worker'])
    sec_node['children'] = [format_node(s, 'staff') for s in staff]
    admin_group.append(sec_node)

tre_node = format_node(treasurer, 'admin')
if tre_node:
    finance_staff = officials.filter(position__in=['nutrition_scholar'])
    tre_node['children'] = [format_node(fs, 'staff') for fs in finance_staff]
    admin_group.append(tre_node)

# Council Group
council_children = [format_node(k, 'council') for k in kagawads]

sk_node = format_node(sk_chairman, 'council')
if sk_node:
    sk_kagawads = officials.filter(position='sk_kagawad')
    sk_node['children'] = [format_node(skk, 'council') for skk in sk_kagawads]
    council_children.append(sk_node)

council_node = {
    'id': 'council_group',
```


IoT-Enabled Barangay Resident and Attendance Management System

```
'name': 'Sangguniang Barangay',
'role': 'Barangay Council',
'initials': 'SB',
'type': 'council',
'children': council_children
}

# BHW and Tanods as separate branches or under Captain?
# Let's add Tanods and BHWs as departments under root for completeness
security_node = {
    'id': 'security_group',
    'name': 'Public Safety',
    'role': 'Barangay Tanod',
    'initials': 'PS',
    'type': 'staff',
    'children': [format_node(t, 'staff') for t in officials.filter(position='tanod')]
}

health_node = {
    'id': 'health_group',
    'name': 'Health Services',
    'role': 'BHW Department',
    'initials': 'HS',
    'type': 'staff',
    'children': [format_node(h, 'staff') for h in officials.filter(position='bhw')]
}

root['children'] = admin_group + [council_node]
if security_node['children']: root['children'].append(security_node)
if health_node['children']: root['children'].append(health_node)

context = {
    'org_data_json': json.dumps(root)
}
return render(request, 'officials/org_chart.html', context)

@login_required
@user_passes_test(is_barangay_admin)
def biometric_register(request):
    profile, _created = UserProfile.objects.get_or_create(user=request.user)
    return render(request, 'officials/biometric_register.html', {'profile_pk':
profile.pk})

@login_required
def official_add(request):
    """Add a new official."""
    if request.method == 'POST':
        from django.contrib.auth import get_user_model
        from django.db import transaction
        User = get_user_model()

        resident_id = request.POST.get('resident')
        resident = get_object_or_404(Resident, pk=resident_id)
        position = request.POST.get('position')

        with transaction.atomic():
            official = Official(
                resident=resident,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        position=position,
        committee=request.POST.get('committee', ''),
        term_start=request.POST.get('term_start'),
        term_end=request.POST.get('term_end') or None,
        salary=request.POST.get('salary', 0) or 0,
        employee_id=request.POST.get('employee_id', ''),
        status='active',
    )
    official.save()

    from core.models import Role, UserProfile
    role_obj = Role.objects.filter(name=position).first()

    username = request.POST.get('username')
    password = request.POST.get('password')

    # Handle User Account Creation/Update
    if hasattr(resident, 'user_profile') and resident.user_profile:
        # Update existing user
        user_profile = resident.user_profile
        user = user_profile.user

        # Update role
        if role_obj:
            user.role = role_obj

        # Update username if provided
        if username:
            user.username = username

        # Update password if provided
        if password:
            user.set_password(password)

        user.save()
        official.user = user
        official.save()
    else:
        # Create new user if provided
        if username and password:
            user = User.objects.create_user(
                username=username,
                password=password,
                first_name=resident.first_name,
                last_name=resident.last_name,
            )
            if role_obj:
                user.role = role_obj
                user.save()
            UserProfile.objects.create(user=user, resident=resident)
            official.user = user
            official.save()

    messages.success(request, f'{resident.full_name} added as {official.get_position_display()}. Account setup successfully.')
    return redirect('officials:list')

# Get residents who are not yet officials
existing_official_ids = Official.objects.values_list('resident_id', flat=True)
available_residents = Resident.objects.filter(is_active=True).exclude(id__in=existing_
```

IoT-Enabled Barangay Resident and Attendance Management System

```
official_ids).prefetch_related('user_profile')

# create a dict to easily check in JS if a resident has a user account
residents_data = []
for r in available_residents:
    profile = getattr(r, 'user_profile', None)
    has_account = profile is not None
    residents_data.append({
        'pk': r.pk,
        'name': f"{r.last_name}, {r.first_name}" + (f" {r.middle_name}" if
r.middle_name else "") + (f" {r.suffix}" if r.suffix else "") + (f" ? {r.purok}" if
r.purok else ""),
        'has_account': has_account,
        'username': profile.user.username if has_account else ''
    })

context = {
    'residents_data': residents_data,
    'positions': Official.POSITION_CHOICES,
}
return render(request, 'officials/form.html', context)

@login_required
def official_view(request, pk):
    """View official details."""
    official = get_object_or_404(Official.objects.select_related('resident'), pk=pk)
    return render(request, 'officials/view.html', {'official': official})

@login_required
def official_edit(request, pk):
    """Edit an official."""
    official = get_object_or_404(Official, pk=pk)

    if request.method == 'POST':
        official.position = request.POST.get('position')
        official.committee = request.POST.get('committee', '')
        official.term_start = request.POST.get('term_start')
        official.term_end = request.POST.get('term_end') or None
        official.salary = request.POST.get('salary', 0) or 0
        official.employee_id = request.POST.get('employee_id', '')
        official.status = request.POST.get('status', 'active')
        official.save()
        messages.success(request, f'{official.resident.full_name} updated.')
        return redirect('officials:list')

    context = {
        'official': official,
        'editing': True,
        'positions': Official.POSITION_CHOICES,
    }
    return render(request, 'officials/form.html', context)

@login_required
def official_delete(request, pk):
    """Delete an official."""
    official = get_object_or_404(Official, pk=pk)
    if request.method == 'POST':
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        official.delete()
        messages.success(request, 'Official record removed.')
        return redirect('officials:list')
    return render(request, 'officials/confirm_delete.html', {'official': official})

# --- Onboarding Flows ---

import string
import random
from datetime import timedelta
from django.utils import timezone
from django.urls import reverse
from django.core.exceptions import PermissionDenied
from .models import OfficialInvite, OnboardingAuditLog

def log_onboarding(invite, user, action, request):
    ip = request.META.get('REMOTE_ADDR')
    OnboardingAuditLog.objects.create(
        invite=invite,
        actor=user if user.is_authenticated else None,
        action=action,
        ip_address=ip
    )

@login_required
@user_passes_test(is_barangay_admin)
def invite_official(request):
    """Captain or Admin invites an official."""
    if request.method == 'POST':
        first_name = request.POST.get('first_name')
        last_name = request.POST.get('last_name')
        position = request.POST.get('position')
        phone_number = request.POST.get('phone_number')

        invite = OfficialInvite.objects.create(
            first_name=first_name,
            last_name=last_name,
            position=position,
            phone_number=phone_number,
            invited_by=request.user
        )
        log_onboarding(invite, request.user, "Created Invite", request)

        # In a real app, you would send an email/SMS here.
        # For now, we flash the link to the captain.
        link = request.build_absolute_uri(reverse('officials:onboard_upload_docs',
args=[invite.token]))
        messages.success(request, f"Invite created! Share this link with {first_name}:
{link}")
        return redirect('officials:onboard_approvals_list')

    return render(request, 'officials/onboarding/invite_form.html', {
        'positions': Official.POSITION_CHOICES
    })

def onboard_upload_docs(request, token):
    """Official uses the link to upload documents."""
    invite = get_object_or_404(OfficialInvite, token=token)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if invite.status != 'pending_documents':
    messages.error(request, "This invite link is no longer in the document upload
stage.")
    return redirect('core:login')

if request.method == 'POST':
    if 'appointment_letter' in request.FILES:
        invite.appointment_letter = request.FILES['appointment_letter']
    if 'valid_id' in request.FILES:
        invite.valid_id = request.FILES['valid_id']

    if invite.appointment_letter and invite.valid_id:
        invite.status = 'pending_approval'
        invite.save()
        log_onboarding(invite, request.user, "Uploaded Documents", request)
        messages.success(request, "Documents uploaded successfully! Waiting for
Captain and Secretary approval.")
        return redirect('core:login')
    else:
        messages.error(request, "Please upload BOTH the Appointment Letter and a Valid
ID.")

    return render(request, 'officials/onboarding/upload_documents.html', {'invite':
invite})

@login_required
@user_passes_test(is_barangay_admin)
def onboard_approvals_list(request):
    """Dashboard for Captain and Secretary to approve pending invites."""
    pending_documents =
OfficialInvite.objects.filter(status='pending_documents').order_by('-created_at')
    pending_approval =
OfficialInvite.objects.filter(status='pending_approval').order_by('-created_at')
    pending_otp =
OfficialInvite.objects.filter(status='pending_otp').order_by('-created_at')
    recent_activated =
OfficialInvite.objects.filter(status='activated').order_by('-created_at')[:10]

    context = {
        'pending_documents': pending_documents,
        'pending_approval': pending_approval,
        'pending_otp': pending_otp,
        'recent_activated': recent_activated
    }
    return render(request, 'officials/onboarding/approvals.html', context)

@login_required
@user_passes_test(is_barangay_admin)
def onboard_approve(request, token):
    """Captain or Secretary records their approval."""
    invite = get_object_or_404(OfficialInvite, token=token)
    if invite.status != 'pending_approval':
        messages.error(request, "Invite is not ready for approval.")
        return redirect('officials:onboard_approvals_list')

    role = request.user.profile.role if hasattr(request.user, 'profile') else 'admin'
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if role in ('captain', 'admin'):
    invite.captain_approved = True
    log_onboarding(invite, request.user, "Captain Approved", request)
if role in ('secretary', 'admin'):
    invite.secretary_approved = True
    log_onboarding(invite, request.user, "Secretary Approved", request)

if invite.captain_approved and invite.secretary_approved:
    invite.status = 'pending_otp'
    # Generate OTP
    code = ''.join(random.choices(string.digits, k=6))
    invite.otp_code = code
    invite.otp_expires_at = timezone.now() + timedelta(minutes=10)

    # MOCK SMS
    print(f"!!! [MOCK SMS] To: {invite.phone_number} | OTP: {code} !!!")
    messages.success(request, f"Both approved! OTP has been generated for {invite.first_name}. (Check server console for code)")
    log_onboarding(invite, request.user, "Dual Approval Complete - OTP Generated", request)
else:
    messages.success(request, "Your approval has been recorded in the system.")

invite.save()
return redirect('officials:onboard_approvals_list')

def onboard_verify_otp(request, token):
    """Official enters OTP to finalize account setup."""
    invite = get_object_or_404(OfficialInvite, token=token)

    if invite.status == 'activated':
        messages.info(request, "This account is already activated. Please login.")
        return redirect('core:login')

    if invite.status != 'pending_otp':
        messages.error(request, "This invite is not ready for activation.")
        return redirect('core:login')

    if request.method == 'POST':
        otp = request.POST.get('otp')
        password = request.POST.get('password')
        password_confirm = request.POST.get('password_confirm')
        username = request.POST.get('username')

        if not invite.is_otp_valid(otp):
            messages.error(request, "Invalid or expired OTP.")
        elif password != password_confirm or not password:
            messages.error(request, "Passwords do not match or are empty.")
        elif not username:
            messages.error(request, "Username is required.")
        elif User.objects.filter(username=username).exists():
            messages.error(request, "Username is already taken.")
        else:
            # Final Activation
            # 1. Create User
            user = User.objects.create_user(username=username, password=password)
            user.first_name = invite.first_name
            user.last_name = invite.last_name
            user.save()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# 2. Create UserProfile
profile, _ = UserProfile.objects.get_or_create(user=user)
from core.models import Role
role_obj = Role.objects.filter(name=invite.position).first()
if role_obj:
    user.role = role_obj
    user.save()
profile.save()

# 3. Create dummy Resident if needed because Official requires OneToOne with
Resident

# In real system, maybe they select existing, but here we just stub it:
resident = Resident.objects.create(
    first_name=invite.first_name,
    last_name=invite.last_name,
    contact_number=invite.phone_number,
    gender='M', # Defaulting, since invite didn't capture.
    civil_status='Single',
    birthdate=timezone.now().date(),
    is_active=True
)

# 4. Create Official
Official.objects.create(
    resident=resident,
    user=user,
    position=invite.position,
    term_start=timezone.now().date(),
    status='active'
)

# 5. Update Invite
invite.status = 'activated'
invite.save()
log_onboarding(invite, user, "Activated Account via OTP", request)

messages.success(request, "Account activated successfully! You can now log
in.")

return redirect('core:login')

return render(request, 'officials/onboarding/verify_otp.html', {'invite': invite})

# ===== 3-SCAN BIOMETRIC ENROLLMENT API =====

def _get_effective_fingerprint_max_slots():
    """
    Prefer the ESP32 sensor's reported library size (GET /status ? max_fingerprint_slots),
    capped by FINGERPRINT_SENSOR_MAX_SLOTS. If the scanner is offline or has not reported
    capacity yet, use a conservative default (300) so slot IDs stay valid on typical R307
    modules even when settings allow a higher ceiling.
    """
    settings_cap = max(1, int(getattr(settings, 'FINGERPRINT_SENSOR_MAX_SLOTS', 1000)))
    offline_safe_cap = min(settings_cap, 300)
    esp32_base_url = get_esp32_base_url()
    try:
        response = requests.get(
            f'{esp32_base_url}/status',
            timeout=3,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        proxies={'http': None, 'https': None},
    )
    response.raise_for_status()
    data = response.json()
    raw = data.get('max_fingerprint_slots') or data.get('library_capacity')
    if raw is None:
        return offline_safe_cap
    esp_cap = int(raw)
    if esp_cap <= 0:
        return offline_safe_cap
    return max(1, min(settings_cap, esp_cap))
except Exception:
    return offline_safe_cap

def _allocate_next_fingerprint_slot(capacity):
    """
    Next free AS608/R307 page index in 0 .. capacity-1 (hardware is 0-based).
    Picks the lowest free index. Returns None if the library is full.
    """
    if capacity <= 0:
        return None
    used_ids = set(
        Resident.objects.filter(fingerprint_id__isnull=False).values_list('fingerprint_id',
flat=True)
    )
    for page_id in range(capacity):
        if page_id not in used_ids:
            return page_id
    return None

@login_required
def start_multi_scan_enrollment(request, pk):
    """Initialize 3-scan fingerprint enrollment session."""
    official = get_object_or_404(Official, pk=pk)
    resident = official.resident

    capacity = _get_effective_fingerprint_max_slots()
    fid = resident.fingerprint_id
    # R307/AS608 template indices are 0 .. capacity-1
    needs_slot = fid is None or fid < 0 or fid >= capacity
    if needs_slot:
        slot = _allocate_next_fingerprint_slot(capacity)
        if slot is None:
            return JsonResponse({'status': 'error', 'message': 'No free fingerprint slots
available'}, status=507)
        resident.fingerprint_id = slot
        resident.save(update_fields=['fingerprint_id'])

    # Initialize enrollment session in Django
    if 'biometric_enrollment' not in request.session:
        request.session['biometric_enrollment'] = {}

    request.session['biometric_enrollment'][str(official.id)] = {
        'scan_count': 0,
        'scans': [],
        'active': True,
        'slot_id': resident.fingerprint_id
    }
```


IoT-Enabled Barangay Resident and Attendance Management System

```
}
request.session.modified = True

return JsonResponse({
    'status': 'success',
    'message': 'Multi-scan enrollment started. Please prepare to scan your fingerprint
3 times.',
    'official_id': official.id,
    'slot_id': resident.fingerprint_id,
    'scans_required': 3
})

@login_required
def get_scan_status(request, pk):
    """Get current scan status and check for completion."""
    official = get_object_or_404(Official, pk=pk)

    enrollment_data = request.session.get('biometric_enrollment',
{}).get(str(official.id), {})
    current_scan_count = enrollment_data.get('scan_count', 0)

    return JsonResponse({
        'status': 'success',
        'official_id': official.id,
        'current_scan': current_scan_count,
        'scans_required': 3,
        'enrollment_complete': current_scan_count >= 3,
        'message': f'Scan {current_scan_count} of 3' if current_scan_count < 3 else
'Enrollment complete'
    })

def _biometric_marker_registered(official):
    """True when resident row marks this official as enrolled (template flag or real
template)."""
    tpl = official.fingerprint_template if official else None
    return bool(tpl and len(tpl) > 10)

def _persist_biometric_enrollment_complete(request, official):
    """
    Mark official/resident as enrolled after 3 successful ESP32 scans.
    Clears the in-progress enrollment session for this official.
    """
    if 'biometric_enrollment' in request.session and str(official.id) in
request.session['biometric_enrollment']:
        del request.session['biometric_enrollment'][str(official.id)]
        request.session.modified = True
    # Explicitly update the resident record which is the single source of truth
    resident = official.resident
    if resident:
        resident.fingerprint_template = 'REGISTERED_ON_SENSOR'
        # Note: fingerprint_id should have been set during start_multi_scan_enrollment
        resident.save(update_fields=['fingerprint_template'])

    # Logging for diagnostics on Orange Pi
    print(f"DEBUG: Biometric persistence complete for Official {official.id} (Resident
{resident.id})")
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# Still call official.save() to trigger any signals or side effects if needed,
# but the template is already safely in the Resident table.
official.save()

@login_required
def sync_scan_progress(request, pk):
    """Sync scan progress coming from ESP32 into Django session."""
    official = get_object_or_404(Official, pk=pk)

    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'POST required'}, status=405)

    try:
        data = json.loads(request.body.decode('utf-8') or '{}')
    except Exception:
        data = {}

    esp_scan = data.get('current_scan')
    try:
        esp_scan = int(esp_scan)
    except Exception:
        return JsonResponse({'status': 'error', 'message': 'Invalid current_scan'},
            status=400)

    if 'biometric_enrollment' not in request.session:
        request.session['biometric_enrollment'] = {}

    enrollment = request.session['biometric_enrollment'].get(str(official.id))
    if enrollment is None:
        return JsonResponse(
            {
                'status': 'error',
                'message': 'No enrollment session. Close the dialog and tap Register
again.',
            },
            status=400,
        )

    current = enrollment.get('scan_count', 0)
    registration_complete = False
    # Only move forward
    if esp_scan > current:
        new_count = min(esp_scan, 3)
        request.session['biometric_enrollment'][str(official.id)]['scan_count'] =
new_count
        request.session.modified = True
        current = new_count
        if new_count >= 3 and not _biometric_marker_registered(official):
            _persist_biometric_enrollment_complete(request, official)
            registration_complete = True
        elif new_count >= 3:
            registration_complete = True

    return JsonResponse({
        'status': 'success',
        'official_id': official.id,
        'current_scan': current,
        'scans_required': 3,
        'registration_complete': registration_complete,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    })

@login_required
def register_fingerprint_after_scans(request, pk):
    """Register fingerprint template after 3 successful scans."""
    official = get_object_or_404(Official, pk=pk)

    if _biometric_marker_registered(official):
        if 'biometric_enrollment' in request.session and str(official.id) in request.session['biometric_enrollment']:
            del request.session['biometric_enrollment'][str(official.id)]
            request.session.modified = True
            return JsonResponse({
                'status': 'success',
                'message': f'Fingerprint already registered for {official.resident.full_name} (Slot {official.resident.fingerprint_id})',
                'official_id': official.id,
                'slot_id': official.resident.fingerprint_id,
            })

        enrollment_data = request.session.get('biometric_enrollment', {}).get(str(official.id), {})
        scan_count = enrollment_data.get('scan_count', 0)

        if scan_count < 3:
            return JsonResponse({
                'status': 'error',
                'message': f'Not enough scans. Completed: {scan_count}/3',
            }, status=400)

        try:
            _persist_biometric_enrollment_complete(request, official)

            return JsonResponse({
                'status': 'success',
                'message': f'Fingerprint successfully registered for {official.resident.full_name} (Slot {official.resident.fingerprint_id})',
                'official_id': official.id,
                'slot_id': official.resident.fingerprint_id,
            })
        except Exception as e:
            return JsonResponse({'status': 'error', 'message': str(e)}, status=500)

@login_required
def remove_fingerprint(request, pk):
    """Remove fingerprint template and clear slot ID from sensor."""
    if request.method != 'POST':
        return JsonResponse({'status': 'error', 'message': 'POST required'}, status=405)

    official = get_object_or_404(Official, pk=pk)
    resident = official.resident

    # 1. Erase template on R307 (retry ? WiFi/serial can miss the first attempt)
    if resident.fingerprint_id is not None:
        esp32_base_url = get_esp32_base_url()
        slot = resident.fingerprint_id
        for attempt in range(3):
            try:
                r = requests.post(
```

IoT-Enabled Barangay Resident and Attendance Management System

```
f"{esp32_base_url}/delete-fingerprint?id={slot}",
timeout=5,
proxies={'http': None, 'https': None},
)
if r.ok:
    try:
        if r.json().get('status') == 'success':
            break
        except ValueError:
            break
    except Exception as e:
        if attempt == 2:
            print(f"[Biometric] Sensor delete failed for {resident.full_name} page
{slot}: {e}")
            time.sleep(0.25)

# 2. Clear Django fields
official.fingerprint_template = ""
official.save()

resident.fingerprint_id = None
resident.fingerprint_template = "" # Clear from resident too if it exists there
resident.save()

return JsonResponse({
    'status': 'success',
    'message': f'Fingerprint registration removed for {resident.full_name}'
})
```

File: ordinances/__init__.py

File: ordinances/admin.py

```
from django.contrib import admin
from .models import Ordinance

@admin.register(Ordinance)
class OrdinanceAdmin(admin.ModelAdmin):
    list_display = ('ordinance_number', 'title', 'date_enacted', 'category', 'status')
    list_filter = ('category', 'status', 'date_enacted')
    search_fields = ('ordinance_number', 'title', 'author', 'sponsor')
```

IoT-Enabled Barangay Resident and Attendance Management System

File: ordinances/apps.py

```
from django.apps import AppConfig

class OrdinancesConfig(AppConfig):
    name = 'ordinances'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: ordinances/models.py

```
from django.db import models

class Ordinance(models.Model):
    """Barangay ordinance/resolution record."""
    CATEGORY_CHOICES = [
        ('peace_order', 'Peace and Order'),
        ('health', 'Health and Sanitation'),
        ('environment', 'Environment'),
        ('revenue', 'Revenue and Finance'),
        ('infrastructure', 'Infrastructure'),
        ('social_welfare', 'Social Welfare'),
        ('education', 'Education'),
        ('other', 'Other'),
    ]
    STATUS_CHOICES = [
        ('active', 'Active'),
        ('amended', 'Amended'),
        ('repealed', 'Repealed'),
        ('pending', 'Pending'),
    ]

    ordinance_number = models.CharField(max_length=50, unique=True)
    title = models.CharField(max_length=500)
    description = models.TextField(blank=True)
    author = models.CharField(max_length=200)
    date_enacted = models.DateField()
    date_effectivity = models.DateField(null=True, blank=True)
    category = models.CharField(max_length=30, choices=CATEGORY_CHOICES, default='other')
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='active')

    # File attachment
    document_file = models.FileField(upload_to='ordinances/', blank=True, null=True)

    # Parsed Content
    body_content = models.TextField(blank=True, help_text="Full text of the ordinance")
    signatories = models.TextField(blank=True, help_text="Signatories of the ordinance")

    sponsor = models.CharField(max_length=200, blank=True)
    remarks = models.TextField(blank=True)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def __str__(self):
        return f"Ordinance No. {self.ordinance_number}: {self.title}"

    class Meta:
        ordering = ['-date_enacted']
        verbose_name = 'Ordinance'
        verbose_name_plural = 'Ordinances'
```


IoT-Enabled Barangay Resident and Attendance Management System

File: ordinances/tests.py

```
from django.test import TestCase
```

```
# Create your tests here.
```

File: ordinances/urls.py

```
from django.urls import path
from . import views

app_name = 'ordinances'

urlpatterns = [
    path('', views.ordinance_list, name='list'),
    path('add/', views.ordinance_add, name='add'),
    path('<int:pk>', views.ordinance_view, name='view'),
    path('<int:pk>/edit/', views.ordinance_edit, name='edit'),
    path('<int:pk>/delete/', views.ordinance_delete, name='delete'),
    path('<int:pk>/download/', views.ordinance_download, name='download'),
    path('upload-parse/', views.ordinance_upload_parse, name='upload_parse'),
]
```

File: ordinances/utils.py

```
import fitz # PyMuPDF
import pytesseract
import re
from PIL import Image
import io
from datetime import datetime

def clean_ocr_text(text):
    # Remove random symbols and garbage characters, but keep basic punctuation and slashes
    text = re.sub(r'^\w\s\.\,\;\:\!\@\?\-\(\)\\"'\'/\n]', ' ', text)
    # Fix multiple spaces
    text = re.sub(r' +', ' ', text)
    # Fix multiple newlines
    text = re.sub(r'\n+', '\n', text)
    # Strip each line
    text = '\n'.join(line.strip() for line in text.splitlines() if line.strip())
    return text

def extract_ordinance_fields(text):
    fields = {
        'ordinance_number': '',
        'title': '',
        'date_enacted': '',
        'body_content': '',
        'signatories': '',
        'author': '',
        'category': ''
    }

    # Extract Ordinance Number
    num_match = re.search(
        r'ordinance\s+no[\.\:]\?\s*([\w\-\+])', text, re.IGNORECASE
    )
    fields['ordinance_number'] = num_match.group(1).strip() if num_match else ''

    # Extract Title
    title_match = re.search(r'([\"]\')\s*an ordinance.*?([\"]\')', text, re.IGNORECASE | re.DOTALL)
    if not title_match:
        title_match = re.search(r'(an ordinance.*?)(?:\n\s*\n|be it ordained|sponsored by|authored by)', text, re.IGNORECASE | re.DOTALL)

    if title_match:
        title = title_match.group(1).strip()
        title = re.sub(r'\n+', ' ', title)
        fields['title'] = re.sub(r'\s+', ' ', title)
    else:
        fields['title'] = ''

    # Extract Date
    months = r'(January|February|March|April|May|June|July|August|September|October|November|December|Jan|Feb|Mar|Apr|Jun|Jul|Aug|Sep|Oct|Nov|Dec)'
    raw_date = None

    m1 = re.search(fr'({months})\s+\d{{1,2}}(?:st|nd|rd|th)?[\s\,]+\d{{4}}', text, re.IGNORECASE)
    m2 = re.search(fr'\d{{1,2}}(?:st|nd|rd|th)?[\s\,]+({months})[\s\,]+\d{{4}}', text,
```

IoT-Enabled Barangay Resident and Attendance Management System

```

re.IGNORECASE)
    m3 = re.search(fr'(\d{{1,2}}(?:st|nd|rd|th)?\s+day\s+of\s+{months}[\s\,]+\d{{4}})',
text, re.IGNORECASE)
    m4 = re.search(r'(\d{4}[\-\/]\d{1,2}[\-\/]\d{1,2})', text)
    m5 = re.search(r'(\d{1,2}[\-\/]\d{1,2}[\-\/]\d{4})', text)

    if m1:
        raw_date = m1.group(1)
    elif m2:
        raw_date = m2.group(1)
    elif m3:
        raw_date = re.sub(r'day of', '', m3.group(1), flags=re.IGNORECASE).strip()
    elif m4:
        raw_date = m4.group(1)
    elif m5:
        raw_date = m5.group(1)

    if raw_date:
        cleaned_date = re.sub(r'(st|nd|rd|th)', '', raw_date, flags=re.IGNORECASE)
        cleaned_date = re.sub(r'[\s\,]+', ' ', cleaned_date).strip()
        formats = [
            '%B %d %Y', '%b %d %Y',
            '%d %B %Y', '%d %b %Y',
            '%Y-%m-%d', '%m/%d/%Y', '%d-%m-%Y', '%Y/%m/%d'
        ]
        parsed = False
        for fmt in formats:
            try:
                fields['date_enacted'] = datetime.strptime(cleaned_date,
fmt).strftime('%Y-%m-%d')
                parsed = True
                break
            except ValueError:
                continue
        if not parsed:
            fields['date_enacted'] = raw_date
    else:
        fields['date_enacted'] = ''

    # Extract Signatories
    sig_match = re.search(
        r'(attested.*(?:secretary|head)[^\n]*)', text, re.IGNORECASE
    )
    fields['signatories'] = sig_match.group(1).strip() if sig_match else ''

    # Extract Author/Sponsor (from original functionality)
    author_match = re.search(r'(?:(SPONSORED BY|AUTHORED BY|INTRODUCED
BY|AUTHOR)\s*:\s*([^\n]+)', text, re.IGNORECASE)
    if author_match:
        author_text = author_match.group(1).strip()
        fields['author'] = author_text[:200]

    # Extract Category by Keywords (from original functionality)
    text_lower = text.lower()
    if any(kw in text_lower for kw in ['budget', 'fund', 'revenue', 'tax',
'appropriation']):
        fields['category'] = 'revenue'
    elif any(kw in text_lower for kw in ['peace', 'order', 'curfew', 'penalty',
'prohibiting', 'tricycle', 'traffic']):
        fields['category'] = 'peace_order'

```

IoT-Enabled Barangay Resident and Attendance Management System

```
elif any(kw in text_lower for kw in ['health', 'sanitation', 'medical', 'hospital']):
    fields['category'] = 'health'
elif any(kw in text_lower for kw in ['environment', 'waste', 'garbage', 'tree']):
    fields['category'] = 'environment'
elif any(kw in text_lower for kw in ['infrastructure', 'road', 'building',
'construction']):
    fields['category'] = 'infrastructure'
elif any(kw in text_lower for kw in ['welfare', 'senior', 'pwd', 'youth']):
    fields['category'] = 'social_welfare'
elif any(kw in text_lower for kw in ['school', 'education', 'scholarship']):
    fields['category'] = 'education'

# Full content is the entire cleaned text
fields['body_content'] = text

return fields

def parse_ordinance_pdf(pdf_path):
    """
    Parses a PDF file containing a barangay ordinance and extracts data using PyMuPDF and
    OCR.
    """
    try:
        # Set tesseract path for Windows
        pytesseract.pytesseract.tesseract_cmd = r'C:\Program Files\Tesseract-
OCR\tesseract.exe'
    except Exception:
        pass

    try:
        doc = fitz.open(pdf_path)
        full_text = ""

        for page_num in range(len(doc)):
            page = doc[page_num]

            # First try native text extraction
            text = page.get_text()

            # If text is too short or garbled, use OCR
            if len(text.strip()) < 50 or 'DDeevveelloopp' in text or '^' in text:
                # Render page as image
                mat = fitz.Matrix(2, 2) # 2x zoom for better OCR accuracy
                clip = page.get_pixmap(matrix=mat)
                img = Image.open(io.BytesIO(clip.tobytes()))

                # Run Tesseract OCR with a fallback
                try:
                    text = pytesseract.image_to_string(img, lang='eng')
                except Exception as ocr_e:
                    print(f"OCR failed for page {page_num}: {ocr_e}")
                    # Keep whatever little text PyMuPDF found
                    pass

            full_text += text + "\n"

        # Clean the extracted text
        cleaned_text = clean_ocr_text(full_text)

        # Extract fields
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        return extract_ordinance_fields(cleaned_text)
except Exception as e:
    print(f"Error parsing PDF: {e}")
    # Return empty dictionary structure on failure
    return {
        'ordinance_number': '',
        'title': '',
        'date_enacted': '',
        'body_content': '',
        'signatories': '',
        'author': '',
        'category': ''
    }
```

File: ordinances/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.contrib import messages
from django.core.paginator import Paginator
from django.http import FileResponse
from django.db.models import Q
from .models import Ordinance

@login_required
def ordinance_list(request):
    """List all ordinances with search and filters."""
    query = request.GET.get('q', '')
    category = request.GET.get('category', '')
    status = request.GET.get('status', '')
    year = request.GET.get('year', '')

    ordinances = Ordinance.objects.all()

    if query:
        ordinances = ordinances.filter(
            Q(title__icontains=query) |
            Q(ordinance_number__icontains=query) |
            Q(author__icontains=query)
        )
    if category:
        ordinances = ordinances.filter(category=category)
    if status:
        ordinances = ordinances.filter(status=status)
    if year:
        ordinances = ordinances.filter(date_enacted__year=year)

    paginator = Paginator(ordinances, 25)
    page = request.GET.get('page')
    ordinances = paginator.get_page(page)

    years = Ordinance.objects.dates('date_enacted', 'year', order='DESC')

    context = {
        'ordinances': ordinances,
        'query': query,
        'selected_category': category,
        'selected_status': status,
        'selected_year': year,
        'categories': Ordinance.CATEGORY_CHOICES,
        'statuses': Ordinance.STATUS_CHOICES,
        'years': years,
    }
    return render(request, 'ordinances/list.html', context)

@login_required
def ordinance_add(request):
    """Add a new ordinance."""
    if request.user.get_permission_level() > 2:
        messages.error(request, 'You do not have permission to add ordinances.')
        return redirect('ordinances:list')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if request.method == 'POST':
    ordinance = Ordinance(
        ordinance_number=request.POST.get('ordinance_number'),
        title=request.POST.get('title'),
        description=request.POST.get('description', ''),
        author=request.POST.get('author'),
        date_enacted=request.POST.get('date_enacted'),
        date_effectivity=request.POST.get('date_effectivity') or None,
        category=request.POST.get('category'),
        status=request.POST.get('status', 'active'),
        sponsor=request.POST.get('sponsor', ''),
        remarks=request.POST.get('remarks', ''),
    )
    if request.FILES.get('document_file'):
        ordinance.document_file = request.FILES['document_file']
    ordinance.save()
    messages.success(request, f'Ordinance {ordinance.ordinance_number} added successfully.')
    return redirect('ordinances:view', pk=ordinance.pk)

return render(request, 'ordinances/form.html', {
    'categories': Ordinance.CATEGORY_CHOICES,
    'statuses': Ordinance.STATUS_CHOICES,
})

@login_required
def ordinance_view(request, pk):
    """View ordinance details."""
    ordinance = get_object_or_404(Ordinance, pk=pk)
    return render(request, 'ordinances/view.html', {'ordinance': ordinance})

@login_required
def ordinance_edit(request, pk):
    """Edit an ordinance."""
    if request.user.get_permission_level() > 2:
        messages.error(request, 'You do not have permission to edit ordinances.')
        return redirect('ordinances:list')
    ordinance = get_object_or_404(Ordinance, pk=pk)

    if request.method == 'POST':
        ordinance.ordinance_number = request.POST.get('ordinance_number')
        ordinance.title = request.POST.get('title')
        ordinance.description = request.POST.get('description', '')
        ordinance.author = request.POST.get('author')
        ordinance.date_enacted = request.POST.get('date_enacted')
        ordinance.date_effectivity = request.POST.get('date_effectivity') or None
        ordinance.category = request.POST.get('category')
        ordinance.status = request.POST.get('status', 'active')
        ordinance.sponsor = request.POST.get('sponsor', '')
        ordinance.remarks = request.POST.get('remarks', '')

    if request.FILES.get('document_file'):
        ordinance.document_file = request.FILES['document_file']

    ordinance.save()
    messages.success(request, f'Ordinance {ordinance.ordinance_number} updated.')
    return redirect('ordinances:view', pk=ordinance.pk)
```


IoT-Enabled Barangay Resident and Attendance Management System

```
return render(request, 'ordinances/form.html', {
    'ordinance': ordinance,
    'editing': True,
    'categories': Ordinance.CATEGORY_CHOICES,
    'statuses': Ordinance.STATUS_CHOICES,
})

@login_required
def ordinance_delete(request, pk):
    """Delete an ordinance."""
    if request.user.get_permission_level() > 2:
        messages.error(request, 'You do not have permission to delete ordinances.')
        return redirect('ordinances:list')
    ordinance = get_object_or_404(Ordinance, pk=pk)
    if request.method == 'POST':
        ordinance.delete()
        messages.success(request, 'Ordinance deleted.')
        return redirect('ordinances:list')
    return render(request, 'ordinances/confirm_delete.html', {'ordinance': ordinance})

@login_required
def ordinance_download(request, pk):
    """Download ordinance document."""
    ordinance = get_object_or_404(Ordinance, pk=pk)
    if ordinance.document_file:
        return FileResponse(ordinance.document_file.open(), as_attachment=True,
                             filename=f"Ordinance_{ordinance.ordinance_number}.pdf")
    messages.error(request, 'No document file attached.')
    return redirect('ordinances:view', pk=pk)

import os
from django.conf import settings
from django.core.files.storage import FileSystemStorage
from .utils import parse_ordinance_pdf

@login_required
def ordinance_upload_parse(request):
    """Handle PDF upload, parse it, and render the review form."""
    if request.user.get_permission_level() > 2:
        messages.error(request, 'You do not have permission to upload ordinances.')
        return redirect('ordinances:list')
    if request.method == 'POST' and request.FILES.get('document_file'):
        pdf_file = request.FILES['document_file']

        # Save file temporarily to parse it
        fs = FileSystemStorage(location=os.path.join(settings.MEDIA_ROOT,
            'temp_ordinances'))
        filename = fs.save(pdf_file.name, pdf_file)
        file_path = fs.path(filename)

        # Parse the PDF
        parsed_data = parse_ordinance_pdf(file_path)

        # Build URL to preview the temp file
        file_url = f"/media/temp_ordinances/{filename}"
```

IoT-Enabled Barangay Resident and Attendance Management System

```
context = {
    'parsed_data': parsed_data,
    'file_url': file_url,
    'temp_filename': filename,
    'categories': Ordinance.CATEGORY_CHOICES,
    'statuses': Ordinance.STATUS_CHOICES,
}
return render(request, 'ordinances/parse_review.html', context)

elif request.method == 'POST' and 'save_ordinance' in request.POST:
    # This is the submission from the parse_review form
    ord_num = request.POST.get('ordinance_number')
    try:
        ordinance = Ordinance.objects.get(ordinance_number=ord_num)
        msg = f'Ordinance {ordinance.ordinance_number} updated successfully.'
    except Ordinance.DoesNotExist:
        ordinance = Ordinance(ordinance_number=ord_num)
        msg = f'Ordinance {ordinance.ordinance_number} parsed and created
successfully.'

    ordinance.title = request.POST.get('title')
    ordinance.description = request.POST.get('description', '')
    ordinance.author = request.POST.get('author')
    ordinance.date_enacted = request.POST.get('date_enacted') or '2024-01-01'
    ordinance.date_effectivity = request.POST.get('date_effectivity') or None
    ordinance.category = request.POST.get('category', 'other')
    ordinance.status = request.POST.get('status', 'active')
    ordinance.sponsor = request.POST.get('sponsor', '')
    ordinance.remarks = request.POST.get('remarks', '')
    ordinance.body_content = request.POST.get('body_content', '')
    ordinance.signatories = request.POST.get('signatories', '')

    # Move the temp file to the permanent location
    temp_filename = request.POST.get('temp_filename')
    if temp_filename:
        temp_path = os.path.join(settings.MEDIA_ROOT, 'temp_ordinances',
temp_filename)
        if os.path.exists(temp_path):
            from django.core.files import File
            with open(temp_path, 'rb') as f:
                ordinance.document_file.save(temp_filename, File(f), save=False)
            ordinance.save()
            os.remove(temp_path)
        else:
            ordinance.save()
    else:
        ordinance.save()

    messages.success(request, msg)
    return redirect('ordinances:view', pk=ordinance.pk)

return render(request, 'ordinances/upload.html')
```

IoT-Enabled Barangay Resident and Attendance Management System

File: payments/apps.py

```
from django.apps import AppConfig

class PaymentsConfig(AppConfig):
    default_auto_field = 'django.db.models.BigAutoField'
    name = 'payments'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: payments/models.py

```
from django.db import models
from django.conf import settings
from datetime import date

class OfficialReceipt(models.Model):
    """Stores information about the issued Official Receipt (OR)."""
    or_number = models.CharField(max_length=50, unique=True, verbose_name="OR Number")
    resident = models.ForeignKey(
        'residents.Resident',
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name='official_receipts',
    )
    amount = models.DecimalField(max_digits=10, decimal_places=2)
    particulars = models.TextField(help_text="Description of what is being paid for (e.g.
Barangay Clearance)")
    issued_by = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
null=True, related_name="issued_receipts")
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    @property
    def resident_name(self):
        """Backward-compat property for templates/code still reading resident_name."""
        return self.resident.full_name if self.resident else "N/A"

    def __str__(self):
        return f"{self.or_number} - {self.resident_name}"

    @staticmethod
    def generate_next_or_number():
        """Generates the next OR number in the format OR-YYYY-XXXXX."""
        year = date.today().year
        last_receipt = OfficialReceipt.objects.filter(or_number__startswith=f"OR-{year}-
").order_by('-or_number').first()

        if last_receipt:
            # Extract the last 5 digits and increment
            last_number = int(last_receipt.or_number.split('-')[-1])
            new_number = last_number + 1
        else:
            new_number = 1

        return f"OR-{year}-{new_number:05d}"

    class Meta:
        verbose_name = "Official Receipt"
        verbose_name_plural = "Official Receipts"
        ordering = ["-created_at"]

class Payment(models.Model):
    """Handles the payment transaction details."""
    PAYMENT_METHOD_CHOICES = [
        ('cash', 'Cash at Hall'),
        ('gcash', 'GCash'),
        ('waived', 'Waived'),
```

IoT-Enabled Barangay Resident and Attendance Management System

```
]

PAYMENT_STATUS_CHOICES = [
    ('unpaid', 'Unpaid'),
    ('pending', 'Pending Verification'),
    ('paid', 'Paid'),
    ('waived', 'Waived'),
]

method = models.CharField(max_length=20, choices=PAYMENT_METHOD_CHOICES,
default='cash')
status = models.CharField(max_length=20, choices=PAYMENT_STATUS_CHOICES,
default='unpaid')
amount = models.DecimalField(max_digits=10, decimal_places=2, default=0.00)

# GCash details
proof_screenshot = models.ImageField(upload_to='payments/proofs/', blank=True,
null=True)
gcash_ref_number = models.CharField(max_length=100, blank=True, null=True,
verbose_name="GCash Reference Number")

# Link to OR if paid via Cash/GCash
official_receipt = models.OneToOneField(OfficialReceipt, on_delete=models.SET_NULL,
null=True, blank=True, related_name="payment_record")

# Waiver details
waive_reason = models.TextField(blank=True, null=True)

# Audit details
verified_by = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.SET_NULL,
null=True, blank=True, related_name="verified_payments")
paid_at = models.DateTimeField(null=True, blank=True)
created_at = models.DateTimeField(auto_now_add=True)
updated_at = models.DateTimeField(auto_now=True)

def __str__(self):
    return f"Payment {self.id} ({self.get_status_display()})"

class Meta:
    verbose_name = "Payment"
    verbose_name_plural = "Payments"
    ordering = ["-created_at"]
```

File: payments/urls.py

```
from django.urls import path
from . import views

app_name = 'payments'

urlpatterns = [
    path('dashboard/', views.payment_dashboard, name='dashboard'),
    path('<int:payment_id>/pay-cash/', views.mark_as_paid, name='mark_paid'),
    path('<int:payment_id>/confirm-gcash/', views.confirm_gcash_payment,
name='confirm_gcash'),
    path('<int:payment_id>/waive/', views.waive_fee, name='waive_fee'),
    path('request/<int:request_id>/pay/', views.resident_choose_payment,
name='choose_payment'),
    path('<int:payment_id>/receipt/', views.download_receipt, name='download_receipt'),
]
```

File: payments/utils.py

```
from decimal import Decimal
from django.utils import timezone
from .models import Payment, OfficialReceipt

def calculate_certificate_fee(resident, cert_type, cert_request=None):
    """
    Calculates the fee for a certificate request based on the resident's status.
    Auto-detects if the resident qualifies for a waiver.
    """
    # Fee Waiver for First Time Jobseekers (RA 11261) is now handled strictly upon admin
    approval
    # of the RA11261Application, which will manually set the payment to 0 and issue an
    EXEMPT receipt.

    # Standard Rates (matching certifications/views.py)
    rates = {
        'clearance': Decimal('100.00'),
        'residency': Decimal('50.00'),
        'indigency': Decimal('0.00'),
        'good_moral': Decimal('50.00'),
        'business_permit': Decimal('500.00'),
        'comelec': Decimal('50.00'),
        'cedula': Decimal('0.00'), # Base rate
        'late_registration': Decimal('50.00'),
    }

    fee = rates.get(cert_type, Decimal('50.00'))

    if cert_type == 'cedula' and cert_request:
        is_indiv = cert_request.taxpayer_type == 'individual'
        basic = Decimal('5.00') if is_indiv else Decimal('500.00')
        cap = Decimal('5000.00') if is_indiv else Decimal('10000.00')

        def to_d(val):
            if val is None: return Decimal('0.00')
            try: return Decimal(str(val))
            except: return Decimal('0.00')

        raw_prop = to_d(cert_request.raw_taxable_property)
        raw_bus = to_d(cert_request.raw_taxable_business)
        raw_inc = to_d(cert_request.raw_taxable_income)

        vProp = Decimal('0.00')
        vBus = Decimal('0.00')
        vInc = Decimal('0.00')

        if is_indiv:
            if raw_prop: vProp = (raw_prop / Decimal('1000')).quantize(Decimal('0.01'))
            if raw_bus: vBus = (raw_bus / Decimal('1000')).quantize(Decimal('0.01'))
            if raw_inc: vInc = (raw_inc / Decimal('1000')).quantize(Decimal('0.01'))
        else:
            if raw_prop: vProp = ((raw_prop / Decimal('5000')) *
Decimal('2')).quantize(Decimal('0.01'))
            if raw_bus: vBus = ((raw_bus / Decimal('5000')) *
Decimal('2')).quantize(Decimal('0.01'))

        additional = vProp + vBus + vInc
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        if additional > cap:
            additional = cap

        fee = basic + additional

    # Only auto-waive if fee is 0 AND it's not a Cedula
    is_waived = (fee == Decimal('0.00') and cert_type != 'cedula')
    return fee, is_waived

def generate_receipt_pdf(payment):
    """
    Generates a PDF official receipt using ReportLab.
    """
    from reportlab.lib import colors
    from reportlab.lib.pagesizes import letter, five_by_seven
    from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle
    from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
    from io import BytesIO

    buffer = BytesIO()
    doc = SimpleDocTemplate(buffer, pagesize=five_by_seven, rightMargin=30, leftMargin=30,
topMargin=30, bottomMargin=30)

    styles = getSampleStyleSheet()
    title_style = ParagraphStyle(
        'TitleStyle',
        parent=styles['Heading1'],
        alignment=1, # Center
        fontSize=14,
        spaceAfter=10
    )

    header_style = ParagraphStyle(
        'HeaderStyle',
        alignment=1, # Center
        fontSize=10,
        spaceAfter=5
    )

    elements = []

    # 1. Letterhead
    elements.append(Paragraph("REPUBLIC OF THE PHILIPPINES", header_style))
    elements.append(Paragraph("PROVINCE OF SURIGAO DEL NORTE", header_style))
    elements.append(Paragraph("MUNICIPALITY OF GIGAQUIT", header_style))
    elements.append(Paragraph("<strong>BARANGAY SICO-SICO</strong>", title_style))
    elements.append(Spacer(1, 12))

    # 2. Receipt Title
    elements.append(Paragraph("OFFICIAL RECEIPT", title_style))
    elements.append(Spacer(1, 12))

    receipt = payment.official_receipt

    # 3. Receipt Details
    data = [
        ["OR Number:", receipt.or_number],
        ["Date:", receipt.created_at.strftime("%B %d, %Y %I:%M %p")],
        ["Payer Name:", receipt.resident_name],
        ["Particulars:", receipt.particulars],
```


IoT-Enabled Barangay Resident and Attendance Management System

```
["Amount Paid:", f"PHP {receipt.amount:,.2f}"],
["Payment Method:", payment.get_method_display()],
["Issued By:", f"{receipt.issued_by.first_name} {receipt.issued_by.last_name}"],
]

table = Table(data, colwidths=[100, 150])
table.setStyle(TableStyle([
    ('FONTNAME', (0, 0), (0, -1), 'Helvetica-Bold'),
    ('BOTTOMPADDING', (0, 0), (-1, -1), 6),
]))
elements.append(table)

elements.append(Spacer(1, 30))
elements.append(Paragraph("--- This serves as your official proof of payment ---",
header_style))

doc.build(elements)

buffer.seek(0)
return buffer
```

File: payments/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.contrib import messages
from django.utils import timezone
from django.http import HttpResponse, FileResponse
from .models import Payment, OfficialReceipt
from certifications.models import CertificateRequest, Certificate
from .utils import calculate_certificate_fee, generate_receipt_pdf
from decimal import Decimal

def is_secretary_or_higher(user):
    try:
        return user.is_superuser or (hasattr(user, 'profile') and user.profile.role in
['captain', 'secretary', 'admin'])
    except:
        return False

@login_required
def payment_dashboard(request):
    """Secretary's dashboard for managing payments."""
    if not is_secretary_or_higher(request.user):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    status_filter = request.GET.get('status', 'pending')
    method_filter = request.GET.get('method', '')

    payments = Payment.objects.all()

    if status_filter:
        payments = payments.filter(status=status_filter)
    if method_filter:
        payments = payments.filter(method=method_filter)

    return render(request, 'payments/dashboard.html', {
        'payments': payments,
        'status_filter': status_filter,
        'method_filter': method_filter
    })

@login_required
def mark_as_paid(request, payment_id):
    """Secretary marks a cash payment as paid and issues an OR."""
    if not is_secretary_or_higher(request.user):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    payment = get_object_or_404(Payment, id=payment_id)

    if request.method == 'POST':
        or_number = request.POST.get('or_number')
        if not or_number:
            or_number = OfficialReceipt.generate_next_or_number()

        # Create Official Receipt
        receipt = OfficialReceipt.objects.create(
            or_number=or_number,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        resident=payment.certificate_request.resident,
        amount=payment.amount,
        particulars=payment.certificate_request.get_cert_type_display(),
        issued_by=request.user
    )

    # Update Payment
    payment.status = 'paid'
    payment.official_receipt = receipt
    payment.verified_by = request.user
    payment.paid_at = timezone.now()
    payment.save()

    messages.success(request, f"Payment marked as PAID. OR Number: {or_number}")

    return redirect('payments:dashboard')

@login_required
def confirm_gcash_payment(request, payment_id):
    """Secretary confirms a GCash payment proof."""
    if not is_secretary_or_higher(request.user):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    payment = get_object_or_404(Payment, id=payment_id)

    if request.method == 'POST':
        gcash_ref = request.POST.get('gcash_ref_number')

        # Create Official Receipt
        receipt = OfficialReceipt.objects.create(
            or_number=OfficialReceipt.generate_next_or_number(),
            resident=payment.certificate_request.resident,
            amount=payment.amount,
            particulars=f"{payment.certificate_request.get_cert_type_display()} (GCash)",
            issued_by=request.user
        )

        # Update Payment
        payment.status = 'paid'
        payment.gcash_ref_number = gcash_ref
        payment.official_receipt = receipt
        payment.verified_by = request.user
        payment.paid_at = timezone.now()
        payment.save()

        messages.success(request, "GCash payment confirmed successfully.")

    return redirect('payments:dashboard')

@login_required
def waive_fee(request, payment_id):
    """Secretary manually waives a fee for a resident."""
    if not is_secretary_or_higher(request.user):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    payment = get_object_or_404(Payment, id=payment_id)

    if request.method == 'POST':
```

IoT-Enabled Barangay Resident and Attendance Management System

```
reason = request.POST.get('reason', 'Manual Waiver')

payment.status = 'waived'
payment.method = 'waived'
payment.waive_reason = reason
payment.verified_by = request.user
payment.paid_at = timezone.now()
payment.save()

messages.warning(request, f"Fee waived for
{payment.certificate_request.resident.full_name}. Reason: {reason}")

return redirect('payments:dashboard')

@login_required
def resident_choose_payment(request, request_id):
    """Resident selects their payment method for a certificate request."""
    cert_request = get_object_or_404(CertificateRequest, id=request_id)

    # Ensure this is the resident's own request
    if cert_request.resident != request.user.profile.resident:
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    if request.method == 'POST':
        method = request.POST.get('method')

        # Calculate fee (pass cert_request for accurate Cedula computation)
        amount, is_waived = calculate_certificate_fee(cert_request.resident,
cert_request.cert_type, cert_request)

        # Create or Get Payment
        payment = cert_request.payment
        if not payment:
            payment = Payment.objects.create(amount=amount)
            cert_request.payment = payment
            cert_request.save()
        else:
            # Always sync the amount in case admin rates changed
            payment.amount = amount
            payment.save()

        payment.method = method
        if is_waived:
            payment.status = 'waived'
            payment.method = 'waived'
            payment.paid_at = timezone.now()
        elif method == 'gcash':
            payment.status = 'pending'
            if request.FILES.get('proof'):
                payment.proof_screenshot = request.FILES['proof']
        else: # Cash
            payment.status = 'unpaid'

        payment.save()

        if is_waived:
            messages.success(request, "Your fee has been automatically waived. Your
request is now ready for processing.")
        else:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        messages.success(request, f"Payment method '{payment.get_method_display()}'
selected. Please proceed as instructed.")

        return redirect('certifications:my_requests')

    amount, is_waived = calculate_certificate_fee(cert_request.resident,
cert_request.cert_type, cert_request)

    # Get the same rate the admin uses for this cert type
    from certifications.views import get_certificate_rate
    standard_rate = get_certificate_rate(cert_request.cert_type)

    return render(request, 'payments/select_payment.html', {
        'cert_request': cert_request,
        'amount': amount,
        'standard_rate': standard_rate,
        'is_waived': is_waived,
    })

@login_required
def download_receipt(request, payment_id):
    """Generates and serves the PDF receipt."""
    payment = get_object_or_404(Payment, id=payment_id)

    # Ensure they have permission (Secretary or the specific Resident)
    is_authorized = False
    if is_secretary_or_higher(request.user):
        is_authorized = True
    elif hasattr(request.user, 'profile') and request.user.profile.resident ==
payment.certificate_request.resident:
        is_authorized = True

    if not is_authorized or not payment.official_receipt:
        messages.error(request, "Unauthorized or no receipt available.")
        return redirect('core:dashboard')

    pdf_buffer = generate_receipt_pdf(payment)
    return FileResponse(pdf_buffer, as_attachment=True,
filename=f"Receipt_{payment.official_receipt.or_number}.pdf")
```

IoT-Enabled Barangay Resident and Attendance Management System

File: philsys/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: philsys/admin.py

```
from django.contrib import admin
from .models import PhilSysVerificationAttempt

@admin.register(PhilSysVerificationAttempt)
class PhilSysVerificationAttemptAdmin(admin.ModelAdmin):
    list_display = [
        "id",
        "user",
        "resident",
        "verification_method",
        "status",
        "philSys_number",
        "ip_address",
        "created_at",
    ]
    list_filter = [
        "status",
        "verification_method",
        "created_at",
    ]
    search_fields = [
        "user__username",
        "philSys_number",
        "ip_address",
    ]
    readonly_fields = [
        "created_at",
        "completed_at",
        "ip_address",
    ]
    ordering = ["-created_at"]
    date_hierarchy = "created_at"

    fieldsets = (
        ("User Information", {"fields": ("user", "resident")}),
        (
            "Verification Details",
            {
                "fields": (
                    "verification_method",
                    "status",
                    "philSys_number",
                    "philSys_card_number",
                )
            },
        ),
        (
            "Response Data",
            {"fields": ("response_data", "error_message"), "classes": ("collapse",)},
        ),
        (
            "Metadata",
            {
                "fields": ("ip_address", "created_at", "completed_at"),
                "classes": ("collapse",),
            },
        ),
    )
```

IoT-Enabled Barangay Resident and Attendance Management System

) ,
)

IoT-Enabled Barangay Resident and Attendance Management System

File: philsys/apps.py

```
from django.apps import AppConfig

class PhilsysConfig(AppConfig):
    default_auto_field = "django.db.models.BigAutoField"
    name = "philsys"
    verbose_name = "PhilSys Verification"
```

IoT-Enabled Barangay Resident and Attendance Management System

File: philsys/models.py

```
from django.db import models
from django.conf import settings

class PhilSysVerificationAttempt(models.Model):
    """Logs PhilSys verification attempts."""

    STATUS_CHOICES = [
        ("pending", "Pending"),
        ("success", "Success"),
        ("failed", "Failed"),
    ]
    VERIFICATION_METHOD_CHOICES = [
        ("qr", "QR Code"),
        ("psn", "PSN"),
        ("biometric", "Biometric"),
    ]

    user = models.ForeignKey(settings.AUTH_USER_MODEL, on_delete=models.CASCADE,
related_name="philsys_attempts"
    )
    resident = models.ForeignKey(
        "residents.Resident",
        on_delete=models.CASCADE,
        null=True,
        blank=True,
        related_name="philsys_attempts",
    )
    philSys_number = models.CharField(max_length=12, blank=True, null=True)
    philSys_card_number = models.CharField(max_length=20, blank=True, null=True)
    verification_method = models.CharField(
        max_length=20, choices=VERIFICATION_METHOD_CHOICES
    )
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default="pending")
    response_payload = models.JSONField(blank=True, null=True, help_text='Raw API response
for audit purposes')
    error_message = models.TextField(blank=True)
    ip_address = models.GenericIPAddressField(blank=True, null=True)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)
    completed_at = models.DateTimeField(blank=True, null=True)

    class Meta:
        ordering = ["-created_at"]
        verbose_name = "PhilSys Verification Attempt"
        verbose_name_plural = "PhilSys Verification Attempts"
        indexes = [
            models.Index(fields=["-created_at"]),
            models.Index(fields=["status"]),
            models.Index(fields=["user", "-created_at"]),
        ]

    def __str__(self):
        return f"Verification {self.id} - {self.user.username} - {self.status}"
```

File: philsys/serializers.py

```
from rest_framework import serializers
from .models import PhilSysVerificationAttempt

class PhilSysVerificationAttemptSerializer(serializers.ModelSerializer):
    resident_name = serializers.CharField(
        source="resident.get_full_name", read_only=True
    )
    user_username = serializers.CharField(source="user.username", read_only=True)

    class Meta:
        model = PhilSysVerificationAttempt
        fields = [
            "id",
            "user",
            "user_username",
            "resident",
            "resident_name",
            "verification_method",
            "status",
            "philSys_number",
            "philSys_card_number",
            "response_data",
            "error_message",
            "ip_address",
            "created_at",
            "completed_at",
        ]
        read_only_fields = ["id", "created_at", "completed_at"]
```

IoT-Enabled Barangay Resident and Attendance Management System

File: philsys/urls.py

```
from django.urls import path
from . import views

app_name = "philsys"

urlpatterns = [
    path("verify/", views.verify_philsys, name="verify"),
    path("verify/qr/", views.verify_qr_code, name="verify_qr"),
    path("verify/psn/", views.verify_psn, name="verify_psn"),
    path("status/<int:attempt_id>/", views.get_verification_status, name="status"),
    path("history/", views.verification_history, name="history"),
    path("check/<int:resident_id>/", views.check_verification, name="check"),
]
```

File: philsys/views.py

```
from django.http import JsonResponse
from django.views.decorators.csrf import csrf_exempt
from django.views.decorators.http import require_http_methods
from django.utils import timezone
from django.contrib.auth.decorators import login_required
import json

from .models import PhilSysVerificationAttempt
from residents.models import Resident
from core.models import UserProfile

@csrf_exempt
@require_http_methods(["POST"])
@login_required
def verify_philsys(request):
    """Verify PhilSys credentials for a resident."""
    try:
        data = json.loads(request.body)
    except json.JSONDecodeError:
        return JsonResponse({"error": "Invalid JSON"}, status=400)

    resident_id = data.get("resident_id")
    philSys_number = data.get("philSys_number")
    philSys_card_number = data.get("philSys_card_number")
    verification_method = data.get("verification_method", "psn")

    if verification_method not in ["qr", "psn", "biometric"]:
        return JsonResponse({"error": "Invalid verification method"}, status=400)

    if not resident_id:
        return JsonResponse({"error": "Resident ID is required"}, status=400)

    try:
        resident = Resident.objects.get(id=resident_id)
    except Resident.DoesNotExist:
        return JsonResponse({"error": "Resident not found"}, status=404)

    attempt = PhilSysVerificationAttempt.objects.create(
        user=request.user,
        resident=resident,
        philSys_number=philSys_number,
        philSys_card_number=philSys_card_number,
        verification_method=verification_method,
        ip_address=get_client_ip(request),
    )

    attempt.status = "success"
    attempt.completed_at = timezone.now()
    attempt.response_data = {
        "verified": True,
        "philSys_number": philSys_number,
        "message": "PhilSys verification successful",
    }
    attempt.save()

    resident.is_philsys_verified = True
```

IoT-Enabled Barangay Resident and Attendance Management System

```
resident.philsys_verified_at = timezone.now()
resident.philsys_verification_method = verification_method
if philSys_number:
    resident.philSys_number = philSys_number
if philSys_card_number:
    resident.philSys_card_number = philSys_card_number
resident.save()

user_profile = UserProfile.objects.filter(user=request.user).first()
if user_profile:
    user_profile.is_philsys_verified = True
    user_profile.philsys_verified_at = timezone.now()
    if philSys_number:
        user_profile.philSys_id = philSys_number
    user_profile.save()

return JsonResponse(
    {
        "success": True,
        "verification_id": attempt.id,
        "message": "PhilSys verification successful",
    }
)

@login_required
def verification_history(request):
    """Get verification history for the current user."""
    attempts = (
        PhilSysVerificationAttempt.objects.filter(user=request.user)
        .select_related("resident")
        .order_by("-created_at")[:50]
    )

    data = [
        {
            "id": attempt.id,
            "resident_name": str(attempt.resident) if attempt.resident else None,
            "philSys_number": attempt.philSys_number,
            "verification_method": attempt.verification_method,
            "status": attempt.status,
            "created_at": attempt.created_at.isoformat(),
            "completed_at": attempt.completed_at.isoformat()
            if attempt.completed_at
            else None,
        }
        for attempt in attempts
    ]

    return JsonResponse({"attempts": data})

@login_required
def check_verification(request, resident_id):
    """Check if a resident is PhilSys verified."""
    try:
        resident = Resident.objects.get(id=resident_id)
    except Resident.DoesNotExist:
        return JsonResponse({"error": "Resident not found"}, status=404)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
return JsonResponse(
    {
        "is_verified": resident.is_philsys_verified,
        "philSys_number": resident.philSys_number,
        "philsys_verified_at": resident.philsys_verified_at.isoformat()
        if resident.philsys_verified_at
        else None,
        "verification_method": resident.philsys_verification_method,
    }
)

@csrf_exempt
@require_http_methods(["POST"])
@login_required
def verify_qr_code(request):
    """Verify PhilID QR code (simulated - validates QR structure)."""
    try:
        data = json.loads(request.body)
    except json.JSONDecodeError:
        return JsonResponse({"error": "Invalid JSON"}, status=400)

    qr_data = data.get("qr_data")
    resident_id = data.get("resident_id")

    if not qr_data:
        return JsonResponse({"error": "QR data is required"}, status=400)

    if not resident_id:
        return JsonResponse({"error": "Resident ID is required"}, status=400)

    try:
        resident = Resident.objects.get(id=resident_id)
    except Resident.DoesNotExist:
        return JsonResponse({"error": "Resident not found"}, status=404)

    attempt = PhilSysVerificationAttempt.objects.create(
        user=request.user,
        resident=resident,
        verification_method="qr",
        ip_address=get_client_ip(request),
    )

    if not qr_data.startswith("PHL-") or len(qr_data) < 10:
        attempt.status = "failed"
        attempt.error_message = (
            "Invalid QR code format. Expected format: PHL-XXXXXXXXXX"
        )
        attempt.save()
        return JsonResponse(
            {
                "success": False,
                "verification_id": attempt.id,
                "message": "Invalid QR code format",
            },
            status=400,
        )

    attempt.status = "success"
    attempt.completed_at = timezone.now()
```

IoT-Enabled Barangay Resident and Attendance Management System

```
attempt.response_data = {
    "verified": True,
    "qr_data": qr_data[:8] + "****",
    "message": "QR code verification successful",
}
attempt.save()

return JsonResponse(
    {
        "success": True,
        "verification_id": attempt.id,
        "message": "QR code verification successful",
    }
)

@csrf_exempt
@require_http_methods(["POST"])
@login_required
def verify_psn(request):
    """Verify via PhilSys Number (simulated)."""
    try:
        data = json.loads(request.body)
    except json.JSONDecodeError:
        return JsonResponse({"error": "Invalid JSON"}, status=400)

    philSys_number = data.get("philSys_number")
    resident_id = data.get("resident_id")

    if not philSys_number:
        return JsonResponse({"error": "PhilSys Number is required"}, status=400)

    if not resident_id:
        return JsonResponse({"error": "Resident ID is required"}, status=400)

    try:
        resident = Resident.objects.get(id=resident_id)
    except Resident.DoesNotExist:
        return JsonResponse({"error": "Resident not found"}, status=404)

    attempt = PhilSysVerificationAttempt.objects.create(
        user=request.user,
        resident=resident,
        philSys_number=philSys_number,
        verification_method="psn",
        ip_address=get_client_ip(request),
    )

    if not philSys_number.isdigit() or len(philSys_number) != 12:
        attempt.status = "failed"
        attempt.error_message = "Invalid PhilSys Number. Must be 12 digits."
        attempt.save()
        return JsonResponse(
            {
                "success": False,
                "verification_id": attempt.id,
                "message": "Invalid PhilSys Number format",
            },
            status=400,
        )
```


IoT-Enabled Barangay Resident and Attendance Management System

```
attempt.status = "success"
attempt.completed_at = timezone.now()
attempt.response_data = {
    "verified": True,
    "philSys_number": philSys_number[:4] + "*****",
    "message": "PhilSys Number verification successful",
}
attempt.save()

return JsonResponse(
    {
        "success": True,
        "verification_id": attempt.id,
        "message": "PhilSys Number verification successful",
    }
)

@login_required
def get_verification_status(request, attempt_id):
    """Get verification status by attempt ID."""
    try:
        attempt = PhilSysVerificationAttempt.objects.get(id=attempt_id)
    except PhilSysVerificationAttempt.DoesNotExist:
        return JsonResponse({"error": "Verification attempt not found"}, status=404)

    return JsonResponse(
        {
            "id": attempt.id,
            "verification_method": attempt.verification_method,
            "status": attempt.status,
            "philSys_number": attempt.philSys_number,
            "error_message": attempt.error_message,
            "created_at": attempt.created_at.isoformat(),
            "completed_at": attempt.completed_at.isoformat()
            if attempt.completed_at
            else None,
        }
    )

def get_client_ip(request):
    """Get client IP address from request."""
    x_forwarded_for = request.META.get("HTTP_X_FORWARDED_FOR")
    if x_forwarded_for:
        ip = x_forwarded_for.split(",")[0]
    else:
        ip = request.META.get("REMOTE_ADDR")
    return ip
```

IoT-Enabled Barangay Resident and Attendance Management System

File: refactor_auth.py

```
import os
import re

directory = '/home/mharkd/.gemini/antigravity/scratch/BrgyManagementSystem'

for root, _, files in os.walk(directory):
    if 'venv' in root or '.git' in root or '__pycache__' in root:
        continue
    for file in files:
        if file.endswith('.py'):
            filepath = os.path.join(root, file)
            with open(filepath, 'r') as f:
                content = f.read()

            new_content = content

            # Handle core/models.py missing import
            if filepath.endswith('core/models.py'):
                if 'from django.conf import settings' not in new_content:
                    new_content = new_content.replace('from django.contrib.auth.models
import AbstractUser', 'from django.contrib.auth.models import AbstractUser\nfrom
django.conf import settings')

            # Replace imports
            if 'from django.contrib.auth import get_user_model
User = get_user_model()' in new_content:
                if 'models.py' in file:
                    new_content = new_content.replace('from django.contrib.auth import
get_user_model
User = get_user_model()', 'from django.conf import settings')
                else:
                    new_content = new_content.replace('from django.contrib.auth import
get_user_model
User = get_user_model()', 'from django.contrib.auth import get_user_model\nUser =
get_user_model()')

            # Fix models.ForeignKey and OneToOneField
            if 'models.py' in file:
                new_content = re.sub(r'models\.ForeignKey\(\s*User\s*,',
r'models.ForeignKey(settings.AUTH_USER_MODEL,', new_content)
                new_content = re.sub(r'models\.ForeignKey\(\s*["']auth\.User["']',
r'models.ForeignKey(settings.AUTH_USER_MODEL"', new_content)
                new_content = re.sub(r'models\.OneToOneField\(\s*User\s*,',
r'models.OneToOneField(settings.AUTH_USER_MODEL,', new_content)

            if new_content != content:
                with open(filepath, 'w') as f:
                    f.write(new_content)
                print(f"Updated {filepath}")
```

IoT-Enabled Barangay Resident and Attendance Management System

File: reports/__init__.py

IoT-Enabled Barangay Resident and Attendance Management System

File: reports/admin.py

```
from django.contrib import admin
```

```
# Register your models here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: reports/apps.py

```
from django.apps import AppConfig
```

```
class ReportsConfig(AppConfig):  
    name = 'reports'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: reports/models.py

```
from django.db import models
```

```
# Create your models here.
```

IoT-Enabled Barangay Resident and Attendance Management System

File: reports/tests.py

```
from django.test import TestCase
```

```
# Create your tests here.
```

File: reports/urls.py

```
from django.urls import path
from . import views

app_name = 'reports'

urlpatterns = [
    path('', views.reports_dashboard, name='dashboard'),
]
```


IoT-Enabled Barangay Resident and Attendance Management System

File: reports/views.py

```
from django.shortcuts import render
from django.contrib.auth.decorators import login_required
from django.db.models import Count, Sum
from residents.models import Resident, Household
from certifications.models import Certificate
from attendance.models import AttendanceLog
from officials.models import Official
from ordinances.models import Ordinance
from datetime import date, timedelta

@login_required
def reports_dashboard(request):
    """Reports dashboard with summary data."""
    today = date.today()
    month_start = today.replace(day=1)

    context = {
        # Population
        'total_population': Resident.objects.filter(is_active=True).count(),
        'total_households': Household.objects.count(),
        'male_count': Resident.objects.filter(is_active=True, gender='M').count(),
        'female_count': Resident.objects.filter(is_active=True, gender='F').count(),
        'voters': Resident.objects.filter(is_registered_voter=True).count(),
        'pwd': Resident.objects.filter(is_pwd=True).count(),
        'senior': Resident.objects.filter(is_senior_citizen=True).count(),
        'fourps': Resident.objects.filter(is_4ps_member=True).count(),

        # Certificates this month
        'certs_this_month':
Certificate.objects.filter(date_issued__gte=month_start).count(),
        'cert_breakdown': [
            {'type':
dict(Certificate.TYPE_CHOICES).get(item['certificate_request__cert_type'],
item['certificate_request__cert_type'].replace('_', ' ').title()), 'count': item['count']}
            for item in Certificate.objects.filter(date_issued__gte=month_start).values('c
ertificate_request__cert_type').annotate(count=Count('id'))
            ],
        'total_revenue': Certificate.objects.filter(date_issued__gte=month_start).aggregat
e(total=Sum('certificate_request__payment__amount'))['total'] or 0,

        # Attendance
        'attendance_today': AttendanceLog.objects.filter(date=today).count(),

        # Officials
        'active_officials': Official.objects.filter(status='active').count(),

        # Ordinances
        'total_ordinances': Ordinance.objects.count(),
        'active_ordinances': Ordinance.objects.filter(status='active').count(),
    }
    return render(request, 'reports/dashboard.html', context)
```

File: residents/__init__.py

File: residents/admin.py

```
from django.contrib import admin
from .models import Resident, Household, Purok

@admin.register(Purok)
class PurokAdmin(admin.ModelAdmin):
    list_display = ('name',)
    search_fields = ('name',)

@admin.register(Household)
class HouseholdAdmin(admin.ModelAdmin):
    list_display = ('household_no', 'purok', 'member_count')
    list_filter = ('purok',)
    search_fields = ('household_no', 'address')

@admin.register(Resident)
class ResidentAdmin(admin.ModelAdmin):
    list_display = ('full_name', 'age', 'gender', 'purok', 'contact_number',
'is_official', 'is_active')
    list_filter = ('purok', 'gender', 'is_official', 'is_active', 'is_registered_voter',
'is_senior_citizen')
    search_fields = ('first_name', 'last_name', 'middle_name', 'philSys_number',
'contact_number')
    readonly_fields = ('created_at', 'updated_at')
    list_per_page = 25
```

IoT-Enabled Barangay Resident and Attendance Management System

File: residents/apps.py

```
from django.apps import AppConfig
```

```
class ResidentsConfig(AppConfig):  
    name = 'residents'
```

IoT-Enabled Barangay Resident and Attendance Management System

File: residents/models.py

```
from django.db import models
from datetime import date

class Purok(models.Model):
    """Purok/zone lookup table for normalized queries."""
    name = models.CharField(max_length=100, unique=True)

    # Geo coordinates for mapping
    latitude = models.FloatField(null=True, blank=True)
    longitude = models.FloatField(null=True, blank=True)

    def __str__(self):
        return self.name

    class Meta:
        ordering = ['name']
        verbose_name = 'Purok'
        verbose_name_plural = 'Puroks'

class Household(models.Model):
    """Household/family unit."""

    household_no = models.CharField(max_length=50, unique=True)
    address = models.TextField()
    purok = models.ForeignKey(
        Purok,
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name='households',
    )
    # Geo coordinates for mapping
    latitude = models.FloatField(null=True, blank=True)
    longitude = models.FloatField(null=True, blank=True)

    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    def __str__(self):
        return f"Household #{self.household_no} - {self.purok or 'N/A'}"

    @property
    def member_count(self):
        return self.members.count()

    @property
    def head(self):
        return self.members.filter(is_household_head=True).first()

    class Meta:
        ordering = ["household_no"]

class Resident(models.Model):
    """Barangay resident record."""
```

IoT-Enabled Barangay Resident and Attendance Management System

```
GENDER_CHOICES = [("M", "Male"), ("F", "Female")]
CIVIL_STATUS_CHOICES = [
    ("single", "Single"),
    ("married", "Married"),
    ("widowed", "Widowed"),
    ("separated", "Separated"),
    ("divorced", "Divorced"),
]
PHILSYS_VERIFICATION_METHOD_CHOICES = [
    ("qr", "QR Code"),
    ("psn", "PSN"),
    ("biometric", "Biometric"),
]

# Personal info
last_name = models.CharField(max_length=100)
first_name = models.CharField(max_length=100)
middle_name = models.CharField(max_length=100, blank=True)
suffix = models.CharField(max_length=20, blank=True)
birthdate = models.DateField()
birthplace = models.CharField(max_length=200, blank=True)
gender = models.CharField(max_length=1, choices=GENDER_CHOICES)
civil_status = models.CharField(
    max_length=20, choices=CIVIL_STATUS_CHOICES, default="single"
)
nationality = models.CharField(max_length=100, default="Filipino")
religion = models.CharField(max_length=100, blank=True)
occupation = models.CharField(max_length=200, blank=True)

# Contact
contact_number = models.CharField(max_length=20, blank=True)
email = models.EmailField(blank=True)

# Address
address = models.TextField()
purok = models.ForeignKey(
    Purok,
    on_delete=models.SET_NULL,
    null=True,
    blank=True,
    related_name='residents',
)

# Household
HOUSEHOLD_RELATIONSHIP_CHOICES = [
    ("head", "Head"),
    ("spouse", "Spouse"),
    ("child", "Child"),
    ("parent", "Parent"),
    ("sibling", "Sibling"),
    ("grandchild", "Grandchild"),
    ("grandparent", "Grandparent"),
    ("other", "Other Relative"),
]

household = models.ForeignKey(
    Household,
    on_delete=models.SET_NULL,
    null=True,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        blank=True,
        related_name="members",
    )
    is_household_head = models.BooleanField(default=False)
    household_relationship = models.CharField(
        max_length=20,
        choices=HOUSEHOLD_RELATIONSHIP_CHOICES,
        blank=True,
        null=True,
        verbose_name="Relationship to Head"
    )
    parent_member = models.ForeignKey(
        'self',
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name='children_members',
        help_text="Custom parent relationship for interactive family tree"
    )

    # PhilSys Integration
    philSys_number = models.CharField(
        max_length=12, blank=True, null=True, verbose_name="PhilSys Number (PSN)"
    )
    philSys_card_number = models.CharField(
        max_length=20, blank=True, null=True, verbose_name="PhilSys Card Number (PCN)"
    )
    is_philsys_verified = models.BooleanField(default=False)
    philsys_verified_at = models.DateTimeField(blank=True, null=True)
    philsys_verification_method = models.CharField(
        max_length=20,
        choices=PHILSYS_VERIFICATION_METHOD_CHOICES,
        blank=True,
        null=True,
    )

    # Status
    is_registered_voter = models.BooleanField(default=False)
    is_pwd = models.BooleanField(default=False, verbose_name="Person with Disability")
    is_senior_citizen = models.BooleanField(default=False)
    is_4ps_member = models.BooleanField(default=False, verbose_name="4Ps Member")
    is_indigent = models.BooleanField(default=False)
    is_solo_parent = models.BooleanField(default=False)
    is_official = models.BooleanField(default=False, verbose_name="Barangay Functionary")

    # Photo
    photo = models.ImageField(upload_to="residents/photos/", blank=True, null=True)

    # Biometrics (single source of truth for fingerprint data)
    fingerprint_template = models.TextField(blank=True, null=True, help_text="Base64
encoded fingerprint template")
    fingerprint_id = models.IntegerField(
        null=True,
        blank=True,
        unique=True,
        help_text="0-based template page index in the R307/AS608 module (0 ..
capacity-1)",
    )

    # Metadata
```

IoT-Enabled Barangay Resident and Attendance Management System

```
is_active = models.BooleanField(default=True)
remarks = models.TextField(blank=True)
created_at = models.DateTimeField(auto_now_add=True)
updated_at = models.DateTimeField(auto_now=True)

@property
def display_occupation(self):
    """Return official position if functionary, else regular occupation."""
    if self.is_official and hasattr(self, 'official_record'):
        return self.official_record.get_position_display()
    return self.occupation or "Unemployed"

def __str__(self):
    name = f"{self.last_name}, {self.first_name}"
    if self.middle_name:
        name += f" {self.middle_name}"
    if self.suffix:
        name += f" {self.suffix}"
    return name

@property
def full_name(self):
    parts = [self.first_name, self.middle_name, self.last_name]
    if self.suffix:
        parts.append(self.suffix)
    return " ".join(p for p in parts if p)

@property
def age(self):
    today = date.today()
    return (
        today.year
        - self.birthdate.year
        - ((today.month, today.day) < (self.birthdate.month, self.birthdate.day))
    )

@property
def age_group(self):
    age = self.age
    if age < 1:
        return "Infant"
    elif age <= 5:
        return "Early Childhood"
    elif age <= 12:
        return "Child"
    elif age <= 17:
        return "Teenager"
    elif age <= 24:
        return "Young Adult"
    elif age <= 59:
        return "Adult"
    else:
        return "Senior Citizen"

class Meta:
    ordering = ["last_name", "first_name"]
    verbose_name = "Resident"
    verbose_name_plural = "Residents"
    constraints = [
        models.UniqueConstraint(
```


IoT-Enabled Barangay Resident and Attendance Management System

```
        fields=['first_name', 'middle_name', 'last_name', 'suffix', 'birthdate'],
        name='unique_resident_name_birthdate'
    )
]

import uuid
import random
import string

def generate_reference_number():
    """Generate a unique tracking reference number (e.g. BRGY-2026-X1Y2Z3)."""
    current_year = date.today().year
    random_str = ''.join(random.choices(string.ascii_uppercase + string.digits, k=6))
    return f"BRGY-{current_year}-{random_str}"

class ResidentRegistration(models.Model):
    """Temporary storage for resident registration before verification."""

    STATUS_CHOICES = [
        ("pending", "Pending Verification"),
        ("approved", "Approved"),
        ("rejected", "Rejected"),
    ]

    # Tracking
    reference_number = models.CharField(max_length=20, unique=True,
default=generate_reference_number)
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default="pending")

    # Step 1: Personal Info
    last_name = models.CharField(max_length=100)
    first_name = models.CharField(max_length=100)
    middle_name = models.CharField(max_length=100, blank=True)
    suffix = models.CharField(max_length=20, blank=True)
    birthdate = models.DateField()
    birthplace = models.CharField(max_length=200, blank=True)
    gender = models.CharField(max_length=1, choices=[("M", "Male"), ("F", "Female")])
    civil_status = models.CharField(max_length=20, default="single")
    nationality = models.CharField(max_length=100, default="Filipino")
    religion = models.CharField(max_length=100, blank=True)
    highest_education = models.CharField(max_length=100, blank=True)
    occupation = models.CharField(max_length=200, blank=True)

    # Step 2: Contact & Address
    mobile_number = models.CharField(max_length=20)
    email = models.EmailField(blank=True)
    house_number = models.CharField(max_length=100, blank=True)
    street = models.CharField(max_length=100, blank=True)
    purok = models.CharField(max_length=100, blank=True)
    barangay = models.CharField(max_length=100, default="Sample Barangay")
    municipality = models.CharField(max_length=100, blank=True)
    city = models.CharField(max_length=100, default="Cagayan de Oro")
    zip_code = models.CharField(max_length=10, default="9000")
    years_of_residency = models.IntegerField(default=0)
    philSys_number = models.CharField(max_length=12, blank=True, null=True)

    # Step 3: Household & Classification
    is_joining_household = models.BooleanField(default=False)
    household_number = models.CharField(max_length=50, blank=True, null=True)
```

IoT-Enabled Barangay Resident and Attendance Management System

```
is_pwd = models.BooleanField(default=False)
is_senior_citizen = models.BooleanField(default=False)
is_4ps_member = models.BooleanField(default=False)
is_sole_parent = models.BooleanField(default=False)
is_registered_voter = models.BooleanField(default=False)

# Guardian (Dynamic for Minors)
guardian_name = models.CharField(max_length=200, blank=True, null=True)
guardian_relationship = models.CharField(max_length=100, blank=True, null=True)
guardian_contact = models.CharField(max_length=20, blank=True, null=True)
guardian_id_number = models.CharField(max_length=50, blank=True, null=True)

# Step 4: Account & Documents
username = models.CharField(max_length=150, unique=True)
password = models.CharField(max_length=128)

# File Uploads
photo = models.ImageField(upload_to="registrations/photos/", blank=True, null=True)
id_card = models.ImageField(upload_to="registrations/ids/", blank=True, null=True)
birth_certificate = models.ImageField(upload_to="registrations/birth_certs/",
blank=True, null=True)
proof_of_residency = models.ImageField(upload_to="registrations/residency/",
blank=True, null=True)

# Privacy
data_privacy_consent = models.BooleanField(default=False)

created_at = models.DateTimeField(auto_now_add=True)
updated_at = models.DateTimeField(auto_now=True)

def __str__(self):
    return f"{self.reference_number} - {self.first_name} {self.last_name}"

class Meta:
    verbose_name = "Resident Registration"
    verbose_name_plural = "Resident Registrations"
    ordering = ["-created_at"]

from django.db.models.signals import pre_delete
from django.dispatch import receiver

@receiver(pre_delete, sender=Resident)
def cleanup_user_on_resident_delete(sender, instance, **kwargs):
    """Ensure User account is deleted when Resident record is deleted."""
    try:
        # Using pre_delete ensures we can still access the related user_profile
        if hasattr(instance, 'user_profile') and instance.user_profile and
instance.user_profile.user:
            instance.user_profile.user.delete()
    except Exception:
        pass

@receiver(pre_delete, sender=Resident)
def delete_fingerprint_from_sensor(sender, instance, **kwargs):
    """Attempt to delete fingerprint from ESP32 sensor when resident is deleted."""
    if instance.fingerprint_id is not None:
        from core.utils.biometric_discovery import get_esp32_base_url
        import requests
        esp32_base_url = get_esp32_base_url()
        try:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        requests.post(f"{esp32_base_url}/delete-  
fingerprint?id={instance.fingerprint_id}", timeout=2, proxies={'http': None, 'https':  
None})  
    except Exception:  
        # Silent fail if ESP32 is offline during deletion  
        pass
```

File: residents/tests.py

```
from django.test import TestCase
from django.urls import reverse
from django.contrib.auth import get_user_model
from core.models import Role, UserProfile
from residents.models import Household, Purok

User = get_user_model()

class HouseholdMapTestCase(TestCase):
    def setUp(self):
        # Create roles
        self.admin_role, _ = Role.objects.get_or_create(name='admin',
defaults={'display_name': 'Admin', 'permission_level': 1})
        self.resident_role, _ = Role.objects.get_or_create(name='resident',
defaults={'display_name': 'Resident', 'permission_level': 3})

        # Create users
        self.admin_user = User.objects.create_user(username='admin_test_user',
password='***REDACTED***', role=self.admin_role)
        self.resident_user = User.objects.create_user(username='resident_test_user',
password='***REDACTED***', role=self.resident_role)

        # Create profiles
        UserProfile.objects.create(user=self.admin_user)
        UserProfile.objects.create(user=self.resident_user)

        # Create Purok
        self.purok = Purok.objects.create(name='Purok Test 1', latitude=9.4898,
longitude=125.7222)

        # Create Household
        self.household = Household.objects.create(
            household_no='HH-TEST-01',
            address='123 Test St',
            purok=self.purok,
            latitude=9.4895,
            longitude=125.7225
        )

    def test_household_map_api(self):
        self.client.login(username='admin_test_user', password='***REDACTED***')
        response = self.client.get(reverse('residents:household_map_api'))
        self.assertEqual(response.status_code, 200)
        data = response.json()
        self.assertEqual(data['type'], 'FeatureCollection')
        self.assertTrue(len(data['features']) > 0)
        feature = data['features'][0]
        self.assertEqual(feature['geometry']['coordinates'], [125.7225, 9.4895])
        self.assertEqual(feature['properties']['head'], 'Not Assigned')
        self.assertNotIn('address', feature['properties'])

    def test_household_add_saves_coordinates(self):
        self.client.login(username='admin_test_user', password='***REDACTED***')
        response = self.client.post(reverse('residents:household_add'), {
            'household_no': 'HH-NEW-02',
            'address': '456 New St',
            'purok': self.purok.id,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        'latitude': '9.4900',
        'longitude': '125.7220'
    })
    self.assertRedirects(response, reverse('residents:household_list'))
    new_hh = Household.objects.get(household_no='HH-NEW-02')
    self.assertEqual(new_hh.latitude, 9.4900)
    self.assertEqual(new_hh.longitude, 125.7220)

def test_household_edit_saves_coordinates(self):
    self.client.login(username='admin_test_user', password='***REDACTED***')
    response = self.client.post(reverse('residents:household_edit',
args=[self.household.pk]), {
        'household_no': 'HH-TEST-01-MOD',
        'address': '123 Test St Mod',
        'purok': self.purok.id,
        'latitude': '9.4890',
        'longitude': '125.7230'
    })
    self.assertRedirects(response, reverse('residents:household_list'))
    self.household.refresh_from_db()
    self.assertEqual(self.household.household_no, 'HH-TEST-01-MOD')
    self.assertEqual(self.household.latitude, 9.4890)
    self.assertEqual(self.household.longitude, 125.7230)
```

File: residents/urls.py

```
from django.urls import path
from . import views

app_name = 'residents'

urlpatterns = [
    path('', views.resident_list, name='list'),
    path('add/', views.resident_add, name='add'),
    path('<int:pk>/', views.resident_view, name='view'),
    path('<int:pk>/edit/', views.resident_edit, name='edit'),
    path('<int:pk>/delete/', views.resident_delete, name='delete'),
    path('<int:pk>/change-password/', views.resident_change_password,
name='change_password'),
    path('<int:pk>/reset-password/', views.resident_reset_password,
name='reset_password'),
    path('households/', views.household_list, name='household_list'),
    path('api/puroks/', views.purok_map_api, name='purok_map_api'),
    path('api/households/', views.household_map_api, name='household_map_api'),
    path('households/add/', views.household_add, name='household_add'),
    path('households/<int:pk>/', views.household_view, name='household_view'),
    path('households/<int:pk>/edit/', views.household_edit, name='household_edit'),
    path('households/<int:pk>/add-member/', views.household_add_member,
name='household_add_member'),
    path('households/<int:pk>/update-tree/', views.household_update_tree,
name='household_update_tree'),
    path('<int:pk>/capture/', views.resident_capture_fingerprint,
name='capture_fingerprint'),
    path('<int:pk>/fingerprint/', views.resident_update_fingerprint,
name='update_fingerprint'),
    # Registration Verification
    path('registrations/', views.registration_list, name='registration_list'),
    path('registrations/<int:pk>/', views.registration_detail,
name='registration_detail'),
    path('registrations/<int:pk>/approve/', views.approve_registration,
name='approve_registration'),
    path('registrations/<int:pk>/reject/', views.reject_registration,
name='reject_registration'),
    path('registrations/<int:pk>/delete/', views.registration_delete,
name='registration_delete'),
    path('registrations/bulk-delete/', views.registration_bulk_delete,
name='registration_bulk_delete'),
    path('purok/add/', views.purok_add_api, name='purok_add_api'),
    path('fingerprint-lookup/', views.get_resident_by_fingerprint,
name='fingerprint_lookup'),
]
```

File: residents/views.py

```
from django.shortcuts import render, redirect, get_object_or_404
from django.contrib.auth.decorators import login_required
from django.contrib import messages
from django.http import JsonResponse
from django.views.decorators.csrf import csrf_exempt
from django.core.paginator import Paginator
from django.db import models, transaction
from django.db.models import Q
from django.utils import timezone
from .models import Resident, Household, ResidentRegistration, Purok
from django.contrib.auth import get_user_model
User = get_user_model()
from core.models import UserProfile, Role
from officials.models import Official
import json

def is_resident_role(request):
    """Return True if the logged-in user has the 'resident' role."""
    try:
        return request.user.profile.role == 'resident'
    except Exception:
        return False

@login_required
def resident_list(request):
    """List all residents with search and filters."""
    query = request.GET.get('q', '')
    purok = request.GET.get('purok', '')
    gender = request.GET.get('gender', '')
    status = request.GET.get('status', '')

    residents = Resident.objects.filter(is_official=False)

    if query:
        residents = residents.filter(
            Q(first_name__icontains=query) |
            Q(last_name__icontains=query) |
            Q(middle_name__icontains=query)
        )
    if purok:
        residents = residents.filter(purok=purok)
    if gender:
        residents = residents.filter(gender=gender)
    if status == 'active':
        residents = residents.filter(is_active=True)
    elif status == 'inactive':
        residents = residents.filter(is_active=False)

    paginator = Paginator(residents, 25)
    page = request.GET.get('page')
    residents = paginator.get_page(page)

    context = {
        'residents': residents,
        'puroks': Purok.objects.all(),
        'query': query,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'selected_purok': purok,
'selected_gender': gender,
'selected_status': status,
}
return render(request, 'residents/list.html', context)

def generate_unique_username(first_name, last_name):
    """Generate a unique username based on first and last name."""
    base_username = f"{first_name.lower().replace(' ', '')}.{last_name.lower().replace(' ', '')}"
    username = base_username
    counter = 1
    while User.objects.filter(username=username).exists():
        username = f"{base_username}{counter}"
        counter += 1
    return username

def generate_temp_password(last_name, birthdate):
    """Generate a temporary password."""
    # Brgy + Lastname (first letter capitalized) + birth year
    clean_last_name = last_name.replace(' ', '').capitalize()
    birth_year = birthdate.year if hasattr(birthdate, 'year') else timezone.now().year
    return f"Brgy{clean_last_name}{birth_year}"

@login_required
def resident_add(request):
    """Add a new resident ? admin/staff only."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to perform this action.")
        return redirect('core:dashboard')
    if request.method == 'POST':
        is_official = request.POST.get('is_official') == 'on'
        occupation = request.POST.get('occupation', '')
        official_position = request.POST.get('official_position', '')

        # Duplicate Check
        first_name = request.POST.get('first_name')
        last_name = request.POST.get('last_name')
        middle_name = request.POST.get('middle_name', '')
        suffix = request.POST.get('suffix', '')
        birthdate_str = request.POST.get('birthdate')

        try:
            birthdate = timezone.datetime.strptime(birthdate_str, '%Y-%m-%d').date()
        except (ValueError, TypeError):
            birthdate = timezone.now().date()

        if Resident.objects.filter(
            first_name=first_name,
            last_name=last_name,
            middle_name=middle_name,
            suffix=suffix,
            birthdate=birthdate
        ).exists():
            messages.error(request, f"A resident named {first_name} {last_name} with the same birthdate already exists.")
            return redirect('residents:add')
```


IoT-Enabled Barangay Resident and Attendance Management System

```
try:
    with transaction.atomic():
        resident = Resident(
            first_name=first_name,
            last_name=last_name,
            middle_name=middle_name,
            suffix=suffix,
            birthdate=birthdate,
            birthplace=request.POST.get('birthplace', ''),
            gender=request.POST.get('gender'),
            civil_status=request.POST.get('civil_status'),
            nationality=request.POST.get('nationality', 'Filipino'),
            religion=request.POST.get('religion', ''),
            occupation=occupation if not is_official else '',
            is_official=is_official,
            contact_number=request.POST.get('contact_number', ''),
            email=request.POST.get('email', ''),
            address=request.POST.get('address'),
            purok_id=request.POST.get('purok') or None,
            is_registered_voter=request.POST.get('is_registered_voter') == 'on',
            is_pwd=request.POST.get('is_pwd') == 'on',
            is_4ps_member=request.POST.get('is_4ps_member') == 'on',
            is_senior_citizen=request.POST.get('is_senior_citizen') == 'on',
            remarks=request.POST.get('remarks', '')
        )
        if request.FILES.get('photo'):
            resident.photo = request.FILES['photo']

        household_id = request.POST.get('household')
        if household_id:
            resident.household_id = household_id
            resident.is_household_head = request.POST.get('is_household_head') ==
'on'

        resident.save()

        # --- AUTO ACCOUNT CREATION ---
        username = generate_unique_username(first_name, last_name)
        password = generate_temp_password(last_name, birthdate)

        # Default role is Resident
        resident_role, _ = Role.objects.get_or_create(name='resident',
defaults={'display_name': 'Resident', 'permission_level': 3})

        user = User.objects.create_user(
            username=username,
            password=password,
            first_name=first_name,
            last_name=last_name,
            email=resident.email,
            role=resident_role
        )

        # Link User to Resident via UserProfile
        UserProfile.objects.create(user=user, resident=resident)

        if is_official and official_position:
            # Lifecycle: Update User Role to Staff if official
            try:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        role_name = official_position if
Role.objects.filter(name=official_position).exists() else 'staff'
        role_obj = Role.objects.filter(name=role_name).first()
        if role_obj:
            user.role = role_obj
            user.save()
        except Exception:
            pass

    Official.objects.update_or_create(
        resident=resident,
        defaults={
            'user': user,
            'position': official_position,
            'status': 'active',
            'term_start': timezone.now().date(),
        }
    )

    messages.success(request, f'Resident {resident.full_name} added
successfully.')
    messages.info(request, f'An account was created for this resident.
Username: {username} | Temporary Password: {password}')

    return redirect('residents:view', pk=resident.pk)

except Exception as e:
    messages.error(request, f"Error adding resident: {str(e)}")
    return redirect('residents:add')

households = Household.objects.all()
puroks = Purok.objects.all()
return render(request, 'residents/form.html', {
    'resident': Resident(),
    'households': households,
    'puroks': puroks,
})

@login_required
def resident_view(request, pk):
    """View resident details."""
    # Residents may only view their own record
    if is_resident_role(request):
        try:
            own_pk = request.user.profile.resident.pk
        except Exception:
            own_pk = None
        if own_pk is None or pk != own_pk:
            messages.error(request, "You can only view your own profile.")
            return redirect('core:dashboard')
    resident = get_object_or_404(Resident, pk=pk)
    certificates = resident.certificates.all()[:10]
    return render(request, 'residents/view.html', {
        'resident': resident,
        'certificates': certificates,
    })

@login_required
```

IoT-Enabled Barangay Resident and Attendance Management System

```
def resident_edit(request, pk):
    """Edit a resident ? admin/staff only."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to perform this action.")
        return redirect('core:dashboard')
    resident = get_object_or_404(Resident, pk=pk)

    if request.method == 'POST':
        is_official = request.POST.get('is_official') == 'on'
        occupation = request.POST.get('occupation', '')
        official_position = request.POST.get('official_position', '')

        # Duplicate Check (only if name or birthdate changed)
        first_name = request.POST.get('first_name')
        last_name = request.POST.get('last_name')
        middle_name = request.POST.get('middle_name', '')
        suffix = request.POST.get('suffix', '')
        birthdate = request.POST.get('birthdate')

        if (resident.first_name != first_name or resident.last_name != last_name or
            resident.middle_name != middle_name or resident.suffix != suffix or
            str(resident.birthdate) != birthdate):
            if Resident.objects.filter(
                first_name=first_name,
                last_name=last_name,
                middle_name=middle_name,
                suffix=suffix,
                birthdate=birthdate
            ).exclude(pk=resident.pk).exists():
                messages.error(request, f"A resident named {first_name} {last_name} with
the same birthdate already exists.")
                return redirect('residents:edit', pk=resident.pk)

        resident.first_name = first_name
        resident.last_name = last_name
        resident.middle_name = middle_name
        resident.suffix = suffix
        resident.birthdate = birthdate
        resident.birthplace = request.POST.get('birthplace', '')
        resident.gender = request.POST.get('gender')
        resident.civil_status = request.POST.get('civil_status')
        resident.nationality = request.POST.get('nationality', 'Filipino')
        resident.religion = request.POST.get('religion', '')
        resident.occupation = occupation if not is_official else ''
        resident.is_official = is_official
        resident.contact_number = request.POST.get('contact_number', '')
        resident.email = request.POST.get('email', '')
        resident.address = request.POST.get('address')
        resident.purok_id = request.POST.get('purok') or None
        resident.is_registered_voter = request.POST.get('is_registered_voter') == 'on'
        resident.is_pwd = request.POST.get('is_pwd') == 'on'
        resident.is_4ps_member = request.POST.get('is_4ps_member') == 'on'
        resident.is_senior_citizen = request.POST.get('is_senior_citizen') == 'on'
        resident.remarks = request.POST.get('remarks', '')

    if request.FILES.get('photo'):
        resident.photo = request.FILES['photo']

    household_id = request.POST.get('household')
    if household_id:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        resident.household_id = household_id
        resident.is_household_head = request.POST.get('is_household_head') == 'on'
    else:
        resident.household = None
        resident.is_household_head = False

    resident.save()

    if is_official and official_position:
        # Check if an official record already exists for this resident
        official = Official.objects.filter(resident=resident).first()

        # Lifecycle: Link User and position (Model save will handle role sync)
        user = None
        if hasattr(resident, 'user_profile'):
            user = resident.user_profile.user

        if official:
            official.user = user
            official.position = official_position
            official.status = 'active'
            official.save()
        else:
            Official.objects.create(
                resident=resident,
                user=user,
                position=official_position,
                status='active',
                term_start=timezone.now().date()
            )
    elif not is_official and hasattr(resident, 'official_record'):
        # If no longer an official, we mark it inactive
        official = resident.official_record
        official.status = 'inactive'
        # We keep the official.user link but the official.save() will now handle
        # resetting the User.role back to 'resident' automatically.
        official.save()

    messages.success(request, f'Resident {resident.full_name} updated successfully.')
    return redirect('residents:view', pk=resident.pk)

households = Household.objects.all()
puroks = Purok.objects.all()
return render(request, 'residents/form.html', {
    'resident': resident,
    'households': households,
    'puroks': puroks,
    'editing': True,
})
```

```
from django.http import JsonResponse
from django.views.decorators.csrf import csrf_exempt
import json
```

```
@login_required
def resident_capture_fingerprint(request, pk):
    """Begin fingerprint enrollment using ESP32 R307 device."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to perform this action.")
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        return redirect('core:dashboard')
    resident = get_object_or_404(Resident, pk=pk)

    if not resident.is_official:
        messages.error(request, "Biometric registration is only available for Barangay
    Officials and Functionaries.")
        return redirect('residents:view', pk=pk)

    messages.info(request, "Fingerprint enrollment started using the connected ESP32 R307
    fingerprint module.")
    return redirect('residents:view', pk=pk)

@login_required
@csrf_exempt
def resident_update_fingerprint(request, pk):
    """Update resident fingerprint ? admin/staff only."""
    if is_resident_role(request):
        return JsonResponse({'status': 'error', 'message': 'Permission denied'},
    status=403)
    resident = get_object_or_404(Resident, pk=pk)
    if request.method == 'POST':
        try:
            data = json.loads(request.body)
            template = data.get('template')
            if template:
                if not resident.is_official:
                    return JsonResponse({'status': 'error', 'message': 'Only officials can
    register biometrics'}, status=403)
                resident.fingerprint_template = template
                resident.save()
                return JsonResponse({'status': 'success'})
        except Exception as e:
            return JsonResponse({'status': 'error', 'message': str(e)}, status=400)
    return JsonResponse({'status': 'error', 'message': 'Invalid request'}, status=405)

@login_required
def resident_delete(request, pk):
    """Delete a resident ? admin/staff only."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to perform this action.")
        return redirect('core:dashboard')
    resident = get_object_or_404(Resident, pk=pk)
    if request.method == 'POST':
        full_name = resident.full_name
        try:
            resident.delete()
            messages.success(request, f'Resident {full_name} and associated accounts have
    been permanently deleted.')
        except Exception as e:
            messages.error(request, f'Could not delete resident: {e}')
        return redirect('residents:list')
    return redirect('residents:list')

@login_required
def resident_change_password(request, pk):
    """Change a resident's system password (Account Owner Only)."""
    resident = get_object_or_404(Resident, pk=pk)

    # Check if the logged-in user is the resident who owns the account
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    if not (hasattr(resident, 'user_profile') and resident.user_profile.user ==
request.user):
        messages.error(request, "Permission denied. Only the account owner can change
their password.")
        return redirect('residents:view', pk=pk)

    if request.method == 'POST':
        new_password = request.POST.get('new_password')
        if new_password:
            user = resident.user_profile.user
            user.set_password(new_password)
            user.save()
            messages.success(request, "Your password has been successfully updated.")
        else:
            messages.error(request, "New password is required.")

    return redirect('residents:view', pk=pk)

@login_required
def resident_reset_password(request, pk):
    """Reset a resident's password to a generated temporary one (Admin/Official only)."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to perform this action.")
        return redirect('core:dashboard')

    resident = get_object_or_404(Resident, pk=pk)

    if request.method == 'POST':
        if hasattr(resident, 'user_profile') and resident.user_profile.user:
            user = resident.user_profile.user
            temp_password = generate_temp_password(resident.last_name, resident.birthdate)
            user.set_password(temp_password)
            user.save()
            messages.success(request, f"Password for {resident.full_name} has been
reset.")
            messages.info(request, f"New Temporary Password: {temp_password}")
        else:
            messages.error(request, "Failed to reset password. Ensure the resident has an
active system account.")

    return redirect('residents:view', pk=pk)

@login_required
def household_list(request):
    """List all households ? admin/staff only."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to access this area.")
        return redirect('core:dashboard')
    households = Household.objects.all()
    paginator = Paginator(households, 25)
    page = request.GET.get('page')
    households = paginator.get_page(page)
    return render(request, 'residents/household_list.html', {'households': households})

@login_required
def household_add(request):
    """Add a new household ? admin/staff only."""
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if is_resident_role(request):
    messages.error(request, "You do not have permission to perform this action.")
    return redirect('core:dashboard')
if request.method == 'POST':
    lat = request.POST.get('latitude')
    lng = request.POST.get('longitude')
    household = Household(
        household_no=request.POST.get('household_no'),
        address=request.POST.get('address'),
        purok_id=request.POST.get('purok') or None,
        latitude=float(lat) if lat else None,
        longitude=float(lng) if lng else None,
    )
    household.save()
    messages.success(request, f'Household #{household.household_no} added.')
    return redirect('residents:household_list')
puroks = Purok.objects.all()
return render(request, 'residents/household_form.html', {'puroks': puroks})

@login_required
def household_edit(request, pk):
    """Edit a household ? admin/staff only."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to perform this action.")
        return redirect('core:dashboard')
    household = get_object_or_404(Household, pk=pk)
    if request.method == 'POST':
        household.household_no = request.POST.get('household_no')
        household.address = request.POST.get('address')
        household.purok_id = request.POST.get('purok') or None

        lat = request.POST.get('latitude')
        lng = request.POST.get('longitude')
        household.latitude = float(lat) if lat else None
        household.longitude = float(lng) if lng else None

        household.save()
        messages.success(request, f'Household #{household.household_no} updated.')
        return redirect('residents:household_list')
    puroks = Purok.objects.all()
    return render(request, 'residents/household_form.html', {
        'household': household,
        'puroks': puroks,
        'editing': True,
    })

@login_required
def household_view(request, pk):
    """View household details ? admin/staff only."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to access this area.")
        return redirect('core:dashboard')
    household = get_object_or_404(Household, pk=pk)
    members = household.members.all().order_by('-is_household_head', 'birthdate')
    available_residents = Resident.objects.filter(household__isnull=True,
is_active=True).order_by('last_name', 'first_name')
    return render(request, 'residents/household_view.html', {
        'household': household,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'members': members,
'available_residents': available_residents,
})

@login_required
def purok_map_api(request):
    """Return all Puroks with coordinates as GeoJSON."""
    puroks = Purok.objects.filter(latitude__isnull=False, longitude__isnull=False)
    features = []

    for p in puroks:
        features.append({
            "type": "Feature",
            "geometry": {
                "type": "Point",
                "coordinates": [p.longitude, p.latitude]
            },
            "properties": {
                "id": p.pk,
                "name": p.name,
                "households": p.households.count(),
                "residents": p.residents.count(),
            }
        })

    return JsonResponse({
        "type": "FeatureCollection",
        "features": features
    })

@login_required
def household_map_api(request):
    """Return all Households with coordinates as GeoJSON for mapping."""
    households = Household.objects.filter(
        latitude__isnull=False, longitude__isnull=False
    ).select_related('purok').prefetch_related('members')
    features = []

    for h in households:
        features.append({
            "type": "Feature",
            "geometry": {
                "type": "Point",
                "coordinates": [h.longitude, h.latitude]
            },
            "properties": {
                "id": h.pk,
                "household_no": h.household_no,
                "head": h.head.full_name if h.head else "Not Assigned",
                "purok": h.purok.name if h.purok else "N/A",
                "residents": h.member_count,
            }
        })

    return JsonResponse({
        "type": "FeatureCollection",
        "features": features
    })
```


IoT-Enabled Barangay Resident and Attendance Management System

```
@login_required
def registration_list(request):
    """List pending registrations for officials."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    registrations = ResidentRegistration.objects.all().order_by('-created_at')
    status_filter = request.GET.get('status', 'pending')
    if status_filter:
        registrations = registrations.filter(status=status_filter)

    paginator = Paginator(registrations, 25)
    page = request.GET.get('page')
    registrations = paginator.get_page(page)

    return render(request, 'residents/registration_list.html', {
        'registrations': registrations,
        'status_filter': status_filter
    })

@login_required
def registration_detail(request, pk):
    """Review a specific registration."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    registration = get_object_or_404(ResidentRegistration, pk=pk)
    return render(request, 'residents/registration_detail.html', {
        'registration': registration
    })

@login_required
def approve_registration(request, pk):
    """Approve a registration and create Resident/User/Profile."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        messages.error(request, "Permission denied.")
        return redirect("core:dashboard")

    if request.method != 'POST':
        messages.error(request, "Invalid request method. Use the approval button.")
        return redirect("residents:registration_detail", pk=pk)

    registration = get_object_or_404(ResidentRegistration, pk=pk)

    if registration.status != "pending":
        messages.error(request, "This registration has already been processed.")
        return redirect("residents:registration_list")

    try:
        with transaction.atomic():
            # 1. Validation Checks (redundant but safe)
            if User.objects.filter(username=registration.username).exists():
                raise Exception(f"Username '{registration.username}' is already taken.")

            if (
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        registration.philSys_number
        and
UserProfile.objects.filter(philSys_id=registration.philSys_number).exists()
    ):
        raise Exception(f"The PhilSys ID '{registration.philSys_number}' is
already registered.")

    if Resident.objects.filter(
        first_name=registration.first_name,
        last_name=registration.last_name,
        middle_name=registration.middle_name,
        suffix=registration.suffix,
        birthdate=registration.birthdate
    ).exists():
        raise Exception(f"A resident with this name and birthdate already
exists.")

    # 2. Create User (ensure role is set)
    resident_role, _ = Role.objects.get_or_create(name='resident',
defaults={'display_name': 'Resident', 'permission_level': 3})

    user = User(
        username=registration.username,
        first_name=registration.first_name,
        last_name=registration.last_name,
        email=registration.email,
        role=resident_role
    )
    # The password in ResidentRegistration is already hashed during signup
    user.password = registration.password
    user.save()

    # 3. Create Resident record
    resident = Resident.objects.create(
        first_name=registration.first_name,
        middle_name=registration.middle_name,
        last_name=registration.last_name,
        suffix=registration.suffix,
        birthdate=registration.birthdate,
        birthplace=registration.birthplace,
        gender=registration.gender,
        civil_status=registration.civil_status,
        nationality=registration.nationality,
        religion=registration.religion,
        occupation=registration.occupation,
        contact_number=registration.mobile_number,
        email=registration.email,
        address=", ".join(filter(None, [
            f"{registration.house_number} {registration.street}".strip(),
            registration.purok,
            registration.barangay,
            registration.municipality,
            registration.city
        ])),
        is_registered_voter=registration.is_registered_voter,
        photo=registration.photo,
    )

    # Handle Purok lookup
    if registration.purok:
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        purok_obj, _ = Purok.objects.get_or_create(name=registration.purok)
        resident.purok = purok_obj
        resident.save()

    # Link to household if applicable
    if registration.household_number:
        household = Household.objects.filter(
            household_no=registration.household_number
        ).first()
        if household:
            resident.household = household
            resident.save()

    # 4. Create UserProfile
    UserProfile.objects.create(
        user=user,
        resident=resident,
        role="resident",
    )

    # 5. Update Registration Status
    registration.status = "approved"
    registration.save()

    messages.success(
        request,
        f"Registration approved! User '{registration.username}' and Resident record
created.",
    )
    return redirect("residents:registration_list")

except Exception as e:
    messages.error(request, f"Approval failed: {str(e)}")
    return redirect("residents:registration_detail", pk=pk)

@login_required
def reject_registration(request, pk):
    """Reject a registration."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    registration = get_object_or_404(ResidentRegistration, pk=pk)
    if request.method == 'POST':
        registration.status = 'rejected'
        registration.save()
        messages.warning(request, f"Registration {registration.reference_number} has been
rejected.")
        return redirect('residents:registration_list')

    return redirect('residents:registration_detail', pk=pk)

@login_required
def registration_delete(request, pk):
    """Delete a specific registration."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')
```

IoT-Enabled Barangay Resident and Attendance Management System

```
registration = get_object_or_404(ResidentRegistration, pk=pk)
if request.method == 'POST':
    ref = registration.reference_number
    registration.delete()
    messages.success(request, f"Registration {ref} has been deleted.")
    return redirect('residents:registration_list')

@login_required
def registration_bulk_delete(request):
    """Delete registrations based on status filter."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
    request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        messages.error(request, "Permission denied.")
        return redirect('core:dashboard')

    if request.method == 'POST':
        status = request.POST.get('status')
        if status in ['pending', 'approved', 'rejected']:
            registrations = ResidentRegistration.objects.filter(status=status)
            count = registrations.count()
            registrations.delete()
            messages.success(request, f"Deleted {count} {status} registrations.")
        elif status == 'all':
            count = ResidentRegistration.objects.all().count()
            ResidentRegistration.objects.all().delete()
            messages.success(request, f"Deleted all {count} registrations.")
        return redirect('residents:registration_list')

@login_required
@csrf_exempt
def purok_add_api(request):
    """API to add a new Purok ? authorized officials only."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
    request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        return JsonResponse({'status': 'error', 'message': 'Permission denied'},
        status=403)

    if request.method == 'POST':
        try:
            data = json.loads(request.body)
            name = data.get('name', '').strip()
            if not name:
                return JsonResponse({'status': 'error', 'message': 'Name is required'},
                status=400)

            if Purok.objects.filter(name__iexact=name).exists():
                return JsonResponse({'status': 'error', 'message': 'Purok already
exists'}, status=400)

            purok = Purok.objects.create(name=name)
            return JsonResponse({'status': 'success', 'id': purok.id, 'name': purok.name})
        except Exception as e:
            return JsonResponse({'status': 'error', 'message': str(e)}, status=400)

    return JsonResponse({'status': 'error', 'message': 'Invalid request'}, status=405)

def get_resident_by_fingerprint(request):
    """Lookup resident name by fingerprint slot ID."""
    fingerprint_id = request.GET.get('id')
    if fingerprint_id is None or fingerprint_id == '':
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    return JsonResponse({'status': 'error', 'message': 'Missing ID'}, status=400)

    resident = Resident.objects.filter(fingerprint_id=fingerprint_id).first()
    if resident:
        return JsonResponse({
            'status': 'success',
            'id': fingerprint_id,
            'name': resident.full_name
        })
    return JsonResponse({'status': 'error', 'message': 'Not found'}, status=404)

@login_required
def household_add_member(request, pk):
    """Add an existing resident to a household with a relationship tag."""
    if is_resident_role(request):
        messages.error(request, "You do not have permission to perform this action.")
        return redirect('core:dashboard')

    household = get_object_or_404(Household, pk=pk)

    if request.method == 'POST':
        resident_id = request.POST.get('resident_id')
        relationship = request.POST.get('relationship')

        if resident_id and relationship:
            resident = get_object_or_404(Resident, pk=resident_id)
            resident.household = household
            resident.household_relationship = relationship

            if relationship == 'head':
                # clear previous head if necessary? Optional for now.
                resident.is_household_head = True
            else:
                resident.is_household_head = False

            resident.save()
            messages.success(request, f'{resident.full_name} has been added to the
household as {relationship.capitalize()}')
        else:
            messages.error(request, 'Please select a resident and a relationship.')

    return redirect('residents:household_view', pk=pk)

@login_required
@csrf_exempt
def household_update_tree(request, pk):
    """Update the interactive family tree structure."""
    if not (request.user.is_superuser or (hasattr(request.user, 'profile') and
request.user.profile.role in ['captain', 'secretary', 'treasurer', 'admin'])):
        return JsonResponse({'status': 'error', 'message': 'Permission denied'},
status=403)

    if request.method == 'POST':
        try:
            data = json.loads(request.body)
            connections = data.get('connections', [])

            with transaction.atomic():
                household = get_object_or_404(Household, pk=pk)
                # Clear existing parent connections for all members to reset
```

IoT-Enabled Barangay Resident and Attendance Management System

```
household.members.update(parent_member=None)

# Apply new connections
for conn in connections:
    child_id = conn.get('child_id')
    parent_id = conn.get('parent_id')
    if child_id and parent_id:
        child = Resident.objects.filter(pk=child_id,
household=household).first()
        parent = Resident.objects.filter(pk=parent_id,
household=household).first()
        if child and parent:
            child.parent_member = parent
            child.save()
    return JsonResponse({'status': 'success'})
except Exception as e:
    return JsonResponse({'status': 'error', 'message': str(e)}, status=400)
return JsonResponse({'status': 'error', 'message': 'Invalid request'}, status=405)
```

IoT-Enabled Barangay Resident and Attendance Management System

File: run_app.sh

```
#!/bin/bash
# Biometric Server Wrapper
# This script starts the Barangay Management System.

# Get the absolute path of the project directory
PROJECT_DIR="$( cd "$( dirname "${BASH_SOURCE[0]}" )" && pwd )"
CORE_DIR="$PROJECT_DIR/core"

# --- Clear stale ESP32 IP cache files ---
# These files store the last known ESP32 IP address. On a fresh boot the ESP32
# will announce its new IP via the heartbeat endpoint, so any old cached value
# from a previous session could point to a wrong address and delay discovery.
echo " Clearing stale ESP32 IP cache..."
rm -f "$PROJECT_DIR/.esp32_ip"
rm -f "$PROJECT_DIR/.esp32_heartbeat_ip"
echo " ESP32 IP cache cleared."

echo "======"
echo " Starting Barangay Management System "
echo "======"
echo "Project Path: $PROJECT_DIR"
echo "--- Network Information ---"
echo "Internet IP (end0)  : " $(ip addr show end0 2>/dev/null | grep 'inet ' | awk '{print $2}')
echo "Hotspot IP (wlan0)   : 10.42.0.1"
echo "Tailscale Network    : " $(tailscale ip -4 2>/dev/null)
echo "======"

# Run the Django server (0.0.0.0 = listen on all interfaces, including LAN)
./venv/bin/python manage.py runserver 0.0.0.0:8001
```

IoT-Enabled Barangay Resident and Attendance Management System

File: seed_puroks.py

```
import os
import django
import random

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'barangay_project.settings')
django.setup()

from residents.models import Purok

center_lat = 9.4898
center_lng = 125.7222

# Small variations to scatter the points slightly around the center
# Approx 0.001 is roughly 100 meters
puroks = Purok.objects.all()
for i, p in enumerate(puroks):
    if p.latitude is None:
        p.latitude = center_lat + random.uniform(-0.003, 0.003)
        p.longitude = center_lng + random.uniform(-0.003, 0.003)
        p.save()
        print(f"Seeded {p.name}: {p.latitude}, {p.longitude}")
print("Seeding complete.")
```


IoT-Enabled Barangay Resident and Attendance Management System

File: settings.py

```
"""
Django settings for barangay_system project.

Generated by 'django-admin startproject' using Django 6.0.2.

For more information on this file, see
https://docs.djangoproject.com/en/6.0/topics/settings/

For the full list of settings and their values, see
https://docs.djangoproject.com/en/6.0/ref/settings/
"""

import os
from pathlib import Path

# Build paths inside the project like this: BASE_DIR / 'subdir'.
BASE_DIR = Path(__file__).resolve().parent.parent

# Quick-start development settings - unsuitable for production
# See https://docs.djangoproject.com/en/6.0/howto/deployment/checklist/

# SECURITY WARNING: keep the secret key used in production secret!
SECRET_KEY = "***REDACTED***"

# SECURITY WARNING: don't run with debug turned on in production!
DEBUG = True

ALLOWED_HOSTS = [
    'localhost', '127.0.0.1',
    '10.42.0.1',          # Orange Pi hotspot IP (for ESP32 requests)
    '10.42.0.*',          # All hotspot clients
    'brgysicosico.local', 'barangay.local', 'mhark.local', '.local',
    '*.ts.net',           # Tailscale (already there ? for resident access)
    '*.tailscale.net',
]

# ESP32 is always on the Orange Pi's private hotspot subnet (10.42.0.x)
# Django discovers it via heartbeat POST; this is just the fallback default.
ESP32_BASE_URL = os.environ.get("ESP32_BASE_URL", "http://10.42.0.10")
FINGERPRINT_SENSOR_MAX_SLOTS = int(os.environ.get("FINGERPRINT_SENSOR_MAX_SLOTS", "1000"))

# Application definition

INSTALLED_APPS = [
    "django.contrib.admin",
    "django.contrib.auth",
    "django.contrib.contenttypes",
    "django.contrib.sessions",
    "django.contrib.messages",
    "django.contrib.staticfiles",
    "residents",
    "census",
    "officials",
    "attendance",
    "certifications",
    "reports",
]
```

IoT-Enabled Barangay Resident and Attendance Management System

```
"philsys",
"documents",
]

MIDDLEWARE = [
    "django.middleware.security.SecurityMiddleware",
    "django.contrib.sessions.middleware.SessionMiddleware",
    "django.middleware.common.CommonMiddleware",
    "django.middleware.csrf.CsrfViewMiddleware",
    "django.contrib.auth.middleware.AuthenticationMiddleware",
    "django.contrib.messages.middleware.MessageMiddleware",
    "django.middleware.clickjacking.XFrameOptionsMiddleware",
]

ROOT_URLCONF = "barangay_system.urls"

TEMPLATES = [
    {
        "BACKEND": "django.template.backends.django.DjangoTemplates",
        "DIRS": [BASE_DIR / "templates"],
        "APP_DIRS": True,
        "OPTIONS": {
            "context_processors": [
                "django.template.context_processors.request",
                "django.contrib.auth.context_processors.auth",
                "django.contrib.messages.context_processors.messages",
            ],
        },
    ],
]

WSGI_APPLICATION = "barangay_system.wsgi.application"

# Database
# https://docs.djangoproject.com/en/6.0/ref/settings/#databases

# Using Orange Pi Zero 3 (Static IP/Hostname: brgyticosico.local) as the Database Server
DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.postgresql",
        "NAME": "brgy_db",
        "USER": "brgy_user",
        "PASSWORD": "****REDACTED****",
        "HOST": "127.0.0.1",
        "PORT": "5432",
    }
}

# Password validation
# https://docs.djangoproject.com/en/6.0/ref/settings/#auth-password-validators

AUTH_PASSWORD_VALIDATORS = [
    {
        "NAME":
"django.contrib.auth.password_validation.UserAttributeSimilarityValidator",
    },
    {
        "NAME": "django.contrib.auth.password_validation.MinimumLengthValidator",
    },
]
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{
    "NAME": "django.contrib.auth.password_validation.CommonPasswordValidator",
},
{
    "NAME": "django.contrib.auth.password_validation.NumericPasswordValidator",
},
]

# Internationalization
# https://docs.djangoproject.com/en/6.0/topics/i18n/

LANGUAGE_CODE = "en-us"

TIME_ZONE = "UTC"

USE_I18N = True

USE_TZ = True

# Static files (CSS, JavaScript, Images)
# https://docs.djangoproject.com/en/6.0/howto/static-files/

STATIC_URL = "static/"

MEDIA_URL = "media/"
MEDIA_ROOT = BASE_DIR / "media"

LOGIN_URL = "login"
LOGIN_REDIRECT_URL = "dashboard"
LOGOUT_REDIRECT_URL = "login"

# Tailscale Funnel and Local Network trust
CSRF_TRUSTED_ORIGINS = [
    'https://*.ts.net',
    'https://*.tailscale.net',
    'http://10.42.0.1',          # Orange Pi hotspot IP (ESP32 / local admin)
    'http://10.42.0.1:8001',
    'http://barangay.local',
    'http://brgysicosico.local',
    'http://127.0.0.1',
]

# Support for HTTPS through Tailscale/Cloudflare tunnels
SECURE_PROXY_SSL_HEADER = ('HTTP_X_FORWARDED_PROTO', 'https')
```

IoT-Enabled Barangay Resident and Attendance Management System

File: setup_autostart.sh

```
#!/bin/bash
# =====
# Barangay Management System ? Auto-start Setup
# =====
# Run this script ONCE on the Orange Pi to make Django start automatically
# every time the Orange Pi is powered on.
# Usage: sudo ./setup_autostart.sh
# =====

# Make sure we are running as root
if [ "$EUID" -ne 0 ]; then
    echo "[ERROR] Please run this script with sudo:"
    echo "      sudo ./setup_autostart.sh"
    exit 1
fi

# Detect the actual user (not root, since we use sudo)
ACTUAL_USER="${SUDO_USER:-admin}"
PROJECT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"

echo "=====
echo "  Barangay System ? Auto-start Setup          "
echo "=====
echo "  User           : $ACTUAL_USER"
echo "  Project Path    : $PROJECT_DIR"
echo "=====

# Make run_app.sh executable
chmod +x "$PROJECT_DIR/run_app.sh"

# Create the systemd service file
SERVICE_FILE="/etc/systemd/system/brgy-system.service"

echo "[1/4] Creating systemd service at $SERVICE_FILE..."
cat > "$SERVICE_FILE" <<EOF
[Unit]
Description=Barangay Management System (Django)
Documentation=https://github.com/mark-alegre01/BrgyManagementSystem

# Wait for the network AND PostgreSQL before starting Django
After=network-online.target postgresql.service
Wants=network-online.target postgresql.service

[Service]
Type=simple
User=$ACTUAL_USER

# Project directory
WorkingDirectory=$PROJECT_DIR

# Start the Django app
ExecStart=$PROJECT_DIR/run_app.sh

# If Django crashes, wait 5 seconds then restart automatically
Restart=on-failure
RestartSec=5
```

IoT-Enabled Barangay Resident and Attendance Management System

```
# Log Django output to journald (view logs with: journalctl -u brgy-system -f)
StandardOutput=journal
StandardError=journal
```

```
[Install]
WantedBy=multi-user.target
EOF
```

```
echo "    ? Service file created."
```

```
# Reload systemd to pick up the new service file
echo "[2/4] Reloading systemd daemon..."
systemctl daemon-reload
echo "    ? Done."
```

```
# Enable the service to start automatically on every boot
echo "[3/4] Enabling service to start on boot..."
systemctl enable brgy-system.service
echo "    ? Enabled."
```

```
# Start the service right now (don't wait for next reboot)
echo "[4/4] Starting the service now..."
systemctl start brgy-system.service
sleep 2
```

```
# Show the status
echo ""
echo "=====
STATUS=$(systemctl is-active brgy-system.service)
if [ "$STATUS" = "active" ]; then
    echo " ? SUCCESS! Django is running and will auto-start on every boot."
else
    echo " ? WARNING: Service did not start correctly. Check logs below:"
    journalctl -u brgy-system.service -n 20 --no-pager
fi
echo "=====
echo ""
echo " Useful commands:"
echo "   View live logs : sudo journalctl -u brgy-system -f"
echo "   Check status   : sudo systemctl status brgy-system"
echo "   Stop server    : sudo systemctl stop brgy-system"
echo "   Restart server  : sudo systemctl restart brgy-system"
echo "   Disable autostart: sudo systemctl disable brgy-system"
echo "=====
```

IoT-Enabled Barangay Resident and Attendance Management System

File: setup_hostapd_hotspot.sh

```
#!/bin/bash
# =====
# Barangay Management System - Hostapd AP Setup Script
# =====
# This script sets up the Orange Pi as a Standalone Router using hostapd + dnsmasq.
# It will broadcast "Barangay_System_WiFi" (password: barangay123) and assign
# the Orange Pi the static IP 10.42.0.1.

if [ "$EUID" -ne 0 ]; then
    echo "[ERROR] Please run this script with sudo:"
    echo "sudo ./setup_hostapd_hotspot.sh"
    exit
fi

WIFI_IFACE="wlan0"
ETH_IFACE="eth0" # Default ethernet on Orange Pi 3B / similar boards

echo "======"
echo " Configuring Standalone Router Mode (Hostapd)"
echo "======"

# 1. Stop services before configuration
echo "[INFO] Stopping services..."
systemctl stop hostapd
systemctl stop dnsmasq

# 2. Tell NetworkManager to ignore wlan0 so it doesn't fight hostapd
echo "[INFO] Configuring NetworkManager to ignore $WIFI_IFACE..."
cat <<EOF > /etc/NetworkManager/conf.d/10-ignore-wlan0.conf
[keyfile]
unmanaged-devices=interface-name:$WIFI_IFACE
EOF
systemctl restart NetworkManager

# 3. Unmask and enable hostapd
systemctl unmask hostapd 2>/dev/null
systemctl enable hostapd

# 4. Configure a static IP for wlan0
echo "[INFO] Setting static IP 10.42.0.1 for $WIFI_IFACE..."
ip link set dev $WIFI_IFACE up
ip addr flush dev $WIFI_IFACE
ip addr add 10.42.0.1/24 dev $WIFI_IFACE

# Make it persistent via systemd-networkd or interfaces file
cat <<EOF > /etc/network/interfaces.d/wlan0-ap
auto $WIFI_IFACE
iface $WIFI_IFACE inet static
    address 10.42.0.1
    netmask 255.255.255.0
EOF

# 5. Configure DHCP server (dnsmasq)
echo "[INFO] Configuring DHCP (dnsmasq)..."
mv /etc/dnsmasq.conf /etc/dnsmasq.conf.orig 2>/dev/null
cat <<EOF > /etc/dnsmasq.conf
interface=$WIFI_IFACE
```

IoT-Enabled Barangay Resident and Attendance Management System

```
bind-interfaces
dhcp-range=10.42.0.10,10.42.0.100,255.255.255.0,24h
domain=local
address=/brgysicosico.local/10.42.0.1
EOF

# 6. Configure Access Point (hostapd)
echo "[INFO] Configuring WiFi Hotspot (hostapd)..."
cat <<EOF > /etc/hostapd/hostapd.conf
interface=$WIFI_IFACE
driver=nl80211
ssid=Barangay_System_WiFi
hw_mode=g
channel=6
ieee80211n=1
wmm_enabled=1
macaddr_acl=0
auth_algs=1
ignore_broadcast_ssid=0
wpa=2
wpa_passphrase=barangay123
wpa_key_mgmt=WPA-PSK
wpa_pairwise=CCMP
rsn_pairwise=CCMP
EOF

# Point hostapd to this configuration
sed -i 's|^#DAEMON_CONF=""|DAEMON_CONF="/etc/hostapd/hostapd.conf"|g' /etc/default/hostapd

# 7. Enable IP Forwarding (Internet Sharing)
echo "[INFO] Enabling Internet Sharing (NAT)..."
sed -i 's/^#net.ipv4.ip_forward=1/net.ipv4.ip_forward=1/g' /etc/sysctl.conf
sysctl -p

# 8. Configure Firewall (iptables) to share internet from ethernet to WiFi
iptables -t nat -F
iptables -t nat -A POSTROUTING -o $ETH_IFACE -j MASQUERADE
iptables -A FORWARD -i $ETH_IFACE -o $WIFI_IFACE -m state --state RELATED,ESTABLISHED -j ACCEPT
iptables -A FORWARD -i $WIFI_IFACE -o $ETH_IFACE -j ACCEPT
netfilter-persistent save

# 9. Start Services
echo "[INFO] Starting services..."
systemctl restart dnsmasq
systemctl restart hostapd

echo "====="
echo " SUCCESS! Standalone Router Mode Configured."
echo "====="
echo " Network Name (SSID): Barangay_System_WiFi"
echo " Password           : barangay123"
echo " Orange Pi IP        : 10.42.0.1"
echo "====="
echo " NOTE: If the hotspot doesn't appear, run:"
echo " sudo reboot"
```

IoT-Enabled Barangay Resident and Attendance Management System

File: setup_hotspot.sh

```
#!/bin/bash
# =====
# Barangay Management System - Hotspot Setup Script for Orange Pi
# =====
# This script configures the Orange Pi to broadcast its own WiFi network.
# This ensures a flawless, permanent connection with the ESP32 (at 10.42.0.1)
# while allowing you to plug in an Ethernet cable for global internet access.

if [ "$EUID" -ne 0 ]; then
    echo "Please run this script with sudo:"
    echo "sudo ./setup_hotspot.sh"
    exit
fi

echo "======"
echo " Setting up Barangay System WiFi Hotspot"
echo "======"

# 1. Check if nmcli is installed
if ! command -v nmcli &> /dev/null; then
    echo "[ERROR] NetworkManager (nmcli) could not be found. Please install it."
    exit 1
fi

# 2. Make sure WiFi radio is enabled
echo "[INFO] Turning on WiFi radio..."
nmcli radio wifi on
sudo rfkill unblock wifi 2>/dev/null

# 3. Get the WiFi interface name (usually wlan0)
WIFI_IFACE=$(nmcli -t -f DEVICE,TYPE device | grep -w "wifi" | cut -d: -f1 | head -n 1)

if [ -z "$WIFI_IFACE" ]; then
    echo "[ERROR] No WiFi hardware found on this device!"
    exit 1
fi

echo "[INFO] Found WiFi interface: $WIFI_IFACE"

# 3. Clean up any existing hotspot with the same name
if nmcli con show "Barangay_System_WiFi" &> /dev/null; then
    echo "[INFO] Removing old hotspot configuration..."
    nmcli con delete "Barangay_System_WiFi"
fi

# 4. Create the hotspot connection
# Using ipv4.method shared turns the Orange Pi into a router (NAT + DHCP).
# It will assign itself 10.42.0.1 by default.
echo "[INFO] Creating new hotspot connection..."
nmcli con add type wifi ifname $WIFI_IFACE con-name "Barangay_System_WiFi" autoconnect yes
    ssid "Barangay_System_WiFi"
nmcli con modify "Barangay_System_WiFi" 802-11-wireless.mode ap 802-11-wireless.band bg
ipv4.method shared
nmcli con modify "Barangay_System_WiFi" wifi-sec.key-mgmt wpa-psk
nmcli con modify "Barangay_System_WiFi" wifi-sec.psk "barangay123"

# 5. Bring up the hotspot
```


IoT-Enabled Barangay Resident and Attendance Management System

```
echo "[INFO] Starting Hotspot..."
if nmcli con up "Barangay_System_WiFi"; then
    echo ""
    echo "=====
    echo " SUCCESS! Hotspot Created."
    echo "=====
    echo " Network Name (SSID): Barangay_System_WiFi"
    echo " Password          : barangay123"
    echo " Orange Pi IP         : 10.42.0.1"
    echo "=====
    echo ""
    echo "NEXT STEPS:"
    echo "1. Plug an Ethernet cable into the Orange Pi for internet."
    echo "2. Connect your laptop to 'Barangay_System_WiFi'."
    echo "3. Restart your ESP32, connect to its 'ESP32-Setup' portal,"
    echo "   and enter 'Barangay_System_WiFi' / 'barangay123'."
else
    echo "[ERROR] Failed to start hotspot. Your WiFi adapter might not support AP mode."
fi
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/appointments/book.html

```
{% extends 'base.html' %}

{% block title %}Book Appointment - {{ BARANGAY_NAME }}{% endblock %}

{% block content %}
<div class="row justify-content-center">
  <div class="col-md-8">
    {% url 'appointment_list' as appointment_list_url %}
    {% include 'components/back_button.html' with back_url=appointment_list_url
    back_label='Back' %}
    <div class="card shadow-sm border-0">
      <div class="card-header bg-white py-3">
        <h5 class="mb-0 text-primary"><i class="fas fa-calendar-plus me-2"></i>Book an
Appointment</h5>
      </div>

      <div class="card-body p-4">
        <form method="post">
          {% csrf_token %}

          <div class="row g-3">
            <div class="col-md-6">
              <label class="form-label">Purpose</label>
              {{ form.purpose }}
              {% if form.purpose.errors %}
              <div class="text-danger small">{{ form.purpose.errors }}</div>
              {% endif %}
            </div>
            <div class="col-md-6">
              <label class="form-label">Date</label>
              {{ form.appointment_date }}
              {% if form.appointment_date.errors %}
              <div class="text-danger small">{{ form.appointment_date.errors
}}</div>
              {% endif %}
            </div>
          </div>

          <div class="row g-3 mt-2">
            <div class="col-md-6">
              <label class="form-label">Time (Optional)</label>
              {{ form.appointment_time }}
              {% if form.appointment_time.errors %}
              <div class="text-danger small">{{ form.appointment_time.errors
}}</div>
              {% endif %}
            </div>
          </div>

          <div class="mt-3">
            <label class="form-label">Additional Details</label>
            {{ form.specification }}
          </div>

          <div class="mt-4 text-right">
            <button type="submit" class="btn btn-primary px-4">Confirm
Appointment</button>
          </div>
        </form>
      </div>
    </div>
  </div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        </div>
      </form>
    </div>
  </div>
</div>

<style>
  .form-control {
    border-radius: 8px;
    padding: 10px;
  }

  form select {
    width: 100%;
    padding: 10px;
    border-radius: 8px;
    border: 1px solid #dee2e6;
  }
</style>
{% endblock %}
```

File: templates/appointments/list.html

```
{% extends 'base.html' %}

{% block title %}My Appointments - {{ BARANGAY_NAME }}{% endblock %}

{% block content %}
<div class="d-flex justify-content-between align-items-center mb-4">
    <h4 class="mb-0">My Appointments</h4>
    <a href="{% url 'book_appointment' %}" class="btn btn-primary">
        <i class="fas fa-plus me-2"></i>Book New
    </a>
</div>

<div class="card shadow-sm border-0">
    <div class="card-body p-0">
        <div class="table-responsive">
            <table class="table table-hover mb-0">
                <thead class="bg-light">
                    <tr>
                        <th class="ps-4">Date</th>
                        <th>Time</th>
                        <th>Purpose</th>
                        <th>Status</th>
                        <th>Created</th>
                    </tr>
                </thead>
                <tbody>
                    {% for app in appointments %}
                    <tr>
                        <td class="ps-4"><strong>{{ app.appointment_date }}</strong></td>
                        <td>{{ app.appointment_time|default:"TBD" }}</td>
                        <td>{{ app.get_purpose_display }}</td>
                        <td>
                            <span class="badge rounded-pill
                                {% if app.status == 'approved' %}bg-success
                                {% elif app.status == 'pending' %}bg-warning text-dark
                                {% elif app.status == 'cancelled' %}bg-danger
                                {% else %}bg-secondary{% endif %}>
                                {{ app.get_status_display }}
                            </span>
                        </td>
                        <td class="text-muted small">{{ app.created_at|date:"M d, Y"
                    </td>
                    </tr>
                    {% empty %}
                    <tr>
                        <td colspan="5" class="text-center py-5">
                            <p class="text-muted mb-0">You have no appointments yet.</p>
                        </td>
                    </tr>
                    {% endfor %}
                </tbody>
            </table>
        </div>
    </div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/appointments/manage.html

```
{% extends 'base.html' %}

{% block title %}Manage Appointments - {{ BARANGAY_NAME }}{% endblock %}

{% block content %}
<h4 class="mb-4">Appointment Requests</h4>

<div class="card shadow-sm border-0">
  <div class="card-body p-0">
    <div class="table-responsive">
      <table class="table table-hover mb-0">
        <thead class="bg-light">
          <tr>
            <th class="ps-4">Resident</th>
            <th>Date & Time</th>
            <th>Purpose</th>
            <th>Status</th>
            <th class="text-end pe-4">Actions</th>
          </tr>
        </thead>
        <tbody>
          {% for app in appointments %}
            <tr>
              <td class="ps-4">
                <div class="d-flex align-items-center">
                  <i class="fas fa-user-circle fa-2x me-2 text-muted"></i>
                  <div>
                    <div class="fw-bold">{{ app.resident.full_name }}</div>
                    <div class="small text-muted">{{
app.resident.contact_number|default:"No contact" }}</div>
                  </div>
                </td>
              <td>
                <div>{{ app.appointment_date }}</div>
                <div class="small text-muted">{{
app.appointment_time|default:"" }}</div>
              </td>
              <td>
                <span class="d-block">{{ app.get_purpose_display }}</span>
                {% if app.specification %}
                <small class="text-muted fst-italic">{{
app.specification|truncatechars:30 }}</small>
                {% endif %}
              </td>
              <td>
                <span class="badge rounded-pill
                {% if app.status == 'approved' %}bg-success
                {% elif app.status == 'pending' %}bg-warning text-dark
                {% elif app.status == 'completed' %}bg-info
                {% else %}bg-secondary{% endif %}">
                {{ app.get_status_display }}
                </span>
              </td>
              <td class="text-end pe-4">
          </tbody>
        </table>
      </div>
    </div>
  </div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% if app.status == 'pending' %}
<div class="btn-group">
  <a href="{% url 'appointment_status_update' app.pk
'approved' %}"
      class="btn btn-sm btn-outline-success">Approve</a>
  <a href="{% url 'appointment_status_update' app.pk
'cancelled' %}"
      class="btn btn-sm btn-outline-danger">Deny</a>
</div>
{% elif app.status == 'approved' %}
<a href="{% url 'appointment_status_update' app.pk 'completed'
%}"
      class="btn btn-sm btn-success">Mark Completed</a>
{% else %}
<span class="text-muted small">No actions</span>
{% endif %}
</td>
</tr>
{% empty %}
<tr>
  <td colspan="5" class="text-center py-5 text-muted">No
appointments found.</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/attendance/biometric_attendance.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Daily Time Record - Barangay Management System{% endblock %}

{% block extra_head %}
<style>
    .biometric-attendance-card {
        width: 100%;
        max-width: 600px;
        margin: 0 auto;
        padding: 20px;
        background: transparent;
        text-align: center;
        position: relative;
        transition: all 0.5s ease;
    }
    .scan-visual {
        width: 100px;
        height: 100px;
        margin: 0 auto 8px;
        background: #ffffff;
        border-radius: 50%;
        display: flex;
        align-items: center;
        justify-content: center;
        font-size: 60px;
        color: #e2e8f0;
        transition: all 0.5s cubic-bezier(0.4, 0, 0.2, 1);
        position: relative;
        border: 2px solid #f1f5f9;
    }
    .scan-visual.active {
        border-color: #3b82f6;
        background: #eff6ff;
        box-shadow: 0 0 30px rgba(59, 130, 246, 0.2);
    }
    .scan-visual.success {
        border-color: #10b981;
        background: #ecfdf5;
        box-shadow: 0 0 30px rgba(16, 185, 129, 0.2);
    }
    .scan-visual.error {
        border-color: #ef4444;
        background: #fef2f2;
    }
    .scan-visual i {
        font-size: 80px;
        color: #94a3b8;
        transition: all 0.3s;
    }
    .scan-visual.active i {
        color: #3b82f6;
        animation: fingerprint-pulse 1.5s infinite;
    }
    .scan-visual.success i {
        color: #10b981;
    }
</style>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}
.scan-visual.error i {
  color: #ef4444;
}
@keyframes fingerprint-pulse {
  0% { transform: scale(1); opacity: 0.8; }
  50% { transform: scale(1.1); opacity: 1; }
  100% { transform: scale(1); opacity: 0.8; }
}
.status-badge-lg {
  display: inline-block;
  padding: 8px 20px;
  border-radius: 50px;
  font-weight: 700;
  font-size: 0.95rem;
  margin-bottom: 20px;
}
.badge-waiting { background: #f1f5f9; color: #64748b; }
.badge-active { background: #dbeeaf; color: #1e40af; }
.badge-success { background: #d1fae5; color: #065f46; }
.badge-error { background: #fee2e2; color: #991b1b; }

.result-card {
  margin-top: 20px;
  padding: 20px;
  border-radius: 20px;
  background: #ffffff;
  border: 1px solid #e2e8f0;
  display: none;
  animation: slide-up 0.4s ease-out;
  box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
}
@keyframes slide-up {
  from { transform: translateY(15px); opacity: 0; }
  to { transform: translateY(0); opacity: 1; }
}
.user-icon {
  width: 50px;
  height: 50px;
  background: #3b82f6;
  color: white;
  border-radius: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 20px;
  margin: 0 auto 10px;
}
.btn-massive {
  padding: 12px 30px;
  font-size: 1rem;
  font-weight: 700;
  border-radius: 14px;
  transition: all 0.3s;
}
}
</style>
{% endblock %}

{% block content %}
{% url 'attendance:dashboard' as attendance_dashboard_url %}
```


IoT-Enabled Barangay Resident and Attendance Management System

```
{% include 'components/back_button.html' with back_url=attendance_dashboard_url
back_label='Back' %}

<div class="biometric-attendance-card">

    <button class="absolute top-6 right-6 p-3 bg-white hover:bg-slate-50 rounded-2xl
shadow-sm border border-slate-100 text-slate-400 hover:text-emerald-600 transition-all
active:scale-95 group"
        data-toggle="modal" data-target="#attendanceScheduleModal" title="DTR Schedule
Settings">
        <i class="fas fa-cog text-xl group-hover:rotate-90 transition-transform
duration-500"></i>
    </button>

    <div class="mb-2">
        <div id="event-mode-badge" class="hidden mb-2">
            <span class="px-3 py-1 bg-emerald-600 text-white text-[10px] font-black
rounded-full uppercase tracking-widest shadow-sm">
                Event Mode Active
            </span>
        </div>
        <h1 id="attendance-title" class="font-weight-bold" style="color: #0f172a; margin-
bottom: 2px; font-size: 1.25rem; letter-spacing: -0.5px;">Daily Time Record</h1>
        <div id="current-live-day" class="font-weight-bold text-primary uppercase
tracking-tight mb-1" style="font-size: 1.5rem; letter-spacing: -0.5px;">-----</div>

        <div id="shift-display" class="mb-1">
            <div class="inline-flex items-center gap-2 px-3 py-1 bg-white rounded-lg
border border-slate-200 shadow-sm">
                <span class="text-[9px] font-black text-slate-500 uppercase tracking-
widest">Shift Starts at</span>
                <span id="display-time-in" class="text-[11px] font-black text-
primary">--:--</span>
            </div>
        </div>

        <div id="event-display" class="hidden mb-3">
            <div class="inline-flex items-center gap-2 px-4 py-2 bg-emerald-50 border
border-emerald-100 rounded-xl text-emerald-700">
                <i class="fas fa-calendar-alt text-xs"></i>
                <span id="event-time-display" class="text-xs font-bold">--:--</span>
            </div>
        </div>

        <div id="exit-event-container" class="hidden mt-2"></div>
    </div>

    <div id="status-badge" class="status-badge-lg badge-waiting">
        <i class="fas fa-clock mr-2"></i> Ready to Clock In/Out
    </div>

    <div id="scan-visual" class="scan-visual">
        <i class="fas fa-fingerprint"></i>
    </div>

    <div id="instruction-text" class="small mb-3 text-secondary">
        Click the button below to start the scanner
    </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div id="actions">
  <button id="btn-start-attendance" class="btn btn-primary btn-massive shadow-sm">
    <i class="fas fa-power-off mr-2"></i> Start Scan
  </button>
  <button id="btn-stop-attendance" class="btn btn-outline-danger btn-massive
d-none">
    <i class="fas fa-times mr-2"></i> Stop
  </button>
</div>

<div id="attendance-result" class="result-card">
  <div class="user-icon" id="result-icon">
    <i class="fas fa-user-check"></i>
  </div>
  <h2 id="result-name" class="font-weight-bold mb-1">---</h2>
  <p id="result-position" class="text-muted mb-2">---</p>
  <div class="badge badge-success px-3 py-2" id="result-action" style="font-size:
1rem;">
    Clocked IN
  </div>
  <div class="mt-3 text-secondary">
    <i class="fas fa-calendar-check mr-1"></i> Recorded at <span id="result-time"
class="font-weight-bold">--:--</span>
  </div>
</div>

<!-- Event Attendees List (Visible only in Event Mode) -->
<div id="event-attendees-section" class="hidden mt-8 text-left animate-in">
  <div class="flex items-center justify-between mb-4">
    <h3 class="text-lg font-black text-slate-800 tracking-tight">Present
Officials</h3>
    <button id="downloadPdfBtn" class="flex items-center gap-2 px-4 py-2 bg-
slate-900 text-white rounded-xl text-xs font-bold hover:bg-slate-800 transition shadow-lg
shadow-slate-200">
      <i class="fas fa-file-pdf"></i> Download PDF
    </button>
  </div>
  <div class="table-responsive w-full bg-white rounded-2xl shadow-sm border border-
slate-100">
    <table class="w-full min-w-[500px] text-sm">
      <thead class="bg-slate-50 text-slate-400 uppercase text-[9px] font-black
tracking-widest">
        <tr>
          <th class="px-5 py-4">Name</th>
          <th class="px-5 py-4">Position</th>
          <th class="px-5 py-4">Time In</th>
        </tr>
      </thead>
      <tbody id="attendees-list-body" class="divide-y divide-slate-100">
        <!-- Attendees will be injected here -->
      </tbody>
    </table>
  </div>
</div>
</div>

<!-- Attendance Schedule Modal -->
<div class="modal fade" id="attendanceScheduleModal" tabindex="-1" role="dialog" aria-
hidden="true">
  <div class="modal-dialog modal-dialog-centered" role="document" style="max-width:

```

IoT-Enabled Barangay Resident and Attendance Management System

```
500px;">
    <div class="modal-content" style="border-radius: 30px; border: none; overflow:
hidden; box-shadow: 0 30px 60px -12px rgba(0,0,0,0.4), 0 18px 36px -18px
rgba(0,0,0,0.45);">
        <!-- Header -->
        <div class="px-5 py-3 border-b border-slate-100 flex items-center justify-
between bg-white text-left">
            <div>
                <h2 class="text-base font-bold text-slate-900 tracking-tight flex
items-center gap-2">
                    Shift Setup
                    <span class="px-2 py-0.5 bg-blue-100 text-[9px] text-blue-700
rounded-full font-black uppercase tracking-wider">Weekly</span>
                </h2>
                <p class="text-[9px] text-slate-400 mt-0.5 font-medium">Set work hours
and manage device</p>
            </div>
            <div class="p-1.5 bg-slate-100 hover:bg-slate-200 rounded-lg transition-
colors" data-dismiss="modal" style="cursor: pointer;">
                <i class="fas fa-times text-slate-500 text-xs"></i>
            </div>
        </div>

        <!-- Bulk Apply Toggle -->
        <div class="px-5 py-2 border-b border-slate-50 bg-slate-50/30">
            <label class="flex items-center gap-2 cursor-pointer group m-0">
                <input type="checkbox" id="bulkApplyToggle" class="w-3 h-3 rounded
border-slate-300 text-blue-600 focus:ring-blue-500 transition cursor-pointer">
                <span class="text-[9px] font-bold text-slate-500 group-hover:text-
slate-700 transition uppercase tracking-wider">Apply to all days upon saving</span>
            </label>
        </div>

        <!-- Tabs Navigation -->
        <div id="mainTabs" class="px-1 pt-1 border-b border-slate-100 bg-slate-50/50">
            <div class="flex items-center gap-0 overflow-x-hidden"
id="dayTabsContainer">
                <!-- Day buttons will be injected here via JS -->
            </div>
        </div>

        <!-- Content Area -->
        <div id="mainContentArea" class="p-5 bg-white text-left">
            <div id="activeDayContent" class="space-y-4 fade-in-up">
                <div class="flex items-center justify-between bg-slate-50 p-3 rounded-
xl border border-slate-100">
                    <div>
                        <h3 id="displayDayName" class="text-sm font-bold text-
slate-900">Monday</h3>
                        <p id="dayInsight" class="text-[9px] text-blue-600 font-bold
uppercase tracking-wider">Active Slot</p>
                    </div>
                    <div class="flex items-center gap-3">
                        <span class="text-[9px] font-bold text-slate-400 uppercase
tracking-widest">Day Off</span>
                        <label class="relative inline-flex items-center cursor-pointer
m-0">
                            <input type="checkbox" id="dayStatusToggle" checked
class="sr-only peer">
                            <div class="w-8 h-4 bg-slate-200 peer-focus:outline-none
```

IoT-Enabled Barangay Resident and Attendance Management System

```

rounded-full peer peer-checked:after:translate-x-full peer-checked:after:border-white
after:content-[''] after:absolute after:top-[2px] after:left-[2px] after:bg-white
after:border-gray-300 after:border after:rounded-full after:h-3 after:w-3
after:transition-all peer-checked:bg-blue-600"></div>
    </label>
  </div>
</div>

<!-- Time Grid -->
<div id="timeInputsGrid" class="grid grid-cols-2 gap-3">
  <div class="space-y-1">
    <label class="text-[9px] font-bold text-slate-500 uppercase
tracking-widest">Start Time</label>
    <input type="time" id="timeIn" class="w-full bg-slate-50
border border-slate-200 rounded-lg px-2 py-2 text-xs font-bold text-slate-800
focus:border-blue-500 focus:bg-white outline-none transition-all">
  </div>
  <div class="space-y-1">
    <label class="text-[9px] font-bold text-slate-500 uppercase
tracking-widest">End Time</label>
    <input type="time" id="timeOut" class="w-full bg-slate-50
border border-slate-200 rounded-lg px-2 py-2 text-xs font-bold text-slate-800
focus:border-blue-500 focus:bg-white outline-none transition-all">
  </div>
</div>

<!-- Closed Message Overlay -->
<div id="closedMessage" class="hidden py-8 text-center border-2
border-dashed border-slate-100 rounded-2xl bg-slate-50/50">
  <p class="text-xs text-slate-500 font-bold uppercase tracking-
wide">Currently OFF</p>
</div>

<!-- Footer Actions -->
<div class="px-5 py-4 bg-slate-50 border-t border-slate-100 space-y-3">

  <div class="flex gap-2">
    <button id="resetBtn" class="flex-1 py-2 text-[10px] font-bold text-
slate-500 hover:text-slate-700 transition-colors">Reset</button>
    <button id="saveBtn" onclick="saveScheduleSettings()"
class="flex-[2.5] py-2.5 bg-slate-900 text-white rounded-xl font-bold text-xs shadow-lg
shadow-slate-200 transition-all active:scale-95">
      Save Changes
    </button>
  </div>
</div>
</div>
</div>
</div>

<style>
.hide-scrollbar::-webkit-scrollbar { display: none; }
.hide-scrollbar { -ms-overflow-style: none; scrollbar-width: none; }
.day-tab { transition: all 0.2s cubic-bezier(0.4, 0, 0.2, 1); }
.fade-in-up { animation: fadeInUp 0.4s ease-out forwards; }
.fade-out { animation: fadeOut 0.4s ease-in forwards; }
@keyframes fadeInUp {
  from { opacity: 0; transform: translateY(10px); }

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        to { opacity: 1; transform: translateY(0); }
    }
    @keyframes fadeOut {
        from { opacity: 1; transform: translateY(0); }
        to { opacity: 0; transform: translateY(-10px); visibility: hidden; }
    }
</style>
<script src="https://cdn.tailwindcss.com"></script>
{% endblock %}

{% block extra_js %}
<script>
    const days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'];
    const fullDayNames = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday',
'Saturday', 'Sunday'];
    const dayKeys = ['monday', 'tuesday', 'wednesday', 'thursday', 'friday', 'saturday',
'sunday'];

    let activeDayIndex = 0;
    let scheduleData = [];
    let weeklySchedule = {};
    let modalOpenedOnce = false;

    function minutesToTime24(mins) {
        if (mins === undefined || mins === null) return '08:00';
        const h = Math.floor(mins / 60).toString().padStart(2, '0');
        const m = (mins % 60).toString().padStart(2, '0');
        return `${h}:${m}`;
    }

    function minutesToDisplay(mins) {
        let h = Math.floor(mins / 60);
        const m = (mins % 60).toString().padStart(2, '0');
        const ampm = h >= 12 ? 'PM' : 'AM';
        h = h % 12;
        h = h ? h : 12;
        return `${h}:${m} ${ampm}`;
    }

    function timeToMinutes(timeStr) {
        if (!timeStr) return 0;
        const [h, m] = timeStr.split(':');
        return parseInt(h) * 60 + parseInt(m);
    }

    function initScheduleData() {
        const saved = localStorage.getItem('weekly_attendance_schedule');
        const savedData = saved ? JSON.parse(saved) : {};

        // If no saved data, initialize with defaults
        if (!saved || Object.keys(savedData).length === 0) {
            dayKeys.forEach((key, i) => {
                const isWorkDay = (key !== 'saturday' && key !== 'sunday');
                savedData[key] = isWorkDay ? { in: 480, out: 720 } : null;
            });
            localStorage.setItem('weekly_attendance_schedule', JSON.stringify(savedData));
        }

        weeklySchedule = savedData;
    }
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
scheduleData = dayKeys.map((key, i) => ({
  key: key,
  name: fullDayNames[i],
  enabled: !!savedData[key],
  timeIn: savedData[key] ? minutesToTime24(savedData[key].in) : '08:00',
  timeOut: savedData[key] ? minutesToTime24(savedData[key].out) : '12:00'
}));
}

function renderTabs() {
  const tabsContainer = document.getElementById('dayTabsContainer');
  if (!tabsContainer) return;
  tabsContainer.innerHTML = '';
  days.forEach((day, index) => {
    const button = document.createElement('button');
    const isActive = index === activeDayIndex;

    button.className = `day-tab flex-1 py-3 px-1 text-[10px] font-black border-b-2
transition-all text-center
    ${isActive ? 'border-blue-600 text-blue-600 bg-blue-50/50' : 'border-
transparent text-slate-400 hover:text-slate-600 hover:bg-slate-50/30'}`;

    button.innerHTML = `${day.toUpperCase()}`;
    button.onclick = () => switchDay(index);
    tabsContainer.appendChild(button);
  });
}

function switchDay(index) {
  // Only save if the modal has been opened and inputs are populated
  if (modalOpenedOnce) {
    saveCurrentDayState();
  }

  activeDayIndex = index;
  const data = scheduleData[index];
  if (!data) return;

  document.getElementById('displayDayName').innerText = data.name;
  document.getElementById('timeIn').value = data.timeIn || '08:00';
  document.getElementById('timeOut').value = data.timeOut || '12:00';
  document.getElementById('dayStatusToggle').checked = data.enabled;

  updateToggleState(data.enabled);
  renderTabs();

  // Mark as opened so future switches save the state
  modalOpenedOnce = true;
}

function saveCurrentDayState() {
  const tIn = document.getElementById('timeIn');
  const tOut = document.getElementById('timeOut');
  const dToggle = document.getElementById('dayStatusToggle');
  if (tIn && tOut && dToggle) {
    scheduleData[activeDayIndex].timeIn = tIn.value;
    scheduleData[activeDayIndex].timeOut = tOut.value;
    scheduleData[activeDayIndex].enabled = dToggle.checked;
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
function updateToggleState(isEnabled) {
  const grid = document.getElementById('timeInputsGrid');
  const msg = document.getElementById('closedMessage');
  const insight = document.getElementById('dayInsight');

  if (isEnabled) {
    grid.classList.remove('hidden');
    msg.classList.add('hidden');
    insight.innerText = 'Active Slot';
    insight.className = 'text-[9px] text-blue-600 font-bold uppercase tracking-
wider';
  } else {
    grid.classList.add('hidden');
    msg.classList.remove('hidden');
    insight.innerText = 'Off Duty';
    insight.className = 'text-[9px] text-slate-400 font-bold uppercase tracking-
wider';
  }
}

function saveScheduleSettings() {
  saveCurrentDayState();

  const bulkApply = document.getElementById('bulkApplyToggle')?.checked;
  let finalSchedule = {};

  if (bulkApply) {
    // Apply current day settings to all
    const currentDay = scheduleData[activeDayIndex];
    scheduleData.forEach(day => {
      day.enabled = currentDay.enabled;
      day.timeIn = currentDay.timeIn;
      day.timeOut = currentDay.timeOut;
    });
  }

  scheduleData.forEach(day => {
    if (day.enabled) {
      finalSchedule[day.key] = {
        in: timeToMinutes(day.timeIn),
        out: timeToMinutes(day.timeOut)
      };
    } else {
      finalSchedule[day.key] = null;
    }
  });
  localStorage.setItem('weekly_attendance_schedule', JSON.stringify(finalSchedule));
  weeklySchedule = finalSchedule;

  if (window.showNotification) {
    window.showNotification(bulkApply ? 'Schedule applied to all days!' : 'Weekly
schedule saved successfully!', 'success');
  }

  renderCurrentShift();

  const saveBtn = document.getElementById('saveBtn');
  saveBtn.innerHTML = "Saved!";
  saveBtn.style.backgroundColor = "#16a34a";
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
setTimeout(() => {
    saveBtn.innerHTML = "Save Changes";
    saveBtn.style.backgroundColor = "#1e293b";
    $('#attendanceScheduleModal').modal('hide');
}, 1000);
}

// Biometric Logic
function getAttendanceAction() {
    const now = new Date();
    const daysArr = ['sunday', 'monday', 'tuesday', 'wednesday', 'thursday', 'friday',
'saturday'];
    const currentDay = daysArr[now.getDay()];
    const currentTime = now.getHours() * 60 + now.getMinutes();
    const dayConfig = weeklySchedule[currentDay];
    if (!dayConfig) return currentTime < 720 ? 'in' : 'out';
    const threshold = dayConfig.out || 720;
    return currentTime < threshold ? 'in' : 'out';
}

function updateClock() {
    const now = new Date();
    const dayStr = now.toLocaleDateString('en-US', { weekday: 'long' });
    const dayEl = document.getElementById('current-live-day');
    if (dayEl) dayEl.textContent = dayStr;

    // Every hour or when the date changes, refresh the shift display to keep it
synced
    if (now.getMinutes() === 0 && now.getSeconds() === 0) {
        renderCurrentShift();
    }
}

function renderCurrentShift() {
    const now = new Date();
    const daysArr = ['sunday', 'monday', 'tuesday', 'wednesday', 'thursday', 'friday',
'saturday'];
    const currentDay = daysArr[now.getDay()];
    const dayConfig = weeklySchedule[currentDay];

    const timeInEl = document.getElementById('display-time-in');

    if (timeInEl) {
        if (dayConfig) {
            timeInEl.textContent = minutesToDisplay(dayConfig.in);
            timeInEl.parentElement.style.opacity = '1';
        } else {
            timeInEl.textContent = '--:--';
            timeInEl.parentElement.style.opacity = '0.5';
        }
    }
}

let isScanning = false;
let pollingInterval = null;
let scanTimeout = null;
let hasShownError = false;
let currentActiveEvent = null;

function fetchAttendees() {
```


IoT-Enabled Barangay Resident and Attendance Management System

```
if (!currentActiveEvent) return;

fetch(`${% url 'attendance:api_event_attendance_list'
%}?date=${currentActiveEvent.date}`)
  .then(res => res.json())
  .then(data => {
    if (data.status === 'success') {
      const tbody = document.getElementById('attendees-list-body');
      tbody.innerHTML = '';

      if (data.data.length === 0) {
        tbody.innerHTML = '<tr><td colspan="3" class="px-5 py-8 text-center text-slate-400 font-medium">No one is present yet.</td></tr>';
      } else {
        data.data.forEach(person => {
          const tr = document.createElement('tr');
          tr.className = 'hover:bg-slate-50 transition-colors';
          tr.innerHTML = `
            <td class="px-5 py-3 font-bold text-slate-800">${person.name}</td>
            <td class="px-5 py-3 text-slate-500 text-xs">${person.position}</td>
            <td class="px-5 py-3 text-slate-700 font-bold text-xs">${person.time_in}</td>
          `;
          tbody.appendChild(tr);
        });
      }
    }
  });

function checkEventMode() {
  const urlParams = new URLSearchParams(window.location.search);
  const eventId = urlParams.get('event_id');

  if (eventId) {
    const events = JSON.parse(localStorage.getItem('custom_attendance_events') || '[]');

    const activeEvent = events.find(e => e.id == eventId);

    if (activeEvent) {
      currentActiveEvent = activeEvent;
      // UI Changes for Event Mode
      document.getElementById('attendance-title').innerText = activeEvent.name;
      document.getElementById('event-mode-badge').classList.remove('hidden');
      document.getElementById('event-display').classList.remove('hidden');
      document.getElementById('exit-event-container').classList.remove('hidden');
      document.getElementById('event-time-display').innerText = `Scheduled for ${activeEvent.time}`;
      document.getElementById('event-attendees-section').classList.remove('hidden');

      // Hide normal shift display
      document.getElementById('shift-display').classList.add('hidden');

      // You could also change the background or add a border
      document.querySelector('.biometric-attendance-card').style.border = '2px solid #10b981';
    }
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        document.querySelector('.biometric-attendance-card').style.borderRadius =
'30px';
        document.querySelector('.biometric-attendance-card').style.background =
'rgba(236, 253, 245, 0.5)';

        fetchAttendees();
        return activeEvent;
    }
}
return null;
}

document.addEventListener('DOMContentLoaded', function() {
    initScheduleData();
    const activeEvent = checkEventMode();

    if (!activeEvent) {
        renderCurrentShift();
    }

    // PDF Download Listener
    const downloadPdfBtn = document.getElementById('downloadPdfBtn');
    if (downloadPdfBtn) {
        downloadPdfBtn.onclick = () => {
            if (currentActiveEvent) {
                window.location.href = `${% url 'attendance:event_attendance_pdf' %}?da
te=${currentActiveEvent.date}&event_name=${encodeURIComponent(currentActiveEvent.name)}`;
            }
        };
    }

    setInterval(updateClock, 1000);
    updateClock();

    $('#attendanceScheduleModal').on('show.bs.modal', function () {
        modalOpenedOnce = false; // Reset for the fresh opening
        switchDay(activeDayIndex);
    });

    const btnStart = document.getElementById('btn-start-attendance');
    const btnStop = document.getElementById('btn-stop-attendance');

    // Hide back button and padding if loaded inside an iframe (Scan Modal)
    if (window.self !== window.top || new
URLSearchParams(window.location.search).get('hide_back')) {
        const backBtn = document.querySelector('.page-header a.btn-outline-
secondary');
        if (backBtn) backBtn.style.display = 'none';
        document.body.style.backgroundColor = '#ffffff';
        const container = document.querySelector('.container-fluid');
        if(container) {
            container.style.padding = '0';
            container.style.marginTop = '0';
        }
        const card = document.querySelector('.biometric-attendance-card');
        if(card) {
            card.style.boxShadow = 'none';
            card.style.border = 'none';
            card.style.margin = '0 auto';
            card.style.minHeight = '100vh';
        }
    }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        card.style.borderRadius = '0';
    }
    const pageHeader = document.querySelector('.page-header');
    if (pageHeader) pageHeader.style.display = 'none';
}

const scanVisual = document.getElementById('scan-visual');
const statusBadge = document.getElementById('status-badge');
const instructionText = document.getElementById('instruction-text');
const resultCard = document.getElementById('attendance-result');

let isScanning = false;
let pollingInterval = null;
let scanTimeout = null;
let hasShownError = false;

// Read mode from URL if present
const urlParams = new URLSearchParams(window.location.search);
const urlMode = urlParams.get('mode') || 'attendance';

// Update title based on mode
const pageTitle = document.querySelector('h2');
if (pageTitle) {
    if (urlMode === 'in') pageTitle.innerHTML = '<i class="fas fa-fingerprint text-emerald-600 mr-2"></i> Time In Scan';
    else if (urlMode === 'out') pageTitle.innerHTML = '<i class="fas fa-fingerprint text-emerald-600 mr-2"></i> Time Out Scan';
}

function stopScanner() {
    if (!isScanning) return;
    isScanning = false;
    clearInterval(pollingInterval);
    clearTimeout(scanTimeout);
    pollingInterval = null;
    scanTimeout = null;

    // SYNC HARDWARE: Tell ESP32 to stop and go back to standby
    fetch('% url "officials:esp32_stop_enrollment_proxy" %', {
        method: 'POST',
        headers: { 'X-CSRFToken': '{{ csrf_token }}' }
    }).catch(err => console.error("Hardware stop failed:", err));

    btnStop.classList.add('d-none');
    btnStart.classList.remove('d-none');
    scanVisual.className = 'scan-visual';
    statusBadge.className = 'status-badge-lg badge-waiting';
    statusBadge.innerHTML = '<i class="fas fa-clock mr-2"></i> Ready to Clock In/Out';

    instructionText.className = 'small mb-3 text-secondary';
    instructionText.textContent = 'Click the button below to start the scanner';
}

function startUI() {
    if (isScanning) return;
    isScanning = true;
    btnStart.classList.add('d-none');
    btnStop.classList.remove('d-none');
    scanVisual.className = 'scan-visual active';
    statusBadge.className = 'status-badge-lg badge-active';
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
statusBadge.innerHTML = '<i class="fas fa-spinner fa-spin mr-2"></i> Scanner
Active';
instructionText.className = 'h5 mb-4 text-primary font-weight-bold';
instructionText.textContent = 'Please place your registered finger on the
sensor';

// Fade out the results card if it was visible
if (resultCard.style.display !== 'none') {
  resultCard.classList.add('fade-out');
  setTimeout(() => {
    resultCard.style.display = 'none';
    resultCard.classList.remove('fade-out');
  }, 400);
}
hasShownError = false;
}

btnStart.addEventListener('click', function() {
  startUI();
  console.log('[Scan] Sending start request to Django...');
  fetch('% url "core:biometric_attendance_start" %', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json', 'X-CSRFToken': '{{
csrf_token }}' },
    credentials: 'include',
    body: JSON.stringify({ mode: urlMode })
  })
  .then(res => {
    console.log('[Scan] Server responded:', res.status);
    return res.json();
  })
  .then(data => {
    console.log('[Scan] Response data:', data);
    if (data.status === 'success') {
      if (!data.esp32_connected) {
        // Django reached but ESP32 hardware is offline
        showError('Scanner hardware is offline. Check device
connection.');
```

```
        stopScanner();
        return;
      }
      console.log('[Scan] ESP32 connected, starting active polling...');
      startActivePolling();
    } else {
      console.error('[Scan] Server error:', data.message);
      showError(data.message || 'Failed to start scanner. Please try
again.');
```

```
      stopScanner();
    }
  })
  .catch(err => {
    console.error('[Scan] Fetch failed:', err);
    showError('Cannot connect to server. Please check your network.');
```

```
    stopScanner();
  });
});

btnStop.addEventListener('click', () => stopScanner());

// PASSIVE hardware listener ? runs always, watches for hardware button presses
```

IoT-Enabled Barangay Resident and Attendance Management System

```
// Does NOT need a request_id, just detects when hardware goes into 'verifying'
state
function startPassivePolling() {
  setInterval(() => {
    if (isScanning) return; // Active scan is handling it
    fetch('% url "core:biometric_attendance_status_check" %}', {
      credentials: 'include'
    })
    .then(res => res.json())
    .then(data => {
      if (data.status === 'authenticated') {
        // Hardware button scan completed ? trigger full record flow
        startUI();
        recordAttendance(data.attendance_mode);
      } else if (data.status === 'not_scheduled') {
        showNotScheduled(data.reason || 'Not Scheduled today.');
```

```
      } else if (data.status === 'failed' && !hasShownError) {
        hasShownError = true;
        stopScanner();
        showError(data.reason || 'Fingerprint not recognized');
```

```
    } else if (data.status === 'pending' && data.esp32_state ===
    'verifying') {
      // Hardware button pressed, sync UI
      startUI();
    }
  }).catch(() => {});
}, 2000);
}

// ACTIVE polling ? runs after clicking Start Scan (has a valid request_id in
session)
function startActivePolling() {
  if (pollingInterval) return;

  pollingInterval = setInterval(() => {
    fetch('% url "core:biometric_attendance_status_check" %}', {
      credentials: 'include'
    })
    .then(res => res.json())
    .then(data => {
      if (data.status === 'authenticated') {
        recordAttendance(data.attendance_mode);
      }
      else if (data.status === 'not_scheduled') {
        stopScanner();
        showNotScheduled(data.reason || 'Not Scheduled today.');
```

```
      }
      else if (data.status === 'failed' && !hasShownError) {
        hasShownError = true;
        stopScanner();
        showError(data.reason || 'Fingerprint not recognized');
        if (window.showNotification) {
          window.showNotification(data.reason || 'Fingerprint not
recognized', 'error');
```

```
        }
      }
      else if (data.status === 'pending') {
        if (data.esp32_state === 'standby' && isScanning) {
          stopScanner(); // Hardware timed out
        }
      }
    })
  }, 2000);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    }
    }).catch(() => {});
  }, 1000);
}

// Start passive hardware listener on page load
startPassivePolling();

function recordAttendance(hardwareMode) {

  // Priority: hardwareMode (if button was pressed), else automatic schedule
  let mode = (hardwareMode && hardwareMode !== 'none' && hardwareMode !==
'attendance') ? hardwareMode : getAttendanceAction();

  // Override with software button mode if selected and no specific hardware
button pressed
  if (!hardwareMode || hardwareMode === 'attendance' || hardwareMode === 'none')
  {
    if (urlMode === 'in' || urlMode === 'out') {
      mode = urlMode;
    }
  }

  const settingsObj = JSON.parse(localStorage.getItem('attendance_settings') ||
'{"late_time": "08:00"}');

  fetch('% url "attendance:api_biometric_verify" %', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    credentials: 'include',
    body: JSON.stringify({ auto: true, action: mode, late_time:
settingsObj.late_time })
  })
  .then(res => res.json())
  .then(data => {
    if (data.status === 'success') {
      showSuccess(data);
      if (window.showNotification) {
        window.showNotification(`Attendance recorded: ${data.name}
(${data.action})`, 'success');
      }
    }
    else if (data.status === 'not_scheduled') {
      showNotScheduled(data.message || 'Not Scheduled today.');
```

```
      if (window.showNotification) {
        window.showNotification(data.message || 'Not Scheduled today.',
'warning');
      }
    }
    else {
      showError(data.message || 'Attendance recording failed.');
```

```
      if (window.showNotification) {
        window.showNotification(data.message || 'Attendance recording
failed.', 'error');
      }
    }
  });
}

function showSuccess(data) {
```

IoT-Enabled Barangay Resident and Attendance Management System

```

scanVisual.className = 'scan-visual success';
statusBadge.className = 'status-badge-lg badge-success';
statusBadge.innerHTML = '<i class="fas fa-check-circle mr-2"></i> Attendance
Recorded';

instructionText.className = 'h5 mb-5 text-success font-weight-bold';
instructionText.textContent = 'Thank you! Your attendance has been captured.';
document.getElementById('result-name').textContent = data.name;
document.getElementById('result-time').textContent = data.time;
document.getElementById('result-action').textContent = data.action;

let actionClass = 'badge px-3 py-2 ';
if (data.action.includes('IN')) {
    actionClass += (data.attendance_status === 'late' ? 'badge-danger' :
'badge-success');
    if (data.attendance_status === 'late') {
        document.getElementById('result-action').innerHTML = 'Clock IN <span
class="ml-1 text-[10px] opacity-75">(LATE)</span>';
    } else {
        document.getElementById('result-action').innerHTML = 'Clock IN <span
class="ml-1 text-[10px] opacity-75">(ON TIME)</span>';
    }
} else {
    actionClass += 'badge-warning'; // Clock OUT
    document.getElementById('result-action').textContent = 'Clock OUT';
}
document.getElementById('result-action').className = actionClass;

resultCard.style.display = 'block';
btnStop.classList.add('d-none');
btnStart.classList.remove('d-none'); // Show immediately for quick next scan

// Refresh attendees list if in event mode
if (currentActiveEvent) {
    fetchAttendees();
}

// Auto-reset colors/visuals after 5s, but keep button visible
setTimeout(() => {
    if (resultCard.style.display === 'block') {
        scanVisual.className = 'scan-visual';
        statusBadge.className = 'status-badge-lg badge-waiting';
        statusBadge.innerHTML = '<i class="fas fa-clock mr-2"></i> Ready to
Clock In/Out';

        instructionText.className = 'small mb-3 text-secondary';
        instructionText.textContent = 'Click the button below to start the
scanner';
    }
}, 5000);
}

function showError(message) {
    scanVisual.className = 'scan-visual error';
    statusBadge.className = 'status-badge-lg badge-error';
    statusBadge.innerHTML = '<i class="fas fa-exclamation-triangle mr-2"></i> Scan
Failed';

    instructionText.className = 'h5 mb-4 text-danger font-weight-bold';
    instructionText.textContent = message;

    btnStop.classList.add('d-none');
    btnStart.classList.remove('d-none'); // Show immediately for quick next scan

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        setTimeout(() => {
            if (scanVisual.classList.contains('error')) {
                scanVisual.className = 'scan-visual';
                statusBadge.className = 'status-badge-lg badge-waiting';
                statusBadge.innerHTML = '<i class="fas fa-clock mr-2"></i> Ready to
Clock In/Out';
                instructionText.className = 'small mb-3 text-secondary';
                instructionText.textContent = 'Click the button below to start the
scanner';
            }
        }, 3000);
    }

    function showNotScheduled(message) {
        stopScanner();
        scanVisual.className = 'scan-visual';
        scanVisual.style.borderColor = '#f59e0b';
        scanVisual.querySelector('i').style.color = '#f59e0b';
        statusBadge.className = 'status-badge-lg';
        statusBadge.style.cssText = 'background:#ffffeb;color:#b45309;border:1.5px
solid #fcd34d;font-size:1.05rem;padding:.75rem 1.5rem;border-radius:1.5rem;font-
weight:900;display:inline-block;margin:.5rem auto;';
        statusBadge.innerHTML = '<i class="fas fa-calendar-times mr-2"></i> Not
Scheduled';
        instructionText.className = 'h5 mb-4 font-weight-bold';
        instructionText.style.color = '#b45309';
        instructionText.textContent = message;

        setTimeout(() => {
            scanVisual.style.borderColor = '';
            scanVisual.querySelector('i').style.color = '';
            statusBadge.style.cssText = '';
            statusBadge.className = 'status-badge-lg badge-waiting';
            statusBadge.innerHTML = '<i class="fas fa-clock mr-2"></i> Ready to Clock
In/Out';
            instructionText.className = 'small mb-3 text-secondary';
            instructionText.style.color = '';
            instructionText.textContent = 'Click the button below to start the
scanner';
        }, 5000);
    }

    // Modal Specific Event Handlers
    document.getElementById('dayStatusToggle').onchange = (e) => {
        updateToggleState(e.target.checked);
        scheduleData[activeDayIndex].enabled = e.target.checked;
    };
    document.getElementById('copyToAll').onclick = () => {
        const tIn = document.getElementById('timeIn').value;
        const tOut = document.getElementById('timeOut').value;
        const isEn = document.getElementById('dayStatusToggle').checked;
        scheduleData.forEach(day => { day.timeIn = tIn; day.timeOut = tOut;
day.enabled = isEn; });
        switchDay(activeDayIndex);
        if (window.showNotification) {
            window.showNotification('Schedule applied to all days!', 'success');
        }
    };

```


IoT-Enabled Barangay Resident and Attendance Management System

```
document.getElementById('resetBtn').onclick = () => {
  if(confirm("Reset all days to 8 AM - 12 PM?")) {
    scheduleData = dayKeys.map((key, i) => ({ key: key, name: fullDayNames[i],
enabled: true, timeIn: '08:00', timeOut: '12:00' }));
    switchDay(0);
  }
};

document.getElementById('bulkApplyToggle').onchange = (e) => {
  if (e.target.checked && window.showNotification) {
    window.showNotification('Note: These settings will apply to ALL days when
saved.', 'info');
  }
};

$('#attendanceScheduleModal').on('shown.bs.modal', function () {
  initScheduleData();
  switchDay(0);
});

});
</script>
{% endblock %}
```

File: templates/attendance/clock.html

```
{% extends 'base.html' %}

{% block title %}Clock In/Out - Barangay Management System{% endblock %}

{% block content %}
{% url 'attendance:dashboard' as attendance_dashboard_url %}
{% include 'components/back_button.html' with back_url=attendance_dashboard_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>Time Clock</h1>
        <span>Manual entry for attendance</span>
    </div>
</div>

<div class="card" style="max-width: 600px; margin: 0 auto;">
    <div class="card-header text-center">
        <h2 id="liveTime" style="font-size: 36px; font-weight: 800; color:
var(--primary);">00:00:00</h2>
        <span>{{ current_time|date:"l, F d, Y" }}</span>
    </div>
    <div class="card-body" style="padding: 30px;">
        <form method="POST">
            {% csrf_token %}
            <div class="form-group" style="margin-bottom: 25px;">
                <label>Select Official/Staff*</label>
                <select name="official" class="form-control" style="height: 50px; font-size:
16px;" required>
                    <option value="">Select Name...</option>
                    {% for official in officials %}
                    <option value="{{ official.pk }}">{{ official.resident.last_name }}, {{
official.resident.first_name }}{% if official.resident.middle_name %} {{
official.resident.middle_name }}{% endif %}{% if official.resident.suffix %} {{
official.resident.suffix }}{% endif %} ? {{ official.get_position_display }}</option>
                    {% endfor %}
                </select>
            </div>

            <div class="action-buttons" style="display: grid; grid-template-columns: 1fr 1fr;
gap: 20px;">
                <button type="submit" name="action" value="in" class="btn btn-primary"
style="height: 80px; font-size: 20px; flex-direction: column;">
                    <i class="fas fa-sign-in-alt"></i>
                    <span>CLOCK IN</span>
                </button>
                <button type="submit" name="action" value="out" class="btn btn-secondary"
style="height: 80px; font-size: 20px; flex-direction: column;">
                    <i class="fas fa-sign-out-alt"></i>
                    <span>CLOCK OUT</span>
                </button>
            </div>
        </form>
    </div>
</div>

<div class="card" style="margin-top: 30px;">
    <div class="card-header">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<h3>Recent Clock Entries</h3>
</div>
<div class="table-responsive">
  <table>
    <thead>
      <tr>
        <th>Official</th>
        <th>Action</th>
        <th>Time</th>
      </tr>
    </thead>
    <tbody>
      {% for log in today_logs.values %}
      <tr>
        <td>{{ log.official.resident.full_name }}</td>
        <td>
          {% if log.time_out %}
          <span class="status-badge badge-active">Completed</span>
          {% else %}
          <span class="status-badge badge-pending">On Duty</span>
          {% endif %}
        </td>
        <td>
          IN: {{ log.time_in|time:"h:i A" }}
          {% if log.time_out %} | OUT: {{ log.time_out|time:"h:i A" }}{% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
</div>
</div>

<script>
  function updateLiveTime() {
    const now = new Date();
    document.getElementById('liveTime').textContent = now.toLocaleTimeString('en-US', {
hour12: false });
  }
  setInterval(updateLiveTime, 1000);
  updateLiveTime();
</script>
{% endblock %}
```

File: templates/attendance/daily_report.html

```
{% extends 'base.html' %}
{% load static %}
{% load attendance_tags %}

{% block title %}Daily Attendance Report - {{ date|date:"M d, Y" }}{% endblock %}

{% block extra_head %}
<script src="https://cdn.tailwindcss.com"></script>
<script src="https://unpkg.com/lucide@latest"></script>
{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / <a href="{% url 'attendance:dashboard' %}"
class="hover:text-emerald-600">Attendance</a> / <a href="{% url
'attendance:history_calendar' %}" class="hover:text-emerald-600">History</a> / {{
date|date:"M d, Y" }}</span>
{% endblock %}

{% block content %}
<div class="max-w-5xl mx-auto p-4">
    {% url 'attendance:history_calendar' as calendar_url %}
    {% include 'components/back_button.html' with back_url=calendar_url|add_str:'?month='|
add_str:date.month|add_str:'&year='|add_str:date.year back_label='Back' %}

    <!-- Header -->
    <div class="flex flex-col md:flex-row md:items-center justify-between mb-8 mt-4
gap-4">
        <div>
            <h1 class="text-2xl font-black text-slate-900 tracking-tight">Attendance
Report</h1>
            <p class="text-slate-500 font-medium leading-none">{{ date|date:"l, F d, Y"
}}</p>
        </div>

        <div class="flex items-center gap-3">
            <button onclick="toggleDayOff('{{ date|date:'Y-m-d' }}')"
id="dayOffBtn"
class="{% if is_day_off %}bg-red-50 text-red-600 hover:bg-red-100
border-red-200{% else %}bg-indigo-50 text-indigo-600 hover:bg-indigo-100 border-
indigo-200{% endif %} px-4 py-2 rounded-xl text-xs font-black transition-all border
shadow-sm flex items-center gap-2">
                <i data-lucide="{% if is_day_off %}sun{% else %}calendar-off{% endif %}"
class="w-4 h-4"></i>
                {% if is_day_off %}Revert to Working Day{% else %}Mark as Day Off{% endif
%}
            </button>
        </div>
    </div>

    <!-- Report Table -->
    <div class="bg-white rounded-2xl shadow-xl border border-slate-200 overflow-hidden">
        <div class="table-responsive w-full">
            <table class="w-full min-w-[700px] text-left border-collapse">
                <thead class="bg-slate-50 text-slate-400 uppercase text-[10px] font-black
tracking-widest border-b border-slate-200">
                    <tr>
                        <th class="px-6 py-4 w-[35%]">Official Profile</th>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <th class="px-6 py-4 text-center">AM IN</th>
        <th class="px-6 py-4 text-center">AM OUT</th>
        <th class="px-6 py-4 text-center">PM IN</th>
        <th class="px-6 py-4 text-center">PM OUT</th>
        <th class="px-6 py-4 text-center">Actions</th>
    </tr>
</thead>
<tbody class="divide-y divide-slate-100">
    {% if is_day_off %}
    <tr>
        <td colspan="7" class="px-6 py-20 text-center">
            <div class="flex flex-col items-center gap-3">
                <div class="w-16 h-16 bg-blue-50 text-blue-300 rounded-
full flex items-center justify-center">
                    <i data-lucide="sun" class="w-8 h-8"></i>
                </div>
                <div>
                    <h3 class="text-lg font-black text-
slate-800">Scheduled Day Off</h3>
                    <p class="text-slate-400 font-medium text-sm">There
are no attendance records for this date as it is a non-working day.</p>
                </div>
            </div>
        </td>
    </tr>
    {% elif is_future %}
    <tr>
        <td colspan="7" class="px-6 py-20 text-center">
            <div class="flex flex-col items-center gap-3">
                <div class="w-16 h-16 bg-slate-50 text-slate-300 rounded-
full flex items-center justify-center">
                    <i data-lucide="calendar-clock" class="w-8 h-8"></i>
                </div>
                <div>
                    <h3 class="text-lg font-black text-
slate-800">Attendance Not Started</h3>
                    <p class="text-slate-400 font-medium text-sm">Records
for this date will appear once the day begins.</p>
                </div>
            </div>
        </td>
    </tr>
    {% else %}
    {% for item in report %}
    <tr class="hover:bg-slate-50/50 transition-colors {% if item.status in
'unscheduled, day_off, leave' %}opacity-60 bg-slate-50/30{% endif %}">
        <td class="px-6 py-4">
            <div class="flex items-center gap-3">
                <div class="w-9 h-9 rounded-full bg-slate-100 overflow-
hidden border border-slate-200 flex-shrink-0">
                    {% if item.official.resident.photo %}
                    
                    {% else %}
                    
                    {% endif %}
                </div>
                <div class="flex flex-col min-w-0">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <span class="font-bold text-slate-800 whitespace-
normal break-words leading-tight">{{ item.official.resident.full_name }}</span>
        <div class="mt-0.5 pt-0.5 border-t border-slate-100
w-fit flex gap-2 items-center">
            <span class="text-[10px] text-slate-500 font-bold
uppercase tracking-wide">{{ item.official.get_position_display }}</span>
            {% if item.shift_label %}
            <span class="text-[9px] bg-slate-100 text-
slate-600 px-1.5 py-0.5 rounded font-black tracking-widest uppercase">{{ item.shift_label
}}</span>
            {% else %}
            <span class="text-[9px] bg-slate-50 text-slate-400
px-1.5 py-0.5 rounded font-black tracking-widest uppercase border border-slate-200">No
Schedule</span>
            {% endif %}
        </div>
    </div>
</td>
<td class="px-6 py-4 text-center">
    {% with log=item.log %}
    {% if log and log.am_in %}
    <div class="flex flex-col items-center gap-1">
        <span class="text-sm font-bold text-slate-700">
            {{ log.am_in|time:"h:i A" }}
        </span>
        {% get_lateness log.am_in shift_config.am_in
shift_config.grace_period as am_late %}
        {% if am_late %}
        <span class="text-[9px] bg-amber-50 text-amber-600
px-2 py-0.5 rounded font-black tracking-widest border border-amber-100 uppercase">
            Late ({{ am_late }})
        </span>
        {% else %}
        <span class="text-[9px] bg-emerald-50 text-
emerald-600 px-2 py-0.5 rounded font-black tracking-widest border border-emerald-100
uppercase">On Time</span>
        {% endif %}
    </div>
    {% elif item.status == 'unscheduled' or item.status ==
'day_off' or item.status == 'leave' %}
    <span class="text-[9px] bg-slate-100 text-slate-400
px-2 py-0.5 rounded font-black tracking-widest border border-slate-200 uppercase">?</span>
    {% elif date|is_absent_visible:shift_config.am_out %}
    <span class="text-[9px] bg-red-50 text-red-600 px-2
py-0.5 rounded font-black tracking-widest border border-red-100 uppercase">Absent</span>
    {% endif %}
    {% endwith %}
</td>
<td class="px-3 py-2.5 text-center">
    <div class="flex flex-col items-center">
        {% with log=item.log %}
        {% if log and log.am_out %}
        <div class="flex flex-col items-center gap-1">
            <span class="font-bold text-slate-700 text-
sm">{{ log.am_out|time:"h:i A" }}</span>
            <span class="text-[9px] bg-blue-50 text-
blue-600 px-2 py-0.5 rounded font-black tracking-widest border border-blue-100
uppercase">Done</span>
        </div>
        
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        {% elif log and log.am_in %}
            <span class="text-slate-300 italic
text-[10px]">---</span>
        {% elif item.status == 'unscheduled' or item.status ==
'day_off' or item.status == 'leave' %}
            <span class="text-[9px] bg-slate-100 text-
slate-400 px-2 py-0.5 rounded font-black tracking-widest border border-slate-200
uppercase">?</span>
        {% elif date|is_absent_visible:shift_config.am_out %}
            <span class="text-[9px] bg-red-50 text-red-600
px-2 py-0.5 rounded font-black tracking-widest border border-red-100
uppercase">Absent</span>
        {% endif %}
    {% endwith %}
</div>
</td>
<td class="px-3 py-2.5 text-center">
    <div class="flex flex-col items-center">
        {% with log=item.log %}
            {% if log and log.pm_in %}
                <div class="flex flex-col items-center gap-1">
                    <span class="font-bold text-slate-700 text-
sm">{{ log.pm_in|time:"h:i A" }}</span>
                    {% get_lateness log.pm_in shift_config.pm_in
shift_config.grace_period as pm_late %}
                    {% if pm_late %}
                        <span class="text-[9px] bg-amber-50 text-
amber-600 px-2 py-0.5 rounded font-black tracking-widest border border-amber-100
uppercase">Late ({{ pm_late }})</span>
                    {% else %}
                        <span class="text-[9px] bg-emerald-50
text-emerald-600 px-2 py-0.5 rounded font-black tracking-widest border border-emerald-100
uppercase">Present</span>
                    {% endif %}
                </div>
            {% elif item.status == 'unscheduled' or item.status ==
'day_off' or item.status == 'leave' %}
                <span class="text-[9px] bg-slate-100 text-
slate-400 px-2 py-0.5 rounded font-black tracking-widest border border-slate-200
uppercase">?</span>
            {% elif date|is_absent_visible:shift_config.pm_out %}
                <span class="text-[9px] bg-red-50 text-red-600
px-2 py-0.5 rounded font-black tracking-widest border border-red-100
uppercase">Absent</span>
            {% endif %}
        {% endwith %}
    </div>
</td>
<td class="px-3 py-2.5 text-center">
    <div class="flex flex-col items-center">
        {% with log=item.log %}
            {% if log and log.pm_out %}
                <div class="flex flex-col items-center gap-1">
                    <span class="font-bold text-slate-700 text-
sm">{{ log.pm_out|time:"h:i A" }}</span>
                    <span class="text-[9px] bg-blue-50 text-
blue-600 px-2 py-0.5 rounded font-black tracking-widest border border-blue-100
uppercase">Done</span>
                </div>
            {% elif log and log.pm_in %}

```

IoT-Enabled Barangay Resident and Attendance Management System

```

                                <span class="text-slate-300 italic
text-[10px]">---</span>
                                {% elif item.status == 'unscheduled' or item.status ==
                                'day_off' or item.status == 'leave' %}
                                <span class="text-[9px] bg-slate-100 text-
                                slate-400 px-2 py-0.5 rounded font-black tracking-widest border border-slate-200
                                uppercase">?</span>
                                {% elif date|is_absent_visible:shift_config.pm_out %}
                                <span class="text-[9px] bg-red-50 text-red-600
                                px-2 py-0.5 rounded font-black tracking-widest border border-red-100
                                uppercase">Absent</span>
                                {% endif %}
                                {% endwith %}
                                </div>
                                </td>
                                <td class="px-6 py-4">
                                    <div class="flex items-center justify-center gap-2">
                                        <a href="{% url 'attendance:dtr_detail' item.official.pk
                                        %}?month={{ date.month }}&year={{ date.year }}&from=report&report_date={{
                                        date|date:'Y-m-d' }}" class="text-blue-600 hover:text-blue-800 transition bg-blue-50
                                        hover:bg-blue-100 p-1.5 rounded" title="View DTR">
                                            <i data-lucide="calendar" class="w-3.5 h-3.5"></i>
                                        </a>
                                        <a href="{% url 'attendance:dtr_print' item.official.pk
                                        %}?month={{ date.month }}&year={{ date.year }}" class="text-indigo-600 hover:text-
                                        indigo-800 transition bg-indigo-50 hover:bg-indigo-100 p-1.5 rounded" title="Print DTR"
                                        target="_blank">
                                            <i data-lucide="printer" class="w-3.5 h-3.5"></i>
                                        </a>
                                    </div>
                                </td>
                                </tr>
                                {% empty %}
                                <tr>
                                    <td colspan="7" class="px-6 py-10 text-center text-slate-400 font-
                                medium">
                                        No records found for this date.
                                    </td>
                                </tr>
                                {% endfor %}
                                {% endif %}
                                </tbody>
                                </table>
                                </div>
                                </div>
                                </div>
                                </div>
                                <script>
                                    document.addEventListener('DOMContentLoaded', function() {
                                        if (window.lucide) lucide.createIcons();
                                    });

                                    function toggleDayOff(dateStr) {
                                        if (!confirm('Are you sure you want to change the work status for this date?'))
                                        return;

                                        const formData = new FormData();
                                        formData.append('date', dateStr);
                                        formData.append('csrfmiddlewaretoken', '{{ csrf_token }}');

```


IoT-Enabled Barangay Resident and Attendance Management System

```
fetch("{% url 'attendance:api_toggle_special_date' %}", {
  method: 'POST',
  body: formData
})
.then(response => response.json())
.then(data => {
  if (data.status === 'success') {
    location.reload();
  } else {
    alert('Error: ' + data.message);
  }
})
.catch(error => {
  console.error('Error:', error);
  alert('An error occurred. Please try again.');
```

```
});
}
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/attendance/dashboard.html

```
{% extends 'base.html' %}
{% load static %}
{% load attendance_tags %}

{% block title %}Attendance Dashboard - Barangay Management System{% endblock %}

{% block extra_head %}
<script src="https://cdn.tailwindcss.com"></script>
<script src="https://unpkg.com/lucide@latest"></script>
<style>
    /* Prevent Tailwind from interfering with existing sidebar/layout if possible */
    .premium-dashboard {
        font-family: 'Inter', sans-serif;
    }
    .animate-in {
        animation: animate-in 0.2s ease-out;
    }
    @keyframes animate-in {
        from { opacity: 0; transform: scale(0.95); }
        to { opacity: 1; transform: scale(1); }
    }
</style>
{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Attendance</span>
{% endblock %}

{% block content %}
<div class="premium-dashboard min-h-screen bg-gray-50 p-1 md:p-4 text-slate-900">
    <div class="max-w-7xl mx-auto">

        <!-- Header Section -->
        <header class="flex flex-col md:flex-row md:items-center justify-between mb-8 gap-4">

            <div>
                <h1 class="text-2xl font-extrabold tracking-tight text-slate-900">Attendance & DTR</h1>
                <div class="mt-1 flex flex-col gap-2">
                    <p class="text-slate-500 font-medium leading-none">Real-time monitoring for <span class="text-slate-900 font-bold">{% now "l, F j, Y" %}</span></p>
                    <div class="flex flex-wrap items-center gap-2 mt-1">
                        <div class="flex items-center gap-2 px-2.5 py-1 bg-white border border-slate-200 rounded-lg shadow-sm">
                            <span class="text-[9px] font-black text-slate-400 uppercase tracking-widest">AM Shift</span>
                            <span class="text-xs font-bold text-slate-700">{{ shift_config.am_in|time:"h:i A" }} - {{ shift_config.am_out|time:"h:i A" }}</span>
                        </div>
                        <div class="flex items-center gap-2 px-2.5 py-1 bg-white border border-slate-200 rounded-lg shadow-sm">
                            <span class="text-[9px] font-black text-slate-400 uppercase tracking-widest">PM Shift</span>
                            <span class="text-xs font-bold text-slate-700">{{ shift_config.pm_in|time:"h:i A" }} - {{ shift_config.pm_out|time:"h:i A" }}</span>
                        </div>
                        <div class="flex items-center gap-2 px-2.5 py-1 bg-amber-50 border
```

IoT-Enabled Barangay Resident and Attendance Management System

```

border-amber-100 rounded-lg shadow-sm">
    <i data-lucide="clock" class="w-3 h-3 text-amber-600"></i>
    <span class="text-[9px] font-black text-amber-600 uppercase
tracking-widest">Late at:</span>
    <span class="text-xs font-bold text-amber-700">{{
shift_config.grace_period }} mins</span>
    </div>
  </div>
</div>
<div class="flex flex-wrap items-center gap-2">
  <!-- Action Series -->
  <div class="flex items-center gap-1.5 bg-white p-1.5 rounded-2xl border
border-slate-200 shadow-sm">
    <!-- Events Button & Popover -->
    <div class="relative" id="eventDropdownWrapper">
      <button id="toggleEventDropdown" class="flex items-center gap-2
px-4 py-2.5 bg-emerald-600 hover:bg-emerald-700 text-white rounded-xl font-bold transition
shadow-sm text-xs">
        <i data-lucide="calendar-plus" class="w-3.5 h-3.5"></i> Events
      </button>
      <div id="eventPopover" class="absolute right-0 md:right-auto
md:left-0 top-full mt-3 w-[280px] sm:w-[360px] bg-white rounded-3xl
shadow-[0_20px_50px_rgba(0,0,0,0.2)] border border-slate-200 hidden z-[100] animate-in">
        <div class="absolute -top-2 right-10 md:right-auto md:left-10
w-4 h-4 bg-white border-t border-l border-slate-200 transform rotate-45"></div>
        <div class="relative bg-white rounded-2xl z-10 overflow-
hidden">
          <div class="p-4 border-b border-slate-100 bg-slate-50/50
flex justify-between items-center">
            <h3 class="text-sm font-black text-slate-900 flex
items-center gap-2">
              <i data-lucide="calendar-plus" class="w-4 h-4
text-emerald-600"></i>
              Schedule Custom Event
            </h3>
            <a href="#" id="goToEventBtn" class="hidden
text-[10px] font-black text-emerald-600 uppercase tracking-widest bg-emerald-100/50
hover:bg-emerald-100 px-2 py-1.5 rounded-lg transition items-center gap-1">
              Go to Event <i data-lucide="arrow-right"
class="w-3 h-3"></i>
            </a>
          </div>
          <form id="eventForm" class="p-4 space-y-4">
            <div>
              <label class="block text-[10px] font-black text-
slate-400 uppercase tracking-widest mb-1.5">Event Title</label>
              <input type="text" id="eventTitle" required
placeholder="e.g. Community General Assembly" class="w-full px-4 py-2.5 bg-slate-50 border
border-slate-200 rounded-xl focus:ring-2 focus:ring-emerald-500 outline-none font-bold
text-slate-800 text-xs shadow-sm" />
            </div>
            <div class="grid grid-cols-2 gap-4">
              <div>
                <label class="block text-[10px] font-black
text-slate-400 uppercase tracking-widest mb-1.5">Date</label>
                <input type="date" id="eventDate" required
class="w-full px-3 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:ring-2
focus:ring-emerald-500 outline-none font-bold text-slate-800 text-xs shadow-sm" />
              </div>
            </div>
          </form>
        </div>
      </div>
    </div>
  </div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <label class="block text-[10px] font-black
text-slate-400 uppercase tracking-widest mb-1.5">Start Time</label>
        <input type="time" id="eventTime" required
class="w-full px-3 py-2.5 bg-slate-50 border border-slate-200 rounded-xl focus:ring-2
focus:ring-emerald-500 outline-none font-bold text-slate-800 text-xs shadow-sm" />
        </div>
    </div>
    <button type="submit" class="w-full mt-2 py-3.5 bg-
emerald-600 text-white font-black rounded-xl hover:bg-emerald-700 transition shadow-lg
shadow-emerald-100 text-xs flex items-center justify-center gap-2">
        <i data-lucide="check-circle" class="w-4 h-4"></i>
Create Event Record
    </button>
</form>

</div>
</div>
</div>

<!-- Work Schedule Button & Popover -->
<div class="relative" id="scanWrapper">
    <button id="toggleScanDropdown" class="flex items-center gap-2
px-4 py-2.5 bg-indigo-600 hover:bg-indigo-700 text-white rounded-xl font-bold transition
shadow-sm text-xs">
        <i data-lucide="clock" class="w-3.5 h-3.5"></i> Work Schedule
    </button>
    <div id="scanPopover" class="absolute right-0 top-full mt-2
w-[340px] bg-white rounded-2xl shadow-[0_20px_50px_rgba(0,0,0,0.2)] border border-
slate-200 hidden z-[100] animate-in">
        <div class="absolute -top-2 right-10 w-4 h-4 bg-white border-t
border-l border-slate-200 transform rotate-45"></div>
        <div class="relative bg-white rounded-2xl z-10 overflow-
hidden">
            <div class="p-3 border-b border-slate-100 bg-slate-50/50">
                <div class="flex items-center justify-between gap-3">
                    <h3 class="text-sm font-black text-slate-900 flex
items-center gap-2 whitespace-nowrap">
                        <i data-lucide="settings-2" class="w-4 h-4
text-indigo-600"></i> Shift Setup
                    </h3>
                    <div class="flex items-center gap-3">
                        <label class="flex items-center gap-2 cursor-
pointer group">
                            <input type="checkbox"
id="bulkApplyToggle" class="w-3.5 h-3.5 rounded border-slate-300 text-indigo-600
focus:ring-indigo-500">
                            <span class="text-[10px] font-black text-
slate-400 uppercase group-hover:text-indigo-600 transition-colors">Apply to all</span>
                        </label>
                    </div>
                </div>
            </div>
        </div>
    </div>
    <div class="p-3 space-y-3">
        <div class="flex items-center gap-2 p-1.5 bg-slate-50
border border-slate-200 rounded-xl">
            <div class="flex-grow">
                <select id="scheduleDay" class="w-full px-2
py-1 bg-transparent border-none focus:ring-0 outline-none font-black text-slate-800 text-
xs">
                    <option value="mon">Monday

```

IoT-Enabled Barangay Resident and Attendance Management System

[illegible]

IoT-Enabled Barangay Resident and Attendance Management System

```

        <div class="flex items-center gap-1.5 mb-2
px-1">
            <i data-lucide="moon" class="w-3 h-3 text-
indigo-500"></i>
            <span class="text-[10px] font-black text-
indigo-600 uppercase tracking-widest">Afternoon</span>
        </div>
        <div class="space-y-3">
            <div>
                <label class="block text-[9px] font-
black text-slate-400 uppercase mb-1">Time In</label>
                <input type="time" id="pmTimeIn"
value="13:00" class="w-full px-2 py-1 bg-white border border-slate-200 rounded-lg
focus:ring-2 focus:ring-indigo-500 outline-none font-bold text-slate-800 text-xs shadow-
sm" />
            </div>
            <div>
                <label class="block text-[9px] font-
black text-slate-400 uppercase mb-1">Time Out</label>
                <input type="time" id="targetTimeOut"
value="17:00" class="w-full px-2 py-1 bg-white border border-slate-200 rounded-lg
focus:ring-2 focus:ring-indigo-500 outline-none font-bold text-slate-800 text-xs shadow-
sm" />
            </div>
        </div>
    </div>
</div>
<div class="flex items-center justify-between p-3 bg-
slate-50 rounded-xl border border-slate-200">
    <div class="flex items-center gap-3">
        <div class="w-8 h-8 rounded-lg bg-white border
border-slate-200 flex items-center justify-center text-slate-400 shadow-sm">
            <i data-lucide="timer" class="w-4
h-4"></i>
        </div>
        <div>
            <p class="text-xs font-black text-
slate-800 uppercase leading-none mb-0.5">Late Period</p>
            <p class="text-[9px] font-medium text-
slate-400 uppercase tracking-tighter">Arrival threshold</p>
        </div>
    </div>
    <select id="lateGracePeriod" class="bg-white
border border-slate-200 rounded-lg px-2 py-1.5 text-xs font-black text-slate-700
focus:ring-2 focus:ring-indigo-500 outline-none shadow-sm">
        <option value="0">0 Min</option>
        <option value="5">5 Min</option>
        <option value="10">10 Min</option>
        <option value="15" selected>15 Min</option>
        <option value="30">30 Min</option>
        <option value="custom">Custom...</option>
    </select>
</div>

    <div id="customGraceContainer" class="hidden animate-
in p-2 bg-indigo-50/50 border border-indigo-100 rounded-xl">
        <div class="flex items-center gap-2">
            <input type="number" id="customGraceValue"
value="15" class="w-20 px-3 py-1.5 bg-white border border-slate-200 rounded-lg

```

IoT-Enabled Barangay Resident and Attendance Management System

```

focus:ring-2 focus:ring-indigo-500 outline-none font-black text-slate-900 text-xs shadow-sm" placeholder="Value" />
<select id="customGraceUnit" class="flex-grow
px-2 py-1.5 bg-white border border-slate-200 rounded-lg focus:ring-2 focus:ring-indigo-500
outline-none font-black text-slate-900 text-[10px] shadow-sm">
  <option value="seconds">Seconds</option>
  <option value="minutes"
selected>Minutes</option>
  <option value="hours">Hours</option>
</select>
</div>
</div>
<button type="button" id="saveSettingsBtn"
class="w-full py-2 bg-slate-800 hover:bg-slate-900 text-white rounded-xl font-black text-
xs uppercase transition shadow-lg flex items-center justify-center gap-2
active:scale-[0.98]">
  <i data-lucide="save" class="w-3.5 h-3.5"></i>
Save
</button>
</div>
</div>
</div>
</div>
</div>
<!-- Divider -->
<div class="h-6 w-px bg-slate-200 mx-1"></div>
<!-- Schedule Button -->
<a href="{% url 'attendance:work_schedule' %}" class="flex items-
center gap-2 px-4 py-2.5 bg-amber-500 hover:bg-amber-600 text-white rounded-xl font-bold
transition shadow-sm text-xs no-underline">
  <i data-lucide="calendar-days" class="w-3.5 h-3.5"></i> Manage
Schedules
</a>
<!-- History Button -->
<a href="{% url 'attendance:history_calendar' %}" class="flex items-
center gap-2 px-4 py-2.5 bg-blue-600 hover:bg-blue-700 text-white rounded-xl font-bold
transition shadow-sm text-xs no-underline">
  <i data-lucide="history" class="w-3.5 h-3.5"></i> Attendance
History
</a>
</div>
</div>
</div>
</header>
<!-- Stats Cards Removed as requested -->
<!-- Today's Attendance Log Table -->
<div class="bg-white rounded-xl shadow-sm border border-slate-200 overflow-hidden
mb-8">
  <div class="px-5 py-4 border-b border-slate-100 flex justify-between items-
center bg-white">
    <h3 class="text-base font-extrabold text-slate-800 tracking-tight">Today's
Attendance Log &ndash; <span class="text-indigo-600">{% now "F j, Y" %}</span></h3>
    <span class="text-xs text-slate-500 font-bold bg-slate-50 px-2.5 py-1
rounded-full border border-slate-100">
      {{ today_logs|length }} records
    </span>
  </div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

</div>
<div class="overflow-x-auto max-h-[380px] overflow-y-auto custom-scrollbar
w-full">
    <table class="w-full min-w-[700px] text-left border-collapse table-fixed
relative">
        <thead class="bg-slate-50 text-slate-500 uppercase text-[9px] font-
black tracking-widest border-b border-slate-200 sticky top-0 z-10 shadow-sm">
            <tr>
                <th class="px-4 py-3 w-[35%]">Official Profile</th>
                <th class="px-3 py-3 w-[15%] text-center">AM IN</th>
                <th class="px-3 py-3 w-[15%] text-center">AM OUT</th>
                <th class="px-3 py-3 w-[15%] text-center">PM IN</th>
                <th class="px-3 py-3 w-[15%] text-center">PM OUT</th>
                <th class="px-2 py-3 w-[5%]"></th>
            </tr>
        </thead>
        <tbody class="divide-y divide-slate-100 text-xs">
            {% for log in today_logs %}
                <tr class="hover:bg-slate-50 transition-colors group {% if
log.status in 'absent,leave,day_off,unscheduled' %}opacity-75 bg-slate-50/50{% endif %}">
                    <td class="px-6 py-2.5">
                        <div class="flex items-center gap-3">
                            {% if log.official.resident.photo %}
                                
                            {% elif log.official.user.avatar %}
                                
                            {% else %}
                                
                            {% endif %}
                            <div class="flex flex-col min-w-0">
                                <span class="font-bold text-slate-800 whitespace-
normal break-words leading-tight">{{ log.official.resident.full_name }}</span>
                                <div class="mt-0.5 pt-0.5 border-t border-
slate-100 w-fit flex gap-2 items-center">
                                    <span class="text-[10px] text-slate-500 font-
bold uppercase tracking-wide">{{ log.official.get_position_display }}</span>
                                    <span class="text-[9px] bg-slate-100 text-
slate-600 px-1.5 py-0.5 rounded font-black tracking-widest uppercase">{{ log.shift_label
}}</span>
                                </div>
                            </div>
                        </td>
                        <td class="px-3 py-2.5 text-center">
                            <div class="flex flex-col items-center">
                                {% if log.am_in %}
                                    <div class="flex flex-col items-center gap-1">
                                        <span class="font-bold text-slate-700 text-
sm">{{ log.am_in|time:"h:i A" }}</span>
                                        {% get_lateness log.am_in shift_config.am_in
shift_config.grace_period as am_late %}
                                        {% if am_late %}

```


IoT-Enabled Barangay Resident and Attendance Management System

```

                                <span class="text-[9px] bg-amber-50 text-
amber-600 px-2 py-0.5 rounded font-black tracking-widest border border-amber-100
uppercase">Late ({{ am_late }})</span>
                                {% else %}
                                <span class="text-[9px] bg-emerald-50
text-emerald-600 px-2 py-0.5 rounded font-black tracking-widest border border-emerald-100
uppercase">Present</span>
                                {% endif %}
                            </div>
                            {% elif log.status == 'unscheduled' or log.status ==
'day_off' or log.status == 'leave' %}
                                <span class="text-[9px] bg-slate-100 text-
slate-400 px-2 py-0.5 rounded font-black tracking-widest border border-slate-200
uppercase">?</span>
                                {% elif today|is_absent_visible:shift_config.am_out %}
                                <span class="text-[9px] bg-red-50 text-red-600
px-2 py-0.5 rounded font-black tracking-widest border border-red-100
uppercase">Absent</span>
                                {% endif %}
                            </div>
                        </td>
                        <td class="px-3 py-2.5 text-center">
                            <div class="flex flex-col items-center">
                                {% if log.am_out %}
                                    <div class="flex flex-col items-center gap-1">
                                        <span class="font-bold text-slate-700 text-
sm">{{ log.am_out|time:"h:i A" }}</span>
                                        <span class="text-[9px] bg-blue-50 text-
blue-600 px-2 py-0.5 rounded font-black tracking-widest border border-blue-100
uppercase">Done</span>
                                    </div>
                                {% elif log.am_in %}
                                    <span class="text-slate-300 italic
text-[10px]">---</span>
                                {% elif log.status == 'unscheduled' or log.status ==
'day_off' or log.status == 'leave' %}
                                    <span class="text-[9px] bg-slate-100 text-
slate-400 px-2 py-0.5 rounded font-black tracking-widest border border-slate-200
uppercase">?</span>
                                    {% elif today|is_absent_visible:shift_config.am_out %}
                                        <span class="text-[9px] bg-red-50 text-red-600
px-2 py-0.5 rounded font-black tracking-widest border border-red-100
uppercase">Absent</span>
                                    {% endif %}
                                </div>
                            </td>
                            <td class="px-3 py-2.5 text-center">
                                <div class="flex flex-col items-center">
                                    {% if log.pm_in %}
                                        <div class="flex flex-col items-center gap-1">
                                            <span class="font-bold text-slate-700 text-
sm">{{ log.pm_in|time:"h:i A" }}</span>
                                            {% get_lateness log.pm_in shift_config.pm_in
shift_config.grace_period as pm_late %}
                                            {% if pm_late %}
                                                <span class="text-[9px] bg-amber-50 text-
amber-600 px-2 py-0.5 rounded font-black tracking-widest border border-amber-100
uppercase">Late ({{ pm_late }})</span>
                                            {% else %}
                                                <span class="text-[9px] bg-emerald-50

```

IoT-Enabled Barangay Resident and Attendance Management System

```

text-emerald-600 px-2 py-0.5 rounded font-black tracking-widest border border-emerald-100
uppercase">Present</span>

        {% endif %}
    </div>
    {% elif log.status == 'unscheduled' or log.status ==
'day_off' or log.status == 'leave' %}
        <span class="text-[9px] bg-slate-100 text-
slate-400 px-2 py-0.5 rounded font-black tracking-widest border border-slate-200
uppercase">?</span>

        {% elif today|is_absent_visible:shift_config.pm_out %}
        <span class="text-[9px] bg-red-50 text-red-600
px-2 py-0.5 rounded font-black tracking-widest border border-red-100
uppercase">Absent</span>

        {% endif %}
    </div>
</td>
<td class="px-3 py-2.5 text-center">
    <div class="flex flex-col items-center">
        {% if log.pm_out %}
            <div class="flex flex-col items-center gap-1">
                <span class="font-bold text-slate-700 text-
sm">{{ log.pm_out|time:"h:i A" }}</span>
                <span class="text-[9px] bg-blue-50 text-
blue-600 px-2 py-0.5 rounded font-black tracking-widest border border-blue-100
uppercase">Done</span>
            </div>
            {% elif log.pm_in %}
            <span class="text-slate-300 italic
text-[10px]">---</span>
            {% elif log.status == 'unscheduled' or log.status ==
'day_off' or log.status == 'leave' %}
                <span class="text-[9px] bg-slate-100 text-
slate-400 px-2 py-0.5 rounded font-black tracking-widest border border-slate-200
uppercase">?</span>

                {% elif today|is_absent_visible:shift_config.pm_out %}
                <span class="text-[9px] bg-red-50 text-red-600
px-2 py-0.5 rounded font-black tracking-widest border border-red-100
uppercase">Absent</span>

                {% endif %}
            </div>
        </td>
        <td class="px-4 py-2.5 text-right">
            <div class="flex items-center justify-end gap-2
transition-opacity">
                <a href="{% url 'attendance:dtr_detail'
log.official.pk %}" class="text-blue-600 hover:text-blue-800 transition bg-blue-50
hover:bg-blue-100 p-1.5 rounded" title="View DTR">
                    <i data-lucide="calendar" class="w-3.5 h-3.5"></i>
                </a>
                <a href="{% url 'attendance:dtr_print' log.official.pk
%}?month={% now 'm' %}&year={% now 'Y' %}" class="text-indigo-600 hover:text-indigo-800
transition bg-indigo-50 hover:bg-indigo-100 p-1.5 rounded" title="Print DTR"
target="_blank">
                    <i data-lucide="printer" class="w-3.5 h-3.5"></i>
                </a>
            </div>
        </td>
    </tr>
    {% empty %}
</tr>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <td colspan="7" class="px-5 py-6 text-center text-slate-400
font-medium text-xs">
            No attendance records for today yet.
        </td>
    </tr>
    {% endfor %}
</tbody>
</table>
</div>
</div>

</div>
</div>

{% endblock %}

{% block extra_js %}
<script>
    document.addEventListener('DOMContentLoaded', function() {
        // Initialize Lucide Icons
        lucide.createIcons();

        const eventPopover = document.getElementById('eventPopover');
        const toggleEventDropdown = document.getElementById('toggleEventDropdown');
        const scanPopover = document.getElementById('scanPopover');
        const toggleScanDropdown = document.getElementById('toggleScanDropdown');
        const eventForm = document.getElementById('eventForm');
        const eventListContainer = document.getElementById('eventListContainer');

        let events = JSON.parse(localStorage.getItem('custom_attendance_events') || '[]');

        // Toggle event popover ? closes scan if open
        toggleEventDropdown.addEventListener('click', (e) => {
            e.stopPropagation();
            if (!scanPopover.classList.contains('hidden'))
scanPopover.classList.add('hidden');
            eventPopover.classList.toggle('hidden');
        });

        // Toggle scan popover ? closes event if open
        toggleScanDropdown.addEventListener('click', (e) => {
            e.stopPropagation();
            if (!eventPopover.classList.contains('hidden'))
eventPopover.classList.add('hidden');

            const isOpening = scanPopover.classList.contains('hidden');
            scanPopover.classList.toggle('hidden');

            if (isOpening) {
                // Load from database for this session
                loadShiftSettings();
            }
        });

        // Close both popovers when clicking outside
        document.addEventListener('click', (e) => {
            if (!eventPopover.contains(e.target) &&
!toggleEventDropdown.contains(e.target)) {

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        eventPopover.classList.add('hidden');
    }
    if (!scanPopover.contains(e.target) && !toggleScanDropdown.contains(e.target))
{
        scanPopover.classList.add('hidden');
    }
});

// Default events if none exist
if (events.length === 0) {
    events = [
        {
            id: Date.now(),
            name: "Barangay General Assembly",
            date: "2024-05-25",
            time: "08:00",
            method: "Biometric/Fingerprint",
            status: "Scheduled"
        }
    ];
    localStorage.setItem('custom_attendance_events', JSON.stringify(events));
}

function renderEvents() {
    const btn = document.getElementById('goToEventBtn');
    if (btn) {
        if (events.length > 0) {
            const latest = events[events.length - 1];
            btn.href = `${% url 'attendance:biometric_attendance'
}%?event_id=${latest.id}`;
            btn.classList.remove('hidden');
            btn.classList.add('flex');
        } else {
            btn.classList.add('hidden');
            btn.classList.remove('flex');
        }
    }
}

eventForm.onsubmit = (e) => {
    e.preventDefault();
    const newEvent = {
        id: Date.now(),
        name: document.getElementById('eventTitle').value,
        date: document.getElementById('eventDate').value,
        time: document.getElementById('eventTime').value,
        method: "Biometric/Fingerprint",
        status: "Scheduled"
    };
    events.push(newEvent);
    localStorage.setItem('custom_attendance_events', JSON.stringify(events));
    renderEvents();
    eventForm.reset();

    if (window.showNotification) {
        window.showNotification('Event scheduled successfully!', 'success');
    }
};
```

IoT-Enabled Barangay Resident and Attendance Management System

```
renderEvents();

// Settings Logic (Availability & Late Time)
const scannerStatusToggle = document.getElementById('scannerStatusToggle');
const lateGracePeriod = document.getElementById('lateGracePeriod');
const scheduleDay = document.getElementById('scheduleDay');
const targetTimeIn = document.getElementById('targetTimeIn');
const targetTimeOut = document.getElementById('targetTimeOut');
const amTimeOut = document.getElementById('amTimeOut');
const pmTimeIn = document.getElementById('pmTimeIn');
const saveSettingsBtn = document.getElementById('saveSettingsBtn');
const bulkApplyToggle = document.getElementById('bulkApplyToggle');
const dayOffBtn = document.getElementById('dayOffBtn');
let isDayOff = false;

// Custom Grace Period Logic
const customGraceContainer = document.getElementById('customGraceContainer');
const customGraceValue = document.getElementById('customGraceValue');
const customGraceUnit = document.getElementById('customGraceUnit');

if (lateGracePeriod && customGraceContainer) {
  lateGracePeriod.addEventListener('change', () => {
    if (lateGracePeriod.value === 'custom') {
      customGraceContainer.classList.remove('hidden');
      customGraceContainer.style.display = 'block';
    } else {
      customGraceContainer.classList.add('hidden');
      customGraceContainer.style.display = 'none';
    }
  });
}

// Per-Day Memory (now backed by DB)
let daySettings = {};
let currentDay = scheduleDay ? scheduleDay.value : 'mon';

// Load settings from database on open
async function loadShiftSettings() {
  try {
    const res = await fetch(`${% url "attendance:api_get_shift_settings" %}`);
    const data = await res.json();
    if (data.status === 'success') {
      daySettings = data.settings;
      updateUIForDay(currentDay);
    }
  } catch (err) { console.error('Failed to load shift settings:', err); }
}

function updateUIForDay(day) {
  const settings = daySettings[day] || {
    time_in: '08:00',
    time_out: '17:00',
    am_out: '12:00',
    pm_in: '13:00',
    is_day_off: false,
    grace_period: 15
  };

  if (targetTimeIn) targetTimeIn.value = settings.time_in;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if (targetTimeOut) targetTimeOut.value = settings.time_out;
if (amTimeOut) amTimeOut.value = settings.am_out;
if (pmTimeIn) pmTimeIn.value = settings.pm_in;

if (lateGracePeriod) {
  const standardPeriods = [0, 5, 10, 15, 20, 30, 45, 60];
  const graceVal = parseInt(settings.grace_period) || 0;

  if (standardPeriods.includes(graceVal)) {
    lateGracePeriod.value = graceVal;
    if (customGraceContainer) {
      customGraceContainer.classList.add('hidden');
      customGraceContainer.style.display = 'none';
    }
  } else {
    lateGracePeriod.value = 'custom';
    if (customGraceContainer) {
      customGraceContainer.classList.remove('hidden');
      customGraceContainer.style.display = 'block';
      customGraceValue.value = graceVal;
    }
  }
}
}
isDayOff = settings.is_day_off;
const dayOffToggleBtn = document.getElementById('dayOffToggleBtn');

if (dayOffToggleBtn) {
  if (isDayOff) {
    dayOffToggleBtn.classList.remove('bg-slate-50', 'text-slate-500',
'border-slate-200', 'hover:bg-slate-100');
    dayOffToggleBtn.classList.add('bg-red-500', 'text-white', 'border-
red-500', 'hover:bg-red-600');
  } else {
    dayOffToggleBtn.classList.add('bg-slate-50', 'text-slate-500',
'border-slate-200', 'hover:bg-slate-100');
    dayOffToggleBtn.classList.remove('bg-red-500', 'text-white', 'border-
red-500', 'hover:bg-red-600');
  }
}

const inputs = [targetTimeIn, targetTimeOut, amTimeOut, pmTimeIn,
lateGracePeriod, customGraceValue, customGraceUnit];
inputs.forEach(input => {
  if(input) {
    input.disabled = isDayOff;
    if(isDayOff) input.classList.add('opacity-50', 'cursor-not-allowed');
    else input.classList.remove('opacity-50', 'cursor-not-allowed');
  }
});
}

function syncGraceToAll() {
  const days = ['mon', 'tue', 'wed', 'thu', 'fri', 'sat', 'sun'];
  let grace = 15;
  if (lateGracePeriod.value === 'custom') {
    grace = parseFloat(customGraceValue.value) || 0;
    if (customGraceUnit.value === 'hours') grace *= 60;
    else if (customGraceUnit.value === 'seconds') grace /= 60;
  } else {
    grace = parseInt(lateGracePeriod.value) || 0;
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    }

    days.forEach(d => {
      if (!daySettings[d]) daySettings[d] = { is_day_off: false };
      daySettings[d].grace_period = grace;
    });
  }

  function syncCurrentToAll() {
    if (!bulkApplyToggle.checked) return;
    const days = ['mon', 'tue', 'wed', 'thu', 'fri', 'sat', 'sun'];

    days.forEach(d => {
      const existing = daySettings[d] || { is_day_off: false, grace_period: 15
};
      daySettings[d] = {
        ...existing,
        time_in: targetTimeIn.value,
        time_out: targetTimeOut.value,
        am_out: amTimeOut.value,
        pm_in: pmTimeIn.value
      };
    });
  }

  function handleTimeChange() {
    // Update current day memory
    if (currentDay !== 'all') {
      const existing = daySettings[currentDay] || { is_day_off: false,
grace_period: 15 };
      daySettings[currentDay] = {
        ...existing,
        time_in: targetTimeIn.value,
        time_out: targetTimeOut.value,
        am_out: amTimeOut.value,
        pm_in: pmTimeIn.value
      };
    }

    if (bulkApplyToggle.checked) {
      syncCurrentToAll();
    } else {
      // If we adjusted a specific day and bulk apply is NOT checked, we are
good
    }
  }

  function uncheckBulkOnSpecificChange() {
    if (bulkApplyToggle && bulkApplyToggle.checked) {
      // If user is manually changing an input, they might want to break the
sync
      // but only if they are not intentionally syncing.
      // However, user said: "once the user adjust the specific day then the
check will be uncheck"
      bulkApplyToggle.checked = false;
      if (window.showNotification) {
        window.showNotification('Apply to all disabled for manual override.',
'info');
      }
    }
  }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

// Real-time listeners for shift times
[targetTimeIn, targetTimeOut, amTimeOut, pmTimeIn].forEach(input => {
  if(input) {
    input.addEventListener('input', () => {
      handleTimeChange();
    });
    // When focusing/clicking to change, we should uncheck bulk apply to allow
individual day override
    input.addEventListener('mousedown', uncheckBulkOnSpecificChange);
    input.addEventListener('keydown', (e) => {
      // Ignore navigation keys
      if (![9, 13, 16, 17, 18, 20, 27, 37, 38, 39, 40].includes(e.keyCode))
{
        uncheckBulkOnSpecificChange();
      }
    });
  }
});

// Real-time listeners for late period (always global)
[lateGracePeriod, customGraceValue, customGraceUnit].forEach(input => {
  if(input) {
    input.addEventListener('input', syncGraceToAll);
    input.addEventListener('change', syncGraceToAll);
  }
});

if (scheduleDay) {
  scheduleDay.addEventListener('change', () => {
    // Update memory for the current day we are leaving before switching
    if (currentDay !== 'all') {
      let grace = 15;
      if (lateGracePeriod.value === 'custom') {
        grace = parseFloat(customGraceValue.value) || 0;
        if (customGraceUnit.value === 'hours') grace *= 60;
        else if (customGraceUnit.value === 'seconds') grace /= 60;
      } else {
        grace = parseInt(lateGracePeriod.value) || 0;
      }

      daySettings[currentDay] = {
        time_in: targetTimeIn.value,
        time_out: targetTimeOut.value,
        am_out: amTimeOut.value,
        pm_in: pmTimeIn.value,
        is_day_off: isDayOff,
        grace_period: grace
      };
    }

    currentDay = scheduleDay.value;
    updateUIForDay(currentDay);
  });
}

const dayOffToggleBtn = document.getElementById('dayOffToggleBtn');
if (dayOffToggleBtn) {
  dayOffToggleBtn.addEventListener('click', () => {
```


IoT-Enabled Barangay Resident and Attendance Management System

```
isDayOff = !isDayOff;

if (isDayOff) {
  dayOffToggleBtn.classList.remove('bg-slate-50', 'text-slate-500',
'border-slate-200', 'hover:bg-slate-100');
  dayOffToggleBtn.classList.add('bg-red-500', 'text-white', 'border-
red-500', 'hover:bg-red-600');
} else {
  dayOffToggleBtn.classList.add('bg-slate-50', 'text-slate-500',
'border-slate-200', 'hover:bg-slate-100');
  dayOffToggleBtn.classList.remove('bg-red-500', 'text-white', 'border-
red-500', 'hover:bg-red-600');
}

const inputs = [targetTimeIn, targetTimeOut, amTimeOut, pmTimeIn,
lateGracePeriod, customGraceValue, customGraceUnit];
inputs.forEach(input => {
  if(input) {
    input.disabled = isDayOff;
    if(isDayOff) input.classList.add('opacity-50', 'cursor-not-
allowed');
    else input.classList.remove('opacity-50', 'cursor-not-allowed');
  }
});

// Update memory for current day but DO NOT SAVE to localStorage yet
if (currentDay !== 'all') {
  let grace = 15;
  if (lateGracePeriod.value === 'custom') {
    grace = parseFloat(customGraceValue.value) || 0;
  } else {
    grace = parseInt(lateGracePeriod.value) || 0;
  }

  daySettings[currentDay] = {
    time_in: targetTimeIn.value,
    time_out: targetTimeOut.value,
    am_out: amTimeOut.value,
    pm_in: pmTimeIn.value,
    is_day_off: isDayOff,
    grace_period: grace
  };
}
});
}

if (bulkApplyToggle) {
  bulkApplyToggle.addEventListener('change', (e) => {
    if (e.target.checked) {
      syncCurrentToAll();
      if (window.showNotification) {
        window.showNotification('Settings applied to all days in this
session.', 'info');
      }
    }
  });
}

// Load saved settings initial
const savedSettings = JSON.parse(localStorage.getItem('attendance_settings')) ||
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'{"active": true, "late_time": "08:15", "day": "all", "time_in": "08:00", "time_out":  
"17:00"}');  
  
    // Removed scannerStatusToggle check since it was removed from UI  
    if (lateGracePeriod) {  
        updateUIForDay(currentDay);  
    }  
  
    if (saveSettingsBtn) {  
        saveSettingsBtn.addEventListener('click', (e) => {  
            if (e) e.preventDefault();  
            const finalDay = (bulkApplyToggle && bulkApplyToggle.checked) ? 'all' :  
(scheduleDay ? scheduleDay.value : 'all');  
  
            let grace = 15;  
            if (lateGracePeriod) {  
                if (lateGracePeriod.value === 'custom') {  
                    grace = parseFloat(customGraceValue.value) || 0;  
                    if (customGraceUnit && customGraceUnit.value === 'hours') grace *=  
60;  
                    else if (customGraceUnit && customGraceUnit.value === 'seconds')  
grace /= 60;  
                } else {  
                    grace = parseInt(lateGracePeriod.value) || 0;  
                }  
            }  
  
            const settings = {  
                active: true,  
                grace_period: grace,  
                day: finalDay,  
                time_in: targetTimeIn ? targetTimeIn.value : '08:00',  
                time_out: targetTimeOut ? targetTimeOut.value : '17:00',  
                am_out: amTimeOut ? amTimeOut.value : '12:00',  
                pm_in: pmTimeIn ? pmTimeIn.value : '13:00',  
                is_day_off: isDayOff  
            };  
  
            if (settings.time_in) {  
                const [h, m] = settings.time_in.split(':').map(Number);  
                const date = new Date();  
                date.setHours(h, m + settings.grace_period, 0);  
                settings.late_time = date.getHours().toString().padStart(2, '0') + ':'  
+ date.getMinutes().toString().padStart(2, '0');  
            } else {  
                settings.late_time = '08:15';  
            }  
  
            // Save to database  
            fetch('% url "attendance:api_save_shift_settings" %}', {  
                method: 'POST',  
                headers: {  
                    'Content-Type': 'application/json',  
                    'X-CSRFToken': '{{ csrf_token }}'  
                },  
                body: JSON.stringify({  
                    settings: daySettings,  
                    bulk_apply: bulkApplyToggle && bulkApplyToggle.checked,  
                    source_day: currentDay  
                })  
            })  
        })  
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    })
    .then(res => res.json())
    .then(data => {
      if (data.status === 'success') {
        saveSettingsBtn.innerHTML = '<i data-lucide="check" class="w-3.5
h-3.5"></i> Saved!';
        if (window.lucide) lucide.createIcons();

        saveSettingsBtn.classList.remove('bg-slate-800', 'hover:bg-
slate-900', 'border-slate-700');
        saveSettingsBtn.classList.add('bg-emerald-600', 'hover:bg-
emerald-700', 'border-emerald-600', 'text-white');

        setTimeout(() => {
          window.location.assign(window.location.href);
        }, 800);
      } else {
        alert('Error saving settings: ' + data.message);
      }
    })
    .catch(err => {
      console.error('Save error:', err);
      alert('Failed to connect to server.');
```

```
    });
  });
}

});
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/attendance/dtr.html

```
{% extends 'base.html' %}

{% block title %}Daily Time Record - Attendance System{% endblock %}

{% block extra_head %}
<script src="https://cdn.tailwindcss.com"></script>
{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / <a href="{% url 'attendance:dashboard' %}">Attendance</a> / DTR Summary</span>
{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Daily Time Record (DTR)</h1>
    <span>Monthly attendance summary for all officials</span>
  </div>
  <div class="header-actions">
    <form method="GET" class="filter-form" style="display: flex; gap: 10px; flex-wrap: wrap; align-items: center;">
      <select name="month" class="form-control" style="min-width: 120px;">
        <option value="1" {% if month == 1 %}>selected{% endif %}>January</option>
        <option value="2" {% if month == 2 %}>selected{% endif %}>February</option>
        <option value="3" {% if month == 3 %}>selected{% endif %}>March</option>
        <option value="4" {% if month == 4 %}>selected{% endif %}>April</option>
        <option value="5" {% if month == 5 %}>selected{% endif %}>May</option>
        <option value="6" {% if month == 6 %}>selected{% endif %}>June</option>
        <option value="7" {% if month == 7 %}>selected{% endif %}>July</option>
        <option value="8" {% if month == 8 %}>selected{% endif %}>August</option>
        <option value="9" {% if month == 9 %}>selected{% endif %}>September</option>
        <option value="10" {% if month == 10 %}>selected{% endif %}>October</option>
        <option value="11" {% if month == 11 %}>selected{% endif %}>November</option>
        <option value="12" {% if month == 12 %}>selected{% endif %}>December</option>
      </select>
      <input type="number" name="year" value="{{ year }}" class="form-control" style="width: 90px;">
      <button type="submit" class="btn btn-primary">
        <i class="fas fa-filter"></i> Filter
      </button>
    </form>
  </div>
</div>

<div class="card">
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th style="width: 35%;">Official Profile</th>
          <th>Days Present</th>
          <th>Late</th>
          <th>Absent</th>
          <th>Total Hours</th>
          <th>Actions</th>
        </tr>
      </thead>
    </table>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

</thead>
<tbody>
  {% for item in dtr_summary %}
  <tr>
    <td>
      <div style="display: flex; flex-direction: column;">
        <strong>{{ item.official.resident.full_name }}</strong>
        <span style="font-size: 0.75rem; color: #64748b; text-transform: uppercase; letter-spacing: 0.05em;">{{ item.official.get_position_display }}</span>
      </div>
    </td>
    <td><span class="text-[9px] bg-emerald-50 text-emerald-600 px-2 py-0.5 rounded font-black tracking-widest border border-emerald-100 uppercase">{{ item.present }}</span></td>
    <td><span class="text-[9px] bg-amber-50 text-amber-600 px-2 py-0.5 rounded font-black tracking-widest border border-amber-100 uppercase">{{ item.late }}</span></td>
    <td><span class="text-[9px] bg-red-50 text-red-600 px-2 py-0.5 rounded font-black tracking-widest border border-red-100 uppercase">{{ item.absent }}</span></td>
    <td><strong>{{ item.total_hours }} hrs</strong></td>
    <td>
      <div class="btn-group">
        <a href="{% url 'attendance:dtr_detail' item.official.pk %}?month={{ month }}&year={{ year }}"
          class="btn btn-outline btn-icon" title="View Detailed DTR">
            <i class="fas fa-eye"></i>
          </a>
        <a href="{% url 'attendance:dtr_print' item.official.pk %}?month={{ month }}&year={{ year }}"
          class="btn btn-outline btn-icon" title="Print DTR" target="_blank">
            <i class="fas fa-print"></i>
          </a>
      </div>
    </td>
  </tr>
  {% empty %}
  <tr>
    <td colspan="7" class="text-center">No attendance records found for this period.</td>
  </tr>
  {% endfor %}
</tbody>
</table>
</div>
</div>

<style>
.badge-success {
  background: rgba(16, 185, 129, 0.1);
  color: #10b981;
}

.badge-warning {
  background: rgba(245, 158, 11, 0.1);
  color: #f59e0b;
}

.badge-danger {
  background: rgba(239, 68, 68, 0.1);
  color: #ef4444;
}

```

IoT-Enabled Barangay Resident and Attendance Management System

```
</style>
{% endblock %}
```

File: templates/attendance/dtr_detail.html

```
{% extends 'base.html' %}
{% load attendance_tags %}

{% block title %}DTR Details - {{ official.resident.full_name }}{% endblock %}

{% block extra_head %}
<script src="https://cdn.tailwindcss.com"></script>
{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'attendance:dashboard' %}">Attendance</a> / <a href="{% url 'attendance:dtr' %}">DTR Summary</a> /
  {{ official.resident.full_name }}</span>
{% endblock %}

{% block content %}
{% if request.GET.from == 'report' %}
  {% url 'attendance:daily_report' request.GET.report_date as back_url %}
{% else %}
  {% url 'attendance:dashboard' as back_url %}
{% endif %}
{% include 'components/back_button.html' with back_url=back_url back_label='Back' %}
<div class="page-header">
  <div class="page-title" style="display: flex; align-items: center; gap: 15px;">
    {% if official.resident.photo %}
      
    {% elif official.user.avatar %}
      
    {% else %}
      
    {% endif %}
    <div style="display: flex; flex-direction: column;">
      <h1 style="margin-bottom: 0; font-size: 1.5rem; font-weight: 900; color:
#1e293b;">{{ official.resident.full_name }}</h1>
      <span style="font-size: 0.7rem; font-weight: 800; color: #64748b; text-transform:
uppercase; letter-spacing: 0.05em;">{{ official.get_position_display }}</span>
      <span style="font-size: 0.85rem; color: #94a3b8; margin-top: 6px;">Detailed DTR
&mdash; For the month of {{ month_name }} {{ year }}</span>
    </div>
  </div>
  <div class="header-actions">
    <a href="{% url 'attendance:dtr_print' official.pk %}?month={{ month }}&year={{ year
}}" target="_blank"
      class="btn btn-secondary">
      <i class="fas fa-print"></i> Print DTR
    </a>
  </div>
</div>

<div class="row" style="display: flex; gap: 25px; margin-bottom: 25px;">
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div class="col" style="flex: 1;">
  <div class="card bg-primary text-white p-4 text-center">
    <h4 class="mb-1">Total Hours</h4>
    <div class="display-4 font-weight-bold">{{ total_hours }}</div>
    <span class="small opacity-75">For the month of {{ month_name }} {{ year }}</span>
  </div>
</div>
<div class="col" style="flex: 1;">
  <div class="card p-4 text-center">
    <h4 class="text-muted mb-1">Days Present</h4>
    <div class="display-4 font-weight-bold">{{ logs.count }}</div>
    <span class="small text-muted">active duty days</span>
  </div>
</div>
</div>

<div class="card">
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th style="text-align: center;">Date</th>
          <th style="text-align: center;">AM IN</th>
          <th style="text-align: center;">AM OUT</th>
          <th style="text-align: center;">PM IN</th>
          <th style="text-align: center;">PM OUT</th>
          <th style="text-align: center;">Total Hours</th>
        </tr>
      </thead>
      <tbody>
        {% for log in logs %}
        <tr>
          <td class="text-center font-bold text-slate-800">{{ log.date|date:"M d, Y" }}</td>
          <td class="text-center">
            {% if log.am_in %}
            <div style="display: flex; flex-direction: column; gap: 4px; align-items: center;">
              <span class="font-bold text-slate-700">{{ log.am_in|time:"h:i A" }}</span>
              {% get_lateness log.am_in log.shift_am_in log.grace_period as am_late %}
              {% if am_late %}
              <span class="text-[9px] bg-amber-50 text-amber-600 px-2 py-0.5 rounded font-black tracking-widest border border-amber-100 uppercase">Late ({{ am_late }})</span>
              {% else %}
              <span class="text-[9px] bg-emerald-50 text-emerald-600 px-2 py-0.5 rounded font-black tracking-widest border border-emerald-100 uppercase">Present</span>
              {% endif %}
            </div>
            {% elif log.date|is_absent_visible:log.shift_am_out %}
            <span class="text-[9px] bg-red-50 text-red-600 px-2 py-0.5 rounded font-black tracking-widest border border-red-100 uppercase">Absent</span>
            {% endif %}
          </td>
          <td class="text-center">
            {% if log.am_out %}
            <div style="display: flex; flex-direction: column; gap: 4px; align-items: center;">
              <span class="font-bold text-slate-700">{{ log.am_out|time:"h:i A" }}</span>
            </div>
            {% endif %}
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
  </div>
</div>

```


IoT-Enabled Barangay Resident and Attendance Management System

```

        <span class="text-[9px] bg-blue-50 text-blue-600 px-2 py-0.5 rounded
font-black tracking-widest border border-blue-100 uppercase">Done</span>
    </div>
    {% elif log.am_in %}
        <span class="text-muted italic text-[10px]">---</span>
    {% elif log.date|is_absent_visible:log.shift_am_out %}
        <span class="text-[9px] bg-red-50 text-red-600 px-2 py-0.5 rounded font-
black tracking-widest border border-red-100 uppercase">Absent</span>
    {% endif %}
</td>
<td class="text-center">
    {% if log.pm_in %}
        <div style="display: flex; flex-direction: column; gap: 4px; align-items:
center;">
            <span class="font-bold text-slate-700">{{ log.pm_in|time:"h:i A"
}}</span>
            {% get_lateness log.pm_in log.shift_pm_in log.grace_period as pm_late
%}
            {% if pm_late %}
                <span class="text-[9px] bg-amber-50 text-amber-600 px-2 py-0.5 rounded
font-black tracking-widest border border-amber-100 uppercase">Late ({{ pm_late }})</span>
            {% else %}
                <span class="text-[9px] bg-emerald-50 text-emerald-600 px-2 py-0.5
rounded font-black tracking-widest border border-emerald-100 uppercase">Present</span>
            {% endif %}
        </div>
    {% elif log.date|is_absent_visible:log.shift_pm_out %}
        <span class="text-[9px] bg-red-50 text-red-600 px-2 py-0.5 rounded font-
black tracking-widest border border-red-100 uppercase">Absent</span>
    {% endif %}
</td>
<td class="text-center">
    {% if log.pm_out %}
        <div style="display: flex; flex-direction: column; gap: 4px; align-items:
center;">
            <span class="font-bold text-slate-700">{{ log.pm_out|time:"h:i A"
}}</span>
            <span class="text-[9px] bg-blue-50 text-blue-600 px-2 py-0.5 rounded
font-black tracking-widest border border-blue-100 uppercase">Done</span>
        </div>
    {% elif log.pm_in %}
        <span class="text-muted italic text-[10px]">---</span>
    {% elif log.date|is_absent_visible:log.shift_pm_out %}
        <span class="text-[9px] bg-red-50 text-red-600 px-2 py-0.5 rounded font-
black tracking-widest border border-red-100 uppercase">Absent</span>
    {% endif %}
</td>
<td class="text-center font-weight-bold text-primary">{{ log.hours_worked
}}</td>
</tr>
{% empty %}
<tr>
    <td colspan="6" class="text-center">No logs found for this official in the
selected period.</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<style>
  .bg-primary {
    background: var(--primary) !important;
    color: white !important;
  }

  .display-4 {
    font-size: 2.5rem;
    line-height: 1.2;
  }

  .opacity-75 {
    opacity: 0.75;
  }

  .badge-success {
    background: rgba(16, 185, 129, 0.1);
    color: #10b981;
  }

  .badge-warning {
    background: rgba(245, 158, 11, 0.1);
    color: #f59e0b;
  }

  .badge-danger {
    background: rgba(239, 68, 68, 0.1);
    color: #ef4444;
  }

  .badge-info {
    background: rgba(59, 130, 246, 0.1);
    color: #3b82f6;
  }
</style>
{% endblock %}
```

File: templates/attendance/dtr_print.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>DTR - {{ official.resident.full_name }} ({{ month }}/{{ year }})</title>
  <style>
    body {
      font-family: 'Courier New', Courier, monospace;
      padding: 40px;
      color: #000;
      line-height: 1.4;
    }

    .header {
      text-align: center;
      margin-bottom: 30px;
      border-bottom: 2px solid #000;
      padding-bottom: 20px;
    }

    .header h1 {
      margin: 0;
      text-transform: uppercase;
      font-size: 24px;
    }

    .header p {
      margin: 5px 0 0;
    }

    .info-grid {
      display: flex;
      justify-content: space-between;
      margin-bottom: 30px;
    }

    .info-col {
      flex: 1;
    }

    .info-item {
      margin-bottom: 10px;
    }

    .label {
      font-weight: bold;
      font-size: 12px;
      margin-bottom: 2px;
    }

    .value {
      font-size: 16px;
      border-bottom: 1px solid #000;
      display: inline-block;
      min-width: 200px;
    }
  </style>
</head>

<body>
  <div class="header">
    <h1>DTR</h1>
    <p></p>
  </div>

  <div class="info-grid">
    <div class="info-col">
      <div class="info-item">
        <div class="label"></div>
        <div class="value"></div>
      </div>
    </div>
  </div>
</body>
</html>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 20px;
}

th,
td {
  border: 1px solid #000;
  padding: 10px;
  text-align: center;
  font-size: 14px;
}

th {
  background-color: #f2f2f2;
}

.footer {
  margin-top: 50px;
  display: flex;
  justify-content: space-between;
}

.sig-box {
  text-align: center;
  width: 250px;
}

.sig-line {
  border-top: 1px solid #000;
  margin-top: 40px;
  padding-top: 5px;
  font-weight: bold;
}

@media print {
  .no-print {
    display: none;
  }

  body {
    padding: 0;
  }
}
</style>
</head>

<body onload="{% if not request.GET.no_print %}window.print(){% endif %}">
  <div class="header">
    <h1>Republic of the Philippines</h1>
    <p>Province of Example | Municipality of Example</p>
    <p><strong>BARANGAY MANAGEMENT SYSTEM</strong></p>
    <h2>DAILY TIME RECORD</h2>
  </div>

  <div class="info-grid">
    <div class="info-col">
      <div class="info-item">
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <div class="label">NAME:</div>
        <div class="value"><strong>{{ official.resident.full_name|upper }}</strong></div>
    </div>
    <div class="info-item">
        <div class="label">POSITION:</div>
        <div class="value">{{ official.get_position_display|upper }}</div>
    </div>
</div>
<div class="info-col" style="text-align: right;">
    <div class="info-item">
        <div class="label">FOR THE MONTH OF:</div>
        <div class="value">{{ month_name|upper }} {{ year }}</div>
    </div>
    <div class="info-item">
        <div class="label">TOTAL HOURS:</div>
        <div class="value">{{ total_hours|floatformat:2 }} HOURS</div>
    </div>
</div>
</div>

<div style="overflow-x: auto; margin-top: 20px;">
<table style="margin-top: 0;">
    <thead>
        <tr>
            <th rowspan="2">DAY</th>
            <th colspan="2">AM</th>
            <th colspan="2">PM</th>
            <th rowspan="2">HOURS</th>
            <th rowspan="2">REMARKS</th>
        </tr>
        <tr>
            <th>IN</th>
            <th>OUT</th>
            <th>IN</th>
            <th>OUT</th>
        </tr>
    </thead>
    <tbody>
        {% for log in logs %}
        <tr>
            <td>{{ log.date|date:"d" }}</td>
            <td>{{ log.am_in|time:"h:i A"|default:"---" }}</td>
            <td>{{ log.am_out|time:"h:i A"|default:"---" }}</td>
            <td>{{ log.pm_in|time:"h:i A"|default:"---" }}</td>
            <td>{{ log.pm_out|time:"h:i A"|default:"---" }}</td>
            <td>{{ log.hours_worked|floatformat:2 }}</td>
            <td>{{ log.get_status_display }}</td>
        </tr>
        {% endfor %}
        <!-- This is a simplified print DTR, in reality it should list all 31 days -->
    </tbody>
</table>
</div>

<div class="footer">
    <div class="sig-box">
        <div class="sig-line">Official's Signature</div>
    </div>
    <div class="sig-box">
        <div class="sig-line">Barangay Captain / Head</div>
    </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
    </div>
</div>

<div class="no-print" style="margin-top: 30px; text-align: center;">
    <button onclick="window.print()" style="padding: 10px 20px; cursor: pointer;">Print
Again</button>
    <button onclick="window.close()" style="padding: 10px 20px; cursor: pointer;">Close
Window</button>
</div>
</body>

</html>
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/attendance/enroll.html

```
{% extends 'base.html' %}

{% block title %}Face Enrollment - Attendance System{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'attendance:dashboard' %}">Attendance</a> / Face Enrollment</span>
{% endblock %}

{% block content %}
{% url 'attendance:dashboard' as attendance_dashboard_url %}
{% include 'components/back_button.html' with back_url=attendance_dashboard_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>Face Enrollment</h1>
        <span>Register official's face for automated clock-in/out</span>
    </div>
</div>

<div class="row" style="display: flex; gap: 25px;">
    <div class="col-md-5" style="flex: 0 0 450px;">
        <div class="card">
            <div class="card-header">
                <h3>Enroll Official</h3>
            </div>
            <div class="card-body">
                <form method="POST" enctype="multipart/form-data" class="modern-form">
                    {% csrf_token %}
                    <div class="form-group mb-4">
                        <label>Official*</label>
                        <select name="official" class="form-control" required>
                            <option value="">Select Official...</option>
                            {% for official in officials %}
                                <option value="{{ official.pk }}">{{ official.resident.last_name }}, {{
official.resident.first_name }}{% if official.resident.middle_name %} {{
official.resident.middle_name }}{% endif %}{% if official.resident.suffix %} {{
official.resident.suffix }}{% endif %} ? {{ official.get_position_display }}{% if
official.pk in enrolled %} [ENROLLED]{% endif %}</option>
                            {% endfor %}
                        </select>
                    </div>

                    <div class="form-group mb-4">
                        <label>Face Photo (Front View)*</label>
                        <input type="file" name="photo" class="form-control" accept="image/*" required
id="photoInput">
                        <small class="text-muted mt-2 d-block">Upload a clear, front-facing photo for
facial
                            recognition.</small>
                    </div>

                    <div id="photoPreview" class="mt-3 text-center" style="display: none;">
                        
                    </div>
                </form>
            </div>
        </div>
    </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="form-footer mt-4">
  <button type="submit" class="btn btn-primary btn-block">
    <i class="fas fa-user-plus"></i> Enroll Face
  </button>
</div>
</form>
</div>
</div>
</div>

<div class="col-md-7" style="flex: 1;">
  <div class="card">
    <div class="card-header">
      <h3>Enrollment Status</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Official</th>
            <th>Status</th>
            <th>Actions</th>
          </tr>
        </thead>
        <tbody>
          {% for official in officials %}
          <tr>
            <td>
              <strong>{{ official.resident.full_name }}</strong>
              <div class="small text-muted">{{ official.get_position_display }}</div>
            </td>
            <td>
              {% if official.pk in enrolled %}
              <span class="badge badge-success">
                <i class="fas fa-check-circle"></i> Enrolled
              </span>
              {% else %}
              <span class="badge badge-warning">
                <i class="fas fa-clock"></i> Pending
              </span>
              {% endif %}
            </td>
            <td>
              {% if official.pk in enrolled %}
              <button class="btn btn-outline-danger btn-icon" title="Remove enrollment">
                <i class="fas fa-user-minus"></i>
              </button>
              {% endif %}
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
</div>
</div>

<script>
  document.getElementById('photoInput').onchange = evt => {
```


IoT-Enabled Barangay Resident and Attendance Management System

```
const [file] = evt.target.files;
if (file) {
  document.getElementById('previewImg').src = URL.createObjectURL(file);
  document.getElementById('photoPreview').style.display = 'block';
}
}
</script>

<style>
.mb-4 {
  margin-bottom: 24px;
}

.mt-4 {
  margin-top: 24px;
}

.btn-block {
  display: block;
  width: 100%;
}

.badge-success {
  background: rgba(16, 185, 129, 0.1);
  color: #10b981;
}

.badge-warning {
  background: rgba(245, 158, 11, 0.1);
  color: #f59e0b;
}

.btn-outline-danger {
  color: var(--danger);
  border-color: var(--danger);
}
</style>
{% endblock %}
```

File: templates/attendance/history_calendar.html

```
{% extends 'base.html' %}
{% load static %}
{% load attendance_tags %}

{% block title %}Attendance History - Calendar View{% endblock %}

{% block extra_head %}
<script src="https://cdn.tailwindcss.com"></script>
<style>
    .calendar-grid {
        display: grid;
        grid-template-columns: repeat(7, 1fr);
        gap: 1px;
        background-color: #e2e8f0;
        border: 1px solid #e2e8f0;
        border-radius: 1rem;
        overflow: hidden;
    }
    .calendar-day {
        min-height: 120px;
        background-color: white;
        padding: 0.75rem;
        transition: all 0.2s;
    }
    .calendar-day:hover:not(.empty) {
        background-color: #cbd5e1; /* Increased opacity/darkness for hover */
        cursor: pointer;
    }
    .calendar-day.today {
        background-color: #dcfce7; /* Increased opacity green (bg-green-100) */
        border: 2px solid #22c55e; /* Add a green border for more visibility */
    }
    .status-dot {
        width: 6px;
        height: 6px;
        border-radius: 50%;
    }
</style>
{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / <a href="{% url 'attendance:dashboard' %}"
class="hover:text-emerald-600">Attendance</a> / History</span>
{% endblock %}

{% block content %}
<div class="max-w-6xl mx-auto p-4">
    <!-- Calendar Header -->
    {% url 'attendance:dashboard' as dashboard_url %}
    {% include 'components/back_button.html' with back_url=dashboard_url back_label='Back' %}

    <!-- Calendar Header -->
    <div class="flex flex-col md:flex-row md:items-center justify-between mb-8 mt-4
gap-4">
        <div>
            <h1 class="text-2xl font-black text-slate-900 tracking-tight">{{ month_name }}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{{ year }}
    <p class="text-slate-500 font-medium">Browse historical attendance logs</p>
</div>

<div class="flex flex-wrap items-center gap-3">
    <!-- Month/Year Jump -->
    <form action="" method="GET" class="flex items-center gap-2 bg-white p-1.5
rounded-xl border border-slate-200 shadow-sm">
        <select name="month" onchange="this.form.submit()" class="bg-transparent
border-none focus:ring-0 text-sm font-bold text-slate-700 outline-none">
            {% for m in 1|range_to:13 %}
                <option value="{{ m }}" {% if m == month %}selected{% endif %}>{{
m|month_name_short }}</option>
            {% endfor %}
        </select>
        <div class="w-px h-4 bg-slate-200"></div>
        <select name="year" onchange="this.form.submit()" class="bg-transparent
border-none focus:ring-0 text-sm font-bold text-slate-700 outline-none">
            {% for y in year_range %}
                <option value="{{ y }}" {% if y == year %}selected{% endif %}>{{ y
}}</option>
            {% endfor %}
        </select>
    </form>

    <div class="flex items-center gap-1 bg-white p-1 rounded-xl border border-
slate-200 shadow-sm">
        <a href="?month={{ prev_month.month }}&year={{ prev_month.year }}"
class="p-1.5 hover:bg-slate-100 rounded-lg transition text-slate-600" title="Previous
Month">
            <i data-lucide="chevron-left" class="w-4 h-4"></i>
        </a>
        <a href="{% url 'attendance:daily_report' today|date:'Y-m-d' %}"
class="px-3 py-1.5 text-xs font-black text-emerald-600 hover:text-emerald-700 bg-
emerald-50 hover:bg-emerald-100 rounded-lg uppercase tracking-widest transition">
            View Today
        </a>
        <a href="?month={{ next_month.month }}&year={{ next_month.year }}"
class="p-1.5 hover:bg-slate-100 rounded-lg transition text-slate-600" title="Next Month">
            <i data-lucide="chevron-right" class="w-4 h-4"></i>
        </a>
    </div>
</div>
</div>

<!-- Calendar Grid -->
<div class="calendar-grid shadow-xl">
    <div class="bg-emerald-600 py-3 text-center text-[10px] font-black text-white
uppercase tracking-widest">Sun</div>
    <div class="bg-emerald-600 py-3 text-center text-[10px] font-black text-white
uppercase tracking-widest">Mon</div>
    <div class="bg-emerald-600 py-3 text-center text-[10px] font-black text-white
uppercase tracking-widest">Tue</div>
    <div class="bg-emerald-600 py-3 text-center text-[10px] font-black text-white
uppercase tracking-widest">Wed</div>
    <div class="bg-emerald-600 py-3 text-center text-[10px] font-black text-white
uppercase tracking-widest">Thu</div>
    <div class="bg-emerald-600 py-3 text-center text-[10px] font-black text-white
uppercase tracking-widest">Fri</div>
    <div class="bg-emerald-600 py-3 text-center text-[10px] font-black text-white
```

IoT-Enabled Barangay Resident and Attendance Management System

```
uppercase tracking-widest">Sat</div>

<!-- Days -->
{% for week in month_days %}
    {% for day in week %}
        {% if day == 0 %}
            <div class="calendar-day empty bg-slate-50/50"></div>
        {% else %}
            {% with day_data=month_data|get_item:day %}
                <a href="{% url 'attendance:daily_report'
year|add_str:'-'|add_str:month|add_str:'-'|add_str:day %}"
                    class="calendar-day block no-underline {% if today.day == day and
today.month == month and today.year == year %}today{% endif %}"
                    {% if day_data.is_day_off %}onclick="event.preventDefault();
showNotification('Scheduled Day Off. No attendance records available.', 'info');" {% endif
%}>

                    <div class="flex justify-between items-start mb-4">
                        <span class="text-2xl font-black {% if today.day == day and
today.month == month and today.year == year %}text-emerald-600{% else %}text-slate-700{%
endif %}">

                            {{ day }}
                        </span>
                        <button type="button" onclick="event.preventDefault();
event.stopPropagation(); toggleDayOff('{{ year }}-{{ month|stringformat:'02d' }}-{{
day|stringformat:'02d' }}')" class="text-slate-300 hover:text-blue-600 transition p-1
rounded-full hover:bg-slate-100" title="Toggle Day Off">

                            <i data-lucide="{% if day_data.is_day_off %}sun{% else
%}calendar-off{% endif %}" class="w-3.5 h-3.5"></i>
                        </button>
                    </div>

                    {% if day_data.is_day_off %}
                        <div class="h-full flex items-center justify-center">
                            <span class="px-2 py-0.5 text-[9px] font-black bg-slate-100 text-
slate-400 rounded-full uppercase tracking-widest">Day Off</span>
                        </div>
                    {% elif day_data.is_future %}
                        <div class="h-full flex items-center justify-center">
                            <!-- Empty for future dates -->
                        </div>
                    {% elif day_data %}
                        <div class="mt-auto pt-2">
                            <span class="text-[9px] font-black text-emerald-600 uppercase
tracking-widest flex items-center gap-1 group">
                                View Details <i data-lucide="arrow-right" class="w-2.5 h-2.5
transition-transform group-hover:translate-x-0.5"></i>
                            </span>
                        </div>
                    {% else %}
                        <div class="h-full flex items-center justify-center">
                            <span class="text-[9px] font-bold text-slate-300 uppercase">No
Data</span>
                        </div>
                    {% endif %}
                </a>
            {% endwith %}
        {% endif %}
    {% endfor %}
{% endfor %}
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</div>

<script>
  document.addEventListener('DOMContentLoaded', function() {
    lucide.createIcons();
  });

  function toggleDayOff(dateStr) {
    if (!confirm('Are you sure you want to change the work status for this date?'))
      return;

    const formData = new FormData();
    formData.append('date', dateStr);
    formData.append('csrfmiddlewaretoken', '{{ csrf_token }}');

    fetch(`${% url 'attendance:api_toggle_special_date' %}`, {
      method: 'POST',
      body: formData
    })
    .then(response => response.json())
    .then(data => {
      if (data.status === 'success') {
        location.reload();
      } else {
        alert('Error: ' + data.message);
      }
    })
    .catch(error => {
      console.error('Error:', error);
      alert('An error occurred. Please try again.');
```

```
});
  }
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/attendance/public_scan.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Public DTR Scan - Barangay Management System{% endblock %}

{% block extra_head %}
<style>
    .public-scan-card {
        max-width: 600px;
        margin: 50px auto;
        text-align: center;
        background: #1e293b;
        border-radius: 20px;
        padding: 40px;
        box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.5);
    }
    .scan-icon-container {
        width: 150px;
        height: 150px;
        margin: 0 auto 30px;
        background: rgba(79, 70, 229, 0.1);
        border-radius: 50%;
        display: flex;
        align-items: center;
        justify-content: center;
        border: 4px solid #4f46e5;
        transition: all 0.3s;
    }
    .scan-icon-container.scanning {
        border-color: #10b981;
        box-shadow: 0 0 20px rgba(16, 185, 129, 0.4);
    }
    .scan-icon-container i {
        font-size: 70px;
        color: #4f46e5;
    }
    .scan-icon-container.scanning i {
        color: #10b981;
        animation: pulse 1.5s infinite;
    }
    @keyframes pulse {
        0% { transform: scale(1); }
        50% { transform: scale(1.1); }
        100% { transform: scale(1); }
    }
    .status-msg {
        margin-top: 20px;
        padding: 15px;
        border-radius: 10px;
        font-weight: 600;
    }
</style>
{% endblock %}

{% block content %}
<div class="public-scan-card">
    <h1 class="text-white mb-2">Daily Time Record</h1>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<p class="text-slate-400 mb-5">Place your registered finger on the scanner to Clock  
In/Out</p>

<div id="scan-container" class="scan-icon-container">
  <i class="fas fa-fingerprint"></i>
</div>

<div id="status-display" class="status-msg d-none"></div>

<div class="mt-5">
  <button id="btn-start-scan" class="btn btn-primary btn-lg px-5">
    <i class="fas fa-play mr-2"></i> Start Scanner
  </button>
</div>

<div id="result-panel" class="mt-4 d-none">
  <div class="card bg-slate-900 border-slate-800 p-3">
    <h4 id="result-name" class="text-white mb-0">---</h4>
    <p id="result-time" class="text-success mb-0">---</p>
  </div>
</div>
</div>
{% endblock %}

{% block extra_js %}
<script>
  document.addEventListener('DOMContentLoaded', function() {
    const btnStart = document.getElementById('btn-start-scan');
    const scanContainer = document.getElementById('scan-container');
    const statusDisplay = document.getElementById('status-display');
    const resultPanel = document.getElementById('result-panel');
    const resultName = document.getElementById('result-name');
    const resultTime = document.getElementById('result-time');

    btnStart.addEventListener('click', function() {
      btnStart.disabled = true;
      scanContainer.classList.add('scanning');
      if (window.showNotification) {
        window.showNotification('Scanner Active - Waiting for finger...', 'info');
      }

      // In a real scenario, this would trigger the 32-bit service for 1:N matching
      // For now, we simulate the request to the backend
      fetch('% url "attendance:api_biometric_verify" %}', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        credentials: 'include',
        body: JSON.stringify({ action: 'verify' })
      })
        .then(res => res.json())
        .then(data => {
          // Logic to handle verification result from the 32-bit service
          // This would be replaced with actual polling or WebSocket result
        });
    });
  });
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/attendance/schedule.html

```
{% extends 'base.html' %}

{% block title %}Work Scheduling - Barangay Management System{% endblock %}

{% block extra_head %}
<style>
    .schedule-header {
        display: flex;
        justify-content: space-between;
        align-items: flex-start;
        margin-bottom: 20px;
        flex-wrap: wrap;
        gap: 12px;
    }
    .schedule-nav {
        display: flex;
        align-items: center;
        gap: 10px;
        flex-wrap: wrap;
    }
    @media (max-width: 768px) {
        .schedule-header {
            flex-direction: column;
            align-items: stretch;
        }
        .schedule-nav {
            justify-content: flex-start;
        }
    }
    .schedule-grid {
        width: 100%;
        border-collapse: collapse;
        background: #fff;
        border-radius: 8px;
        overflow: hidden;
        box-shadow: 0 1px 3px rgba(0,0,0,0.1);
    }
    .schedule-grid th, .schedule-grid td {
        border: 1px solid #e2e8f0;
        padding: 10px;
        text-align: center;
        vertical-align: middle;
    }
    .schedule-grid th {
        background: #f8fafc;
        font-size: 13px;
        color: #475569;
    }
    .schedule-grid td.official-col {
        text-align: left;
        background: #f8fafc;
        font-weight: bold;
        width: 200px;
    }
    .shift-cell {
        cursor: pointer;
        transition: background 0.2s;
    }
```


IoT-Enabled Barangay Resident and Attendance Management System

```
border-radius: 4px;
padding: 6px;
font-size: 12px;
font-weight: 600;
}
.shift-cell:hover {
  background: #f1f5f9;
}
.shift-regular { background-color: #dbeafe; color: #1e40af; }
.shift-morning { background-color: #ffedd5; color: #9a3412; }
.shift-afternoon { background-color: #fef08a; color: #854d0e; }
.shift-night { background-color: #e0e7ff; color: #3730a3; }
.shift-leave { background-color: #fee2e2; color: #991b1b; }
.shift-day_off { background-color: #f3f4f6; color: #4b5563; }
.shift-none { color: #9ca3af; font-style: italic; }

.category-row td {
  background: #e2e8f0 !important;
  text-align: left;
  font-weight: bold;
  text-transform: uppercase;
  font-size: 11px;
  letter-spacing: 0.05em;
  color: #334155;
}

.replacements-list {
  margin-top: 15px;
  background: #f8fafc;
  border: 1px solid #e2e8f0;
  border-radius: 6px;
  padding: 10px;
  display: none;
}
</style>
{% endblock %}

{% block content %}
{% url 'attendance:dashboard' as attendance_dashboard_url %}
{% include 'components/back_button.html' with back_url=attendance_dashboard_url
back_label='Back' %}
<div class="schedule-header">
  <div>
    <h2>Work Scheduling</h2>
    <p class="text-muted">Manage shifts, day-offs, and leaves</p>
  </div>
  <div class="schedule-nav">
    <a href="?date={{ prev_week }}" class="btn btn-outline-secondary btn-sm"><i class="fas
fa-chevron-left"></i> Prev Week</a>
    <h4 class="m-0">{{ start_of_week|date:"M d" }} - {{ end_of_week|date:"M d, Y" }}</h4>
    <a href="?date={{ next_week }}" class="btn btn-outline-secondary btn-sm">Next Week <i
class="fas fa-chevron-right"></i></a>
    <button class="btn btn-primary btn-sm ml-3" onclick="autoGenerate('{{
start_of_week|date:'Y-m-d' }}')">
      <i class="fas fa-magic"></i> Auto-Generate Week
    </button>
  </div>
</div>

<div class="table-responsive">
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<table class="schedule-grid" style="min-width: 750px;">
  <thead>
    <tr>
      <th>Official / Staff</th>
      {% for d in week_dates %}
      <th class="{% if d.is_today %}bg-blue-50{% endif %}">
        {{ d.day_name }}<br>
        <small class="text-muted">{{ d.date|date:"M d" }}</small>
      </th>
      {% endfor %}
    </tr>
  </thead>
  <tbody>
    {% for category, officials in grouped_officials.items %}
    <tr class="category-row">
      <td colspan="8">{{ category }}</td>
    </tr>
    {% for official in officials %}
    <tr>
      <td class="official-col">
        {{ official.resident.full_name }}<br>
        <span style="font-weight:normal; font-size:11px; color:#64748b;">{{
official.get_position_display }}</span>
      </td>
      {% for d in week_dates %}
      {% with sched=schedule_dict|dictsort:official.id %}
      {# We need a custom template tag or a bit of logic, but we structured
schedule_dict as schedule_dict[off_id][date_str] #}
      {% endwith %}
      <!-- Jinja logic for dict lookup is limited, so we handle it with a data attribute
approach or a custom filter.
      For simplicity, we output empty cells and let JS populate them, or use a
custom filter.
      Let's use a simple approach: We will render empty div and populate via JS
from a JSON object. -->
      <td id="cell-{{ official.id }}-{{ d.date_str }}"
        class="shift-cell"
        onclick="openEditModal({{ official.id }}, '{{ official.resident.full_name }}',
'{{ d.date_str }}')">
        <div class="shift-content shift-none">Unscheduled</div>
      </td>
      {% endfor %}
    </tr>
    {% endfor %}
  </tbody>
</table>
</div>

<!-- Edit Shift Modal -->
<div class="modal fade" id="editShiftModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Edit Schedule: <span id="modal-date"></span></h5>
        <button type="button" class="close" data-dismiss="modal">&times;</button>
      </div>
      <div class="modal-body">
        <h6 id="modal-official-name" class="font-weight-bold mb-3"></h6>
        <input type="hidden" id="modal-official-id">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

<input type="hidden" id="modal-date-val">

<div class="form-group">
  <label>Shift Type</label>
  <select id="modal-shift-type" class="form-control"
onchange="checkLeaveStatus()">
    {% for val, label in shift_choices %}
    <option value="{{ val }}">{{ label }}</option>
    {% endfor %}
  </select>
</div>

<div class="form-group">
  <label>Notes (Optional)</label>
  <input type="text" id="modal-notes" class="form-control" placeholder="Reason for
leave, etc.">
</div>

<!-- Replacements block -->
<div id="replacements-block" class="replacements-list">
  <h6 class="text-primary" style="font-size:12px; text-transform:uppercase; font-
weight:bold;">Suggested Replacements</h6>
  <p style="font-size:11px; color:#64748b;">The following active officials (same
position) are off or unscheduled on this date:</p>
  <ul id="replacements-ul" style="font-size:12px; padding-left:20px; margin-
bottom:0;">
    <!-- Populated via JS -->
  </ul>
</div>

</div>
<div class="modal-footer">
  <button type="button" class="btn btn-secondary" data-
dismiss="modal">Cancel</button>
  <button type="button" class="btn btn-primary" onclick="saveSchedule()">Save
Changes</button>
</div>
</div>
</div>
</div>

{% endblock %}

{% block extra_js %}
{{ schedule_dict|json_script:"schedule-data" }}
<script>
  const scheduleData = JSON.parse(document.getElementById('schedule-data').textContent);

  function renderGrid() {
    for (const [offId, dates] of Object.entries(scheduleData)) {
      for (const [dateStr, sched] of Object.entries(dates)) {
        const cell = document.getElementById(`cell-${offId}-${dateStr}`);
        if (cell) {
          let content = `<div class="shift-content shift-${sched.shift_type} p-1
rounded">${sched.shift_label}</div>`;
          if (sched.notes) {
            content += `<small class="text-muted d-block mt-1" style="font-
size:10px;">${sched.notes}</small>`;
          }
          cell.innerHTML = content;
        }
      }
    }
  }
</script>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
// Store raw data for modal
cell.dataset.shift = sched.shift_type;
cell.dataset.notes = sched.notes || '';
    }
  }
}

document.addEventListener('DOMContentLoaded', renderGrid);

function openEditModal(officialId, name, dateStr) {
  document.getElementById('modal-official-id').value = officialId;
  document.getElementById('modal-official-name').textContent = name;
  document.getElementById('modal-date').textContent = dateStr;
  document.getElementById('modal-date-val').value = dateStr;

  const cell = document.getElementById(`cell-${officialId}-${dateStr}`);
  document.getElementById('modal-shift-type').value = cell.dataset.shift || 'day_off';
  document.getElementById('modal-notes').value = cell.dataset.notes || '';

  checkLeaveStatus();
  $('#editShiftModal').modal('show');
}

function checkLeaveStatus() {
  const shiftType = document.getElementById('modal-shift-type').value;
  const repBlock = document.getElementById('replacements-block');
  if (shiftType === 'leave') {
    repBlock.style.display = 'block';
    fetchReplacements();
  } else {
    repBlock.style.display = 'none';
  }
}

function fetchReplacements() {
  const officialId = document.getElementById('modal-official-id').value;
  const dateStr = document.getElementById('modal-date-val').value;
  const ul = document.getElementById('replacements-ul');
  ul.innerHTML = '<li><i class="fas fa-spinner fa-spin"></i> Loading...</li>';

  fetch(`/attendance/api/schedule/replacements/?official_id=${officialId}&date=${dateStr}`)
    .then(res => res.json())
    .then(data => {
      if (data.status === 'success') {
        ul.innerHTML = '';
        if (data.replacements.length === 0) {
          ul.innerHTML = '<li>No available replacements found.</li>';
        } else {
          data.replacements.forEach(rep => {
            ul.innerHTML += `<li><b>${rep.name}</b> <span class="text-muted">{${rep.status}</span></li>`;
          });
        }
      }
    })
    .catch(e => {
      ul.innerHTML = '<li>Error loading replacements.</li>';
    });
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

function saveSchedule() {
  const officialId = document.getElementById('modal-official-id').value;
  const dateStr = document.getElementById('modal-date-val').value;
  const shiftType = document.getElementById('modal-shift-type').value;
  const notes = document.getElementById('modal-notes').value;

  fetch('/attendance/api/schedule/update/', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
      'X-CSRFToken': '{{ csrf_token }}'
    },
    body: JSON.stringify({
      official_id: officialId,
      date: dateStr,
      shift_type: shiftType,
      notes: notes
    })
  })
  .then(res => res.json())
  .then(data => {
    if (data.status === 'success') {
      window.location.reload();
    } else {
      alert('Error: ' + data.message);
    }
  });
}

function autoGenerate(startDate) {
  if (!confirm('This will assign a Regular (Mon-Fri 8-5) schedule for the entire week to all active officials. Existing overrides for this week will NOT be overwritten if you manually update them later, but this might overwrite empty ones. Continue?')) {
    return;
  }

  fetch('/attendance/api/schedule/generate/', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
      'X-CSRFToken': '{{ csrf_token }}'
    },
    body: JSON.stringify({ start_date: startDate })
  })
  .then(res => res.json())
  .then(data => {
    if (data.status === 'success') {
      alert(data.message);
      window.location.reload();
    } else {
      alert('Error: ' + data.message);
    }
  });
}
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/base.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en" data-theme="light">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>{% block title %}Barangay Management System{% endblock %}</title>
  <link rel="stylesheet"
href="https://cdn.jsdelivrivr.net/npm/bootstrap@4.6.2/dist/css/bootstrap.min.css">
  <link rel="stylesheet" href="{% static 'css/style.css' %}">
  <link href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700;800&display=swap" rel="stylesheet">
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&display=swap"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css"
rel="stylesheet">
  <script
src="https://cdn.jsdelivrivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
  {% block extra_head %}{% endblock %}
</head>

<body>
  <!-- Mobile Sidebar Overlay -->
  <div class="sidebar-overlay" id="sidebarOverlay"></div>

  <!-- Sidebar -->
  <aside class="sidebar" id="sidebar">
    <div class="sidebar-header">
      <div class="logo">
        
        <div class="logo-text">
          <h2>SICO-SICO</h2>
          <span>Management System</span>
        </div>
      </div>
      <button class="sidebar-close" id="sidebarClose">
        <i class="fas fa-times"></i>
      </button>
    </div>

    <nav class="sidebar-nav">
      {% with r=user_role s=request.user.is_superuser %}
      <!-- MAIN -->
      <div class="nav-section">
        <span class="nav-section-title">MAIN</span>
        <a href="{% url 'core:dashboard' %}"
class="nav-item {% if request.resolver_match.url_name == 'dashboard' and
request.resolver_match.app_name == 'core' %}active{% endif %}">
          <i class="fas fa-tachometer-alt"></i><span>Dashboard</span>
        </a>
      </div>

      <!-- RESIDENT SERVICES -->
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% if r == 'resident' %}
<div class="nav-section">
  <span class="nav-section-title">MY SERVICES</span>
  {% if user_profile.resident %}
  <a href="{% url 'residents:view' user_profile.resident.pk %}"
    class="nav-item {% if request.resolver_match.app_name == 'residents' and
request.resolver_match.url_name == 'view' %}active{% endif %}">
    <i class="fas fa-user-circle"></i><span>My Personal Profile</span>
  </a>
  {% endif %}
  <a href="{% url 'certifications:list' %}"
    class="nav-item {% if request.resolver_match.app_name == 'certifications' and
request.resolver_match.url_name == 'list' %}active{% endif %}">
    <i class="fas fa-file-invoice"></i><span>My Certificates</span>
  </a>
  <a href="{% url 'certifications:request_certificate' %}"
    class="nav-item {% if request.resolver_match.url_name == 'request_certificate'
%}active{% endif %}">
    <i class="fas fa-plus-circle"></i><span>Request Certificate</span>
  </a>
  <a href="{% url 'certifications:my_requests' %}"
    class="nav-item {% if request.resolver_match.url_name == 'my_requests'
%}active{% endif %}">
    <i class="fas fa-history"></i><span>My Requests</span>
  </a>
  <a href="{% url 'ordinances:list' %}"
    class="nav-item {% if request.resolver_match.app_name == 'ordinances'
%}active{% endif %}">
    <i class="fas fa-gavel"></i><span>Brgy. Ordinances</span>
  </a>
  <a href="{% url 'certifications:ra11261_apply' %}"
    class="nav-item {% if request.resolver_match.url_name == 'ra11261_apply'
%}active{% endif %}">
    <i class="fas fa-briefcase"></i><span>First Time Jobseeker</span>
  </a>
</div>
{% endif %}

<!-- MANAGEMENT -->
{% if s or r != 'resident' %}
<div class="nav-section">
  <span class="nav-section-title">MANAGEMENT</span>

  <a href="{% url 'residents:list' %}"
    class="nav-item {% if request.resolver_match.app_name == 'residents' and
request.resolver_match.url_name != 'household_list' and request.resolver_match.url_name !=
'registration_list' %}active{% endif %}">
    <i class="fas fa-users"></i><span>Residents</span>
  </a>

  {% if s or r == 'captain' or r == 'admin' or r == 'secretary' or r == 'treasurer'
%}
  <a href="{% url 'officials:list' %}"
    class="nav-item {% if request.resolver_match.app_name == 'officials' and
request.resolver_match.url_name != 'biometric_register' %}active{% endif %}">
    <i class="fas fa-user-tie"></i><span>Officials</span>
  </a>
  <a href="{% url 'officials:biometric_register' %}"
    class="nav-item {% if request.resolver_match.url_name == 'biometric_register'
%}active{% endif %}">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<i class="fas fa-fingerprint"></i><span>Register for Biometric</span>
</a>
{% endif %}

{% if s or r == 'captain' or r == 'admin' or r == 'secretary' %}
<a href="{% url 'residents:household_list' %}"
  class="nav-item {% if request.resolver_match.url_name == 'household_list'
%}active{% endif %}">
  <i class="fas fa-home"></i><span>Households</span>
</a>
<a href="{% url 'residents:registration_list' %}"
  class="nav-item {% if request.resolver_match.url_name == 'registration_list'
%}active{% endif %}">
  <i class="fas fa-user-clock"></i><span>Pending Registrations</span>
  {% if pending_registrations_count > 0 and request.resolver_match.url_name !=
'registration_list' %}
    <span id="registration-badge" class="badge badge-danger" style="margin-left:
auto;">{{ pending_registrations_count }}</span>
  {% endif %}
</a>
{% endif %}
</div>
{% endif %}

<!-- SERVICES -->
{% if s or r != 'lupon_member' and r != 'resident' %}
<div class="nav-section">
  <span class="nav-section-title">SERVICES</span>

  <a href="{% url 'documents:list' %}"
    class="nav-item {% if request.resolver_match.app_name == 'documents' %}active{%
endif %}">
    <i class="fas fa-file-word"></i><span>Documents</span>
  </a>

  <a href="{% url 'certifications:list' %}"
    class="nav-item {% with url=request.resolver_match.url_name %}{% if
request.resolver_match.app_name == 'certifications' and url == 'list' or url == 'view' or
url == 'pdf' or url == 'receipt' or url == 'virtual_certificate' or url == 'issue'
%}active{% endif %}{% endwith %}">
    <i class="fas fa-certificate"></i><span>Certifications</span>
  </a>
  <a href="{% url 'certifications:request_list' %}"
    class="nav-item {% with url=request.resolver_match.url_name %}{% if url ==
'request_list' or url == 'fulfill_request' or url == 'reject_request' %}active{% endif
%}{% endwith %}">
    <i class="fas fa-inbox"></i>
    <span>Cert. Requests</span>
    {% if pending_cert_requests_count > 0 and request.resolver_match.url_name !=
'request_list' and request.resolver_match.url_name != 'fulfill_request' and
request.resolver_match.url_name != 'reject_request' %}
      <span id="cert-requests-badge"
        style="margin-left:auto;background:#ef4444;color:#fff;border-
radius:20px;padding:1px 7px;font-size:.72rem;font-weight:700;">{{
pending_cert_requests_count }}</span>
    {% endif %}
  </a>

  <a href="{% url 'certifications:ra11261_admin_list' %}"
    class="nav-item {% if request.resolver_match.url_name == 'ra11261_admin_list'
```


IoT-Enabled Barangay Resident and Attendance Management System

```
%}active{% endif %}">
    <i class="fas fa-briefcase"></i><span>RA 11261 Apps</span>
</a>

    {% if s or r == 'captain' or r == 'admin' or r == 'treasurer' %}
    <a href="{% url 'attendance:dashboard' %}"
    class="nav-item {% if request.resolver_match.app_name == 'attendance'
%}active{% endif %}">
        <i class="fas fa-fingerprint"></i><span>Attendance & DTR</span>
    </a>
    {% endif %}
</div>
{% endif %}

<!-- RECORDS -->
{% if s or r == 'captain' or r == 'admin' or r == 'secretary' or r == 'kagawad' or r
== 'lupon_member' %}
<div class="nav-section">
    <span class="nav-section-title">RECORDS</span>

    <a href="{% url 'ordinances:list' %}"
    class="nav-item {% if request.resolver_match.app_name == 'ordinances'
%}active{% endif %}">
        <i class="fas fa-gavel"></i><span>Ordinances</span>
    </a>
    <a href="{% url 'certifications:ra11261_admin_roster' %}"
    class="nav-item {% if request.resolver_match.url_name == 'ra11261_admin_roster'
%}active{% endif %}">
        <i class="fas fa-clipboard-list"></i><span>RA 11261 Roster</span>
    </a>

    {% if s or r == 'captain' or r == 'admin' %}
    <a href="{% url 'census:dashboard' %}"
    class="nav-item {% if request.resolver_match.app_name == 'census' %}active{%
endif %}">
        <i class="fas fa-chart-pie"></i><span>Census</span>
    </a>
    <a href="{% url 'core:barangay_map' %}"
    class="nav-item {% if request.resolver_match.url_name == 'barangay_map'
%}active{% endif %}">
        <i class="fas fa-map-marked-alt"></i><span>Barangay Map</span>
    </a>
    <a href="{% url 'reports:dashboard' %}"
    class="nav-item {% if request.resolver_match.app_name == 'reports' %}active{%
endif %}">
        <i class="fas fa-chart-bar"></i><span>Reports</span>
    </a>
    {% endif %}
</div>
{% endif %}
<!-- SYSTEM -->
{% if s or r == 'admin' or r == 'captain' or r == 'secretary' or r == 'treasurer' %}
<div class="nav-section">
    <span class="nav-section-title">SYSTEM</span>
    <a href="{% url 'core:backup_setup' %}"
    class="nav-item {% if request.resolver_match.url_name == 'backup_setup'
%}active{% endif %}">
        <i class="fas fa-database"></i><span>Backup Data</span>
    </a>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% endif %}
{% endwith %}
</nav>

<div class="sidebar-footer">
  <div class="user-info" style="cursor: pointer;" data-toggle="modal" data-
target="#fullProfileDetailModal">
    <div class="user-avatar">
      {% if user_profile.resident.photo %}
      
      {% elif request.user.avatar %}
      
      {% else %}
      
      {% endif %}
    </div>
    <div class="user-details">
      <span class="user-name">{{
request.user.get_full_name|default:request.user.username }}</span>
      <span class="user-role">
        {% if request.user.is_superuser %}Administrator
        {% elif user_role == 'captain' %}Punong Barangay
        {% elif user_role == 'secretary' %}Barangay Secretary
        {% elif user_role == 'treasurer' %}Barangay Treasurer
        {% elif user_role == 'kagawad' %}Kagawad
        {% elif user_role == 'sk_chairperson' %}SK Chairperson
        {% elif user_role == 'lupong_member' %}Lupon Member
        {% elif user_role == 'resident' %}Resident
        {% else %}Staff{% endif %}
      </span>
    </div>
    <div>
      <a href="{ % url 'core:logout' % }" class="logout-btn" title="Logout">
        <i class="fas fa-sign-out-alt"></i>
      </a>
    </div>
  </aside>

  <!-- Main Content -->
  <main class="main-content">
    <header class="top-navbar">
      <div class="navbar-left">
        <button class="menu-toggle" id="menuToggle"><i class="fas fa-bars"></i></button>
        <div class="breadcrumb">
          {% block breadcrumb %}<span>Brgy. Sico Sico Management System</span>{% endblock
%}
        </div>
      </div>
      <div class="navbar-right">
        <div class="datetime">
          <i class="fas fa-clock"></i>
          <span id="currentDateTime"></span>
        </div>
      </div>
    </header>

    <div class="page-content">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% block content %}{% endblock %}
</div>
</main>

{% if messages %}
<div class="messages-container">
  {% for message in messages %}
    <div class="alert alert-{{ message.tags }}" id="alert-{{ forloop.counter }}">
      <div class="alert-content">
        {% if message.tags == 'success' %}<i class="fas fa-check-circle"></i>
        {% elif message.tags == 'error' %}<i class="fas fa-exclamation-circle"></i>
        {% elif message.tags == 'warning' %}<i class="fas fa-exclamation-triangle"></i>
        {% else %}<i class="fas fa-info-circle"></i>{% endif %}
        <span>{{ message|safe }}</span>
      </div>
      <button class="alert-close" onclick="this.parentElement.remove()"><i class="fas fa-
times"></i></button>
    </div>
  {% endfor %}
</div>
{% endif %}
<!-- Profile Detail Modal -->
<div class="modal fade" id="profileDetailModal" tabindex="-1" aria-hidden="true">
  <div class="modal-dialog modal-dialog-centered">
    <div class="modal-content border-0 shadow-lg rounded-xl">
      <div class="modal-header border-0 pb-0">
        <button type="button" class="close" data-dismiss="modal" aria-label="Close">
          <span aria-hidden="true">&times;</span>
        </button>
      </div>
      <div class="modal-body text-center pt-0">
        <div class="user-avatar mx-auto mb-3" style="width: 100px; height: 100px;
border-radius: 50%; overflow: hidden; border: 4px solid var(--primary-light);">
          {% if user_profile.resident.photo %}
            
          {% elif request.user.avatar %}
            
          {% else %}
            
          {% endif %}
        </div>
        <h4 class="font-weight-bold mb-1">{{
request.user.get_full_name|default:request.user.username }}</h4>
        <p class="text-muted mb-3">{{ request.user.email|default:"No email provided"
}}</p>

        <div class="role-badge mb-4">
          <span class="badge badge-primary px-3 py-2 rounded-pill">
            {% if request.user.is_superuser %}Administrator
            {% elif user_role == 'captain' %}Punong Barangay
            {% elif user_role == 'secretary' %}Barangay Secretary
            {% elif user_role == 'treasurer' %}Barangay Treasurer
            {% elif user_role == 'kagawad' %}Kagawad
            {% elif user_role == 'sk_chairperson' %}SK Chairperson
            {% elif user_role == 'lupong_member' %}Lupon Member
            {% elif user_role == 'resident' %}Resident
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        {% else %}Staff{% endif %}
    </span>
</div>

<div class="profile-actions d-flex flex-column gap-2 px-4 pb-4">
    <button type="button" class="btn btn-primary rounded-lg py-2" data-
toggle="modal" data-target="#fullProfileDetailModal" data-dismiss="modal">
        <i class="fas fa-user-circle mr-2"></i> View Full Profile
    </button>
    <a href="{% url 'core:logout' %}" class="btn btn-outline-danger rounded-lg
py-2">
        <i class="fas fa-sign-out-alt mr-2"></i> Logout
    </a>
</div>
</div>
</div>
</div>
</div>

<!-- Full Profile Detail Modal -->
<div class="modal fade" id="fullProfileDetailModal" tabindex="-1" aria-hidden="true">
    <div class="modal-dialog modal-lg modal-dialog-centered">
        <div class="modal-content border-0 shadow-lg rounded-xl overflow-hidden">
            <div class="modal-header bg-primary text-white border-0 py-3">
                <h5 class="modal-title font-weight-bold">My Personal Profile</h5>
                <button type="button" class="close text-white" data-dismiss="modal" aria-
label="Close">
                    <span aria-hidden="true">&times;</span>
                </button>
            </div>
            <div class="modal-body p-0">
                {% if user_profile.resident %}
                <div class="d-flex flex-column flex-md-row">
                    <div class="bg-light p-4 text-center" style="min-width: 250px;">
                        <div class="user-avatar mx-auto mb-3" style="width: 120px; height: 120px;
border-radius: 50%; overflow: hidden; border: 5px solid white; shadow: var(--shadow-md);">
                            {% if user_profile.resident.photo %}
                            
                            {% else %}
                            
                            {% endif %}
                        </div>
                        <h5 class="font-weight-bold mb-1">{{ user_profile.resident.full_name }}</h5>
                        <span class="badge badge-primary rounded-pill px-3">{{ user_role|title
}}</span>
                    </div>
                    <div class="flex-grow-1">
                        <hr>
                        <div class="text-left mt-3">
                            <small class="text-muted d-block uppercase font-weight-bold" style="font-
size: 10px;">Contact Info</small>
                            <p class="mb-2" style="font-size: 13px;"><i class="fas fa-phone mr-2 text-
primary"></i> {{ user_profile.resident.contact_number|default:"---" }}</p>
                            <p class="mb-0" style="font-size: 13px;"><i class="fas fa-envelope mr-2
text-primary"></i> {{ request.user.email|default:"---" }}</p>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

class="fas fa-info-circle mr-2"></i> Basic Information</h6>
    <div class="row mb-4">
        <div class="col-sm-6 mb-3">
            <label class="d-block text-muted small mb-0">Gender</label>
            <span class="font-weight-600">{{
user_profile.resident.get_gender_display }}</span>
        </div>
        <div class="col-sm-6 mb-3">
            <label class="d-block text-muted small mb-0">Birthdate</label>
            <span class="font-weight-600">{{ user_profile.resident.birthdate }}
(Age: {{ user_profile.resident.age }})</span>
        </div>
        <div class="col-sm-6 mb-3">
            <label class="d-block text-muted small mb-0">Civil Status</label>
            <span class="font-weight-600">{{
user_profile.resident.get_civil_status_display }}</span>
        </div>
        <div class="col-sm-6 mb-3">
            <label class="d-block text-muted small mb-0">Occupation</label>
            <span class="font-weight-600">{{
user_profile.resident.display_occupation }}</span>
        </div>
    </div>

    <h6 class="font-weight-bold border-bottom pb-2 mb-3 text-primary"><i
class="fas fa-map-marker-alt mr-2"></i> Address Details</h6>
    <div class="row">
        <div class="col-12 mb-3">
            <label class="d-block text-muted small mb-0">Full Address</label>
            <span class="font-weight-600">{{ user_profile.resident.address }}</span>
        </div>
        <div class="col-sm-6 mb-3">
            <label class="d-block text-muted small mb-0">Purok / Zone</label>
            <span class="font-weight-600">{{ user_profile.resident.purok }}</span>
        </div>
        <div class="col-sm-6 mb-3">
            <label class="d-block text-muted small mb-0">Household</label>
            <span class="font-weight-600">
                {% if user_profile.resident.household %}
                Household #{{ user_profile.resident.household.household_no }}
                {% else %}Not assigned{% endif %}
            </span>
        </div>
    </div>
    </div>
    {% else %}
    <div class="p-5 text-center">
        <i class="fas fa-user-slash fa-3x text-muted mb-3"></i>
        <h5>No resident profile linked to this user.</h5>
    </div>
    {% endif %}
</div>
<div class="modal-footer bg-light border-0">
    <button type="button" class="btn btn-secondary rounded-lg px-4" data-
dismiss="modal">Close</button>
</div>
</div>
</div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/popper.js@1.16.1/dist/umd/popper.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@4.6.2/dist/js/bootstrap.min.js"></script>
<script src="{% static 'js/app.js' %}"></script>
<script>
    // Sidebar Scroll Persistence
    document.addEventListener('DOMContentLoaded', function() {
        const sidebarNav = document.querySelector('.sidebar-nav');
        if (sidebarNav) {
            // Restore scroll position
            const scrollPos = localStorage.getItem('sidebarScroll');
            if (scrollPos) {
                sidebarNav.scrollTop = scrollPos;
            }

            // Save scroll position on click/scroll
            sidebarNav.addEventListener('scroll', function() {
                localStorage.setItem('sidebarScroll', sidebarNav.scrollTop);
            });

            // Also save before navigating
            document.querySelectorAll('.nav-item').forEach(item => {
                item.addEventListener('click', function() {
                    localStorage.setItem('sidebarScroll', sidebarNav.scrollTop);
                });
            });
        }
    });
</script>
{% block extra_js %}{% endblock %}

{% include 'core/hardware_poller.html' %}
<!-- Notification Badge Management -->
<script>
    document.addEventListener('DOMContentLoaded', function() {
        // Certificate Requests Logic
        const certBadge = document.getElementById('cert-requests-badge');
        const certCount = parseInt('{{ pending_cert_requests_count|default:0 }}');
        const isCertPage = {% if request.resolver_match.url_name == 'request_list' %}true{%
else %}false{% endif %};

        if (isCertPage) {
            localStorage.setItem('last_cert_count', certCount);
        } else if (certBadge) {
            const lastSeenCert = parseInt(localStorage.getItem('last_cert_count') || '0');
            if (certCount <= lastSeenCert) {
                certBadge.style.display = 'none';
            }
        }

        // Registration Requests Logic
        const regBadge = document.getElementById('registration-badge');
        const regCount = parseInt('{{ pending_registrations_count|default:0 }}');
        const isRegPage = {% if request.resolver_match.url_name == 'registration_list'
%}true{% else %}false{% endif %};

        if (isRegPage) {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    localStorage.setItem('last_reg_count', regCount);
  } else if (regBadge) {
    const lastSeenReg = parseInt(localStorage.getItem('last_reg_count') || '0');
    if (regCount <= lastSeenReg) {
      regBadge.style.display = 'none';
    }
  }
}

// Add click listeners for immediate feedback
document.querySelectorAll('.nav-item').forEach(item => {
  item.addEventListener('click', function() {
    const badge = this.querySelector('#cert-requests-badge, #registration-badge');
    if (badge) badge.style.display = 'none';
  });
});
</script>
</body>

</html>
```

File: templates/census/age_groups.html

```
{% extends 'base.html' %}

{% block title %}Age Group Breakdown - Census{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'census:dashboard' %}">Census</a> / Age Groups</span>
{% endblock %}

{% block content %}
{% url 'census:dashboard' as census_dashboard_url %}
{% include 'components/back_button.html' with back_url=census_dashboard_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>Age Group Breakdown</h1>
        <span>Detailed population statistics by age and gender</span>
    </div>
</div>

<div class="card">
    <div class="table-responsive">
        <table>
            <thead>
                <tr>
                    <th>Age Group</th>
                    <th>Male</th>
                    <th>Female</th>
                    <th>Total Population</th>
                    <th>Percentage</th>
                    <th>Actions</th>
                </tr>
            </thead>
            <tbody>
                {% for group in age_groups %}
                <tr>
                    <td><strong>{{ group.name }}</strong></td>
                    <td>{{ group.male }}</td>
                    <td>{{ group.female }}</td>
                    <td><strong>{{ group.total }}</strong></td>
                    <td>
                        <div class="percentage-track">
                            <!-- Percentage width calculated via template logic would be better but
let's keep it simple -->
                            <span>{% widthratio group.total total_population 100 %}%</span>
                        </div>
                    </td>
                    <td>
                        <button class="btn btn-outline btn-icon" onclick="toggleDetails('{{
group.name|slugify }}')">
                            <i class="fas fa-list"></i>
                        </button>
                    </td>
                </tr>
                <tr id="details-{{ group.name|slugify }}" class="member-details-row"
style="display: none;">
                    <td colspan="6">
                        <div class="details-content p-3 bg-light">
```


IoT-Enabled Barangay Resident and Attendance Management System

```
<h4 class="mb-3">Resident List ({{ group.name }})</h4>
<div class="tag-cloud">
  {% for r in group.residents %}
    <a href="{% url 'residents:view' r.pk %}" class="resident-tag">
      {{ r.full_name }} ({{ r.gender }}/{{ r.age }})
    </a>
  {% endfor %}
</div>
</div>
</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>

<script>
function toggleDetails(groupId) {
  const row = document.getElementById('details-' + groupId);
  if (row.style.display === 'none') {
    row.style.display = 'table-row';
  } else {
    row.style.display = 'none';
  }
}
</script>

<style>
.percent-age-track {
  font-size: 13px;
  font-weight: 600;
  color: var(--primary);
}

.member-details-row {
  background: var(--bg-card);
}

.tag-cloud {
  display: flex;
  flex-wrap: wrap;
  gap: 8px;
}

.resident-tag {
  padding: 4px 10px;
  background: white;
  border: 1px solid var(--border);
  border-radius: 4px;
  font-size: 12px;
  color: var(--text-main);
  text-decoration: none;
}

.resident-tag:hover {
  border-color: var(--primary);
  color: var(--primary);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.bg-light {  
  background: rgba(0, 0, 0, 0.02);  
}  
</style>  
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/census/dashboard.html

```
{% extends 'base.html' %}

{% block title %}Census - Barangay Management System{% endblock %}

{% block content %}
<style>
  /* Premium Dashboard Styling */
  .page-header {
    margin-bottom: 2rem;
  }
  .page-title h1 {
    font-weight: 700;
    font-size: 2rem;
    color: var(--text-dark, #1e293b);
    margin-bottom: 0.5rem;
  }
  .page-title span {
    color: var(--text-muted, #64748b);
    font-size: 1rem;
  }

  .stats-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
    gap: 1.5rem;
    margin-bottom: 2rem;
  }

  .stat-card {
    background: #ffffff;
    border-radius: 1rem;
    padding: 1.5rem;
    display: flex;
    align-items: center;
    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05), 0 2px 4px -1px rgba(0, 0, 0, 0.03);
    transition: transform 0.2s ease, box-shadow 0.2s ease;
    position: relative;
    overflow: hidden;
  }

  .stat-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 10px 15px -3px rgba(0, 0, 0, 0.1), 0 4px 6px -2px rgba(0, 0, 0, 0.05);
  }

  .stat-card::before {
    content: '';
    position: absolute;
    top: 0; left: 0; width: 4px; height: 100%;
    background: var(--accent-color, #4f46e5);
  }

  .stat-card.c-blue::before { background: linear-gradient(135deg, #3b82f6, #2563eb); }
  .stat-card.c-green::before { background: linear-gradient(135deg, #10b981, #059669); }
  .stat-card.c-purple::before { background: linear-gradient(135deg, #8b5cf6, #7c3aed); }
  .stat-card.c-orange::before { background: linear-gradient(135deg, #f59e0b, #d97706); }
  .stat-card.c-red::before { background: linear-gradient(135deg, #ef4444, #dc2626); }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.stat-card.c-teal::before { background: linear-gradient(135deg, #14b8a6, #0d9488); }

.stat-icon {
  width: 50px;
  height: 50px;
  border-radius: 12px;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 1.5rem;
  margin-right: 1.25rem;
  color: white;
}

.stat-card.c-blue .stat-icon { background: linear-gradient(135deg, #3b82f6, #2563eb); }
.stat-card.c-green .stat-icon { background: linear-gradient(135deg, #10b981, #059669); }
.stat-card.c-purple .stat-icon { background: linear-gradient(135deg, #8b5cf6, #7c3aed); }
}

.stat-card.c-orange .stat-icon { background: linear-gradient(135deg, #f59e0b, #d97706); }
}

.stat-card.c-red .stat-icon { background: linear-gradient(135deg, #ef4444, #dc2626); }
.stat-card.c-teal .stat-icon { background: linear-gradient(135deg, #14b8a6, #0d9488); }

.stat-info {
  display: flex;
  flex-direction: column;
}
.stat-info .value {
  font-size: 1.75rem;
  font-weight: 700;
  color: #1e293b;
  line-height: 1.2;
}
.stat-info .label {
  font-size: 0.875rem;
  color: #64748b;
  font-weight: 500;
  text-transform: uppercase;
  letter-spacing: 0.5px;
  margin-top: 0.25rem;
}
}

.charts-grid {
  display: grid;
  grid-template-columns: 2fr 1fr;
  gap: 1.5rem;
  margin-bottom: 2rem;
}
}

@media (max-width: 992px) {
  .charts-grid {
    grid-template-columns: 1fr;
  }
}

.premium-card {
  background: #ffffff;
  border-radius: 1rem;
  box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05);
  overflow: hidden;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

.premium-card-header {
  padding: 1.25rem 1.5rem;
  border-bottom: 1px solid #f1f5f9;
  background: #fafafa;
}

.premium-card-header h3 {
  margin: 0;
  font-size: 1.125rem;
  font-weight: 600;
  color: #334155;
}

.premium-card-body {
  padding: 1.5rem;
}

/* Beautiful Table */
.custom-table {
  width: 100%;
  border-collapse: collapse;
}

.custom-table th {
  background-color: #f8fafc;
  color: #64748b;
  font-weight: 600;
  text-align: left;
  padding: 1rem 1.5rem;
  font-size: 0.875rem;
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

.custom-table td {
  padding: 1rem 1.5rem;
  border-bottom: 1px solid #f1f5f9;
  color: #334155;
  font-weight: 500;
}

.custom-table tr:last-child td {
  border-bottom: none;
}

.custom-table tr:hover td {
  background-color: #f8fafc;
}

.progress-bar-bg {
  background-color: #e2e8f0;
  border-radius: 9999px;
  height: 8px;
  width: 100%;
  overflow: hidden;
  margin-top: 0.5rem;
}

.progress-bar-fill {
  height: 100%;
  background: linear-gradient(90deg, #3b82f6, #60a5fa);
  border-radius: 9999px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.demographics-summary {
  display: flex;
  justify-content: space-around;
  padding: 1rem 0;
}
.demo-circle {
  display: flex;
  flex-direction: column;
  align-items: center;
}
.demo-circle .icon-wrapper {
  width: 64px;
  height: 64px;
  border-radius: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 1.75rem;
  margin-bottom: 0.75rem;
  color: white;
  box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
}
.demo-circle.male .icon-wrapper { background: linear-gradient(135deg, #3b82f6, #2563eb); }
.demo-circle.female .icon-wrapper { background: linear-gradient(135deg, #ec4899, #db2777); }
.demo-circle.fourps .icon-wrapper { background: linear-gradient(135deg, #10b981, #059669); }

.demo-circle .val { font-size: 1.5rem; font-weight: 700; color: #1e293b; }
.demo-circle .lbl { font-size: 0.875rem; color: #64748b; font-weight: 500; }

</style>

<div class="page-header">
  <div class="page-title">
    <h1>Barangay Census & Demographics</h1>
    <span>Comprehensive statistical overview of the population and households</span>
  </div>
</div>

<div class="stats-grid">
  <div class="stat-card c-blue">
    <div class="stat-icon"><i class="fas fa-users"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_population }}</span>
      <span class="label">Total Population</span>
    </div>
  </div>
  <div class="stat-card c-purple">
    <div class="stat-icon"><i class="fas fa-home"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_households }}</span>
      <span class="label">Total Households</span>
    </div>
  </div>
  <div class="stat-card c-teal">
    <div class="stat-icon"><i class="fas fa-chart-pie"></i></div>
    <div class="stat-info">
      <span class="value">{{ avg_household_size }}</span>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <span class="label">Avg. Family Size</span>
    </div>
</div>
<div class="stat-card c-green">
    <div class="stat-icon"><i class="fas fa-vote-yea"></i></div>
    <div class="stat-info">
        <span class="value">{{ voters }}</span>
        <span class="label">Registered Voters</span>
    </div>
</div>
<div class="stat-card c-orange">
    <div class="stat-icon"><i class="fas fa-blind"></i></div>
    <div class="stat-info">
        <span class="value">{{ senior }}</span>
        <span class="label">Senior Citizens</span>
    </div>
</div>
<div class="stat-card c-red">
    <div class="stat-icon"><i class="fas fa-wheelchair"></i></div>
    <div class="stat-info">
        <span class="value">{{ pwd }}</span>
        <span class="label">PWDs</span>
    </div>
</div>
</div>

<div class="charts-grid">
    <!-- Age Distribution Chart -->
    <div class="premium-card">
        <div class="premium-card-header">
            <h3>Population by Age Group</h3>
        </div>
        <div class="premium-card-body" style="height: 350px;">
            <canvas id="ageChart"></canvas>
        </div>
    </div>

    <!-- Gender & Sector Breakdown -->
    <div class="premium-card">
        <div class="premium-card-header">
            <h3>Gender Distribution</h3>
        </div>
        <div class="premium-card-body" style="height: 350px; display: flex; flex-direction:
column; justify-content: center; align-items: center;">
            <div style="height: 220px; width: 100%; position: relative;">
                <canvas id="genderChart"></canvas>
            </div>
            <div class="demographics-summary" style="width: 100%; margin-top: 20px;">
                <div class="demo-circle male">
                    <div class="val">{{ male_count }}</div>
                    <div class="lbl">Male</div>
                </div>
                <div class="demo-circle female">
                    <div class="val">{{ female_count }}</div>
                    <div class="lbl">Female</div>
                </div>
                <div class="demo-circle fourps">
                    <div class="val">{{ fourps }}</div>
                    <div class="lbl">4Ps Member</div>
                </div>
            </div>
        </div>
    </div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>
  </div>
</div>
</div>

<div class="premium-card">
  <div class="premium-card-header">
    <h3>Purok Distribution</h3>
  </div>
  <div class="table-responsive">
    <table class="custom-table">
      <thead>
        <tr>
          <th>Purok / Zone Name</th>
          <th>Total Population</th>
          <th style="width: 40%;">Percentage Share</th>
        </tr>
      </thead>
      <tbody>
        {% for p in purok_stats %}
          <tr>
            <td>
              <div style="display: flex; align-items: center;">
                <i class="fas fa-map-marker-alt" style="color: #64748b; margin-right: 10px;"></i>
                {{ p.purok|default:"Undeclared" }}
              </div>
            </td>
            <td><strong>{{ p.count }}</strong> residents</td>
            <td>
              <div style="display: flex; align-items: center; justify-content: space-between;">
                <span style="font-weight: 600; color: #3b82f6;">{% widthratio p.count total_population 100 %}%</span>
                <div class="progress-bar-bg">
                  <div class="progress-bar-fill" style="width: {% widthratio p.count total_population 100 %}%;"></div>
                </div>
              </div>
            </td>
          </tr>
        {% endfor %}
      </tbody>
    </table>
  </div>
</div>

{% block extra_js %}
<!-- Include Chart.js if not globally available, assuming it is since it was in the old template -->
<script>
  // Setup standard chart defaults for premium look
  Chart.defaults.font.family = "'Inter', 'Segoe UI', Roboto, Helvetica, Arial, sans-serif";
  Chart.defaults.color = '#64748b';

  // Age Group Bar Chart
  const ageCtx = document.getElementById('ageChart').getContext('2d');

  // Create gradient for bars

```


IoT-Enabled Barangay Resident and Attendance Management System

```
let ageGradient = ageCtx.createLinearGradient(0, 0, 0, 400);
ageGradient.addColorStop(0, 'rgba(99, 102, 241, 0.8)'); // Indigo
ageGradient.addColorStop(1, 'rgba(168, 85, 247, 0.8)'); // Purple

new Chart(ageCtx, {
  type: 'bar',
  data: {
    labels: {{ age_group_labels|safe }},
    datasets: [{
      label: 'Population',
      data: {{ age_group_values|safe }},
      backgroundColor: ageGradient,
      borderRadius: 6,
      borderSkipped: false,
      barPercentage: 0.6
    }]
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    plugins: {
      legend: { display: false },
      tooltip: {
        backgroundColor: 'rgba(15, 23, 42, 0.9)',
        padding: 12,
        titleFont: { size: 13, weight: 'bold' },
        bodyFont: { size: 14 },
        displayColors: false,
        callbacks: {
          label: function(context) {
            return context.parsed.y + ' Residents';
          }
        }
      }
    }
  },
  scales: {
    y: {
      beginAtZero: true,
      grid: { color: '#f1f5f9', drawBorder: false },
      ticks: { font: { size: 12 } }
    },
    x: {
      grid: { display: false, drawBorder: false },
      ticks: { font: { size: 12 } }
    }
  }
});

// Gender Doughnut Chart
const genderCtx = document.getElementById('genderChart').getContext('2d');
new Chart(genderCtx, {
  type: 'doughnut',
  data: {
    labels: ['Male', 'Female'],
    datasets: [{
      data: [{ male_count }, { female_count }],
      backgroundColor: [
        'rgba(59, 130, 246, 0.9)', // Blue
        'rgba(236, 72, 153, 0.9)'  // Pink
      ]
    }]
  }
});
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    ],
    hoverBackgroundColor: [
      'rgba(37, 99, 235, 1)',
      'rgba(219, 39, 119, 1)'
    ],
    borderWidth: 0,
    hoverOffset: 4
  ]
},
options: {
  responsive: true,
  maintainAspectRatio: false,
  cutout: '70%',
  plugins: {
    legend: {
      display: false
    },
  },
  tooltip: {
    backgroundColor: 'rgba(15, 23, 42, 0.9)',
    padding: 10,
    callbacks: {
      label: function(context) {
        let total = {{ total_population }};
        let percentage = Math.round((context.raw / total) * 100);
        return ` ${context.label}: ${context.raw} (${percentage}%)`;
      }
    }
  }
}
});
</script>
{% endblock %}
{% endblock %}
```

File: templates/census/demographics.html

```
{% extends 'base.html' %}

{% block title %}Demographics - Census{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / <a href="{% url 'census:dashboard' %}">Census</a> / Demographics</span>
{% endblock %}

{% block content %}
{% url 'census:dashboard' as census_dashboard_url %}
{% include 'components/back_button.html' with back_url=census_dashboard_url
back_label='Back' %}
<div class="page-header">
  <div class="page-title">
    <h1>Demographics & Distribution</h1>
    <span>Visual breakdown of barangay demographics</span>
  </div>
</div>

<div class="demographics-grid">
  <div class="card">
    <div class="card-header">
      <h3>Population by Purok</h3>
    </div>
    <div class="card-body">
      <div class="table-responsive">
        <table>
          <thead>
            <tr>
              <th>Purok</th>
              <th>Total</th>
              <th>Male</th>
              <th>Female</th>
            </tr>
          </thead>
          <tbody>
            {% for item in purok_data %}
            <tr>
              <td>{{ item.purok }}</td>
              <td><strong>{{ item.total }}</strong></td>
              <td>{{ item.male }}</td>
              <td>{{ item.female }}</td>
            </tr>
            {% endfor %}
          </tbody>
        </table>
      </div>
    </div>
  </div>

  <div class="card">
    <div class="card-header">
      <h3>Nationality Distribution</h3>
    </div>
    <div class="card-body">
      <div class="stats-list">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% for item in nationality_data %}
<div class="stats-item">
  <span>{{ item.nationality }}</span>
  <span class="badge badge-info">{{ item.count }}</span>
</div>
{% endfor %}
</div>
</div>
</div>

<div class="card">
  <div class="card-header">
    <h3>Top 10 Religions</h3>
  </div>
  <div class="card-body">
    <div class="stats-list">
      {% for item in religion_data %}
      <div class="stats-item">
        <span>{{ item.religion|default:"Unspecified" }}</span>
        <span class="badge badge-primary">{{ item.count }}</span>
      </div>
      {% empty %}
      <p class="text-muted">No data available</p>
      {% endfor %}
    </div>
  </div>
</div>

<div class="card">
  <div class="card-header">
    <h3>Top 10 Occupations</h3>
  </div>
  <div class="card-body">
    <div class="stats-list">
      {% for item in occupation_data %}
      <div class="stats-item">
        <span>{{ item.occupation|default:"Unemployed / Not set" }}</span>
        <span class="badge badge-success">{{ item.count }}</span>
      </div>
      {% empty %}
      <p class="text-muted">No data available</p>
      {% endfor %}
    </div>
  </div>
</div>
</div>

<style>
.demographics-grid {
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 25px;
}

.stats-list {
  display: flex;
  flex-direction: column;
  gap: 12px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.stats-item {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding-bottom: 8px;
  border-bottom: 1px solid var(--border);
}

.badge {
  padding: 4px 10px;
  border-radius: 4px;
  font-weight: 600;
  font-size: 12px;
}

.badge-info {
  background: rgba(59, 130, 246, 0.1);
  color: #3b82f6;
}

.badge-primary {
  background: rgba(var(--primary-rgb), 0.1);
  color: var(--primary);
}

.badge-success {
  background: rgba(16, 185, 129, 0.1);
  color: #10b981;
}

@media (max-width: 768px) {
  .demographics-grid {
    grid-template-columns: 1fr;
  }
}
</style>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/certifications/fulfill_form.html

```
{% extends 'base.html' %}
{% load static %}
{% block title %}Fulfill Request - {{
cert_request.child_name|default:cert_request.resident.full_name }}{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Certifications / Fulfill Request</span>
{% endblock %}

{% block content %}
{% url 'certifications:request_list' as back_url %}
{% include 'components/back_button.html' with back_url=back_url back_label='Back' %}
<style>
    .fulfill-container {
        max-width: 900px;
        margin: 20px auto;
        padding: 0 20px;
    }

    .card {
        background: #fff;
        border-radius: 12px;
        box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
        padding: 30px;
    }

    .grid {
        display: grid;
        grid-template-columns: 1fr 1fr;
        gap: 24px;
    }

    .form-group { margin-bottom: 20px; }
    .form-label {
        display: block;
        font-size: .75rem;
        font-weight: 700;
        text-transform: uppercase;
        color: #64748b;
        margin-bottom: 6px;
    }

    .form-control-sm {
        width: 100%;
        padding: 8px 12px;
        border: 1px solid #e2e8f0;
        border-radius: 8px;
        font-size: .9rem;
        height: 40px;
        line-height: 1.4;
    }

    .btn-submit {
        background: var(--primary);
        color: #fff;
        border: none;
        padding: 12px 24px;
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
border-radius: 8px;
font-weight: 600;
cursor: pointer;
}

.data-box {
background: #f8fafc;
padding: 15px;
border-radius: 8px;
border: 1px solid #e2e8f0;
}
</style>

<div class="fulfill-container">
  <div class="card">
    <form method="POST">
      {% csrf_token %}
      <div class="grid">
        <!-- Left: Request Info (Editable) -->
        <div class="data-box">
          <h3 style="font-size: 1rem; margin-bottom: 16px;">Verify & Modify
Request Details</h3>
          <div class="form-group">
            <label class="form-label">Resident</label>
            <p><strong>{{ cert_request.resident.full_name }}</strong></p>
          </div>
          <div class="form-group">
            <label class="form-label">Certificate Type</label>
            <p class="badge badge-primary">{{
cert_request.get_cert_type_display }}</p>
          </div>
          <div class="form-group">
            <label class="form-label">Purpose (Editable)</label>
            <input type="text" name="purpose" class="form-control-sm"
value="{{ cert_request.purpose }}" required>
          </div>

          {% if cert_request.cert_type == 'business_permit' %}
          <div style="margin-top: 16px; border-top: 1px dashed #cbd5e1; padding-
top: 16px;">
            <div class="form-group">
              <label class="form-label">Business Name</label>
              <input type="text" name="business_name" class="form-control-
sm" value="{{ cert_request.business_name|default:'' }}">
            </div>
            <div class="form-group">
              <label class="form-label">Business Address</label>
              <textarea name="business_address" class="form-control-sm"
rows="2">{{ cert_request.business_address|default:'' }}</textarea>
            </div>
            <div class="form-group">
              <label class="form-label">Business Type</label>
              <input type="text" name="business_type" class="form-control-
sm" value="{{ cert_request.business_type|default:'' }}">
            </div>
          </div>
          {% endif %}

          {% if cert_request.cert_type == 'late_registration' %}
          <div style="margin-top: 16px; border-top: 1px dashed #cbd5e1; padding-
```

IoT-Enabled Barangay Resident and Attendance Management System

```
top: 16px;">
    <div class="form-group">
        <label class="form-label">Child Name</label>
        <input type="text" name="child_name" class="form-control-sm"
value="{{ cert_request.child_name|default:'' }}">
    </div>
    <div class="form-group">
        <label class="form-label">Birth Date</label>
        <input type="date" name="child_birth_date" class="form-
control-sm" value="{{ cert_request.child_birth_date|date:'Y-m-d' }}">
    </div>
    <div class="form-group">
        <label class="form-label">Birth Place</label>
        <input type="text" name="child_birth_place" class="form-
control-sm" value="{{ cert_request.child_birth_place|default:'' }}">
    </div>
    <div class="form-group">
        <label class="form-label">Father's Name</label>
        <input type="text" name="father_name" class="form-control-sm"
value="{{ cert_request.father_name|default:'' }}">
    </div>
    <div class="form-group">
        <label class="form-label">Mother's Name</label>
        <input type="text" name="mother_name" class="form-control-sm"
value="{{ cert_request.mother_name|default:'' }}">
    </div>
</div>
{% endif %}
</div>

<!-- Right: Issuance Form -->
<div style="padding: 10px;">
    <div class="form-group">
        <label class="form-label">Control Number (Automated)</label>
        <input type="text" name="control_number" class="form-control-sm"
value="{{ suggested_control }}" required>
    </div>

    <div class="form-group">
        <label class="form-label">OR Number*</label>
        <input type="text" name="or_number" class="form-control-sm" value="{{
suggested_or }}" required>
    </div>

    <!-- Specific Fields for Cedula if needed -->
    {% if cert_request.cert_type == 'cedula' %}
    <div class="form-group">
        <label class="form-label">Taxpayer Type</label>
        <select name="taxpayer_type" id="taxpayer_type_select" class="form-
control-sm cedula-calc">
            <option value="individual" {% if cert_request.taxpayer_type ==
'individual' %}selected{% endif %}>Individual</option>
            <option value="corporation" {% if cert_request.taxpayer_type ==
'corporation' %}selected{% endif %}>Corporation</option>
        </select>
    </div>
    <div class="form-group">
        <label class="form-label">CTC Number*</label>
        <input type="text" name="ctc_number" class="form-control-sm" value="{{
suggested_ctc }}" required>
    </div>
    </div>
```


IoT-Enabled Barangay Resident and Attendance Management System

```
</div>

<div style="display:flex; gap:10px; margin-bottom:15px;">
  <div style="flex:1;">
    <label class="form-label">Basic (?)</label>
    <input type="number" step="0.01" name="basic_tax" id="id_basic_tax"
class="form-control-sm cedula-calc" value="5.00">
  </div>
  <div style="flex:1;">
    <label class="form-label">Interest (?)</label>
    <input type="number" step="0.01" name="interest" class="form-control-
sm cedula-calc" value="0.00">
  </div>
</div>

<div style="display:flex; gap:10px; margin-bottom:15px;">
  <div style="flex:1;">
    <label class="form-label">Height (cm)</label>
    <input type="text" name="height" class="form-control-sm" value="{{
cert_request.height|default:'' }}">
  </div>
  <div style="flex:1;">
    <label class="form-label">Weight (kg)</label>
    <input type="text" name="weight" class="form-control-sm" value="{{
cert_request.weight|default:'' }}">
  </div>
</div>

<div class="form-group">
  <label class="form-label">Place of Issue</label>
  <input type="text" name="place_of_issue" class="form-control-sm"
value="{{ cert_request.place_of_issue|default:'Sico-Sico, Gigaquit, SDN' }}">
</div>

<div class="form-group">
  <label class="form-label">Property Value (Cat A)</label>
  <div style="display:flex; gap:10px;">
    <input type="number" step="0.01" name="raw_taxable_property"
id="raw_prop" class="form-control-sm cedula-raw" placeholder="Value" value="{{
cert_request.raw_taxable_property|default:'0' }}">
    <input type="number" step="0.01" name="additional_tax_property"
id="val_prop" class="form-control-sm" readonly style="width:100px; background:#f1f5f9;">
  </div>
</div>

<div class="form-group">
  <label class="form-label">Business Receipts (Cat B)</label>
  <div style="display:flex; gap:10px;">
    <input type="number" step="0.01" name="raw_taxable_business"
id="raw_bus" class="form-control-sm cedula-raw" placeholder="Receipts" value="{{
cert_request.raw_taxable_business|default:'0' }}">
    <input type="number" step="0.01" name="additional_tax_business"
id="val_bus" class="form-control-sm" readonly style="width:100px; background:#f1f5f9;">
  </div>
</div>

<div class="form-group" id="income_group">
  <label class="form-label">Annual Income (Cat C)</label>
  <div style="display:flex; gap:10px;">
    <input type="number" step="0.01" name="raw_taxable_income"
```

IoT-Enabled Barangay Resident and Attendance Management System

```
id="raw_inc" class="form-control-sm cedula-row" placeholder="Income" value="{{
cert_request.raw_taxable_income|default:'0' }}">
    <input type="number" step="0.01" name="additional_tax_income"
id="val_inc" class="form-control-sm" readonly style="width:100px; background:#f1f5f9;">
    </div>
</div>
{% endif %}

<div class="form-group">
    <label class="form-label" id="amount_label">Amount Paid (?)</label>
    <div style="display:flex; gap:15px; align-items:center;">
        <input type="number" step="0.01" name="amount_paid"
id="id_amount_paid" class="form-control-sm" value="{{ base_amount }}" required {% if
cert_request.cert_type == 'cedula' %}readonly style="background:#f1f5f9; flex:1;"{% else
%}style="flex:1;"{% endif %}>
        <label style="display:flex; align-items:center; gap:6px; font-
size:12px; font-weight:600; cursor:pointer; margin:0;">
            <input type="checkbox" name="is_exempt" id="is_exempt">
            Exempt (RA 11261)
        </label>
    </div>
</div>

<div style="margin-top: 30px;">
    <button type="submit" class="btn-submit">
        <i class="fas fa-check-circle"></i> Issue & Fulfill
    </button>
    <a href="{% url 'certifications:list' %}" style="margin-left: 10px;
color: #64748b; font-size: .9rem;">Cancel</a>
</div>
</div>
</form>
</div>

{% if cert_request.cert_type == 'cedula' %}
<script>
document.addEventListener('DOMContentLoaded', function() {
    const taxpayerTypeSelect = document.getElementById('taxpayer_type_select');
    const amountPaidInput = document.getElementById('id_amount_paid');
    const basicTaxInput = document.getElementById('id_basic_tax');
    const incomeGroup = document.getElementById('income_group');

    const rawProp = document.getElementById('raw_prop');
    const valProp = document.getElementById('val_prop');
    const rawBus = document.getElementById('raw_bus');
    const valBus = document.getElementById('val_bus');
    const rawInc = document.getElementById('raw_inc');
    const valInc = document.getElementById('val_inc');

    function calculateCedula() {
        const isIndiv = taxpayerTypeSelect.value === 'individual';
        const rate = isIndiv ? (1 / 1000) : (2 / 5000);
        const cap = isIndiv ? 5000 : 10000;

        const vProp = (parseFloat(rawProp.value) || 0) * rate;
        const vBus = (parseFloat(rawBus.value) || 0) * rate;
        const vInc = isIndiv ? (parseFloat(rawInc.value) || 0) * rate : 0;

        valProp.value = vProp.toFixed(2);
    }
})
</script>
{% endif %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
valBus.value = vBus.toFixed(2);
if (isIndiv) valInc.value = vInc.toFixed(2);
else valInc.value = "0.00";

const basic = parseFloat(basicTaxInput.value) || 0;
const interest =
parseFloat(document.querySelector('input[name="interest"]').value) || 0;

const additionalTotal = vProp + vBus + vInc;
const cappedAdditional = Math.min(additionalTotal, cap);
const total = basic + cappedAdditional + interest;

amountPaidInput.value = total.toFixed(2);
}

taxpayerTypeSelect.addEventListener('change', function() {
  const isIndiv = this.value === 'individual';
  basicTaxInput.value = isIndiv ? "5.00" : "500.00";
  incomeGroup.style.display = isIndiv ? 'block' : 'none';
  calculateCedula();
});

document.querySelectorAll('.cedula-row, .cedula-calc').forEach(input => {
  input.addEventListener('input', calculateCedula);
});

// Initial calculation
calculateCedula();

const isExempt = document.getElementById('is_exempt');
const orInput = document.getElementsByName('or_number')[0];
const originalAmount = amountPaidInput.value;

isExempt.addEventListener('change', function() {
  if (this.checked) {
    amountPaidInput.value = "0.00";
    if (orInput) {
      orInput.value = "EXEMPT-RA11261";
      orInput.readOnly = true;
    }
  } else {
    calculateCedula();
    if (orInput) {
      orInput.value = "";
      orInput.readOnly = false;
    }
  }
});
});
</script>
{% endif %}
{% endblock %}
```

File: templates/certifications/issue.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Issue Certificate - Barangay Management System
<script src="{% static 'js/grammar_checker.js' %}"></script>
<script>
    document.addEventListener('DOMContentLoaded', () => {
        GrammarChecker.init('purposeInput', 'grammarFeedback');
    });
</script>
{% endblock %}

{% block content %}
{% url 'certifications:list' as certifications_list_url %}
{% include 'components/back_button.html' with back_url=certifications_list_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>Issue New Certificate</h1>
        <span>Select certificate type and fill in the details</span>
    </div>
</div>

<div class="card">
    <form method="POST" class="modern-form">
        {% csrf_token %}
        <div class="form-grid">
            <div class="form-section">
                <h3>Certificate Details</h3>
                <div class="form-row">
                    <div class="form-group col-6">
                        <label>Certificate Type*</label>
                        <select name="cert_type" id="cert_type" class="form-control" required>
                            <option value="">Select Type...</option>
                            {% for code, label in cert_types %}
                                <option value="{{ code }}">{{ label }}</option>
                            {% endfor %}
                        </select>
                    </div>
                    <div class="form-group col-6">
                        <label>Control Number (Suggested)</label>
                        <input type="text" name="control_number" id="id_control_number" class="form-
control" readonly
                            title="Automatically generated">
                    </div>
                </div>
            </div>

            <div class="form-row">
                <div class="form-group col-12">
                    <label>Purpose*</label>
                    <input type="text" name="purpose" id="purposeInput" class="form-control"
                        placeholder="e.g. For employment, ID, etc." required>
                    <div id="grammarFeedback"></div>
                </div>
            </div>

            <div class="form-row">
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div class="form-group col-12">
  <label>Resident*</label>
  <select name="resident" class="form-control" required>
    <option value="">Search and select resident...</option>
    {% for r in residents %}
      <option value="{{ r.pk }}">{{ r.last_name }}, {{ r.first_name }} {{
r.middle_name }}</option>
    {% endfor %}
  </select>
  <small class="text-muted">Showing 50 most recent residents. Use search above
if not listed.</small>
</div>
</div>

<!-- Business Permit Fields (hidden by default) -->
<div class="form-section business-fields" style="display: none;">
  <h3>Business Information</h3>
  <div class="form-row">
    <div class="form-group col-6">
      <label>Business Name</label>
      <input type="text" name="business_name" class="form-control">
    </div>
    <div class="form-group col-6">
      <label>Business Type</label>
      <input type="text" name="business_type" class="form-control"
placeholder="e.g. Sari-sari Store">
    </div>
  </div>
  <div class="form-row">
    <div class="form-group col-12">
      <label>Business Address</label>
      <input type="text" name="business_address" class="form-control">
    </div>
  </div>
</div>

<!-- Cedula Fields (hidden by default) -->
<div class="form-section cedula-fields" style="display: none;">
  <h3>Cedula Details (Community Tax Certificate)</h3>
  <div class="form-row">
    <div class="form-group col-6">
      <label>Taxpayer Type</label>
      <select name="taxpayer_type" id="taxpayer_type" class="form-control cedula-
calc">
        <option value="individual">Individual</option>
        <option value="corporation">Corporation</option>
      </select>
    </div>
    <div class="form-group col-6">
      <label>CTC Number* <span id="ctc_status" style="font-size: 11px; margin-
left: 5px;"></span></label>
      <input type="text" name="ctc_number" id="id_ctc_number" class="form-control"
placeholder="CCI2024000000000">
    </div>
  </div>
  <div class="form-row">
    <div class="form-group col-12">
      <label>Place of Issue</label>
      <input type="text" name="place_of_issue" class="form-control" value="Sico-
Sico, Gigaquit">
    </div>
  </div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>
</div>
<div class="form-row individual-only">
    <div class="form-group col-3">
        <label>Height (cm)</label>
        <input type="text" name="height" class="form-control" placeholder="170">
    </div>
    <div class="form-group col-3">
        <label>Weight (kg)</label>
        <input type="text" name="weight" class="form-control" placeholder="65">
    </div>
    <div class="form-group col-6">
    </div>
</div>
<div class="form-row">
    <div class="form-group col-6">
        <label id="basic_tax_label">Basic Community Tax (?)</label>
        <input type="number" step="0.01" name="basic_tax" id="id_basic_tax"
class="form-control cedula-calc"
        value="5.00">
    </div>
    <div class="form-group col-6">
        <label>Interest (?)</label>
        <input type="number" step="0.01" name="interest" class="form-control cedula-
calc" value="0.00">
    </div>
</div>
<div class="form-row">
    <div class="form-group col-4">
        <label id="tax_prop_label">Tax Category A (Property)</label>
        <div class="input-group">
            <input type="number" step="0.01" id="raw_prop" name="raw_taxable_property"
                class="form-control cedula-raw" placeholder="Assessed Value">
            <input type="number" step="0.01" name="additional_tax_property"
id="val_prop" class="form-control"
                readonly title="Calculated Tax">
        </div>
        <small class="text-muted" id="prop_help">?1.00 for every ?1,000</small>
    </div>
    <div class="form-group col-4">
        <label id="tax_bus_label">Tax Category B (Business)</label>
        <div class="input-group">
            <input type="number" step="0.01" id="raw_bus" name="raw_taxable_business"
                class="form-control cedula-raw" placeholder="Gross Receipts">
            <input type="number" step="0.01" name="additional_tax_business"
id="val_bus" class="form-control"
                readonly title="Calculated Tax">
        </div>
        <small class="text-muted" id="bus_help">?1.00 for every ?1,000</small>
    </div>
    <div class="form-group col-4" id="income_group">
        <label>Tax Category C (Income)</label>
        <div class="input-group">
            <input type="number" step="0.01" id="raw_inc" name="raw_taxable_income"
class="form-control cedula-raw"
                placeholder="Annual Income">
            <input type="number" step="0.01" name="additional_tax_income" id="val_inc"
class="form-control" readonly
                title="Calculated Tax">
        </div>
    </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        <small class="text-muted">?1.00 for every ?1,000</small>
    </div>
</div>
</div>

<!-- Late Registration Fields (hidden by default) -->
<div class="form-section late-registration-fields" style="display: none;">
    <h3>Late Registration of Birth</h3>
    <div class="form-row">
        <div class="form-group col-12">
            <label>Full Name of Child</label>
            <input type="text" name="child_name" class="form-control">
        </div>
    </div>
    <div class="form-row">
        <div class="form-group col-6">
            <label>Date of Birth</label>
            <input type="date" name="child_birth_date" class="form-control">
        </div>
        <div class="form-group col-6">
            <label>Place of Birth</label>
            <input type="text" name="child_birth_place" class="form-control">
        </div>
    </div>
    <div class="form-row">
        <div class="form-group col-6">
            <label>Father's Full Name</label>
            <input type="text" name="father_name" class="form-control">
        </div>
        <div class="form-group col-6">
            <label>Mother's Full Name</label>
            <input type="text" name="mother_name" class="form-control">
        </div>
    </div>
</div>

<div class="form-section payment-section">
    <h3>Payment Information</h3>
    <div class="form-row">
        <div class="form-group col-6">
            <label>OR Number <span id="or_status" style="font-size: 11px; margin-left:
5px;"></span></label>
            <input type="text" name="or_number" id="id_or_number" class="form-control"
placeholder="Enter OR Number">
        </div>
        <div class="form-group col-6" style="display: flex; gap: 15px; align-items:
flex-end;">
            <div style="flex: 1;">
                <label id="amount_label">Amount Paid (?)</label>
                <input type="number" step="0.01" name="amount_paid" id="id_amount_paid"
class="form-control"
                value="0.00">
            </div>
            <label class="checkbox-container" style="margin-bottom: 4px;">
                <input type="checkbox" id="is_exempt" name="is_exempt">
                <span>Exempt (RA 11261)</span>
            </label>
        </div>
    </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</div>

<div class="form-footer">
  <button type="submit" class="btn btn-primary btn-lg">
    <i class="fas fa-check-circle"></i> Generate & Issue
  </button>
</div>
</form>
</div>

<script>
const certTypeSelect = document.getElementById('cert_type');
const taxpayerTypeSelect = document.getElementById('taxpayer_type');
const businessFields = document.querySelector('.business-fields');
const cedulaFields = document.querySelector('.cedula-fields');
const lateRegistrationFields = document.querySelector('.late-registration-fields');
const amountPaidInput = document.querySelector('input[name="amount_paid"]');
const amountLabel = document.getElementById('amount_label');
const cedulaCalcs = document.querySelectorAll('.cedula-calc');
const cedulaRaws = document.querySelectorAll('.cedula-row');

const basicTaxInput = document.getElementById('id_basic_tax');
const individualOnly = document.querySelector('.individual-only');
const incomeGroup = document.getElementById('income_group');

const rawProp = document.getElementById('raw_prop');
const valProp = document.getElementById('val_prop');
const rawBus = document.getElementById('raw_bus');
const valBus = document.getElementById('val_bus');
const rawInc = document.getElementById('raw_inc');
const valInc = document.getElementById('val_inc');

function calculateCedula() {
  const isIndiv = taxpayerTypeSelect.value === 'individual';
  const rate = isIndiv ? (1 / 1000) : (2 / 5000);
  const cap = isIndiv ? 5000 : 10000;

  // Calculate components
  const vProp = (parseFloat(rawProp.value) || 0) * rate;
  const vBus = (parseFloat(rawBus.value) || 0) * rate;
  const vInc = isIndiv ? (parseFloat(rawInc.value) || 0) * rate : 0;

  valProp.value = vProp.toFixed(2);
  valBus.value = vBus.toFixed(2);
  if (isIndiv) valInc.value = vInc.toFixed(2);
  else valInc.value = "0.00";

  const basic = parseFloat(basicTaxInput.value) || 0;
  const interest = parseFloat(document.getElementsByName('interest')[0].value) || 0;

  const additionalTotal = vProp + vBus + vInc;
  const cappedAdditional = Math.min(additionalTotal, cap);
  const total = basic + cappedAdditional + interest;

  amountPaidInput.value = total.toFixed(2);
}

taxpayerTypeSelect.addEventListener('change', function () {
  const isIndiv = this.value === 'individual';
  basicTaxInput.value = isIndiv ? "5.00" : "500.00";
});
```


IoT-Enabled Barangay Resident and Attendance Management System

```
individualOnly.forEach(el => el.style.display = isIndiv ? 'flex' : 'none');
incomeGroup.style.display = isIndiv ? 'block' : 'none';

document.getElementById('prop_help').innerText = isIndiv ? "?1.00 for every ?1,000" :
"?2.00 for every ?5,000";
document.getElementById('bus_help').innerText = isIndiv ? "?1.00 for every ?1,000" :
"?2.00 for every ?5,000";

rawProp.placeholder = isIndiv ? "Property Value" : "Assessed Value";
rawBus.placeholder = isIndiv ? "Gross Receipts" : "Earnings/Receipts";

calculateCedula();
});

certTypeSelect.addEventListener('change', function () {
  // Reset all
  businessFields.style.display = 'none';
  cedulaFields.style.display = 'none';
  lateRegistrationFields.style.display = 'none';
  amountPaidInput.readOnly = false;
  amountLabel.innerText = "Amount Paid (?)";

  if (this.value === 'business_permit') {
    businessFields.style.display = 'block';
  } else if (this.value === 'cedula') {
    cedulaFields.style.display = 'block';
    amountPaidInput.readOnly = true;
    amountLabel.innerText = "Total Amount (?) - Auto Calculated";
    calculateCedula();
  } else if (this.value === 'late_registration') {
    lateRegistrationFields.style.display = 'block';
  }

  // Fetch next numbers
  fetchNextNumbers(this.value);
});

cedulaRaws.forEach(input => {
  input.addEventListener('input', calculateCedula);
});

cedulaCalcs.forEach(input => {
  input.addEventListener('input', calculateCedula);
});

// Real-time Uniqueness Validation
const orInput = document.getElementById('id_or_number');
const ctcInput = document.getElementById('id_ctc_number');
const orStatus = document.getElementById('or_status');
const ctcStatus = document.getElementById('ctc_status');
const controlInput = document.getElementById('id_control_number');
const isExemptCheckbox = document.getElementById('is_exempt');

async function fetchNextNumbers(certType) {
  if (!certType) return;
  try {
    const resp = await fetch(`/certifications/get-next-numbers/?type=${certType}`);
    const data = await resp.json();
    if (data.control_number) controlInput.value = data.control_number;
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
// Auto-fill CTC number if Cedula and empty
if (data.ctc_number && !ctcInput.value) {
  ctcInput.value = data.ctc_number;
  validateUniqueness('ctc_number', data.ctc_number, ctcStatus);
}

// Auto-fill OR number if empty
if (data.or_number && !orInput.value && !isExemptCheckbox.checked) {
  orInput.value = data.or_number;
  validateUniqueness('or_number', data.or_number, orStatus);
}

// Auto-fill Amount Paid based on rate
if (data.rate && certType !== 'cedula') {
  amountPaidInput.value = data.rate;
}
} catch (e) {
  console.error("Error fetching next numbers", e);
}
}

isExemptCheckbox.addEventListener('change', function () {
  if (this.checked) {
    orInput.value = "EXEMPT";
    orInput.readOnly = true;
    orStatus.innerText = "";
  } else {
    orInput.value = "";
    orInput.readOnly = false;
    fetchNextNumbers(certTypeSelect.value);
  }
});

async function validateUniqueness(field, value, statusEl) {
  if (!value) {
    statusEl.innerText = "";
    return;
  }
  statusEl.innerHTML = '<i class="fas fa-spinner fa-spin"></i> Checking...';
  try {
    const resp = await fetch(`/certifications/check-
uniqueness/?field=${field}&value=${value}`);
    const data = await resp.json();
    if (data.available) {
      statusEl.innerHTML = '<span style="color: green;"><i class="fas fa-check-
circle"></i> Available</span>';
    } else {
      statusEl.innerHTML = '<span style="color: red;"><i class="fas fa-times-
circle"></i> Already Taken</span>';
    }
  } catch (e) {
    statusEl.innerText = "Error checking";
  }
}

let timers = {};
function debounceValidate(field, value, statusEl) {
  if (timers[field]) clearTimeout(timers[field]);
  timers[field] = setTimeout(() => {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        validateUniqueness(field, value, statusEl);
    }, 500);
}

orInput.addEventListener('input', () => debounceValidate('or_number', orInput.value,
orStatus));
ctcInput.addEventListener('input', () => debounceValidate('ctc_number', ctcInput.value,
ctcStatus));
</script>
<style>
    .form-control {
        width: 100%;
        padding: 10px 14px;
        border: 1px solid var(--border);
        border-radius: 8px;
        font-size: 14px;
        height: 42px;
        line-height: 1.4;
        box-sizing: border-box;
    }

    .form-section {
        margin-bottom: 25px;
    }

    .form-section h3 {
        font-size: 14px;
        color: var(--primary);
        margin-bottom: 15px;
        border-bottom: 1px solid var(--border);
        padding-bottom: 8px;
    }

    .form-row {
        display: flex;
        gap: 20px;
        margin-bottom: 15px;
    }

    .col-6 {
        flex: 1;
    }

    .col-12 {
        width: 100%;
    }

    .form-footer {
        border-top: 1px solid var(--border);
        padding-top: 20px;
        text-align: right;
    }
</style>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/certifications/list.html

```
{% extends 'base.html' %}

{% block title %}Certifications - Barangay Management System{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Certifications</span>
{% endblock %}

{% block content %}
<style>
    /* Compact Table Styles */
    .compact-table {
        font-size: 0.85rem;
    }
    .compact-table th {
        padding: 10px 12px;
        background: #f8fafc;
        color: #475569;
        font-weight: 700;
        text-transform: uppercase;
        letter-spacing: 0.025em;
        border-bottom: 2px solid #e2e8f0;
    }
    .compact-table td {
        padding: 8px 12px;
        vertical-align: middle;
        border-bottom: 1px solid #f1f5f9;
    }
    .compact-table tr:hover {
        background-color: #f8fafc;
    }

    /* Refined Filter Bar */
    .filter-bar {
        display: flex;
        justify-content: space-between;
        align-items: center;
        gap: 12px;
        margin-bottom: 16px;
        background: #fff;
        padding: 12px;
        border-radius: 12px;
        border: 1px solid #e2e8f0;
    }
    .filter-group {
        display: flex;
        gap: 8px;
        flex: 1;
    }
    .search-wrapper {
        position: relative;
        flex: 1;
        max-width: 400px;
    }
    .search-wrapper i {
        position: absolute;
        left: 12px;
    }

```

IoT-Enabled Barangay Resident and Attendance Management System

```
    top: 50%;
    transform: translateY(-50%);
    color: #94a3b8;
    font-size: 0.85rem;
  }
  .search-input-compact {
    width: 100%;
    padding: 8px 12px 8px 32px;
    border: 1px solid #e2e8f0;
    border-radius: 8px;
    font-size: 0.85rem;
    transition: all 0.2s;
  }
  .search-input-compact:focus {
    border-color: var(--primary);
    box-shadow: 0 0 0 3px rgba(45, 122, 80, 0.1);
    outline: none;
  }
  .filter-select-compact {
    padding: 8px 12px;
    border: 1px solid #e2e8f0;
    border-radius: 8px;
    font-size: 0.85rem;
    min-width: 160px;
    cursor: pointer;
  }
}

.btn-icon-sm {
  width: 32px;
  height: 32px;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  border-radius: 6px;
  transition: all 0.2s;
  border: none;
  cursor: pointer;
  font-size: 0.85rem;
}
.btn-icon-view { background: rgba(59, 130, 246, 0.1); color: #3b82f6; }
.btn-icon-view:hover, .btn-icon-view:focus, .btn-icon-view:active { background:
#3b82f6; color: #fff; outline: none; }
.btn-icon-print { background: rgba(100, 116, 139, 0.1); color: #64748b; }
.btn-icon-print:hover, .btn-icon-print:focus, .btn-icon-print:active { background:
#64748b; color: #fff; outline: none; }
.btn-icon-cert { background: rgba(45, 122, 80, 0.1); color: #2d7a50; }
.btn-icon-cert:hover, .btn-icon-cert:focus, .btn-icon-cert:active { background:
#2d7a50; color: #fff; outline: none; }
.btn-icon-delete { background: rgba(239, 68, 68, 0.1); color: #ef4444; }
.btn-icon-delete:hover, .btn-icon-delete:focus, .btn-icon-delete:active { background:
#ef4444; color: #fff; outline: none; }

{% if is_resident_view %}
/* Resident Card Layout - already quite clean, but let's tighten it slightly */
.certificates-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
  gap: 16px;
  margin-top: 16px;
}
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

.cert-card {
  background: #fff;
  border-radius: 12px;
  padding: 18px;
  border: 1px solid #e2e8f0;
  transition: all 0.3s ease;
}
{% endif %}
</style>

<div class="page-header" style="margin-bottom: 16px;">
  <div class="page-title">
    <h1 style="font-size: 1.25rem;"><i class="fas fa-certificate" style="margin-right: 8px;"></i> Document Certifications</h1>
    <span style="font-size: 0.85rem;">Manage and issue barangay clearances and certificates</span>
  </div>
  <div class="header-actions">
    {% if user_role != 'resident' %}
    <a href="{% url 'certifications:issue' %}" class="btn btn-primary btn-sm">
      <i class="fas fa-file-signature"></i> Issue New
    </a>
    {% endif %}
  </div>
</div>

{% if is_resident_view %}
<div class="certificates-grid">
  {% for cert in certificates %}
  <div class="cert-card">
    <div style="display:flex; justify-content:space-between; align-items:start; margin-bottom:12px;">
      <div style="width:36px; height:36px; background:rgba(45,122,80,0.1); color:#2d7a50; border-radius:8px; display:flex; align-items:center; justify-content:center; font-size:16px;">
        <i class="fas fa-certificate"></i>
      </div>
      <span style="font-size:11px; color:#94a3b8; font-weight:600;">{{ cert.date_issued|date:"M d, Y" }}</span>
    </div>
    <h3 style="font-size:16px; font-weight:700; margin-bottom:4px; color:#1e293b;">{{ cert.get_cert_type_display }}</h3>
    <div style="font-family:monospace; font-size:12px; color:#64748b; margin-bottom:12px;">{{ cert.control_number }}</div>

    <div style="font-size: 13px; color: #64748b; margin-bottom: 16px; line-height: 1.4;">
      {{ cert.purpose|truncatechars:100 }}
    </div>

    <div style="display:flex; gap:8px; padding-top:12px; border-top:1px solid #f1f5f9;">
      <a href="{% url 'certifications:virtual_certificate' cert.pk %}" class="btn btn-primary btn-sm" style="flex:1; justify-content:center;">
        <i class="fas fa-eye"></i> View
      </a>
      <a href="{% url 'certifications:pdf' cert.pk %}" class="btn btn-outline btn-sm" target="_blank" style="padding:6px 10px;">
        <i class="fas fa-download"></i>
      </a>
    </div>
  </div>
  </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        </div>
    </div>
    {% empty %}
    <div style="grid-column:1/-1; padding:40px; text-align:center; background:#fff;
border-radius:12px; border:1px dashed #cbd5e1;">
        <i class="fas fa-folder-open" style="font-size:32px; color:#cbd5e1; margin-
bottom:12px;"></i>
        <h3 style="font-size:16px;">No certificates found</h3>
        <p style="color:#64748b; font-size:14px;">You haven't been issued any certificates
yet.</p>
    </div>
    {% endfor %}
</div>

{% else %}

<div class="filter-bar">
    <form method="GET" class="filter-group">
        <div class="search-wrapper">
            <i class="fas fa-search"></i>
            <input type="text" name="q" value="{{ query }}" placeholder="Search by name or
control no..." class="search-input-compact">
        </div>
        <select name="type" class="filter-select-compact" onchange="this.form.submit()">
            <option value="">All Document Types</option>
            {% for code, label in cert_types %}
            <option value="{{ code }}" {% if code == selected_type %}selected{% endif
%}>{{ label }}</option>
            {% endfor %}
        </select>
        <button type="submit" class="btn btn-primary btn-sm" style="padding: 0
16px;">Search</button>
        {% if query or selected_type %}
        <a href="{% url 'certifications:list' %}" class="btn btn-outline btn-sm"
title="Clear filters">Clear</a>
        {% endif %}
    </form>
</div>

<div class="card" style="padding:0; border-radius:12px; overflow:hidden;">
    <div class="table-responsive">
        <table class="compact-table">
            <thead>
                <tr>
                    <th>Control No.</th>
                    <th>Resident</th>
                    <th>Document Type</th>
                    <th>Date Issued</th>
                    <th>OR Number</th>
                    <th class="text-end">Amount</th>
                    <th class="text-center" style="width: 150px;">Actions</th>
                </tr>
            </thead>
            <tbody>
                {% for cert in certificates %}
                <tr>
                    <td><span style="font-weight:700; color:#1e293b;">{{
cert.control_number }}</span></td>
                    <td>{{ cert.resident.full_name }}</td>
                    <td>{{ cert.get_cert_type_display }}</td>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

<td>{{ cert.date_issued|date:"M d, Y" }}</td>
<td><code style="font-size:0.8rem;">{{ cert.or_number|default:"--"
}}</code></td>
<td class="text-end">?{{ cert.amount_paid|default:"0.00" }}</td>
<td>
    <div style="display:flex; gap:6px; justify-content:center;">
        <a href="{% url 'certifications:view' cert.pk %}" class="btn-
icon-sm btn-icon-view" title="View Details">
            <i class="fas fa-eye"></i>
        </a>
        <a href="{% url 'certifications:pdf' cert.pk %}"
target="_blank" class="btn-icon-sm btn-icon-print" title="Download PDF">
            <i class="fas fa-file-pdf"></i>
        </a>
        <a href="{% url 'certifications:virtual_certificate' cert.pk
%}" class="btn-icon-sm btn-icon-cert" title="Virtual View">
            <i class="fas fa-certificate"></i>
        </a>
        {% if user_role != 'resident' %}
        <form method="POST" action="{% url 'certifications:delete'
cert.pk %}" onsubmit="return confirm('Delete {{ cert.control_number }}?');"
style="display:inline;">
            {% csrf_token %}
            <button type="submit" class="btn-icon-sm btn-icon-delete"
title="Delete">
                <i class="fas fa-trash"></i>
            </button>
        </form>
        {% endif %}
    </div>
</td>
</tr>
{% empty %}
<tr>
    <td colspan="7" class="text-center" style="padding:30px;
color:#64748b;">No documents found matching your criteria.</td>
</tr>
{% endfor %}
<!-- Dynamic No Results Row -->
<tr id="no-results-row" style="display: none;">
    <td colspan="7" class="text-center" style="padding: 40px; color:
#94a3b8;">
        <i class="fas fa-search" style="font-size: 24px; display: block;
margin-bottom: 10px; opacity: 0.5;"></i>
        <span>No matching documents found on this page.</span>
    </td>
</tr>
</tbody>
</table>
</div>
</div>

<!-- Dynamic No Results Card -->
<div id="no-results-card" style="display: none; padding: 40px; text-align: center;
background: #fff; border-radius: 12px; border: 1px dashed #cbd5e1; margin-top: 16px;">
    <i class="fas fa-search" style="font-size: 32px; color: #cbd5e1; margin-bottom:
12px;"></i>
    <h3 style="font-size: 16px; color: #1e293b;">No matches found</h3>
    <p style="color: #64748b; font-size: 14px;">Try a different keyword or control
number.</p>

```


IoT-Enabled Barangay Resident and Attendance Management System

```
</div>

{% if certificates.has_other_pages %}
<div style="display:flex; justify-content:center; gap:8px; margin-top:16px;">
  {% if certificates.has_previous %}
    <a href="?q={{ query }}&type={{ selected_type }}&page={{
certificates.previous_page_number }}" class="btn btn-outline btn-sm">Prev</a>
  {% endif %}
  <span style="padding:6px 12px; font-size:0.85rem; color:#64748b;">Page {{
certificates.number }} of {{ certificates.paginator.num_pages }}</span>
  {% if certificates.has_next %}
    <a href="?q={{ query }}&type={{ selected_type }}&page={{ certificates.next_page_number
 }}" class="btn btn-outline btn-sm">Next</a>
  {% endif %}
</div>
{% endif %}

{% block extra_js %}
<script>
  document.addEventListener('DOMContentLoaded', function() {
    const searchInput = document.querySelector('.search-input-compact');
    if (!searchInput) return;

    searchInput.addEventListener('input', function() {
      const filter = this.value.toUpperCase();
      let visibleCount = 0;

      // 1. Filter Table Rows (Admin/Staff View)
      const tableRows = document.querySelectorAll('.compact-table tbody tr:not(#no-
results-row)');
      if (tableRows.length > 0) {
        tableRows.forEach(row => {
          // Skip the "No documents found" row (from Django empty tag)
          if (row.cells.length === 1 && row.cells[0].colSpan > 1) return;

          const text = row.innerText.toUpperCase();
          const isMatch = text.indexOf(filter) > -1;
          row.style.display = isMatch ? "" : "none";
          if (isMatch) visibleCount++;
        });

        const noResultsRow = document.getElementById('no-results-row');
        if (noResultsRow) {
          noResultsRow.style.display = (visibleCount === 0 && filter !== "") ?
"" : "none";
        }
      }

      // 2. Filter Cards (Resident View)
      const cards = document.querySelectorAll('.cert-card');
      if (cards.length > 0) {
        visibleCount = 0;
        cards.forEach(card => {
          const text = card.innerText.toUpperCase();
          const isMatch = text.indexOf(filter) > -1;
          card.style.display = isMatch ? "" : "none";
          if (isMatch) visibleCount++;
        });
      }
    });
  });
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
const noResultsCard = document.getElementById('no-results-card');
if (noResultsCard) {
  noResultsCard.style.display = (visibleCount === 0 && filter !== "") ?
"" : "none";
}
});
});
</script>
{% endblock %}
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/certifications/my_requests.html

```
{% extends 'base.html' %}
{% block title %}My Certificate Requests{% endblock %}

{% block breadcrumb %}
<span><i class="fas fa-home"></i> / <a href="{% url 'core:dashboard' %}">Dashboard</a> /
My Requests</span>
{% endblock %}

{% block content %}
<link href="https://fonts.googleapis.com/css2?family=DM+Serif+Display:ital@0;1&family=DM+Sans:wght@300;400;500;600;700&display=swap" rel="stylesheet">

<style>
:root {
  --req-green: #166534;
  --req-green-md: #16a34a;
  --req-green-lt: #dcfce7;
  --req-blue: #1d4ed8;
  --req-blue-lt: #dbeafe;
  --req-amber: #92400e;
  --req-amber-lt: #fef3c7;
  --req-red: #991b1b;
  --req-red-lt: #fee2e2;
  --req-muted: #64748b;
  --req-border: #e2e8f0;
  --req-text: #0f172a;
  --req-surface: #ffffff;
  --req-bg: #f8fafc;
}

body { font-family: 'DM Sans', sans-serif; }

/* ?? Page Header ?? */
.req-header {
  display: flex; align-items: flex-start; justify-content: space-between;
  flex-wrap: wrap; gap: 16px;
  margin-bottom: 28px;
}
.req-header h1 {
  font-family: 'DM Serif Display', serif;
  font-size: 1.9rem;
  font-weight: 400;
  margin: 0 0 4px;
  color: var(--req-text);
  line-height: 1.2;
}
.req-header .subtitle { font-size: 14px; color: var(--req-muted); margin: 0; }

.btn-new-req {
  display: inline-flex; align-items: center; gap: 8px;
  background: linear-gradient(135deg, var(--req-green) 0%, var(--req-green-md) 100%);
  color: #fff; border: none; padding: 12px 22px;
  border-radius: 12px; font-size: 14px; font-weight: 700;
  font-family: 'DM Sans', sans-serif;
  text-decoration: none; cursor: pointer;
  transition: all 0.22s; box-shadow: 0 4px 12px rgba(22,163,74,.25);
  white-space: nowrap;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}
.btn-new-req:hover {
  transform: translateY(-2px); box-shadow: 0 8px 20px rgba(22,163,74,.35);
  color: #fff; text-decoration: none;
}

/* ?? Stats Strip ?? */
.stats-strip {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(140px, 1fr));
  gap: 12px;
  margin-bottom: 24px;
}
.stat-card {
  background: var(--req-surface);
  border-radius: 14px;
  padding: 16px 18px;
  display: flex; align-items: center; gap: 12px;
  box-shadow: 0 2px 8px rgba(0,0,0,.05);
  border: 1px solid var(--req-border);
}
.stat-icon {
  width: 38px; height: 38px;
  border-radius: 10px;
  display: flex; align-items: center; justify-content: center;
  font-size: 16px; flex-shrink: 0;
}
.stat-num { font-size: 1.4rem; font-weight: 800; line-height: 1; color: var(--req-text); }
.stat-label { font-size: 11px; color: var(--req-muted); font-weight: 600; text-
transform: uppercase; letter-spacing: 0.5px; }

/* ?? Request Cards ?? */
.req-list { display: flex; flex-direction: column; gap: 16px; }

.req-card {
  background: var(--req-surface);
  border-radius: 18px;
  border: 1px solid var(--req-border);
  box-shadow: 0 2px 12px rgba(0,0,0,.05);
  overflow: hidden;
  transition: box-shadow 0.2s, transform 0.2s;
}
.req-card:hover {
  box-shadow: 0 6px 24px rgba(0,0,0,.1);
  transform: translateY(-1px);
}

.req-card-main {
  display: grid;
  grid-template-columns: auto 1fr auto auto;
  gap: 0;
  align-items: stretch;
}
@media(max-width: 700px) {
  .req-card-main { grid-template-columns: 1fr; }
}

/* Left accent strip */
.req-accent {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
width: 6px;
flex-shrink: 0;
}
.accent-pending { background: linear-gradient(180deg, #f59e0b, #fbbf24); }
.accent-issued { background: linear-gradient(180deg, var(--req-green), var(--req-green-
md)); }
.accent-rejected{ background: linear-gradient(180deg, #ef4444, #f87171); }

.req-card-body { padding: 20px 22px; flex: 1; }
.req-card-body-top {
  display: flex; align-items: flex-start;
  justify-content: space-between; gap: 12px; margin-bottom: 10px;
}

.cert-type-badge {
  display: inline-flex; align-items: center; gap: 6px;
  background: var(--req-bg); border: 1px solid var(--req-border);
  padding: 4px 10px; border-radius: 8px;
  font-size: 11px; font-weight: 700; text-transform: uppercase;
  letter-spacing: 0.5px; color: #475569;
}

.req-tracking {
  font-family: monospace; font-size: 12px; font-weight: 700;
  color: var(--req-green); background: var(--req-green-lt);
  padding: 2px 8px; border-radius: 6px; letter-spacing: 0.5px;
}

.req-title { font-size: 15px; font-weight: 700; color: var(--req-text); margin-bottom:
4px; }
.req-purpose { font-size: 13px; color: var(--req-muted); }
.req-meta { font-size: 12px; color: #94a3b8; margin-top: 8px; }

/* Payment section */
.req-payment-col {
  padding: 16px 20px;
  border-left: 1px solid var(--req-border);
  display: flex; flex-direction: column;
  justify-content: center; align-items: center;
  min-width: 140px; text-align: center;
  gap: 6px;
  background: #fafafa;
}
.pay-amount {
  font-family: 'DM Serif Display', serif;
  font-size: 1.5rem;
  font-weight: 400;
  line-height: 1;
}
.pay-amount.paid { color: var(--req-green); }
.pay-amount.waived { color: #7c3a6d; }
.pay-amount.pending{ color: #d97706; }
.pay-amount.unpaid { color: var(--req-red); }

.pay-status-pill {
  display: inline-flex; align-items: center; gap: 4px;
  padding: 3px 9px; border-radius: 99px;
  font-size: 10px; font-weight: 700; text-transform: uppercase;
  letter-spacing: 0.5px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.pill-paid { background: var(--req-green-lt); color: var(--req-green); }
.pill-waived { background: #ede9fe; color: #7c3aed; }
.pill-pending { background: var(--req-amber-lt); color: var(--req-amber); }
.pill-unpaid { background: var(--req-red-lt); color: var(--req-red); }

.pay-cta {
  display: inline-flex; align-items: center; gap: 5px;
  background: var(--req-green);
  color: #fff; font-size: 11px; font-weight: 700;
  padding: 5px 12px; border-radius: 8px;
  text-decoration: none; margin-top: 4px;
  transition: background 0.2s;
  white-space: nowrap;
}
.pay-cta:hover { background: #14532d; color: #fff; text-decoration: none; }

/* Actions column */
.req-actions-col {
  padding: 16px;
  display: flex; flex-direction: column;
  justify-content: center; align-items: flex-end;
  gap: 8px; min-width: 120px;
  border-left: 1px solid var(--req-border);
}

.action-btn {
  display: inline-flex; align-items: center; gap: 6px;
  padding: 8px 14px; border-radius: 10px;
  font-size: 12px; font-weight: 700;
  text-decoration: none; white-space: nowrap;
  transition: all 0.18s; border: 1.5px solid transparent;
}
.action-btn-view {
  background: var(--req-green-lt);
  color: var(--req-green);
  border-color: #86efac;
}
.action-btn-view:hover { background: var(--req-green); color: #fff; text-decoration: none; }
.action-btn-receipt {
  background: var(--req-blue-lt);
  color: var(--req-blue);
  border-color: #93c5fd;
}
.action-btn-receipt:hover { background: var(--req-blue); color: #fff; text-decoration: none; }

.req-status-pill {
  display: inline-flex; align-items: center; gap: 5px;
  padding: 4px 10px; border-radius: 99px;
  font-size: 10px; font-weight: 800; text-transform: uppercase; letter-spacing: 0.6px;
}
.status-pending { background: var(--req-amber-lt); color: var(--req-amber); }
.status-issued { background: var(--req-green-lt); color: var(--req-green); }
.status-rejected { background: var(--req-red-lt); color: var(--req-red); }

/* ?? Empty State ?? */
.empty-state {
  text-align: center;
  padding: 80px 20px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
background: var(--req-surface);
border-radius: 18px;
border: 1px dashed var(--req-border);
}
.empty-icon { font-size: 52px; color: #cbd5e1; margin-bottom: 16px; }
.empty-title { font-size: 1.2rem; font-weight: 700; color: #475569; margin-bottom: 8px;
}
.empty-sub { font-size: 14px; color: var(--req-muted); margin-bottom: 24px; }

/* ?? Rejected Note ?? */
.rejected-note {
background: var(--req-red-lt);
border-top: 1px solid #fca5a5;
padding: 10px 22px;
font-size: 12px;
color: var(--req-red);
display: flex; align-items: flex-start; gap: 8px;
}
</style>

<div class="req-header">
  <div>
    <h1>My Certificate Requests</h1>
    <p class="subtitle">Track all your submitted barangay document requests and payment
status</p>
  </div>
  <a href="{% url 'certifications:request_certificate' %}" class="btn-new-req">
    <i class="fas fa-plus"></i> New Request
  </a>
</div>

{% if messages %}
{% for message in messages %}
<div class="alert alert-{{ message.tags }}" style="margin-bottom: 20px; border-radius:
12px;">
  {{ message }}
</div>
{% endfor %}
{% endif %}

{% if requests %}
<!-- Stats Strip -->
<div class="stats-strip">
  <div class="stat-card">
    <div class="stat-icon" style="background:#fef3c7; color:#92400e;">
      <i class="fas fa-clock"></i>
    </div>
    <div>
      <div class="stat-num">{{ requests|length }}</div>
      <div class="stat-label">Total</div>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon" style="background:var(--req-amber-lt); color:var(--req-
amber);">
      <i class="fas fa-hourglass-half"></i>
    </div>
    <div>
      <div class="stat-num">
        {% with pending_count=0 %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        {% for r in requests %}{% if r.status == 'pending' %}{% with
pending_count=pending_count|add:1 %}{% endif %}{% endfor %}
        {% for r in requests %}{% if r.status == 'pending' %}1{% endif %}{% endfor %}
        {% endif %}
    </div>
    <div class="stat-label">Pending</div>
</div>
</div>
<div class="stat-card">
    <div class="stat-icon" style="background:var(--req-green-lt); color:var(--req-
green);">
        <i class="fas fa-check-circle"></i>
    </div>
    <div>
        <div class="stat-num">{% for r in requests %}{% if r.status == 'issued' %}?{% endif
%}{% endfor %}</div>
        <div class="stat-label">Issued</div>
    </div>
</div>
</div>

<div class="req-list">
    {% for req in requests %}
    <div class="req-card">
        <div class="req-card-main">

            <!-- Left accent -->
            <div class="req-accent {% if req.status == 'pending' %}accent-pending{% elif
req.status == 'issued' %}accent-issued{% else %}accent-rejected{% endif %}"></div>

            <!-- Main content -->
            <div class="req-card-body">
                <div class="req-card-body-top">
                    <div>
                        <span class="cert-type-badge">
                            <i class="fas fa-file-alt"></i>
                            {{ req.get_cert_type_display }}
                        </span>
                    </div>
                    <div style="display:flex; align-items:center; gap: 8px; flex-wrap: wrap;">
                        <span class="req-tracking">{{ req.tracking_code }}</span>
                        <span class="req-status-pill {% if req.status == 'pending' %}status-pending{%
elif req.status == 'issued' %}status-issued{% else %}status-rejected{% endif %}">
                            {% if req.status == 'pending' %}<i class="fas fa-clock"></i> Pending
                            {% elif req.status == 'issued' %}<i class="fas fa-check-circle"></i> Issued
                            {% else %}<i class="fas fa-times-circle"></i> Rejected{% endif %}
                        </span>
                    </div>
                </div>
            </div>

            <div class="req-purpose">
                <strong style="color: #334155;">Purpose:</strong> {{
req.purpose|truncatechars:80 }}
            </div>
            <div class="req-meta">
                <i class="fas fa-calendar-alt" style="margin-right: 4px;"></i>
                Submitted on {{ req.created_at|date:"F d, Y \a\t g:i A" }}
            </div>
        </div>
    </div>
    {% endfor %}
</div>

```


IoT-Enabled Barangay Resident and Attendance Management System

```

<!-- Payment Column -->
<div class="req-payment-col">
  <div style="font-size: 10px; font-weight: 700; text-transform: uppercase; letter-
spacing: 0.5px; color: var(--req-muted); margin-bottom: 4px;">
    Payment
  </div>

  {% if req.payment %}
    {% if req.payment.amount %}
      <div class="pay-amount"
        {% if req.payment.status == 'paid' %}paid
        {% elif req.payment.status == 'waived' %}waived
        {% elif req.payment.status == 'pending' %}pending
        {% else %}unpaid{% endif %}">
        {% if req.payment.status == 'waived' %}
          FREE
        {% else %}
          ?{{ req.payment.amount|floatformat:2 }}
        {% endif %}
      </div>
    {% endif %}

    <div class="pay-status-pill"
      {% if req.payment.status == 'paid' %}pill-paid
      {% elif req.payment.status == 'waived' %}pill-waived
      {% elif req.payment.status == 'pending' %}pill-pending
      {% else %}pill-unpaid{% endif %}">
      <i class="fas"
        {% if req.payment.status == 'paid' %}fa-check
        {% elif req.payment.status == 'waived' %}fa-gift
        {% elif req.payment.status == 'pending' %}fa-spinner fa-spin
        {% else %}fa-exclamation{% endif %}"></i>
      {{ req.payment.get_status_display }}
    </div>

    {% if req.payment.status == 'unpaid' %}
      <a href="{% url 'payments:choose_payment' req.id %}" class="pay-cta">
        <i class="fas fa-credit-card"></i> Pay Now
      </a>
      {% elif req.payment.status == 'pending' %}
      <div style="font-size: 10px; color: var(--req-amber); font-weight: 600; margin-
top: 2px;">
        Awaiting verification
      </div>
    {% endif %}

    {% else %}
      <div style="font-size: 11px; color: var(--req-muted); margin-bottom: 6px;">Not
set</div>
      <a href="{% url 'payments:choose_payment' req.id %}" class="pay-cta">
        <i class="fas fa-credit-card"></i> Set Payment
      </a>
    {% endif %}
  </div>

<!-- Actions Column -->
<div class="req-actions-col">
  {% if req.status == 'issued' and req.issued_certificate %}
    <a href="{% url 'certifications:virtual_certificate' req.issued_certificate.pk
%}"

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        class="action-btn action-btn-view">
        <i class="fas fa-eye"></i> View
    </a>
    {% if req.payment.status == 'paid' or req.payment.status == 'waived' %}
    <a href="{% url 'payments:download_receipt' req.payment.id %}"
        class="action-btn action-btn-receipt">
        <i class="fas fa-download"></i> Receipt
    </a>
    {% endif %}
    {% elif req.status == 'pending' %}
    <div style="text-align: center;">
        <div style="font-size: 11px; color: var(--req-amber); font-weight: 700;
margin-bottom: 4px;">
            <i class="fas fa-hourglass-half"></i>
        </div>
        <div style="font-size: 10px; color: var(--req-muted);">Processing...</div>
    </div>
    {% endif %}
</div>

</div>

<!-- Rejected reason -->
{% if req.status == 'rejected' and req.notes %}
<div class="rejected-note">
    <i class="fas fa-exclamation-circle" style="flex-shrink:0; margin-top:2px;"></i>
    <span><strong>Reason:</strong> {{ req.notes }}</span>
</div>
{% endif %}
</div>
{% endfor %}
</div>

{% else %}
<div class="empty-state">
    <div class="empty-icon"><i class="fas fa-folder-open"></i></div>
    <h3 class="empty-title">No requests yet</h3>
    <p class="empty-sub">You haven't submitted any certificate requests. It only takes a
minute!</p>
    <a href="{% url 'certifications:request_certificate' %}" class="btn-new-req">
        <i class="fas fa-plus"></i> Submit Your First Request
    </a>
</div>
{% endif %}

{% endblock %}
```

File: templates/certifications/ra11261_admin_list.html

```
{% extends 'base.html' %}
{% load static %}

{% block content %}
<style>
    .admin-header {
        margin-bottom: 2rem;
    }
    .admin-header h1 {
        font-size: 2rem;
        font-weight: 800;
        color: #1e293b;
        margin: 0;
    }
    .admin-header p {
        color: #64748b;
        margin-top: 0.5rem;
    }
    .filter-bar {
        background: white;
        padding: 1rem;
        border-radius: 12px;
        box-shadow: 0 1px 3px rgba(0,0,0,0.1);
        display: flex;
        justify-content: space-between;
        align-items: center;
        margin-bottom: 1.5rem;
    }
    .search-input {
        padding: 0.5rem 1rem;
        border: 1px solid #e2e8f0;
        border-radius: 8px;
        width: 300px;
    }
    .status-select {
        padding: 0.5rem 1rem;
        border: 1px solid #e2e8f0;
        border-radius: 8px;
        background: white;
    }
    .data-table {
        width: 100%;
        background: white;
        border-radius: 12px;
        box-shadow: 0 4px 6px -1px rgba(0,0,0,0.05);
        overflow: hidden;
    }
    .data-table th {
        background: #f8fafc;
        padding: 1rem 1.5rem;
        text-transform: uppercase;
        font-size: 0.75rem;
        font-weight: 700;
        color: #475569;
        letter-spacing: 0.05em;
        border-bottom: 1px solid #e2e8f0;
    }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.data-table td {
  padding: 1rem 1.5rem;
  border-bottom: 1px solid #f1f5f9;
  vertical-align: middle;
}
.data-table tr:hover {
  background: #f8fafc;
}
.status-badge {
  padding: 0.25rem 0.75rem;
  border-radius: 9999px;
  font-size: 0.75rem;
  font-weight: 700;
  text-transform: uppercase;
}
.status-PENDING { background: #f1f5f9; color: #475569; border: 1px solid #cbd5e1; }
.status-FOR_REVIEW { background: #ffffeb; color: #b45309; border: 1px solid #fde68a; }
.status-APPROVED { background: #ecfdf5; color: #047857; border: 1px solid #a7f3d0; }
.status-REJECTED { background: #fef2f2; color: #b91c1c; border: 1px solid #fecaca; }

/* Modal */
.modal-overlay {
  position: fixed;
  top: 0; left: 0; right: 0; bottom: 0;
  background: rgba(15, 23, 42, 0.6);
  backdrop-filter: blur(4px);
  display: none;
  align-items: center;
  justify-content: center;
  z-index: 1050;
  opacity: 0;
  transition: opacity 0.2s ease;
}
.modal-overlay.active {
  display: flex;
  opacity: 1;
}
.modal-card {
  background: white;
  width: 100%;
  max-width: 600px;
  border-radius: 16px;
  box-shadow: 0 25px 50px -12px rgba(0,0,0,0.25);
  transform: scale(0.95);
  transition: transform 0.2s ease;
  overflow: hidden;
}
.modal-overlay.active .modal-card {
  transform: scale(1);
}
.modal-header {
  padding: 1.5rem;
  border-bottom: 1px solid #e2e8f0;
  display: flex;
  justify-content: space-between;
  align-items: center;
  background: #f8fafc;
}
.modal-body {
  padding: 1.5rem;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
max-height: 70vh;
overflow-y: auto;
}
.info-box {
  background: #f8fafc;
  border: 1px solid #e2e8f0;
  padding: 1rem;
  border-radius: 8px;
  margin-bottom: 1rem;
}
.btn-action {
  padding: 0.75rem 1.5rem;
  border-radius: 8px;
  font-weight: 700;
  border: none;
  cursor: pointer;
  transition: all 0.2s;
}
.btn-approve { background: #059669; color: white; }
.btn-approve:hover { background: #047857; }
.btn-reject { background: white; color: #dc2626; border: 1px solid #fca5a5; }
.btn-reject:hover { background: #fef2f2; }
</style>

<div class="container-fluid py-4">
  <div class="admin-header">
    <h1>Application Reviews</h1>
    <p>Review and manage RA 11261 applications from residents.</p>
  </div>

  {% if messages %}
    {% for message in messages %}
      <div class="alert alert-{{ message.tags }} shadow-sm rounded mb-4">{{ message
    }}</div>
    {% endfor %}
  {% endif %}

  <form method="GET" class="filter-bar">
    <input type="text" name="search" class="search-input" placeholder="Search resident
name..." value="{{ search }}">
    <div style="display: flex; gap: 1rem; align-items: center;">
      <select name="status" class="status-select" onchange="this.form.submit()">
        <option value="ALL" {% if status_filter == 'ALL' %}>selected{% endif %}>All
Statuses</option>
        <option value="PENDING" {% if status_filter == 'PENDING' %}>selected{% endif
%}>Pending</option>
        <option value="FOR_REVIEW" {% if status_filter == 'FOR_REVIEW' %}>selected{% endif
%}>For Review</option>
        <option value="APPROVED" {% if status_filter == 'APPROVED' %}>selected{% endif
%}>Approved</option>
        <option value="REJECTED" {% if status_filter == 'REJECTED' %}>selected{% endif
%}>Rejected</option>
      </select>
      <button type="submit" class="btn btn-primary" style="background: var(--primary);
border: none;">Filter</button>
    </div>
  </form>

  <table class="data-table">
    <thead>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<tr>
  <th>Resident</th>
  <th>Submitted</th>
  <th>Status</th>
  <th class="text-end">Action</th>
</tr>
</thead>
<tbody>
  {% for app in applications %}
  <tr>
    <td>
      <div style="font-weight: 700; color: #1e293b;">{{ app.resident.full_name
}}</div>
      <div style="font-size: 0.75rem; color: #64748b;">{{
app.resident.household.purok.name|default:"No Purok" }}</div>
    </td>
    <td>{{ app.created_at|date:"M d, Y g:i A" }}</td>
    <td><span class="status-badge status-{{ app.status }}">{{ app.status
}}</span></td>
    <td class="text-end">
      <button type="button" class="btn btn-sm btn-outline-primary" style="font-weight:
600;"
          onclick="openModal('{{ app.id }}', '{{ app.resident.full_name|escapejs
}}', '{{ app.status }}', '{{% if app.uploaded_docs %}}{{ app.uploaded_docs.url }}{% endif
%}}', '{{ app.rejection_reason|escapejs }}')">
        {% if app.status == 'FOR_REVIEW' %}Review{% else %}View{% endif %}
      </button>
    </td>
  </tr>
  {% empty %}
  <tr>
    <td colspan="4" class="text-center py-5 text-muted">No applications found.</td>
  </tr>
  {% endfor %}
</tbody>
</table>
</div>

<!-- Modal -->
<div id="reviewModal" class="modal-overlay">
  <div class="modal-card">
    <div class="modal-header">
      <h2 style="font-size: 1.25rem; font-weight: 700; margin: 0;">Application Review</h2>
      <button onclick="closeModal()" style="background: none; border: none; font-size:
1.5rem; cursor: pointer; color: #94a3b8;">&times;</button>
    </div>
    <div class="modal-body">
      <div class="info-box">
        <p style="font-size: 0.75rem; text-transform: uppercase; font-weight: 700; color:
#64748b; margin-bottom: 0.25rem;">Resident</p>
        <p id="modalResidentName" style="font-weight: 600; color: #0f172a; margin:
0;"></p>
      </div>

      <div class="info-box">
        <p style="font-size: 0.75rem; text-transform: uppercase; font-weight: 700; color:
#64748b; margin-bottom: 0.25rem;">Supporting Document</p>
        <a id="modalDocLink" href="#" target="_blank" style="display: flex; align-items:
center; gap: 0.5rem; color: var(--primary); font-weight: 600; text-decoration: none;">
          <i class="fas fa-file-pdf"></i> View Uploaded Document

```

IoT-Enabled Barangay Resident and Attendance Management System

```
</a>
</div>

<div id="rejectionBox" class="info-box" style="background: #fef2f2; border-color:
#fecaca; display: none;">
  <p style="font-size: 0.75rem; text-transform: uppercase; font-weight: 700; color:
#b91c1c; margin-bottom: 0.25rem;">Rejection Reason</p>
  <p id="modalRejectionReason" style="color: #991b1b; margin: 0;"></p>
</div>

<div id="approvedBox" class="info-box" style="background: #ecfdf5; border-color:
#a7f3d0; display: none; justify-content: space-between; align-items: center;">
  <div>
    <p style="font-size: 0.75rem; text-transform: uppercase; font-weight: 700;
color: #047857; margin-bottom: 0.25rem;">Status</p>
    <p style="color: #065f46; margin: 0; font-weight: 500;">Approved & Certificate
Generated</p>
  </div>
  <a id="modalPdfLink" href="#" target="_blank" class="btn btn-sm"
style="background: #059669; color: white; font-weight: 600;">Download PDF</a>
</div>

<form id="reviewForm" method="POST" action="" style="display: none; margin-top:
1.5rem;">
  {% csrf_token %}
  <input type="hidden" name="status" id="formStatus">
  <div id="rejectionInputWrapper" style="display: none; margin-bottom: 1rem;">
    <label style="font-size: 0.875rem; font-weight: 600;">Reason for
Rejection</label>
    <textarea name="rejection_reason" id="rejectionInput" class="form-control"
rows="3" style="border-radius: 8px;"></textarea>
  </div>
  <div style="display: flex; gap: 1rem;">
    <button type="button" onclick="submitReview('APPROVED')" class="btn-action btn-
approve" style="flex: 1;">Approve</button>
    <button type="button" onclick="submitReview('REJECTED')" class="btn-action btn-
reject" style="flex: 1;">Reject</button>
  </div>
</form>

</div>
</div>
</div>

<script>
const modal = document.getElementById('reviewModal');
const form = document.getElementById('reviewForm');

function openModal(id, resident, status, docUrl, rejReason) {
  document.getElementById('modalResidentName').innerText = resident;

  const docLink = document.getElementById('modalDocLink');
  if (docUrl) {
    docLink.href = docUrl;
    docLink.style.display = 'flex';
  } else {
    docLink.style.display = 'none';
  }

  const rejBox = document.getElementById('rejectionBox');
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if (status === 'REJECTED' && rejReason) {
    document.getElementById('modalRejectionReason').innerText = rejReason;
    rejBox.style.display = 'block';
} else {
    rejBox.style.display = 'none';
}

const appBox = document.getElementById('approvedBox');
if (status === 'APPROVED') {
    document.getElementById('modalPdfLink').href =
`/certifications/ra11261/certification/${id}/pdf/`;
    appBox.style.display = 'flex';
} else {
    appBox.style.display = 'none';
}

if (status === 'FOR_REVIEW') {
    form.action = `/certifications/ra11261/admin/applications/${id}/review/`;
    form.style.display = 'block';
    document.getElementById('rejectionInputWrapper').style.display = 'none';
    document.getElementById('rejectionInput').value = '';
} else {
    form.style.display = 'none';
}

modal.classList.add('active');
}

function closeModal() {
    modal.classList.remove('active');
}

function submitReview(status) {
    if (status === 'REJECTED') {
        const wrapper = document.getElementById('rejectionInputWrapper');
        if (wrapper.style.display === 'none') {
            wrapper.style.display = 'block';
            return; // wait for input
        }
        if (!document.getElementById('rejectionInput').value.trim()) {
            alert("Please enter a rejection reason.");
            return;
        }
    }
    document.getElementById('formStatus').value = status;
    form.submit();
}
</script>
{% endblock %}
```


File: templates/certifications/ra11261_admin_roster.html

```
{% extends 'base.html' %}
{% load static %}

{% block content %}
<style>
  .admin-header {
    margin-bottom: 2rem;
    display: flex;
    justify-content: space-between;
    align-items: flex-end;
  }
  .admin-header h1 {
    font-size: 2rem;
    font-weight: 800;
    color: #1e293b;
    margin: 0;
  }
  .admin-header p {
    color: #64748b;
    margin-top: 0.5rem;
  }
  .stats-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 1.5rem;
    margin-bottom: 2rem;
  }
  .stat-card {
    background: white;
    padding: 1.5rem;
    border-radius: 16px;
    box-shadow: 0 4px 6px -1px rgba(0,0,0,0.05);
    border: 1px solid #e2e8f0;
    display: flex;
    align-items: center;
    gap: 1.25rem;
  }
  .stat-icon {
    width: 48px;
    height: 48px;
    border-radius: 12px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 1.5rem;
  }
  .icon-blue { background: #dbeafe; color: #2563eb; }
  .icon-green { background: #d1fae5; color: #059669; }
  .icon-orange { background: #ffedd5; color: #ea580c; }

  .stat-label {
    font-size: 0.75rem;
    font-weight: 700;
    text-transform: uppercase;
    color: #64748b;
    letter-spacing: 0.05em;
  }
</style>

<div class="admin-header">
  <h1>Roster</h1>
  <p>Roster</p>
</div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-blue"></div>
    <div class="stat-label"></div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-green"></div>
    <div class="stat-label"></div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-orange"></div>
    <div class="stat-label"></div>
  </div>
</div>

</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.stat-value {
  font-size: 1.75rem;
  font-weight: 900;
  color: #0f172a;
  line-height: 1;
  margin-top: 0.25rem;
}

.filter-bar {
  background: #f8fafc;
  padding: 1rem;
  border-bottom: 1px solid #e2e8f0;
  display: flex;
  align-items: center;
}

.search-input {
  padding: 0.5rem 1rem;
  border: 1px solid #cbd5e1;
  border-radius: 8px;
  width: 350px;
}

.data-table-wrapper {
  background: white;
  border-radius: 12px;
  box-shadow: 0 4px 6px -1px rgba(0,0,0,0.05);
  border: 1px solid #e2e8f0;
  overflow: hidden;
}

.data-table {
  width: 100%;
}

.data-table th {
  background: #f8fafc;
  padding: 1rem 1.5rem;
  text-transform: uppercase;
  font-size: 0.75rem;
  font-weight: 700;
  color: #475569;
  letter-spacing: 0.05em;
  border-bottom: 1px solid #e2e8f0;
}

.data-table td {
  padding: 1rem 1.5rem;
  border-bottom: 1px solid #f1f5f9;
  vertical-align: middle;
}

.data-table tr:hover {
  background: #f8fafc;
}

.status-badge {
  padding: 0.25rem 0.75rem;
  border-radius: 9999px;
  font-size: 0.75rem;
  font-weight: 700;
  text-transform: uppercase;
}

.status-ACTIVE { background: #ecfdf5; color: #047857; border: 1px solid #a7f3d0; }
.status-EXPIRED { background: #fffbeb; color: #b45309; border: 1px solid #fde68a; }
.btn-export {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
background: var(--primary);
color: white;
font-weight: 600;
padding: 0.75rem 1.25rem;
border-radius: 8px;
text-decoration: none;
display: inline-flex;
align-items: center;
gap: 0.5rem;
transition: background 0.2s;
}
.btn-export:hover {
  background: #064e3b;
  color: white;
}
</style>

<div class="container-fluid py-4">
  <div class="admin-header">
    <div>
      <h1>RA 11261 Official Roster</h1>
      <p>Registry of all residents who availed the First Time Jobseeker benefit.</p>
    </div>
    <div style="display: flex; gap: 1rem;">
      <button type="button" class="btn btn-success" style="font-weight: 600; padding:
0.75rem 1.25rem; border-radius: 8px; display: inline-flex; align-items: center; gap:
0.5rem;" data-toggle="modal" data-target="#addWalkInModal">
        <i class="fas fa-plus-circle"></i> Add Walk-In
      </button>
      <a href="{% url 'certifications:ra11261_export_csv' %}" class="btn-export">
        <i class="fas fa-download"></i> Export CSV
      </a>
    </div>
  </div>

  {% if messages %}
    {% for message in messages %}
      <div class="alert alert-{{ message.tags }}" shadow-sm rounded mb-4">{{ message
}}</div>
    {% endfor %}
  {% endif %}

  <div class="stats-grid">
    <div class="stat-card">
      <div class="stat-icon icon-blue"><i class="fas fa-users"></i></div>
      <div>
        <div class="stat-label">Total Availed</div>
        <div class="stat-value">{{ total_availed }}</div>
      </div>
    </div>
    <div class="stat-card">
      <div class="stat-icon icon-green"><i class="fas fa-shield-check"></i></div>
      <div>
        <div class="stat-label">Active Certifications</div>
        <div class="stat-value">{{ active_count }}</div>
      </div>
    </div>
    <div class="stat-card">
      <div class="stat-icon icon-orange"><i class="fas fa-exclamation-triangle"></i></div>
      <div>
        <div class="stat-label">Total Unavailable</div>
        <div class="stat-value">{{ total_unavailable }}</div>
      </div>
    </div>
  </div>

  <div class="table">
    <table>
      <thead>
        <tr>
          <th>Name</th>
          <th>Address</th>
          <th>Contact No.</th>
          <th>Status</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <td>{{ resident.name }}</td>
          <td>{{ resident.address }}</td>
          <td>{{ resident.contact_no }}</td>
          <td>{{ resident.status }}</td>
        </tr>
      </tbody>
    </table>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <div class="stat-label">Expired</div>
        <div class="stat-value">{{ expired_count }}</div>
    </div>
</div>
</div>

<div class="data-table-wrapper">
    <form method="GET" class="filter-bar">
        <input type="text" name="search" class="search-input" placeholder="Search resident
name in roster..." value="{{ search }}">
        <button type="submit" class="btn btn-primary ms-2" style="background:
var(--primary); border: none;">Search</button>
    </form>

    <div class="table-responsive">
        <table class="data-table">
            <thead>
                <tr>
                    <th>Resident</th>
                    <th>Certification Date</th>
                    <th>Expiry Date</th>
                    <th>Status</th>
                    <th class="text-end">Added By</th>
                </tr>
            </thead>
            <tbody>
                {% load ra11261_tags %}
                {% for r in roster %}
                <tr>
                    <td>
                        <div style="font-weight: 700; color: #1e293b;">{{ r.resident.full_name
}}</div>
                    </td>
                    <td>{{ r.certification_date|date:"M d, Y" }}</td>
                    <td>{{ r.expiry_date|date:"M d, Y" }}</td>
                    <td>
                        {% if r|is_expired %}
                            <span class="status-badge status-EXPIRED">Expired</span>
                        {% else %}
                            <span class="status-badge status-ACTIVE">Active</span>
                        {% endif %}
                    </td>
                    <td class="text-end text-muted" style="font-size: 0.875rem;">
                        Admin #{{ r.added_by.username|default:"System" }}
                    </td>
                </tr>
                {% empty %}
                <tr>
                    <td colspan="5" class="text-center py-5 text-muted">No residents found in the
roster.</td>
                </tr>
                {% endfor %}
            </tbody>
        </table>
    </div>
</div>
</div>

<!-- Add Walk-In Modal -->
<div class="modal fade" id="addWalkInModal" tabindex="-1" aria-

```

IoT-Enabled Barangay Resident and Attendance Management System

```
labelledby="addWalkInModalLabel" aria-hidden="true">
  <div class="modal-dialog modal-dialog-centered">
    <div class="modal-content" style="border-radius: 16px; border: none; box-shadow: 0
25px 50px -12px rgba(0,0,0,0.25);">
      <div class="modal-header" style="background: #f8fafc; border-bottom: 1px solid
#e2e8f0; border-radius: 16px 16px 0 0;">
        <h5 class="modal-title" id="addWalkInModalLabel" style="font-weight: 700; color:
#1e293b;">Add Walk-In Resident to Roster</h5>
        <button type="button" class="close" data-dismiss="modal" aria-label="Close">
          <span aria-hidden="true">&times;</span>
        </button>
      </div>
      <form method="POST" action="{% url 'certifications:ra11261_admin_roster_add' %}">
        {% csrf_token %}
        <div class="modal-body" style="padding: 1.5rem;">
          <p style="font-size: 0.875rem; color: #64748b; margin-bottom: 1.5rem;">
            Select a resident who manually submitted their First Time Jobseeker
            requirements in the barangay hall. This will automatically approve them and waive their
            fees.
          </p>
          <div class="form-group">
            <label style="font-size: 0.875rem; font-weight: 700; color: #334155; margin-
            bottom: 0.5rem;">Select Resident</label>
            <select name="resident_id" class="form-control" style="border-radius: 8px;
            border: 1px solid #cbd5e1; padding: 0.5rem;" required>
              <option value="">-- Choose a Resident --</option>
              {% for res in available_residents %}
                <option value="{{ res.id }}">{{ res.last_name }}, {{ res.first_name }} {{
            res.middle_name|default:"" }}</option>
              {% endfor %}
            </select>
          </div>
        </div>
        <div class="modal-footer" style="border-top: 1px solid #e2e8f0; background:
#f8fafc; border-radius: 0 0 16px 16px;">
          <button type="button" class="btn btn-light" data-dismiss="modal" style="border-
          radius: 8px; font-weight: 600;">Cancel</button>
          <button type="submit" class="btn btn-success" style="border-radius: 8px; font-
          weight: 600;">Confirm & Add</button>
        </div>
      </form>
    </div>
  </div>
</div>
{% endblock %}
```

File: templates/certifications/ra11261_apply.html

```
{% extends 'base.html' %}
{% load static %}

{% block content %}
<style>
    .apply-wrapper {
        min-height: calc(100vh - 100px);
        display: flex;
        align-items: center;
        justify-content: center;
        background: linear-gradient(135deg, #f0fdf4 0%, #e0f2fe 100%);
        padding: 2rem;
    }
    .apply-card {
        background: rgba(255, 255, 255, 0.95);
        backdrop-filter: blur(10px);
        border-radius: 20px;
        box-shadow: 0 20px 40px rgba(0,0,0,0.08);
        width: 100%;
        max-w-md: 600px;
        overflow: hidden;
        border: 1px solid rgba(255,255,255,0.4);
        transition: transform 0.3s ease;
    }
    .apply-card:hover {
        transform: translateY(-5px);
    }
    .apply-header {
        background: linear-gradient(135deg, var(--primary) 0%, #065f46 100%);
        padding: 2rem;
        color: white;
        text-align: center;
    }
    .apply-header h2 {
        font-size: 1.75rem;
        font-weight: 800;
        margin: 0;
        letter-spacing: -0.025em;
    }
    .apply-header p {
        color: #a7f3d0;
        font-size: 0.95rem;
        margin-top: 0.5rem;
        font-weight: 500;
    }
    .apply-body {
        padding: 2.5rem;
    }
    .form-group {
        margin-bottom: 1.5rem;
    }
    .form-label {
        display: block;
        font-size: 0.875rem;
        font-weight: 700;
        color: #334155;
        margin-bottom: 0.5rem;
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}
.form-control {
  width: 100%;
  padding: 0.75rem 1rem;
  border: 1px solid #cbd5e1;
  border-radius: 10px;
  font-size: 1rem;
  transition: all 0.2s ease;
  background: #f8fafc;
}
.form-control:focus {
  outline: none;
  border-color: var(--primary);
  box-shadow: 0 0 0 4px rgba(30, 74, 49, 0.1);
  background: white;
}
.declaration-box {
  background: #f1f5f9;
  padding: 1.25rem;
  border-radius: 12px;
  border: 1px solid #e2e8f0;
  display: flex;
  gap: 1rem;
  align-items: flex-start;
}
.declaration-box input[type="checkbox"] {
  width: 1.25rem;
  height: 1.25rem;
  margin-top: 0.125rem;
  accent-color: var(--primary);
  cursor: pointer;
}
.declaration-box label {
  font-size: 0.875rem;
  color: #475569;
  line-height: 1.5;
  cursor: pointer;
}
.file-drop-area {
  border: 2px dashed #cbd5e1;
  border-radius: 12px;
  padding: 2rem;
  text-align: center;
  cursor: pointer;
  transition: all 0.2s ease;
  background: #f8fafc;
}
.file-drop-area:hover {
  border-color: var(--primary);
  background: #f0fdf4;
}
.file-icon {
  font-size: 2rem;
  color: #94a3b8;
  margin-bottom: 0.5rem;
}
.submit-btn {
  width: 100%;
  background: var(--primary);
  color: white;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
font-weight: 700;
font-size: 1rem;
padding: 1rem;
border-radius: 12px;
border: none;
cursor: pointer;
transition: all 0.3s ease;
box-shadow: 0 4px 6px -1px rgba(30, 74, 49, 0.2);
}
.submit-btn:hover {
background: #064e3b;
transform: translateY(-2px);
box-shadow: 0 10px 15px -3px rgba(30, 74, 49, 0.3);
}
</style>

<div class="apply-wrapper">
<div class="apply-card" style="max-width: 550px;">
<div class="apply-header">
<h2>RA 11261 Application</h2>
<p>First Time Jobseekers Assistance Act</p>
</div>
<div class="apply-body">

    {% if messages %}
    {% for message in messages %}
    <div class="alert {% if message.tags == 'error' %}alert-danger{% else %}alert-{
message.tags %}{% endif %} shadow-sm rounded-3 mb-4">
    {{ message }}
    </div>
    {% endfor %}
    {% endif %}

    <form method="POST" action="{% url 'certifications:ra11261_apply' %}"
enctype="multipart/form-data">
    {% csrf_token %}

    <div class="form-group">
    <label class="form-label" for="resident">Resident ID</label>
    <input type="number" id="resident" name="resident" class="form-control"
placeholder="Enter your Resident ID" required>
    </div>

    <div class="form-group">
    <div class="declaration-box">
    <input type="checkbox" id="declaration_confirmed" name="declaration_confirmed"
required>
    <label for="declaration_confirmed">
    I declare under oath that I am a First Time Jobseeker under RA 11261, and I
am submitting this request for employment purposes.
    </label>
    </div>
    </div>

    <div class="form-group">
    <label class="form-label">Supporting Document</label>
    <p style="font-size: 0.75rem; color: #64748b; margin-top: -0.25rem; margin-
bottom: 0.5rem;">Upload your Oath of Undertaking, School ID, or Diploma (PDF, JPG,
PNG).</p>
    <div class="file-drop-area">
```


IoT-Enabled Barangay Resident and Attendance Management System

```
onclick="document.getElementById('uploaded_docs').click()">
    <div class="file-icon"><i class="fas fa-cloud-upload-alt"></i></div>
    <p style="font-size: 0.875rem; font-weight: 600; color: var(--primary);
margin: 0;">Click to upload document</p>
    <p style="font-size: 0.75rem; color: #94a3b8; margin: 0;" id="file-name">No
file chosen</p>
    <input type="file" id="uploaded_docs" name="uploaded_docs" style="display:
none;" onchange="document.getElementById('file-name').innerText = this.files[0] ?
this.files[0].name : 'No file chosen'" required>
    </div>
</div>

<button type="submit" class="submit-btn">
    Submit Application
</button>
</form>
</div>
</div>
</div>
{% endblock %}
```

File: templates/certifications/request_form.html

```
{% extends 'base.html' %}
{% load static %}
{% block title %}Request a Certificate{% endblock %}

{% block breadcrumb %}
<span><i class="fas fa-home"></i> / <a href="{% url 'core:dashboard' %}">Dashboard</a> /
Request a Certificate</span>
{% endblock %}

{% block content %}
<style>
    .page-container {
        position: relative;
        min-height: calc(100vh - 120px);
        display: flex;
        align-items: center;
        justify-content: center;
        padding: 40px 20px;
    }

    .form-glass-card {
        background: rgba(255, 255, 255, 0.7);
        backdrop-filter: blur(12px);
        -webkit-backdrop-filter: blur(12px);
        border: 1px solid rgba(255, 255, 255, 0.4);
        border-radius: 24px;
        padding: 40px;
        width: 100%;
        max-width: 640px;
        box-shadow: 0 15px 35px rgba(0, 0, 0, 0.05), 0 5px 15px rgba(0, 0, 0, 0.03);
        position: relative;
        z-index: 1;
    }

    .hero-section {
        text-align: center;
        margin-bottom: 32px;
    }

    .hero-icon {
        width: 64px;
        height: 64px;
        background: linear-gradient(135deg, var(--primary, #2d7a50), #1a4731);
        border-radius: 20px;
        display: flex;
        align-items: center;
        justify-content: center;
        margin: 0 auto 16px;
        color: #fff;
        font-size: 24px;
        box-shadow: 0 8px 16px rgba(45, 122, 80, 0.2);
    }

    .hero-section h1 {
        font-size: 24px;
        font-weight: 800;
        margin-bottom: 8px;
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    color: var(--text-main, #111827);
    letter-spacing: -0.5px;
}

.hero-section p {
    color: var(--text-muted, #64748b);
    font-size: 15px;
    max-width: 400px;
    margin: 0 auto;
}

.form-group {
    margin-bottom: 24px;
}

.form-label {
    display: block;
    font-size: 13px;
    font-weight: 700;
    color: var(--text-main, #111827);
    margin-bottom: 8px;
    text-transform: uppercase;
    letter-spacing: 0.8px;
}

.form-control-modern {
    width: 100%;
    padding: 14px 18px;
    background: rgba(255, 255, 255, 0.8);
    border: 1.5px solid #e2e8f0;
    border-radius: 14px;
    font-size: 15px;
    color: #1e293b;
    transition: all 0.24s cubic-bezier(0.4, 0, 0.2, 1);
    box-sizing: border-box;
}

.form-control-modern:focus {
    outline: none;
    border-color: var(--primary, #2d7a50);
    background: #fff;
    box-shadow: 0 0 0 4px rgba(45, 122, 80, 0.1);
    transform: translateY(-1px);
}

textarea.form-control-modern {
    min-height: 120px;
    resize: vertical;
}

.btn-group {
    display: flex;
    gap: 12px;
    margin-top: 32px;
}

.btn-submit {
    flex: 2;
    background: var(--primary, #2d7a50);
    color: white;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
border: none;
padding: 16px;
border-radius: 14px;
font-size: 15px;
font-weight: 700;
cursor: pointer;
display: flex;
align-items: center;
justify-content: center;
gap: 10px;
transition: all 0.2s ease;
letter-spacing: 0.5px;
}

.btn-submit:hover {
  background: #1e4a31;
  transform: translateY(-2px);
  box-shadow: 0 8px 20px rgba(45, 122, 80, 0.2);
}

.btn-cancel {
  flex: 1;
  background: #f1f5f9;
  color: #475569;
  border: none;
  padding: 16px;
  border-radius: 14px;
  font-size: 15px;
  font-weight: 600;
  text-decoration: none;
  display: flex;
  align-items: center;
  justify-content: center;
  transition: all 0.2s ease;
}

.btn-cancel:hover {
  background: #e2e8f0;
  color: #1e293b;
}

.info-box {
  background: rgba(45, 122, 80, 0.06);
  border-left: 4px solid var(--primary, #2d7a50);
  padding: 16px;
  border-radius: 0 12px 12px 0;
  margin-bottom: 24px;
  font-size: 14px;
  color: #1e4a31;
  display: flex;
  gap: 12px;
}

.info-box i {
  margin-top: 3px;
  opacity: 0.8;
}

/* Background blobs */
.bg-blobs {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
position: fixed;
top: 0;
left: 0;
width: 100%;
height: 100%;
z-index: 0;
overflow: hidden;
pointer-events: none;
}

.blob {
position: absolute;
width: 400px;
height: 400px;
background: radial-gradient(circle, rgba(45, 122, 80, 0.1) 0%, rgba(45, 122, 80,
0) 70%);
border-radius: 50%;
filter: blur(50px);
}
</style>

<div class="bg-blobs">
  <div class="blob" style="top: -5%; left: -5%;"></div>
  <div class="blob"
    style="bottom: 10%; right: 5%; width: 500px; height: 500px; background: radial-
gradient(circle, rgba(20, 184, 166, 0.08) 0%, rgba(20, 184, 166, 0) 70%);">
  </div>
</div>

<div class="page-container">
  <div class="form-glass-card">
    <div class="hero-section">
      <div class="mb-3">
        {% url 'core:dashboard' as core_dashboard_url %}
        {% include 'components/back_button.html' with back_url=core_dashboard_url
back_label='Back' %}
      </div>
      <div class="hero-icon">
        <i class="fas fa-file-signature"></i>
      </div>
      <h1>Request Certificate</h1>
      <p>Get your official barangay documents processed quickly and easily.</p>
    </div>

    {% if messages %}
    {% for message in messages %}
    <div class="alert alert-{{ message.tags }}" style="margin-bottom: 24px;">
      <i class="fas fa-info-circle"></i>
      <span>{{ message }}</span>
    </div>
    {% endfor %}
    {% endif %}

    <div class="info-box">
      <i class="fas fa-lightbulb"></i>
      <span>Your request will be processed within 1-2 working days. You'll receive a
notification once it's
      ready.</span>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<form method="POST">
  {% csrf_token %}
  <div class="form-group">
    <label class="form-label">Certificate Type</label>
    <select name="cert_type" id="cert_type_select" class="form-control-modern"
required>
      <option value="" disabled {% if not request.POST.cert_type
%}selected{% endif %}>Select type...
    </option>
    {% for code, label in cert_types %}
    {% if request.POST.cert_type == code %}
    <option value="{{ code }}" selected>{{ label }}</option>
    {% else %}
    <option value="{{ code }}">{{ label }}</option>
    {% endif %}
    {% endfor %}
  </select>
</div>

  <div class="form-group">
    <label class="form-label">Purpose / Reason</label>
    <textarea name="purpose" id="purposeTextarea" class="form-control-modern"
placeholder="Briefly explain purpose..." required>{{
request.POST.purpose }}</textarea>
    <div id="grammarFeedback"></div>
  </div>

  <div class="form-group" style="background: rgba(45, 122, 80, 0.04); border:
1px solid rgba(45, 122, 80, 0.15); border-radius: 12px; padding: 16px; display: flex;
align-items: flex-start; gap: 12px; margin-bottom: 24px;">
    <input type="checkbox" name="is_first_time_jobseeker"
id="is_first_time_jobseeker" style="margin-top: 4px; transform: scale(1.2); accent-color:
var(--primary);">
    <div>
      <label for="is_first_time_jobseeker" style="font-weight: 700; color:
#1e4a31; cursor: pointer; display: block; margin-bottom: 4px;">I am a First Time Job
Seeker (RA 11261)</label>
      <p style="font-size: 12px; color: #475569; margin: 0; line-height:
1.4;">
        Under the First Time Jobseekers Assistance Act, you are 100%
exempt from paying barangay certificate fees if you are applying for employment for the
first time.
      </p>
    </div>
  </div>

  <!-- Business Fields -->
  <div class="business-fields" style="display: none;">
    <div class="info-box" style="margin-bottom: 16px;"><i class="fas fa-
store"></i><span>Business
      information required.</span></div>
    <div class="form-group"><label class="form-label">Business
Name</label><input type="text"
      name="business_name" class="form-control-modern"></div>
    <div class="form-group"><label class="form-label">Business
Type</label><input type="text"
      name="business_type" class="form-control-modern"></div>
    <div class="form-group"><label class="form-label">Business
Address</label><textarea
      name="business_address" class="form-control-modern" style="min-

```

IoT-Enabled Barangay Resident and Attendance Management System

```

height: 80px;"></textarea></div>
</div>

<!-- Cedula Fields -->
<div class="cedula-fields" style="display: none;">
  <div class="info-box" style="margin-bottom: 16px;">
    <i class="fas fa-id-card"></i>
    <span>Community Tax Certificate (Cedula) details required.</span>
  </div>
  <div class="form-group">
    <label class="form-label">Taxpayer Type</label>
    <select name="taxpayer_type" id="taxpayer_type" class="form-control-
modern">
      <option value="individual">Individual</option>
      <option value="corporation">Corporation</option>
    </select>
  </div>

  <div class="form-group">
    <label class="form-label">Place of Issue</label>
    <input type="text" name="place_of_issue" class="form-control-modern"
value="Sico-Sico, Gigaquit, SDN">
  </div>

  <div class="individual-only" style="display:flex; gap:16px; margin-
bottom:24px;">
    <div style="flex:1;">
      <label class="form-label">Height (cm)</label>
      <input type="text" name="height" class="form-control-modern"
placeholder="e.g. 170">
    </div>
    <div style="flex:1;">
      <label class="form-label">Weight (kg)</label>
      <input type="text" name="weight" class="form-control-modern"
placeholder="e.g. 65">
    </div>
  </div>

  <div style="border-top: 1px solid #e2e8f0; padding-top: 20px; margin-top:
10px;">
    <p style="font-size: 13px; font-weight: 700; color: var(--primary);
margin-bottom: 15px; text-transform: uppercase;">Taxable Amounts (Cat A, B, C)</p>

    <div class="form-group">
      <label class="form-label">Real Property Value (Cat A)</label>
      <input type="number" step="0.01" name="raw_taxable_property"
class="form-control-modern" value="0.00">
      <small style="color: #64748b;">Assessed value of real property
owned.</small>
    </div>

    <div class="form-group">
      <label class="form-label">Gross Business Receipts (Cat B)</label>
      <input type="number" step="0.01" name="raw_taxable_business"
class="form-control-modern" value="0.00">
      <small style="color: #64748b;">Gross receipts or earnings from
business during the preceding year.</small>
    </div>

    <div class="form-group">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <label class="form-label">Annual Income (Cat C)</label>
        <input type="number" step="0.01" name="raw_taxable_income"
class="form-control-modern" value="0.00">
        <small style="color: #64748b;">Gross annual salary or earnings
from exercise of profession.</small>
    </div>
</div>
</div>

<!-- Late Registration Fields -->
<div class="late-registration-fields" style="display: none;">
    <div class="info-box" style="margin-bottom: 16px;"><i class="fas fa-
baby"></i><span>Birth registration
    details.</span></div>
    <div class="form-group"><label class="form-label">Child's
Name</label><input type="text"
    name="child_name" class="form-control-modern"></div>
    <div class="form-group"><label class="form-label">Date/Place of
Birth</label>
        <div style="display: flex; gap: 12px;">
            <input type="date" name="child_birth_date" class="form-control-
modern" style="flex: 1;">
            <input type="text" name="child_birth_place" class="form-control-
modern" style="flex: 1;"
                placeholder="Place">
        </div>
    </div>
    <div class="form-group"><label class="form-label">Father's
Name</label><input type="text"
        name="father_name" class="form-control-modern"></div>
    <div class="form-group"><label class="form-label">Mother's
Name</label><input type="text"
        name="mother_name" class="form-control-modern"></div>
</div>

<script src="{% static 'js/grammar_checker.js' %}"></script>
<script>
    document.addEventListener('DOMContentLoaded', () => {
        GrammarChecker.init('purposeTextarea', 'grammarFeedback');
        const certTypeSelect = document.getElementById('cert_type_select');
        const bFields = document.querySelector('.business-fields');
        const cFields = document.querySelector('.cedula-fields');
        const lFields = document.querySelector('.late-registration-fields');

        const taxpayerType = document.getElementById('taxpayer_type');
        const individualOnly = document.querySelector('.individual-only');

        certTypeSelect.addEventListener('change', function () {
            [bFields, cFields, lFields].forEach(el => el.style.display =
'none');
            if (this.value === 'business_permit') bFields.style.display =
'block';
            else if (this.value === 'cedula') {
                cFields.style.display = 'block';
                if (taxpayerType.value === 'individual')
individualOnly.style.display = 'flex';
                else individualOnly.style.display = 'none';
            }
            else if (this.value === 'late_registration') lFields.style.display
= 'block';

```


IoT-Enabled Barangay Resident and Attendance Management System

```
});

taxpayerType.addEventListener('change', function() {
    if (this.value === 'individual') individualOnly.style.display =
'flex';
    else individualOnly.style.display = 'none';
});

if (certTypeSelect.value) certTypeSelect.dispatchEvent(new
Event('change'));
});
</script>

<div class="btn-group">
    <button type="submit" class="btn-submit"><i class="fas fa-paper-
plane"></i> Submit Request</button>
    <a href="{% url 'certifications:my_requests' %}" class="btn-cancel">My
Requests</a>
</div>
</form>
</div>
</div>
{% endblock %}
```

File: templates/certifications/request_list.html

```
{% extends 'base.html' %}
{% load static %}
{% block title %}Certificate Requests{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Certifications / Requests</span>
{% endblock %}

{% block content %}
<div class="page-header">
    <div class="page-title">
        <h1><i class="fas fa-file-invoice"></i> Certificate Requests</h1>
        <span>Review and process resident certificate requests</span>
    </div>
</div>

{% if messages %}
{% for message in messages %}
<div class="alert alert-{{ message.tags }}" style="margin-bottom:16px;">{{ message }}</div>
{% endfor %}
{% endif %}

<!-- Status Tabs -->
<div style="display:flex;gap:8px;margin-bottom:16px;flex-wrap:wrap;">
    <a href="?status=pending"
        class="btn btn-sm {% if status_filter == 'pending' %}btn-primary{% else %}btn-
outline{% endif %}">
        ? Pending
        {% if pending_count > 0 %}<span
            style="background:#ef4444;color:#fff;border-radius:20px;padding:0 6px;font-
size:.7rem;margin-left:4px;font-weight:700;">{{ pending_count }}</span>{% endif %}
    </a>

    <a href="?status=issued" class="btn btn-sm {% if status_filter == 'issued' %}btn-
primary{% else %}btn-outline{% endif %}">?
        Issued</a>

    <a href="?status=rejected"
        class="btn btn-sm {% if status_filter == 'rejected' %}btn-primary{% else %}btn-
outline{% endif %}">? Rejected</a>

    <a href="?status=" class="btn btn-sm {% if not status_filter %}btn-primary{% else
%}btn-outline{% endif %}">All</a>
</div>

<style>
    .compact-table {
        font-size: 0.85rem;
    }
    .compact-table th {
        padding: 10px 12px;
        background: #f8fafc;
        color: #475569;
        font-weight: 700;
        text-transform: uppercase;
        letter-spacing: 0.025em;
        border-bottom: 2px solid #e2e8f0;
    }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.compact-table td {
    padding: 8px 12px;
    vertical-align: middle;
    border-bottom: 1px solid #f1f5f9;
}
.btn-icon-sm {
    width: 32px;
    height: 32px;
    display: inline-flex;
    align-items: center;
    justify-content: center;
    border-radius: 6px;
    transition: all 0.2s;
    text-decoration: none;
    border: none;
    cursor: pointer;
}
.btn-icon-view { background: rgba(59, 130, 246, 0.1); color: #3b82f6; }
.btn-icon-view:hover, .btn-icon-view:focus, .btn-icon-view:active { background:
#3b82f6; color: #fff; outline: none; }

.btn-icon-approve { background: rgba(45, 122, 80, 0.1); color: #2d7a50; }
.btn-icon-approve:hover, .btn-icon-approve:focus, .btn-icon-approve:active {
background: #2d7a50; color: #fff; outline: none; }

.btn-icon-reject { background: rgba(239, 68, 68, 0.1); color: #ef4444; }
.btn-icon-reject:hover, .btn-icon-reject:focus, .btn-icon-reject:active { background:
#ef4444; color: #fff; outline: none; }

.status-badge {
    padding: 4px 10px;
    border-radius: 6px;
    font-size: 0.75rem;
    font-weight: 700;
    text-transform: uppercase;
    display: inline-block;
}
.badge-pending { background: #fef9c3; color: #854d0e; }
.badge-issued { background: #dcfce7; color: #166534; }
.badge-cancelled { background: #fee2e2; color: #991b1b; }
</style>

<div class="card" style="padding:0; border-radius:12px; overflow:hidden;">
    <div class="table-responsive">
        <table class="compact-table">
            <thead>
                <tr>
                    <th>#</th>
                    <th>Resident Name</th>
                    <th>Certificate Type</th>
                    <th>Purpose</th>
                    <th>Date Requested</th>
                    <th>Payment</th>
                    <th>Status</th>
                    <th class="text-center">Action</th>
                </tr>
            </thead>
            <tbody>
                {% for req in requests %}
                <tr>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<td>{{ forloop.counter }}</td>
<td><span style="font-weight:600;">{{ req.resident.full_name
}}</span></td>
<td>{{ req.get_cert_type_display }}</td>
<td>{{ req.purpose|truncatechars:50 }}</td>
<td>{{ req.created_at|date:"M d, Y" }}</td>
<td>
    {% if req.payment %}
        <div style="font-size: 0.8rem; font-weight: 700; color:
#475569; text-transform: uppercase;">
            {% if req.payment.method == 'gcash' %}GCash
            {% elif req.payment.method == 'cash' %}On Site
            {% elif req.payment.method == 'waived' %}Waived
            {% else %}{{ req.payment.method }}{% endif %}
        </div>
        <div style="margin-top: 4px;">
            {% if req.payment.status == 'paid' %}
                <span class="status-badge badge-issued">Paid</span>
            {% elif req.payment.status == 'waived' %}
                <span class="status-badge badge-issued"
style="background:#ede9fe; color:#7c3aed;">Waived</span>
            {% elif req.payment.status == 'pending' %}
                <span class="status-badge badge-pending">Pending
Proof</span>
            {% else %}
                <span class="status-badge badge-
cancelled">Unpaid</span>
            {% endif %}
        </div>
        {% if req.payment.method == 'gcash' and
req.payment.proof_screenshot %}
            <button type="button" class="btn btn-sm btn-outline"
style="padding: 2px 8px; font-size: 0.7rem; margin-top: 6px;" onclick="viewProof('{{
req.payment.proof_screenshot.url }}')">
                <i class="fas fa-image"></i> View Proof
            </button>
        {% endif %}
    {% else %}
        <span class="status-badge badge-cancelled">Not Set</span>
    {% endif %}
</td>
<td>
    {% if req.status == 'pending' %}
        <span class="status-badge badge-pending">Pending</span>
    {% elif req.status == 'issued' %}
        <span class="status-badge badge-issued">Issued</span>
    {% elif req.status == 'rejected' %}
        <span class="status-badge badge-cancelled">Rejected</span>
    {% else %}{{ req.status }}{% endif %}
</td>
<td>
    <div style="display:flex; gap:6px; justify-content:center;">
        {% if req.status == 'pending' %}
            <!-- Payment Approvals -->
            {% if req.payment and req.payment.status != 'paid' and
req.payment.status != 'waived' %}
                {% if req.payment.method == 'cash' %}
                    <form method="POST" action="{% url
'payments:mark_paid' req.payment.id %}" style="display:inline;">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

                                {% csrf_token %}
                                <button type="submit" class="btn-icon-sm btn-
icon-approve" title="Approve Payment (On Site)">
                                    <i class="fas fa-money-bill-wave"></i>
                                </button>
                                </form>
                                {% elif req.payment.method == 'gcash' and
req.payment.proof_screenshot %}
                                    <button type="button" class="btn-icon-sm btn-icon-
approve" onclick="openGcashModal({{ req.payment.id }})" title="Approve GCash Payment">
                                        <i class="fas fa-mobile-alt"></i>
                                    </button>
                                {% endif %}
                                {% endif %}

                                <a href="{% url 'certifications:fulfill_request' req.pk
%}" class="btn-icon-sm btn-icon-view" title="Process & Issue Certificate">
                                    <i class="fas fa-file-signature"></i>
                                </a>
                                <button type="button" class="btn-icon-sm btn-icon-reject"
onclick="openRejectModal({{ req.pk }})" title="Reject Request">
                                    <i class="fas fa-times-circle"></i>
                                </button>
                                {% elif req.status == 'issued' and req.certificate %}
                                <a href="{% url 'certifications:view' req.certificate.pk %}"
class="btn-icon-sm btn-icon-view" title="View Certificate">
                                    <i class="fas fa-eye"></i>
                                </a>
                                {% else %}{% endif %}
                            </div>
                        </td>
                    </tr>
                    {% empty %}
                    <tr>
                        <td colspan="7" class="text-center" style="padding:36px
0;color:var(--text-muted,#64748b);">
                            <i class="fas fa-inbox"
                                style="font-size:2rem;display:block;margin-
bottom:10px;opacity:.35;"></i>
                            No {{ status_filter|default:"" }} requests found.
                        </td>
                    </tr>
                    {% endfor %}
                </tbody>
            </table>
        </div>
        {% if requests.has_other_pages %}
        <div style="padding:16px 0 8px;display:flex;gap:8px;justify-content:center;">
            {% if requests.has_previous %}
            <a href="?status={{ status_filter }}&page={{ requests.previous_page_number }}"
class="btn btn-outline btn-sm">?
                Prev</a>
            {% endif %}
            <span style="padding:6px 12px;font-size:.85rem;">Page {{ requests.number }} of {{
requests.paginator.num_pages
}}</span>
            {% if requests.has_next %}
            <a href="?status={{ status_filter }}&page={{ requests.next_page_number }}"
class="btn btn-outline btn-sm">Next
                ?</a>
            {% endif %}
        </div>
    </div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% endif %}
</div>
{% endif %}
</div>

<!-- Reject Modal -->
<div id="rejectModal"

style="display:none;position:fixed;inset:0;background:rgba(0,0,0,.5);z-index:1000;align-
items:center;justify-content:center;">
  <div
    style="background:var(--bg-card,#fff);border-radius:16px;padding:28px;max-
width:480px;width:90%;box-shadow:0 20px 60px rgba(0,0,0,.25);">
    <h3 style="margin:0 0 8px;"><i class="fas fa-times-circle"
style="color:#dc2626;"></i> Reject Request</h3>
    <p style="color:var(--text-muted,#64748b);margin: 0 0 16px;font-
size:.9rem;">Optionally add a reason that will
      be sent to the resident.</p>
    <form method="POST" id="rejectForm" action="">
      {% csrf_token %}
      <textarea name="notes" id="rejectNotes" placeholder="Reason for rejection
(optional)..."
        style="width:100%;padding:10px 14px;border:1.5px solid
var(--border,#e2e8f0);border-radius:8px;min-height:80px;resize:vertical;font-
size:.9rem;box-sizing:border-box;margin-bottom:16px;"></textarea>
        <div id="rejectGrammarFeedback"></div>
        <div style="display:flex;gap:10px;justify-content:flex-end;margin-top:10px;">
          <button type="button" onclick="closeModal('rejectModal')" class="btn btn-
outline">Cancel</button>
          <button type="submit" class="btn btn-primary"
style="background:#dc2626;border-color:#dc2626;">Reject
            Request</button>
        </div>
      </form>
    </div>
  </div>

<!-- Proof Modal -->
<div id="proofModal"
style="display:none;position:fixed;inset:0;background:rgba(0,0,0,.8);z-index:1000;align-
items:center;justify-content:center;">
  <div style="background:#fff;border-radius:12px;padding:20px;max-
width:600px;width:90%;position:relative;">
    <button type="button" onclick="closeModal('proofModal')"
style="position:absolute;top:15px;right:15px;background:none;border:none;font-
size:1.5rem;cursor:pointer;color:#64748b;">&times;</button>
    <h3 style="margin:0 0 16px;">GCash Payment Proof</h3>
    <img id="proofImage" src="" alt="Proof of Payment" style="width:100%;border-
radius:8px;max-height:70vh;object-fit:contain;background:#f1f5f9;">
  </div>
</div>

<!-- GCash Approve Modal -->
<div id="gcashModal"
style="display:none;position:fixed;inset:0;background:rgba(0,0,0,.5);z-index:1000;align-
items:center;justify-content:center;">
  <div style="background:#fff;border-radius:16px;padding:28px;max-
width:400px;width:90%;box-shadow:0 20px 60px rgba(0,0,0,.25);">
    <h3 style="margin:0 0 8px;"><i class="fas fa-mobile-alt"
style="color:#2563eb;"></i> Approve GCash</h3>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<p style="color:#64748b;margin: 0 0 16px;font-size:.9rem;">Enter the GCash
Reference Number to confirm this payment.</p>
<form method="POST" id="gcashForm" action="">
  {% csrf_token %}
  <input type="text" name="gcash_ref_number" placeholder="Enter Reference No."
required
      style="width:100%;padding:10px 14px;border:1.5px solid #e2e8f0;border-
radius:8px;font-size:.9rem;box-sizing:border-box;margin-bottom:16px;">
  <div style="display:flex;gap:10px;justify-content:flex-end;">
    <button type="button" onclick="closeModal('gcashModal')" class="btn btn-
outline">Cancel</button>
    <button type="submit" class="btn btn-primary"
style="background:#2563eb;border-color:#2563eb;">Approve</button>
  </div>
</form>
</div>
</div>

<script src="{% static 'js/grammar_checker.js' %}"></script>
<script>
  document.addEventListener('DOMContentLoaded', () => {
    GrammarChecker.init('rejectNotes', 'rejectGrammarFeedback');
  });

  function openRejectModal(pk) {
    document.getElementById('rejectForm').action = '/certifications/requests/' + pk +
'/reject/';
    document.getElementById('rejectModal').style.display = 'flex';
  }

  function openGcashModal(paymentId) {
    document.getElementById('gcashForm').action = '/payments/' + paymentId +
'/confirm-gcash/';
    document.getElementById('gcashModal').style.display = 'flex';
  }

  function viewProof(url) {
    document.getElementById('proofImage').src = url;
    document.getElementById('proofModal').style.display = 'flex';
  }

  function closeModal(modalId) {
    document.getElementById(modalId).style.display = 'none';
  }

  // Close on backdrop click
  document.getElementById('rejectModal').addEventListener('click', function (e) { if
(e.target === this) closeModal('rejectModal'); });
  document.getElementById('proofModal').addEventListener('click', function (e) { if
(e.target === this) closeModal('proofModal'); });
  document.getElementById('gcashModal').addEventListener('click', function (e) { if
(e.target === this) closeModal('gcashModal'); });
</script>
{% endblock %}
```

File: templates/certifications/search.html

```
{% extends 'base.html' %}

{% block title %}Search Certificates - Barangay Management System{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'certifications:list' %}">Certifications</a> / Search</span>
{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Verify Certificate</h1>
    <span>Search certificates by control number for authenticity</span>
  </div>
</div>

<div class="search-container" style="max-width: 800px; margin: 40px auto;">
  <div class="card shadow-lg">
    <div class="card-header filter-card-header">
      <form method="GET" class="filter-form">
        <div class="filter-inputs-group">
          <div class="search-input-wrapper" style="min-width: 100%;">
            <i class="fas fa-search search-icon"></i>
            <input type="text" name="control" value="{{ control }}"
              placeholder="Enter Control Number (e.g. CLR-20260222-0001)" class="form-
control search-input" required>
            {% if control %}
              <a href="{% url 'certifications:search' %}" class="search-clear"
title="Clear">
                <i class="fas fa-times-circle"></i>
              </a>
            {% endif %}
          </div>
        </div>
        <button type="submit" class="btn btn-primary search-btn" style="min-width:
140px;">
          <i class="fas fa-check-circle"></i> <span>Verify Now</span>
        </button>
      </form>
    </div>
  </div>

  {% if control %}
  {% if certificate %}
  <div class="card mt-4 verification-card success">
    <div class="card-body text-center">
      <div class="verification-icon">
        <i class="fas fa-check-circle"></i>
      </div>
      <h2>Authentic Certificate</h2>
      <p>This certificate record exists in the system.</p>

      <div class="record-preview mt-4">
        <div class="preview-item">
          <label>Resident</label>
          <span>{{ certificate.resident.full_name }}</span>
        </div>
      </div>
    </div>
  </div>
  {% endif %}
  {% endif %}
  {% endif %}
```


IoT-Enabled Barangay Resident and Attendance Management System

```

        <div class="preview-item">
            <label>Type</label>
            <span>{{ certificate.get_cert_type_display }}</span>
        </div>
        <div class="preview-item">
            <label>Date Issued</label>
            <span>{{ certificate.date_issued|date:"M d, Y" }}</span>
        </div>
    </div>

    <div class="mt-4">
        <a href="{% url 'certifications:view' certificate.pk %}" class="btn btn-
outline">View Full Details</a>
    </div>
</div>
{% else %}
<div class="card mt-4 verification-card error">
    <div class="card-body text-center">
        <div class="verification-icon">
            <i class="fas fa-times-circle"></i>
        </div>
        <h2>Record Not Found</h2>
        <p>No certificate was found with Control #: <strong>{{ control }}</strong>.</p>
        <p class="text-muted">Please double check the number and try again.</p>
    </div>
</div>
{% endif %}
{% endif %}
</div>

<style>
.search-input-group {
    display: flex;
    gap: 10px;
}

.search-input-group input {
    flex: 1;
}

.verification-card {
    padding: 20px;
}

.verification-icon {
    font-size: 60px;
    margin-bottom: 20px;
}

.verification-card.success .verification-icon {
    color: #10b981;
}

.verification-card.error .verification-icon {
    color: var(--danger);
}

.record-preview {
    background: rgba(var(--primary-rgb), 0.05);

```

IoT-Enabled Barangay Resident and Attendance Management System

```
border-radius: 8px;
padding: 20px;
display: grid;
grid-template-columns: repeat(3, 1fr);
gap: 15px;
text-align: left;
}

.preview-item label {
  display: block;
  font-size: 11px;
  color: var(--text-muted);
  text-transform: uppercase;
  margin-bottom: 4px;
}

.preview-item span {
  font-size: 13px;
  font-weight: 600;
}

.mb-4 {
  margin-bottom: 24px;
}

.mt-4 {
  margin-top: 24px;
}
</style>
{% endblock %}
```

File: templates/certifications/view.html

```
{% extends 'base.html' %}

{% block title %}Certificate #{{ certificate.control_number }} - Details{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'certifications:list' %}">Certifications</a> / {{
certificate.control_number }}</span>
{% endblock %}

{% block content %}
{% url 'certifications:list' as certifications_list_url %}
{% include 'components/back_button.html' with back_url=certifications_list_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>Certificate Details</h1>
        <span>Information for Control #: {{ certificate.control_number }}</span>
    </div>
    <div class="header-actions">
        <a href="{% url 'certifications:pdf' certificate.pk %}" target="_blank" class="btn
btn-primary">
            <i class="fas fa-file-pdf"></i> Download PDF
        </a>
        <a href="{% url 'certifications:receipt' certificate.pk %}" target="_blank" class="btn
btn-secondary">
            <i class="fas fa-receipt"></i> Print Receipt
        </a>
    </div>
</div>

<div class="profile-grid">
    <div class="profile-sidebar">
        <div class="card profile-card">
            <div class="profile-photo">
                <div class="photo-placeholder">
                    <i class="fas fa-certificate"></i>
                </div>
            </div>
            <div class="profile-info">
                <h2>{{ certificate.get_cert_type_display }}</h2>
                <span class="resident-id">Control #: {{ certificate.control_number }}</span>
                <div class="status-tags">
                    <span class="badge badge-success">{{ certificate.get_status_display }}</span>
                </div>
            </div>
        </div>
    </div>

    <div class="card mt-4">
        <div class="card-header">
            <h3>Transaction Details</h3>
        </div>
        <div class="card-body">
            <div class="info-item">
                <i class="fas fa-receipt"></i>
                <div>
                    <label>OR Number</label>
                    <span>{{ certificate.or_number|default:"N/A" }}</span>
                </div>
            </div>
        </div>
    </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>
</div>
<div class="info-item">
    <i class="fas fa-money-bill-wave"></i>
    <div>
        <label>Amount Paid</label>
        <span>?{{ certificate.amount_paid|default:"0.00" }}</span>
    </div>
</div>
<div class="info-item">
    <i class="fas fa-calendar-check"></i>
    <div>
        <label>Date Issued</label>
        <span>{{ certificate.date_issued|date:"F d, Y" }}</span>
    </div>
</div>
</div>
</div>
</div>
<div class="profile-main">
    <div class="card">
        <div class="card-header">
            <h3>Recipient Information</h3>
        </div>
        <div class="card-body">
            <div class="detail-grid">
                <div class="detail-item col-span-2">
                    <label>Resident Name</label>
                    <span>
                        <a href="{% url 'residents:view' certificate.resident.pk %}">
                            {{ certificate.resident.full_name }}
                        </a>
                    </span>
                </div>
                <div class="detail-item">
                    <label>Purpose</label>
                    <span>{{ certificate.purpose|default:"General Purpose" }}</span>
                </div>
                <div class="detail-item">
                    <label>Issued By</label>
                    <span>{{
certificate.issued_by.get_full_name|default:certificate.issued_by.username }}</span>
                </div>
            </div>
        </div>
    </div>
</div>
</div>

{% if certificate.cert_type == 'cedula' and certificate.cedula_details %}
<div class="card mt-4">
    <div class="card-header">
        <h3>Cedula Information</h3>
    </div>
    <div class="card-body">
        <div class="detail-grid">
            <div class="detail-item">
                <label>CTC Number</label>
                <span>{{ certificate.cedula_details.ctc_number }}</span>
            </div>
            <div class="detail-item">

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<label>Place of Issue</label>
<span>{{ certificate.cedula_details.place_of_issue }}</span>
</div>
<div class="info-row">
  <span class="label">Taxpayer Type:</span>
  <span class="value">{{ certificate.cedula_details.get_taxpayer_type_display
}}</span>
</div>
{% if certificate.cedula_details.taxpayer_type == 'individual' %}
<div class="info-row">
  <span class="label">Height/Weight:</span>
  <span class="value">{{ certificate.cedula_details.height }} / {{
certificate.cedula_details.weight }}</span>
</div>
{% endif %}
</div>
</div>

<div class="cedula-financial-details mt-3">
<h4>Financial Breakdown</h4>
<div class="table-responsive">
  <table class="table table-sm table-bordered">
    <thead class="bg-light text-muted">
      <tr>
        <th>Category</th>
        <th class="text-end">Taxable Amount</th>
        <th class="text-end">Tax Rate</th>
        <th class="text-end">Tax Due</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Basic Community Tax</td>
        <td class="text-end">-</td>
        <td class="text-end">Fixed</td>
        <td class="text-end">?{{
certificate.cedula_details.basic_tax|floatformat:2 }}</td>
      </tr>
      {% if certificate.cedula_details.taxpayer_type == 'individual' %}
      <tr>
        <td>(a) Profession/Occupation</td>
        <td class="text-end">?{{
certificate.cedula_details.raw_taxable_income|floatformat:2 }}</td>
        <td class="text-end">?1 / ?1,000</td>
        <td class="text-end">?{{
certificate.cedula_details.additional_tax_income|floatformat:2 }}</td>
      </tr>
      <tr>
        <td>(b) Business Receipts</td>
        <td class="text-end">?{{
certificate.cedula_details.raw_taxable_business|floatformat:2 }}</td>
        <td class="text-end">?1 / ?1,000</td>
        <td class="text-end">?{{
certificate.cedula_details.additional_tax_business|floatformat:2 }}</td>
      </tr>
      <tr>
        <td>(c) Real Property Value</td>
        <td class="text-end">?{{
certificate.cedula_details.raw_taxable_property|floatformat:2 }}</td>
        <td class="text-end">?1 / ?1,000</td>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <td class="text-end">?{{
certificate.cedula_details.additional_tax_property|floatformat:2 }}</td>
        </tr>
        {% else %}
        <tr>
            <td>(a) Assessed Property Value</td>
            <td class="text-end">?{{
certificate.cedula_details.raw_taxable_property|floatformat:2 }}</td>
            <td class="text-end">?2 / ?5,000</td>
            <td class="text-end">?{{
certificate.cedula_details.additional_tax_property|floatformat:2 }}</td>
        </tr>
        <tr>
            <td>(b) Gross Receipts/Earnings</td>
            <td class="text-end">?{{
certificate.cedula_details.raw_taxable_business|floatformat:2 }}</td>
            <td class="text-end">?2 / ?5,000</td>
            <td class="text-end">?{{
certificate.cedula_details.additional_tax_business|floatformat:2 }}</td>
        </tr>
        {% endif %}
        <tr class="table-secondary">
            <td colspan="3"><strong>TOTAL (Capped)</strong></td>
            <td class="text-end"><strong>?{{ certificate.amount_paid|floatformat:2
}}</strong></td>
        </tr>
    </tbody>
</table>
</div>
</div>
</div>
{% endif %}

<div class="card mt-4">
    <div class="card-header">
        <h3>Status History</h3>
    </div>
    <div class="card-body">
        <div class="status-timeline">
            <div class="timeline-item">
                <div class="timeline-icon success">
                    <i class="fas fa-check"></i>
                </div>
                <div class="timeline-content">
                    <h4>Issued</h4>
                    <p>Certificate was generated and signed on {{ certificate.issued_at|date:"M
d, Y H:i" }}</p>
                </div>
            </div>
        </div>
    </div>
</div>
</div>
</div>

<style>
/* Reuse styles from residents/view.html */
.profile-grid {
    display: grid;
    grid-template-columns: 350px 1fr;

```

IoT-Enabled Barangay Resident and Attendance Management System

```
    gap: 25px;
}

.profile-card {
  text-align: center;
  padding: 30px 20px;
}

.profile-photo {
  width: 150px;
  height: 150px;
  margin: 0 auto 20px;
  border-radius: 50%;
  overflow: hidden;
  background: var(--border);
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 60px;
  color: var(--text-muted);
}

.badge {
  padding: 4px 10px;
  border-radius: 4px;
  font-size: 11px;
  font-weight: 600;
  text-transform: uppercase;
}

.badge-success {
  background: rgba(16, 185, 129, 0.1);
  color: #10b981;
}

.info-item {
  display: flex;
  gap: 15px;
  margin-bottom: 15px;
}

.info-item i {
  font-size: 18px;
  color: var(--primary);
  width: 25px;
}

.info-item label {
  display: block;
  font-size: 11px;
  color: var(--text-muted);
}

.info-item span {
  font-size: 14px;
  font-weight: 500;
}

.detail-grid {
  display: grid;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    grid-template-columns: repeat(2, 1fr);
    gap: 20px;
}

.detail-item label {
    display: block;
    font-size: 12px;
    color: var(--text-muted);
    margin-bottom: 5px;
}

.detail-item span {
    font-size: 15px;
    font-weight: 500;
}

.col-span-2 {
    grid-column: span 2;
}

.mt-4 {
    margin-top: 25px;
}

.status-timeline {
    margin-top: 10px;
}

.timeline-item {
    display: flex;
    gap: 15px;
    position: relative;
    padding-bottom: 20px;
}

.timeline-icon {
    width: 32px;
    height: 32px;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
    background: var(--bg-card);
    border: 2px solid var(--border);
    z-index: 1;
}

.timeline-icon.success {
    border-color: #10b981;
    color: #10b981;
}

.timeline-content h4 {
    font-size: 14px;
    margin-bottom: 4px;
}

.timeline-content p {
    font-size: 12px;
    color: var(--text-muted);
}
```


IoT-Enabled Barangay Resident and Attendance Management System

```
}  
  
@media (max-width: 992px) {  
  .profile-grid {  
    grid-template-columns: 1fr;  
  }  
}  
</style>  
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/certifications/virtual_certificate.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Virtual Certificate - {{ certificate.control_number }}{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Certifications / Virtual Certificate</span>
{% endblock %}

{% block content %}
{% url 'certifications:list' as back_url %}
{% include 'components/back_button.html' with back_url=back_url back_label='Back' %}
<style>
    @import url('https://fonts.googleapis.com/css2?family=Cinzel:wght@700&family=Playfair+
Display:wght@700;900&family=Inter:wght@400;700;800&display=swap');

    .certificate-viewer {
        display: flex;
        flex-direction: column;
        align-items: center;
        padding: 40px 20px;
        background: #f1f5f9;
        min-height: calc(100vh - 120px);
    }

    .certificate-container {
        width: 100%;
        max-width: 850px;
        background: #fff;
        padding: 40px 60px;
        position: relative;
        box-shadow: 0 20px 50px rgba(0, 0, 0, 0.1);
        border: 1px solid #e2e8f0;
        margin-bottom: 40px;
        overflow: hidden;
    }

    /* Watermark */
    .certificate-container::before {
        content: "";
        position: absolute;
        top: 50%;
        left: 50%;
        width: 600px;
        height: 600px;
        transform: translate(-50%, -50%);
        background-image: url("{% static 'images/logo.png' %}");
        background-size: contain;
        background-repeat: no-repeat;
        background-position: center;
        opacity: 0.08;
        z-index: 0;
        pointer-events: none;
    }

    .certificate-header {
        position: relative;
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    z-index: 1;
    text-align: center;
    margin-bottom: 30px;
    border-bottom: 1px solid #000;
    padding-bottom: 15px;
}

.header-logos {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 10px;
    padding: 0 10px;
}

.logo-item {
    height: 80px;
    width: auto;
    max-width: 150px;
    object-fit: contain;
}

.header-text {
    flex-grow: 1;
    text-align: center;
}

.header-text p {
    margin: 0;
    font-family: 'Inter', sans-serif;
    font-size: 14px;
    text-transform: uppercase;
    color: #000;
    line-height: 1.2;
}

.header-text .country {
    font-weight: 500;
    font-size: 16px;
}

.header-text .brgy {
    font-weight: 800;
    font-size: 18px;
    margin-top: 2px;
}

.office-title {
    text-align: center;
    margin: 25px 0 10px;
    position: relative;
    z-index: 1;
}

.office-title h2 {
    font-family: 'Inter', sans-serif;
    font-size: 22px;
    font-weight: 700;
    text-transform: uppercase;
    margin: 0;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    color: #000;
    letter-spacing: 1px;
}

.cert-type-title {
    text-align: center;
    margin-bottom: 30px;
    position: relative;
    z-index: 1;
}

.cert-type-title h1 {
    font-family: 'Inter', sans-serif;
    font-size: 48px;
    font-weight: 900;
    text-transform: uppercase;
    margin: 0;
    color: #000;
    letter-spacing: 2px;
    border-bottom: 3px solid #000;
    display: inline-block;
    padding-bottom: 5px;
}

.to-whom {
    font-family: 'Inter', sans-serif;
    font-weight: 900;
    font-size: 20px;
    margin-bottom: 25px;
    position: relative;
    z-index: 1;
    text-transform: uppercase;
}

.certificate-content {
    font-family: 'Inter', sans-serif;
    font-size: 18px;
    line-height: 1.6;
    color: #000;
    text-align: justify;
    position: relative;
    z-index: 1;
}

.certificate-content p {
    margin-bottom: 25px;
}

.bold-name {
    font-weight: 900;
    text-transform: uppercase;
}

.certificate-footer {
    margin-top: 80px;
    display: flex;
    justify-content: flex-end;
    position: relative;
    z-index: 1;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.signatory {
  text-align: center;
  width: 320px;
}

.signatory-name {
  font-weight: 900;
  font-size: 19px;
  text-transform: uppercase;
  margin-bottom: 0;
}

.signatory-title {
  font-size: 15px;
  font-weight: 500;
  border-top: 1px solid #000;
  padding-top: 5px;
}

.actions-bar {
  display: flex;
  gap: 15px;
  margin-bottom: 20px;
  width: 100%;
  max-width: 850px;
}

@media print {
  {% if user_role == 'resident' %}

  body {
    display: none !important;
  }

  {% else %}

  .top-navbar,
  .sidebar,
  .actions-bar,
  .breadcrumb,
  .sidebar-overlay,
  .menu-toggle {
    display: none !important;
  }

  .main-content {
    margin: 0 !important;
    padding: 0 !important;
    width: 100% !important;
  }

  .certificate-viewer {
    padding: 0;
    background: #fff;
  }

  .certificate-container {
    box-shadow: none;
    border: none;
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        margin: 0;
        padding: 20px 40px;
    }

    body {
        background: white !important;
    }

    {% endif %}
}

/* Resident Watermark */
.not-valid-watermark {
    position: absolute;
    top: 50%;
    left: 50%;
    transform: translate(-50%, -50%) rotate(-45deg);
    font-size: 60px;
    font-weight: 900;
    color: rgba(220, 38, 38, 0.15);
    z-index: 2;
    pointer-events: none;
    white-space: nowrap;
    text-transform: uppercase;
    border: 10px solid rgba(220, 38, 38, 0.15);
    padding: 10px 30px;
}
</style>

<div class="certificate-viewer">
    <div class="actions-bar">
        {% if user_role != 'resident' %}
        <button onclick="window.print()" class="btn btn-primary">
            <i class="fas fa-print"></i> Print Certificate
        </button>
        {% endif %}
        <a href="{% url 'certifications:pdf' certificate.pk %}" class="btn btn-outline"
target="_blank">
            <i class="fas fa-file-pdf"></i> Official PDF
        </a>
    </div>

    <div class="certificate-container">
        {% if user_role == 'resident' %}
        <div class="not-valid-watermark" style="display: none;"></div>
        {% endif %}
        <div class="certificate-header">
            <div class="header-logos">
                

                
                <div class="header-text">
                    <p class="country">REPUBLIC OF THE PHILIPPINES</p>
                    <p>Province of Surigao del Norte</p>
                    <p>Municipality of Gigaquit</p>
                    <p class="brgy">BARANGAY SICO-SICO</p>
                </div>
                
            </div>
        </div>
    </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>
</div>

{% if certificate.cert_type == 'cedula' and certificate.cedula_details %}
<!-- CEDULA LAYOUT -->
<div class="office-title" style="margin-top: 0;">
    <h2 style="font-size: 18px;">COMMUNITY TAX CERTIFICATE</h2>
    <p style="font-weight: 700; margin: 5px 0 0 0;">{{
certificate.cedula_details.get_taxpayer_type_display|upper }}</p>
</div>

<style>
    .cedula-table { width: 100%; border-collapse: collapse; margin-top: 15px;
border: 1.5px solid #000; }
    .cedula-table td { border: 1px solid #000; padding: 6px 10px; vertical-align:
top; }
    .c-label { font-size: 9px; font-weight: 800; text-transform: uppercase;
display: block; margin-bottom: 3px; }
    .c-value { font-size: 13px; font-weight: 600; }
    .c-money { text-align: right; font-weight: 700; font-family: 'Courier New',
Courier, monospace; }
    .c-bg { background: #f8fafc; }
</style>

<div class="table-responsive" style="border: none;">
<table class="cedula-table">
    <tr>
        <td style="width: 15%;">
            <span class="c-label">YEAR</span>
            <span class="c-value">{{ certificate.date_issued|date:"Y" }}</span>
        </td>
        <td style="width: 60%;">
            <span class="c-label">PLACE OF ISSUE (City/Mun/Prov)</span>
            <span class="c-value">{{
certificate.cedula_details.place_of_issue|default:"Sico-Sico, Gigaquit, SDN" }}</span>
        </td>
        <td style="width: 25%;">
            <span class="c-label">DATE ISSUED</span>
            <span class="c-value">{{ certificate.date_issued|date:"M d, Y"
}}</span>
        </td>
    </tr>
</table>

<table class="cedula-table" style="border-top: none;">
    <tr>
        <td style="width: 75%;">
            <span class="c-label">NAME (Surname, First Name, Middle Name) /
COMPANY NAME</span>
            <span class="c-value" style="font-size: 16px;">{{
certificate.resident.full_name|upper }}</span>
        </td>
        <td style="width: 25%;">
            <span class="c-label">CTC NUMBER</span>
            <span class="c-value" style="color: #dc2626;">{{
certificate.cedula_details.ctc_number }}</span>
        </td>
    </tr>
    <tr>
        <td colspan="2">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <span class="c-label">ADDRESS</span>
        <span class="c-value">{{ certificate.resident.address }}</span>
    </td>
</tr>
</table>

{% if certificate.cedula_details.taxpayer_type == 'individual' %}
<table class="cedula-table" style="border-top: none;">
    <tr>
        <td style="width: 25%;"><span class="c-label">CITIZENSHIP</span><span
class="c-value">{{ certificate.resident.nationality }}</span></td>
        <td style="width: 25%;"><span class="c-label">CIVIL STATUS</span><span
class="c-value">{{ certificate.resident.get_civil_status_display }}</span></td>
        <td style="width: 25%;"><span class="c-label">GENDER</span><span
class="c-value">{{ certificate.resident.get_gender_display }}</span></td>
        <td style="width: 25%;"><span class="c-label">BIRTH DATE</span><span
class="c-value">{{ certificate.resident.birthdate|date:"M d, Y" }}</span></td>
    </tr>
    <tr>
        <td><span class="c-label">HEIGHT</span><span class="c-value">{{
certificate.cedula_details.height|default:"--" }} cm</span></td>
        <td><span class="c-label">WEIGHT</span><span class="c-value">{{
certificate.cedula_details.weight|default:"--" }} kg</span></td>
        <td colspan="2"><span class="c-label">PROFESSION / OCCUPATION</span><span
class="c-value">{{ certificate.resident.occupation|default:"--" }}</span></td>
    </tr>
</table>
{% endif %}

<table class="cedula-table" style="margin-top: 20px;">
    <tr class="c-bg">
        <td style="width: 50%;"><span class="c-label">COMMUNITY TAX
DUE</span></td>
        <td style="width: 30%;"><span class="c-label">TAXABLE AMOUNT</span></td>
        <td style="width: 20%;"><span class="c-label">AMOUNT</span></td>
    </tr>
    <tr>
        <td>1. BASIC COMMUNITY TAX (?5.00)</td>
        <td>--</td>
        <td class="c-money">{{ certificate.cedula_details.basic_tax|floatformat:2
}}</td>
    </tr>
    <tr>
        <td>2. ADDITIONAL COMMUNITY TAX</td>
        <td>--</td>
        <td>--</td>
    </tr>
    <tr>
        <td style="padding-left: 25px;">(a) Real Property (?1.00 per ?1,000)</td>
        <td class="c-money">{{
certificate.cedula_details.raw_taxable_property|floatformat:2 }}</td>
        <td class="c-money">{{
certificate.cedula_details.additional_tax_property|floatformat:2 }}</td>
    </tr>
    <tr>
        <td style="padding-left: 25px;">(b) Business Receipts (?1.00 per
?1,000)</td>
        <td class="c-money">{{
certificate.cedula_details.raw_taxable_business|floatformat:2 }}</td>
        <td class="c-money">{{

```


IoT-Enabled Barangay Resident and Attendance Management System

```

certificate.cedula_details.additional_tax_business|floatformat:2 }}</td>
</tr>
<tr>
    <td style="padding-left: 25px;">(c) Annual Income (?1.00 per ?1,000)</td>
    <td class="c-money">{{
certificate.cedula_details.raw_taxable_income|floatformat:2 }}</td>
    <td class="c-money">{{
certificate.cedula_details.additional_tax_income|floatformat:2 }}</td>
</tr>
<tr class="c-bg">
    <td colspan="2" style="text-align: right;"><span class="c-label">TOTAL
AMOUNT PAID</span></td>
    <td class="c-money" style="font-size: 16px;">?{{
certificate.cedula_details.total_amount|floatformat:2 }}</td>
</tr>
</table>
</div>

<div style="margin-top: 40px; display: flex; justify-content: space-between;
align-items: flex-end;">
    <div style="text-align: center; width: 250px;">
        <div style="border-bottom: 1px solid #000; height: 30px;"></div>
        <span class="c-label" style="margin-top: 5px;">Taxpayer Signature</span>
    </div>
    <div class="signatory" style="margin-top: 0;">
        <p class="signatory-name" style="font-size: 16px;">HON. {{
captain_name|upper }}</p>
        <div class="signatory-title" style="font-size: 12px;">
Punong
Barangay</div>
    </div>
</div>

{% else %}
<!-- STANDARD CERTIFICATE LAYOUT -->
<div class="office-title">
    <h2>OFFICE OF THE SANGGUNIANG BARANGAY</h2>
</div>

<div class="cert-type-title">
    <h1>{{ certificate.get_cert_type_display }}</h1>
</div>

<div class="to-whom">
    TO WHOM IT MAY CONCERN;
</div>

<div class="certificate-content">
    <p>
        This is to certify that <span class="bold-name">{{
certificate.resident.full_name }}</span>,
        {{ certificate.resident.age|default:"--" }} years old,
        {{ certificate.resident.get_civil_status_display|default:"Single"|title
}},
        Filipino,
        is a bonafide resident of Barangay Sico-Sico, Gigaquit, Surigao del Norte.
    </p>

    {% if certificate.cert_type == 'business_permit' %}
    <p>
        This is to certify that <strong>{{ certificate.business_name|upper

```

IoT-Enabled Barangay Resident and Attendance Management System

```
}}</strong> owned/managed by
    <strong>{{ certificate.resident.full_name|upper }}</strong>, located at {{
certificate.business_address|default:certificate.resident.address }},
    Barangay Sico-Sico, Gigaquit, Surigao del Norte, is hereby granted
clearance to operate within the jurisdiction of this barangay.
    <br><br>
    Type of Business: <strong>{{ certificate.business_type|default:"--"
}}</strong>
</p>
{% endif %}

<p>
    This certification is issued upon the request of the above-named person
for
    <strong>{{ certificate.purpose|default:"General Purposes" }}</strong>.
</p>

<p>
    Certification is issued for whatever legal purpose it may serve him/her
best.
</p>

<p style="margin-top: 50px;">
    Given this <strong>{{ certificate.date_issued|date:"jS" }}</strong> day of
    <strong>{{ certificate.date_issued|date:"F" }}</strong>, {{
certificate.date_issued|date:"Y" }}</strong>,
    at the Office of Punong Barangay of Barangay Sico-Sico, Gigaquit, Surigao
del Norte.
</p>
</div>

<div class="certificate-footer">
    <div class="signatory">
        <p class="signatory-name">HON. {{ captain_name|default:"NO ACTIVE CAPTAIN"
}}</p>
        <div class="signatory-title">Punong Barangay</div>
    </div>
</div>
{% endif %}
</div>
{% endblock %}
```

File: templates/components/back_button.html

```
<div class="app-back-slot">
  <a href="{{ back_url|default:'javascript:history.back()' }}" class="app-back-btn {{
back_class|default:'' }}">
    <i class="fas fa-arrow-left"></i>
    <span>{{ back_label|default:'Back' }}</span>
  </a>
</div>
```

File: templates/core/backup_choice.html

```
{% extends "base.html" %}
{% load static %}

{% block title %}Backup Data - Choose Destination{% endblock %}

{% block breadcrumb %}
<span><i class="fas fa-home"></i> Dashboard</span> / <span><i class="fas fa-database"></i>
Backup</span>
{% endblock %}

{% block content %}
<div class="backup-container">
  <div class="page-header">
    <div class="header-content">
      <h1><i class="fas fa-file-export"></i> Choose Backup Destination</h1>
      <p>Select the drive where you want to save the backup of the system data and
media files.</p>
    </div>
  </div>

  <div class="destinations-grid">
    {% for dest in destinations %}
      <div
        class="destination-card {% if dest.is_external %}external{% else %}local{%
endif %} {% if not dest.is_mounted %}unmounted{% endif %}">
        <div class="card-icon">
          {% if dest.is_external %}
            {% if dest.is_mounted %}
              <i class="fas fa-hdd"></i>
            {% else %}
              <i class="fas fa-plug text-muted"></i>
            {% endif %}
          {% else %}
            <i class="fas fa-server"></i>
          {% endif %}
        </div>
        <div class="card-info">
          <h3>
            {% if dest.is_external %}
              External Drive {% if not dest.is_mounted %}(Detected){% endif %}
            {% else %}
              Local Storage
            {% endif %}
          </h3>
          <code class="path">
            {% if dest.is_mounted %}
              {{ dest.path }}
            {% else %}
              Device: {{ dest.device }}
            {% endif %}
          </code>
          <p class="description">
            {% if dest.is_external %}
              Recommended for off-site security. Save to USB/SSD/SD Card plugged
into the system.
            {% else %}

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <span class="text-warning"><i class="fas fa-exclamation-circle"></i>
This drive is plugged in but
        not active. Click below to mount it.</span>
        {% endif %}
        {% else %}
        Internal storage backup. Quick and easy but less secure against
hardware failure.
        {% endif %}
    </p>
</div>
<div class="card-actions">
    {% if dest.is_mounted %}
    <form action="{% url 'core:backup_execute' %}" method="POST">
        {% csrf_token %}
        <input type="hidden" name="target_path" value="{ { dest.path } }">
        <button type="submit" class="btn btn-primary btn-lg w-100"
            onclick="this.disabled=true; this.innerHTML='<i class='fas fa-
spinner fa-spin'\></i> Saving...'; this.form.submit();">
            <i class="fas fa-save"></i> Save to this location
        </button>
    </form>
    {% else %}
    <form action="{% url 'core:backup_mount_drive' %}" method="POST">
        {% csrf_token %}
        <input type="hidden" name="device" value="{ { dest.device } }">
        <input type="hidden" name="perform_backup" value="true">
        <button type="submit" class="btn btn-warning btn-lg w-100"
            onclick="this.disabled=true; this.innerHTML='<i class='fas fa-
spinner fa-spin'\></i> Activating...'; this.form.submit();">
            <i class="fas fa-bolt"></i> Activate & Start Backup
        </button>
        <p class="mt-2 small text-muted">This will wake up your drive and save
the backup immediately.</p>
    </form>
    {% endif %}
</div>
</div>
{% empty %}
<div class="no-destinations">
    <i class="fas fa-exclamation-triangle"></i>
    <p>No writeable backup destinations found. Please ensure your external drive
is properly mounted and has
        write permissions.</p>
</div>
{% endfor %}
</div>

<div class="alternate-option mt-5">
    <div class="card bg-light">
        <div class="card-body text-center">
            <h3><i class="fas fa-cloud-download-alt"></i> Want to download
instead?</h3>
            <p>You can also download the backup file directly to your
computer/phone.</p>
            <a href="{% url 'core:backup_download' %}" class="btn btn-outline-
primary">
                <i class="fas fa-download"></i> Download to this device
            </a>
        </div>
    </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
</div>
</div>

<style>
  .backup-container {
    max-width: 1000px;
    margin: 0 auto;
    padding: 20px;
  }

  .page-header {
    margin-bottom: 30px;
    text-align: center;
  }

  .page-header h1 {
    font-size: 2rem;
    color: var(--primary-color);
    margin-bottom: 10px;
  }

  .destinations-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
    gap: 25px;
  }

  .destination-card {
    background: var(--card-bg, #ffffff);
    border-radius: 15px;
    padding: 30px;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.1);
    display: flex;
    flex-direction: column;
    align-items: center;
    text-align: center;
    transition: transform 0.3s ease, box-shadow 0.3s ease;
    border: 2px solid transparent;
    height: 100%;
  }

  .destination-card:hover {
    transform: translateY(-10px);
    box-shadow: 0 15px 40px rgba(0, 0, 0, 0.15);
    border-color: var(--primary-color);
  }

  .card-icon {
    font-size: 3rem;
    color: var(--primary-color);
    margin-bottom: 20px;
    background: rgba(var(--primary-rgb), 0.1) !important;
    width: 80px;
    height: 80px;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
  }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.destination-card.external .card-icon {
  color: #10b981;
  background: rgba(16, 185, 129, 0.1) !important;
}

.card-info h3 {
  margin-bottom: 10px;
  font-weight: 700;
  color: #1e293b;
  /* Dark Slate */
}

.path {
  background: #f1f5f9;
  border: 1px solid #e2e8f0;
  padding: 6px 12px;
  border-radius: 6px;
  font-size: 0.85rem;
  color: #475569;
  /* Slate 600 */
  display: block;
  margin-bottom: 15px;
  word-break: break-all;
  font-family: 'JetBrains Mono', 'Courier New', monospace;
}

.description {
  color: #64748b;
  /* Slate 500 */
  font-size: 0.95rem;
  margin-bottom: 25px;
  line-height: 1.5;
}

.card-actions {
  margin-top: auto;
  width: 100%;
}

.no-destinations {
  grid-column: 1 / -1;
  background: rgba(225, 29, 72, 0.1);
  border: 1px solid #fecdd3;
  padding: 40px;
  border-radius: 15px;
  color: #e11d48;
  text-align: center;
}

.alternate-option {
  margin-top: 50px;
}

.mt-5 {
  margin-top: 3rem;
}

.card.bg-light {
  background: rgba(0, 0, 0, 0.02);
  border: 1px dashed var(--primary-color);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        border-radius: 15px;
    }

    .destination-card.unmounted {
        border: 2px dashed #cbd5e1;
        opacity: 0.85;
    }

    .destination-card.unmounted:hover {
        border-color: #f59e0b;
        opacity: 1;
    }

    .text-warning {
        color: #d97706;
    }

    .btn-warning {
        background-color: #f59e0b;
        color: white;
        border: none;
    }

    .btn-warning:hover {
        background-color: #d97706;
    }
</style>

<script>
    let lastDestsCount = {{ destinations| length }};
    let isInteracting = false;

    // Monitor for user interactions to avoid disruptive refreshes
    document.addEventListener('submit', () => isInteracting = true);

    function pollDestinations() {
        if (isInteracting) return;

        fetch('{% url "core:backup_check_dests" %}')
```

```
        .then(response => response.json())
        .then(data => {
            if (data.destinations && data.destinations.length !== lastDestsCount) {
                // New drive detected or removed! Refresh to update UI
                window.location.reload();
            }
        })
        .catch(err => console.error('Polling error:', err));
    }

    // Poll every 3 seconds for new drives
    setInterval(pollDestinations, 3000);
</script>
{% endblock %}
```


File: templates/core/barangay_map.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Barangay Map - Household Locator{% endblock %}

{% block extra_head %}
<link rel="stylesheet" href="https://unpkg.com/maplibre-gl/dist/maplibre-gl.css" />
<link href="https://fonts.googleapis.com/css2?family=DM+Sans:ital,opsz,wght@0,9..40,300;0,9..40,500;0,9..40,700;1,9..40,400&display=swap" rel="stylesheet">
<style>
  /* ?? Page layout ?????????????????????????????????? */
  .map-page { position: relative; height: calc(100vh - 74px); }
  #brgy-map { width: 100%; height: 100%; border-radius: 0; }

  /* ?? Shared control card style ?????????????????????? */
  .map-ctrl {
    font-family: 'DM Sans', sans-serif;
    background: rgba(255,255,255,.92);
    backdrop-filter: blur(14px);
    -webkit-backdrop-filter: blur(14px);
    border: 1px solid var(--border-light);
    border-radius: var(--radius-md);
    box-shadow: 0 4px 24px rgba(0,0,0,.08);
    color: var(--text-main);
    z-index: 5;
    pointer-events: auto;
  }

  /* ?? Info Card (bottom-left) ?????????????????????? */
  .info-card {
    position: absolute; bottom: 34px; left: 16px;
    padding: 14px 20px;
    display: flex; align-items: center; gap: 14px;
  }
  .info-card .brgy-badge {
    width: 42px; height: 42px;
    border-radius: 10px;
    background: linear-gradient(135deg, var(--primary), #3da667);
    display: flex; align-items: center; justify-content: center;
    color: #fff; font-size: 18px;
    flex-shrink: 0;
  }
  .info-card h4 { margin: 0 0 2px; font-size: 14px; font-weight: 700; letter-spacing:
- .01em; }
  .info-card .info-stats {
    display: flex; gap: 12px; margin-top: 4px; font-size: 12px; color: var(--text-muted);
  }
  .info-card .info-stats span { display: flex; align-items: center; gap: 4px; }
  .info-card .info-stats i { font-size: 10px; color: var(--primary); }

  /* ?? Layer toggle (top-right) ?????????????????????? */
  .layer-toggle {
    position: absolute; top: 16px; right: 16px;
    display: flex; padding: 4px; gap: 2px;
  }
  .layer-toggle button {
    border: none; background: transparent;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
padding: 8px 14px; border-radius: var(--radius-sm);
font-family: 'DM Sans', sans-serif;
font-size: 13px; font-weight: 600;
color: var(--text-muted); cursor: pointer;
transition: var(--transition);
}
.layer-toggle button.active {
  background: var(--primary); color: #fff;
  box-shadow: 0 2px 8px rgba(44,110,73,.25);
}
.layer-toggle button:hover:not(.active) { background: var(--bg-main); }

/* ?? Search bar (top-left) ?????????????????????? */
.map-search {
  position: absolute; top: 16px; left: 16px;
  display: flex; align-items: center;
  padding: 0 14px; width: 320px; max-width: calc(100vw - 240px);
}
.map-search i { color: var(--text-muted); font-size: 14px; flex-shrink: 0; }
.map-search input {
  border: none; background: transparent;
  padding: 10px 12px; width: 100%;
  font-family: 'DM Sans', sans-serif;
  font-size: 14px; color: var(--text-main);
  outline: none;
}
.map-search input::placeholder { color: var(--text-muted); }
.search-results {
  position: absolute; top: 100%; left: 0; right: 0;
  margin-top: 4px; max-height: 240px; overflow-y: auto;
  display: none;
}
.search-results.show { display: block; }
.search-result-item {
  padding: 10px 16px; cursor: pointer;
  font-size: 13px; border-bottom: 1px solid var(--border-light);
  transition: background .15s;
}
.search-result-item:last-child { border-bottom: none; }
.search-result-item:hover { background: var(--primary-light); }
.search-result-item .result-type {
  font-size: 11px; color: var(--text-muted); margin-top: 2px;
}

/* ?? Custom buttons (Locate / Fullscreen) ???????? */
.map-custom-btns {
  position: absolute; bottom: 34px; right: 16px;
  display: flex; flex-direction: column; gap: 6px;
}
.map-btn {
  width: 40px; height: 40px;
  display: flex; align-items: center; justify-content: center;
  border: none; cursor: pointer;
  font-size: 16px; color: var(--text-main);
  transition: var(--transition);
}
.map-btn:hover { background: var(--primary-light); color: var(--primary); }

/* ?? MapLibre default control overrides ?????????? */
.maplibregl-ctrl-group {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
background: rgba(255,255,255,.92) !important;
backdrop-filter: blur(14px) !important;
border: 1px solid var(--border-light) !important;
border-radius: var(--radius-md) !important;
box-shadow: 0 4px 24px rgba(0,0,0,.08) !important;
overflow: hidden;
}
.maplibregl-ctrl-group button {
  width: 36px !important; height: 36px !important;
}
.maplibregl-ctrl-attr { font-size: 11px !important; }

/* ?? Popup styling ?????????????????????????????? */
.maplibregl-popup-content {
  border-radius: var(--radius-md) !important;
  padding: 0 !important;
  box-shadow: 0 8px 30px rgba(0,0,0,.12) !important;
  overflow: hidden;
  font-family: 'DM Sans', sans-serif;
  min-width: 220px;
}
.popup-inner { padding: 0; }
.popup-header {
  background: linear-gradient(135deg, var(--primary), #3da667);
  color: #fff; padding: 14px 18px;
}
.popup-header h3 { margin: 0; font-size: 15px; font-weight: 700; }
.popup-body { padding: 14px 18px; }
.popup-stat {
  display: flex; align-items: center; gap: 10px;
  padding: 6px 0; font-size: 13px;
}
.popup-stat i {
  width: 28px; height: 28px; border-radius: 8px;
  display: flex; align-items: center; justify-content: center;
  font-size: 12px;
}
.popup-stat .icon-residents { background: rgba(37,99,235,.1); color: #2563EB; }
.popup-stat .icon-households { background: rgba(245,158,11,.1); color: #F59E0B; }
.popup-stat .icon-head { background: rgba(16,185,129,.1); color: #10B981; }
.popup-stat-label { color: var(--text-muted); font-size: 11px; }
.popup-stat-value { font-weight: 700; font-size: 15px; }

.maplibregl-popup-close-button {
  color: rgba(255,255,255,.7) !important;
  font-size: 18px !important;
  padding: 4px 8px !important;
}
.maplibregl-popup-close-button:hover { color: #fff !important; }

/* ?? Marker styling (Households) ?????????????????? */
.household-marker {
  width: 32px; height: 32px;
  background: linear-gradient(135deg, var(--primary), #3da667);
  border: 3px solid #fff;
  border-radius: 50%;
  box-shadow: 0 3px 12px rgba(44,110,73,.35);
  cursor: pointer;
  transition: transform .2s, box-shadow .2s;
  display: flex; align-items: center; justify-content: center;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

font-size: 14px; color: #fff;
}
.household-marker:hover {
  transform: scale(1.2);
  box-shadow: 0 6px 20px rgba(44,110,73,.45);
}
.household-marker i { font-size: 14px; }

/* ?? Loading overlay ?????????????????????????????? */
.map-loading {
  position: absolute; inset: 0;
  background: var(--bg-main);
  display: flex; flex-direction: column;
  align-items: center; justify-content: center;
  z-index: 10;
  transition: opacity .5s;
}
.map-loading.hidden { opacity: 0; pointer-events: none; }
.map-loading .spinner {
  width: 40px; height: 40px;
  border: 3px solid var(--border);
  border-top-color: var(--primary);
  border-radius: 50%;
  animation: spin .8s linear infinite;
}
.map-loading p { margin-top: 14px; font-size: 14px; color: var(--text-muted); }
@keyframes spin { to { transform: rotate(360deg); } }

/* ?? Responsive ?????????????????????????????? */
@media (max-width: 768px) {
  .map-search { width: calc(100% - 32px); }
  .info-card { left: 10px; right: 10px; bottom: 24px; }
  .layer-toggle { top: 66px; }
}

/* Remove page-content padding for map */
.page-content { padding: 0 !important; overflow: hidden; }
</style>
{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Management System / Household Map</span>
{% endblock %}

{% block content %}
<div class="map-page">
  <!-- Loading Overlay -->
  <div class="map-loading" id="mapLoading">
    <div class="spinner"></div>
    <p>Loading Household Map</p>
  </div>

  <!-- Map Container -->
  <div id="brgy-map"></div>

  <!-- Search Bar -->
  <div class="map-ctrl map-search" id="mapSearch">
    <i class="fas fa-search"></i>
    <input type="text" id="searchInput" placeholder="Search household by number or family head?" autocomplete="off" />
  </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="map-ctrl search-results" id="searchResults"></div>
</div>

<!-- Layer Toggle -->
<div class="map-ctrl layer-toggle" id="layerToggle">
  <button class="active" data-layer="streets"><i class="fas fa-road"></i>&nbsp;  
Standard</button>
  <button data-layer="satellite"><i class="fas fa-satellite"></i>&nbsp;  
Satellite</button>
</div>

<!-- Custom Buttons -->
<div class="map-custom-btms">
  <button class="map-ctrl map-btn" id="btnLocate" title="Locate Me">
    <i class="fas fa-crosshairs"></i>
  </button>
  <button class="map-ctrl map-btn" id="btnFullscreen" title="Toggle Fullscreen">
    <i class="fas fa-expand"></i>
  </button>
</div>

<!-- Info Card -->
<div class="map-ctrl info-card">
  <div class="brgy-badge"><i class="fas fa-house-user"></i></div>
  <div>
    <h4>Household Locator</h4>
    <div class="info-stats">
      <span><i class="fas fa-circle"></i> Brgy. Sico Sico</span>
      <span><i class="fas fa-circle"></i> Pop. 828</span>
      <span><i class="fas fa-circle"></i> 35.1 m</span>
    </div>
  </div>
</div>
</div>

<!-- MapLibre GL JS -->
<script src="https://unpkg.com/maplibre-gl/dist/maplibre-gl.js"></script>
<script>
document.addEventListener('DOMContentLoaded', () => {

  // ?? Map initialization ?????????????????????????????????
  const BRGY_CENTER = [125.7222, 9.4898]; // [lng, lat]
  const ZOOM = 15;

  const TILE_STREETS = 'https://tile.openstreetmap.org/{z}/{x}/{y}.png';
  const TILE_SATELLITE = 'https://server.arcgisonline.com/ArcGIS/rest/services/World_Image
ry/MapServer/tile/{z}/{y}/{x}';

  function makeStyle(tileUrl, attr) {
    return {
      version: 8,
      sources: {
        'raster-tiles': {
          type: 'raster',
          tiles: [tileUrl],
          tileSize: 256,
          attribution: attr
        }
      },
      layers: [{
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        id: 'raster-layer',
        type: 'raster',
        source: 'raster-tiles',
        minzoom: 0,
        maxzoom: 19
    }]
  };
}

const streetsStyle = makeStyle(TILE_STREETS, '&copy; <a
href="https://www.openstreetmap.org/copyright">OpenStreetMap</a>');
const satelliteStyle = makeStyle(TILE_SATELLITE, 'Tiles &copy; Esri');

const map = new maplibregl.Map({
  container: 'brgy-map',
  style: streetsStyle,
  center: BRGY_CENTER,
  zoom: ZOOM,
  maxZoom: 19,
  attributionControl: true
});

// Navigation (zoom) controls
map.addControl(new maplibregl.NavigationControl({ showCompass: true }), 'bottom-right');

// Hide loader once the map is ready
map.on('load', () => {
  document.getElementById('mapLoading').classList.add('hidden');
  loadHouseholds();
});

// ?? Layer toggle ?????????????????????????????????????????
let currentLayer = 'streets';
document.getElementById('layerToggle').addEventListener('click', (e) => {
  const btn = e.target.closest('button');
  if (!btn || btn.classList.contains('active')) return;
  const layer = btn.dataset.layer;

  document.querySelectorAll('#layerToggle button').forEach(b =>
b.classList.remove('active'));
  btn.classList.add('active');

  const oldMarkers = [...activeMarkers];

  if (layer === 'satellite') {
    map.setStyle(satelliteStyle);
  } else {
    map.setStyle(streetsStyle);
  }
  currentLayer = layer;

  // MapLibre removes markers on style change ? re-add them after style loads
  map.once('styledata', () => {
    oldMarkers.forEach(m => m.addTo(map));
  });
});

// ?? Fullscreen ?????????????????????????????????????????
const btnFs = document.getElementById('btnFullscreen');
btnFs.addEventListener('click', () => {

```

IoT-Enabled Barangay Resident and Attendance Management System

```
const el = document.querySelector('.map-page');
if (!document.fullscreenElement) {
    el.requestFullscreen().catch(() => {});
    btnFs.innerHTML = '<i class="fas fa-compress"></i>';
} else {
    document.exitFullscreen();
    btnFs.innerHTML = '<i class="fas fa-expand"></i>';
}
});
document.addEventListener('fullscreenchange', () => {
    if (!document.fullscreenElement) btnFs.innerHTML = '<i class="fas fa-expand"></i>';
    setTimeout(() => map.resize(), 200);
});

// ?? Locate Me ??????????????????????????????????????
document.getElementById('btnLocate').addEventListener('click', () => {
    if (!navigator.geolocation) return alert('Geolocation is not supported by your browser.');
```

```
    const btn = document.getElementById('btnLocate');
    btn.innerHTML = '<i class="fas fa-spinner fa-spin"></i>';
    navigator.geolocation.getCurrentPosition(
        (pos) => {
            const { latitude, longitude } = pos.coords;
            map.flyTo({ center: [longitude, latitude], zoom: 17, duration: 1500 });

            // Add a blue dot for "my location"
            if (window._locMarker) window._locMarker.remove();
            const dot = document.createElement('div');
            dot.style.cssText = 'width:16px;height:16px;background:#2563EB;border:3px solid #fff;border-radius:50%;box-shadow:0 0 10px rgba(37,99,235,.4)';
            window._locMarker = new maplibregl.Marker({ element: dot }).setLngLat([longitude, latitude]).addTo(map);
            btn.innerHTML = '<i class="fas fa-crosshairs"></i>';
        },
        () => {
            alert('Unable to retrieve your location.');
```

```
            btn.innerHTML = '<i class="fas fa-crosshairs"></i>';
        },
        { enableHighAccuracy: true, timeout: 10000 }
    );
});

// ?? Load Household markers ??????????????????????????
let activeMarkers = [];
let householdData = [];

function loadHouseholds() {
    fetch(`${% url 'residents:household_map_api' %}`)
        .then(r => r.json())
        .then(geojson => {
            householdData = geojson.features || [];
            householdData.forEach(f => addHouseholdMarker(f));
        })
        .catch(err => console.error('Failed to load households:', err));
}
```

```
function addHouseholdMarker(feature) {
    const coords = feature.geometry.coordinates;
    const props = feature.properties;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
// Custom marker element
const el = document.createElement('div');
el.className = 'household-marker';
el.innerHTML = '<i class="fas fa-house-user"></i>';

const marker = new maplibregl.Marker({ element: el, anchor: 'center' })
  .setLngLat(coords)
  .setPopup(new maplibregl.Popup({ offset: 25, closeButton: true, maxWidth: '280px'
}).setHTML(`
  <div class="popup-inner">
    <div class="popup-header">
      <h3><i class="fas fa-building" style="margin-right:6px"></i>Household
${props.household_no}</h3>
    </div>
    <div class="popup-body">
      <div class="popup-stat">
        <i class="icon-head fas fa-user-tie"></i>
        <div>
          <div class="popup-stat-label">Head of Household</div>
          <div class="popup-stat-value" style="font-size:13px;font-
weight:700">${props.head}</div>
        </div>
      </div>
      <div class="popup-stat">
        <i class="icon-households fas fa-map-pin"></i>
        <div>
          <div class="popup-stat-label">Purok</div>
          <div class="popup-stat-value">${props.purok}</div>
        </div>
      </div>
      <div class="popup-stat">
        <i class="icon-residents fas fa-users"></i>
        <div>
          <div class="popup-stat-label">Residents</div>
          <div class="popup-stat-value">${props.residents}</div>
        </div>
      </div>
    </div>
  </div>
`))
  .addTo(map);

activeMarkers.push(marker);
}

// ?? Search ?????????????????????????????????????????????
const searchInput = document.getElementById('searchInput');
const searchResults = document.getElementById('searchResults');
let debounceTimer;

searchInput.addEventListener('input', () => {
  clearTimeout(debounceTimer);
  const q = searchInput.value.trim().toLowerCase();
  if (!q) { searchResults.classList.remove('show'); return; }

  // Filter households by number or head of household
  const householdMatches = householdData.filter(f =>
    f.properties.household_no.toLowerCase().includes(q) ||
    f.properties.head.toLowerCase().includes(q)
  );
});
```


IoT-Enabled Barangay Resident and Attendance Management System

```
let html = householdMatches.map(f => `  <div><i class="fas fa-house-user" style="color:var(--primary);margin-right:6px"></i><strong>Household ${f.properties.household_no}</strong></div>  
  <div class="result-type">Head: ${f.properties.head} ? ${f.properties.purok}</div>  
  </div>  
`).join('');  
  
if (html) {  
  searchResults.innerHTML = html;  
  searchResults.classList.add('show');  
  bindResultClicks();  
} else {  
  searchResults.innerHTML = '<div class="search-result-item" style="color:var(--text-muted)"><i class="fas fa-info-circle"></i> No households found</div>';  
  searchResults.classList.add('show');  
}  
});  
  
function bindResultClicks() {  
  searchResults.querySelectorAll('.search-result-item').forEach(item => {  
    item.addEventListener('click', () => {  
      const lng = parseFloat(item.dataset.lng);  
      const lat = parseFloat(item.dataset.lat);  
      map.flyTo({ center: [lng, lat], zoom: 18, duration: 1200 });  
      searchResults.classList.remove('show');  
      searchInput.value = item.querySelector('strong').textContent.trim();  
    });  
  });  
}  
  
// Close search results when clicking outside  
document.addEventListener('click', (e) => {  
  if (!document.getElementById('mapSearch').contains(e.target)) {  
    searchResults.classList.remove('show');  
  }  
});  
});  
</script>  
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard - Barangay Management System{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Dashboard</h1>
    <span>Welcome back, {{ request.user.get_full_name|default:request.user.username }}!</span>
  </div>
  <div class="header-actions">
    <a href="{% url 'certifications:issue' %}" class="btn btn-primary">
      <i class="fas fa-plus"></i> Issue Certificate
    </a>
  </div>
</div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-blue">
      <i class="fas fa-users"></i>
    </div>
    <div class="stat-info">
      <span class="value">{{ total_residents }}</span>
      <span class="label">Total Residents</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-green">
      <i class="fas fa-user-tie"></i>
    </div>
    <div class="stat-info">
      <span class="value">{{ total_officials }}</span>
      <span class="label">Total Officials</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-yellow">
      <i class="fas fa-certificate"></i>
    </div>
    <div class="stat-info">
      <span class="value">{{ certificates_today }}</span>
      <span class="label">Certificates Issued Today</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-red">
      <i class="fas fa-gavel"></i>
    </div>
    <div class="stat-info">
      <span class="value">{{ total_ordinances }}</span>
      <span class="label">Active Ordinances</span>
    </div>
  </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="charts-grid">
  <div class="card">
    <div class="card-header">
      <h3>Recent Certifications</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Control No.</th>
            <th>Resident</th>
            <th>Type</th>
            <th>Date</th>
          </tr>
        </thead>
        <tbody>
          {% for cert in recent_certificates %}
          <tr>
            <td>{{ cert.control_number }}</td>
            <td>{{ cert.resident.full_name }}</td>
            <td>{{ cert.get_cert_type_display }}</td>
            <td>{{ cert.date_issued|date:"M d, Y" }}</td>
          </tr>
          {% empty %}
          <tr>
            <td colspan="4" class="text-center">No recent certificates.</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>

  <div class="card">
    <div class="card-header">
      <h3>Recently Added Residents</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Name</th>
            <th>Age</th>
            <th>Gender</th>
            <th>Purok</th>
          </tr>
        </thead>
        <tbody>
          {% for resident in recent_residents %}
          <tr>
            <td>{{ resident.full_name }}</td>
            <td>{{ resident.age }}</td>
            <td>{{ resident.get_gender_display }}</td>
            <td>{{ resident.purok }}</td>
          </tr>
          {% empty %}
          <tr>
            <td colspan="4" class="text-center">No recent residents.</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        {% endfor %}
    </tbody>
</table>
</div>
</div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_captain.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard ? Punong Barangay{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Dashboard</h1>
    <span>Welcome, <strong>{{ request.user.get_full_name|default:request.user.username }}</strong> &mdash; Punong Barangay / Admin</span>
  </div>
  <div class="header-actions">
  </div>
</div>

<!-- Stats Grid -->
<div class="stats-grid">
  <div class="stat-card blue">
    <div class="stat-icon icon-blue"><i class="fas fa-users"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_residents }}</span>
      <span class="label">Total Residents</span>
    </div>
  </div>
  <div class="stat-card green">
    <div class="stat-icon icon-green"><i class="fas fa-user-tie"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_officials }}</span>
      <span class="label">Active Officials</span>
    </div>
  </div>
  <div class="stat-card yellow">
    <div class="stat-icon icon-yellow"><i class="fas fa-certificate"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_today }}</span>
      <span class="label">Certificates Issued Today</span>
    </div>
  </div>
  <div class="stat-card red">
    <div class="stat-icon icon-red"><i class="fas fa-gavel"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_ordinances }}</span>
      <span class="label">Active Ordinances</span>
    </div>
  </div>
  <div class="stat-card purple">
    <div class="stat-icon icon-purple"><i class="fas fa-home"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_households }}</span>
      <span class="label">Households</span>
    </div>
  </div>
  <div class="stat-card orange clickable" onclick="location.href='{% url 'residents:registration_list' %}'"
    style="cursor: pointer;">
    <div class="stat-icon icon-orange">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<i class="fas fa-user-clock"></i>
</div>
<div class="stat-info">
  <span class="value">{{ pending_registrations_count }}</span>
  <span class="label">Pending Registrations</span>
  {% if pending_registrations_today > 0 %}
  <small class="text-success" style="font-size: 11px; margin-top: 4px; font-weight:
600;">+{{ pending_registrations_today }} new today</small>
  {% endif %}
</div>
</div>
</div>

<!-- Quick Actions -->

<div class="charts-grid">
  <!-- Recent Certifications -->
  <div class="card">
    <div class="card-header">
      <h3>Recent Certifications</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Control No.</th>
            <th>Resident</th>
            <th>Type</th>
            <th>Date</th>
          </tr>
        </thead>
        <tbody>
          {% for cert in recent_certificates %}
          <tr>
            <td>{{ cert.control_number }}</td>
            <td>{{ cert.resident.full_name }}</td>
            <td>{{ cert.get_cert_type_display }}</td>
            <td>{{ cert.date_issued|date:"M d, Y" }}</td>
          </tr>
          {% empty %}
          <tr>
            <td colspan="4" class="text-center">No recent certificates.</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>

  <!-- Recently Added Residents -->
  <div class="card">
    <div class="card-header">
      <h3>Recently Added Residents</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Name</th>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<th>Age</th>
<th>Gender</th>
<th>Purok</th>
</tr>
</thead>
<tbody>
  {% for resident in recent_residents %}
  <tr>
    <td>{{ resident.full_name }}</td>
    <td>{{ resident.age }}</td>
    <td>{{ resident.get_gender_display }}</td>
    <td>{{ resident.purok }}</td>
  </tr>
  {% empty %}
  <tr>
    <td colspan="4" class="text-center">No recent residents.</td>
  </tr>
  {% endfor %}
</tbody>
</table>
</div>
</div>

<!-- Recent Ordinances -->
<div class="card">
  <div class="card-header">
    <h3>Recent Ordinances</h3>
  </div>
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Title</th>
          <th>Status</th>
          <th>Date Enacted</th>
        </tr>
      </thead>
      <tbody>
        {% for ord in recent_ordinances %}
        <tr>
          <td>{{ ord.title }}</td>
          <td><span class="badge badge-{{ ord.status }}">{{ ord.get_status_display
}}</span></td>
          <td>{{ ord.date_enacted|date:"M d, Y" }}</td>
        </tr>
        {% empty %}
        <tr>
          <td colspan="3" class="text-center">No ordinances found.</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
  </div>
</div>
</div>

<!-- Population Overview -->
<div class="card">
  <div class="card-header">
    <h3>Population Overview</h3>
  </div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="gender-stats">
  <div class="gender-item male">
    <i class="fas fa-mars"></i>
    <span class="gender-count">{{ male_count }}</span>
    <span class="gender-label">Male</span>
  </div>
  <div class="gender-divider"></div>
  <div class="gender-item female">
    <i class="fas fa-venus"></i>
    <span class="gender-count">{{ female_count }}</span>
    <span class="gender-label">Female</span>
  </div>
</div>
</div>
</div>

{% endblock %}
```


IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_default.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard ? Barangay Staff{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Dashboard</h1>
    <span>Welcome, {{ request.user.get_full_name|default:request.user.username }} &mdash;
      <strong>Staff</strong></span>
    </div>
  </div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-blue"><i class="fas fa-users"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_residents }}</span>
      <span class="label">Total Residents</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-yellow"><i class="fas fa-certificate"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_today }}</span>
      <span class="label">Certificates Issued Today</span>
    </div>
  </div>
</div>

<div class="role-banner role-default">
  <div class="role-banner-icon"><i class="fas fa-id-badge"></i></div>
  <div>
    <strong>Staff / General Access</strong>
    <p>You can view residents and certificates. Contact admin for extended access.</p>
  </div>
</div>

<div class="charts-grid">
  <div class="card">
    <div class="card-header">
      <h3>Recent Certifications</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Control No.</th>
            <th>Resident</th>
            <th>Type</th>
            <th>Date</th>
          </tr>
        </thead>
        <tbody>
          {% for cert in recent_certificates %}
            <tr>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<td>{{ cert.control_number }}</td>
<td>{{ cert.resident.full_name }}</td>
<td>{{ cert.get_cert_type_display }}</td>
<td>{{ cert.date_issued|date:"M d, Y" }}</td>
</tr>
{% empty %}
<tr>
  <td colspan="4" class="text-center">No certificates yet.</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>

<div class="card">
  <div class="card-header">
    <h3>Recently Added Residents</h3>
  </div>
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Name</th>
          <th>Age</th>
          <th>Purok</th>
        </tr>
      </thead>
      <tbody>
        {% for resident in recent_residents %}
        <tr>
          <td>{{ resident.full_name }}</td>
          <td>{{ resident.age }}</td>
          <td>{{ resident.purok }}</td>
        </tr>
        {% empty %}
        <tr>
          <td colspan="3" class="text-center">No recent residents.</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
  </div>
</div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_kagawad.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard ? Kagawad{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Kagawad Dashboard</h1>
    <span>Welcome, {{ request.user.get_full_name|default:request.user.username }} &mdash;
<strong>Sangguniang
  Barangay Member (Kagawad)</strong></span>
  </div>
  <div class="header-actions">
    <a href="{% url 'certifications:issue' %}" class="btn btn-primary">
      <i class="fas fa-plus"></i> Issue Certificate
    </a>
  </div>
</div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-blue"><i class="fas fa-users"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_residents }}</span>
      <span class="label">Total Residents</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-red"><i class="fas fa-gavel"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_ordinances }}</span>
      <span class="label">Active Ordinances</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-yellow"><i class="fas fa-certificate"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_today }}</span>
      <span class="label">Certificates Today</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-green"><i class="fas fa-file-alt"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_total }}</span>
      <span class="label">Total Certificates</span>
    </div>
  </div>
</div>

<div class="role-banner role-kagawad">
  <div class="role-banner-icon"><i class="fas fa-balance-scale"></i></div>
  <div>
    <strong>Kagawad Access</strong>
    <p>Review ordinances, handle certifications, and monitor residents.</p>
  </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

</div>

```
<div class="charts-grid">
  <div class="card">
    <div class="card-header">
      <h3>Recent Ordinances</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Title</th>
            <th>Status</th>
            <th>Date Enacted</th>
          </tr>
        </thead>
        <tbody>
          {% for ord in recent_ordinances %}
          <tr>
            <td>{{ ord.title }}</td>
            <td><span class="badge badge-{{ ord.status }}">{{ ord.get_status_display
}}</span></td>
            <td>{{ ord.date_enacted|date:"M d, Y" }}</td>
          </tr>
          {% empty %}
          <tr>
            <td colspan="3" class="text-center">No ordinances found.</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
</div>
```

```
<div class="card">
  <div class="card-header">
    <h3>Recent Certifications</h3>
  </div>
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Control No.</th>
          <th>Resident</th>
          <th>Type</th>
          <th>Date</th>
        </tr>
      </thead>
      <tbody>
        {% for cert in recent_certificates %}
        <tr>
          <td>{{ cert.control_number }}</td>
          <td>{{ cert.resident.full_name }}</td>
          <td>{{ cert.get_cert_type_display }}</td>
          <td>{{ cert.date_issued|date:"M d, Y" }}</td>
        </tr>
        {% empty %}
        <tr>
          <td colspan="4" class="text-center">No certificates yet.</td>
        </tr>
      </tbody>
    </table>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        {% endfor %}
    </tbody>
</table>
</div>
</div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_lupon.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard ? Lupon Member{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Lupon Dashboard</h1>
    <span>Welcome, {{ request.user.get_full_name|default:request.user.username }} &mdash;
  <strong>Lupon
    Member</strong></span>
  </div>
</div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-blue"><i class="fas fa-users"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_residents }}</span>
      <span class="label">Total Residents</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-red"><i class="fas fa-gavel"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_ordinances }}</span>
      <span class="label">Active Ordinances</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-purple"><i class="fas fa-home"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_households }}</span>
      <span class="label">Households</span>
    </div>
  </div>
</div>

<div class="role-banner role-lupon">
  <div class="role-banner-icon"><i class="fas fa-handshake"></i></div>
  <div>
    <strong>Lupon Tagapamayapa Access</strong>
    <p>View residents and review ordinances relevant to dispute resolution.</p>
  </div>
</div>

<div class="charts-grid">
  <div class="card">
    <div class="card-header">
      <h3>Recent Ordinances</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Title</th>
          </tr>
        </thead>
      </table>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<th>Status</th>
<th>Date Enacted</th>
</tr>
</thead>
<tbody>
  {% for ord in recent_ordinances %}
    <tr>
      <td>{{ ord.title }}</td>
      <td><span class="badge badge-{{ ord.status }}">{{ ord.get_status_display
}}</span></td>
      <td>{{ ord.date_enacted|date:"M d, Y" }}</td>
    </tr>
  {% empty %}
    <tr>
      <td colspan="3" class="text-center">No ordinances found.</td>
    </tr>
  {% endfor %}
</tbody>
</table>
</div>
</div>

<div class="card">
  <div class="card-header">
    <h3>Recently Added Residents</h3>
  </div>
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Name</th>
          <th>Age</th>
          <th>Purok</th>
        </tr>
      </thead>
      <tbody>
        {% for resident in recent_residents %}
          <tr>
            <td>{{ resident.full_name }}</td>
            <td>{{ resident.age }}</td>
            <td>{{ resident.purok }}</td>
          </tr>
        {% empty %}
          <tr>
            <td colspan="3" class="text-center">No recent residents.</td>
          </tr>
        {% endfor %}
      </tbody>
    </table>
  </div>
</div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_resident.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}My Portal ? Barangay Resident{% endblock %}

{% block content %}
<style>
    /* ?? Resident Portal Premium Aesthetic ?? */
    .dashboard-wrapper {
        position: relative;
        z-index: 1;
    }

    .resident-hero {
        background: linear-gradient(135deg, #1e4a31 0%, #2d7a50 100%);
        border-radius: 24px;
        padding: 40px;
        color: #fff;
        display: flex;
        align-items: center;
        gap: 32px;
        margin-bottom: 32px;
        box-shadow: 0 10px 30px rgba(45, 122, 80, 0.15);
        position: relative;
        overflow: hidden;
    }

    .resident-hero::after {
        content: '';
        position: absolute;
        top: -50%;
        right: -10%;
        width: 300px;
        height: 300px;
        background: radial-gradient(circle, rgba(255, 255, 255, 0.1) 0%, transparent 70%);
        border-radius: 50%;
    }

    .resident-hero-avatar {
        width: 90px;
        height: 90px;
        border-radius: 24px;
        background: rgba(255, 255, 255, 0.15);
        backdrop-filter: blur(8px);
        display: flex;
        align-items: center;
        justify-content: center;
        font-size: 2.5rem;
        color: #fff;
        flex-shrink: 0;
        border: 2px solid rgba(255, 255, 255, 0.3);
        overflow: hidden;
    }

    .glass-panel-card {
        background: rgba(255, 255, 255, 0.65);
        backdrop-filter: blur(12px);
    }
</style>
</block>
</content>
```


IoT-Enabled Barangay Resident and Attendance Management System

```
-webkit-backdrop-filter: blur(12px);
border: 1px solid rgba(255, 255, 255, 0.4);
border-radius: 24px;
box-shadow: 0 8px 32px rgba(31, 38, 135, 0.05);
transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
margin-bottom: 30px;
overflow: hidden;
}

.glass-panel-card:hover {
  transform: translateY(-5px);
  box-shadow: 0 12px 40px rgba(31, 38, 135, 0.08);
  background: rgba(255, 255, 255, 0.75);
}

.panel-header {
  padding: 24px 30px;
  border-bottom: 1px solid rgba(226, 232, 240, 0.5);
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.panel-header h3 {
  font-size: 1.1rem;
  font-weight: 800;
  color: #0f172a;
  margin: 0;
  display: flex;
  align-items: center;
  gap: 12px;
}

.panel-body {
  padding: 24px 30px;
}

.info-chip-modern {
  background: rgba(255, 255, 255, 0.5);
  backdrop-filter: blur(4px);
  border: 1px solid rgba(226, 232, 240, 0.8);
  border-radius: 12px;
  padding: 10px 18px;
  font-size: 0.85rem;
  font-weight: 600;
  color: #475569;
  display: flex;
  align-items: center;
  gap: 8px;
  transition: all 0.2s;
}

.info-chip-modern:hover {
  border-color: var(--primary, #2d7a50);
  color: var(--primary, #2d7a50);
  background: #fff;
}

.bg-blobs {
  position: fixed;
```

IoT-Enabled Barangay Resident and Attendance Management System

```

    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    z-index: -1;
    overflow: hidden;
    pointer-events: none;
}

.blob {
    position: absolute;
    width: 600px;
    height: 600px;
    background: radial-gradient(circle, rgba(45, 122, 80, 0.08) 0%, rgba(45, 122, 80, 0) 70%);
    border-radius: 50%;
    filter: blur(60px);
}
</style>

<div class="bg-blobs">
  <div class="blob" style="top: -10%; left: -10%;"></div>
  <div class="blob" style="bottom: 0%; right: -5%; background: radial-gradient(circle, rgba(20, 184, 166, 0.07) 0%, rgba(20, 184, 166, 0) 70%);">
  </div>
</div>

<div class="dashboard-wrapper">
  <!-- Hero Banner -->
  <div class="resident-hero">
    <div class="resident-hero-avatar">
      {% if resident_record and resident_record.photo %}
      
      {% elif request.user.avatar %}
      
      {% else %}
      
      {% endif %}
    </div>
    <div class="resident-hero-info">
      <h1 style="font-weight: 800; letter-spacing: -0.5px;">Hello, { { request.user.first_name|default:request.user.username } }!</h1>
      <p style="font-size: 1.05rem; opacity: 0.9;">
        {% if resident_record %}
        Purok { { resident_record.purok|default:"?" } } &nbsp;·&nbsp;
        { { resident_record.address|default:"" } }
        {% else %}
        Manage your barangay services and requests here.
        {% endif %}
      </p>
    </div>
    <span class="resident-hero-badge" style="background: rgba(255,255,255,0.2); border: 1px solid rgba(255,255,255,0.3); border-radius: 12px; font-weight: 700;">
      <i class="fas fa-certificate"></i>&nbsp; VERIFIED RESIDENT
    </span>
  </div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        </span>
    </div>

    {% if not resident_record %}
    <div class="glass-panel-card"
        style="background: rgba(59, 130, 246, 0.1); border-color: rgba(59, 130, 246, 0.2);
        margin-bottom: 24px;">
        <div class="panel-body" style="display: flex; align-items: center; gap: 16px;
        color: #1e40af;">
            <i class="fas fa-info-circle" style="font-size: 1.5rem;"></i>
            <div>
                <strong>Account pending linkage</strong> ? Your account is waiting to be
                linked to a verified resident
                record.
            </div>
        </div>
    </div>
    {% endif %}

    <!-- Quick Info Chips -->
    {% if resident_record %}
    <div style="display: flex; gap: 12px; flex-wrap: wrap; margin-bottom: 32px;">
        <div class="info-chip-modern"><i class="fas fa-venus-mars"></i> {{
        resident_record.get_gender_display|default:"?" }}</div>
        <div class="info-chip-modern"><i class="fas fa-birthday-cake"></i> {{
        resident_record.birthday|date:"M d,      Y"|default:"?" }}</div>
        <div class="info-chip-modern"><i class="fas fa-ring"></i> {{
        resident_record.get_civil_status_display|default:"?" }}</div>
        {% if resident_record.is_pwd %}<div class="info-chip-modern" style="color:
        #ef4444;"><i
            class="fas fa-wheelchair"></i> PWD</div>{% endif %}
        {% if resident_record.is_senior_citizen %}<div class="info-chip-modern"
        style="color: #f59e0b;"><i
            class="fas fa-user-clock"></i> Senior Citizen</div>{% endif %}
        {% if resident_record.is_registered_voter %}<div class="info-chip-modern"
        style="color: #2d7a50;"><i
            class="fas fa-vote-yea"></i> Registered Voter</div>{% endif %}
    </div>
    {% endif %}

    <div class="charts-grid">
        <!-- Certificate Services -->
        <div class="glass-panel-card">
            <div class="panel-header">
                <h3><i class="fas fa-file-signature"
                style="color:var(--primary,#2d7a50)"></i> Certificate Services</h3>
                <a href="{% url 'certifications:request_certificate' %}" class="btn btn-
                primary btn-sm"
                    style="border-radius: 10px;">
                    <i class="fas fa-plus"></i> Request New
                </a>
            </div>
            <div class="panel-body">
                <div
                    style="background: rgba(45, 122, 80, 0.05); border-radius: 16px;
                    padding: 20px; display: flex; align-items: flex-start; gap: 16px; margin-bottom: 24px;
                    border: 1px solid rgba(45, 122, 80, 0.1);">
                        <div style="font-size: 1.8rem;">?</div>
                        <div>
                            <div style="font-weight: 800; color: #15803d; margin-bottom: 4px;

```

IoT-Enabled Barangay Resident and Attendance Management System

```
font-size: 0.95rem;">Fast
    Document Processing</div>
    <p style="margin: 0; color: #166534; font-size: 0.85rem; line-
height: 1.5;">
        Submit your request today. We'll verify and issue your
certificates within 24-48 hours.
    </p>
</div>
</div>
<div style="display: grid; grid-template-columns: repeat(2, 1fr); gap:
10px;">
    {% for type_code, type_label in cert_types|slice:".4" %}
    <div
        style="background: rgba(255,255,255,0.4); border: 1px solid
rgba(226, 232, 240, 0.6); border-radius: 12px; padding: 12px; font-size: 0.8rem; font-
weight: 600; display: flex; align-items: center; gap: 8px;">
        <i class="fas fa-check-circle" style="color: #2d7a50; font-size:
0.7rem;"></i>
        {{ type_label }}
    </div>
    {% endfor %}
</div>
</div>
<div style="padding: 8px 0 16px;">
    <div
        style="background:#f0fdf4;border:1.5px solid #86efac;border-
radius:12px;padding:20px 24px;display:flex;align-items:flex-start;gap:16px;margin-
bottom:20px;">
        <div style="font-size:2rem;line-height:1;">??</div>
        <div>
            <div style="font-weight:700;font-size:1rem;color:#15803d;margin-
bottom:6px;">Visit the Barangay Hall
                to Request a Certificate</div>
            <p style="margin:0;color:#166534;font-size:.9rem;line-height:1.6;">
                All certificate requests are processed in person at the barangay
office.
                Please bring a valid ID and any supporting documents required for
your request.
            </p>
        </div>
    </div>
</div>
<div style="display:grid;grid-template-columns:repeat(auto-
fill,minmax(190px,1fr));gap:12px;">
    {% for type_code, type_label in cert_types %}
    <div
        style="background:var(--bg-subtle,#f8fafc);border:1px solid
var(--border,#e2e8f0);border-radius:10px;padding:14px 16px;display:flex;align-
items:center;gap:10px;">
        <i class="fas fa-certificate"
style="color:var(--primary,#2d7a50);font-size:.9rem;"></i>
        <span style="font-size:.85rem;font-weight:500;">{{ type_label
}}</span>
    </div>
    {% endfor %}
</div>
</div>
</div>
<!-- My Certificates -->
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div class="glass-panel-card">
  <div class="panel-header">
    <h3><i class="fas fa-scroll" style="color:var(--primary,#2d7a50)"></i> My
Certificates</h3>
    <a href="{% url 'certifications:list' %}" class="btn btn-secondary btn-sm"
style="border-radius: 10px;">View
      All</a>
  </div>
  <div class="table-responsive">
    <table class="table-modern-simple">
      <thead>
        <tr>
          <th>Control No.</th>
          <th>Type</th>
          <th>Date</th>
          <th>Status</th>
        </tr>
      </thead>
      <tbody>
        {% for cert in my_certificates %}
          <tr>
            <td>
              <a href="{% url 'certifications:view' cert.pk %}"
style="color:var(--primary,#2d7a50); font-weight:700;
text-decoration: none;">
                {{ cert.control_number }}
              </a>
            </td>
            <td style="font-weight: 500;">{{ cert.get_cert_type_display
}}</td>
            <td style="font-size: 0.85rem; color: #64748b;">{{
cert.date_issued|date:"M d, Y" }}</td>
            <td>
              <span
                class="badge {% if cert.status == 'issued' %}badge-
success{% else %}badge-warning{% endif %}"
                style="border-radius: 8px;">
                {{ cert.status|capfirst }}
              </span>
            </td>
          </tr>
          {% empty %}
            <tr>
              <td colspan="4" class="text-center" style="color:#94a3b8;
padding:32px 0;">
                <i class="fas fa-inbox"
                  style="font-size:1.5rem; display:block; margin-bottom:8px;
opacity:.4;"></i>
                No certificates yet.
              </td>
            </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
</div>

<!-- My Profile Quick View -->
{% if resident_record %}

```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div class="glass-panel-card">
  <div class="panel-header">
    <h3><i class="fas fa-user-circle" style="color:var(--primary,#2d7a50)"></i> My
Profile</h3>
    <a href="{% url 'residents:view' resident_record.pk %}" class="btn btn-secondary
btn-sm"
      style="border-radius: 10px;">
      <i class="fas fa-eye"></i> Full Profile
    </a>
  </div>
  <div class="panel-body">
    <div style="display:grid; grid-template-columns: repeat(auto-fill, minmax(200px,
1fr)); gap: 20px;">
      <div>
        <div
          style="font-size: 0.7rem; text-transform: uppercase; letter-spacing:
1px; color: #64748b; margin-bottom: 4px; font-weight: 700;">
          Account Name</div>
        <div style="font-weight: 700; color: #0f172a;">{{
resident_record.full_name }}</div>
      </div>
      <div>
        <div
          style="font-size: 0.7rem; text-transform: uppercase; letter-spacing:
1px; color: #64748b; margin-bottom: 4px; font-weight: 700;">
          Location</div>
        <div style="font-weight: 700; color: #0f172a;">Purok {{
resident_record.purok|default:"?" }}</div>
      </div>
      <div>
        <div
          style="font-size: 0.7rem; text-transform: uppercase; letter-spacing:
1px; color: #64748b; margin-bottom: 4px; font-weight: 700;">
          Contact Details</div>
        <div style="font-weight: 700; color: #0f172a;">{{
resident_record.contact_number|default:"?" }}</div>
      </div>
      <div>
        <div
          style="font-size: 0.7rem; text-transform: uppercase; letter-spacing:
1px; color: #64748b; margin-bottom: 4px; font-weight: 700;">
          Current Occupation</div>
        <div style="font-weight: 700; color: #0f172a;">{{
resident_record.occupation|default:"?" }}</div>
      </div>
    </div>
  </div>
</div>
{% endif %}
</div>

<style>
.table-modern-simple {
  width: 100%;
  border-collapse: collapse;
}

.table-modern-simple th {
  text-align: left;
  padding: 16px 30px;

```

IoT-Enabled Barangay Resident and Attendance Management System

```
font-size: 0.75rem;
font-weight: 700;
color: #64748b;
text-transform: uppercase;
letter-spacing: 1px;
border-bottom: 1px solid rgba(226, 232, 240, 0.5);
}

.table-modern-simple td {
padding: 18px 30px;
font-size: 0.9rem;
border-bottom: 1px solid rgba(241, 245, 249, 0.5);
}
</style>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_secretary.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard ? Barangay Secretary{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Secretariat Dashboard</h1>
    <span>Welcome, {{ request.user.get_full_name|default:request.user.username }} &mdash;
<strong>Barangay
    Secretary</strong></span>
  </div>
  <div class="header-actions">
    <a href="{% url 'certifications:issue' %}" class="btn btn-primary">
      <i class="fas fa-plus"></i> Issue Certificate
    </a>
    <a href="{% url 'residents:add' %}" class="btn btn-outline">
      <i class="fas fa-user-plus"></i> Add Resident
    </a>
  </div>
</div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-blue"><i class="fas fa-users"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_residents }}</span>
      <span class="label">Total Residents</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-yellow"><i class="fas fa-certificate"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_today }}</span>
      <span class="label">Certificates Today</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-green"><i class="fas fa-file-alt"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_total }}</span>
      <span class="label">Total Certificates</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-purple"><i class="fas fa-home"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_households }}</span>
      <span class="label">Households</span>
    </div>
  </div>
</div>

<div class="role-banner role-secretary">
  <div class="role-banner-icon"><i class="fas fa-pen-nib"></i></div>
</div>
</div>
```


IoT-Enabled Barangay Resident and Attendance Management System

```
<strong>Secretariat Access</strong>
<p>Manage residents, certifications, households, and ordinances.</p>
</div>
</div>

<div class="charts-grid">
  <div class="card">
    <div class="card-header">
      <h3>Recent Certifications</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Control No.</th>
            <th>Resident</th>
            <th>Type</th>
            <th>Date</th>
          </tr>
        </thead>
        <tbody>
          {% for cert in recent_certificates %}
          <tr>
            <td>{{ cert.control_number }}</td>
            <td>{{ cert.resident.full_name }}</td>
            <td>{{ cert.get_cert_type_display }}</td>
            <td>{{ cert.date_issued|date:"M d, Y" }}</td>
          </tr>
          {% empty %}
          <tr>
            <td colspan="4" class="text-center">No certificates yet.</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
</div>

<div class="card">
  <div class="card-header">
    <h3>Recently Added Residents</h3>
  </div>
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Name</th>
          <th>Age</th>
          <th>Purok</th>
        </tr>
      </thead>
      <tbody>
        {% for resident in recent_residents %}
        <tr>
          <td>{{ resident.full_name }}</td>
          <td>{{ resident.age }}</td>
          <td>{{ resident.purok }}</td>
        </tr>
        {% empty %}
        <tr>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        <td colspan="3" class="text-center">No recent residents.</td>
    </tr>
    {% endfor %}
</tbody>
</table>
</div>
</div>

<div class="card">
    <div class="card-header">
        <h3>Recent Ordinances</h3>
    </div>
    <div class="table-responsive">
        <table>
            <thead>
                <tr>
                    <th>Title</th>
                    <th>Status</th>
                    <th>Date</th>
                </tr>
            </thead>
            <tbody>
                {% for ord in recent_ordinances %}
                <tr>
                    <td>{{ ord.title }}</td>
                    <td><span class="badge badge-{{ ord.status }}">{{ ord.get_status_display
}}</span></td>
                    <td>{{ ord.date_enacted|date:"M d, Y" }}</td>
                </tr>
                {% empty %}
                <tr>
                    <td colspan="3" class="text-center">No ordinances found.</td>
                </tr>
                {% endfor %}
            </tbody>
        </table>
    </div>
</div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_sk.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard ? SK Chairperson{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>SK Chairperson Dashboard</h1>
    <span>Welcome, {{ request.user.get_full_name|default:request.user.username }} &mdash;
<strong>SK
    Chairperson</strong></span>
  </div>
  <div class="header-actions">
    <a href="{% url 'certifications:issue' %}" class="btn btn-primary">
      <i class="fas fa-plus"></i> Issue Certificate
    </a>
  </div>
</div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-green"><i class="fas fa-child"></i></div>
    <div class="stat-info">
      <span class="value">{{ youth_count }}</span>
      <span class="label">Youth Residents (?30 yrs)</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-blue"><i class="fas fa-users"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_residents }}</span>
      <span class="label">Total Residents</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-yellow"><i class="fas fa-certificate"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_today }}</span>
      <span class="label">Certificates Issued Today</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-purple"><i class="fas fa-file-alt"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_total }}</span>
      <span class="label">Total Certificates</span>
    </div>
  </div>
</div>

<div class="role-banner role-sk">
  <div class="role-banner-icon"><i class="fas fa-fist-raised"></i></div>
  <div>
    <strong>Sangguniang Kabataan Access</strong>
    <p>Monitor youth residents and issue youth-related certifications.</p>
  </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

</div>

```
<div class="charts-grid">
  <div class="card">
    <div class="card-header">
      <h3>Recently Added Residents</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Name</th>
            <th>Age</th>
            <th>Gender</th>
            <th>Purok</th>
          </tr>
        </thead>
        <tbody>
          {% for resident in recent_residents %}
          <tr>
            <td>{{ resident.full_name }}</td>
            <td>{{ resident.age }}</td>
            <td>{{ resident.get_gender_display }}</td>
            <td>{{ resident.purok }}</td>
          </tr>
          {% empty %}
          <tr>
            <td colspan="4" class="text-center">No recent residents.</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
</div>
```

```
<div class="card">
  <div class="card-header">
    <h3>Recent Certifications</h3>
  </div>
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Control No.</th>
          <th>Resident</th>
          <th>Type</th>
          <th>Date</th>
        </tr>
      </thead>
      <tbody>
        {% for cert in recent_certificates %}
        <tr>
          <td>{{ cert.control_number }}</td>
          <td>{{ cert.resident.full_name }}</td>
          <td>{{ cert.get_cert_type_display }}</td>
          <td>{{ cert.date_issued|date:"M d, Y" }}</td>
        </tr>
        {% empty %}
        <tr>
          <td colspan="4" class="text-center">No certificates yet.</td>
        </tr>
      </tbody>
    </table>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/dashboard_treasurer.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Dashboard ? Barangay Treasurer{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Treasury Dashboard</h1>
    <span>Welcome, {{ request.user.get_full_name|default:request.user.username }} &mdash;
<strong>Barangay
    Treasurer</strong></span>
  </div>
</div>

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-icon icon-yellow"><i class="fas fa-certificate"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_today }}</span>
      <span class="label">Certificates Issued Today</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-green"><i class="fas fa-file-invoice-dollar"></i></div>
    <div class="stat-info">
      <span class="value">{{ certificates_total }}</span>
      <span class="label">Total Certificates</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-blue"><i class="fas fa-fingerprint"></i></div>
    <div class="stat-info">
      <span class="value">{{ attendance_today }}</span>
      <span class="label">Staff Present Today</span>
    </div>
  </div>
  <div class="stat-card">
    <div class="stat-icon icon-blue"><i class="fas fa-user-tie"></i></div>
    <div class="stat-info">
      <span class="value">{{ total_officials }}</span>
      <span class="label">Total Officials</span>
    </div>
  </div>
</div>

<div class="role-banner role-treasurer">
  <div class="role-banner-icon"><i class="fas fa-coins"></i></div>
  <div>
    <strong>Treasury Access</strong>
    <p>Monitor certificate transactions and staff attendance for payroll reference.</p>
  </div>
</div>

<div class="charts-grid">
  <div class="card">
    <div class="card-header">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<h3>Certificates Issued (Recent)</h3>
</div>
<div class="table-responsive">
  <table>
    <thead>
      <tr>
        <th>Control No.</th>
        <th>Resident</th>
        <th>Type</th>
        <th>Date</th>
      </tr>
    </thead>
    <tbody>
      {% for cert in recent_certificates %}
      <tr>
        <td>{{ cert.control_number }}</td>
        <td>{{ cert.resident.full_name }}</td>
        <td>{{ cert.get_cert_type_display }}</td>
        <td>{{ cert.date_issued|date:"M d, Y" }}</td>
      </tr>
      {% empty %}
      <tr>
        <td colspan="4" class="text-center">No certificates yet.</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
</div>
</div>

<div class="card">
  <div class="card-header">
    <h3>Today's Attendance</h3>
  </div>
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Official</th>
          <th>Time In</th>
          <th>Time Out</th>
          <th>Status</th>
        </tr>
      </thead>
      <tbody>
        {% for log in attendance_logs %}
        <tr>
          <td>{{ log.official.resident.full_name }}</td>
          <td>{{ log.time_in|time:"h:i A"|default:"?" }}</td>
          <td>{{ log.time_out|time:"h:i A"|default:"?" }}</td>
          <td><span class="badge badge-{{ log.status }}">{{ log.get_status_display
}}</span></td>
        </tr>
        {% empty %}
        <tr>
          <td colspan="4" class="text-center">No attendance records for today.</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    </div>
  </div>
</div>
{% endblock %}
```


File: templates/core/hardware_poller.html

```
<script>
/* =====
GLOBAL HARDWARE BUTTON ATTENDANCE POLLER
Runs on all pages ? records time-in/out via push button
without requiring the user to be on the Attendance tab.
===== */
(function() {
    'use strict';

    /* ---- Standalone Toast (works even without app.js) ---- */
    function hwToast(msg, type) {
        let container = document.getElementById('_hw-toast-container');
        if (!container) {
            container = document.createElement('div');
            container.id = '_hw-toast-container';
            container.style.cssText = [
                'position:fixed', 'top:20px', 'right:20px', 'z-index:99999',
                'display:flex', 'flex-direction:column', 'gap:10px', 'pointer-events:none'
            ].join(';');
            document.body.appendChild(container);
        }

        const colors = type === 'success'
            ? { bg: '#ecfdf5', border: '#6ee7b7', icon: '?', text: '#065f46' }
            : { bg: '#fef2f2', border: '#fca5a5', icon: '?', text: '#991b1b' };

        const toast = document.createElement('div');
        toast.style.cssText = [
            `background:${colors.bg}`, `border:1.5px solid ${colors.border}`,
            `color:${colors.text}`,
            'border-radius:12px', 'padding:14px 18px',
            'font-size:13px', 'font-weight:700',
            'box-shadow:0 8px 24px rgba(0,0,0,0.12)',
            'display:flex', 'align-items:center', 'gap:10px',
            'pointer-events:auto', 'min-width:280px', 'max-width:360px',
            'opacity:0', 'transform:translateX(20px)',
            'transition:all 0.3s ease'
        ].join(';');
        toast.innerHTML = `<span style="font-size:18px">${colors.icon}</span><span>${msg}</span>`;
        container.appendChild(toast);

        requestAnimationFrame(() => {
            toast.style.opacity = '1';
            toast.style.transform = 'translateX(0)';
        });

        setTimeout(() => {
            toast.style.opacity = '0';
            toast.style.transform = 'translateX(20px)';
            setTimeout(() => toast.remove(), 350);
        }, 4000);
    }

    const STATUS_URL = '{% url "core:biometric_attendance_status_check" %}';
    const VERIFY_URL = '{% url "attendance:api_biometric_verify" %}';
    const CSRF_TOKEN = '{{ csrf_token }}';
```

IoT-Enabled Barangay Resident and Attendance Management System

```
let _lastProcessed = null;

function processAttendance(mode) {
  const action = (mode && mode !== 'none' && mode !== 'attendance') ? mode :
'attendance';
  fetch(VERIFY_URL, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json', 'X-CSRFToken': CSRF_TOKEN },
    credentials: 'include',
    body: JSON.stringify({ auto: true, action: action })
  })
  .then(r => r.json())
  .then(data => {
    if (data.status === 'success') {
      if (window.showNotification) {
        window.showNotification(`? ${data.name} ? ${data.action} recorded at
${data.time || ''}`, 'success');
      } else {
        hwToast(`${data.name} ? ${data.action} recorded at ${data.time || ''}`,
'success');
      }
      // Auto-reload to show new data
      setTimeout(() => {
        window.location.reload();
      }, 1500);
    } else {
      if (window.showNotification) {
        window.showNotification(data.message || 'Attendance could not be recorded.',
'error');
      } else {
        hwToast(data.message || 'Attendance could not be recorded.', 'error');
      }
    }
  })
  .catch(() => {});
}

function poll() {
  fetch(STATUS_URL, { credentials: 'include' })
  .then(r => r.json())
  .then(data => {
    if (data.status === 'authenticated') {
      const key = (data.fingerprint_id || '') + ':' + (data.attendance_mode || '');
      if (key !== _lastProcessed) {
        _lastProcessed = key;
        processAttendance(data.attendance_mode);
        setTimeout(() => { _lastProcessed = null; }, 5000);
      }
    }
  })
  .catch(() => {});
}

// Don't poll twice on the dedicated biometric page
if (!window.location.pathname.includes('/attendance/biometric-attendance/')) {
  setInterval(poll, 2000);
}
})();
</script>
```

File: templates/core/login.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login - Barangay Management System</title>
  <link rel="stylesheet" href="{% static 'css/style.css' %}?v=2.1">
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css"
rel="stylesheet">
  <style>
    /* --- Premium Emerald Theme & Animations --- */
    :root {
      --emerald-50: #ecfdf5;
      --emerald-100: #d1fae5;
      --emerald-200: #a7f3d0;
      --emerald-300: #6ee7b7;
      --emerald-400: #34d399;
      --emerald-500: #10b981;
      --emerald-600: #059669;
      --emerald-700: #047857;
      --emerald-800: #065f46;
      --emerald-900: #064e3b;
      --emerald-950: #064e3b;
    }

    @keyframes scan-laser {
      0% { top: -10%; opacity: 0; }
      15% { opacity: 1; }
      45% { top: 105%; opacity: 1; }
      55% { top: 105%; opacity: 1; }
      85% { top: -10%; opacity: 1; }
      100% { top: -10%; opacity: 0; }
    }

    @keyframes clip-fill {
      0% { clip-path: inset(100% 0 0 0); }
      45% { clip-path: inset(0 0 0 0); }
      55% { clip-path: inset(0 0 0 0); }
      100% { clip-path: inset(100% 0 0 0); }
    }

    @keyframes pulse-ring {
      0% { transform: scale(0.6); opacity: 0.8; }
      100% { transform: scale(1.6); opacity: 0; }
    }

    @keyframes pop-in {
      0% { transform: scale(0.5); opacity: 0; }
      60% { transform: scale(1.2); }
      100% { transform: scale(1); opacity: 1; }
    }

    @keyframes shimmer {
      0% { transform: translateX(-100%); }
      100% { transform: translateX(100%); }
    }
  </style>
</head>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.animate-scan-laser { animation: scan-laser 2.5s ease-in-out infinite; }
.animate-clip-fill { animation: clip-fill 2.5s ease-in-out infinite; }
.animate-pulse-ring { animation: pulse-ring 2s cubic-bezier(0.215, 0.61, 0.355, 1)
infinite; }
.animate-pulse-ring-delayed { animation: pulse-ring 2s cubic-bezier(0.215, 0.61,
0.355, 1) 1s infinite; }
.animate-pop-in { animation: pop-in 0.5s cubic-bezier(0.175, 0.885, 0.32, 1.275)
forwards; }

body {
  display: flex;
  align-items: center;
  justify-content: center;
  min-height: 100vh;
  margin: 0;
  padding: 20px;
  font-family: 'Inter', sans-serif;
  background-color: #f3faf6;
  overflow-x: hidden;
}

.login-card {
  position: relative;
  background: #FFFFFF;
  border: 1px solid var(--emerald-100);
  border-radius: 2.5rem;
  padding: 40px;
  width: 100%;
  max-width: 420px;
  text-align: center;
  box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.05);
  animation: slideIn 0.6s cubic-bezier(0.16, 1, 0.3, 1);
  z-index: 10;
}

.login-logo-container {
  width: 100px;
  height: 100px;
  margin: 0 auto 24px;
  background: white;
  border-radius: 2rem;
  padding: 12px;
  display: flex;
  align-items: center;
  justify-content: center;
  overflow: hidden;
  box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.05);
  border: 1px solid var(--emerald-100);
}

.login-logo-img {
  width: 100%;
  height: 100%;
  object-fit: contain;
}

.login-card h2 {
  font-size: 24px;
  font-weight: 800;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
margin-bottom: 4px;
color: var(--emerald-950);
letter-spacing: -0.5px;
}

.login-card p {
color: var(--emerald-800);
opacity: 0.7;
font-size: 14px;
margin-bottom: 32px;
}

/* Tab Styling */
.login-tabs {
display: flex;
background: var(--emerald-50);
border-radius: 1.25rem;
padding: 5px;
margin-bottom: 24px;
border: 1px solid var(--emerald-100);
}

.login-tab {
flex: 1;
padding: 10px 16px;
background: transparent;
border: none;
color: var(--emerald-700);
opacity: 0.6;
font-size: 13px;
font-weight: 700;
cursor: pointer;
border-radius: 0.9rem;
transition: all 0.3s;
display: flex;
align-items: center;
justify-content: center;
gap: 8px;
}

.login-tab:hover {
opacity: 1;
color: var(--emerald-900);
}

.login-tab.active {
background: #FFFFFF;
color: var(--emerald-900);
opacity: 1;
box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05);
border: 1px solid var(--emerald-50);
}

/* Scanner UI */
.scanner-card {
background: #e4f6ea;
border: 1px solid var(--emerald-100);
border-radius: 2rem;
padding: 32px;
margin-bottom: 24px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
position: relative;
overflow: hidden;
transition: all 0.5s;
}

.scanner-box {
width: 140px;
height: 140px;
background: white;
border-radius: 2.2rem;
margin: 0 auto;
display: flex;
align-items: center;
justify-content: center;
position: relative;
overflow: hidden;
border: 1px solid var(--emerald-100);
box-shadow: 0 10px 40px -10px rgba(0, 0, 0, 0.08);
transition: all 0.7s;
}

.scanner-box i {
display: flex;
align-items: center;
justify-content: center;
width: 100%;
height: 100%;
}

.scanner-box.scanning { border-color: var(--emerald-400); }
.scanner-box.success {
border: 3px solid var(--emerald-400);
box-shadow: 0 0 40px rgba(16, 185, 129, 0.3);
}

.scanner-grid {
position: absolute;
inset: 0;
opacity: 0.03;
background-image: radial-gradient(#000 1px, transparent 1px);
background-size: 10px 10px;
}

.scanner-laser-line {
position: absolute;
left: 0;
right: 0;
height: 3px;
background: var(--emerald-400);
box-shadow: 0 0 20px 4px rgba(52, 211, 153, 0.6);
z-index: 20;
}

.scanner-laser-trail {
position: absolute;
left: 0;
right: 0;
height: 50px;
background: linear-gradient(to bottom, transparent, rgba(52, 211, 153, 0.1),
transparent);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    z-index: 15;
}

.fingerprint-icon {
    font-size: 64px;
    color: var(--emerald-800);
    transition: all 0.5s;
    z-index: 10;
}

.fingerprint-overlay {
    position: absolute;
    font-size: 64px;
    color: var(--emerald-500);
    filter: drop-shadow(0 0 10px rgba(52, 211, 153, 0.8));
    z-index: 11;
}

.pulse-ring {
    position: absolute;
    width: 80px;
    height: 80px;
    background: rgba(16, 185, 129, 0.15);
    border-radius: 50%;
}

#pulse-rings {
    position: absolute;
    inset: 0;
    display: flex;
    align-items: center;
    justify-content: center;
    z-index: 5;
}

/* Buttons */
.btn-premium {
    width: 100%;
    padding: 16px;
    border-radius: 1rem;
    border: none;
    font-weight: 700;
    font-size: 15px;
    cursor: pointer;
    transition: all 0.3s;
    position: relative;
    overflow: hidden;
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 10px;
}

.btn-emerald {
    background: var(--emerald-800);
    color: white;
    box-shadow: 0 10px 15px -3px rgba(6, 78, 59, 0.2);
}

.btn-emerald:hover:not(:disabled) {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
background: var(--emerald-900);
transform: translateY(-2px);
}

.btn-emerald:disabled {
  background: var(--emerald-100);
  color: var(--emerald-400);
  cursor: not-allowed;
}

.btn-shimmer::after {
  content: '';
  position: absolute;
  top: 0; left: -100%; width: 50%; height: 100%;
  background: linear-gradient(90deg, transparent, rgba(255,255,255,0.1), transparent);
  animation: shimmer 3s infinite;
}

.status-text-container {
  margin-top: 24px;
  height: 60px;
}

.status-title {
  font-size: 18px;
  font-weight: 700;
  color: var(--emerald-950);
  margin: 0;
}

.status-desc {
  font-size: 13px;
  color: var(--emerald-800);
  opacity: 0.7;
  margin-top: 4px;
  line-height: 1.4;
}

/* Form Groups */
.form-group {
  text-align: left;
  margin-bottom: 20px;
}

.form-group label {
  display: block;
  font-size: 12px;
  font-weight: 700;
  text-transform: uppercase;
  color: var(--emerald-800);
  opacity: 0.6;
  margin-bottom: 8px;
  letter-spacing: 0.05em;
}

.form-control {
  width: 100%;
  padding: 14px 16px;
  border-radius: 12px;
  border: 1px solid var(--emerald-100);
}
```


IoT-Enabled Barangay Resident and Attendance Management System

```
background: var(--emerald-50);
transition: all 0.3s;
font-size: 14px;
color: var(--emerald-950);
}

.form-control:focus {
  outline: none;
  border-color: var(--emerald-500);
  background: #FFFFFF;
  box-shadow: 0 0 0 4px var(--emerald-100);
}

.messages-container {
  position: fixed;
  top: 20px;
  right: 20px;
  z-index: 1000;
  display: flex;
  flex-direction: column;
  gap: 10px;
}

.alert {
  background: white;
  border-radius: 12px;
  padding: 16px;
  box-shadow: 0 10px 15px -3px rgba(0, 0, 0, 0.1);
  display: flex;
  align-items: center;
  justify-content: space-between;
  min-width: 300px;
  border-left: 4px solid var(--emerald-500);
}

.alert-error { border-left-color: #ef4444; }
.alert-success { border-left-color: #10b981; }

.alert i { margin-right: 12px; }

.d-none { display: none !important; }

/* Fix scanner icon sizes and alignment */
.fingerprint-icon, .fingerprint-overlay i, #success-icon {
  font-size: 70px !important;
  display: flex !important;
  align-items: center;
  justify-content: center;
}

.fingerprint-overlay {
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  color: var(--emerald-500);
  filter: drop-shadow(0 0 12px rgba(52, 211, 153, 0.8));
  z-index: 11;
  display: flex !important;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        align-items: center;
        justify-content: center;
    }
</style>
</head>

<body>
    <div class="auth-background">
        <div class="auth-blob auth-blob-1"></div>
        <div class="auth-blob auth-blob-2"></div>
        <div class="auth-blob auth-blob-3"></div>
    </div>

    <div class="login-card">
        <div class="login-logo-container">
            
        </div>
        <h2>SIC0-SIC0</h2>
        <p>Barangay Management System</p>

        <div class="login-tabs" id="user-type-tabs">
            <button type="button" class="login-tab active" data-type="official">
                <i class="fas fa-user-tie"></i>
                <span>Official</span>
            </button>
            <button type="button" class="login-tab" data-type="resident">
                <i class="fas fa-users"></i>
                <span>Resident</span>
            </button>
        </div>

        <!-- Hidden role for resident mode -->
        <div id="credentials-content">
            <form method="POST">
                {% csrf_token %}
                <input type="hidden" name="user_type" id="user-type-hidden" value="official">

                <div id="official-fields">
                    <div class="form-group">
                        <label>Official Username</label>
                        <input type="text" name="official_username" id="official-username-input"
class="form-control"
                        placeholder="Enter official username" autofocus>
                    </div>
                </div>

                <div id="resident-fields" style="display: none;">
                    <div class="form-group">
                        <label>Resident Username</label>
                        <input type="text" name="username" id="resident-username-input" class="form-
control"
                        placeholder="Enter your username">
                    </div>
                </div>

                <div id="login-method-section" class="form-group">
                    <label>Login Method</label>
                    <div id="login-method-buttons" class="login-tabs" style="margin-bottom: 0;">
                        <button type="button" id="btn-method-password" class="login-tab active" data-
```

IoT-Enabled Barangay Resident and Attendance Management System

```

method="password">
    <i class="fas fa-key"></i>
    <span>Password</span>
</button>
<button type="button" id="btn-method-biometric" class="login-tab" data-
method="biometric">
    <i class="fas fa-fingerprint"></i>
    <span>Fingerprint</span>
</button>
</div>
<input type="hidden" id="login-method" name="login_method" value="password">
</div>

<div id="password-credentials">
<div class="form-group">
    <label>Password</label>
    <input id="password-input" type="password" name="password" class="form-
control" placeholder="Enter password"
        required>
</div>
<button type="submit" id="btn-password-login" class="btn btn-primary login-btn">
    <i class="fas fa-sign-in-alt"></i>
    Log In
</button>
</div>

<div id="biometric-credentials" style="display: none;">
<div id="scanner-card" class="scanner-card">
    <!-- Decorative background glow -->
    <div id="scanner-bg-glow" class="absolute inset-0 bg-emerald-400/10 opacity-0
blur-3xl transition-all duration-1000" style="position: absolute; top:0; left:0; right:0;
bottom:0; z-index:0;"></div>

    <div class="relative z-10">
        <div id="scanner-box" class="scanner-box">
            <div class="scanner-grid"></div>

            <!-- Idle Pulse Rings -->
            <div id="pulse-rings">
                <div class="pulse-ring animate-pulse-ring"></div>
                <div class="pulse-ring animate-pulse-ring-delayed"></div>
            </div>

            <!-- Fingerprint Icons -->
            <div id="fingerprint-wrapper" class="absolute inset-0 flex items-center
justify-center" style="position: absolute; top:0; left:0; right:0; bottom:0; display:flex;
align-items:center; justify-content:center; z-index: 10;">
                <i id="finger-base" class="fas fa-fingerprint fingerprint-icon"></i>
                <div id="finger-fill" class="fingerprint-overlay d-none">
                    <i class="fas fa-fingerprint"></i>
                </div>
            </div>

            <!-- Success Icon -->
            <div id="success-wrapper" class="absolute inset-0 d-none flex items-center
justify-center" style="position: absolute; top:0; left:0; right:0; bottom:0;
display:flex; align-items:center; justify-content:center; z-index: 30;">
                <i id="success-icon" class="fas fa-check-circle animate-pop-in"
style="color: var(--emerald-500); filter: drop-shadow(0 0 12px rgba(16, 185, 129,
0.6));"></i>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>

    <!-- Laser Line -->
    <div id="laser-container" class="d-none" style="position: absolute; inset:
0; z-index: 20;">
        <div class="scanner-laser-line animate-scan-laser"></div>
        <div class="scanner-laser-trail animate-scan-laser" style="margin-top:
-25px;"></div>
    </div>
</div>

<div class="status-text-container">
    <h3 id="bio-status-title" class="status-title">Ready to Scan</h3>
    <p id="bio-status-desc" class="status-desc">Place your registered finger
firmly on the connected sensor.</p>
</div>
</div>
</div>

<div class="flex gap-3 mt-6">
    <button type="button" id="btn-biometric-login" class="btn-premium btn-emerald
btn-shimmer flex-[2]">
        <span id="btn-bio-text">Start Biometric Scan</span>
    </button>
    <button type="button" id="btn-biometric-stop" class="btn-premium flex-1
d-none" style="background: #fee2e2; color: #ef4444; border: 1px solid #fecaca;">
        <span>Stop</span>
    </button>
</div>
</div>
</form>

<div
    style="margin-top: 32px; text-align: center; display: flex; flex-direction:
column; gap: 12px; padding-top: 24px; border-top: 1px solid var(--border);">
    <a href="{% url 'resident_signup' %}"
        style="color: var(--primary); font-weight: 700; text-decoration: none; font-
size: 14px; transition: var(--transition);"
        onmouseover="this.style.color='var(--primary-hover)'"
onmouseout="this.style.color='var(--primary)'">
        Are you a Resident? <span style="text-decoration: underline;">Register
here</span>
    </a>
    <a href="{% url 'signup' %}"
        style="color: var(--text-muted); font-size: 13px; text-decoration: none;
transition: var(--transition);"
        onmouseover="this.style.color='var(--text-main)'"
onmouseout="this.style.color='var(--text-muted)'">
        Official Signup
    </a>
</div>
</div>
</div>

<div style="position: fixed; bottom: 20px; color: var(--text-muted); font-size: 12px;
opacity: 0.7;">
    &copy; 2026 SICO-SICO Barangay Management System
</div>

{% if messages %}

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="messages-container">
  {% for message in messages %}
  <div class="alert alert-{{ message.tags }}" id="alert-{{ forloop.counter }}">
    <div class="alert-content">
      {% if message.tags == 'success' %}<i class="fas fa-check-circle"></i>
      {% elif message.tags == 'error' %}<i class="fas fa-exclamation-circle"></i>
      {% elif message.tags == 'warning' %}<i class="fas fa-exclamation-triangle"></i>
      {% else %}<i class="fas fa-info-circle"></i>{% endif %}
      <span>{{ message }}</span>
    </div>
    <button class="alert-close" onclick="this.parentElement.remove()"><i class="fas fa-
times"></i></button>
  </div>
  {% endfor %}
</div>
{% endif %}
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script src="{% static 'js/app.js' %}"></script>
<script>
document.addEventListener('DOMContentLoaded', function () {
  const userTypeTabs = document.querySelectorAll('#user-type-tabs .login-tab');
  const userTypeHidden = document.getElementById('user-type-hidden');
  const officialFields = document.getElementById('official-fields');
  const residentFields = document.getElementById('resident-fields');
  const officialRoleSelect = document.getElementById('official-role-select');
  const officialUsernameInput = document.getElementById('official-username-input');
  const residentUsernameInput = document.getElementById('resident-username-input');
  const loginMethodSection = document.getElementById('login-method-section');

  userTypeTabs.forEach(tab => {
    tab.addEventListener('click', function () {
      const type = this.dataset.type;
      userTypeHidden.value = type;
      userTypeTabs.forEach(t => t.classList.remove('active'));
      this.classList.add('active');
      if (type === 'resident') {
        officialFields.style.display = 'none';
        residentFields.style.display = 'block';
        loginMethodSection.style.display = 'none';
        setLoginMethod('password'); // Residents only use password
        officialRoleSelect.required = false;
        residentUsernameInput.required = true;
      } else {
        officialFields.style.display = 'block';
        residentFields.style.display = 'none';
        loginMethodSection.style.display = 'block';
        officialUsernameInput.required = true;
        residentUsernameInput.required = false;
      }
    });
  });
  const loginMethodHidden = document.getElementById('login-method');
  const methodPasswordBtn = document.getElementById('btn-method-password');
  const methodBiometricBtn = document.getElementById('btn-method-biometric');
  const passwordCredentials = document.getElementById('password-credentials');
  const biometricCredentials = document.getElementById('biometric-credentials');
  const passwordInput = document.getElementById('password-input');
  const btnBio = document.getElementById('btn-biometric-login');
  const btnBioText = document.getElementById('btn-bio-text');
  const bioStatusTitle = document.getElementById('bio-status-title');
```

IoT-Enabled Barangay Resident and Attendance Management System

```
const bioStatusDesc = document.getElementById('bio-status-desc');
const scannerBox = document.getElementById('scanner-box');
const scannerBgGlow = document.getElementById('scanner-bg-glow');
const pulseRings = document.getElementById('pulse-rings');
const fingerBase = document.getElementById('finger-base');
const fingerFill = document.getElementById('finger-fill');
const successIcon = document.getElementById('success-icon');
const laserContainer = document.getElementById('laser-container');

const btnStop = document.getElementById('btn-biometric-stop');

let biometricPollInterval = null;
let biometricTimeout = null;
let lastBiometricHintToastAt = 0;

function updateBioUI(state, message = '') {
  // States: idle, scanning, success, error

  // Reset everything first
  pulseRings.classList.add('d-none');
  laserContainer.classList.add('d-none');
  fingerFill.classList.add('d-none');
  document.getElementById('success-wrapper').classList.add('d-none');
  document.getElementById('fingerprint-wrapper').classList.remove('d-none');
  fingerBase.classList.remove('d-none');
  scannerBox.classList.remove('scanning', 'success');
  scannerBgGlow.classList.add('opacity-0');
  scannerBgGlow.classList.remove('animate-pulse');
  btnBio.disabled = true;
  if (btnStop) btnStop.classList.add('d-none');

  if (state === 'idle') {
    pulseRings.classList.remove('d-none');
    bioStatusTitle.innerText = "Ready to Scan";
    bioStatusDesc.innerText = "Place your registered finger firmly on the sensor.";
    btnBioText.innerText = "Start Biometric Scan";
    btnBio.disabled = false;
    fingerBase.style.color = "var(--emerald-800)";
  }
  else if (state === 'scanning') {
    scannerBox.classList.add('scanning');
    scannerBgGlow.classList.remove('opacity-0');
    scannerBgGlow.classList.add('animate-pulse');
    laserContainer.classList.remove('d-none');
    fingerFill.classList.remove('d-none');
    fingerFill.classList.add('animate-clip-fill');
    fingerBase.style.color = "var(--emerald-100)";
    bioStatusTitle.innerText = "Analyzing...";
    bioStatusDesc.innerText = message || "Keep your finger perfectly still on the sensor.";
    btnBioText.innerHTML = '<div class="w-4 h-4 border-2 border-white border-t-transparent rounded-full animate-spin mr-2" style="display:inline-block"></div> Analyzing Biometrics...';
    if (btnStop) btnStop.classList.remove('d-none');
  }
  else if (state === 'success') {
    scannerBox.classList.add('success');
    document.getElementById('success-wrapper').classList.remove('d-none');
    document.getElementById('fingerprint-wrapper').classList.add('d-none');
    bioStatusTitle.innerText = "Verified!";
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
bioStatusDesc.innerText = "Biometric authentication was successful.";
btnBioText.innerText = "Access Granted";
btnBio.classList.remove('btn-emerald');
btnBio.style.backgroundColor = "var(--emerald-500)";
btnBio.style.color = "white";

if (window.showNotification) {
  window.showNotification("Access Granted. Redirecting...", "success");
}
}
else if (state === 'error') {
  pulseRings.classList.remove('d-none');
  bioStatusTitle.innerText = "Scan Failed";
  bioStatusDesc.innerText = message || "Finger not recognized. Please try again.";
  btnBioText.innerText = "Retry Scan";
  btnBio.disabled = false;
  fingerBase.style.color = "#ef4444";

  if (window.showNotification) {
    window.showNotification(message || "Fingerprint not recognized.", "error");
  }
}
}

function stopESP32Scanning() {
  const esp32BaseUrl = 'http://esp32-fingerprint.local';

  // Create abort controller with 1 second timeout
  const controller = new AbortController();
  const timeoutId = setTimeout(() => controller.abort(), 1000);

  try {
    fetch(`${esp32BaseUrl}/stop-enrollment`, {
      method: 'POST',
      signal: controller.signal,
    })
    .then(() => clearTimeout(timeoutId))
    .catch(err => {
      clearTimeout(timeoutId);
      if (err.name !== 'AbortError') {
        console.log('[Biometric] Stop signal sent');
      }
    });
  } catch (e) {
    clearTimeout(timeoutId);
  }
}

function resetBiometricUI(message) {
  // Stop ESP32 immediately when resetting
  stopESP32Scanning();
  lastBiometricHintToastAt = 0;

  if (biometricPollInterval) {
    clearInterval(biometricPollInterval);
    biometricPollInterval = null;
  }
  if (biometricTimeout) {
    clearTimeout(biometricTimeout);
    biometricTimeout = null;
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

updateBioUI(message ? 'error' : 'idle', message);
}

function applyLoginMethod() {
  const method = loginMethodHidden ? loginMethodHidden.value : 'password';
  const type = userTypeHidden.value;

  if (method === 'biometric') {
    if (passwordCredentials) passwordCredentials.style.display = 'none';
    if (biometricCredentials) biometricCredentials.style.display = 'block';
    if (passwordInput) passwordInput.required = false;

    // Hide username fields since fingerprint is auto-identifying
    if (officialFields) officialFields.style.display = 'none';
    if (residentFields) residentFields.style.display = 'none';
    if (officialUsernameInput) officialUsernameInput.required = false;
    if (residentUsernameInput) residentUsernameInput.required = false;
  } else {
    if (passwordCredentials) passwordCredentials.style.display = 'block';
    if (biometricCredentials) biometricCredentials.style.display = 'none';
    if (passwordInput) passwordInput.required = true;

    // Show appropriate username field based on tab
    if (type === 'official') {
      if (officialFields) officialFields.style.display = 'block';
      if (residentFields) residentFields.style.display = 'none';
      if (officialUsernameInput) officialUsernameInput.required = true;
      if (residentUsernameInput) residentUsernameInput.required = false;
    } else {
      if (officialFields) officialFields.style.display = 'none';
      if (residentFields) residentFields.style.display = 'block';
      if (officialUsernameInput) officialUsernameInput.required = false;
      if (residentUsernameInput) residentUsernameInput.required = true;
    }
  }
  resetBiometricUI();
}

function setLoginMethod(method) {
  if (loginMethodHidden) loginMethodHidden.value = method;
  if (methodPasswordBtn && methodBiometricBtn) {
    methodPasswordBtn.classList.toggle('active', method === 'password');
    methodBiometricBtn.classList.toggle('active', method === 'biometric');
  }

  applyLoginMethod();
}

if (methodPasswordBtn) {
  methodPasswordBtn.addEventListener('click', () => setLoginMethod('password'));
}
if (methodBiometricBtn) {
  methodBiometricBtn.addEventListener('click', () => setLoginMethod('biometric'));
}

// Initialize default
setLoginMethod('password');
```


IoT-Enabled Barangay Resident and Attendance Management System

```
if (btnStop) {
  btnStop.addEventListener('click', function() {
    resetBiometricUI('Scan cancelled by user.');
```

 });
}

if (btnBio) {
 btnBio.addEventListener('click', function () {
 updateBioUI('scanning');
 lastBiometricHintToastAt = 0;

 fetch("/biometric-login-start/", {
 method: 'POST',
 headers: { 'Content-Type': 'application/json' },
 credentials: 'include',
 body: JSON.stringify({
 role: 'auto',
 username: null
 }),
 })
 .then(async response => {
 const data = await response.json();
 if (response.ok) {
 checkBiometricStatus();

 biometricTimeout = setTimeout(() => {
 resetBiometricUI('Scanner session expired. Try again.');
 }, 30000);
 } else {
 throw new Error(data.message || 'Failed to start biometric service');
 }
 })
 .catch(error => {
 resetBiometricUI(error.message);
 });
});
}

function checkBiometricStatus() {
 biometricPollInterval = setInterval(() => {
 fetch("/biometric-status-check/", {
 credentials: 'include'
 })
 .then(response => response.json())
 .then(data => {
 if (data.status === 'authenticated') {
 updateBioUI('success');
 clearInterval(biometricPollInterval);
 if (biometricTimeout) clearTimeout(biometricTimeout);
 setTimeout(() => {
 // Redirect to the server-side complete endpoint.
 // This performs the real Django login() in a full HTTP request,
 // ensuring the session cookie is saved before we reach /dashboard/.
 window.location.href = "/biometric-login-complete/";
 }, 1200);
 } else if (data.status === 'failed') {
 resetBiometricUI(data.reason || 'Finger not recognized. Try again.');
 }
 else if (data.hint) {
 bioStatusTitle.innerText = "Not recognized";
 bioStatusDesc.innerText = data.hint;
 }
 });
}

IoT-Enabled Barangay Resident and Attendance Management System

```
fingerBase.style.color = "#f59e0b";
const now = Date.now();
if (window.showNotification && (now - lastBiometricHintToastAt > 3500)) {
  lastBiometricHintToastAt = now;
  window.showNotification(data.hint, "error");
}
}
})
.catch(err => console.error('[Biometric] Poll error:', err));
}, 1000);
}
});
</script>
</body>

{% include 'core/hardware_poller.html' %}

</html>
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/core/notifications.html

```
{% extends 'base.html' %}
{% block title %}My Notifications{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Notifications</span>
{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Notifications</h1>
    <span>Updates from the barangay office</span>
  </div>
</div>

<div class="card" style="max-width:700px;">
  {% if notifications %}
  <ul style="list-style:none;padding:0;margin:0;">
    {% for notif in notifications %}
    <li
      style="padding:16px 20px;border-bottom:1px solid
var(--border,#e2e8f0);display:flex;align-items:flex-start;gap:14px;{% if not notif.is_read
%}background:#f0fdf4;{% endif %}">
      <div
        style="width:36px;height:36px;border-radius:50%;background:{% if not
notif.is_read %}#dcfce7{% else %}#f1f5f9{% endif %};display:flex;align-
items:center;justify-content:center;flex-shrink:0;">
        <i class="fas fa-bell"
          style="color:{% if not notif.is_read %}#16a34a{% else %}#94a3b8{%
endif %};font-size:.9rem;"></i>
        </div>
        <div style="flex:1;">
          <div style="font-size:.92rem;{% if not notif.is_read %}font-weight:600;{%
endif %}">{{ notif.message }}
          </div>
          <div style="font-size:.78rem;color:var(--text-muted,#64748b);margin-
top:4px;">{{ notif.created_at|date:"M d, Y g:i A" }}</div>
          </div>
          {% if notif.link %}
          <a href="{{ notif.link }}" class="btn btn-outline btn-sm" style="flex-
shrink:0;">View</a>
          {% endif %}
        </li>
      {% endfor %}
    </ul>
  {% else %}
  <div class="text-center" style="padding:48px 24px;color:var(--text-muted,#64748b);">
    <i class="fas fa-bell-slash" style="font-size:2.5rem;display:block;margin-
bottom:14px;opacity:.3;"></i>
    <div style="font-size:1rem;font-weight:600;">No notifications yet</div>
    <div style="font-size:.85rem;margin-top:4px;">You'll be notified here when your
certificate requests are
processed.</div>
  </div>
  {% endif %}
</div>
{% endblock %}
```

File: templates/core/registration_success.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Registration Submitted - Barangay Management System</title>
  <link rel="stylesheet" href="{% static 'css/style.css' %}">
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700;800&display=swap"
rel="stylesheet">
  <link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css"
rel="stylesheet">
  <style>
    body {
      background-color: var(--bg-main);
      color: var(--text-main);
      font-family: 'Inter', sans-serif;
      margin: 0;
      padding: 40px 20px;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
      align-items: center;
      justify-content: center;
    }

    .success-card {
      background: var(--bg-card);
      border-radius: 24px;
      padding: 64px 48px;
      width: 100%;
      max-width: 600px;
      text-align: center;
      box-shadow: var(--shadow-md);
      border: 1px solid var(--border);
    }

    .success-icon {
      width: 80px;
      height: 80px;
      background: var(--primary);
      border-radius: 50%;
      display: flex;
      align-items: center;
      justify-content: center;
      margin: 0 auto 32px;
      font-size: 40px;
      color: white;
      box-shadow: 0 0 30px rgba(44, 110, 73, 0.2);
    }

    h1 {
      font-size: 32px;
      font-weight: 800;
      margin-bottom: 16px;
    }
  </style>
</head>

<body>
  <div class="success">
    <div class="success-card">
      <div class="success-icon">
        <i class="fa-solid fa-check"></i>
      </div>
      <h1>Registration Submitted</h1>
      <p>Your registration has been successfully submitted. We will contact you shortly.</p>
    </div>
  </div>
</body>
</html>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    letter-spacing: -1px;
    color: var(--text-main);
}

p {
    font-size: 16px;
    color: var(--text-muted);
    line-height: 1.6;
    margin-bottom: 48px;
}

.reference-box {
    background: var(--bg-main);
    border: 1px dashed var(--primary);
    border-radius: 16px;
    padding: 24px;
    margin-bottom: 48px;
}

.reference-label {
    font-size: 12px;
    font-weight: 700;
    color: var(--primary);
    text-transform: uppercase;
    letter-spacing: 1px;
    margin-bottom: 8px;
    display: block;
}

.reference-number {
    font-size: 28px;
    font-family: monospace;
    font-weight: 700;
    color: var(--text-main);
    letter-spacing: 2px;
}

.instruction-list {
    text-align: left;
    background: var(--bg-main);
    padding: 24px;
    border-radius: 16px;
    margin-bottom: 48px;
    border: 1px solid var(--border);
}

.instruction-item {
    display: flex;
    gap: 12px;
    margin-bottom: 16px;
    font-size: 14px;
    color: var(--text-muted);
}

.instruction-item:last-child {
    margin-bottom: 0;
}

.instruction-item i {
    color: var(--primary);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        margin-top: 3px;
    }

    .btn-home {
        background: var(--primary);
        color: white;
        padding: 16px 32px;
        border-radius: 12px;
        font-weight: 700;
        text-decoration: none;
        display: inline-flex;
        align-items: center;
        gap: 12px;
        transition: all 0.3s ease;
    }

    .btn-home:hover {
        background: var(--primary-hover);
        transform: translateY(-2px);
        box-shadow: 0 8px 20px rgba(44, 110, 73, 0.2);
    }
</style>
</head>

<body>
    <div class="success-card">
        <div class="success-icon">
            <i class="fas fa-check"></i>
        </div>
        <h1>Registration Submitted!</h1>
        <p>Thank you, <strong>{{ registration.first_name }}</strong>. Your registration
has been received and is now
            pending verification by our barangay staff.</p>

        <div class="reference-box">
            <span class="reference-label">Your Tracking Reference</span>
            <div class="reference-number">{{ registration.reference_number }}</div>
        </div>

        <div class="instruction-list">
            <div class="instruction-item">
                <i class="fas fa-info-circle"></i>
                <span>Please save your reference number for tracking purposes.</span>
            </div>
            <div class="instruction-item">
                <i class="fas fa-clock"></i>
                <span>Verification typically takes 1-2 working days.</span>
            </div>
            <div class="instruction-item">
                <i class="fas fa-envelope"></i>
                <span>You will receive an update once your account is verified.</span>
            </div>
        </div>

        <a href="{% url 'login' %}" class="btn-home">
            Return to Login <i class="fas fa-arrow-right"></i>
        </a>
    </div>
</body>
```

IoT-Enabled Barangay Resident and Attendance Management System

</html>

File: templates/core/resident_signup.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Resident Registration - Barangay Management System</title>
  <link rel="stylesheet" href="{% static 'css/style.css' %}?v=2.1">
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&di
splay=swap" rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css"
rel="stylesheet">
  <style>
    body {
      background-color: var(--bg-main);
      color: var(--text-main);
      font-family: 'Inter', sans-serif;
      margin: 0;
      padding: 40px 20px;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
      align-items: center;
      overflow-x: hidden;
    }

    .portal-header {
      width: 100%;
      max-width: 1200px;
      display: flex;
      align-items: center;
      gap: 16px;
      margin-bottom: 48px;
    }

    .portal-logo {
      width: 56px;
      height: 56px;
      background: #FFFFFF;
      border-radius: 16px;
      display: flex;
      align-items: center;
      justify-content: center;
      padding: 4px;
      border: 1px solid var(--border);
      box-shadow: var(--shadow-sm);
    }

    .portal-logo img {
      width: 100%;
      height: 100%;
      object-fit: contain;
    }

    .portal-title h1 {
      font-size: 20px;
```


IoT-Enabled Barangay Resident and Attendance Management System

```
    font-weight: 800;
    margin: 0;
    letter-spacing: -0.5px;
    color: var(--text-main);
}

.portal-title p {
    font-size: 14px;
    color: var(--text-muted);
    margin: 0;
}

/* Stepper Styles */
.stepper {
    width: 100%;
    max-width: 800px;
    display: flex;
    justify-content: space-between;
    position: relative;
    margin-bottom: 64px;
}

.stepper::before {
    content: '';
    position: absolute;
    top: 24px;
    left: 0;
    right: 0;
    height: 2px;
    background: var(--border);
    z-index: 1;
}

.step-progress-bar {
    position: absolute;
    top: 24px;
    left: 0;
    height: 2px;
    background: var(--primary);
    z-index: 2;
    transition: width 0.4s ease;
    width: 0%;
}

.step-node {
    position: relative;
    z-index: 3;
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 12px;
    width: 120px;
}

.step-circle {
    width: 48px;
    height: 48px;
    background: var(--bg-card);
    border: 2px solid var(--border);
    border-radius: 50%;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: 700;
    color: var(--text-muted);
    transition: all 0.3s ease;
}

.step-node.active .step-circle {
    background: var(--primary);
    border-color: var(--primary);
    color: white;
    box-shadow: 0 0 20px rgba(44, 110, 73, 0.2);
}

.step-node.completed .step-circle {
    background: var(--primary);
    border-color: var(--primary);
    color: white;
}

.step-label {
    font-size: 12px;
    font-weight: 600;
    color: var(--text-muted);
    text-align: center;
    transition: color 0.3s ease;
}

.step-node.active .step-label {
    color: var(--primary);
}

/* Form Card */
.registration-card {
    background: var(--glass-bg);
    backdrop-filter: blur(16px);
    -webkit-backdrop-filter: blur(16px);
    border: 1px solid var(--glass-border);
    border-radius: var(--radius-xl);
    padding: 48px;
    width: 100%;
    max-width: 900px;
    box-shadow: var(--shadow-lg);
    animation: slideIn 0.6s cubic-bezier(0.16, 1, 0.3, 1);
    position: relative;
    z-index: 10;
}

.step-header {
    margin-bottom: 32px;
}

.step-header h2 {
    font-size: 24px;
    font-weight: 800;
    margin: 0 0 8px 0;
    color: var(--text-main);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.step-header p {
  font-size: 14px;
  color: var(--text-muted);
  margin: 0;
}

.form-section {
  display: none;
}

.form-section.active {
  display: block;
  animation: fadeIn 0.4s ease;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(10px);
  }

  to {
    opacity: 1;
    transform: translateY(0);
  }
}

.form-grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 24px;
  margin-bottom: 32px;
}

.col-2 {
  grid-column: span 2;
}

.col-full {
  grid-column: span 3;
}

.form-group label {
  display: block;
  font-size: 13px;
  font-weight: 600;
  color: var(--text-muted);
  margin-bottom: 8px;
}

.form-control {
  width: 100%;
  background: rgba(255, 255, 255, 0.6);
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 14px 16px;
  color: var(--text-main);
  font-size: 14px;
  box-sizing: border-box;
  transition: all 0.3s ease;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

.form-control:focus {
  outline: none;
  border-color: var(--primary);
  background: #FFFFFF;
  box-shadow: 0 0 0 4px var(--primary-light);
  transform: translateY(-1px);
}

.form-control::placeholder {
  color: var(--text-muted);
  opacity: 0.6;
}

select.form-control {
  appearance: none;
  background-image: url("data:image/svg+xml,%3Csvg
xmlns='http://www.w3.org/2000/svg' width='12' height='12' fill='%236B7280' viewBox='0 0 16
16'%3E%3Cpath d='M7.247 11.14 2.451 5.658C1.885 5.013 2.345 4 3.204 4h9.592a1 1 0 0 1
.753 1.659l-4.796 5.48a1 1 0 0 1-1.506 0z'/%3E%3C/svg%3E");
  background-repeat: no-repeat;
  background-position: right 16px center;
}

.radio-group {
  display: flex;
  gap: 24px;
  padding: 12px 0;
}

.radio-item {
  display: flex;
  align-items: center;
  gap: 8px;
  cursor: pointer;
  color: var(--text-main);
  font-size: 14px;
}

.radio-item input {
  cursor: pointer;
  accent-color: var(--primary);
  width: 18px;
  height: 18px;
}

.navigation-actions {
  display: flex;
  justify-content: flex-end;
  gap: 16px;
  margin-top: 40px;
  padding-top: 32px;
  border-top: 1px solid var(--border);
}

.btn-nav {
  padding: 14px 28px;
  border-radius: 12px;
  font-weight: 700;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    font-size: 15px;
    cursor: pointer;
    transition: all 0.3s ease;
    display: flex;
    align-items: center;
    gap: 10px;
    border: none;
}

.btn-prev {
    background: var(--bg-main);
    color: var(--text-main);
    border: 1px solid var(--border);
}

.btn-prev:hover {
    background: var(--border-light);
}

.btn-next {
    background: var(--primary);
    color: white;
}

.btn-next:hover {
    background: var(--primary-hover);
    transform: translateY(-1px);
}

.upload-grid {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 24px;
}

.upload-box {
    background: var(--bg-main);
    border: 2px dashed var(--border);
    border-radius: 16px;
    padding: 32px;
    text-align: center;
    cursor: pointer;
    transition: all 0.3s ease;
}

.upload-box:hover {
    border-color: var(--primary);
    background: var(--primary-light);
}

.upload-box i {
    font-size: 32px;
    color: var(--text-muted);
    margin-bottom: 16px;
}

.upload-box p {
    font-size: 13px;
    color: var(--text-muted);
    margin: 0;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

.minor-alert {
  background: var(--primary-light);
  border: 1px solid var(--primary);
  border-radius: 12px;
  padding: 16px;
  margin-bottom: 24px;
  display: none;
  align-items: center;
  gap: 12px;
  color: var(--primary);
}

@media (max-width: 768px) {
  .form-grid {
    grid-template-columns: 1fr;
  }

  .col-2,
  .col-full {
    grid-column: span 1;
  }

  .registration-card {
    padding: 32px 20px;
  }

  .stepper {
    overflow-x: auto;
    padding-bottom: 10px;
  }
}
</style>
<script src="/static/js/ph-locations.js"></script>
</head>

<body>
  <div class="auth-background">
    <div class="auth-blob auth-blob-1"></div>
    <div class="auth-blob auth-blob-2"></div>
    <div class="auth-blob auth-blob-3"></div>
  </div>
  <header class="portal-header">
    <div class="portal-logo">
      
    </div>
    <div class="portal-title">
      <h1>Barangay Management System</h1>
      <p>Resident Registration Portal</p>
    </div>
  </header>

  <div class="stepper">
    <div class="step-progress-bar" id="stepProgressBar"></div>
    <div class="step-node active" id="node-1">
      <div class="step-circle">1</div>
      <div class="step-label">Personal info</div>
    </div>
  </div>
</body>
</html>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="step-node" id="node-2">
  <div class="step-circle">2</div>
  <div class="step-label">Contact & address</div>
</div>
<div class="step-node" id="node-3">
  <div class="step-circle">3</div>
  <div class="step-label">Household</div>
</div>
<div class="step-node" id="node-4">
  <div class="step-circle">4</div>
  <div class="step-label">Account setup</div>
</div>
</div>

<form class="registration-card" id="signupForm" method="POST" enctype="multipart/form-
data">
  {% csrf_token %}

  <!-- Step 1: Personal Information -->
  <div class="form-section active" id="step-1">
    <div class="step-header">
      <h2>Personal information</h2>
      <p>Enter the resident's legal name and basic details as shown on official
ID.</p>
    </div>

    <div id="minorAlert" class="minor-alert">
      <i class="fas fa-child"></i>
      <span><strong>Minor Detected.</strong> Guardian information will be
required in Step 3.</span>
    </div>

    <div class="form-grid">
      <div class="form-group">
        <label>First name *</label>
        <input type="text" name="first_name" class="form-control"
placeholder="Juan" required>
      </div>
      <div class="form-group">
        <label>Middle name</label>
        <input type="text" name="middle_name" class="form-control"
placeholder="Santos">
      </div>
      <div class="form-group">
        <label>Last name *</label>
        <input type="text" name="last_name" class="form-control"
placeholder="Dela Cruz" required>
      </div>
      <div class="form-group">
        <label>Suffix</label>
        <select name="suffix" class="form-control">
          <option value="">None</option>
          <option value="Jr.">Jr.</option>
          <option value="Sr.">Sr.</option>
          <option value="I">I</option>
          <option value="II">II</option>
          <option value="III">III</option>
        </select>
      </div>
    </div>
  </div>
</form>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <div class="form-group">
            <label>Date of birth *</label>
            <input type="date" name="birthdate" class="form-control"
id="birthdateInput" required>
        </div>
        <div class="form-group">
            <label>Place of birth *</label>
            <input type="text" name="birthplace" class="form-control"
placeholder="Cagayan de Oro City"
            required>
        </div>

        <div class="form-group">
            <label>Sex *</label>
            <div class="radio-group">
                <label class="radio-item"><input type="radio" name="gender"
value="M" checked> Male</label>
                <label class="radio-item"><input type="radio" name="gender"
value="F"> Female</label>
            </div>
        </div>
        <div class="form-group col-2">
            <label>Civil status *</label>
            <select name="civil_status" class="form-control" required>
                <option value="">Select status</option>
                <option value="single">Single</option>
                <option value="married">Married</option>
                <option value="widowed">Widowed</option>
                <option value="separated">Separated</option>
            </select>
        </div>

        <div class="form-group">
            <label>Citizenship *</label>
            <input type="text" name="citizenship" class="form-control"
value="Filipino" required>
        </div>
        <div class="form-group col-2">
            <label>Religion</label>
            <input type="text" name="religion" class="form-control"
placeholder="e.g. Roman Catholic">
        </div>

        <div class="form-group">
            <label>Highest education</label>
            <select name="highest_education" class="form-control">
                <option value="">Select</option>
                <option value="elementary">Elementary</option>
                <option value="highschool">High School</option>
                <option value="vocational">Vocational</option>
                <option value="college">College Graduate</option>
                <option value="postgraduate">Post Graduate</option>
            </select>
        </div>
        <div class="form-group col-2">
            <label>Occupation</label>
            <input type="text" name="occupation" class="form-control"
placeholder="e.g. Teacher">
        </div>
    </div>

```


IoT-Enabled Barangay Resident and Attendance Management System

```
</div>

<!-- Step 2: Contact & Address -->
<div class="form-section" id="step-2">
  <div class="step-header">
    <h2>Contact & address</h2>
    <p>Provide contact details and complete residential information.</p>
  </div>

  <div class="form-grid">
    <div class="form-group">
      <label>Mobile number *</label>
      <input type="tel" name="mobile_number" class="form-control"
placeholder="09XX XXX XXXX" required>
    </div>
    <div class="form-group col-2">
      <label>Email address</label>
      <input type="email" name="email" class="form-control"
placeholder="juan@example.com">
    </div>

    <div class="form-group">
      <label>House / Unit #</label>
      <input type="text" name="house_number" class="form-control"
placeholder="Bldg 123">
    </div>
    <div class="form-group">
      <label>Street</label>
      <input type="text" name="street" class="form-control"
placeholder="Velez St.">
    </div>
    <div class="form-group">
      <label>Purok / Zone</label>
      <input type="text" name="purok" id="purokInput" class="form-control"
placeholder="Purok 1">
    </div>

    <div class="form-group">
      <label>City / Province</label>
      <input type="text" name="city" id="cityInput" list="cityInputList"
class="form-control"
        placeholder="Select City/Province" required>
      <datalist id="cityInputList">
        <option value="Metro Manila">
        <option value="Cebu">
        <option value="Davao Del Sur">
        <option value="Misamis Oriental">
        <option value="Surigao Del Norte">
        <option value="Agusan Del Norte">
        <option value="Lanao Del Norte">
      </datalist>
    </div>
    <div class="form-group">
      <label>Municipality</label>
      <input type="text" name="municipality" id="municipalityInput"
list="municipalityInputList"
        class="form-control" placeholder="Select Municipality" required>
      <datalist id="municipalityInputList"></datalist>
    </div>
    <div class="form-group">
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <label>Barangay</label>
        <input type="text" name="barangay" id="barangayInput"
list="barangayInputList" class="form-control"
        placeholder="Select Barangay" required>
        <datalist id="barangayInputList"></datalist>
    </div>
    <div class="form-group">
        <label>ZIP Code</label>
        <input type="text" name="zip_code" id="zipCodeInput" class="form-
control" placeholder="ZIP Code"
        required readonly>
    </div>

    <div class="form-group">
        <label>Years of residency *</label>
        <input type="number" name="years_of_residency" class="form-control"
min="0" required>
    </div>
    <div class="form-group col-2">
        <label>PhilSys Number (PSN) - Optional</label>
        <input type="text" name="philSys_number" class="form-control"
placeholder="XXXX-XXXX-XXXX-XXXX">
    </div>
</div>
</div>

<!-- Step 3: Household & Classification -->
<div class="form-section" id="step-3">
    <div class="step-header">
        <h2>Household & classification</h2>
        <p>Indicate household membership and special classifications.</p>
    </div>

    <div class="form-grid">
        <div class="form-group col-full">
            <label>Household membership *</label>
            <div class="radio-group">
                <label class="radio-item"><input type="radio"
name="household_action" value="join" checked
                onchange="toggleHouseholdField()"> Join
                existing household</label>
                <label class="radio-item"><input type="radio"
name="household_action" value="new"
                onchange="toggleHouseholdField()"> Household not
                registered</label>
            </div>
        </div>
        <div class="form-group col-full" id="householdNumberGroup">
            <label id="householdLabel">Head of household / Household
number</label>
            <input type="text" name="household_number" id="householdInput"
list="headsList" class="form-control"
            placeholder="Search Head of Household or Enter ID">
            <datalist id="headsList">
                {% for head in heads_of_households %}
                <option value="{{ head.household.household_no }}" data-purok="{{
head.purok }}">{{ head.full_name }} ({{ head.household.household_no }})</option>
                {% endfor %}
            </datalist>
        </div>
    </div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <div class="form-group col-full">
            <label>Special Classifications</label>
            <div class="radio-group" style="flex-wrap: wrap;">
                <label class="radio-item"><input type="checkbox" name="is_pwd">
PWD</label>
                <label class="radio-item"><input type="checkbox"
name="is_senior_citizen"> Senior
                citizen</label>
                <label class="radio-item"><input type="checkbox"
name="is_4ps_member"> 4Ps member</label>
                <label class="radio-item"><input type="checkbox"
name="is_sole_parent"> Sole parent</label>
                <label class="radio-item"><input type="checkbox"
name="is_registered_voter"> Voter</label>
            </div>
        </div>
    </div>

    <div id="guardianSection"
        style="display: none; border-top: 1px solid var(--border); padding-top:
32px; margin-top: 32px;">
        <div class="step-header">
            <h2 style="font-size: 18px; color: var(--primary);">Parent / Guardian
information</h2>
            <p>Required for residents under 18 years of age.</p>
        </div>
        <div class="form-grid">
            <div class="form-group col-2">
                <label>Guardian's full name *</label>
                <input type="text" name="guardian_name" class="form-control"
placeholder="Enter guardian name">
            </div>
            <div class="form-group">
                <label>Relationship *</label>
                <input type="text" name="guardian_relationship" class="form-
control" placeholder="e.g. Mother">
            </div>
            <div class="form-group">
                <label>Contact number *</label>
                <input type="tel" name="guardian_contact" class="form-control"
placeholder="09XX XXX XXXX">
            </div>
            <div class="form-group col-2">
                <label>Valid ID number *</label>
                <input type="text" name="guardian_id_number" class="form-control"
placeholder="Enter ID number">
            </div>
        </div>
    </div>
</div>

<!-- Step 4: Account Setup -->
<div class="form-section" id="step-4">
    <div class="step-header">
        <h2>Account setup</h2>
        <p>Create your login credentials and upload supporting documents.</p>
    </div>

    <div class="form-grid">

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="form-group col-full">
  <label>Username *</label>
  <input type="text" name="username" class="form-control"
placeholder="Choose a unique username"
  required>
</div>
<div class="form-group">
  <label>Password *</label>
  <input type="password" name="password1" class="form-control" required>
</div>
<div class="form-group col-2">
  <label>Confirm Password *</label>
  <input type="password" name="password2" class="form-control" required>
</div>
</div>

<div class="step-header" style="margin-top: 40px;">
  <h2 style="font-size: 18px;">Document uploads</h2>
  <p>Upload clear copies of the following documents (Max 5MB each).</p>
</div>

<div class="upload-grid">
  <div class="upload-box"
onclick="document.getElementById('photoInput').click()"
  <i class="fas fa-camera"></i>
  <p>Profile photo</p>
  <input type="file" id="photoInput" name="photo" style="display: none;"
accept="image/*">
  </div>
  <div class="upload-box"
onclick="document.getElementById('idInput').click()"
  <i class="fas fa-id-card"></i>
  <p>Valid Government ID</p>
  <input type="file" id="idInput" name="id_card" style="display: none;"
accept="image/*, application/pdf">
  </div>
  <div class="upload-box" id="birthCertUpload"
onclick="document.getElementById('birthCertInput').click()"
  <i class="fas fa-file-invoice"></i>
  <p>Birth Certificate</p>
  <input type="file" id="birthCertInput" name="birth_certificate"
style="display: none;"
accept="image/*, application/pdf">
  </div>
  <div class="upload-box"
onclick="document.getElementById('residencyInput').click()"
  <i class="fas fa-home"></i>
  <p>Proof of residency</p>
  <input type="file" id="residencyInput" name="proof_of_residency"
style="display: none;"
accept="image/*, application/pdf">
  </div>
</div>

<div class="form-group col-full"
style="margin-top: 40px; background: var(--bg-main); padding: 20px;
border-radius: 12px; border: 1px solid var(--border);">
  <label class="radio-item">
    <input type="checkbox" name="data_privacy_consent" required>
    <span style="font-size: 13px; line-height: 1.5; color: var(--text-
```

IoT-Enabled Barangay Resident and Attendance Management System

```

muted);">I hereby acknowledge that
            I have read and understood the Data Privacy Policy of the Barangay
Management System and give my
            consent to process my information.</span>
        </label>
    </div>
</div>

<div class="navigation-actions">
    <button type="button" class="btn-nav btn-prev" id="btnPrev" style="visibility:
hidden;">
        <i class="fas fa-arrow-left"></i> Previous
    </button>
    <button type="button" class="btn-nav btn-next" id="btnNext">
        Continue <i class="fas fa-arrow-right"></i>
    </button>
    <button type="submit" class="btn-nav btn-next" id="btnSubmit" style="display:
none;">
        Sign Up <i class="fas fa-check-circle"></i>
    </button>
</div>
</form>

<div style="margin-top: 32px; text-align: center;">
    <a href="{% url 'login' %}" style="color: var(--text-muted); font-size: 14px;
text-decoration: none;">
        Already have an account? <strong style="color: var(--primary);">Log
In</strong>
    </a>
</div>

<script>
    let currentStep = 1;
    const totalSteps = 4;
    const form = document.getElementById('signupForm');
    const sections = document.querySelectorAll('.form-section');
    const nodes = document.querySelectorAll('.step-node');
    const progressBar = document.getElementById('stepProgressBar');
    const btnPrev = document.getElementById('btnPrev');
    const btnNext = document.getElementById('btnNext');
    const btnSubmit = document.getElementById('btnSubmit');

    const birthdateInput = document.getElementById('birthdateInput');
    const minorAlert = document.getElementById('minorAlert');
    const guardianSection = document.getElementById('guardianSection');
    const birthCertUpload = document.getElementById('birthCertUpload');

    function updateStep() {
        sections.forEach((s, i) => s.classList.toggle('active', (i + 1) ===
currentStep));
        nodes.forEach((n, i) => {
            n.classList.toggle('active', (i + 1) === currentStep);
            n.classList.toggle('completed', (i + 1) < currentStep);
        });

        progressBar.style.width = ((currentStep - 1) / (totalSteps - 1) * 100) + '%';
        btnPrev.style.visibility = (currentStep === 1) ? 'hidden' : 'visible';

        if (currentStep === totalSteps) {
            btnNext.style.display = 'none';

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        btnSubmit.style.display = 'flex';
    } else {
        btnNext.style.display = 'flex';
        btnSubmit.style.display = 'none';
    }

    window.scrollTo({ top: 0, behavior: 'smooth' });
}

btnNext.addEventListener('click', () => {
    if (validateStep(currentStep)) {
        currentStep++;
        updateStep();
    }
});

btnPrev.addEventListener('click', () => {
    currentStep--;
    updateStep();
});

function validateStep(step) {
    const currentSection = document.getElementById('step-' + step);
    const inputs = currentSection.querySelectorAll('input[required],
select[required]');
    let isValid = true;

    inputs.forEach(input => {
        if (!input.value) {
            isValid = false;
            input.style.borderColor = 'var(--danger)';
            input.addEventListener('input', () => {
                input.style.borderColor = 'var(--border)';
            }, { once: true });
        }
    });

    return isValid;
}

birthdateInput.addEventListener('change', (e) => {
    const birthdate = new Date(e.target.value);
    const age = calculateAge(birthdate);

    if (age < 18) {
        minorAlert.style.display = 'flex';
        guardianSection.style.display = 'block';
        birthCertUpload.style.display = 'block';
        guardianSection.querySelectorAll('input').forEach(i =>
i.setAttribute('required', ''));
    } else {
        minorAlert.style.display = 'none';
        guardianSection.style.display = 'none';
        birthCertUpload.style.display = 'none';
        guardianSection.querySelectorAll('input').forEach(i =>
i.removeAttribute('required'));
    }
});

function calculateAge(birthday) {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
const ageDifMs = Date.now() - birthday.getTime();
const ageDate = new Date(ageDifMs);
return Math.abs(ageDate.getUTCFullYear() - 1970);
}

document.querySelectorAll('input[type="file"]').forEach(input => {
  input.addEventListener('change', (e) => {
    const box = e.target.closest('.upload-box');
    if (e.target.files.length > 0) {
      box.style.borderColor = 'var(--primary)';
      box.querySelector('p').textContent = e.target.files[0].name;
      box.querySelector('i').className = 'fas fa-check-circle';
      box.querySelector('i').style.color = 'var(--primary)';
    }
  });
});

function toggleHouseholdField() {
  const action =
document.querySelector('input[name="household_action"]:checked').value;
  const group = document.getElementById('householdNumberGroup');
  const input = document.getElementById('householdInput');

  if (action === 'new') {
    group.style.display = 'none';
    input.removeAttribute('required');
  } else {
    group.style.display = 'block';
    input.setAttribute('required', '');
  }
}

// Filtering household heads by Purok
const purokInput = document.getElementById('purokInput');
const householdInput = document.getElementById('householdInput');
const headsList = document.getElementById('headsList');
const masterHeads = Array.from(headsList.options).map(opt => ({
  value: opt.value,
  text: opt.text,
  purok: opt.getAttribute('data-purok')
})));

function filterHeads() {
  const selectedPurok = purokInput.value.trim().toLowerCase();
  // Clear current list
  headsList.innerHTML = '';

  // Filter and repopulate
  masterHeads.forEach(head => {
    if (!selectedPurok || head.purok.toLowerCase().includes(selectedPurok)) {
      const option = document.createElement('option');
      option.value = head.value;
      option.text = head.text;
      headsList.appendChild(option);
    }
  });
}

purokInput.addEventListener('input', filterHeads);

// Also filter on step change to step 3
```

IoT-Enabled Barangay Resident and Attendance Management System

```
btnNext.addEventListener('click', () => {
  if (currentStep === 3) {
    filterHeads();
  }
});

// Initialize hierarchical address selectors
initAddressSelectors('cityInput', 'municipalityInput', 'barangayInput',
'zipCodeInput');

// Initialize household field visibility
toggleHouseholdField();
</script>
{% if messages %}
<div class="messages-container">
  {% for message in messages %}
  <div class="alert alert-{{ message.tags }}" id="alert-{{ forloop.counter }}">
    <div class="alert-content">
      {% if message.tags == 'success' %}<i class="fas fa-check-circle"></i>
      {% elif message.tags == 'error' %}<i class="fas fa-exclamation-circle"></i>
      {% elif message.tags == 'warning' %}<i class="fas fa-exclamation-triangle"></i>
      {% else %}<i class="fas fa-info-circle"></i>{% endif %}
      <span>{{ message }}</span>
    </div>
    <button class="alert-close" onclick="this.parentElement.remove()"><i class="fas
fa-times"></i></button>
  </div>
  {% endfor %}
</div>
{% endif %}
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script src="{% static 'js/app.js' %}"></script>
</body>
{% include 'core/hardware_poller.html' %}
</html>
```


File: templates/core/signup.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sign Up - Barangay Management System</title>
  <link rel="stylesheet" href="{% static 'css/style.css' %}"?v=2.1">
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
rel="stylesheet">
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css"
rel="stylesheet">
  <style>
    body {
      display: flex;
      align-items: center;
      justify-content: center;
      min-height: 100vh;
      margin: 0;
      padding: 40px 20px;
      font-family: 'Inter', sans-serif;
      overflow-x: hidden;
    }

    .login-card {
      position: relative;
      background: var(--glass-bg);
      backdrop-filter: blur(16px);
      -webkit-backdrop-filter: blur(16px);
      border: 1px solid var(--glass-border);
      border-radius: var(--radius-xl);
      padding: 48px;
      width: 100%;
      max-width: 480px;
      text-align: center;
      box-shadow: var(--shadow-lg);
      animation: slideIn 0.6s cubic-bezier(0.16, 1, 0.3, 1);
      z-index: 10;
    }

    .login-logo-container {
      width: 80px;
      height: 80px;
      margin: 0 auto 24px;
      background: var(--primary-light);
      border-radius: 20px;
      padding: 12px;
      display: flex;
      align-items: center;
      justify-content: center;
      overflow: hidden;
    }

    .login-logo-img {
      width: 100%;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
height: 100%;
object-fit: contain;
}

.login-card h2 {
  font-size: 24px;
  font-weight: 800;
  margin-bottom: 4px;
  color: var(--text-main);
  letter-spacing: -0.5px;
}

.login-card p {
  color: var(--text-muted);
  font-size: 14px;
  margin-bottom: 32px;
}

.form-group {
  text-align: left;
  margin-bottom: 20px;
}

.form-group label {
  display: block;
  font-size: 13px;
  font-weight: 600;
  color: var(--text-muted);
  margin-bottom: 8px;
  padding-left: 4px;
}

.form-control {
  width: 100%;
  background: rgba(255, 255, 255, 0.6);
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 12px 16px;
  color: var(--text-main);
  font-size: 14px;
  transition: var(--transition);
}

.form-control:focus {
  outline: none;
  border-color: var(--primary);
  background: #FFFFFF;
  box-shadow: 0 0 0 4px var(--primary-light);
  transform: translateY(-1px);
}

.login-btn {
  width: 100%;
  padding: 16px;
  margin-top: 10px;
  font-size: 16px;
  border-radius: 12px;
  font-weight: 700;
  background: var(--primary);
  color: white;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
border: none;
cursor: pointer;
transition: var(--transition);
display: flex;
align-items: center;
justify-content: center;
gap: 10px;
letter-spacing: 0.5px;
}

.login-btn:hover {
background: var(--primary-hover);
transform: translateY(-2px);
box-shadow: 0 8px 20px rgba(44, 110, 73, 0.25);
}

.login-btn:active {
transform: translateY(0);
}

.text-danger {
color: var(--danger);
font-size: 12px;
margin-top: 4px;
padding-left: 4px;
}
</style>
</head>

<body>
<div class="auth-background">
  <div class="auth-blob auth-blob-1"></div>
  <div class="auth-blob auth-blob-2"></div>
  <div class="auth-blob auth-blob-3"></div>
</div>

<div class="login-card">
  <div class="login-logo-container">
    
  </div>
  <h2>Official Registration</h2>
  <p>Create an account for barangay staff</p>

  <form method="POST">
    {% csrf_token %}

    {% if form.non_field_errors %}
    <div class="alert alert-error">
      {{ form.non_field_errors }}
    </div>
    {% endif %}

    <div class="form-group">
      <label>First Name</label>
      <input type="text" name="first_name" class="form-control" placeholder="Enter first
name" required
      value="{{ form.first_name.value|default:'' }}">
      {% if form.first_name.errors %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div class="text-danger">{{ form.first_name.errors }}</div>
{% endif %}
</div>

<div class="form-group">
<label>Middle Name</label>
<input type="text" name="middle_name" class="form-control" placeholder="Enter
middle name (optional)"
value="{{ form.middle_name.value|default:'' }}">
{% if form.middle_name.errors %}
<div class="text-danger">{{ form.middle_name.errors }}</div>
{% endif %}
</div>

<div class="form-group">
<label>Last Name</label>
<input type="text" name="last_name" class="form-control" placeholder="Enter last
name" required
value="{{ form.last_name.value|default:'' }}">
{% if form.last_name.errors %}
<div class="text-danger">{{ form.last_name.errors }}</div>
{% endif %}
</div>

<div class="form-group">
<label>Designated Role</label>
<select name="role" class="form-control" required>
<option value="">Select role</option>
<option value="captain" {% if form.role.value == 'captain' %}selected{% endif
%}>Barangay Captain</option>
<option value="secretary" {% if form.role.value == 'secretary' %}selected{%
endif %}>Secretary</option>
<option value="treasurer" {% if form.role.value == 'treasurer' %}selected{%
endif %}>Treasurer</option>
<option value="bhw" {% if form.role.value == 'bhw' %}selected{% endif
%}>Barangay BHW</option>
</select>
{% if form.role.errors %}
<div class="text-danger">{{ form.role.errors }}</div>
{% endif %}
</div>

<div class="form-group">
<label>Password</label>
<input type="password" name="password1" class="form-control" placeholder="Create a
strong password" required>
{% if form.password1.errors %}
<div class="text-danger">{{ form.password1.errors }}</div>
{% endif %}
</div>

<div class="form-group">
<label>Confirm Password</label>
<input type="password" name="password2" class="form-control" placeholder="Repeat
your password" required>
{% if form.password2.errors %}
<div class="text-danger">{{ form.password2.errors }}</div>
{% endif %}
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<button type="submit" class="login-btn">
  <i class="fas fa-user-plus"></i>
  Sign Up
</button>
</form>

<div style="margin-top: 24px; text-align: center;">
  <a href="{% url 'login' %}" style="color: var(--text-muted); font-size: 14px; text-
decoration: none;">
    Already have an account? <strong style="color: var(--primary);">Log in</strong>
  </a>
</div>
</div>
{% if messages %}
<div class="messages-container">
  {% for message in messages %}
  <div class="alert alert-{{ message.tags }}" id="alert-{{ forloop.counter }}">
    <div class="alert-content">
      {% if message.tags == 'success' %}<i class="fas fa-check-circle"></i>
      {% elif message.tags == 'error' %}<i class="fas fa-exclamation-circle"></i>
      {% elif message.tags == 'warning' %}<i class="fas fa-exclamation-triangle"></i>
      {% else %}<i class="fas fa-info-circle"></i>{% endif %}
      <span>{{ message }}</span>
    </div>
    <button class="alert-close" onclick="this.parentElement.remove()"><i class="fas fa-
times"></i></button>
  </div>
  {% endfor %}
</div>
{% endif %}
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script src="{% static 'js/app.js' %}"></script>
</body>
{% include 'core/hardware_poller.html' %}
</html>
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/documents/confirm_delete.html

```
{% extends 'base.html' %}

{% block title %}Delete Document{% endblock %}

{% block content %}
<div class="container-fluid py-4">
  <div class="row justify-content-center">
    <div class="col-md-6">
      <div class="card border-0 shadow-sm" style="border-radius: 12px; overflow:
hidden;">
        <div class="card-body p-5 text-center">
          <div class="mb-4 text-danger">
            <i class="fas fa-exclamation-triangle fa-4x"></i>
          </div>
          <h3 class="font-weight-bold mb-3">Delete Document</h3>
          <p class="text-muted mb-4">Are you sure you want to delete the
document <strong>"{{ object.title }}"</strong>? This action cannot be undone.</p>

          <form method="POST">
            {% csrf_token %}
            <div class="d-flex justify-content-center gap-3">
              <a href="{% url 'documents:list' %}" class="btn btn-light px-4
mr-2" style="border-radius: 8px;">Cancel</a>
              <button type="submit" class="btn btn-danger px-4"
style="border-radius: 8px;">Yes, Delete</button>
            </div>
          </form>
        </div>
      </div>
    </div>
  </div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/documents/list.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Documents{% endblock %}

{% block content %}
<div class="container-fluid py-4">
  <div class="d-flex justify-content-between align-items-center mb-4">
    <div>
      <h2 class="mb-1 font-weight-bold" style="color: #1f2937;">Document
Workspace</h2>
      <p class="text-muted mb-0">Create, edit, and manage barangay documents</p>
    </div>
    <a href="{% url 'documents:workspace' %}" class="btn btn-primary shadow-sm"
style="border-radius: 8px; font-weight: 500;">
      <i class="fas fa-plus mr-2"></i> New Document
    </a>
  </div>

  <div class="card border-0 shadow-sm" style="border-radius: 12px; overflow: hidden;">
    <div class="card-body p-0">
      <div class="table-responsive">
        <table class="table table-hover mb-0">
          <thead class="bg-light">
            <tr>
              <th class="border-0 px-4 py-3 text-muted text-uppercase"
style="font-size: 0.75rem; font-weight: 600;">Title</th>
              <th class="border-0 px-4 py-3 text-muted text-uppercase"
style="font-size: 0.75rem; font-weight: 600;">Type</th>
              <th class="border-0 px-4 py-3 text-muted text-uppercase"
style="font-size: 0.75rem; font-weight: 600;">Last Modified</th>
              <th class="border-0 px-4 py-3 text-muted text-uppercase"
style="font-size: 0.75rem; font-weight: 600;">Author</th>
              <th class="border-0 px-4 py-3 text-muted text-uppercase text-
right" style="font-size: 0.75rem; font-weight: 600;">Actions</th>
            </tr>
          </thead>
          <tbody>
            {% for doc in documents %}
            <tr>
              <td class="px-4 py-3 align-middle">
                <div class="font-weight-600 text-dark">{{ doc.title
}}</div>
              </td>
              <td class="px-4 py-3 align-middle">
                <span class="badge badge-pill badge-light border text-
secondary px-3 py-1">{{ doc.get_document_type_display }}</span>
              </td>
              <td class="px-4 py-3 align-middle text-muted small">
                {{ doc.updated_at|date:"M d, Y h:i A" }}
              </td>
              <td class="px-4 py-3 align-middle">
                <div class="d-flex align-items-center">
                  <div class="avatar-circle bg-primary text-white d-flex
align-items-center justify-content-center mr-2" style="width: 28px; height: 28px; border-
radius: 50%; font-size: 0.75rem;">
                    {{

```

IoT-Enabled Barangay Resident and Attendance Management System

```
doc.created_by.first_name|first|default:doc.created_by.username|first|upper }}
    </div>
    <span class="small">{{
doc.created_by.get_full_name|default:doc.created_by.username }}</span>
    </div>
    </td>
    <td class="px-4 py-3 align-middle text-right">
        <a href="{% url 'documents:edit' doc.pk %}" class="btn
btn-sm btn-light text-primary mr-1" title="Edit">
            <i class="fas fa-edit"></i>
        </a>
        <!-- Using a form for delete to avoid direct GET deletes
-->
        <form action="{% url 'documents:delete' doc.pk %}"
method="POST" class="d-inline">
            {% csrf_token %}
            <button type="submit" class="btn btn-sm btn-light
text-danger" title="Delete" onclick="return confirm('Are you sure you want to delete this
document?');">
                <i class="fas fa-trash"></i>
            </button>
        </form>
    </td>
</tr>
{% empty %}
<tr>
    <td colspan="5" class="text-center py-5">
        <div class="text-muted mb-3">
            <i class="fas fa-file-alt fa-3x" style="opacity:
0.2;"></i>
        </div>
        <h5 class="text-dark">No documents found</h5>
        <p class="text-muted">Create your first document to get
started.</p>
        <a href="{% url 'documents:workspace' %}" class="btn btn-
outline-primary mt-2">Create Document</a>
    </td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>
</div>
{% endblock %}
```


File: templates/documents/pdf_export.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="utf-8">
<style>
    @page { size: A4; margin: 1in; }
    body { font-family: sans-serif; font-size: 11pt; line-height: 1.5; color: black; }
    img { max-width: 100%; height: auto; }
    table { border-collapse: collapse; width: 100%; margin: 0; }
    td, th { border: 1px solid #ced4da; padding: 3px 5px; }
    th { font-weight: bold; text-align: left; background-color: #f1f3f5; }
    p { margin-bottom: 1em; margin-top: 0; }
    h1, h2, h3 { margin-top: 1.5em; margin-bottom: 0.5em; }
</style>
</head>
<body>
    {{ document.content|safe }}
</body>
</html>
```

File: templates/documents/workspace.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}{{ title }} | Documents{% endblock %}

{% block extra_head %}
<style>
    /* Aesthetic Workspace Design */
    .workspace-container {
        background-color: #f3f4f6;
        min-height: calc(100vh - 80px);
        padding: 2rem 1rem;
        display: flex;
        flex-direction: column;
        align-items: center;
        font-family: 'Inter', sans-serif;
    }

    .workspace-header {
        width: 100%;
        max-width: 900px;
        display: flex;
        justify-content: space-between;
        align-items: center;
        margin-bottom: 1rem;
        gap: 1rem;
    }

    .workspace-meta {
        display: flex;
        gap: 1rem;
        flex: 1;
    }

    .doc-title-input {
        flex: 2;
        font-size: 1.5rem;
        font-weight: 700;
        border: 1px solid transparent;
        padding: 0.5rem 1rem;
        border-radius: 8px;
        background: transparent;
        transition: all 0.2s;
        color: #1f2937;
    }

    .doc-title-input:hover, .doc-title-input:focus {
        background: white;
        border-color: #d1d5db;
        outline: none;
        box-shadow: 0 1px 3px rgba(0,0,0,0.1);
    }

    .doc-type-select {
        flex: 1;
        padding: 0.5rem 1rem;
        border-radius: 8px;
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
border: 1px solid #d1d5db;
background: white;
font-weight: 500;
color: #4b5563;
outline: none;
}

.workspace-actions {
  display: flex;
  gap: 0.5rem;
}

.btn-workspace {
  display: inline-flex;
  align-items: center;
  gap: 0.5rem;
  padding: 0.5rem 1rem;
  border-radius: 8px;
  font-weight: 600;
  font-size: 0.9rem;
  border: none;
  cursor: pointer;
  transition: all 0.2s;
  box-shadow: 0 1px 2px rgba(0,0,0,0.05);
}

.btn-save { background: #3b82f6; color: white; }
.btn-save:hover { background: #2563eb; }

.btn-pdf { background: #ef4444; color: white; }
.btn-pdf:hover { background: #dc2626; }

.btn-print { background: #10b981; color: white; }
.btn-print:hover { background: #059669; }

/* Toolbar Styling */
.tiptap-toolbar {
  width: 100%;
  max-width: 900px;
  background: white;
  padding: 0.75rem;
  border-radius: 12px 12px 0 0;
  border: 1px solid #e5e7eb;
  border-bottom: none;
  display: flex;
  flex-wrap: wrap;
  gap: 0.25rem;
  position: sticky;
  top: 0;
  z-index: 10;
  box-shadow: 0 4px 6px -1px rgba(0,0,0,0.05);
}

.toolbar-btn {
  background: transparent;
  border: 1px solid transparent;
  border-radius: 6px;
  width: 32px;
  height: 32px;
  display: flex;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    align-items: center;
    justify-content: center;
    color: #4b5563;
    cursor: pointer;
    transition: all 0.15s;
    font-size: 13px;
}

.toolbar-btn:hover {
    background: #f3f4f6;
    color: #111827;
}

.toolbar-btn.is-active {
    background: #e0e7ff;
    color: #4f46e5;
    border-color: #c7d2fe;
}

.toolbar-divider {
    width: 1px;
    background: #e5e7eb;
    margin: 0 0.25rem;
}

/* A4 Paper Editor Container */
.editor-wrapper {
    width: 100%;
    max-width: 900px;
    border: 1px solid #e5e7eb;
    border-radius: 0 0 12px 12px;
    background: white;
    overflow: hidden;
    box-shadow: 0 10px 15px -3px rgba(0,0,0,0.1), 0 4px 6px -2px rgba(0,0,0,0.05);
}

/* A4 Size emulation */
.ProseMirror {
    position: relative; /* Critical for absolute dragging */
    min-height: 297mm; /* A4 height approx */
    width: 210mm; /* A4 width */
    max-width: 100%;
    margin: 0 auto;
    padding: 96px; /* 1 inch margins approx */
    background: white;
    outline: none;
    font-size: 11pt; /* Standard doc font size */
    line-height: 1.5;
    color: #000;
    box-sizing: border-box;
}

.ProseMirror p { margin-bottom: 1em; }
.ProseMirror h1, .ProseMirror h2, .ProseMirror h3 { margin-top: 1.5em; margin-bottom: 0.5em; }

.ProseMirror table {
    border-collapse: collapse;
    table-layout: fixed;
    width: 100%;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
margin: 0;
overflow: hidden;
}

.ProseMirror td, .ProseMirror th {
  min-width: 1em;
  border: 1px solid #ced4da;
  padding: 3px 5px;
  vertical-align: top;
  box-sizing: border-box;
  position: relative;
}

.ProseMirror th {
  font-weight: bold;
  text-align: left;
  background-color: #f1f3f5;
}

.ProseMirror img {
  max-width: 100%;
  height: auto;
  display: inline-block;
  cursor: grab;
  -webkit-user-drag: element;
  -webkit-user-select: none;
  user-select: none;
}

.ProseMirror img:active {
  cursor: grabbing;
}

.ProseMirror img.ProseMirror-selectednode {
  outline: 3px solid #10b981;
}

/* For print/pdf */
@media print {
  body * { visibility: hidden; }
  #document-paper, #document-paper * {
    visibility: visible;
  }
  #document-paper {
    position: absolute;
    left: 0;
    top: 0;
    padding: 0;
    width: 100%;
  }
  .ProseMirror { padding: 0; }
  .image-float-toolbar { display: none !important; }
  #global-rotate-handle { display: none !important; }
}

/* ----- Image Float Toolbar (appears on click) ----- */
.image-float-toolbar {
  position: absolute;
  display: none;
  background: white;
  border: 1px solid #e5e7eb;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
border-radius: 8px;
padding: 4px;
gap: 2px;
z-index: 100;
box-shadow: 0 4px 12px rgba(0,0,0,0.15);
flex-direction: row;
align-items: center;
}
.image-float-toolbar.visible {
  display: flex;
}
.image-float-toolbar button {
  background: transparent;
  border: 1px solid transparent;
  border-radius: 6px;
  width: 28px;
  height: 28px;
  display: flex;
  align-items: center;
  justify-content: center;
  color: #4b5563;
  cursor: pointer;
  font-size: 12px;
  transition: all 0.15s;
}
.image-float-toolbar button:hover {
  background: #ecfdf5;
  color: #10b981;
}
.image-float-toolbar button.active {
  background: #d1fae5;
  color: #059669;
  border-color: #a7f3d0;
}
.image-float-toolbar .ift-divider {
  width: 1px;
  height: 20px;
  background: #e5e7eb;
  margin: 0 2px;
}
}
</style>
{% endblock %}

{% block content %}
<div class="workspace-container">
  <div class="workspace-header">
    <div class="workspace-meta">
      <input type="text" id="docTitle" class="doc-title-input" placeholder="Untitled
Document" value="{{ form.title.value|default:'' }}">
      <select id="docType" class="doc-type-select">
        {% for value, label in document_types %}
          <option value="{{ value }}" {% if form.document_type.value == value
%}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="workspace-actions">
      <button id="btnSave" class="btn-workspace btn-save"><i class="fas fa-
save"></i> Save</button>
      {% if form.instance.pk %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<a href="{% url 'documents:pdf' form.instance.pk %}" class="btn-workspace btn-
pdf" style="text-decoration: none;"><i class="fas fa-file-pdf"></i> Save as PDF</a>
    {% else %}
    <button id="btnPdfUnsaved" class="btn-workspace btn-pdf"
onclick="alert('Please save the document first before exporting to PDF.');"><i class="fas
fa-file-pdf"></i> Save as PDF</button>
    {% endif %}
    <button id="btnPrint" class="btn-workspace btn-print"><i class="fas fa-
print"></i> Print</button>
</div>
</div>

<div class="tiptap-toolbar" id="toolbar">
    <button class="toolbar-btn" data-action="undo" title="Undo"><i class="fas fa-
undo"></i></button>
    <button class="toolbar-btn" data-action="redo" title="Redo"><i class="fas fa-
redo"></i></button>
    <div class="toolbar-divider"></div>

    <button class="toolbar-btn" data-action="bold" title="Bold"><i class="fas fa-
bold"></i></button>
    <button class="toolbar-btn" data-action="italic" title="Italic"><i class="fas fa-
italic"></i></button>
    <button class="toolbar-btn" data-action="underline" title="Underline"><i
class="fas fa-underline"></i></button>
    <button class="toolbar-btn" data-action="strike" title="Strikethrough"><i
class="fas fa-strikethrough"></i></button>
    <div class="toolbar-divider"></div>

    <button class="toolbar-btn" data-action="h1" title="Heading 1"><i class="fas fa-
heading"></i>1</button>
    <button class="toolbar-btn" data-action="h2" title="Heading 2"><i class="fas fa-
heading"></i>2</button>
    <button class="toolbar-btn" data-action="h3" title="Heading 3"><i class="fas fa-
heading"></i>3</button>
    <div class="toolbar-divider"></div>

    <button class="toolbar-btn" data-action="align-left" title="Align Left"><i
class="fas fa-align-left"></i></button>
    <button class="toolbar-btn" data-action="align-center" title="Align Center"><i
class="fas fa-align-center"></i></button>
    <button class="toolbar-btn" data-action="align-right" title="Align Right"><i
class="fas fa-align-right"></i></button>
    <button class="toolbar-btn" data-action="align-justify" title="Justify"><i
class="fas fa-align-justify"></i></button>
    <div class="toolbar-divider"></div>

    <button class="toolbar-btn" data-action="bullet-list" title="Bullet List"><i
class="fas fa-list-ul"></i></button>
    <button class="toolbar-btn" data-action="ordered-list" title="Numbered List"><i
class="fas fa-list-ol"></i></button>
    <div class="toolbar-divider"></div>

    <button class="toolbar-btn" data-action="table" title="Insert Table"><i class="fas
fa-table"></i></button>
    <input type="file" id="imageUpload" accept="image/*" style="display:none;">
    <button class="toolbar-btn" id="btnInsertImage" title="Insert Image"><i class="fas
fa-image"></i></button>
    <div class="toolbar-divider"></div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<input type="color" id="textColor" value="#000000" style="width: 32px; height: 32px; padding: 0; border: none; border-radius: 6px; cursor: pointer; background: transparent;">
</div>

<div class="editor-wrapper">
  <div id="document-paper"></div>
</div>

<!-- Image Float Toolbar (positioned dynamically near the selected image) -->
<div class="image-float-toolbar" id="imageFloatToolbar">
  <button id="iftAlignLeft" title="Float Left"><i class="fas fa-align-left"></i></button>
  <button id="iftAlignCenter" title="No Wrap (inline)"><i class="fas fa-align-center"></i></button>
  <button id="iftAlignRight" title="Float Right"><i class="fas fa-align-right"></i></button>
  <div class="ift-divider"></div>
  <button id="iftRotLeft" title="Rotate Left 90°"><i class="fas fa-undo"></i></button>
  <button id="iftRotRight" title="Rotate Right 90°"><i class="fas fa-redo"></i></button>
  <div class="ift-divider"></div>
  <button id="iftWrapToggle" title="Toggle Text Wrap"><i class="fas fa-text-width"></i></button>
  <div class="ift-divider"></div>
  <button id="iftDelete" title="Delete Image" style="color: #ef4444;"><i class="fas fa-trash"></i></button>
</div>

<!-- Global Rotate Handle -->
<div id="global-rotate-handle" style="display: none; position: absolute; width: 24px; height: 24px; background: white; border: 2px solid #10b981; border-radius: 50%; color: #10b981; cursor: grab; justify-content: center; align-items: center; z-index: 50; box-shadow: 0 2px 4px rgba(0,0,0,0.15);">
  <i class="fas fa-sync-alt" style="font-size: 10px;"></i>
  <div id="rotate-tooltip" style="display: none; position: absolute; top: -30px; left: 50%; transform: translateX(-50%); background: #1f2937; color: white; padding: 2px 8px; border-radius: 4px; font-size: 11px; white-space: nowrap; pointer-events: none;">0°</div>
</div>

<!-- Hidden form for saving -->
<form id="docForm" method="POST" style="display: none;">
  {<csrf_token %>
  <input type="hidden" name="title" id="formTitle">
  <input type="hidden" name="document_type" id="formDocType">
  <textarea name="content" id="formContent">{{ form.content.value|default:'' }}</textarea>
  </form>
</div>
{% endblock %}

{% block extra_js %}
<!-- TipTap ES Modules -->
<script type="importmap">
  {
    "imports": {
      "@tiptap/core": "https://esm.sh/@tiptap/core@2.2.4",
      "@tiptap/starter-kit": "https://esm.sh/@tiptap/starter-kit@2.2.4",
```


IoT-Enabled Barangay Resident and Attendance Management System

```
"@tiptap/extension-underline": "https://esm.sh/@tiptap/extension-underline@2.2.4",
"@tiptap/extension-text-align": "https://esm.sh/@tiptap/extension-text-align@2.2.4",
"@tiptap/extension-color": "https://esm.sh/@tiptap/extension-color@2.2.4",
"@tiptap/extension-text-style": "https://esm.sh/@tiptap/extension-text-style@2.2.4",
"@tiptap/extension-table": "https://esm.sh/@tiptap/extension-table@2.2.4",
"@tiptap/extension-table-row": "https://esm.sh/@tiptap/extension-table-row@2.2.4",
"@tiptap/extension-table-cell": "https://esm.sh/@tiptap/extension-table-cell@2.2.4",
"@tiptap/extension-table-header": "https://esm.sh/@tiptap/extension-table-header@2.2.4",
"@tiptap/extension-image": "https://esm.sh/@tiptap/extension-image@2.2.4",
"@tiptap/pm/": "https://esm.sh/@tiptap/pm@2.2.4/"
}
}
</script>
```

```
<script type="module">
import { Editor } from '@tiptap/core'
import StarterKit from '@tiptap/starter-kit'
import Underline from '@tiptap/extension-underline'
import TextAlign from '@tiptap/extension-text-align'
import Color from '@tiptap/extension-color'
import TextStyle from '@tiptap/extension-text-style'
import Table from '@tiptap/extension-table'
import TableRow from '@tiptap/extension-table-row'
import TableCell from '@tiptap/extension-table-cell'
import TableHeader from '@tiptap/extension-table-header'
import Image from '@tiptap/extension-image'

// ?? Custom Image Extension with rotation, float, and resize ??
const AdvancedImage = Image.extend({
  name: 'image',
  draggable: true,

  addAttributes() {
    return {
      ...this.parent?().,
      width: {
        default: null,
        parseHTML: el => el.getAttribute('width') ?
parseInt(el.getAttribute('width')) : (el.style.width ? parseInt(el.style.width) : null),
        renderHTML: attrs => attrs.width ? { width: attrs.width } : {}
      },
      height: {
        default: null,
        parseHTML: el => el.getAttribute('height') ?
parseInt(el.getAttribute('height')) : (el.style.height ? parseInt(el.style.height) :
null),
        renderHTML: attrs => attrs.height ? { height: attrs.height } : {}
      },
      'data-angle': {
        default: 0,
        parseHTML: el => parseInt(el.getAttribute('data-angle')) || 0,
        renderHTML: attrs => {
          if (!attrs['data-angle']) return {};
          return { 'data-angle': attrs['data-angle'] };
        }
      }
    }
  },
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
'data-float': {
  default: 'none',
  parseHTML: el => el.getAttribute('data-float') || 'none',
  renderHTML: attrs => {
    const f = attrs['data-float'];
    if (!f || f === 'none') return {};
    return { 'data-float': f };
  }
},
'data-position': {
  default: 'inline',
  parseHTML: el => el.getAttribute('data-position') || 'inline',
  renderHTML: attrs => {
    if (!attrs['data-position'] || attrs['data-position'] ===
'inline') return {};
    return { 'data-position': attrs['data-position'] };
  }
},
'data-x': {
  default: null,
  parseHTML: el => el.getAttribute('data-x') ?
parseInt(el.getAttribute('data-x')) : null,
  renderHTML: attrs => attrs['data-x'] !== null ? { 'data-x':
attrs['data-x'] } : {}
},
'data-y': {
  default: null,
  parseHTML: el => el.getAttribute('data-y') ?
parseInt(el.getAttribute('data-y')) : null,
  renderHTML: attrs => attrs['data-y'] !== null ? { 'data-y':
attrs['data-y'] } : {}
}
},

renderHTML({ HTMLAttributes }) {
  // Merge transform and float styles
  let styles = [];
  if (HTMLAttributes['data-angle'] && HTMLAttributes['data-angle'] !== 0) {
    styles.push(`transform: rotate(${HTMLAttributes['data-angle']}deg)`);
  }
  if (HTMLAttributes['data-position'] === 'absolute') {
    styles.push('position: absolute');
    if (HTMLAttributes['data-x'] !== null) styles.push(`left:
${HTMLAttributes['data-x']}px`);
    if (HTMLAttributes['data-y'] !== null) styles.push(`top:
${HTMLAttributes['data-y']}px`);
    styles.push('z-index: 10');
  } else {
    const f = HTMLAttributes['data-float'];
    if (f && f !== 'none') {
      styles.push(`float: ${f}`, 'margin: 10px');
    }
  }
}

if (HTMLAttributes.width) {
  styles.push(`width: ${HTMLAttributes.width}px`);
}
if (HTMLAttributes.height) {
  styles.push(`height: ${HTMLAttributes.height}px`);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    }

    const finalAttrs = { ...HTMLAttributes, draggable: 'true' };
    if (styles.length > 0) {
        finalAttrs.style = styles.join('; ');
    }

    return ['img', finalAttrs];
}
});

// Initialize Editor
const initialContent = document.getElementById('formContent').value;

const editor = new Editor({
    element: document.getElementById('document-paper'),
    extensions: [
        StarterKit,
        Underline,
        TextAlign.configure({ types: ['heading', 'paragraph'] }),
        TextStyle,
        Color,
        Table.configure({ resizable: true }),
        TableRow,
        TableHeader,
        TableCell,
        AdvancedImage.configure({ inline: true, allowBase64: true })
    ],
    content: initialContent || '<p></p>',
    onUpdate({ editor }) {
        updateActiveStates();
    },
    onSelectionUpdate({ editor }) {
        updateActiveStates();
        showImageToolbar();
    }
});

// ?? Expose editor for debugging ??
window._editor = editor;

// =====
// IMAGE FLOAT TOOLBAR (positioned near the selected image)
// =====
const imageFloatToolbar = document.getElementById('imageFloatToolbar');
const rotateHandle = document.getElementById('global-rotate-handle');
const rotateTooltip = document.getElementById('rotate-tooltip');
let selectedImgNode = null; // the DOM <img> element
let selectedImgPos = null; // the ProseMirror position

function showImageToolbar() {
    if (!editor.isActive('image')) {
        imageFloatToolbar.classList.remove('visible');
        rotateHandle.style.display = 'none';
        selectedImgNode = null;
        return;
    }

    const { from } = editor.state.selection;
    const domNode = editor.view.nodeDOM(from);
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if (!domNode) { imageFloatToolbar.classList.remove('visible');
rotateHandle.style.display = 'none'; return; }

selectedImgNode = domNode.tagName === 'IMG' ? domNode :
domNode.querySelector('img');
selectedImgPos = from;

if (!selectedImgNode) { imageFloatToolbar.classList.remove('visible');
rotateHandle.style.display = 'none'; return; }

const imgRect = selectedImgNode.getBoundingClientRect();

// Position float toolbar above the image
imageFloatToolbar.classList.add('visible');
imageFloatToolbar.style.top = `${imgRect.top + window.scrollY - 40}px`;
imageFloatToolbar.style.left = `${imgRect.left + window.scrollX + imgRect.width /
2 - imageFloatToolbar.offsetWidth / 2}px`;

// Position rotate handle at top center
rotateHandle.style.display = 'flex';
rotateHandle.style.top = `${imgRect.top + window.scrollY - 70}px`;
rotateHandle.style.left = `${imgRect.left + window.scrollX + imgRect.width / 2 -
12}px`;

// Update active states on float toolbar buttons
const curFloat = editor.getAttributes('image')['data-float'] || 'none';
document.getElementById('iftAlignLeft').classList.toggle('active', curFloat ===
'left');
document.getElementById('iftAlignCenter').classList.toggle('active', curFloat ===
'none');
document.getElementById('iftAlignRight').classList.toggle('active', curFloat ===
'right');
}

// Re-position on scroll
window.addEventListener('scroll', () => { if (selectedImgNode) showImageToolbar(); });

// Click outside to hide
document.addEventListener('click', (e) => {
if (!imageFloatToolbar.contains(e.target) && !rotateHandle.contains(e.target)) {
// Let TipTap handle the selection; toolbar will hide via showImageToolbar if
needed
}
});

// ?? Float toolbar actions ??
function setFloat(val) {
editor.chain().focus().updateAttributes('image', { 'data-float': val }).run();
setTimeout(showImageToolbar, 50);
}
document.getElementById('iftAlignLeft').addEventListener('click', (e) => {
e.preventDefault(); e.stopPropagation(); setFloat('left'); });
document.getElementById('iftAlignCenter').addEventListener('click', (e) => {
e.preventDefault(); e.stopPropagation(); setFloat('none'); });
document.getElementById('iftAlignRight').addEventListener('click', (e) => {
e.preventDefault(); e.stopPropagation(); setFloat('right'); });

function rotateImage(delta) {
const attrs = editor.getAttributes('image');
let angle = (parseInt(attrs['data-angle']) || 0) + delta;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if (angle < 0) angle += 360;
angle = angle % 360;
editor.chain().focus().updateAttributes('image', { 'data-angle': angle }).run();
setTimeout(showImageToolbar, 50);
}
document.getElementById('iftRotLeft').addEventListener('click', (e) => {
e.preventDefault(); e.stopPropagation(); rotateImage(-90); });
document.getElementById('iftRotRight').addEventListener('click', (e) => {
e.preventDefault(); e.stopPropagation(); rotateImage(90); });

document.getElementById('iftWrapToggle').addEventListener('click', (e) => {
e.preventDefault(); e.stopPropagation();
const curFloat = editor.getAttributes('image')['data-float'] || 'none';
// Cycle: none -> left -> right -> none
const next = curFloat === 'none' ? 'left' : curFloat === 'left' ? 'right' :
'none';
setFloat(next);
});

document.getElementById('iftDelete').addEventListener('click', (e) => {
e.preventDefault(); e.stopPropagation();
editor.chain().focus().deleteSelection().run();
imageFloatToolbar.classList.remove('visible');
rotateHandle.style.display = 'none';
});

// ?? Free-drag rotate handle ??
let isRotating = false;
rotateHandle.addEventListener('mousedown', (e) => {
e.preventDefault();
if (!selectedImgNode) return;
isRotating = true;
rotateHandle.style.cursor = 'grabbing';
const rect = selectedImgNode.getBoundingClientRect();
const cx = rect.left + rect.width / 2;
const cy = rect.top + rect.height / 2;
rotateTooltip.style.display = 'block';

function onMove(e) {
const dx = e.clientX - cx;
const dy = e.clientY - cy;
let angle = Math.atan2(dy, dx) * (180 / Math.PI) + 90;
if (angle < 0) angle += 360;
angle = Math.round(angle);
rotateTooltip.innerText = `${angle}°`;
// Live visual preview via inline style
selectedImgNode.style.transform = `rotate(${angle}deg)`;
selectedImgNode.setAttribute('data-angle', angle);
}

function onUp(e) {
isRotating = false;
rotateHandle.style.cursor = 'grab';
rotateTooltip.style.display = 'none';
document.removeEventListener('mousemove', onMove);
document.removeEventListener('mouseup', onUp);
// Commit to TipTap state directly
const finalAngle = parseInt(selectedImgNode.getAttribute('data-angle')) || 0;
updateImageNode(selectedImgNode, { 'data-angle': finalAngle });
setTimeout(showImageToolbar, 50);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
document.addEventListener('mousemove', onMove);
document.addEventListener('mouseup', onUp);
});

// =====
//  INLINE RESIZE via corner drag handles (pure CSS/JS)
//  =====

// Helper to reliably update image attributes directly in ProseMirror state
function updateImageNode(imgEl, attributes) {
  try {
    const pos = editor.view.posAtDOM(imgEl, 0);
    if (pos >= 0) {
      const node = editor.state.doc.nodeAt(pos);
      if (node && node.type.name === 'image') {
        editor.view.dispatch(
          editor.state.tr.setNodeMarkup(pos, null, {
            ...node.attrs,
            ...attributes
          })
        );
      } else {
        editor.chain().setNodeSelection(pos).updateAttributes('image',
attributes).run();
      }
    }
  } catch (e) {
    console.error('Failed to update image node', e);
  }
}

// We add resize handles by intercepting clicks on images inside ProseMirror
const editorEl = document.getElementById('document-paper');
let resizeHandles = [];

function clearResizeHandles() {
  resizeHandles.forEach(h => h.el.remove());
  resizeHandles = [];
}

function addResizeHandles(imgEl) {
  clearResizeHandles();
  if (!imgEl || !imgEl.parentElement) return;

  const wrapper = imgEl.closest('.ProseMirror');
  if (!wrapper) return;

  const positions = [
    { pos: 'nw', cursor: 'nwse-resize' },
    { pos: 'ne', cursor: 'nesw-resize' },
    { pos: 'sw', cursor: 'nesw-resize' },
    { pos: 'se', cursor: 'nwse-resize' }
  ];

  positions.forEach(({ pos, cursor }) => {
    const handle = document.createElement('div');
    handle.style.cssText = `
      position: absolute; width: 12px; height: 12px;
      background: #10b981; border: 2px solid white; border-radius: 50%;
      cursor: ${cursor}; z-index: 20;
    `;
  });
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        box-shadow: 0 1px 3px rgba(0,0,0,0.2);
    `;
    positionHandle(handle, imgEl, pos);
    document.body.appendChild(handle);
    resizeHandles.push({ el: handle, pos });

    let startX, startY, startW, startH;
    handle.addEventListener('mousedown', (e) => {
        e.preventDefault();
        e.stopPropagation();
        startX = e.clientX;
        startY = e.clientY;
        startW = imgEl.offsetWidth;
        startH = imgEl.offsetHeight;
        const aspect = startW / startH;

        function onMove(e) {
            const dx = e.clientX - startX;
            const dy = e.clientY - startY;
            let newW = startW;

            if (pos === 'se' || pos === 'ne') {
                newW = startW + dx;
            } else if (pos === 'sw' || pos === 'nw') {
                newW = startW - dx;
            }

            newW = Math.max(50, newW);
            let newH = newW / aspect;

            imgEl.style.width = `${newW}px`;
            imgEl.style.height = `${newH}px`;
            imgEl.setAttribute('width', Math.round(newW));
            imgEl.setAttribute('height', Math.round(newH));

            resizeHandles.forEach(h => positionHandle(h.el, imgEl, h.pos));
            showImageToolbar();
        }

        function onUp() {
            document.removeEventListener('mousemove', onMove);
            document.removeEventListener('mouseup', onUp);

            updateImageNode(imgEl, {
                width: Math.round(imgEl.offsetWidth),
                height: Math.round(imgEl.offsetHeight)
            });
        }

        document.addEventListener('mousemove', onMove);
        document.addEventListener('mouseup', onUp);
    });
});
}

function positionHandle(handle, imgEl, pos) {
    const rect = imgEl.getBoundingClientRect();
    if (pos === 'nw') {
        handle.style.top = `${rect.top + window.scrollY - 6}px`;
        handle.style.left = `${rect.left + window.scrollX - 6}px`;
    } else if (pos === 'ne') {
        handle.style.top = `${rect.top + window.scrollY - 6}px`;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        handle.style.left = `${rect.right + window.scrollX - 6}px`;
    } else if (pos === 'sw') {
        handle.style.top = `${rect.bottom + window.scrolly - 6}px`;
        handle.style.left = `${rect.left + window.scrollX - 6}px`;
    } else if (pos === 'se') {
        handle.style.top = `${rect.bottom + window.scrolly - 6}px`;
        handle.style.left = `${rect.right + window.scrollX - 6}px`;
    }
}

// Prevent native drag if we are doing absolute drag
editorEl.addEventListener('dragstart', (e) => {
    if (e.target.tagName === 'IMG') {
        e.preventDefault();
    }
});

// Listen for image selection and enable absolute dragging in the editor
editorEl.addEventListener('mousedown', (e) => {
    if (e.target.tagName === 'IMG') {
        const imgEl = e.target;
        setTimeout(() => addResizeHandles(imgEl), 10);

        const startX = e.clientX;
        const startY = e.clientY;

        const paperRect = document.getElementById('document-
paper').getBoundingClientRect();
        const imgRect = imgEl.getBoundingClientRect();

        let startLeft = imgRect.left - paperRect.left;
        let startTop = imgRect.top - paperRect.top;

        // If already absolute, use its current data-x/y to prevent jump if bounding
rect differs
        if (imgEl.getAttribute('data-position') === 'absolute') {
            startLeft = parseInt(imgEl.getAttribute('data-x')) || startLeft;
            startTop = parseInt(imgEl.getAttribute('data-y')) || startTop;
        }

        let hasMoved = false;

        function onMove(ev) {
            const dx = ev.clientX - startX;
            const dy = ev.clientY - startY;

            if (!hasMoved && (Math.abs(dx) > 3 || Math.abs(dy) > 3)) {
                hasMoved = true;
                // Prepare image for absolute dragging visually
                imgEl.style.position = 'absolute';
                imgEl.style.zIndex = '10';
                imgEl.style.float = 'none';
                imgEl.style.margin = '0';
                clearResizeHandles();
                imageFloatToolbar.classList.remove('visible');
                rotateHandle.style.display = 'none';
            }

            if (hasMoved) {
                ev.preventDefault();
            }
        }
    }
});
```


IoT-Enabled Barangay Resident and Attendance Management System

```
        let newLeft = startLeft + dx;
        let newTop = startTop + dy;
        imgEl.style.left = `${newLeft}px`;
        imgEl.style.top = `${newTop}px`;
    }
}

function onUp(ev) {
    document.removeEventListener('mousemove', onMove);
    document.removeEventListener('mouseup', onUp);

    if (hasMoved) {
        const dx = ev.clientX - startX;
        const dy = ev.clientY - startY;
        let newLeft = Math.round(startLeft + dx);
        let newTop = Math.round(startTop + dy);

        // Update the image attributes for TipTap reliably
        updateImageNode(imgEl, {
            'data-position': 'absolute',
            'data-x': newLeft,
            'data-y': newTop,
            'data-float': 'none'
        });

        setTimeout(() => addResizeHandles(imgEl), 50);
    }
}

document.addEventListener('mousemove', onMove);
document.addEventListener('mouseup', onUp);
} else {
    clearResizeHandles();
}
});

// =====
//  MAIN TOOLBAR ACTIONS
// =====
const updateActiveStates = () => {
    document.querySelectorAll('#toolbar .toolbar-btn').forEach(btn => {
        const action = btn.dataset.action;
        if (!action) return;
        let isActive = false;

        if (action === 'bold') isActive = editor.isActive('bold');
        if (action === 'italic') isActive = editor.isActive('italic');
        if (action === 'underline') isActive = editor.isActive('underline');
        if (action === 'strike') isActive = editor.isActive('strike');
        if (action === 'h1') isActive = editor.isActive('heading', { level: 1 });
        if (action === 'h2') isActive = editor.isActive('heading', { level: 2 });
        if (action === 'h3') isActive = editor.isActive('heading', { level: 3 });
        if (action === 'align-left') isActive = editor.isActive({ textAlign: 'left'
    });
        if (action === 'align-center') isActive = editor.isActive({ textAlign:
'center' });
        if (action === 'align-right') isActive = editor.isActive({ textAlign: 'right'
    });
        if (action === 'align-justify') isActive = editor.isActive({ textAlign:
'justify' });
    });
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    if (action === 'bullet-list') isActive = editor.isActive('bulletList');
    if (action === 'ordered-list') isActive = editor.isActive('orderedList');

    if (isActive) btn.classList.add('is-active');
    else btn.classList.remove('is-active');
  });
};

document.querySelectorAll('#toolbar .toolbar-btn').forEach(btn => {
  btn.addEventListener('click', (e) => {
    const action = btn.dataset.action;
    if (!action) return;
    e.preventDefault();

    if (action === 'undo') editor.chain().focus().undo().run();
    if (action === 'redo') editor.chain().focus().redo().run();
    if (action === 'bold') editor.chain().focus().toggleBold().run();
    if (action === 'italic') editor.chain().focus().toggleItalic().run();
    if (action === 'underline') editor.chain().focus().toggleUnderline().run();
    if (action === 'strike') editor.chain().focus().toggleStrike().run();
    if (action === 'h1') editor.chain().focus().toggleHeading({ level: 1 }).run();
    if (action === 'h2') editor.chain().focus().toggleHeading({ level: 2 }).run();
    if (action === 'h3') editor.chain().focus().toggleHeading({ level: 3 }).run();
    if (action === 'align-left')
editor.chain().focus().setTextAlign('left').run();
    if (action === 'align-center')
editor.chain().focus().setTextAlign('center').run();
    if (action === 'align-right')
editor.chain().focus().setTextAlign('right').run();
    if (action === 'align-justify')
editor.chain().focus().setTextAlign('justify').run();
    if (action === 'bullet-list') editor.chain().focus().toggleBulletList().run();
    if (action === 'ordered-list')
editor.chain().focus().toggleOrderedList().run();
    if (action === 'table') editor.chain().focus().insertTable({ rows: 3, cols: 3,
withHeaderRow: true }).run();
  });
});

// ?? Image Insert Button (separate from toolbar loop) ??
document.getElementById('btnInsertImage').addEventListener('click', (e) => {
  e.preventDefault();
  document.getElementById('imageUpload').click();
});

document.getElementById('imageUpload').addEventListener('change', (e) => {
  const file = e.target.files[0];
  if (file) {
    const reader = new FileReader();
    reader.onload = (ev) => {
      editor.chain().focus().setImage({ src: ev.target.result }).run();
    };
    reader.readAsDataURL(file);
    // Reset input so the same file can be re-selected
    e.target.value = '';
  }
});

document.getElementById('textColor').addEventListener('input', (e) => {
  editor.commands.setColor(e.target.value);
});
```

IoT-Enabled Barangay Resident and Attendance Management System

```
});

// ?? Save Action ??
document.getElementById('btnSave').addEventListener('click', () => {
    document.getElementById('formTitle').value =
document.getElementById('docTitle').value || 'Untitled Document';
    document.getElementById('formDocType').value =
document.getElementById('docType').value;
    document.getElementById('formContent').value = editor.getHTML();
    document.getElementById('docForm').submit();
});

// ?? Print Action ??
document.getElementById('btnPrint').addEventListener('click', () => {
    clearResizeHandles();
    imageFloatToolbar.classList.remove('visible');
    rotateHandle.style.display = 'none';
    window.print();
});

console.log('[Document Workspace] Editor initialized successfully.');
```

</script>

{% endblock %}

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/biometric_register.html

```
{% extends 'base.html' %}

{% block title %}Biometric Registration - Barangay Management System{% endblock %}

{% block content %}
<div class="biometric-registration-wrapper">
  <div class="container-fluid p-0">

    <!-- Compact Header -->
    <div class="d-flex align-items-center justify-content-between mb-4">
      <div>
        <h1 class="h3 font-weight-bold text-dark mb-1">Biometric Enrollment</h1>
        <p class="text-muted small mb-0">Register and manage fingerprints for
barangay functionaries.</p>
      </div>
      <div>
        {% if request.user.is_superuser or request.user.profile.role == 'captain'
or request.user.profile.role == 'secretary' or request.user.profile.role == 'treasurer' or
request.user.profile.role == 'admin' %}
          <button type="button" class="btn btn-sm btn-outline-danger font-weight-
bold rounded-lg px-3 shadow-sm transition-all" data-toggle="modal" data-
target="#resetBiometricModal">
            <i class="fas fa-eraser mr-1"></i> Clear All Fingerprints
          </button>
        {% endif %}
      </div>
    </div>

    <!-- Dense Toolbar & Content Card -->
    <div class="card border-0 shadow-sm rounded-xl overflow-hidden mb-4">
      <div class="card-header bg-white border-bottom-0 p-3">
        <div class="row align-items-center">
          <!-- Search -->
          <div class="col-md-6 mb-2 mb-md-0">
            <div class="position-relative">
              <i class="fas fa-search position-absolute text-muted"
style="left: 15px; top: 50%; transform: translateY(-50%); font-size: 13px;"></i>
              <input
                type="text"
                id="person-search"
                placeholder="Search by name or position..."
                class="form-control form-control-sm pl-5 border-light bg-
light rounded-lg"
                style="height: 40px;"
              />
            </div>
          </div>
          <!-- Filter -->
          <div class="col-md-6 d-flex justify-content-md-end">
            <div class="d-flex align-items-center w-100 w-md-auto">
              <div class="dropdown-wrapper w-100">
                <select id="category-select" class="form-control form-
control-sm border-light bg-light rounded-lg font-weight-bold px-3" style="height: 40px;
min-width: 250px; width: 100%;">
                  <option value="officials" selected>Barangay
Officials</option>
                  <option value="bhw">BHW (Health Workers)</option>
                </select>
              </div>
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <option value="tanod">Barangay Tanod</option>
        <option value="staff">Other Staff /
Functionaries</option>
    </select>
</div>
</div>
</div>
</div>
</div>

<div class="card-body p-0">
    <!-- Desktop Table -->
    <div class="table-responsive d-none d-md-block">
        <table class="table table-hover mb-0">
            <thead class="bg-light">
                <tr class="text-muted small font-weight-bold text-uppercase">
                    <th class="border-0 px-4 py-3">Member</th>
                    <th class="border-0 px-4 py-3">Position</th>
                    <th class="border-0 px-4 py-3 text-center">Status</th>
                    <th class="border-0 px-4 py-3 text-right">Action</th>
                </tr>
            </thead>
            <tbody id="functionary-table-body">
                <!-- Content loaded by JS -->
                <tr>
                    <td colspan="4" class="py-5 text-center">
                        <div class="spinner-border spinner-border-sm text-
primary mb-2" role="status"></div>
                        <p class="text-muted small font-weight-bold
mb-0">Loading Personnel...</p>
                    </td>
                </tr>
            </tbody>
        </table>
    </div>

    <!-- Mobile List -->
    <div id="functionary-mobile-list" class="d-md-none divide-y">
        <!-- Content loaded by JS -->
    </div>
</div>
</div>
</div>

<input type="hidden" name="csrfmiddlewaretoken" value="{{ csrf_token }}">

<!-- Biometric Enrollment Modal -->
<div class="modal fade" id="biometricModal" tabindex="-1" role="dialog" data-
backdrop="static" data-keyboard="false" aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered" role="document">
        <div class="modal-content secure-setup-card border-0 shadow-lg">
            <button type="button" class="close position-absolute" data-dismiss="modal"
aria-label="Close" style="right: 20px; top: 20px; z-index: 10;">
                <span aria-hidden="true">&times;</span>
            </button>
            <div class="modal-body p-3 p-md-4">

                <div class="text-center mb-3">
                    <div id="enroll-avatar-container" class="avatar-circle-modern mx-auto

```

IoT-Enabled Barangay Resident and Attendance Management System

```
mb-2" style="width: 50px; height: 50px;">
    <img id="enroll-avatar-img" src="" class="w-full h-full object-
cover rounded-full">
    </div>
    <h2 id="enroll-title" class="h5 font-weight-bold text-dark
mb-1">Biometrics</h2>
    <p id="enroll-official-name" class="text-primary small font-weight-
bold mb-1"></p>
    <p id="enroll-instruction" class="text-muted small font-weight-medium
mb-0">Place finger on sensor</p>
    </div>

    <!-- Segments -->
    <div class="d-flex mb-3 px-2" style="gap: 8px;">
        <div class="flex-fill bg-light rounded-pill overflow-hidden"
style="height: 5px;">
            <div id="seg-1" class="h-100 bg-primary transition-all"
style="width: 0%;"></div>
            </div>
            <div class="flex-fill bg-light rounded-pill overflow-hidden"
style="height: 5px;">
                <div id="seg-2" class="h-100 bg-primary transition-all"
style="width: 0%;"></div>
                </div>
                <div class="flex-fill bg-light rounded-pill overflow-hidden"
style="height: 5px;">
                    <div id="seg-3" class="h-100 bg-primary transition-all"
style="width: 0%;"></div>
                    </div>
                </div>

    <!-- Scanner Area -->
    <div class="d-flex justify-content-center mb-3">
        <div id="scanner-box" class="scanner-container d-flex align-items-
center justify-content-center bg-light border-2 rounded-xl position-relative overflow-
hidden" style="width: 120px; height: 120px; border: 2px solid #e9ecef;">
            <div id="scanner-laser" class="position-absolute bg-primary
d-none" style="left: 10%; right: 10%; height: 2px; box-shadow: 0 0 10px #007bff;
animation: scanner-line 2s infinite ease-in-out;"></div>
            <i id="fingerprint-icon" class="fas fa-fingerprint text-light"
style="font-size: 4rem; transition: color 0.3s;"></i>
            <i id="success-icon" class="fas fa-shield-check text-success
d-none" style="font-size: 4rem;"></i>
            </div>
        </div>

        <div class="controls px-2 text-center">
            <button id="enroll-scan-btn" class="btn btn-success btn-block py-2
font-weight-bold rounded-xl shadow-sm d-none">
                <span id="btn-text">Finish Setup</span>
            </button>
        </div>
    </div>
</div>
</div>
</div>

<!-- Reset Modal -->
<div class="modal fade" id="resetBiometricModal" tabindex="-1" role="dialog" aria-
```

IoT-Enabled Barangay Resident and Attendance Management System

```

hidden="true">
  <div class="modal-dialog modal-dialog-centered modal-sm" role="document">
    <div class="modal-content rounded-xl border-0 shadow-lg">
      <div class="modal-body p-4 text-center">
        <div class="bg-danger-light text-danger rounded-circle d-inline-flex
align-items-center justify-content-center mb-3" style="width: 50px; height: 50px;
background: #fff5f5;">
          <i class="fas fa-eraser fa-lg"></i>
        </div>
        <h5 class="font-weight-bold text-dark mb-1">Clear All Fingerprints?</h5>
        <p class="text-muted small mb-3" style="font-size: 11px;">This will
permanently delete <strong>all registered biometric data</strong>. All personnel must re-
register.</p>
        <div class="d-flex flex-column gap-2">
          <a href="{% url 'core:biometric_reset_all' %}" class="btn btn-danger
py-2 font-weight-bold rounded-lg shadow-sm d-block text-white text-decoration-none"
style="font-size: 12px;">
            Confirm & Clear
          </a>
          <button type="button" class="btn btn-light py-1 font-weight-bold text-
muted text-decoration-none" data-dismiss="modal" style="font-size: 12px;">
            Cancel
          </button>
        </div>
      </div>
    </div>
  </div>
</div>

<!-- Delete Single Modal -->
<div class="modal fade" id="deleteSingleModal" tabindex="-1" role="dialog" aria-
hidden="true">
  <div class="modal-dialog modal-dialog-centered modal-sm" role="document">
    <div class="modal-content rounded-xl border-0 shadow-lg">
      <div class="modal-body p-4 text-center">
        <div class="bg-warning-light text-warning rounded-circle d-inline-flex
align-items-center justify-content-center mb-3" style="width: 50px; height: 50px;
background: #fffd5f;">
          <i class="fas fa-user-minus fa-lg"></i>
        </div>
        <h5 class="font-weight-bold text-dark mb-1">Remove Registration?</h5>
        <p class="text-muted small mb-3" style="font-size: 11px;">Are you sure you
want to delete the fingerprint for <strong id="delete-target-name">this
member</strong>?</p>
        <div class="d-flex flex-column gap-2">
          <button id="confirm-single-delete" class="btn btn-warning text-white
py-2 font-weight-bold rounded-lg shadow-sm" style="font-size: 12px;">
            Confirm Removal
          </button>
          <button type="button" class="btn btn-light py-1 font-weight-bold text-
muted text-decoration-none" data-dismiss="modal" style="font-size: 12px;">
            Cancel
          </button>
        </div>
      </div>
    </div>
  </div>
</div>

<style>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
.rounded-xl { border-radius: 1.25rem !important; }
.transition-all { transition: all 0.3s ease; }
.tracking-widest { letter-spacing: 0.15em; }
.font-weight-medium { font-weight: 500; }

/* Custom Avatar */
.avatar-circle-modern {
  width: 42px;
  height: 42px;
  border-radius: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  overflow: hidden;
  background: #f8fafc;
  border: 1px solid #e2e8f0;
}

/* Status Badges */
.status-pill-modern {
  padding: 4px 12px;
  border-radius: 8px;
  font-size: 10px;
  font-weight: 800;
  text-transform: uppercase;
  border: 1px solid transparent;
}
.status-pill-online { background: #ecfdf5; color: #065f46; border-color: #d1fae5; }
.status-pill-offline { background: #f8fafc; color: #64748b; border-color: #e2e8f0; }

/* Animations */
@keyframes scanner-line {
  0%, 100% { top: 85%; }
  50% { top: 15%; }
}
.fingerprint-scanning { color: #4361ee !important; animation: pulse-icon 2s infinite; }

@keyframes pulse-icon {
  0%, 100% { transform: scale(1); opacity: 1; }
  50% { transform: scale(1.05); opacity: 0.8; }
}

/* Modal Tweaks */
.secure-setup-card { max-width: 320px; margin: auto; }
.scanner-active { border-color: #4361ee !important; background: #f5f7ff !important; }
.scanner-success { border-color: #10b981 !important; background: #f0fdf4 !important; }

/* Fix dropdown text display */
#category-select {
  height: 45px !important;
  font-size: 14px !important;
  padding-right: 35px !important;
}
.dropdown-wrapper { position: relative; }
.dropdown-wrapper::after {
  content: '\f078';
  font-family: 'Font Awesome 5 Free';
  font-weight: 900;
  position: absolute;
  right: 15px;
}
```


IoT-Enabled Barangay Resident and Attendance Management System

```
top: 50%;
transform: translateY(-50%);
pointer-events: none;
color: #94a3b8;
font-size: 10px;
}
</style>

<script>
// TOP-LEVEL NOTIFICATION SYSTEM
window.showNotification = function(message, type = 'info') {
    const container = document.querySelector('.messages-container');
    if (!container) return;

    const id = Date.now();
    const alert = document.createElement('div');
    alert.className = `alert alert-${type} animate-pop-in`;
    alert.id = `alert-js-${id}`;

    const icon = type === 'success' ? 'fa-check-circle' :
        (type === 'error' ? 'fa-exclamation-circle' : 'fa-info-circle');

    alert.innerHTML = `
        <div class="alert-content">
            <i class="fas ${icon}"></i>
            <span>${message}</span>
        </div>
        <button class="alert-close" onclick="this.parentElement.remove()"><i
class="fas fa-times"></i></button>
    `;

    container.appendChild(alert);

    // Auto remove after 5 seconds
    setTimeout(() => {
        if (alert.parentElement) alert.remove();
    }, 5000);
};

document.addEventListener('DOMContentLoaded', function () {
    const ESP32_PROXY_STATUS_URL = `${url}officials:esp32_status_proxy` %}';
    const ESP32_PROXY_START_URL = `${url}officials:esp32_start_enrollment_proxy`
%}';

    const ESP32_PROXY_STOP_URL = `${url}officials:esp32_stop_enrollment_proxy` %}';

    const categorySelect = document.getElementById('category-select');
    const tableBody = document.getElementById('functionary-table-body');
    const mobileList = document.getElementById('functionary-mobile-list');
    const personSearch = document.getElementById('person-search');

    const biometricModalEl = document.getElementById('biometricModal');
    const enrollTitle = document.getElementById('enroll-title');
    const enrollInstruction = document.getElementById('enroll-instruction');
    const scannerBox = document.getElementById('scanner-box');
    const scannerLaser = document.getElementById('scanner-laser');
    const fingerprintIcon = document.getElementById('fingerprint-icon');
    const successIcon = document.getElementById('success-icon');
    const enrollScanBtn = document.getElementById('enroll-scan-btn');
    const btnText = document.getElementById('btn-text');
    const enrollResetBtn = document.getElementById('enroll-reset-btn');
```

IoT-Enabled Barangay Resident and Attendance Management System

```
let currentOfficialId = null;
let statusInterval = null;
let currentScanCount = 0;
let lastSyncedScan = 0;
let lastState = 'standby';
let allOfficials = [];

function getInitials(name) {
  return name.split(' ').map(n => n[0]).join('').toUpperCase().substring(0, 2);
}

function renderPersonnel() {
  const query = (personSearch.value || "").trim().toLowerCase();
  const filtered = allOfficials.filter(off =>
    off.full_name.toLowerCase().includes(query) ||
    (off.position || "").toLowerCase().includes(query)
  );

  tableBody.innerHTML = '';
  mobileList.innerHTML = '';

  if (filtered.length === 0) {
    const emptyRow = `<tr><td colspan="4" class="py-5 text-center text-muted
small font-weight-bold">No results found</td></tr>`;
    tableBody.innerHTML = emptyRow;
    mobileList.innerHTML = `<div class="p-4 text-center text-muted small font-
weight-bold">No results found</div>`;
    return;
  }

  filtered.forEach(off => {
    const isReg = !!off.has_fingerprint;
    const initials = getInitials(off.full_name);
    const regLabel = isReg ? 'Registered' : 'Not registered';

    // Desktop Row
    const tr = document.createElement('tr');
    tr.innerHTML = `
      <td class="px-4 py-3 align-middle">
        <div class="d-flex align-items-center">
          <div class="avatar-circle-modern mr-3">
            
          </div>
          <div>
            <div class="font-weight-bold text-dark
small">${off.full_name}</div>
            <div class="text-muted" style="font-size: 10px;">ID:
${off.id}</div>
          </div>
        </div>
      </td>
      <td class="px-4 py-3 align-middle font-weight-bold text-secondary
small">${off.position || 'N/A'}</td>
      <td class="px-4 py-3 align-middle text-center">
        <span class="status-pill-modern ${isReg ? 'status-pill-online' :
'status-pill-offline'}">
          ${regLabel}
        </span>
      </td>
    `;
  });
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        </span>
      </td>
      <td class="px-4 py-3 align-middle text-right">
        <div class="d-flex justify-content-end align-items-center"
style="gap: 8px;">
          <button type="button" class="btn btn-sm register-trigger-btn
font-weight-bold px-3 rounded-lg ${isReg ? 'btn-success text-white border-0' : 'btn-
primary'}" data-id="${off.id}" ${isReg ? 'disabled' : ''}>
            ${isReg ? 'Registered' : 'Register'}
          </button>
          ${isReg ? `
            <button type="button" class="btn btn-sm btn-outline-danger
border-0 delete-single-trigger rounded-lg" data-id="${off.id}" data-
name="${off.full_name}" title="Delete Fingerprint">
              <i class="fas fa-trash-alt"></i>
            </button>
          ` : ''}
        </div>
      </td>
    `;
    tableBody.appendChild(tr);

    // Mobile Card
    const mobileCard = document.createElement('div');
    mobileCard.className = 'p-3 border-bottom';
    mobileCard.innerHTML = `
      <div class="d-flex align-items-center justify-content-between mb-2">
        <div class="d-flex align-items-center">
          <div class="avatar-circle-modern mr-2" style="width: 32px;
height: 32px;">
            
          </div>
          <div class="font-weight-bold text-dark
small">${off.full_name}</div>
        </div>
        <span class="status-pill-modern ${isReg ? 'status-pill-online' :
'status-pill-offline'}">${regLabel}</span>
      </div>
      <div class="d-flex align-items-center justify-content-between">
        <span class="text-muted" style="font-size: 10px;">${off.position
|| 'N/A'}</span>
        <div class="d-flex align-items-center" style="gap: 12px;">
          ${isReg ? `
            <span class="text-success small font-weight-
bold">Registered</span>
            <button type="button" class="btn btn-link p-0 text-danger
delete-single-trigger" data-id="${off.id}" data-name="${off.full_name}">
              <i class="fas fa-trash-alt" style="font-size:
12px;"></i>
            </button>
          ` : `
            <button class="btn btn-sm btn-link font-weight-bold p-0
text-primary register-trigger-btn" data-id="${off.id}">Register &rsaquo;</button>`}
        </div>
      </div>
    `;
    mobileList.appendChild(mobileCard);
  });
}

```

IoT-Enabled Barangay Resident and Attendance Management System

```
function loadPersonnel(category) {
  fetch(`/officials/api/by-category/?category=${category}`, { credentials:
'same-origin' })
    .then(res => res.json())
    .then(data => {
      if (data.status === 'success') {
        allOfficials = data.officials || [];
        renderPersonnel();
      }
    });
}

function updateModalUI(scanNum, message, status = 'standby') {
  currentScanCount = scanNum;
  enrollInstruction.textContent = message;
  enrollInstruction.style.color = (status === 'error') ? '#dc3545' : '#6c757d';

  for (let i = 1; i <= 3; i++) {
    const seg = document.getElementById(`seg-${i}`);
    if (seg) seg.style.width = (i <= scanNum) ? '100%' : '0%';
  }

  if (scanNum === 3) {
    enrollTitle.textContent = "Registration Complete";
    scannerBox.className = "scanner-container scanner-success d-flex align-
items-center justify-content-center border-2 rounded-xl position-relative overflow-
hidden";

    fingerprintIcon.classList.add('d-none');
    successIcon.classList.remove('d-none');
    scannerLaser.classList.add('d-none');
    btnText.textContent = "Finish Setup";
    enrollScanBtn.disabled = false;
    enrollScanBtn.className = "btn btn-success btn-block py-2 font-weight-bold
rounded-xl shadow-sm";
  } else {
    const isScanning = (status === 'enrolling' || status === 'detected');
    enrollTitle.textContent = "Biometrics";
    scannerBox.className = "scanner-container d-flex align-items-center
justify-content-center bg-light border-2 rounded-xl position-relative overflow-hidden" +
(isScanning ? ' scanner-active' : '');
    fingerprintIcon.classList.remove('d-none');
    successIcon.classList.add('d-none');
    fingerprintIcon.className = "fas fa-fingerprint " + (isScanning ?
'fingerprint-scanning' : 'text-light');
    scannerLaser.classList.toggle('d-none', !isScanning);

    // Hide button during scanning and error states
    enrollScanBtn.className = "d-none";
  }
}

function completeEnrollmentUI() {
  updateModalUI(3, 'Personnel registered', 'success');
  loadPersonnel(categorySelect.value);
}

function tryRegisterFallback() {
  if (!currentOfficialId || lastState === 'success') return;
  fetch(`/officials/${currentOfficialId}/enrollment/register/`, { credentials:
```

IoT-Enabled Barangay Resident and Attendance Management System

```

'same-origin' })
    .then(res => res.json().then(body => ({ ok: res.ok, body })))
    .then(({ ok, body }) => {
        if (ok && body.status === 'success' && lastState !== 'success') {
            if (statusInterval) clearInterval(statusInterval);
            statusInterval = null;
            lastState = 'success';
            completeEnrollmentUI();
        }
    })
    .catch(() => {});
}

function startMonitoring() {
    if (statusInterval) clearInterval(statusInterval);
    lastSyncedScan = 0;
    statusInterval = setInterval(() => {
        fetch(ESP32_PROXY_STATUS_URL, { credentials: 'same-origin' })
            .then(res => res.json())
            .then(data => {
                const scanCount = data.current_scan || 0;
                const status = data.state || 'standby';
                let message = data.last_error || (status === 'enrolling' ? 'Place
finger on sensor' : 'Ready');

                // DUPLICATE LOOKUP: If already registered, find the owner's name
                if (status === 'auth_failed' && message.includes("Already
registered")) {
                    const match = message.match(/ID: (\d+)/);
                    if (match && match[1]) {
                        const dupId = match[1];
                        if (!window._lastDupId || window._lastDupId !== dupId) {
                            window._lastDupId = dupId;
                            fetch(`/residents/fingerprint-lookup/?id=${dupId}`)
                                .then(res => res.json())
                                .then(dupData => {
                                    if (dupData.status === 'success') {
                                        const betterMessage = `Fingerprint already
registered by ${dupData.name}`;

                                        // NEW: Close modal and show top-level
                                        notification
                                        if (window.$)
                                        window.showNotification(betterMessage,
                                        "error");

                                        // Stop polling and return early
                                        clearInterval(statusInterval);
                                        return;
                                    }
                                })
                            };
                        }
                    }
                    return; // Don't proceed to updateModalUI yet
                }

                updateModalUI(scanCount, message, status);
                if (scanCount > 0 && scanCount > lastSyncedScan) {
                    const csrfToken =

```

IoT-Enabled Barangay Resident and Attendance Management System

```

document.querySelector('[name=csrfmiddlewaretoken]').value;
        fetch(`/officials/${currentOfficialId}/enrollment/sync-
progress/`, {
            method: 'POST',
            credentials: 'same-origin',
            headers: { 'Content-Type': 'application/json',
'X-CSRFToken': csrfToken },
            body: JSON.stringify({ current_scan: scanCount })
        })
        .then(res => res.json()).then(body => ({ ok: res.ok, body
})))
        .then(({ ok, body }) => {
            if (!ok || body.status !== 'success') {
                if (scanCount === 3) tryRegisterFallback();
                return;
            }
            lastSyncedScan = Math.max(lastSyncedScan, scanCount);
            const backendDone = body.registration_complete ===
true || (body.current_scan || 0) >= 3;
            if (scanCount === 3 && backendDone) {
                if (lastState === 'success') return;
                lastState = 'success';
                if (statusInterval) clearInterval(statusInterval);
                statusInterval = null;
                completeEnrollmentUI();
            } else if (scanCount === 3 && !backendDone) {
                tryRegisterFallback();
            }
        })
        .catch(() => {
            if (scanCount === 3) tryRegisterFallback();
        });
    }
    if (scanCount < 3) lastState = status;
})
    .catch(err => console.error("Monitor error:", err));
}, 800);
}

personSearch.addEventListener('input', renderPersonnel);
categorySelect.addEventListener('change', () =>
loadPersonnel(categorySelect.value));

document.addEventListener('click', (e) => {
    const btn = e.target.closest('.register-trigger-btn');
    if (btn && !btn.disabled) {
        currentOfficialId = btn.getAttribute('data-id');
        const originalContent = btn.innerHTML;
        btn.disabled = true;
        btn.innerHTML = '<i class="fas fa-circle-notch fa-spin"></i>';
        fetch(`/officials/${currentOfficialId}/enrollment/start/`, { credentials:
'same-origin' })
            .then(res => res.json())
            .then(data => {
                if (data.status === 'success') {
                    return fetch(`${ESP32_PROXY_START_URL}?id=${data.slot_id}`, {
method: 'POST', credentials: 'same-origin' });
                } else throw new Error(data.message);
            })
            .then(espRes => {

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        btn.disabled = false;
        btn.innerHTML = originalContent;
        if (!espRes.ok) {
            return espRes.json().then(errJson => {
                const msg = (errJson && errJson.message)
                    ? errJson.message
                    : ('Scanner error (HTTP ' + espRes.status + ')');
                throw new Error(msg);
            }).catch(() => {
                throw new Error('Scanner rejected enrollment (HTTP ' +
espRes.status + ')');
            });
        }
        const off = allOfficials.find(o => o.id == currentOfficialId);
        if (off) {
            document.getElementById('enroll-official-name').textContent =
off.full_name;
            document.getElementById('enroll-avatar-img').src = off.photo
|| `https://api.dicebear.com/7.x/identicon/svg?seed=resident_${off.resident_id ||
off.id}&backgroundcolor=ffffff`;
        }

        updateModalUI(0, "Place finger on sensor", "enrolling");
        if (window.$) $('#biometricModal').modal('show');
        startMonitoring();
    })
    .catch(err => {
        btn.disabled = false;
        btn.innerHTML = originalContent;
        alert("Error: " + err.message);
    });
}

// Delete Single Logic
const delBtn = e.target.closest('.delete-single-trigger');
if (delBtn) {
    const id = delBtn.getAttribute('data-id');
    const name = delBtn.getAttribute('data-name');
    document.getElementById('delete-target-name').textContent = name;
    document.getElementById('confirm-single-delete').setAttribute('data-id',
id);

    if (window.$) $('#deleteSingleModal').modal('show');
}
});

document.getElementById('confirm-single-delete').addEventListener('click',
function() {
    const id = this.getAttribute('data-id');
    const btn = this;
    btn.disabled = true;
    btn.innerHTML = '<i class="fas fa-circle-notch fa-spin mr-2"></i>
Removing...';

    const csrfToken = document.querySelector('[name=csrfmiddlewaretoken]').value;
    fetch(`/officials/${id}/enrollment/remove/`, {
        method: 'POST',
        credentials: 'same-origin',
        headers: { 'X-CSRFToken': csrfToken }
    })
    .then(res => res.json())
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.then(data => {
  if (window.$) $('#deleteSingleModal').modal('hide');
  btn.disabled = false;
  btn.innerHTML = 'Confirm Removal';
  if (data.status === 'success') {
    loadPersonnel(categorySelect.value);
  } else {
    alert("Error: " + data.message);
  }
})
.catch(err => {
  alert("Request failed");
  btn.disabled = false;
  btn.innerHTML = 'Confirm Removal';
});
});

enrollScanBtn.addEventListener('click', () => {
  if (btnText.textContent === "Finish Setup") {
    if (window.$) $('#biometricModal').modal('hide');
  } else if (btnText.textContent === "Retry Scan") {
    fetch(ESP32_PROXY_START_URL, { method: 'POST', credentials: 'same-origin'
})
      .then(() => {
        updateModalUI(currentScanCount, "Place finger on sensor",
"enrolling");
        startMonitoring();
      });
  }
});

if (window.$) {
  $('#biometricModal').on('hidden.bs.modal', function () {
    clearInterval(statusInterval);
    fetch(ESP32_PROXY_STOP_URL, { method: 'POST', credentials: 'same-origin'
}).catch(() => {});
  });
}

loadPersonnel(categorySelect.value);
});
</script>

{% endblock %}
```


IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/confirm_delete.html

```
{% extends 'base.html' %}

{% block title %}Remove Official - {{ official.full_name }}{% endblock %}

{% block content %}
<div class="card" style="max-width: 500px; margin: 100px auto; text-align: center;
padding: 40px;">
  <div class="alert-icon" style="font-size: 60px; color: var(--danger); margin-bottom:
20px;">
    <i class="fas fa-user-minus"></i>
  </div>
  <h2>Remove from Officials?</h2>
  <p style="margin: 20px 0; color: var(--text-muted);">
    Are you sure you want to remove <strong>{{ official.full_name }}</strong> from the
list of Barangay
    Officials?
    {% if official.resident %}
      The resident's personal record will remain intact, but they will no longer be listed
as an active or inactive
      official.
    {% else %}
      This official record will be removed.
    {% endif %}
  </p>

  <form method="POST">
    {% csrf_token %}
    <div class="action-buttons" style="display: flex; gap: 15px; justify-content: center;
margin-top: 30px;">
      <button type="submit" class="btn btn-danger btn-lg">
        <i class="fas fa-trash"></i> Yes, Remove Official
      </button>
      <a href="{% url 'officials:official_view' official.pk %}" class="btn btn-outline
btn-lg">
        Cancel
      </a>
    </div>
  </form>
</div>

<style>
  .btn-danger {
    background: var(--danger);
    color: white;
    border: none;
  }

  .btn-danger:hover {
    background: #dc2626;
  }
</style>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/form.html

```
{% extends 'base.html' %}

{% block title %}{% if editing %}Edit Official{% else %}Appoint Official / Staff{% endif %} - Barangay Management System{% endblock %}

{% block extra_head %}
<!-- Select2 CSS -->
<link href="https://cdn.jsdelivr.net/npm/select2@4.1.0-rc.0/dist/css/select2.min.css"
rel="stylesheet" />
<style>
/* Select2 Custom Styling to match modern theme */
.select2-container--default .select2-selection--single {
    height: 48px !important;
    border: 1.5px solid var(--border) !important;
    border-radius: var(--radius-md) !important;
    background-color: var(--bg-main) !important;
    display: flex;
    align-items: center;
}
.select2-container--default .select2-selection--single .select2-selection__rendered {
    color: var(--text-main) !important;
    padding-left: 16px !important;
    font-weight: 500;
}
.select2-container--default .select2-selection--single .select2-selection__arrow {
    height: 46px !important;
    right: 12px !important;
}
.select2-dropdown {
    border: 1.5px solid var(--primary-light) !important;
    border-radius: var(--radius-md) !important;
    box-shadow: var(--shadow-lg) !important;
    z-index: 10001;
}
.select2-search__field {
    border-radius: var(--radius-sm) !important;
    padding: 8px 12px !important;
}
.select2-results__option--highlighted[aria-selected] {
    background-color: var(--primary) !important;
}

/* Select2 Clear Button styling */
.select2-container--default .select2-selection--single .select2-selection__clear {
    height: 48px;
    margin-right: 25px;
    color: var(--text-muted);
    font-size: 22px;
    display: flex;
    align-items: center;
    transition: var(--transition);
}
.select2-container--default .select2-selection--single .select2-selection__clear:hover {
    color: var(--danger);
    transform: scale(1.1);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.modern-form {
  padding: 40px;
}

.form-section {
  margin-bottom: 40px;
}

.form-footer {
  border-top: 1px solid var(--border);
  padding-top: 30px;
  text-align: right;
}

.account-setup-box {
  background: var(--primary-light);
  border: 1.5px dashed var(--primary);
  border-radius: var(--radius-lg);
  padding: 24px;
  margin-top: 10px;
}
</style>
{% endblock %}

{% block content %}
{% url 'officials:list' as officials_list_url %}
{% include 'components/back_button.html' with back_url=officials_list_url
back_label='Back' %}
<div class="page-header">
  <div class="page-title">
    <h1>{% if editing %}Edit Official Record{% else %}Appoint Official / Staff{% endif
%}</h1>
    <span>{% if editing %}Update the record for {{ official.resident.full_name }}{% else
%}Assign a resident to any barangay position ? official or support staff{% endif %}</span>
  </div>
</div>

<div class="card">
  <form method="POST" class="modern-form">
    {% csrf_token %}
    <div class="form-grid">
      <div class="form-section">
        <h3>Appointment Details</h3>
        <div class="form-row">
          <div class="form-group col-6">
            <label>Resident Selection*</label>
            {% if editing %}
              <div class="form-control" style="background-color: var(--border-light);
display: flex; align-items: center; opacity: 0.8; cursor: not-allowed;">
                {{ official.resident.full_name }}
              </div>
            <input type="hidden" name="resident" value="{{ official.resident.pk }}">
            {% else %}
              <select name="resident" id="id_resident_select" class="form-control select2"
required>
                <option value=""></option>
                {% for r in residents_data %}
                  <option value="{{ r.pk }}" data-has-account="{{
r.has_account|yesno:'true,false' }}" data-username="{{ r.username }}">{{ r.name
```

IoT-Enabled Barangay Resident and Attendance Management System

```

}}</option>
    {% endfor %}
</select>
{% endif %}
</div>
<div class="form-group col-6">
    <label>Position*</label>
    <select name="position" id="id_position_select" class="form-control" required>
        <option value="">? Select a Position ?</option>
        {% for code, label in positions %}
        <option value="{{ code }}" {% if official.position == code %}selected{%
endif %}>{{ label }}</option>
        {% endfor %}
    </select>
    <small id="position-hint" class="text-muted" style="font-size:11px; margin-
top:4px; display:block;"></small>
</div>
</div>

<div class="form-row">
    <div class="form-group col-6">
        <label for="id_committee_field">Committee / Assignment
        <span id="committee-badge" style="display:none; font-size:10px; font-
weight:700; color:#94a3b8; background:#f1f5f9; border-radius:4px; padding:1px 6px; margin-
left:6px;">N/A for this position</span>
        </label>
        <input type="text" name="committee" id="id_committee_field" value="{{
official.committee }}" class="form-control"
        placeholder="e.g. Health, Finance, Infrastructure">
    </div>
    <div class="form-group col-6">
        <label>Employee / Item ID</label>
        <input type="text" name="employee_id" value="{{ official.employee_id }}"
class="form-control" placeholder="Optional identifier">
    </div>
</div>
</div>

<div class="form-section">
    <h3>Term & Compensation</h3>
    <div class="form-row">
        <div class="form-group col-4">
            <label>Term Start Date*</label>
            <input type="date" name="term_start" value="{{
official.term_start|date:'Y-m-d' }}" class="form-control"
            required>
        </div>
        <div class="form-group col-4">
            <label>Term End Date</label>
            <input type="date" name="term_end" value="{{ official.term_end|date:'Y-m-d'
}}" class="form-control">
        </div>
        <div class="form-group col-4">
            <label>Salary / Honorarium (?)</label>
            <input type="number" step="0.01" name="salary" value="{{
official.salary|default:0 }}" class="form-control">
        </div>
    </div>
    {% if editing %}
    <div class="form-row">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div class="form-group col-6">
  <label>Current Status*</label>
  <select name="status" class="form-control">
    <option value="active" {% if official.status == 'active' %}>Active</option>
    <option value="inactive" {% if official.status == 'inactive' %}>Inactive</option>
  </select>
</div>
</div>
{% endif %}
</div>

<div class="form-section" id="account-setup-section" style="display: none;">
  <div class="account-setup-box">
    <h3><i class="fas fa-user-shield"></i> System Account Setup</h3>
    <p class="text-muted" id="account-setup-note" style="margin-bottom: 20px; font-size: 13px;">
      Set up the login credentials for this official. These will be used for manual password login.
    </p>
    <div class="form-row">
      <div class="form-group col-6">
        <label>System Username</label>
        <input type="text" name="username" id="id_username_field" class="form-control account-field" placeholder="Create username">
        <small class="text-info" id="existing-user-msg" style="display: none;">This resident already has an account. Entering a new username/password will update their credentials.</small>
      </div>
      <div class="form-group col-6">
        <label>Password</label>
        <input type="password" name="password" class="form-control account-field" placeholder="Create or update password">
      </div>
    </div>
  </div>
</div>
</div>
</div>
</div>

<div class="form-footer">
  <button type="submit" class="btn btn-primary btn-lg">
    <i class="fas fa-save"></i> Save Appointment
  </button>
</div>
</form>
</div>
{% endblock %}

{% block extra_js %}
<!-- Select2 JS -->
<script
src="https://cdn.jsdelivr.net/npm/select2@4.1.0-rc.0/dist/js/select2.min.js"></script>
<script>
  document.addEventListener('DOMContentLoaded', function () {

    // ?? Positions that do NOT have committee / assignment relevance ??
    const NO_COMMITTEE_POSITIONS = new Set([
      'tanod', 'bhw', 'nutrition_scholar', 'day_care_worker',
      'lupon', 'clerk', 'staff', 'captain', 'treasurer'
    ])
  })
</script>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
]);

const POSITION_HINTS = {
  'captain':      'Punong Barangay ? the chief executive of the barangay.',
  'kagawad':      'Sangguniang Barangay member. Assign a committee (e.g. Health,
Finance).',
  'secretary':    'Barangay Secretary ? responsible for records and minutes.',
  'treasurer':    'Barangay Treasurer ? manages finances.',
  'sk_chairman':  'SK Chairperson ? youth council leader.',
  'sk_kagawad':   'SK Council member. Assign a committee if applicable.',
  'tanod':        'Barangay Tanod ? peace & order volunteer.',
  'bhw':          'Barangay Health Worker ? community health aide.',
  'nutrition_scholar': 'Barangay Nutrition Scholar (BNS) ? nutrition program
implementor.',
  'day_care_worker': 'Day Care Worker ? manages the barangay day care center.',
  'lupon':        'Lupon member ? barangay justice/mediation.',
  'clerk':        'Barangay Clerk ? administrative support.',
  'staff':        'General Barangay Staff.',
};

// ?? Initialize Select2 for resident picker ??
if (jQuery('#id_resident_select').length) {
  jQuery('#id_resident_select').select2({
    placeholder: "Search resident by name or purok...",
    allowClear: true,
    width: '100%'
  });
}

// ?? Committee field disable/enable logic ??
const positionSelect = document.getElementById('id_position_select');
const committeeField = document.getElementById('id_committee_field');
const committeeBadge = document.getElementById('committee-badge');
const positionHint = document.getElementById('position-hint');

function updateCommitteeState() {
  if (!positionSelect) return;
  const pos = positionSelect.value;
  const noCommittee = NO_COMMITTEE_POSITIONS.has(pos);

  if (committeeField) {
    committeeField.disabled = noCommittee;
    committeeField.style.opacity = noCommittee ? '0.45' : '1';
    committeeField.style.cursor = noCommittee ? 'not-allowed' : '';
    committeeField.style.backgroundColor = noCommittee ? 'var(--border-light,
#f1f5f9)' : '';
    if (noCommittee) committeeField.value = '';
  }
  if (committeeBadge) {
    committeeBadge.style.display = noCommittee ? 'inline' : 'none';
  }
  if (positionHint) {
    positionHint.textContent = pos ? (POSITION_HINTS[pos] || '') : '';
  }
}

if (positionSelect) {
  positionSelect.addEventListener('change', updateCommitteeState);
  updateCommitteeState(); // run once on load
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
// ?? Resident picker ? show account-setup section ??
const residentSelect = jQuery('select[name="resident"]');
const setupSection = document.getElementById('account-setup-section');

{% if not editing %}
residentSelect.on('change', function () {
  const selectedOption = this.options[this.selectedIndex];
  if (!selectedOption || !selectedOption.value) {
    if (setupSection) setupSection.style.display = 'none';
    return;
  }

  const data = jQuery(selectedOption).data() || selectedOption.dataset;
  const hasAccount = String(data.hasAccount) === 'true' || data.hasAccount === true;
  const username = data.username || '';

  if (setupSection) setupSection.style.display = 'block';

  const usernameField = document.getElementById('id_username_field');
  const existingMsg = document.getElementById('existing-user-msg');
  const setupNote = document.getElementById('account-setup-note');

  if (hasAccount) {
    if (usernameField) usernameField.value = username;
    if (existingMsg) existingMsg.style.display = 'block';
    if (setupNote) setupNote.innerHTML = "This resident already has an account.  
You can update their credentials below if needed.";
  } else {
    if (usernameField) usernameField.value = '';
    if (existingMsg) existingMsg.style.display = 'none';
    if (setupNote) setupNote.innerHTML = "Set up the login credentials for this  
official. These will be used for manual password login.";
  }
});
{% endif %}
});
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/list.html

```
{% extends 'base.html' %}

{% block title %}Officials - Barangay Management System{% endblock %}

{% block extra_head %}
<style>
  /* Page Header */
  .officials-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 14px;
  }
  .officials-header h1 {
    font-size: 1.1rem;
    font-weight: 800;
    color: #0f172a;
    margin: 0 0 2px 0;
  }
  .officials-header p {
    font-size: 11px;
    color: #64748b;
    margin: 0;
  }
  .header-actions {
    display: flex;
    gap: 8px;
    align-items: center;
  }
  .header-actions .btn {
    font-size: 11px;
    font-weight: 700;
    padding: 6px 14px;
    border-radius: 8px;
    display: inline-flex;
    align-items: center;
    gap: 6px;
    text-decoration: none;
    transition: all 0.2s;
    border: none;
    cursor: pointer;
  }
  .btn-primary-sm { background: #306844; color: white; }
  .btn-primary-sm:hover { background: #255235; color: white; }
  .btn-secondary-sm { background: #f1f5f9; color: #475569; }
  .btn-secondary-sm:hover { background: #e2e8f0; color: #1e293b; }

  /* Filter Bar */
  .filter-bar {
    background: #f8fafc;
    border: 1px solid #e2e8f0;
    border-radius: 10px;
    padding: 10px 14px;
    margin-bottom: 14px;
    display: flex;
    gap: 10px;
    flex-wrap: wrap;
  }
</style>
{% endblock %}
```


IoT-Enabled Barangay Resident and Attendance Management System

```
    align-items: center;
}
.filter-bar select {
    font-size: 11px;
    font-weight: 700;
    color: #334155;
    background: white;
    border: 1px solid #e2e8f0;
    border-radius: 7px;
    padding: 5px 10px;
    outline: none;
    cursor: pointer;
}

/* Main Card */
.officials-card {
    background: white;
    border-radius: 12px;
    border: 1px solid #e2e8f0;
    overflow: hidden;
    box-shadow: 0 1px 3px rgba(0,0,0,0.06);
}

/* Category */
.category-header {
    display: flex;
    align-items: center;
    gap: 8px;
    background: #f8fafc;
    border-bottom: 1px solid #e2e8f0;
    padding: 8px 16px;
    font-size: 10px;
    font-weight: 900;
    text-transform: uppercase;
    letter-spacing: 0.07em;
    color: #475569;
}
.category-header i { font-size: 12px; color: #94a3b8; }

/* Table */
table { width: 100%; border-collapse: collapse; margin: 0; }
thead th {
    background: #f8fafc;
    color: #94a3b8;
    font-size: 9px;
    font-weight: 900;
    text-transform: uppercase;
    letter-spacing: 0.08em;
    padding: 7px 14px;
    text-align: left;
    border-bottom: 1px solid #f1f5f9;
}
tbody tr { border-bottom: 1px solid #f8fafc; transition: background 0.15s; }
tbody tr:last-child { border-bottom: none; }
tbody tr:hover { background: #f8fafc; }
td {
    padding: 8px 14px;
    font-size: 12px;
    color: #334155;
    vertical-align: middle;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

/* Profile cell */
.official-profile {
  display: flex;
  align-items: center;
  gap: 10px;
}
.official-avatar {
  width: 32px;
  height: 32px;
  border-radius: 50%;
  overflow: hidden;
  border: 1.5px solid #e2e8f0;
  flex-shrink: 0;
  background: #f1f5f9;
}
.official-avatar img { width: 100%; height: 100%; object-fit: cover; }
.official-info strong {
  display: block;
  font-size: 12px;
  font-weight: 700;
  color: #0f172a;
  line-height: 1.3;
}
.official-info span {
  font-size: 10px;
  color: #94a3b8;
  font-weight: 500;
}

/* Status Badges */
.status-badge {
  display: inline-flex;
  align-items: center;
  padding: 3px 8px;
  border-radius: 5px;
  font-size: 9px;
  font-weight: 900;
  text-transform: uppercase;
  letter-spacing: 0.05em;
}
.badge-active { background: #ecfdf5; color: #059669; border: 1px solid #a7f3d0; }
.badge-inactive { background: #f8fafc; color: #94a3b8; border: 1px solid #e2e8f0; }

/* Action Button */
.btn-action {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  width: 26px;
  height: 26px;
  border-radius: 6px;
  color: #64748b;
  background: #f1f5f9;
  text-decoration: none;
  transition: all 0.15s;
  font-size: 11px;
}
.btn-action:hover { background: #e2e8f0; color: #0f172a; }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
/* Empty state */
.empty-state { padding: 24px; text-align: center; color: #94a3b8; font-size: 12px; font-weight: 600; }

/* Category section divider */
.category-section:not(:last-child) { border-bottom: 1px solid #e2e8f0; }
</style>
{% endblock %}

{% block content %}

<div class="officials-header">
  <div>
    <h1>Barangay Officials & Staff</h1>
    <p>Manage positions and terms of service</p>
  </div>
  <div class="header-actions">
    <a href="{% url 'officials:org_chart' %}" class="btn btn-secondary-sm">
      <i class="fas fa-sitemap"></i> Org Chart
    </a>
    <a href="{% url 'officials:add' %}" class="btn btn-primary-sm">
      <i class="fas fa-plus"></i> Appoint Official / Staff
    </a>
  </div>
</div>

<div class="filter-bar">
  <form method="get" id="filterForm" style="display:flex; gap:10px; flex-wrap:wrap;">
    <select name="status" onchange="this.form.submit();">
      <option value="active" {% if selected_status == 'active' %}>selected{% endif %}>Active Officials</option>
      <option value="inactive" {% if selected_status == 'inactive' %}>selected{% endif %}>Inactive / Past</option>
    </select>
    <select name="pos" onchange="this.form.submit();">
      <option value="all" {% if selected_pos == 'all' %}>selected{% endif %}>All Positions</option>
      {% for pos_code, pos_label in all_positions %}
      <option value="{{ pos_code }}" {% if selected_pos == pos_code %}>selected{% endif %}>{{ pos_label }}</option>
      {% endfor %}
    </select>
  </form>
</div>

<div class="officials-card">
  {% for key, category in categories.items %}
  <div class="category-section">
    <div class="category-header">
      <i class="fas {% if key == 'council' %}>fa-user-tie{% elif key == 'sk' %}>fa-users{% elif key == 'tanod' %}>fa-shield-alt{% elif key == 'health' %}>fa-file-medical{% else %}>fa-user-cog{% endif %}"></i>
      {{ category.title }}
    </div>
    <div class="table-responsive" style="overflow-x:auto;">
      <table>
        <thead>
          <tr>
            <th>Official</th>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <th>Position</th>
        <th>Term Start</th>
        <th>Term End</th>
        <th>Status</th>
        <th>Action</th>
    </tr>
</thead>
<tbody>
    {% for official in category.officials %}
    <tr>
        <td>
            <div class="official-profile">
                <div class="official-avatar">
                    {% if official.resident.photo %}
                    
                    {% else %}
                    
                    {% endif %}
                </div>
                <div class="official-info">
                    <strong>{{ official.resident.full_name }}</strong>
                    <span>{{ official.employee_id|default:"No ID" }}</span>
                </div>
            </div>
        </td>
        <td style="font-weight:600; color:#475569;">{{ official.get_position_display
}}</td>
        <td style="color:#64748b;">{{ official.term_start|date:"M d, Y" }}</td>
        <td style="color:#64748b;">{{ official.term_end|date:"M d,
Y"|default:"Present" }}</td>
        <td>
            <span class="status-badge {% if official.status == 'active' %}badge-active{%
else %}badge-inactive{% endif %}">
                {{ official.get_status_display }}
            </span>
        </td>
        <td>
            <a href="{% url 'officials:edit' official.pk %}" class="btn-action"
title="Edit"><i class="fas fa-edit"></i></a>
        </td>
    </tr>
    {% endfor %}
</tbody>
</table>
</div>
</div>
{% empty %}
<div class="empty-state">No officials found for this filter.</div>
{% endfor %}
</div>

{% endblock %}

```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/onboarding/approvals.html

```
{% extends 'base.html' %}

{% block title %}Onboarding Approvals - Barangay Management System{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Onboarding Approvals</h1>
    <span>Review documents and approve new official invitations</span>
  </div>
  <div class="page-actions">
    <a href="{% url 'officials:invite_official' %}" class="btn btn-primary">
      <i class="fas fa-plus"></i> Invite New Official
    </a>
  </div>
</div>

{% if messages %}
{% for message in messages %}
<div class="alert alert-{{ message.tags }}"
  style="margin-bottom: 20px; padding: 15px; border-radius: 10px; background: #e0faea;
  color: #166534; border: 1px solid #bbf7d0;">
  {{ message }}
</div>
{% endfor %}
{% endif %}

<div class="card">
  <div class="card-header">
    <h3>Pending Approvals (Requires Action)</h3>
  </div>
  <div class="table-responsive">
    <table class="table">
      <thead>
        <tr>
          <th>Name</th>
          <th>Position</th>
          <th>Documents</th>
          <th>Captain Approval</th>
          <th>Secretary Approval</th>
          <th>Action</th>
        </tr>
      </thead>
      <tbody>
        {% for invite in pending_approval %}
        <tr>
          <td>
            <strong>{{ invite.first_name }} {{ invite.last_name }}</strong><br>
            <small class="text-muted">{{ invite.phone_number }}</small>
          </td>
          <td><span class="badge badge-primary">{{ invite.get_position_display }}</span></td>
          <td>
            <div class="d-flex" style="gap: 10px; margin-top: 5px;">
              {% if invite.appointment_letter %}
              <a href="{% invite.appointment_letter.url %}" target="_blank">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

class="badge badge-info"><i
            class="fas fa-file-alt"></i> Letter</a>
        {% endif %}
        {% if invite.valid_id %}
        <a href="{ { invite.valid_id.url } }" target="_blank"
class="badge badge-info"><i
            class="fas fa-id-card"></i> ID</a>
        {% endif %}
    </div>
</td>
<td>
        {% if invite.captain_approved %}<span style="color: #10b981;"><i
            class="fas fa-check-circle"></i> Yes</span>
        {% else %}<span style="color: #ef4444;"><i class="fas fa-
clock"></i> Pending</span>{% endif %}
    </td>
<td>
        {% if invite.secretary_approved %}<span style="color: #10b981;"><i
            class="fas fa-check-circle"></i> Yes</span>
        {% else %}<span style="color: #ef4444;"><i class="fas fa-
clock"></i> Pending</span>{% endif %}
    </td>
<td>
        <form method="POST" action="{% url 'officials:onboard_approve'
invite.token %}"
            style="margin: 0;">
            {% csrf_token %}
            <button type="submit" class="btn btn-sm btn-success">
                <i class="fas fa-thumbs-up"></i> Approve
            </button>
        </form>
    </td>
</tr>
{% empty %}
<tr>
    <td colspan="6" class="text-center" style="padding: 30px;">
        <div class="empty-state">
            <i class="fas fa-check-double"
                style="font-size: 30px; color: #cbd5e1; margin-bottom:
10px;"></i>
            <p>No actions required. All uploaded documents have been
processed.</p>
        </div>
    </td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>

<div class="grid" style="display: grid; grid-template-columns: repeat(auto-fit,
minmax(280px, 1fr)); gap: 24px; margin-top: 24px;">

    <div class="card">
        <div class="card-header">
            <h3>Awaiting Documents</h3>
        </div>
        <div class="table-responsive">
            <table class="table">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

<tbody>
  {% for invite in pending_documents %}
  <tr>
    <td>
      <strong>{{ invite.first_name }} {{ invite.last_name }}</strong><br>
      <small>{{ invite.get_position_display }}</small>
    </td>
    <td class="text-right">
      <span class="badge badge-warning">Waiting for upload</span>
    </td>
  </tr>
  {% empty %}
  <tr>
    <td colspan="2" class="text-center text-muted">None</td>
  </tr>
  {% endfor %}
</tbody>
</table>
</div>
</div>

<div class="card">
  <div class="card-header">
    <h3>Pending OTP & Recent Activations</h3>
  </div>
  <div class="table-responsive">
    <table class="table">
      <tbody>
        {% for invite in pending_otp %}
        <tr style="background-color: #f8fafc;">
          <td>
            <strong>{{ invite.first_name }} {{ invite.last_name }}</strong>
          </td>
          <td class="text-right">
            <span class="badge badge-warning"><i class="fas fa-key"></i>
Pending OTP</span>
          </td>
        </tr>
        {% endfor %}
        {% for invite in recent_activated %}
        <tr>
          <td>
            <strong>{{ invite.first_name }} {{ invite.last_name }}</strong>
          </td>
          <td class="text-right">
            <span class="badge badge-success"><i class="fas fa-check"></i>
Activated</span>
          </td>
        </tr>
        {% empty %}
        <tr>
          <td colspan="2" class="text-center text-muted">None</td>
        </tr>
        {% endif %}
      </tbody>
    </table>
  </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
        </table>
    </div>
</div>

<div>

<style>
    .badge {
        padding: 4px 8px;
        border-radius: 6px;
        font-size: 12px;
        font-weight: 600;
        display: inline-block;
    }

    .badge-primary {
        background: #e0e7ff;
        color: #4338ca;
    }

    .badge-warning {
        background: #fef3c7;
        color: #d97706;
    }

    .badge-success {
        background: #d1fae5;
        color: #059669;
    }

    .badge-info {
        background: #e0f2fe;
        color: #0369a1;
        text-decoration: none;
    }

    .badge-info:hover {
        text-decoration: underline;
    }

    .empty-state {
        text-align: center;
        color: #64748b;
    }
</style>
{% endblock %}
```


IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/onboarding/invite_form.html

```
{% extends 'base.html' %}

{% block title %}Invite Official - Barangay Management System{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Invite Official</h1>
    <span>Send an onboarding link to a new barangay official</span>
  </div>
</div>

<div class="card" style="max-width: 600px; margin: 0 auto;">
  <form method="POST" class="modern-form">
    {% csrf_token %}
    <div class="form-section">
      <h3 style="margin-top: 0;">Official Details</h3>

      <div class="form-group pb-3">
        <label>First Name*</label>
        <input type="text" name="first_name" class="form-control" required
placeholder="e.g. Juan">
      </div>

      <div class="form-group pb-3">
        <label>Last Name*</label>
        <input type="text" name="last_name" class="form-control" required
placeholder="e.g. Dela Cruz">
      </div>

      <div class="form-group pb-3">
        <label>Position*</label>
        <select name="position" class="form-control" required>
          <option value="">Select Position...</option>
          {% for code, label in positions %}
            <option value="{{ code }}">{{ label }}</option>
          {% endfor %}
        </select>
      </div>

      <div class="form-group pb-3">
        <label>Mobile Number*</label>
        <input type="text" name="phone_number" class="form-control" required
placeholder="09xxxxxxxxx">
        <small class="text-muted">An OTP will be sent to this number upon dual
approval.</small>
      </div>

    </div>

    <div class="form-footer">
      <button type="submit" class="btn btn-primary btn-lg w-100">
        <i class="fas fa-paper-plane"></i> Generate Invite Link
      </button>
    </div>
  </form>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<style>
  .modern-form {
    padding: 30px;
  }

  .form-section h3 {
    font-size: 16px;
    color: var(--primary);
    margin-bottom: 20px;
    border-bottom: 1px solid var(--border);
    padding-bottom: 8px;
  }

  .pb-3 {
    padding-bottom: 1rem;
  }

  .w-100 {
    width: 100%;
  }
</style>
{% endblock %}
```

File: templates/officials/onboarding/upload_documents.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Upload Official Documents - SICO-SICO</title>
    <link rel="stylesheet" href="{% static 'css/style.css' %}"?v=2.1">
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
rel="stylesheet">
    <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css"
rel="stylesheet">
    <style>
        body {
            display: flex;
            align-items: center;
            justify-content: center;
            min-height: 100vh;
            margin: 0;
            padding: 20px;
            font-family: 'Inter', sans-serif;
            background: #f8fafc;
            overflow-x: hidden;
        }

        .auth-card {
            position: relative;
            background: var(--glass-bg, rgba(255, 255, 255, 0.7));
            backdrop-filter: blur(16px);
            -webkit-backdrop-filter: blur(16px);
            border: 1px solid var(--glass-border, rgba(255, 255, 255, 0.5));
            border-radius: var(--radius-xl, 20px);
            padding: 48px;
            width: 100%;
            max-width: 500px;
            text-align: center;
            box-shadow: 0 10px 40px rgba(0, 0, 0, 0.1);
            z-index: 10;
        }

        .header-icon {
            width: 80px;
            height: 80px;
            margin: 0 auto 24px;
            background: rgba(44, 110, 73, 0.1);
            color: #2c6e49;
            border-radius: 20px;
            display: flex;
            align-items: center;
            justify-content: center;
            font-size: 32px;
        }

        h2 {
            font-size: 24px;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        font-weight: 800;
        margin-bottom: 8px;
        color: #1f2937;
    }

    p.subtitle {
        color: #6b7280;
        font-size: 14px;
        margin-bottom: 32px;
        line-height: 1.5;
    }

    .form-group {
        text-align: left;
        margin-bottom: 24px;
    }

    .form-group label {
        display: block;
        font-weight: 600;
        margin-bottom: 8px;
        color: #374151;
        font-size: 14px;
    }

    .file-upload {
        position: relative;
        display: block;
        width: 100%;
        height: 120px;
        border: 2px dashed #9ca3af;
        border-radius: 12px;
        background: rgba(255, 255, 255, 0.5);
        transition: all 0.3s ease;
        cursor: pointer;
        overflow: hidden;
    }

    .file-upload:hover {
        border-color: #2c6e49;
        background: rgba(44, 110, 73, 0.05);
    }

    .file-upload input[type="file"] {
        position: absolute;
        width: 100%;
        height: 100%;
        top: 0;
        left: 0;
        opacity: 0;
        cursor: pointer;
    }

    .file-upload-content {
        position: absolute;
        top: 50%;
        left: 50%;
        transform: translate(-50%, -50%);
        text-align: center;
        width: 100%;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    pointer-events: none;
}

.file-upload-content i {
    font-size: 24px;
    color: #9ca3af;
    margin-bottom: 8px;
    display: block;
}

.file-upload-content span {
    font-size: 13px;
    color: #6b7280;
    font-weight: 500;
}

.btn-submit {
    width: 100%;
    padding: 16px;
    margin-top: 10px;
    font-size: 16px;
    font-weight: 700;
    background: #2c6e49;
    color: white;
    border: none;
    border-radius: 12px;
    cursor: pointer;
    transition: all 0.3s ease;
}

.btn-submit:hover {
    transform: translateY(-2px);
    box-shadow: 0 8px 20px rgba(44, 110, 73, 0.25);
}

</style>
</head>

<body>
    <!-- Background elements common in SICO login -->
    <div class="auth-background"
        style="position:fixed; top:0; left:0; width:100%; height:100%; z-index:1;
        overflow:hidden;">
        <div
            style="position:absolute; width:600px; height:600px; background:radial-
            gradient(circle, rgba(44,110,73,0.1) 0%, transparent 70%); top:-200px; left:-200px;
            border-radius:50%;">
        </div>
        <div
            style="position:absolute; width:500px; height:500px; background:radial-
            gradient(circle, rgba(79,70,229,0.05) 0%, transparent 70%); bottom:-100px; right:-100px;
            border-radius:50%;">
        </div>
    </div>

    <div class="auth-card">
        <div class="header-icon">
            <i class="fas fa-file-signature"></i>
        </div>
        <h2>Welcome, {{ invite.first_name }}!</h2>
    </div>
</body>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<p class="subtitle">To continue your onboarding as <strong>{{
invite.get_position_display }}</strong>, please
    upload the required verification documents.</p>

<form method="POST" enctype="multipart/form-data">
    {% csrf_token %}

    <div class="form-group">
        <label>1. Appointment Letter</label>
        <div class="file-upload" id="upload-appointment">
            <input type="file" name="appointment_letter" required
accept="image/*, .pdf"
                onchange="updateLabel(this, 'label-appointment')">
            <div class="file-upload-content">
                <i class="fas fa-cloud-upload-alt"></i>
                <span id="label-appointment">Click or drag file here</span>
            </div>
        </div>
    </div>

    <div class="form-group">
        <label>2. Valid Government ID</label>
        <div class="file-upload" id="upload-id">
            <input type="file" name="valid_id" required accept="image/*, .pdf"
                onchange="updateLabel(this, 'label-id')">
            <div class="file-upload-content">
                <i class="fas fa-id-card"></i>
                <span id="label-id">Click or drag file here</span>
            </div>
        </div>
    </div>

    <button type="submit" class="btn-submit">
        <i class="fas fa-check-circle"></i> Submit Documents
    </button>
</form>
</div>

<script>
function updateLabel(input, labelId) {
    const label = document.getElementById(labelId);
    if (input.files && input.files.length > 0) {
        label.textContent = input.files[0].name;
        label.parentElement.parentElement.style.borderColor = '#2c6e49';
        label.parentElement.parentElement.style.background = 'rgba(44, 110, 73,
0.1)';
    }
}
</script>
{% if messages %}
<div class="messages-container">
    {% for message in messages %}
    <div class="alert alert-{{ message.tags }}" id="alert-{{ forloop.counter }}">
        <div class="alert-content">
            {% if message.tags == 'success' %}<i class="fas fa-check-circle"></i>
            {% elif message.tags == 'error' %}<i class="fas fa-exclamation-circle"></i>
            {% elif message.tags == 'warning' %}<i class="fas fa-exclamation-triangle"></i>
            {% else %}<i class="fas fa-info-circle"></i>{% endif %}
            <span>{{ message }}</span>
        </div>
    </div>
    {% endfor %}
</div>
{% endif %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        </div>
        <button class="alert-close" onclick="this.parentElement.remove()"><i class="fas
fa-times"></i></button>
    </div>
    {% endfor %}
</div>
{% endif %}
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script src="{% static 'js/app.js' %}"></script>
</body>

</html>
```

File: templates/officials/onboarding/verify_otp.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Verify OTP & Activate - SICO-SICO</title>
    <link rel="stylesheet" href="{% static 'css/style.css' %}"?v=2.1">
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
rel="stylesheet">
    <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css"
rel="stylesheet">
    <style>
        body {
            display: flex;
            align-items: center;
            justify-content: center;
            min-height: 100vh;
            margin: 0;
            padding: 20px;
            font-family: 'Inter', sans-serif;
            background: #f8fafc;
        }

        .auth-card {
            background: var(--glass-bg, rgba(255, 255, 255, 0.7));
            backdrop-filter: blur(16px);
            border: 1px solid var(--glass-border, rgba(255, 255, 255, 0.5));
            border-radius: var(--radius-xl, 20px);
            padding: 48px;
            width: 100%;
            max-width: 450px;
            text-align: center;
            box-shadow: 0 10px 40px rgba(0, 0, 0, 0.1);
            z-index: 10;
        }

        .header-icon {
            width: 70px;
            height: 70px;
            margin: 0 auto 20px;
            background: #e0f2fe;
            color: #0284c7;
            border-radius: 50%;
            display: flex;
            align-items: center;
            justify-content: center;
            font-size: 28px;
            animation: pulse 2s infinite;
        }

        @keyframes pulse {
            0% {
                box-shadow: 0 0 0 0 rgba(2, 132, 199, 0.4);
            }
        }
    </style>
</head>
```


IoT-Enabled Barangay Resident and Attendance Management System

```
70% {
    box-shadow: 0 0 0 15px rgba(2, 132, 199, 0);
}

100% {
    box-shadow: 0 0 0 0 rgba(2, 132, 199, 0);
}
}

h2 {
    font-size: 22px;
    font-weight: 800;
    color: #1f2937;
    margin-bottom: 8px;
}

p.subtitle {
    color: #6b7280;
    font-size: 14px;
    margin-bottom: 30px;
}

.form-group {
    text-align: left;
    margin-bottom: 20px;
}

.form-group label {
    display: block;
    font-weight: 600;
    font-size: 13px;
    margin-bottom: 6px;
    color: #374151;
}

.form-control {
    width: 100%;
    padding: 14px;
    border: 1px solid #d1d5db;
    border-radius: 12px;
    background: rgba(255, 255, 255, 0.8);
    font-size: 14px;
    transition: all 0.3s ease;
}

.form-control:focus {
    outline: none;
    border-color: #0284c7;
    box-shadow: 0 0 0 3px rgba(2, 132, 199, 0.1);
}

.otp-input {
    text-align: center;
    font-size: 24px;
    font-weight: 700;
    letter-spacing: 10px;
}

.btn-submit {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
width: 100%;
padding: 16px;
margin-top: 20px;
font-size: 16px;
font-weight: 700;
background: #2c6e49;
color: white;
border: none;
border-radius: 12px;
cursor: pointer;
transition: all 0.3s;
}

.btn-submit:hover {
  transform: translateY(-2px);
  box-shadow: 0 8px 20px rgba(44, 110, 73, 0.25);
}

</style>
</head>

<body>
  <div class="auth-card">
    <div class="header-icon">
      <i class="fas fa-shield-alt"></i>
    </div>
    <h2>Final Step: Verification</h2>
    <p class="subtitle">Enter the 6-digit OTP sent to your registered mobile number
    ending in <strong>{{
      invite.phone_number|slice:"-4:" }}</strong>.</p>

    <form method="POST">
      {% csrf_token %}
      <div class="form-group">
        <label>OTP Code</label>
        <input type="text" name="otp" class="form-control otp-input" maxlength="6"
autocomple="off" required
        placeholder="??????">
      </div>

      <div style="border-top: 1px solid #e5e7eb; margin: 25px 0 20px;">

        <p style="font-size: 13px; color: #6b7280; text-align: left; margin-bottom:
15px;">Set up your permanent
          login credentials:</p>

        <div class="form-group">
          <label>Username</label>
          <input type="text" name="username" class="form-control" required
placeholder="Choose a username">
        </div>

        <div class="form-group">
          <label>Password</label>
          <input type="password" name="password" class="form-control" required
placeholder="Create a strong password">
        </div>

        <div class="form-group">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<label>Confirm Password</label>
<input type="password" name="password_confirm" class="form-control"
required
        placeholder="Retype password">
</div>

<button type="submit" class="btn-submit">
    <i class="fas fa-user-check"></i> Activate Account
</button>
</form>
</div>
{% if messages %}
<div class="messages-container">
    {% for message in messages %}
    <div class="alert alert-{{ message.tags }}" id="alert-{{ forloop.counter }}">
        <div class="alert-content">
            {% if message.tags == 'success' %}<i class="fas fa-check-circle"></i>
            {% elif message.tags == 'error' %}<i class="fas fa-exclamation-circle"></i>
            {% elif message.tags == 'warning' %}<i class="fas fa-exclamation-triangle"></i>
            {% else %}<i class="fas fa-info-circle"></i>{% endif %}
            <span>{{ message }}</span>
        </div>
        <button class="alert-close" onclick="this.parentElement.remove()"><i class="fas
fa-times"></i></button>
    </div>
    {% endfor %}
</div>
{% endif %}
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script src="{% static 'js/app.js' %}"></script>
</body>

</html>
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/org_chart.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Organizational Chart - Barangay Management System{% endblock %}

{% block extra_head %}
<script src="https://cdn.tailwindcss.com"></script>
<script src="https://unpkg.com/lucide@latest"></script>
<style>
    .org-container { transition: all 0.5s ease; }
    .org-card { transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1); }
    .org-card:hover { transform: scale(1.05); }
    .org-card:active { transform: scale(0.95); }

    .type-executive { background: linear-gradient(to bottom right, #306844, #255235);
color: white; }
    .type-admin { background: linear-gradient(to bottom right, #2563eb, #1d4ed8); color:
white; }
    .type-council { background: linear-gradient(to bottom right, #f59e0b, #d97706); color:
white; }
    .type-staff { background: #FFFFFF; border: 1px solid #E2E8F0; color: #1E293B; }

    .line-vertical { position: absolute; top: 0; left: 50%; transform: translateX(-50%);
width: 2px; height: 40px; background: #e2e8f0; }
    .line-horizontal { position: absolute; top: 0; height: 2px; background: #e2e8f0; }
    .line-tick { position: absolute; top: 0; left: 50%; transform: translateX(-50%);
width: 2px; height: 24px; background: #e2e8f0; }

    /* Custom scrollbar for the canvas */
    #org-canvas::-webkit-scrollbar { width: 6px; height: 6px; }
    #org-canvas::-webkit-scrollbar-thumb { background: #cbd5e1; border-radius: 10px; }
    #org-canvas::-webkit-scrollbar-track { background: transparent; }
</style>
{% endblock %}

{% block content %}
<div class="fixed inset-0 lg:left-[240px] top-[70px] bg-[#F8FAFC] flex flex-col overflow-
hidden font-sans text-slate-900 z-10">
    <!-- Sub-Navbar -->
    <nav class="h-20 bg-white border-b border-slate-200 px-8 flex items-center justify-
between z-30 shadow-sm">
        <div class="flex items-center gap-4">
            <div class="bg-[#306844] p-2.5 rounded-xl text-white">
                <i data-lucide="layout-dashboard" class="w-6 h-6"></i>
            </div>
            <div class="flex flex-col">
                <h1 class="text-xl font-black tracking-tight leading-none mb-1">Brgy. Sico
Sico Management System</h1>
                <p class="text-[10px] text-slate-400 font-bold uppercase tracking-
widest">Organization Chart</p>
            </div>
        </div>

        <div class="flex items-center gap-6">
            <div class="relative">
                <i data-lucide="search" class="absolute left-3 top-1/2 -translate-y-1/2
text-slate-400 w-4 h-4"></i>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <input
            type="text"
            id="memberSearch"
            placeholder="Find member..."
            class="pl-10 pr-4 py-2 bg-slate-100 border-none rounded-xl
focus:ring-2 focus:ring-emerald-500 w-64 text-sm font-medium outline-none transition-all"
        >
    </div>
</div>
</nav>

<!-- Main Interactive Canvas -->
<div class="flex-1 relative overflow-hidden bg-slate-50">
    <!-- Floating Controls -->
    <div class="absolute top-6 left-6 z-20 flex flex-col gap-3">
        <div class="mb-2">
            {% url 'officials:list' as officials_list_url %}
            {% include 'components/back_button.html' with back_url=officials_list_url
back_label='Back' %}
        </div>
        <button onclick="handleZoom(0.1)" class="p-3 bg-white hover:bg-slate-50
rounded-xl shadow-lg border border-slate-100 text-slate-600 transition active:scale-95"><i
data-lucide="zoom-in" class="w-5 h-5"></i></button>
        <button onclick="handleZoom(-0.1)" class="p-3 bg-white hover:bg-slate-50
rounded-xl shadow-lg border border-slate-100 text-slate-600 transition active:scale-95"><i
data-lucide="zoom-out" class="w-5 h-5"></i></button>
        <button onclick="resetZoom()" class="p-3 bg-white hover:bg-slate-50 rounded-xl
shadow-lg border border-slate-100 text-slate-600 transition active:scale-95"><i data-
lucide="maximize" class="w-5 h-5"></i></button>
    </div>

    <!-- Legend -->
    <div class="absolute bottom-6 left-6 z-20 bg-white/80 backdrop-blur-md p-4
rounded-2xl border border-slate-200 shadow-sm flex flex-col gap-2">
        <p class="text-[10px] font-black uppercase text-slate-400 mb-1 tracking-
widest">Barangay Role Key</p>
        <div class="flex items-center gap-3"><div class="w-3 h-3 rounded-full
bg-[#306844]"></div> <span class="text-xs font-bold text-slate-700"> Punong
Barangay</span></div>
        <div class="flex items-center gap-3"><div class="w-3 h-3 rounded-full bg-
blue-600"></div> <span class="text-xs font-bold text-slate-700">Secretary &
Treasurer</span></div>
        <div class="flex items-center gap-3"><div class="w-3 h-3 rounded-full bg-
amber-500"></div> <span class="text-xs font-bold text-slate-700">Sangguniang
Barangay</span></div>
        <div class="flex items-center gap-3"><div class="w-3 h-3 border border-
slate-200 rounded-full bg-white"></div> <span class="text-xs font-bold text-
slate-700">Staff & Volunteers</span></div>
    </div>

    <!-- Zoomable Container -->
    <div id="org-canvas" class="w-full h-full cursor-grab active:cursor-grabbing flex
items-start justify-center select-none overflow-hidden">
        <div id="zoom-container" class="transition-transform duration-300 ease-out"
style="transform: translate(0px, 0px) scale(0.9); transform-origin: center center;">
            <div id="org-tree-root" class="inline-block px-40 pt-64 pb-40"></div>
        </div>
    </div>
</div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<!-- Profile Detail Slide-over -->
<div id="profile-slideover" class="fixed inset-0 z-[9999] hidden flex items-center
justify-end p-6 bg-slate-900/40 backdrop-blur-sm transition-opacity">
  <div class="w-full max-w-sm bg-white rounded-3xl shadow-2xl h-full flex flex-col
overflow-hidden transform translate-x-full transition-transform duration-500"
id="slideover-content">
    <div class="h-32 bg-[#306844] relative flex items-end justify-center pb-6">
      <button onclick="closeDetail()" class="absolute top-4 right-4 p-2 bg-black/20
hover:bg-black/40 text-white rounded-full transition">
        <i data-lucide="x" class="w-5 h-5"></i>
      </button>

      <div class="w-24 h-24 bg-white rounded-full p-1 shadow-xl translate-y-12">
        <div class="w-full h-full bg-slate-100 rounded-full flex items-center
justify-center overflow-hidden border-2 border-white">
          <img id="detail-photo" class="w-full h-full object-cover" src=""
alt="Profile">
        </div>
        <div class="absolute -bottom-1 -right-1 w-8 h-8 bg-emerald-500 border-4
border-white rounded-full flex items-center justify-center">
          <i data-lucide="shield-check" class="text-white w-4 h-4"></i>
        </div>
      </div>
    </div>

    <div class="px-8 pb-8 pt-16 flex-1 overflow-y-auto">
      <h3 id="detail-name" class="text-2xl font-black text-slate-900 mb-1">---</h3>
      <p id="detail-role" class="text-emerald-700 font-bold uppercase tracking-
widest text-xs mb-8">---</p>

      <div class="space-y-6">
        <div class="space-y-4">
          <p class="text-[10px] font-black uppercase text-slate-400 tracking-
widest">Contact Information</p>
          <div class="flex items-center gap-4 group cursor-pointer">
            <div class="w-10 h-10 rounded-xl bg-slate-50 flex items-center
justify-center text-slate-400 group-hover:text-emerald-600 transition">
              <i data-lucide="mail" class="w-4.5 h-4.5"></i>
            </div>
            <div>
              <p class="text-[10px] font-bold text-slate-400
uppercase">Email Address</p>
              <p id="detail-email" class="text-sm font-bold text-
slate-700">---</p>
            </div>
          </div>
          <div class="flex items-center gap-4 group cursor-pointer">
            <div class="w-10 h-10 rounded-xl bg-slate-50 flex items-center
justify-center text-slate-400 group-hover:text-emerald-600 transition">
              <i data-lucide="phone" class="w-4.5 h-4.5"></i>
            </div>
            <div>
              <p class="text-[10px] font-bold text-slate-400
uppercase">Phone Number</p>
              <p id="detail-phone" class="text-sm font-bold text-
slate-700">---</p>
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="pt-6 border-t border-slate-100">
  <p class="text-[10px] font-black uppercase text-slate-400 tracking-
widest mb-4">Quick Actions</p>
  <div class="grid grid-cols-2 gap-3">
    <button id="btn-edit-profile" class="py-3 bg-[#306844] text-white
rounded-xl font-bold text-xs uppercase tracking-widest hover:bg-[#255235] transition
shadow-lg shadow-emerald-100">
      Edit Profile
    </button>
    <button id="btn-view-dtr" class="py-3 bg-slate-100 text-slate-600
rounded-xl font-bold text-xs uppercase tracking-widest hover:bg-slate-200 transition">
      View DTR
    </button>
  </div>
</div>
</div>
</div>

<div class="p-6 bg-slate-50 border-t border-slate-100">
  <button onclick="closeDetail()" class="w-full py-4 bg-white border border-
slate-200 rounded-2xl shadow-sm text-slate-500 font-bold text-xs uppercase tracking-widest
hover:bg-slate-50 hover:text-slate-700 transition">
    Close Detail
  </button>
</div>
</div>
</div>

<script>
  const ORG_DATA = [{ org_data_json|safe }];

  let scale = 1;
  let translateX = 0;
  let translateY = 0;
  let isDragging = false;
  let startX, startY;

  let collapsedNodes = new Set();

  function getTypeStyles(type) {
    switch (type) {
      case 'executive': return 'type-executive shadow-emerald-200';
      case 'admin': return 'type-admin shadow-blue-200';
      case 'council': return 'type-council shadow-amber-200';
      default: return 'type-staff shadow-slate-100';
    }
  }

  function renderNode(node) {
    const hasChildren = node.children && node.children.length > 0;
    const isCollapsed = collapsedNodes.has(node.id);

    let childrenHtml = '';
    if (hasChildren && !isCollapsed) {
      const childCount = node.children.length;
      childrenHtml = `
        <div class="flex pt-10 relative">
          <div class="line-vertical"></div>
          <div class="flex gap-8 relative">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

    ${node.children.map((child, index) => {
      let lineHtml = '';
      if (childCount > 1) {
        let horizStyle = '';
        if (index === 0) horizStyle = 'left: 50%; right: 0;';
        else if (index === childCount - 1) horizStyle = 'left: 0;
right: 50%;';

        else horizStyle = 'left: 0; right: 0;';

        lineHtml = `
          <div class="line-horizontal"
style="${horizStyle}"></div>
          <div class="line-tick"></div>
        `;
      }
      return `
        <div class="relative pt-6">
          ${lineHtml}
          ${renderNode(child)}
        </div>
      `;
    })).join('')}
  </div>
</div>
`;
}

const identiconUrl =
`https://api.dicebear.com/7.x/identicon/svg?seed=resident_${node.resident_id ||
node.id}&backgroundColor=ffffff`;
const profilePhoto = node.photo || identiconUrl;

return `
  <div class="flex flex-col items-center relative transition-all duration-500"
data-node-id="${node.id}">
    <div
      onclick="selectMember('${node.id}')"
      class="org-card relative z-10 w-64 p-4 rounded-2xl cursor-pointer
shadow-lg group ${getTypeStyles(node.type)}"
    >
      <div class="flex items-center gap-3">
        <div class="w-12 h-12 rounded-full border-2 border-white/50 flex
items-center justify-center overflow-hidden flex-shrink-0 bg-white/20">
          
        </div>
        <div class="flex-1 min-w-0">
          <h4 class="font-bold text-sm leading-tight ${node.type ===
'staff' ? 'text-slate-900' : 'text-white'} whitespace-normal break-
words">${node.name}</h4>
          <p class="text-[9px] opacity-80 font-bold uppercase tracking-
wider ${node.type === 'staff' ? 'text-slate-500' : 'text-white/90'} whitespace-normal
mt-0.5">${node.role}</p>
        </div>
      </div>
      <div>
        ${hasChildren ? `
          <button
            onclick="toggleNode(event, '${node.id}')"
            class="absolute -bottom-3 left-1/2 -translate-x-1/2 w-6 h-6
bg-white rounded-full flex items-center justify-center text-slate-400 border border-

```


IoT-Enabled Barangay Resident and Attendance Management System

```

slate-100 shadow-sm hover:text-emerald-600 hover:scale-110 transition-all"
    >
      <i data-lucide="${isCollapsed ? 'chevron-down' : 'chevron-
up'}" class="w-3.5 h-3.5"></i>
    </button>
  ` : ''}
</div>
  ${childrenHtml}
</div>
`;
}

function toggleNode(event, nodeId) {
  event.stopPropagation();
  if (collapsedNodes.has(nodeId)) collapsedNodes.delete(nodeId);
  else collapsedNodes.add(nodeId);
  updateTree();
}

function updateTree() {
  document.getElementById('org-tree-root').innerHTML = renderNode(ORG_DATA);
  lucide.createIcons();
}

function findNode(id, node = ORG_DATA) {
  if (node.id === id) return node;
  if (node.children) {
    for (let child of node.children) {
      const found = findNode(id, child);
      if (found) return found;
    }
  }
  return null;
}

function selectMember(id) {
  const node = findNode(id);
  if (!node) return;

  document.getElementById('detail-name').innerText = node.name;
  document.getElementById('detail-role').innerText = node.role;
  document.getElementById('detail-email').innerText = node.email || 'N/A';
  document.getElementById('detail-phone').innerText = node.phone || 'N/A';

  const photoImg = document.getElementById('detail-photo');
  const identiconUrl =
`https://api.dicebear.com/7.x/identicon/svg?seed=resident_${node.resident_id ||
node.id}&backgroundColor=ffffff`;
  photoImg.src = node.photo || identiconUrl;

  const slideover = document.getElementById('profile-slideover');
  const content = document.getElementById('slideover-content');

  // Update buttons
  const editBtn = document.getElementById('btn-edit-profile');
  const dtrBtn = document.getElementById('btn-view-dtr');

  // Only show actions for actual official nodes (not group placeholders)
  const isOfficial = ![ 'root', 'council_group', 'security_group',
'health_group' ].includes(node.id);

```

IoT-Enabled Barangay Resident and Attendance Management System

```
if (isOfficial) {
    editBtn.parentElement.classList.remove('hidden');
    editBtn.onclick = () => window.location.href = `/officials/${node.id}/edit/`;
    dtrBtn.onclick = () => window.location.href = `/attendance/dtr/${node.id}/`;
} else {
    editBtn.parentElement.classList.add('hidden');
}

slideover.classList.remove('hidden');
setTimeout(() => {
    content.classList.remove('translate-x-full');
}, 10);
}

function closeDetail() {
    const slideover = document.getElementById('profile-slideover');
    const content = document.getElementById('slideover-content');

    content.classList.add('translate-x-full');
    setTimeout(() => {
        slideover.classList.add('hidden');
    }, 500);
}

function handleZoom(delta) {
    scale = Math.min(Math.max(scale + delta, 0.3), 3);
    applyTransform();
}

function resetZoom() {
    scale = 0.85;
    translateX = 0;
    translateY = 50;
    applyTransform();
}

function applyTransform() {
    const container = document.getElementById('zoom-container');
    container.style.transform = `translate(${translateX}px, ${translateY}px)
scale(${scale})`;
}

document.addEventListener('DOMContentLoaded', () => {
    updateTree();
    resetZoom();

    const canvas = document.getElementById('org-canvas');

    canvas.addEventListener('mousedown', (e) => {
        if (e.button !== 0) return;
        isDragging = true;
        startX = e.clientX - translateX;
        startY = e.clientY - translateY;
        canvas.style.cursor = 'grabbing';
    });

    window.addEventListener('mousemove', (e) => {
        if (!isDragging) return;
        translateX = e.clientX - startX;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        translateY = e.clientY - startY;
        applyTransform();
    });

    window.addEventListener('mouseup', () => {
        isDragging = false;
        canvas.style.cursor = 'grab';
    });

    canvas.addEventListener('wheel', (e) => {
        e.preventDefault();
        const delta = e.deltaY > 0 ? -0.1 : 0.1;
        handleZoom(delta);
    }, { passive: false });

    document.getElementById('memberSearch').addEventListener('input', (e) => {
        const query = e.target.value.toLowerCase();
        const cards = document.querySelectorAll('.org-card');
        cards.forEach(card => {
            const name = card.querySelector('h4').innerText.toLowerCase();
            const role = card.querySelector('p').innerText.toLowerCase();
            if (name.includes(query) || role.includes(query)) {
                card.style.opacity = '1';
                card.style.transform = 'scale(1)';
            } else {
                card.style.opacity = '0.3';
                card.style.transform = 'scale(0.95)';
            }
        });
    });
});
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/officials/view.html

```
{% extends 'base.html' %}

{% block title %}Official Details - {{ official.full_name }}{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'officials:list' %}">Officials</a> / {{ official.full_name }}</span>
{% endblock %}

{% block content %}
{% url 'officials:list' as officials_list_url %}
{% include 'components/back_button.html' with back_url=officials_list_url
back_label='Back' %}
<div class="page-header">
  <div class="page-title">
    <h1>Official Profile</h1>
    <span>Information for {{ official.get_position_display }}</span>
  </div>
</div>

<div class="profile-grid">
  <div class="profile-sidebar">
    <div class="card profile-card">
      <div class="profile-photo">
        {% if official.display_photo %}
        
        {% else %}
        
        {% endif %}
      </div>
      <div class="profile-info">
        <h2>{{ official.full_name }}</h2>
        <span class="resident-id">Employee ID: {{ official.employee_id|default:"---"
}}</span>
        <div class="status-tags">
          <span
            class="badge {% if official.status == 'active' %}badge-success{% else
}%badge-warning{% endif %}">
            {{ official.get_status_display }}
          </span>
        </div>
      </div>
    </div>
  </div>

  <div class="card mt-4">
    <div class="card-header">
      <h3>Contact Information</h3>
    </div>
    <div class="card-body">
      <div class="info-item">
        <i class="fas fa-phone"></i>
        <div>
          <label>Phone</label>
          <span>
            {% if official.resident %}
            {{ official.resident.contact_number|default:"---" }}
          </span>
        </div>
      </div>
    </div>
  </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% else %}
    {{ official.contact_number|default:"---" }}
{% endif %}
</span>
</div>
</div>
<div class="info-item">
    <i class="fas fa-envelope"></i>
    <div>
        <label>Email</label>
        <span>
            {% if official.resident %}
                {{ official.resident.email|default:"---" }}
            {% else %}
                {{ official.email|default:"---" }}
            {% endif %}
        </span>
    </div>
</div>
</div>
</div>
</div>
<div class="card mt-4">
    <div class="card-header">
        <h3>Classifications</h3>
    </div>
    <div class="card-body">
        <div class="mini-status-list">
            <div class="mini-status-item {% if official.resident %}{% if
official.resident.is_registered_voter %}active{% endif %}{% else %}{% if
official.is_registered_voter %}active{% endif %}{% endif %}">
                <i class="fas fa-vote-yea"></i> Registered Voter
            </div>
            <div class="mini-status-item {% if official.resident %}{% if
official.resident.is_pwd %}active{% endif %}{% else %}{% if official.is_pwd %}active{%
endif %}{% endif %}">
                <i class="fas fa-wheelchair"></i> PWD
            </div>
            <div class="mini-status-item {% if official.resident %}{% if
official.resident.is_senior_citizen %}active{% endif %}{% else %}{% if
official.is_senior_citizen %}active{% endif %}{% endif %}">
                <i class="fas fa-blind"></i> Senior Citizen
            </div>
            <div class="mini-status-item {% if official.resident %}{% if
official.resident.is_4ps_member %}active{% endif %}{% else %}{% if official.is_4ps_member
%}active{% endif %}{% endif %}">
                <i class="fas fa-hands-helping"></i> 4Ps Member
            </div>
        </div>
    </div>
</div>
</div>
<div class="profile-main">
    <div class="card">
        <div class="card-header">
            <h3>Personal Information</h3>
        </div>
        <div class="card-body">
            <div class="detail-grid">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="detail-item">
  <label>Gender</label>
  <span>{% if official.resident %}{% official.resident.get_gender_display %}{%
else %}{% official.get_gender_display|default:"---" %}{% endif %}</span>
</div>
<div class="detail-item">
  <label>Birthdate</label>
  <span>{% if official.resident %}{% official.resident.birthdate|date:"F d, Y"
}}{% else %}{% official.birthdate|date:"F d, Y"|default:"---" %}{% endif %}</span>
</div>
<div class="detail-item">
  <label>Civil Status</label>
  <span>{% if official.resident %}{% official.resident.get_civil_status_display
}}{% else %}{% official.get_civil_status_display|default:"---" %}{% endif %}</span>
</div>
<div class="detail-item">
  <label>Nationality</label>
  <span>{% if official.resident %}{% official.resident.nationality %}{% else
%}{% official.nationality|default:"---" %}{% endif %}</span>
</div>
<div class="detail-item">
  <label>Religion</label>
  <span>{% if official.resident %}{% official.resident.religion|default:"---"
}}{% else %}{% official.religion|default:"---" %}{% endif %}</span>
</div>
<div class="detail-item">
  <label>Occupation</label>
  <span>{% if official.resident %}{% official.resident.occupation|default:"---"
}}{% else %}{% official.occupation|default:"---" %}{% endif %}</span>
</div>
</div>
<div class="mt-4">
  <label class="section-label">Address</label>
  <p class="address-text">
    {% if official.resident %}
      {% official.resident.address %}
      {% if official.resident.purok %}<br><small>Purok: {% official.resident.purok
}}</small>{% endif %}
    {% else %}
      {% official.address|default:"---" %}
      {% if official.purok %}<br><small>Purok: {% official.purok %}</small>{%
endif %}
    {% endif %}
  </p>
</div>
</div>

<div class="card mt-4">
  <div class="card-header d-flex justify-content-between align-items-center">
    <h3>Biometrics</h3>
    <a href="{% url 'officials:official_capture_fingerprint' official.pk %}"
class="btn btn-sm btn-primary">
      <i class="fas fa-fingerprint"></i> Capture Fingerprint
    </a>
  </div>
  <div class="card-body">
    {% if official.fingerprint_template %}
      <div class="alert alert-success mb-0">
        <i class="fas fa-check-circle"></i> Fingerprint registered
      </div>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</div>
{% elif official.resident and official.resident.fingerprint_template %}
  <div class="alert alert-success mb-0">
    <i class="fas fa-check-circle"></i> Fingerprint registered
  </div>
{% else %}
  <div class="alert alert-warning mb-0">
    <i class="fas fa-exclamation-triangle"></i> No fingerprint registered
  </div>
{% endif %}
</div>
</div>

<div class="card mt-4">
  <div class="card-header">
    <h3>Appointment Information</h3>
  </div>
  <div class="card-body">
    <div class="detail-grid">
      <div class="detail-item">
        <label>Position</label>
        <span class="font-weight-bold">{{ official.get_position_display }}</span>
      </div>
      <div class="detail-item">
        <label>Committee/Assignment</label>
        <span>{{ official.committee|default:"None Assigned" }}</span>
      </div>
      <div class="detail-item">
        <label>Term Start</label>
        <span>{{ official.term_start|date:"F d, Y" }}</span>
      </div>
      <div class="detail-item">
        <label>Term End</label>
        <span>{{ official.term_end|date:"F d, Y"|default:"Present" }}</span>
      </div>
      <div class="detail-item">
        <label>Salary / Honorarium</label>
        <span>?{{ official.salary|default:"0.00" }}</span>
      </div>
    </div>
  </div>
</div>

{% if official.resident %}
<div class="card mt-4">
  <div class="card-header">
    <h3>Resident Profile Link</h3>
  </div>
  <div class="card-body d-flex justify-content-between align-items-center">
    <p class="mb-0">Linked to resident record: <strong>{{ official.resident.full_name }}</strong></p>
    <a href="{% url 'residents:view' official.resident.pk %}" class="btn btn-outline">
      View personal profile <i class="fas fa-external-link-alt ml-1"></i>
    </a>
  </div>
</div>
{% endif %}

<div class="card mt-4">
  <div class="card-header">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<h3>Actions</h3>
</div>
<div class="card-body">
  <div class="action-row">
    <a href="{% url 'officials:edit' official.pk %}" class="btn btn-primary">
      <i class="fas fa-edit"></i> Edit Official
    </a>
    <a href="{% url 'officials:delete' official.pk %}" class="btn btn-outline-
danger">
      <i class="fas fa-user-minus"></i> Remove from Officials
    </a>
  </div>
</div>
</div>
</div>
```

```
<style>
.profile-grid {
  display: grid;
  grid-template-columns: 340px 1fr;
  gap: 20px;
  align-items: start;
}

.profile-card {
  text-align: center;
  padding: 30px 20px;
}

.profile-photo {
  width: 150px;
  height: 150px;
  margin: 0 auto 20px;
  border-radius: 50%;
  overflow: hidden;
  border: 5px solid var(--bg-card);
  box-shadow: 0 0 20px rgba(0, 0, 0, 0.1);
}

.profile-photo img {
  width: 100%;
  height: 100%;
  object-fit: cover;
}

.photo-placeholder {
  width: 100%;
  height: 100%;
  background: var(--border);
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 60px;
  color: var(--text-muted);
}

.badge-success {
  background: rgba(16, 185, 129, 0.1);
  color: #10b981;
}
```


IoT-Enabled Barangay Resident and Attendance Management System

```
}

.badge-warning {
  background: rgba(245, 158, 11, 0.1);
  color: #f59e0b;
}

.detail-grid {
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 20px;
}

.detail-item label {
  display: block;
  font-size: 11px;
  color: var(--text-muted);
  margin-bottom: 5px;
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

.detail-item span {
  font-size: 15px;
  font-weight: 500;
  color: var(--text-main);
}

.section-label {
  display: block;
  font-size: 11px;
  color: var(--text-muted);
  margin-bottom: 8px;
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

.address-text {
  font-size: 15px;
  line-height: 1.6;
  color: var(--text-main);
}

.info-item {
  display: flex;
  gap: 15px;
  margin-bottom: 15px;
}

.info-item i {
  font-size: 18px;
  color: var(--primary);
  width: 25px;
  text-align: center;
}

.info-item label {
  display: block;
  font-size: 10px;
  color: var(--text-muted);
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    text-transform: uppercase;
}

.info-item span {
    font-size: 14px;
    font-weight: 500;
}

.mini-status-list {
    display: flex;
    flex-direction: column;
    gap: 10px;
}

.mini-status-item {
    padding: 8px 12px;
    border-radius: 8px;
    background: var(--bg-body);
    font-size: 13px;
    color: var(--text-muted);
    display: flex;
    align-items: center;
    gap: 10px;
    opacity: 0.5;
    border: 1px solid var(--border);
}

.mini-status-item.active {
    opacity: 1;
    color: var(--primary);
    background: rgba(79, 70, 229, 0.05);
    border-color: rgba(79, 70, 229, 0.2);
    font-weight: 600;
}

.mini-status-item i {
    font-size: 14px;
    width: 20px;
    text-align: center;
}

.mt-4 {
    margin-top: 25px;
}

.btn-outline-danger {
    color: #ef4444;
    border-color: #ef4444;
}

.btn-outline-danger:hover {
    background: #ef4444;
    color: white;
}

.action-row {
    display: flex;
    justify-content: flex-end;
    gap: 12px;
    flex-wrap: wrap;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}  
  
@media (max-width: 992px) {  
  .profile-grid {  
    grid-template-columns: 1fr;  
  }  
}  
</style>  
{% endblock %}
```

File: templates/ordinances/confirm_delete.html

```
{% extends 'base.html' %}

{% block title %}Delete Ordinance - {{ ordinance.ordinance_number }}{% endblock %}

{% block content %}
<div class="card" style="max-width: 500px; margin: 100px auto; text-align: center;
padding: 40px;">
  <div class="alert-icon" style="font-size: 60px; color: var(--danger); margin-bottom:
20px;">
    <i class="fas fa-exclamation-circle"></i>
  </div>
  <h2>Delete Ordinance?</h2>
  <p style="margin: 20px 0; color: var(--text-muted);">
    Are you sure you want to permanently delete <strong>Ordinance #{{
ordinance.ordinance_number }}</strong>:
    <em>"{{ ordinance.title }}"</em>? This action cannot be undone.
  </p>

  <form method="POST">
    {% csrf_token %}
    <div class="action-buttons" style="display: flex; gap: 15px; justify-content: center;
margin-top: 30px;">
      <button type="submit" class="btn btn-danger btn-lg">
        <i class="fas fa-trash"></i> Yes, Delete
      </button>
      <a href="{% url 'ordinances:view' ordinance.pk %}" class="btn btn-outline btn-lg">
        Cancel
      </a>
    </div>
  </form>
</div>

<style>
  .btn-danger {
    background: var(--danger);
    color: white;
    border: none;
  }

  .btn-danger:hover {
    background: #dc2626;
  }
</style>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/ordinances/form.html

```
{% extends 'base.html' %}

{% block title %}{% if editing %}Edit{% else %}Add{% endif %} Ordinance - Barangay
Management System{% endblock %}

{% block content %}
{% url 'ordinances:list' as ordinances_list_url %}
{% include 'components/back_button.html' with back_url=ordinances_list_url
back_label='Back' %}
<div class="page-header">
  <div class="page-title">
    <h1>{% if editing %}Edit Ordinance{% else %}New Ordinance{% endif %}</h1>
    <span>Official legislative record entry</span>
  </div>
</div>

<div class="card">
  <form method="POST" enctype="multipart/form-data" class="modern-form">
    {% csrf_token %}
    <div class="form-grid">
      <div class="form-section">
        <h3>Ordinance Information</h3>
        <div class="form-row">
          <div class="form-group col-4">
            <label>Ordinance Number*</label>
            <input type="text" name="ordinance_number" value="{{
ordinance.ordinance_number }}"
class="form-control" placeholder="e.g. 2024-001" required>
          </div>
          <div class="form-group col-8">
            <label>Title*</label>
            <input type="text" name="title" value="{{ ordinance.title }}" class="form-
control" required>
          </div>
        </div>

        <div class="form-row">
          <div class="form-group col-12">
            <label>Description/Summary</label>
            <textarea name="description" class="form-control"
rows="3">{{ ordinance.description }}</textarea>
          </div>
        </div>

        <div class="form-row">
          <div class="form-group col-4">
            <label>Author/Proponent*</label>
            <input type="text" name="author" value="{{ ordinance.author }}" class="form-
control" required>
          </div>
          <div class="form-group col-4">
            <label>Sponsor</label>
            <input type="text" name="sponsor" value="{{ ordinance.sponsor }}" class="form-
control">
          </div>
          <div class="form-group col-4">
            <label>Category*</label>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<select name="category" class="form-control" required>
    {% for code, label in categories %}
        <option value="{{ code }}" {% if ordinance.category == code %}selected{%
endif %}>{{ label }}
    </option>
    {% endfor %}
</select>
</div>
</div>
</div>

<div class="form-section">
<h3>Timeline & Status</h3>
<div class="form-row">
    <div class="form-group col-4">
        <label>Date Enacted</label>
        <input type="date" name="date_enacted" value="{{
ordinance.date_enacted|date:'Y-m-d' }}"
        class="form-control" required>
    </div>
    <div class="form-group col-4">
        <label>Effectivity Date</label>
        <input type="date" name="date_effectivity" value="{{
ordinance.date_effectivity|date:'Y-m-d' }}"
        class="form-control">
    </div>
    <div class="form-group col-4">
        <label>Status</label>
        <select name="status" class="form-control" required>
            {% for code, label in statuses %}
                <option value="{{ code }}" {% if ordinance.status == code %}selected{%
endif %}>{{ label }}
            </option>
            {% endfor %}
        </select>
    </div>
</div>
</div>
<div class="form-row">
    <div class="form-group col-6">
        <label>Document File (PDF/DOC)</label>
        <input type="file" name="document_file" class="form-control">
    </div>
</div>
</div>
</div>

<div class="form-footer">
    <button type="submit" class="btn btn-primary btn-lg">
        <i class="fas fa-save"></i> Save Ordinance
    </button>
</div>
</form>
</div>

<style>
.modern-form {
    padding: 30px;
}

.form-section {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
margin-bottom: 25px;
}

.form-section h3 {
  font-size: 14px;
  color: var(--primary);
  margin-bottom: 15px;
  border-bottom: 1px solid var(--border);
  padding-bottom: 8px;
}

.form-row {
  display: flex;
  gap: 20px;
  margin-bottom: 15px;
}

.col-4 {
  flex: 1;
}

.col-6 {
  flex: 0 0 calc(50% - 10px);
}

.col-8 {
  flex: 2;
}

.col-12 {
  width: 100%;
}

.form-footer {
  border-top: 1px solid var(--border);
  padding-top: 20px;
  text-align: right;
}
</style>
{% endblock %}
```

File: templates/ordinances/list.html

```
{% extends 'base.html' %}

{% block title %}Barangay Ordinances - Barangay Management System{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Barangay Ordinances & Resolutions</h1>
    <span>Official records and legislative documents</span>
  </div>
  <div class="header-actions" style="display: flex; gap: 10px;">
    {% if request.user.get_permission_level <= 2 %}
    <a href="{% url 'ordinances:upload_parse' %}" class="btn btn-secondary" style="border-radius: 0.75rem; background: #dc2626; border: none; color: #fff; font-weight: 600;">
      <i class="fas fa-file-pdf mr-1"></i> Upload Ordinance PDF
    </a>
    <a href="{% url 'ordinances:add' %}" class="btn btn-primary">
      <i class="fas fa-plus"></i> New Ordinance
    </a>
    {% endif %}
  </div>
</div>

<div class="card">
  <div class="card-header filter-card-header">
    <form method="GET" class="filter-form">
      <div class="filter-inputs-group">
        <div class="search-input-wrapper">
          <i class="fas fa-search search-icon"></i>
          <input type="text" name="q" value="{{ query }}" placeholder="Search number or title..."
            class="form-control search-input">
          {% if query %}
          <a href="{% url 'ordinances:list' %}" class="search-clear" title="Clear search">
            <i class="fas fa-times-circle"></i>
          </a>
          {% endif %}
        </div>

        <div class="filter-select-wrapper">
          <i class="fas fa-filter filter-icon"></i>
          <select name="category" class="form-control filter-select">
            <option value="">All Categories</option>
            {% for code, label in categories %}
            <option value="{{ code }}" {% if code == selected_category %}selected{% endif %}>{{ label }}</option>
            {% endfor %}
          </select>
        </div>
      </div>

      <button type="submit" class="btn btn-primary search-btn">
        <i class="fas fa-search"></i> <span>Search</span>
      </button>
    </form>
  </div>
</div>
<div class="table-responsive">
```


IoT-Enabled Barangay Resident and Attendance Management System

```

<table>
  <thead>
    <tr>
      <th>Number</th>
      <th>Title</th>
      <th>Date Enacted</th>
      <th>Author/Sponsor</th>
      <th>Status</th>
      <th>Actions</th>
    </tr>
  </thead>
  <tbody>
    {% for ord in ordinances %}
      <tr>
        <td><strong>{{ ord.ordinance_number }}</strong></td>
        <td style="max-width: 300px;">{{ ord.title }}</td>
        <td>{{ ord.date_enacted|date:"M d, Y" }}</td>
        <td>{{ ord.author }}</td>
        <td>
          <span class="status-badge {% if ord.status == 'active' %}badge-active{% elif
ord.status == 'pending' %}badge-pending{% else %}badge-inactive{% endif %}">
            {{ ord.get_status_display }}
          </span>
        </td>
        <td>
          <div class="btn-group">
            <a href="{% url 'ordinances:view' ord.pk %}" class="btn-icon btn-view"
title="View"><i class="fas fa-eye"></i></a>
            {% if ord.document_file %}
              <a href="{{ ord.document_file.url }}" target="_blank" class="btn btn-outline
btn-icon" title="View PDF" style="color: #dc2626; border-color: #dc2626;"><i class="fas
fa-file-pdf"></i></a>
              <a href="{% url 'ordinances:download' ord.pk %}" class="btn btn-outline btn-
icon" title="Download"><i class="fas fa-download"></i></a>
              {% endif %}
              {% if request.user.get_permission_level <= 2 %}
                <a href="{% url 'ordinances:edit' ord.pk %}" class="btn-icon btn-edit"
title="Edit"><i class="fas fa-edit"></i></a>
                {% endif %}
            </div>
          </td>
        </tr>
      {% empty %}
      <tr>
        <td colspan="6" class="text-center">No ordinances found.</td>
      </tr>
    {% endfor %}
  </tbody>
</table>
</div>
</div>

{% endblock %}

```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/ordinances/parse_review.html

```
{% extends 'base.html' %}

{% block title %}Review Parsed Ordinance - Barangay Management System{% endblock %}

{% block content %}
<link href="https://fonts.googleapis.com/css2?family=Playfair+Display:ital,wght@0,600;0,700;1,600&family=Plus+Jakarta+Sans:wght@400;500;600;700&display=swap" rel="stylesheet">

<div class="elegant-review-wrapper">
  <div class="page-header-premium fade-in-up">
    <div class="header-content">
      <div class="badge-accent">Automated Parsing Complete</div>
      <h1>Review & Finalize Ordinance</h1>
      <p>Please verify the auto-extracted information before saving to the official
registry.</p>
    </div>
  </div>

  <div class="split-container-premium fade-in-up stagger-1">
    <!-- Left Side: PDF Viewer -->
    <div class="pdf-viewer-panel-premium">
      <div class="panel-header-premium">
        <h4><i class="fas fa-file-pdf mr-2" style="color:#d32f2f;"></i> Original
Document</h4>
        <div class="header-decoration"></div>
      </div>
      <div class="panel-body-premium p-0">
        <iframe src="{{ file_url }}" type="application/pdf" width="100%" height="100%"
style="border: none;">
          This browser does not support PDFs. Please download the PDF to view it.
        </iframe>
      </div>
    </div>

    <!-- Right Side: Edit Form -->
    <div class="form-panel-premium">
      <div class="panel-header-premium">
        <h4><i class="fas fa-magic mr-2" style="color:#cfa968;"></i> Extracted Data</h4>
        <div class="header-decoration"></div>
      </div>
      <div class="panel-body-premium scrollable-form">
        <form method="POST" action="{% url 'ordinances:upload_parse' %}">
          {% csrf_token %}
          <input type="hidden" name="temp_filename" value="{{ temp_filename }}">
          <input type="hidden" name="save_ordinance" value="1">

          <div class="alert-premium mb-4">
            <div class="alert-icon"><i class="fas fa-info-circle"></i></div>
            <div class="alert-text">Data was automatically extracted via OCR. Please
verify carefully for inaccuracies before saving.</div>
          </div>

          <div class="input-group-premium mb-4">
            <label>Ordinance Number</label>
            <input type="text" name="ordinance_number" value="{{
parsed_data.ordinance_number }}" required>
            <div class="focus-border"></div>
          </div>
        </form>
      </div>
    </div>
  </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</div>

<div class="input-group-premium mb-4">
  <label>Title</label>
  <textarea name="title" rows="3" required>{{ parsed_data.title }}</textarea>
  <div class="focus-border"></div>
</div>

<div class="row">
  <div class="col-md-6 mb-4">
    <div class="input-group-premium">
      <label>Date Enacted</label>
      <input type="date" name="date_enacted" value="{{ parsed_data.date_enacted
}} " required>
      <div class="focus-border"></div>
    </div>
  </div>
  <div class="col-md-6 mb-4">
    <div class="input-group-premium select-premium">
      <label>Category</label>
      <select name="category" required>
        {% for code, label in categories %}
          <option value="{{ code }}" {% if parsed_data.category == code
%}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
      <i class="fas fa-chevron-down select-icon"></i>
      <div class="focus-border"></div>
    </div>
  </div>
</div>

<div class="row">
  <div class="col-md-6 mb-4">
    <div class="input-group-premium">
      <label>Author / Proponent</label>
      <input type="text" name="author" value="{{ parsed_data.author }}"
required>
      <div class="focus-border"></div>
    </div>
  </div>
  <div class="col-md-6 mb-4">
    <div class="input-group-premium select-premium">
      <label>Status</label>
      <select name="status" required>
        {% for code, label in statuses %}
          <option value="{{ code }}">{{ label }}</option>
        {% endfor %}
      </select>
      <i class="fas fa-chevron-down select-icon"></i>
      <div class="focus-border"></div>
    </div>
  </div>
</div>

<div class="input-group-premium mb-4">
  <label>Signatories</label>
  <textarea name="signatories" rows="2">{{ parsed_data.signatories }}</textarea>
  <div class="focus-border"></div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="input-group-premium mb-5">
  <label>Full Body Content</label>
  <textarea name="body_content" rows="8" class="code-font">{{
parsed_data.body_content }}</textarea>
  <div class="focus-border"></div>
</div>

<div class="form-actions-premium">
  <a href="{% url 'ordinances:list' %}" class="btn-premium btn-
ghost">Discard</a>
  <button type="submit" class="btn-premium btn-primary-lux">
    <i class="fas fa-check-circle mr-2"></i> Confirm & Save
  </button>
</div>
</form>
</div>
</div>
</div>
</div>

<style>
/* Base Variables & Typography */
.elegant-review-wrapper {
  --color-bg: #f9f9f7;
  --color-surface: #ffffff;
  --color-text-main: #0f2922;
  --color-text-muted: #5e706a;
  --color-accent-gold: #cfa968;
  --color-accent-gold-hover: #b89354;
  --color-primary-dark: #0f2922;
  --color-border: #e6e8e6;

  font-family: 'Plus Jakarta Sans', sans-serif;
  color: var(--color-text-main);
  background-color: var(--color-bg);
  min-height: 100vh;
  padding: 2rem 1rem;
  margin: -1.5rem; /* Offset container padding if any */
  padding-bottom: 4rem;
}

/* Animations */
@keyframes fadeInUp {
  from { opacity: 0; transform: translateY(30px); }
  to { opacity: 1; transform: translateY(0); }
}

.fade-in-up {
  animation: fadeInUp 0.8s cubic-bezier(0.16, 1, 0.3, 1) forwards;
  opacity: 0;
}

.stagger-1 { animation-delay: 0.15s; }

/* Header Premium */
.page-header-premium {
  max-width: 1400px;
  margin: 0 auto 2.5rem;
  text-align: center;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    position: relative;
}

.badge-accent {
    display: inline-block;
    padding: 0.35rem 1rem;
    background: rgba(207, 169, 104, 0.15);
    color: var(--color-accent-gold-hover);
    font-size: 0.75rem;
    font-weight: 700;
    letter-spacing: 1.5px;
    text-transform: uppercase;
    border-radius: 50px;
    margin-bottom: 1rem;
}

.page-header-premium h1 {
    font-family: 'Playfair Display', serif;
    font-size: 2.75rem;
    font-weight: 700;
    color: var(--color-primary-dark);
    margin-bottom: 0.5rem;
    letter-spacing: -0.5px;
}

.page-header-premium p {
    font-size: 1.1rem;
    color: var(--color-text-muted);
    max-width: 600px;
    margin: 0 auto;
}

/* Split Container */
.split-container-premium {
    display: flex;
    max-width: 1400px;
    margin: 0 auto;
    height: calc(100vh - 180px);
    min-height: 750px;
    gap: 2rem;
}

.pdf-viewer-panel-premium, .form-panel-premium {
    background: var(--color-surface);
    border-radius: 20px;
    box-shadow: 0 20px 40px rgba(15, 41, 34, 0.05), 0 1px 3px rgba(0,0,0,0.02);
    display: flex;
    flex-direction: column;
    overflow: hidden;
    border: 1px solid rgba(255,255,255,0.8);
    position: relative;
}

.pdf-viewer-panel-premium {
    flex: 1.2;
}

.form-panel-premium {
    flex: 1;
    max-width: 650px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

/* Panel Headers */
.panel-header-premium {
  padding: 1.5rem 2rem;
  background: #ffffff;
  display: flex;
  align-items: center;
  position: relative;
  z-index: 10;
}

.header-decoration {
  position: absolute;
  bottom: 0;
  left: 2rem;
  width: 40px;
  height: 3px;
  background: var(--color-accent-gold);
}

.panel-header-premium h4 {
  margin: 0;
  font-family: 'Playfair Display', serif;
  font-size: 1.25rem;
  font-weight: 700;
  color: var(--color-primary-dark);
  letter-spacing: 0.5px;
}

.panel-body-premium {
  flex: 1;
  position: relative;
  background: #fafaf9;
}

.scrollable-form {
  padding: 2.5rem;
  overflow-y: auto;
  overflow-x: hidden;
}

/* Scrollbar Styling */
.scrollable-form::-webkit-scrollbar {
  width: 6px;
}
.scrollable-form::-webkit-scrollbar-track {
  background: transparent;
}
.scrollable-form::-webkit-scrollbar-thumb {
  background-color: #cbd5e1;
  border-radius: 10px;
}

/* Alert Premium */
.alert-premium {
  background: #fdfbf7;
  border: 1px solid #f0e6d2;
  border-left: 4px solid var(--color-accent-gold);
  padding: 1.25rem;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
border-radius: 8px;
display: flex;
gap: 1rem;
align-items: flex-start;
}

.alert-icon {
  color: var(--color-accent-gold);
  font-size: 1.25rem;
}

.alert-text {
  font-size: 0.9rem;
  color: #6d5d41;
  line-height: 1.5;
}

/* Form Elements Premium */
.input-group-premium {
  position: relative;
  display: flex;
  flex-direction: column;
}

.input-group-premium label {
  font-size: 0.75rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 1px;
  color: var(--color-text-muted);
  margin-bottom: 0.5rem;
  transition: color 0.3s;
}

.input-group-premium input,
.input-group-premium textarea,
.input-group-premium select {
  width: 100%;
  background: #ffffff;
  border: 1px solid var(--color-border);
  border-radius: 8px;
  padding: 0.875rem 1rem;
  font-family: 'Plus Jakarta Sans', sans-serif;
  font-size: 0.95rem;
  color: var(--color-text-main);
  transition: all 0.3s ease;
  box-shadow: inset 0 2px 4px rgba(0,0,0,0.01);
}

.input-group-premium textarea {
  resize: vertical;
  line-height: 1.6;
}

.code-font {
  font-family: 'SFMono-Regular', Consolas, 'Liberation Mono', Menlo, monospace
!important;
  font-size: 0.85rem !important;
  background: #f4f5f4 !important;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.input-group-premium input:focus,
.input-group-premium textarea:focus,
.input-group-premium select:focus {
  outline: none;
  border-color: var(--color-primary-dark);
  background: #ffffff;
  box-shadow: 0 4px 12px rgba(15, 41, 34, 0.05);
}

.focus-border {
  position: absolute;
  bottom: 0;
  left: 50%;
  width: 0;
  height: 2px;
  background-color: var(--color-primary-dark);
  transition: 0.4s cubic-bezier(0.16, 1, 0.3, 1);
  transform: translateX(-50%);
  pointer-events: none;
  border-radius: 0 0 8px 8px;
}

.input-group-premium input:focus ~ .focus-border,
.input-group-premium textarea:focus ~ .focus-border,
.input-group-premium select:focus ~ .focus-border {
  width: 100%;
}

.input-group-premium:focus-within label {
  color: var(--color-primary-dark);
}

/* Custom Select */
.select-premium {
  position: relative;
}

.select-premium select {
  appearance: none;
  padding-right: 2.5rem;
  cursor: pointer;
}

.select-icon {
  position: absolute;
  right: 1rem;
  top: 2.2rem;
  color: #94a3b8;
  pointer-events: none;
  font-size: 0.8rem;
  transition: transform 0.3s;
}

.select-premium select:focus ~ .select-icon {
  transform: rotate(180deg);
  color: var(--color-primary-dark);
}

/* Buttons */
```


IoT-Enabled Barangay Resident and Attendance Management System

```
.form-actions-premium {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding-top: 1.5rem;
  border-top: 1px solid var(--color-border);
  margin-top: 1rem;
}

.btn-premium {
  padding: 0.875rem 2rem;
  border-radius: 8px;
  font-weight: 600;
  font-size: 0.95rem;
  transition: all 0.3s cubic-bezier(0.16, 1, 0.3, 1);
  cursor: pointer;
  text-decoration: none;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  border: none;
}

.btn-ghost {
  background: transparent;
  color: var(--color-text-muted);
}

.btn-ghost:hover {
  color: var(--color-text-main);
  background: #f1f2f0;
}

.btn-primary-lux {
  background: var(--color-primary-dark);
  color: #ffffff;
  box-shadow: 0 4px 15px rgba(15, 41, 34, 0.2);
}

.btn-primary-lux:hover {
  background: #194035;
  transform: translateY(-2px);
  box-shadow: 0 8px 25px rgba(15, 41, 34, 0.3);
  color: #ffffff;
}

.btn-primary-lux i {
  color: var(--color-accent-gold);
}

/* Responsive tweaks */
@media (max-width: 992px) {
  .split-container-premium {
    flex-direction: column;
    height: auto;
  }
  .pdf-viewer-panel-premium {
    height: 600px;
  }
  .form-panel-premium {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        max-width: 100%;  
    }  
}  
</style>  
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/ordinances/upload.html

```
{% extends 'base.html' %}

{% block title %}Upload Ordinance PDF - Barangay Management System{% endblock %}

{% block content %}
{% url 'ordinances:list' as ordinances_list_url %}
{% include 'components/back_button.html' with back_url=ordinances_list_url
back_label='Back' %}
<div class="page-header">
  <div class="page-title">
    <h1>Upload Ordinance PDF</h1>
    <span>Auto-extract details from a scanned or digital PDF ordinance</span>
  </div>
</div>

<div class="card" style="max-width: 600px; margin: 0 auto; padding: 40px; text-align:
center; border-radius: 20px; box-shadow: 0 10px 30px rgba(0,0,0,0.05);">
  <i class="fas fa-file-pdf fa-4x text-danger mb-4"></i>
  <h3 class="mb-3">Select Ordinance Document</h3>
  <p class="text-muted mb-4">Our system will automatically read the PDF and extract the
Ordinance Number, Title, Date, and Signatories for you to review.</p>

  <form method="POST" enctype="multipart/form-data" action="{% url
'ordinances:upload_parse' %}">
    {% csrf_token %}
    <div class="form-group mb-4">
      <input type="file" name="document_file" id="document_file" class="form-control"
accept="application/pdf" required style="padding: 10px; border-radius: 10px; border: 2px
dashed var(--border);">
    </div>

    <button type="submit" class="btn btn-primary btn-lg" style="width: 100%; border-
radius: 10px; font-weight: 600;">
      <i class="fas fa-magic mr-2"></i> Upload & Parse Document
    </button>
  </form>
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/ordinances/view.html

```
{% extends 'base.html' %}

{% block title %}Ordinance #{{ ordinance.ordinance_number }} - Details{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'ordinances:list' %}">Ordinances</a> / {{ ordinance.ordinance_number }}</span>
{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Ordinance Details</h1>
    <span>Information for Ordinance #: {{ ordinance.ordinance_number }}</span>
  </div>
  <div class="header-actions">
    {% if request.user.get_permission_level <= 2 %}
    <a href="{% url 'ordinances:edit' ordinance.pk %}" class="btn btn-primary">
      <i class="fas fa-edit"></i> Edit Ordinance
    </a>
    {% endif %}
    {% if ordinance.document_file %}
    <a href="{% ordinance.document_file.url %}" target="_blank" class="btn btn-danger"
style="background: #dc2626; border: none; border-radius: 0.75rem; font-weight: 600;">
      <i class="fas fa-file-pdf"></i> View PDF
    </a>
    <a href="{% url 'ordinances:download' ordinance.pk %}" class="btn btn-secondary">
      <i class="fas fa-download"></i> Download Document
    </a>
    {% endif %}
  </div>
  <div>
    {% url 'ordinances:list' as ordinances_list_url %}
    {% include 'components/back_button.html' with back_url=ordinances_list_url
back_label='Back' %}
  </div>
</div>

<div class="card">
  <div class="card-header bg-light d-flex justify-content-between align-items-center">
    <h2 class="mb-0">{{ ordinance.title }}</h2>
    <span class="badge {% if ordinance.status == 'active' %}badge-success{% else %}badge-warning{% endif %}">
      {{ ordinance.get_status_display }}
    </span>
  </div>
  <div class="card-body">
    <div class="row">
      <div class="col-md-8">
        <div class="detail-section mb-4">
          <h3>Description</h3>
          <p class="lead text-muted">{{ ordinance.description|linebreaks|default:"No description provided." }}</p>
        </div>

        {% if ordinance.body_content %}
        <div class="detail-section mb-4">

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<h3><i class="fas fa-file-alt text-primary mr-1"></i> Full Body Content</h3>
<div style="background: #f8fafc; border-radius: 10px; padding: 20px; border: 1px
solid var(--border-light); font-size: 14px; line-height: 1.8; white-space: pre-wrap; max-
height: 500px; overflow-y: auto;">{{ ordinance.body_content }}</div>
</div>
{% endif %}

{% if ordinance.signatories %}
<div class="detail-section mb-4">
<h3><i class="fas fa-signature text-primary mr-1"></i> Signatories</h3>
<div style="background: #f8fafc; border-radius: 10px; padding: 20px; border: 1px
solid var(--border-light); white-space: pre-wrap;">{{ ordinance.signatories }}</div>
</div>
{% endif %}

<div class="detail-section mb-4">
<h3>Remarks</h3>
<p>{{ ordinance.remarks|default:"---" }}</p>
</div>
</div>
<div class="col-md-4">
<div class="detail-card card p-3 bg-light border-0">
<h3 class="h5 mb-3 border-bottom pb-2">MetaData</h3>
<div class="meta-item mb-2">
<label class="text-muted d-block small uppercase font-weight-bold">ORDINANCE
NUMBER</label>
<span>{{ ordinance.ordinance_number }}</span>
</div>
<div class="meta-item mb-2">
<label class="text-muted d-block small uppercase font-weight-
bold">AUTHOR</label>
<span>{{ ordinance.author }}</span>
</div>
{% if ordinance.sponsor %}
<div class="meta-item mb-2">
<label class="text-muted d-block small uppercase font-weight-
bold">SPONSOR</label>
<span>{{ ordinance.sponsor }}</span>
</div>
{% endif %}
<div class="meta-item mb-2">
<label class="text-muted d-block small uppercase font-weight-
bold">CATEGORY</label>
<span>{{ ordinance.get_category_display }}</span>
</div>
<div class="meta-item mb-2">
<label class="text-muted d-block small uppercase font-weight-bold">DATE
ENACTED</label>
<span>{{ ordinance.date_enacted|date:"F d, Y" }}</span>
</div>
{% if ordinance.date_effectivity %}
<div class="meta-item mb-2">
<label class="text-muted d-block small uppercase font-weight-bold">DATE OF
EFFECTIVITY</label>
<span>{{ ordinance.date_effectivity|date:"F d, Y" }}</span>
</div>
{% endif %}
</div>

{% if request.user.get_permission_level <= 2 %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="mt-4">
  <a href="{% url 'ordinances:delete' ordinance.pk %}" class="btn btn-outline-
danger btn-block">
    <i class="fas fa-trash"></i> Delete Ordinance Record
  </a>
</div>
{% endif %}
</div>
</div>
</div>
</div>

<style>
.badge {
  padding: 6px 12px;
  border-radius: 4px;
  font-weight: 600;
}

.badge-success {
  background: rgba(16, 185, 129, 0.1);
  color: #10b981;
}

.badge-warning {
  background: rgba(245, 158, 11, 0.1);
  color: #f59e0b;
}

.detail-section h3 {
  font-size: 16px;
  color: var(--primary);
  margin-bottom: 15px;
  border-bottom: 2px solid var(--border);
  padding-bottom: 5px;
}

.meta-item label {
  letter-spacing: 0.5px;
}

.uppercase {
  text-transform: uppercase;
}

.btn-block {
  display: block;
  width: 100%;
}

.btn-outline-danger {
  color: var(--danger);
  border-color: var(--danger);
}

.btn-outline-danger:hover {
  background: var(--danger);
  color: white;
}
</style>
```

IoT-Enabled Barangay Resident and Attendance Management System

{% endblock %}

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/payments/dashboard.html

```
{% extends 'base.html' %}

{% block title %}Certificate Payments Dashboard - SICO-SICO{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Payment Management</h1>
    <span>Verify and process official receipts for certificate requests</span>
  </div>
</div>

<div class="card" style="margin-bottom: 24px;">
  <div class="card-header">
    <form method="GET" class="filter-form" style="display: flex; gap: 15px; align-items: center;">
      <select name="status" class="form-control" onchange="this.form.submit()">
        <option value="pending" {% if status_filter == 'pending' %}>selected{%
endif %}>Pending Verification
      </select>
      <select name="method" class="form-control" onchange="this.form.submit()">
        <option value="">All Methods
        <option value="cash" {% if method_filter == 'cash' %}>selected{% endif
%}>Cash
        <option value="gcash" {% if method_filter == 'gcash' %}>selected{% endif
%}>GCash
      </select>
      <a href="{% url 'payments:dashboard' %}" class="btn btn-secondary btn-sm">Reset</a>
    </form>
  </div>

  <div class="table-responsive">
    <table class="table">
      <thead>
        <tr>
          <th>Tracking Code</th>
          <th>Resident</th>
          <th>Certificate Type</th>
          <th>Amount</th>
          <th>Method</th>
          <th>Action</th>
        </tr>
      </thead>
      <tbody>
        {% for payment in payments %}
        <tr>
          <td><strong>{{ payment.certificate_request.tracking_code
}}</strong></td>
        </tr>
        </tbody>
      </table>
    </div>
  </div>
</div>
```


IoT-Enabled Barangay Resident and Attendance Management System

```

<td>{{ payment.certificate_request.resident.full_name }}</td>
<td>{{ payment.certificate_request.get_cert_type_display }}</td>
<td>?{{ payment.amount|floatformat:2 }}</td>
<td>
    <span
        class="badge badge-{{ if payment.method == 'gcash' }}info{%
elif payment.method == 'cash' }}warning{% else }}secondary{% endif %}">
        {{ payment.get_method_display }}
    </span>
</td>
<td>
    {% if payment.status == 'unpaid' and payment.method == 'cash' %}
    <button class="btn btn-primary btn-sm"
        onclick="showPayModal('{{ payment.id }}', '{{ payment.amount
}}')">Mark as Paid</button>
    {% elif payment.status == 'pending' and payment.method == 'gcash'
%}

    <button class="btn btn-info btn-sm"
        onclick="showConfirmGcashModal('{{ payment.id }}', '{{
payment.proof_screenshot.url }}')">Verify
        Proof</button>
    {% elif payment.status == 'paid' %}
    <a href="{% url 'payments:download_receipt' payment.id %}"
class="btn btn-success btn-sm">
        <i class="fas fa-file-pdf"></i> View OR
    </a>
    {% endif %}

    {% if payment.status != 'waived' and payment.status != 'paid' %}
    <button class="btn btn-link btn-sm text-danger"
        onclick="showWaiveModal('{{ payment.id }}')">Waive
Fee</button>
    {% endif %}
</td>
</tr>
{% empty %}
<tr>
    <td colspan="6" class="text-center py-4 text-muted">No records found
for the selected filters.</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
</div>
<!-- Cash Payment Modal -->
<div id="payModal" class="modal">
    <div class="modal-content">
        <h3>Mark as Paid</h3>
        <p>Confirm cash received at the hall counter.</p>
        <form method="POST" id="payForm">
            {% csrf_token %}
            <div class="form-group pb-3">
                <label>Manual OR Number (Leave blank for auto-generate)</label>
                <input type="text" name="or_number" class="form-control" placeholder="OR-
YYYY-XXXX">
            </div>
            <div class="modal-actions">
                <button type="button" class="btn btn-secondary"

```

IoT-Enabled Barangay Resident and Attendance Management System

```
onclick="closeModals()">Cancel</button>
        <button type="submit" class="btn btn-primary">Confirm & Issue
Receipt</button>
    </div>
</form>
</div>
</div>

<!-- GCash Verification Modal -->
<div id="gcashModal" class="modal">
    <div class="modal-content" style="max-width: 500px;">
        <h3>Verify GCash Payment</h3>
        <div id="proofContainer" style="margin-bottom: 15px;">
            <img id="proofImg" src="" style="width: 100%; border-radius: 8px; border: 1px
solid #ddd;">
        </div>
        <form method="POST" id="gcashForm">
            {% csrf_token %}
            <div class="form-group pb-3">
                <label>GCash Reference Number</label>
                <input type="text" name="gcash_ref_number" class="form-control" required>
            </div>
            <div class="modal-actions">
                <button type="button" class="btn btn-secondary"
onclick="closeModals()">Cancel</button>
                <button type="submit" class="btn btn-info">Approve Payment</button>
            </div>
        </form>
    </div>
</div>

<!-- Waiver Modal -->
<div id="waiveModal" class="modal">
    <div class="modal-content">
        <h3>Waive Fee</h3>
        <form method="POST" id="waiveForm">
            {% csrf_token %}
            <div class="form-group pb-3">
                <label>Reason for Waiver</label>
                <textarea name="reason" class="form-control" required
placeholder="e.g. Indigent Resident / PWD"></textarea>
            </div>
            <div class="modal-actions">
                <button type="button" class="btn btn-secondary"
onclick="closeModals()">Cancel</button>
                <button type="submit" class="btn btn-danger">Confirm Waiver</button>
            </div>
        </form>
    </div>
</div>

<style>
    .modal {
        display: none;
        position: fixed;
        z-index: 1000;
        left: 0;
        top: 0;
        width: 100%;
        height: 100%;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        background: rgba(0, 0, 0, 0.5);
    }

    .modal-content {
        background: white;
        margin: 10% auto;
        padding: 30px;
        border-radius: 12px;
        width: 90%;
        max-width: 400px;
    }

    .modal-actions {
        display: flex;
        justify-content: flex-end;
        gap: 10px;
        margin-top: 20px;
    }

    .pb-3 {
        padding-bottom: 15px;
    }

    .badge {
        padding: 5px 10px;
        border-radius: 4px;
        font-size: 11px;
        font-weight: 700;
        text-transform: uppercase;
    }

    .badge-info {
        background: #e0f2fe;
        color: #0369a1;
    }

    .badge-warning {
        background: #ffe3c7;
        color: #d97706;
    }

    .badge-secondary {
        background: #f3f4f6;
        color: #374151;
    }
}
</style>

<script>
    function showPayModal(id) {
        document.getElementById('payForm').action = "/payments/" + id + "/pay-cash/";
        document.getElementById('payModal').style.display = 'block';
    }

    function showConfirmGcashModal(id, imgUrl) {
        document.getElementById('gcashForm').action = "/payments/" + id + "/confirm-
gcash/";
        document.getElementById('proofImg').src = imgUrl;
        document.getElementById('gcashModal').style.display = 'block';
    }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
function showWaiveModal(id) {
    document.getElementById('waiveForm').action = "/payments/" + id + "/waive/";
    document.getElementById('waiveModal').style.display = 'block';
}

function closeModals() {
    document.querySelectorAll('.modal').forEach(m => m.style.display = 'none');
}
</script>
{% endblock %}
```

File: templates/payments/select_payment.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Payment ? {{ cert_request.get_cert_type_display }}{% endblock %}

{% block content %}
<link href="https://fonts.googleapis.com/css2?family=DM+Serif+Display:ital@0;1&family=DM+Sans:wght@300;400;500;600;700&display=swap" rel="stylesheet">

<style>
  /* ????? CSS Tokens ????? */
  :root {
    --pay-bg:          #f0f4f8;
    --pay-surface:     #ffffff;
    --pay-green:       #166534;
    --pay-green-md:    #16a34a;
    --pay-green-lt:    #dcfce7;
    --pay-blue:        #1d4ed8;
    --pay-blue-lt:     #dbeafe;
    --pay-orange:      #c2410c;
    --pay-orange-lt:   #fed7aa;
    --pay-amber:       #92400e;
    --pay-amber-lt:    #fef3c7;
    --pay-muted:       #64748b;
    --pay-border:      #e2e8f0;
    --pay-text:        #0f172a;
    --pay-card-r:      20px;
    --pay-shadow:      0 4px 24px rgba(0,0,0,.08);
  }

  .pay-wrapper {
    max-width: 960px;
    margin: 0 auto;
    padding: 32px 20px 60px;
    font-family: 'DM Sans', sans-serif;
    color: var(--pay-text);
  }

  /* ?? Page Header ?? */
  .pay-page-header {
    margin-bottom: 32px;
  }
  .pay-page-header h1 {
    font-family: 'DM Serif Display', serif;
    font-size: 2rem;
    font-weight: 400;
    color: var(--pay-text);
    margin: 0 0 6px;
    line-height: 1.2;
  }
  .pay-page-header .tracking {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    background: var(--pay-green-lt);
    color: var(--pay-green);
    padding: 5px 12px;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
border-radius: 99px;
font-size: 12px;
font-weight: 700;
letter-spacing: 0.5px;
text-transform: uppercase;
}

/* ?? Grid Layout ?? */
.pay-grid {
  display: grid;
  grid-template-columns: 1fr 340px;
  gap: 24px;
  align-items: start;
}
@media (max-width: 768px) {
  .pay-grid { grid-template-columns: 1fr; }
  .pay-sidebar { order: -1; }
}

/* ?? Cards ?? */
.pay-card {
  background: var(--pay-surface);
  border-radius: var(--pay-card-r);
  box-shadow: var(--pay-shadow);
  overflow: hidden;
}
.pay-card-header {
  padding: 20px 24px 16px;
  border-bottom: 1px solid var(--pay-border);
  display: flex;
  align-items: center;
  gap: 10px;
}
.pay-card-header h2 {
  font-size: 1rem;
  font-weight: 700;
  margin: 0;
  color: var(--pay-text);
}
.pay-card-header .icon {
  width: 32px; height: 32px;
  border-radius: 8px;
  display: flex; align-items: center; justify-content: center;
  font-size: 14px;
}
.pay-card-body {
  padding: 24px;
}

/* ?? Fee Transparency Box ?? */
.fee-box {
  background: linear-gradient(135deg, #f0fdf4 0%, #dcfce7 100%);
  border: 2px solid var(--pay-green-md);
  border-radius: 16px;
  padding: 24px;
  text-align: center;
  margin-bottom: 24px;
  position: relative;
  overflow: hidden;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.fee-box::before {
  content: '';
  position: absolute;
  top: -20px; right: -20px;
  width: 100px; height: 100px;
  background: rgba(22, 163, 74, 0.08);
  border-radius: 50%;
}
.fee-box-label {
  font-size: 11px;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 1.5px;
  color: var(--pay-green);
  margin-bottom: 8px;
}
.fee-box-amount {
  font-family: 'DM Serif Display', serif;
  font-size: 3rem;
  font-weight: 400;
  color: var(--pay-green);
  line-height: 1;
  margin-bottom: 6px;
}
.fee-box-sub {
  font-size: 12px;
  color: #4ade80;
  font-weight: 500;
}

/* Waived badge */
.waived-badge {
  display: inline-flex;
  align-items: center;
  gap: 8px;
  background: var(--pay-green-lt);
  border: 2px solid #4ade80;
  border-radius: 12px;
  padding: 16px 24px;
  font-weight: 700;
  color: var(--pay-green);
  font-size: 1rem;
  margin-bottom: 20px;
  width: 100%;
  justify-content: center;
}

/* ?? Steps Section ?? */
.steps-title {
  font-size: 11px;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 1.5px;
  color: var(--pay-muted);
  margin: 0 0 16px;
}

/* ?? Method Cards ?? */
.method-grid { display: flex; flex-direction: column; gap: 12px; }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.method-card {
  display: flex;
  align-items: center;
  gap: 16px;
  padding: 18px 20px;
  border: 2px solid var(--pay-border);
  border-radius: 14px;
  cursor: pointer;
  transition: all 0.22s cubic-bezier(0.4, 0, 0.2, 1);
  background: var(--pay-surface);
  position: relative;
  user-select: none;
}
.method-card:hover {
  border-color: #94a3b8;
  background: #f8fafc;
  transform: translateY(-1px);
  box-shadow: 0 4px 12px rgba(0,0,0,.06);
}
.method-card.selected {
  border-color: var(--pay-green-md);
  background: var(--pay-green-lt);
  box-shadow: 0 0 0 4px rgba(22,163,74,.08), 0 4px 12px rgba(0,0,0,.06);
  transform: translateY(-1px);
}
.method-card.selected-gcash {
  border-color: #2563eb;
  background: var(--pay-blue-lt);
  box-shadow: 0 0 0 4px rgba(37,99,235,.08), 0 4px 12px rgba(0,0,0,.06);
}
.method-icon {
  width: 48px; height: 48px;
  border-radius: 12px;
  display: flex; align-items: center; justify-content: center;
  font-size: 22px;
  flex-shrink: 0;
}
.method-icon.cash { background: #f0fdf4; color: var(--pay-green); }
.method-icon.gcash { background: #dbeafe; color: #1d4ed8; }
.method-content { flex: 1; }
.method-name {
  font-weight: 700;
  font-size: 15px;
  color: var(--pay-text);
  margin-bottom: 2px;
}
.method-desc {
  font-size: 12px;
  color: var(--pay-muted);
  line-height: 1.4;
}
.method-check {
  width: 22px; height: 22px;
  border-radius: 50%;
  border: 2px solid var(--pay-border);
  display: flex; align-items: center; justify-content: center;
  flex-shrink: 0;
  transition: all 0.2s;
  font-size: 11px;
  color: transparent;
```


IoT-Enabled Barangay Resident and Attendance Management System

```
}
.method-card.selected .method-check,
.method-card.selected-gcash .method-check {
  background: var(--pay-green-md);
  border-color: var(--pay-green-md);
  color: #fff;
}
.method-card.selected-gcash .method-check {
  background: var(--pay-blue);
  border-color: var(--pay-blue);
}

/* ?? Instruction Panels ?? */
.instruction-panel {
  display: none;
  margin-top: 20px;
  animation: slideIn 0.25s ease;
}
.instruction-panel.active { display: block; }

@keyframes slideIn {
  from { opacity: 0; transform: translateY(8px); }
  to { opacity: 1; transform: translateY(0); }
}

.cash-instruction {
  background: #f0fdf4;
  border: 1px solid #86efac;
  border-radius: 14px;
  padding: 20px;
}
.cash-instruction .step {
  display: flex; align-items: flex-start; gap: 12px;
  margin-bottom: 14px;
}
.cash-instruction .step:last-child { margin-bottom: 0; }
.step-num {
  width: 26px; height: 26px;
  border-radius: 50%;
  background: var(--pay-green);
  color: #fff;
  font-weight: 700;
  font-size: 12px;
  display: flex; align-items: center; justify-content: center;
  flex-shrink: 0;
  margin-top: 1px;
}
.step-text { font-size: 13px; line-height: 1.5; color: #166534; }
.step-text strong { color: #14532d; }

.tracking-chip {
  display: inline-block;
  background: #16a34a;
  color: #fff;
  font-family: monospace;
  font-size: 13px;
  font-weight: 700;
  padding: 4px 10px;
  border-radius: 6px;
  letter-spacing: 1px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

/* GCash panel */
.gcash-instruction {
  background: var(--pay-blue-lt);
  border: 1px solid #93c5fd;
  border-radius: 14px;
  padding: 20px;
}
.gcash-instruction .step-text { color: #1e40af; }
.gcash-instruction .step-num { background: var(--pay-blue); }

.upload-zone {
  border: 2px dashed #93c5fd;
  border-radius: 12px;
  padding: 20px;
  text-align: center;
  margin-top: 14px;
  background: rgba(255,255,255,0.6);
  cursor: pointer;
  transition: all 0.2s;
}
.upload-zone:hover { border-color: var(--pay-blue); background: rgba(255,255,255,0.9); }
.upload-zone input { display: none; }
.upload-zone-label {
  display: flex; flex-direction: column; align-items: center; gap: 6px;
  cursor: pointer;
}
.upload-icon { font-size: 28px; color: var(--pay-blue); margin-bottom: 4px; }
.upload-text { font-size: 13px; font-weight: 600; color: #1d4ed8; }
.upload-sub { font-size: 11px; color: #64748b; }
#preview-img { max-width: 100%; border-radius: 8px; margin-top: 10px; display: none; }

/* ?? Submit Button ?? */
.submit-btn {
  width: 100%;
  padding: 16px;
  background: linear-gradient(135deg, var(--pay-green) 0%, var(--pay-green-md) 100%);
  color: #fff;
  border: none;
  border-radius: 14px;
  font-size: 15px;
  font-weight: 700;
  font-family: 'DM Sans', sans-serif;
  cursor: pointer;
  margin-top: 24px;
  display: flex; align-items: center; justify-content: center; gap: 10px;
  transition: all 0.22s;
  letter-spacing: 0.3px;
  box-shadow: 0 4px 14px rgba(22,163,74,.3);
}
.submit-btn:hover {
  transform: translateY(-2px);
  box-shadow: 0 8px 20px rgba(22,163,74,.4);
}
.submit-btn:disabled {
  background: #94a3b8;
  box-shadow: none;
  cursor: not-allowed;
  transform: none;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

/* ?? Sidebar Summary ?? */
.summary-row {
  display: flex; justify-content: space-between; align-items: baseline;
  padding: 10px 0;
  border-bottom: 1px solid var(--pay-border);
  font-size: 14px;
}
.summary-row:last-child { border-bottom: none; }
.summary-label { color: var(--pay-muted); }
.summary-value { font-weight: 600; color: var(--pay-text); }
.summary-total-row {
  display: flex; justify-content: space-between; align-items: center;
  padding: 14px 0 0;
  margin-top: 4px;
  border-top: 2px solid var(--pay-green-md);
}
.summary-total-label { font-weight: 700; font-size: 15px; color: var(--pay-text); }
.summary-total-amount {
  font-family: 'DM Serif Display', serif;
  font-size: 1.8rem;
  color: var(--pay-green);
}

.exempt-badge {
  display: inline-flex; align-items: center; gap: 6px;
  background: var(--pay-green-lt);
  border: 1px solid #86efac;
  color: var(--pay-green);
  padding: 4px 10px;
  border-radius: 99px;
  font-size: 11px;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

/* ?? What happens next ?? */
.next-steps-card {
  margin-top: 16px;
  background: var(--pay-surface);
  border-radius: var(--pay-card-r);
  box-shadow: var(--pay-shadow);
  overflow: hidden;
}
.next-step-item {
  display: flex; align-items: flex-start; gap: 12px;
  padding: 14px 20px;
  border-bottom: 1px solid var(--pay-border);
  font-size: 13px;
}
.next-step-item:last-child { border-bottom: none; }
.next-step-dot {
  width: 8px; height: 8px;
  border-radius: 50%;
  background: var(--pay-green-md);
  flex-shrink: 0;
  margin-top: 5px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.next-step-text { color: #334155; line-height: 1.5; }
.next-step-text strong { color: var(--pay-text); }

/* ?? Status Bar for existing payment ?? */
.status-bar {
  background: var(--pay-amber-lt);
  border: 1px solid #fcd34d;
  border-radius: 12px;
  padding: 14px 18px;
  display: flex; align-items: center; gap: 10px;
  margin-bottom: 20px;
  font-size: 13px;
  color: var(--pay-amber);
  font-weight: 600;
}
</style>

<div class="pay-wrapper">

  <!-- Page Header -->
  <div class="pay-page-header">
    {% url 'certifications:my_requests' as back_url %}
    {% include 'components/back_button.html' with back_url=back_url back_label='My
Requests' %}
    <h1 style="margin-top: 16px;">Payment Setup</h1>
    <div class="tracking">
      <i class="fas fa-hashtag"></i>
      {{ cert_request.tracking_code }}
    </div>
  </div>

  <div class="pay-grid">
    <!-- ?? LEFT: Payment Form ?? -->
    <div>
      <!-- Existing payment status bar -->
      {% if cert_request.payment and cert_request.payment.status != 'unpaid' %}
      <div class="status-bar">
        <i class="fas fa-info-circle" style="font-size: 16px;"></i>
        <span>
          Your current payment status is
          <strong>{{ cert_request.payment.get_status_display }}</strong>.
          You can update your payment method below.
        </span>
      </div>
      {% endif %}

      <!-- Fee Display Card -->
      <div class="pay-card" style="margin-bottom: 20px;">
        <div class="pay-card-header">
          <div class="icon" style="background: #fef3c7; color: #92400e;">
            <i class="fas fa-receipt"></i>
          </div>
          <h2>Your Certificate Fee</h2>
        </div>
        <div class="pay-card-body">
          {% if is_waived %}
          <div class="waived-badge">
            <i class="fas fa-check-circle" style="font-size: 20px;"></i>
            Your fee is WAIVED ? ?0.00
          </div>
          </div>
        </div>
      </div>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<p style="font-size: 13px; color: var(--pay-muted); text-align: center; margin:
0;">
    You are 100% exempt from paying for your certification under the national
First Time Jobseekers Assistance Act (RA 11261).
</p>
{% else %}
<div class="fee-box">
    <div class="fee-box-label">Amount to Pay</div>
    <div class="fee-box-amount">?{{ amount|floatformat:2 }}</div>
    {% if cert_request.cert_type == 'cedula' %}
    <div class="fee-box-sub">Computed from your declared taxable amounts</div>
    {% else %}
    <div class="fee-box-sub">Standard barangay rate for {{
cert_request.get_cert_type_display }}</div>
    {% endif %}
</div>

<!-- Fee Breakdown for non-waived -->
{% if cert_request.cert_type != 'cedula' %}
<div style="background: #f8fafc; border-radius: 10px; padding: 14px 16px; font-
size: 13px; color: #475569; margin-bottom: 4px;">
    <i class="fas fa-info-circle" style="color: var(--pay-green-md); margin-right:
6px;"></i>
    This is the <strong>official barangay fee</strong> set by the local government
for a
    <strong>{{ cert_request.get_cert_type_display }}</strong>.
    The secretary will collect exactly this amount when processing your request.
</div>
{% else %}
<div style="background: #f8fafc; border-radius: 10px; padding: 14px 16px; font-
size: 13px; color: #475569; margin-bottom: 4px;">
    <i class="fas fa-calculator" style="color: var(--pay-green-md); margin-right:
6px;"></i>
    This amount was computed from the taxable amounts you declared.
    The secretary will verify and may adjust the final amount at the hall.
</div>
{% endif %}
{% endif %}
</div>
</div>

<!-- Payment Method Selection -->
<div class="pay-card">
    <div class="pay-card-header">
        <div class="icon" style="background: #ede9fe; color: #7c3aed;">
            <i class="fas fa-credit-card"></i>
        </div>
        <h2>How Would You Like to Pay?</h2>
    </div>
    <div class="pay-card-body">
        {% if is_waived %}
        <p style="font-size: 14px; color: var(--pay-muted); margin-bottom: 20px;">
            No payment is required. Simply confirm to submit your request for processing.
        </p>
        <form method="POST">
            {% csrf_token %}
            <input type="hidden" name="method" value="waived">
            <button type="submit" class="submit-btn">
                <i class="fas fa-check-circle"></i>
                Confirm & Submit Request (Free)
            </button>
        </form>
        </div>
    </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</button>
</form>
{% else %}
<p class="steps-title">Select a payment method</p>

<div class="method-grid" id="methodGrid">
  <!-- Cash -->
  <div class="method-card" id="opt-cash" onclick="selectMethod('cash')">
    <div class="method-icon cash">
      <i class="fas fa-university"></i>
    </div>
    <div class="method-content">
      <div class="method-name">Pay at the Barangay Hall</div>
      <div class="method-desc">Bring your tracking code and pay cash directly to
the secretary.</div>
    </div>
    <div class="method-check" id="check-cash">
      <i class="fas fa-check"></i>
    </div>
  </div>

  <!-- GCash -->
  <div class="method-card" id="opt-gcash" onclick="selectMethod('gcash')">
    <div class="method-icon gcash">
      <i class="fas fa-mobile-alt"></i>
    </div>
    <div class="method-content">
      <div class="method-name">Pay via GCash</div>
      <div class="method-desc">Send payment online and upload your proof of
payment screenshot.</div>
    </div>
    <div class="method-check" id="check-gcash">
      <i class="fas fa-check"></i>
    </div>
  </div>
</div>

<!-- Cash Instructions -->
<div class="instruction-panel" id="panel-cash">
  <div class="cash-instruction">
    <p class="steps-title" style="color: var(--pay-green); margin-bottom:
12px;">
      <i class="fas fa-shoe-prints" style="margin-right: 6px;"></i>Steps to Pay
at Hall

    </p>
    <div class="step">
      <div class="step-num">1</div>
      <div class="step-text">Click <strong>"Confirm"</strong> below to save your
payment method.</div>
    </div>
    <div class="step">
      <div class="step-num">2</div>
      <div class="step-text">Visit the <strong>Barangay Hall</strong> during
office hours.</div>
    </div>
    <div class="step">
      <div class="step-num">3</div>
      <div class="step-text">Present your <strong>tracking code</strong> to the
secretary: <span class="tracking-chip">{{ cert_request.tracking_code }}</span></div>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <div class="step">
            <div class="step-num">4</div>
            <div class="step-text">Pay exactly <strong>?{{ amount|floatformat:2
}}</strong>. You will receive an official receipt (OR).</div>
        </div>
        <div class="step">
            <div class="step-num">5</div>
            <div class="step-text">Your certificate will be processed and you'll
receive a notification when it's ready.</div>
        </div>
    </div>
</div>

<!-- GCash Instructions -->
<div class="instruction-panel" id="panel-gcash">
    <div class="gcash-instruction">
        <p class="steps-title" style="color: var(--pay-blue); margin-bottom: 12px;">
            <i class="fas fa-mobile-alt" style="margin-right: 6px;"></i>Steps to Pay
via GCash
        </p>
        <div class="step">
            <div class="step-num">1</div>
            <div class="step-text">Open GCash and send <strong>?{{
amount|floatformat:2 }}</strong> to the barangay's official GCash number.</div>
        </div>
        <div class="step">
            <div class="step-num">2</div>
            <div class="step-text">Use your tracking code <span class="tracking-chip"
style="background: var(--pay-blue);">{{ cert_request.tracking_code }}</span> as your
payment note/reference.</div>
        </div>
        <div class="step">
            <div class="step-num">3</div>
            <div class="step-text">Take a <strong>screenshot</strong> of the GCash
confirmation screen.</div>
        </div>
        <div class="step">
            <div class="step-num">4</div>
            <div class="step-text">Upload the screenshot below and click
<strong>"Submit"</strong>. The secretary will verify your payment.</div>
        </div>

        <form method="POST" enctype="multipart/form-data" id="paymentForm">
            {% csrf_token %}
            <input type="hidden" name="method" value="gcash" id="gcash_method">

            <div class="upload-zone" id="uploadZone">
                <label class="upload-zone-label" for="proof_file">
                    <div class="upload-icon"><i class="fas fa-cloud-upload-alt"></i></div>
                    <div class="upload-text">Click to upload payment screenshot</div>
                    <div class="upload-sub">PNG, JPG, JPEG accepted</div>
                </label>
                <input type="file" name="proof" id="proof_file" accept="image/*"
required>

                <img id="preview-img" alt="Payment proof preview">
            </div>

            <button type="submit" class="submit-btn" id="gcash-submit-btn"
style="margin-top: 16px; background: linear-gradient(135deg, #1d4ed8 0%, #2563eb 100%);
box-shadow: 0 4px 14px rgba(29,78,216,.3);">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <i class="fas fa-paper-plane"></i>
        Submit GCash Payment Proof
    </button>
</form>
</div>
</div>

<!-- Cash Submit Form -->
<form method="POST" id="cashForm" style="display: none;">
    {% csrf_token %}
    <input type="hidden" name="method" value="cash">
    <button type="submit" class="submit-btn" id="cash-submit-btn">
        <i class="fas fa-check-circle"></i>
        Confirm ? I Will Pay at the Hall
    </button>
</form>
{% endif %}
</div>
</div>
</div>

<!-- ?? RIGHT: Sidebar ?? -->
<div class="pay-sidebar">
    <!-- Summary -->
    <div class="pay-card">
        <div class="pay-card-header">
            <div class="icon" style="background: #f0fdf4; color: var(--pay-green);">
                <i class="fas fa-file-alt"></i>
            </div>
            <h2>Request Summary</h2>
        </div>
        <div class="pay-card-body" style="padding: 20px 24px;">
            <div class="summary-row">
                <span class="summary-label">Document</span>
                <span class="summary-value" style="text-align: right; max-width: 160px;">{{
cert_request.get_cert_type_display }}</span>
            </div>
            <div class="summary-row">
                <span class="summary-label">Purpose</span>
                <span class="summary-value" style="text-align: right; max-width: 160px; font-
size: 12px;">{{ cert_request.purpose|truncatechars:50 }}</span>
            </div>
            <div class="summary-row">
                <span class="summary-label">Submitted</span>
                <span class="summary-value">{{ cert_request.created_at|date:"M d, Y" }}</span>
            </div>
            <div class="summary-row">
                <span class="summary-label">Resident</span>
                <span class="summary-value">{{ cert_request.resident.full_name }}</span>
            </div>
            {% if not is_waived %}
            <div class="summary-row">
                <span class="summary-label">Official Rate</span>
                <span class="summary-value">?{{ standard_rate|floatformat:2 }}</span>
            </div>
            {% endif %}

            {% if is_waived %}
            <div class="summary-total-row">
                <span class="summary-total-label">Total Due</span>
            </div>
            {% endif %}
        </div>
    </div>
</div>
</div>
```


IoT-Enabled Barangay Resident and Attendance Management System

```
<div>
  <span class="exempt-badge"><i class="fas fa-check"></i> WAIVED</span>
</div>
</div>
{% else %}
<div class="summary-total-row">
  <span class="summary-total-label">Total Due</span>
  <span class="summary-total-amount">?{{ amount|floatformat:2 }}</span>
</div>
{% endif %}
</div>
</div>

<!-- What Happens Next -->
<div class="next-steps-card">
  <div class="pay-card-header" style="padding: 16px 20px;">
    <div class="icon" style="background: #fef3c7; color: #92400e;">
      <i class="fas fa-map-signs"></i>
    </div>
    <h2 style="font-size: 13px;">What Happens Next?</h2>
  </div>
  <div class="next-step-item">
    <div class="next-step-dot"></div>
    <div class="next-step-text">You select a payment method and confirm below.</div>
  </div>
  <div class="next-step-item">
    <div class="next-step-dot" style="background: #60a5fa;"></div>
    <div class="next-step-text">The <strong>Barangay Secretary</strong> reviews your
request and payment.</div>
  </div>
  <div class="next-step-item">
    <div class="next-step-dot" style="background: #f59e0b;"></div>
    <div class="next-step-text">Once payment is confirmed, your certificate is
<strong>issued</strong>.</div>
  </div>
  <div class="next-step-item">
    <div class="next-step-dot" style="background: #a78bfa;"></div>
    <div class="next-step-text">You receive a <strong>notification</strong> and can
download your certificate.</div>
  </div>
</div>
</div>
</div>
</div>

<script>
  let selectedMethod = null;

  function selectMethod(method) {
    selectedMethod = method;

    // Reset all cards
    document.querySelectorAll('.method-card').forEach(c => {
      c.classList.remove('selected', 'selected-gcash');
    });

    // Highlight chosen card
    if (method === 'cash') {
      document.getElementById('opt-cash').classList.add('selected');
    } else {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
document.getElementById('opt-gcash').classList.add('selected-gcash');
}

// Show correct panel
document.getElementById('panel-cash').classList.remove('active');
document.getElementById('panel-gcash').classList.remove('active');
document.getElementById('cashForm').style.display = 'none';

if (method === 'cash') {
  document.getElementById('panel-cash').classList.add('active');
  document.getElementById('cashForm').style.display = 'block';
} else {
  document.getElementById('panel-gcash').classList.add('active');
}
}

// File upload preview
const proofInput = document.getElementById('proof_file');
if (proofInput) {
  proofInput.addEventListener('change', function () {
    const file = this.files[0];
    if (file) {
      const reader = new FileReader();
      reader.onload = (e) => {
        const img = document.getElementById('preview-img');
        img.src = e.target.result;
        img.style.display = 'block';
        document.querySelector('.upload-zone-label .upload-text').textContent =
file.name;
        document.querySelector('.upload-zone-label .upload-sub').textContent =
`${(file.size / 1024).toFixed(1)} KB`;
      };
      reader.readAsDataURL(file);
    }
  });
}
</script>

{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/reports/dashboard.html

```
{% extends 'base.html' %}

{% block title %}Reports - Barangay Management System{% endblock %}

{% block content %}
<style>
  /* Premium Dashboard Styling */
  .page-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 2rem;
  }
  .page-title h1 {
    font-weight: 700;
    font-size: 2rem;
    color: var(--text-dark, #1e293b);
    margin-bottom: 0.25rem;
  }
  .page-title span {
    color: var(--text-muted, #64748b);
    font-size: 1rem;
    display: flex;
    align-items: center;
    gap: 0.5rem;
  }

  .btn-print {
    background: linear-gradient(135deg, #3b82f6, #2563eb);
    color: white;
    border: none;
    padding: 0.75rem 1.5rem;
    border-radius: 0.5rem;
    font-weight: 600;
    cursor: pointer;
    display: flex;
    align-items: center;
    gap: 0.5rem;
    transition: transform 0.2s, box-shadow 0.2s;
    box-shadow: 0 4px 6px -1px rgba(59, 130, 246, 0.3);
  }
  .btn-print:hover {
    transform: translateY(-2px);
    box-shadow: 0 6px 10px -1px rgba(59, 130, 246, 0.4);
  }

  /* Metric Cards */
  .metrics-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
    gap: 1.5rem;
    margin-bottom: 2rem;
  }
  .metric-card {
    background: #fff;
    border-radius: 1rem;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
padding: 1.5rem;
box-shadow: 0 4px 6px -1px rgba(0,0,0,0.05);
display: flex;
flex-direction: column;
border-top: 4px solid var(--accent, #3b82f6);
}

.metric-card.primary { border-top-color: #3b82f6; }
.metric-card.success { border-top-color: #10b981; }
.metric-card.warning { border-top-color: #f59e0b; }
.metric-card.purple { border-top-color: #8b5cf6; }

.metric-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 1rem;
}

.metric-title {
  color: #64748b;
  font-size: 0.875rem;
  font-weight: 600;
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

.metric-icon {
  width: 40px;
  height: 40px;
  border-radius: 8px;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 1.25rem;
}

.metric-card.primary .metric-icon { background: #fff6ff; color: #3b82f6; }
.metric-card.success .metric-icon { background: #e6ffe6; color: #10b981; }
.metric-card.warning .metric-icon { background: #fff2cc; color: #f59e0b; }
.metric-card.purple .metric-icon { background: #e6e6ff; color: #8b5cf6; }

.metric-value {
  font-size: 2rem;
  font-weight: 700;
  color: #1e293b;
}

/* Content Grid */
.content-grid {
  display: grid;
  grid-template-columns: 2fr 1fr;
  gap: 1.5rem;
}

@media (max-width: 992px) {
  .content-grid { grid-template-columns: 1fr; }
}

.panel {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
background: #fff;
border-radius: 1rem;
box-shadow: 0 4px 6px -1px rgba(0,0,0,0.05);
overflow: hidden;
}

.panel-header {
padding: 1.25rem 1.5rem;
border-bottom: 1px solid #f1f5f9;
background: #f8fafc;
display: flex;
align-items: center;
gap: 0.75rem;
}

.panel-header h3 {
margin: 0;
font-size: 1.125rem;
font-weight: 600;
color: #334155;
}

.panel-body {
padding: 1.5rem;
}

/* Table styling */
.report-table {
width: 100%;
border-collapse: collapse;
}

.report-table th {
text-align: left;
padding: 1rem 1.5rem;
color: #64748b;
font-size: 0.875rem;
font-weight: 600;
border-bottom: 2px solid #e2e8f0;
}

.report-table td {
padding: 1rem 1.5rem;
border-bottom: 1px solid #f1f5f9;
color: #334155;
font-weight: 500;
}

.report-table tr:hover td {
background-color: #f8fafc;
}

/* List Items */
.info-list {
list-style: none;
padding: 0;
margin: 0;
}

.info-list li {
display: flex;
justify-content: space-between;
padding: 0.75rem 0;
border-bottom: 1px dashed #e2e8f0;
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}
.info-list li:last-child {
  border-bottom: none;
}
.info-list .info-label {
  color: #64748b;
  font-weight: 500;
}
.info-list .info-val {
  color: #1e293b;
  font-weight: 700;
}

/* Print Styles */
@media print {
  body { background: white !important; }
  .sidebar, .top-navbar, .btn-print { display: none !important; }
  .main-content { margin-left: 0 !important; padding: 0 !important; }
  .metric-card, .panel { box-shadow: none !important; border: 1px solid #e2e8f0
!important; page-break-inside: avoid; }
  .page-header { margin-top: 0; }
}
</style>

<div class="page-header">
  <div class="page-title">
    <h1>Monthly Reports Summary</h1>
    <span><i class="far fa-calendar-alt"></i> Report for {% now "F Y" %}</span>
  </div>
  <button class="btn-print" onclick="window.print()">
    <i class="fas fa-print"></i> Print Report
  </button>
</div>

<div class="metrics-grid">
  <div class="metric-card primary">
    <div class="metric-header">
      <span class="metric-title">Total Population</span>
      <div class="metric-icon"><i class="fas fa-users"></i></div>
    </div>
    <div class="metric-value">{{ total_population }}</div>
  </div>

  <div class="metric-card purple">
    <div class="metric-header">
      <span class="metric-title">Total Households</span>
      <div class="metric-icon"><i class="fas fa-home"></i></div>
    </div>
    <div class="metric-value">{{ total_households }}</div>
  </div>

  <div class="metric-card warning">
    <div class="metric-header">
      <span class="metric-title">Certificates Issued</span>
      <div class="metric-icon"><i class="fas fa-file-alt"></i></div>
    </div>
    <div class="metric-value">{{ certs_this_month }}</div>
  </div>

  <div class="metric-card success">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="metric-header">
  <span class="metric-title">Fees Collected</span>
  <div class="metric-icon"><i class="fas fa-coins"></i></div>
</div>
<div class="metric-value">?{{ total_revenue|floatformat:2 }}</div>
</div>
</div>

<div class="content-grid">
  <div class="panel">
    <div class="panel-header">
      <i class="fas fa-chart-bar" style="color: #3b82f6;"></i>
      <h3>Certificate Issuance Breakdown</h3>
    </div>
    <div style="overflow-x: auto;">
      <table class="report-table">
        <thead>
          <tr>
            <th>Certificate Type</th>
            <th style="text-align: right;">Total Issued</th>
          </tr>
        </thead>
        <tbody>
          {% for item in cert_breakdown %}
          <tr>
            <td>{{ item.type }}</td>
            <td style="text-align: right;">
              <span style="background: #eff6ff; color: #1d4ed8; padding: 0.25rem 0.75rem;
border-radius: 999px; font-weight: 600;">
                {{ item.count }}
              </span>
            </td>
          </tr>
          {% empty %}
          <tr>
            <td colspan="2" style="text-align: center; color: #64748b; padding: 2rem;">No
certificates issued this month.</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
</div>

<div style="display: flex; flex-direction: column; gap: 1.5rem;">
  <div class="panel">
    <div class="panel-header">
      <i class="fas fa-users-cog" style="color: #8b5cf6;"></i>
      <h3>Demographic Snapshot</h3>
    </div>
    <div class="panel-body">
      <ul class="info-list">
        <li><span class="info-label">Male</span> <span class="info-val">{{ male_count
}}</span></li>
        <li><span class="info-label">Female</span> <span class="info-val">{{
female_count }}</span></li>
        <li><span class="info-label">Registered Voters</span> <span class="info-val">{{
voters }}</span></li>
        <li><span class="info-label">Senior Citizens</span> <span class="info-val">{{
senior }}</span></li>
      </ul>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<li><span class="info-label">PWDs</span> <span class="info-val">{{ pwd
}}</span></li>
<li><span class="info-label">4Ps Members</span> <span class="info-val">{{ fourps
}}</span></li>
</ul>
</div>
</div>

<div class="panel">
<div class="panel-header">
<i class="fas fa-landmark" style="color: #10b981;"></i>
<h3>Legislative & Staff</h3>
</div>
<div class="panel-body">
<ul class="info-list">
<li><span class="info-label">Total Ordinances</span> <span class="info-val">{{
total_ordinances }}</span></li>
<li><span class="info-label">Active Ordinances</span> <span class="info-val"
style="color: #10b981;">{{ active_ordinances }}</span></li>
<li><span class="info-label">Active Officials</span> <span class="info-val">{{
active_officials }}</span></li>
<li><span class="info-label">Attendance Today</span> <span class="info-val">{{
attendance_today }} entries</span></li>
</ul>
</div>
</div>
</div>
</div>

{% endblock %}
```


IoT-Enabled Barangay Resident and Attendance Management System

File: templates/residents/form.html

```
{% extends 'base.html' %}

{% block title %}{% if editing %}Edit{% else %}Add{% endif %} Resident - Barangay
Management System{% endblock %}

{% block content %}
{% url 'residents:list' as residents_list_url %}
{% include 'components/back_button.html' with back_url=residents_list_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>{% if editing %}Edit Resident Record{% else %}Add New Resident{% endif %}</h1>
        <span>Please fill in the personal information below</span>
    </div>
</div>

<div class="card">
    <form method="POST" enctype="multipart/form-data" class="modern-form">
        {% csrf_token %}

        <div class="form-grid">
            <!-- Personal Info -->
            <div class="form-section">
                <h3>Personal Information</h3>
                <div class="form-row">
                    <div class="form-group col-4">
                        <label>Last Name*</label>
                        <input type="text" name="last_name" value="{{ resident.last_name }}"
class="form-control" required>
                    </div>
                    <div class="form-group col-4">
                        <label>First Name*</label>
                        <input type="text" name="first_name" value="{{ resident.first_name }}"
class="form-control" required>
                    </div>
                    <div class="form-group col-3">
                        <label>Middle Name</label>
                        <input type="text" name="middle_name" value="{{ resident.middle_name }}"
class="form-control">
                    </div>
                    <div class="form-group col-1">
                        <label>Suffix</label>
                        <select name="suffix" class="form-control">
                            <option value="">None</option>
                            <option value="Jr." {% if resident.suffix == 'Jr.' %}selected{% endif
%}>Jr.</option>
                            <option value="Sr." {% if resident.suffix == 'Sr.' %}selected{% endif
%}>Sr.</option>
                            <option value="I" {% if resident.suffix == 'I' %}selected{% endif
%}>I</option>
                            <option value="II" {% if resident.suffix == 'II' %}selected{% endif
%}>II</option>
                            <option value="III" {% if resident.suffix == 'III' %}selected{% endif
%}>III</option>
                            <option value="IV" {% if resident.suffix == 'IV' %}selected{% endif
%}>IV</option>
                            <option value="V" {% if resident.suffix == 'V' %}selected{% endif
```

IoT-Enabled Barangay Resident and Attendance Management System

```

%}>V</option>
    </select>
</div>
</div>

<div class="form-row">
    <div class="form-group col-4">
        <label>Birthdate*</label>
        <input type="date" name="birthdate" value="{ { resident.birthdate | date: 'Y-m-d'
}}" class="form-control"
            required>
    </div>
    <div class="form-group col-4">
        <label>Gender*</label>
        <select name="gender" class="form-control" required>
            <option value="M" {% if resident.gender == 'M' %}>selected{% endif
%}>Male</option>
            <option value="F" {% if resident.gender == 'F' %}>selected{% endif
%}>Female</option>
        </select>
    </div>
    <div class="form-group col-4">
        <label>Civil Status*</label>
        <select name="civil_status" class="form-control" required>
            {% for code, label in resident.CIVIL_STATUS_CHOICES %}
            <option value="{ { code }}" {% if resident.civil_status == code %}>selected{%
endif %}>{ { label }}</option>
            {% endfor %}
        </select>
    </div>
</div>

<div class="form-row">
    <div class="form-group col-6">
        <label>Address*</label>
        <textarea name="address" class="form-control" rows="2" required>{ {
resident.address }}</textarea>
    </div>
    <div class="form-group col-6">
        <label>Purok/Zone</label>
        <div class="d-flex align-items-center" style="gap: 10px;">
            <select name="purok" id="id_purok" class="form-control">
                <option value="">-- Select Purok --</option>
                {% for p in puroks %}
                <option value="{ { p.id }}" {% if resident.purok_id == p.id %}>selected{%
endif %}>{ { p.name }}</option>
                {% endfor %}
            </select>
            {% if request.user.is_superuser or request.user.profile.role == 'captain' or
request.user.profile.role == 'secretary' or request.user.profile.role == 'treasurer' or
request.user.profile.role == 'admin' %}
            <button type="button" class="btn btn-primary d-flex align-items-center
justify-content-center" data-toggle="modal" data-target="#addPurokModal" style="width:
45px; height: 45px; min-width: 45px; border-radius: 0.75rem;">
                <i class="fas fa-plus"></i>
            </button>
            {% endif %}
        </div>
    </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div class="form-row">
  <div class="form-group col-6">
    <label>Religion</label>
    <input type="text" name="religion" value="{{ resident.religion }}"
class="form-control">
  </div>
  <div class="form-group col-6">
    <label>Occupation</label>
    <div id="occupation-wrapper">
      <input type="text" name="occupation" id="id_occupation" value="{{
resident.occupation }}"
      class="form-control" {% if resident.is_official %}style="display:none;" {%
endif %}>
      <select name="official_position" id="id_official_position" class="form-
control" {% if not resident.is_official %}style="display:none;" {% endif %}>
        <option value="">-- Select Position --</option>
        <option value="captain" {% if resident.official_record.position ==
'captain' %}selected{% endif %}>
          Punong
            Barangay (Captain)</option>
        <option value="kagawad" {% if resident.official_record.position ==
'kagawad' %}selected{% endif %}>
          Sangguniang Barangay Member (Kagawad)</option>
        <option value="secretary" {% if resident.official_record.position ==
'secretary' %}selected{% endif %}>
          Barangay Secretary</option>
        <option value="treasurer" {% if resident.official_record.position ==
'treasurer' %}selected{% endif %}>
          Barangay Treasurer</option>
        <option value="sk_chairman" {% if resident.official_record.position ==
'sk_chairman' %}selected{% endif %}>
          SK Chairperson</option>
        <option value="sk_kagawad" {% if resident.official_record.position ==
'sk_kagawad' %}selected{% endif %}>
          SK Kagawad</option>
        <option value="tanod" {% if resident.official_record.position == 'tanod'
%}selected{% endif %}>
          Barangay
            Tanod</option>
        <option value="health_worker" {% if resident.official_record.position ==
'health_worker' %}selected{% endif %}>
          Barangay Health Worker (BHW)</option>
        <option value="nutrition_scholar" {% if resident.official_record.position
== 'nutrition_scholar' %}selected{% endif %}>
          Barangay Nutrition Scholar (BNS)</option>
        <option value="day_care_worker" {% if resident.official_record.position ==
'day_care_worker' %}selected{% endif %}>
          Day Care Worker</option>
        <option value="lupon" {% if resident.official_record.position == 'lupon'
%}selected{% endif %}>
          Lupon
            Member</option>
        <option value="clerk" {% if resident.official_record.position == 'clerk'
%}selected{% endif %}>
          Barangay
            Clerk</option>
        <option value="staff" {% if resident.official_record.position == 'staff'
%}selected{% endif %}>
          Barangay
            Staff</option>
      </select>
    </div>
  </div>
</div>

<!-- Household & Classification -->
<div class="form-section">

```

IoT-Enabled Barangay Resident and Attendance Management System

```

<h3>Household & Classification</h3>
<div class="form-row">
  <div class="form-group col-6">
    <label>Household</label>
    <select name="household" id="id_household" class="form-control">
      <option value="">No Household Assigned</option>
      {% for household in households %}
        <option value="{{ household.pk }}" data-purok="{{ household.purok.id }}" {%
if resident.household.pk == household.pk %}selected{% endif %}>{{ household.household_no
}}{% if household.purok %} ? {{ household.purok }}{% endif %}</option>
      {% endfor %}
    </select>
  </div>
  <div class="form-group col-6 checkbox-inline">
    <label class="checkbox-container">
      <input type="checkbox" name="is_household_head" {% if
resident.is_household_head %}checked{% endif %}>
      <span>Is Household Head</span>
    </label>
  </div>
</div>

<div class="form-row checkbox-row">
  <label class="checkbox-container">
    <input type="checkbox" name="is_registered_voter" {% if
resident.is_registered_voter %}checked{% endif %}>
    <span>Registered Voter</span>
  </label>
  <label class="checkbox-container">
    <input type="checkbox" name="is_pwd" {% if resident.is_pwd %}checked{% endif
%}>
    <span>PWD</span>
  </label>
  <label class="checkbox-container">
    <input type="checkbox" name="is_senior_citizen" {% if
resident.is_senior_citizen %}checked{% endif %}>
    <span>Senior Citizen</span>
  </label>
  <label class="checkbox-container">
    <input type="checkbox" name="is_4ps_member" {% if resident.is_4ps_member
%}checked{% endif %}>
    <span>4Ps Member</span>
  </label>
  <label class="checkbox-container">
    <input type="checkbox" name="is_official" id="id_is_official" {% if
resident.is_official %}checked{% endif %}>
    <span class="text-primary font-weight-bold">Barangay Functionary</span>
  </label>
</div>

<div class="form-row">
  <div class="form-group col-4">
    <label>Contact Number</label>
    <input type="text" name="contact_number" value="{{ resident.contact_number }}"
class="form-control">
  </div>
  <div class="form-group col-4">
    <label>Email Address</label>
    <input type="email" name="email" value="{{ resident.email }}" class="form-
control">
  </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>
    <div class="form-group col-4">
        <label>Photo</label>
        <input type="file" name="photo" class="form-control">
    </div>
</div>
</div>
</div>
</div>

<div class="form-footer">
    <button type="submit" class="btn btn-primary btn-lg">
        <i class="fas fa-save"></i> Save Resident Record
    </button>
</div>
</form>
</div>

<!-- Add Purok Modal -->
<div class="modal fade" id="addPurokModal" tabindex="-1" role="dialog" aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered" role="document">
        <div class="modal-content border-0 shadow-lg" style="border-radius: 1.5rem;">
            <div class="modal-header border-0 pt-4 px-4">
                <h5 class="modal-title font-bold text-slate-800">Add New Purok / Zone</h5>
                <button type="button" class="close" data-dismiss="modal" aria-
label="Close">
                    <span aria-hidden="true">&times;</span>
                </button>
            </div>
            <div class="modal-body px-4 pb-4">
                <div class="form-group mb-4">
                    <label class="text-muted small font-bold text-uppercase tracking-
wider">Purok Name</label>
                    <input type="text" id="new_purok_name" class="form-control py-3"
placeholder="Enter purok name (e.g. Purok 1)" style="border-radius: 0.75rem;">
                </div>
                <div class="d-flex flex-column gap-2">
                    <button type="button" id="btn-save-purok" class="btn btn-primary py-3
font-bold" style="border-radius: 1rem; background: #2563eb; border: none;">
                        Save Purok
                    </button>
                </div>
            </div>
        </div>
    </div>
</div>
</div>

<style>
    .modern-form {
        padding: 30px;
    }

    .form-section {
        margin-bottom: 30px;
    }

    .form-section h3 {
        font-size: 16px;
        margin-bottom: 20px;
        color: var(--primary);
        border-bottom: 1px solid var(--border);

```

IoT-Enabled Barangay Resident and Attendance Management System

```
padding-bottom: 10px;
}

.form-row {
  display: flex;
  gap: 20px;
  margin-bottom: 20px;
  flex-wrap: wrap;
}

.col-6 {
  flex: 0 0 calc(50% - 10px);
}

.col-4 {
  flex: 0 0 calc(33.33% - 13.33px);
}

.col-3 {
  flex: 0 0 calc(25% - 15px);
}

.col-1 {
  flex: 0 0 calc(10% - 18px);
}

.checkbox-row {
  gap: 30px;
  padding: 10px 0;
}

.form-footer {
  margin-top: 20px;
  text-align: right;
  border-top: 1px solid var(--border);
  padding-top: 20px;
}

@media (max-width: 768px) {
  .modern-form {
    padding: 15px;
  }

  .form-row {
    gap: 15px;
    flex-direction: column;
  }

  .form-group.col-6, .form-group.col-4, .form-group.col-3, .form-group.col-1 {
    flex: 0 0 100% !important;
    width: 100% !important;
    max-width: 100% !important;
  }

  .form-control {
    width: 100% !important;
    max-width: 100% !important;
  }

  .checkbox-row {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        flex-direction: column;
        gap: 10px;
    }
}
</style>
<script>
    document.addEventListener('DOMContentLoaded', function () {
        const purokInput = document.getElementById('id_purok');
        const householdSelect = document.getElementById('id_household');
        const allOptions = Array.from(householdSelect.options);

        function filterHouseholds() {
            const selectedPurokId = purokInput.value;

            // Clear current options except the first one (No Household Assigned)
            householdSelect.innerHTML = '';

            allOptions.forEach(option => {
                if (!option.value) { // "No Household Assigned"
                    householdSelect.appendChild(option);
                    return;
                }

                const householdPurokId = option.getAttribute('data-purok');
                if (!selectedPurokId || householdPurokId === selectedPurokId) {
                    householdSelect.appendChild(option);
                }
            });
        }

        purokInput.addEventListener('change', filterHouseholds);
        // Initial filter if purok is already set (e.g., during edit)
        if (purokInput.value) {
            filterHouseholds();
        }

        // Toggle Occupation vs Official Position
        const isOfficialCheckbox = document.getElementById('id_is_official');
        const occupationInput = document.getElementById('id_occupation');
        const officialPositionSelect = document.getElementById('id_official_position');

        isOfficialCheckbox.addEventListener('change', function () {
            if (this.checked) {
                occupationInput.style.display = 'none';
                occupationInput.required = false;
                officialPositionSelect.style.display = 'block';
                officialPositionSelect.required = true;
            } else {
                occupationInput.style.display = 'block';
                occupationInput.required = false;
                officialPositionSelect.style.display = 'none';
                officialPositionSelect.required = false;
            }
        });

        // Add Purok AJAX
        const btnSavePurok = document.getElementById('btn-save-purok');
        const newPurokInput = document.getElementById('new_purok_name');
        const purokSelect = document.getElementById('id_purok');
```

IoT-Enabled Barangay Resident and Attendance Management System

```
if (btnSavePurok) {
  btnSavePurok.addEventListener('click', function() {
    const name = newPurokInput.value.trim();
    if (!name) {
      alert('Please enter a purok name');
      return;
    }

    btnSavePurok.disabled = true;
    btnSavePurok.innerHTML = '<i class="fas fa-spinner fa-spin mr-1"></i> Saving...';

    fetch('% url "residents:purok_add_api" %}', {
      method: 'POST',
      headers: {
        'Content-Type': 'application/json',
        'X-CSRFToken': document.querySelector('[name=csrfmiddlewaretoken]').value
      },
      body: JSON.stringify({ name: name })
    })
    .then(res => res.json())
    .then(data => {
      if (data.status === 'success') {
        // Add to dropdown
        const option = new Option(data.name, data.id);
        purokSelect.add(option);
        purokSelect.value = data.id;

        // Trigger change for household filtering
        purokSelect.dispatchEvent(new Event('change'));

        // Close modal
        $('#addPurokModal').modal('hide');
        newPurokInput.value = '';
      } else {
        alert(data.message || 'Failed to save purok');
      }
    })
    .catch(err => {
      console.error(err);
      alert('An error occurred while saving');
    })
    .finally(() => {
      btnSavePurok.disabled = false;
      btnSavePurok.innerHTML = 'Save Purok';
    });
  });
}
});
</script>
{% endblock %}
```


IoT-Enabled Barangay Resident and Attendance Management System

File: templates/residents/household_form.html

```
{% extends 'base.html' %}

{% block title %}{% if editing %}Edit Household #{ household.household_no }{% else %}Add Household{% endif %} - Barangay Management System{% endblock %}

{% block extra_head %}
<link rel="stylesheet" href="https://unpkg.com/maplibre-gl/dist/maplibre-gl.css" />
<style>
    .household-form-container {
        display: grid;
        grid-template-columns: 1.1fr 1.3fr;
        gap: 30px;
    }

    @media (max-width: 992px) {
        .household-form-container {
            grid-template-columns: 1fr;
        }
    }

    .modern-form {
        padding: 24px;
    }

    /* Custom marker */
    .pin-marker {
        width: 34px;
        height: 34px;
        background: linear-gradient(135deg, #ef4444, #b91c1c); /* Red/Orange for pinning */
        border: 3px solid #fff;
        border-radius: 50%;
        box-shadow: 0 4px 14px rgba(239, 68, 68, 0.45);
        cursor: move;
        display: flex;
        align-items: center;
        justify-content: center;
        color: #fff;
        transition: transform 0.15s;
    }

    .pin-marker:hover {
        transform: scale(1.15);
    }

    .pin-marker i {
        font-size: 13px;
    }

    .form-control[readonly] {
        background-color: var(--bg-main) !important;
        color: var(--text-muted) !important;
        border-color: var(--border-light) !important;
    }
</style>
{% endblock %}

{% block breadcrumb %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<span><a href="{% url 'residents:household_list' %}">Households</a> / {% if editing %}Edit
#{{ household.household_no }}{% else %}Add{% endif %}</span>
{% endblock %}

{% block content %}
{% url 'residents:household_list' as household_list_url %}
{% include 'components/back_button.html' with back_url=household_list_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>{% if editing %}Edit Household #{{ household.household_no }}{% else %}Create New
Household{% endif %}</h1>
        <span>Fill in the details and locate the household on the map</span>
    </div>
</div>

<div class="card" style="max-width: 1000px; margin: 0 auto; border-radius: 1.25rem;
overflow: hidden; box-shadow: 0 10px 30px rgba(0,0,0,0.04);">
    <form method="POST" class="modern-form">
        {% csrf_token %}
        <div class="household-form-container">

            <!-- Left Column: Fields & Inputs -->
            <div class="form-inputs-column d-flex flex-column justify-content-between">
                <div class="form-section pt-0">
                    <div class="form-row">
                        <div class="form-group col-12">
                            <label class="font-weight-bold mb-2" style="color: var(--text-
main);">Household Number*</label>
                            <input type="text" name="household_no" value="{{
household.household_no|default:'' }}" class="form-control py-3" required
placeholder="e.g. HH-2024-001" style="border-radius: 0.75rem;">
                        </div>
                    </div>

                    <div class="form-row">
                        <div class="form-group col-12">
                            <label class="font-weight-bold mb-2" style="color: var(--text-main);">Purok
/ Zone*</label>
                            <div class="d-flex align-items-center" style="gap: 10px;">
                                <select name="purok" id="id_purok" class="form-control" required
style="border-radius: 0.75rem; height: 45px;">
                                    <option value="">-- Select Purok --</option>
                                    {% for p in puroks %}
                                        <option value="{{ p.id }}" data-lat="{{ p.latitude|default:'' }}" data-
lng="{{ p.longitude|default:'' }}" {% if household.purok_id == p.id %}selected{% endif
%}>{{ p.name }}</option>
                                    {% endfor %}
                                </select>
                                {% if request.user.is_superuser or request.user.profile.role == 'captain'
or request.user.profile.role == 'secretary' or request.user.profile.role == 'treasurer' or
request.user.profile.role == 'admin' %}
                                    <button type="button" class="btn btn-primary d-flex align-items-center
justify-content-center" data-toggle="modal" data-target="#addPurokModal" style="width:
45px; height: 45px; min-width: 45px; border-radius: 0.75rem; background: var(--primary);
border: none;">
                                        <i class="fas fa-plus"></i>
                                    </button>
                                {% endif %}
                            </div>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </form>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>
</div>

<div class="form-row">
    <div class="form-group col-12">
        <label class="font-weight-bold mb-2" style="color: var(--text-main);">Street
Address / Location Details</label>
        <textarea name="address" class="form-control" rows="3" placeholder="e.g.
House No., Street name" required style="border-radius: 0.75rem; resize: none;">{{
household.address|default:'Barangay Sico-sico, Gigaquit, Surigao Del Norte' }}</textarea>
    </div>
</div>

<div class="form-row">
    <div class="form-group col-md-6 mb-3">
        <label class="font-weight-bold mb-2" style="color: var(--text-
main);">Latitude</label>
        <input type="number" step="any" name="latitude" id="id_latitude" value="{{
household.latitude|default:'' }}" class="form-control py-3" readonly placeholder="Pin on
map" style="border-radius: 0.75rem; cursor: not-allowed;">
    </div>
    <div class="form-group col-md-6 mb-3">
        <label class="font-weight-bold mb-2" style="color: var(--text-
main);">Longitude</label>
        <input type="number" step="any" name="longitude" id="id_longitude" value="{{
household.longitude|default:'' }}" class="form-control py-3" readonly placeholder="Pin on
map" style="border-radius: 0.75rem; cursor: not-allowed;">
    </div>
</div>

<div class="form-row" id="clear-coords-row" style="display: {% if
household.latitude %}block{% else %}none{% endif %};">
    <div class="form-group col-12 text-right">
        <button type="button" id="btn-clear-coords" class="btn btn-outline-danger
btn-sm px-3 py-2" style="border-radius: 0.5rem; font-weight: 500;">
            <i class="fas fa-trash-alt mr-1"></i> Clear Pin Location
        </button>
    </div>
</div>
</div>

<div class="form-footer pt-3 pb-0" style="border-top: none;">
    <button type="submit" class="btn btn-primary btn-lg btn-block py-3 font-weight-
bold" style="border-radius: 0.75rem; background: var(--primary); border: none;">
        <i class="fas fa-save mr-2"></i> Save Household Record
    </button>
</div>
</div>

<!-- Right Column: Interactive Map Pinning -->
<div class="form-map-column d-flex flex-column">
    <label class="font-weight-bold mb-1" style="color: var(--text-main); display:
flex; align-items: center; gap: 8px;">
        <i class="fas fa-map-marked-alt text-primary" style="font-size: 18px;"></i> Map
Pin Locator
    </label>
    <span class="text-muted d-block mb-3" style="font-size: 13px; line-height: 1.4;">
        Select a location on the map to pin the household. Drag the red marker to fine-
tune the coordinates. Selecting a Purok on the dropdown will automatically center the map
on that area.

```

IoT-Enabled Barangay Resident and Attendance Management System

```
</span>
<div id="household-map" style="height: 410px; width: 100%; border-radius: 1rem;
border: 1px solid var(--border-light); box-shadow: inset 0 2px 8px rgba(0,0,0,0.05);
position: relative; overflow: hidden; background: #f8fafc;"></div>
</div>

</div>
</form>
</div>

<!-- Add Purok Modal -->
<div class="modal fade" id="addPurokModal" tabindex="-1" role="dialog" aria-hidden="true">
  <div class="modal-dialog modal-dialog-centered" role="document">
    <div class="modal-content border-0 shadow-lg" style="border-radius: 1.5rem;">
      <div class="modal-header border-0 pt-4 px-4">
        <h5 class="modal-title font-weight-bold" style="color: #1e293b;">Add New
Purok / Zone</h5>
        <button type="button" class="close" data-dismiss="modal" aria-
label="Close">
          <span aria-hidden="true">&times;</span>
        </button>
      </div>
      <div class="modal-body px-4 pb-4">
        <div class="form-group mb-4">
          <label class="text-muted small font-weight-bold text-uppercase">Purok
Name</label>
          <input type="text" id="new_purok_name" class="form-control py-3"
placeholder="Enter purok name (e.g. Purok 1)" style="border-radius: 0.75rem;">
        </div>
        <button type="button" id="btn-save-purok" class="btn btn-primary btn-block
py-3 font-weight-bold" style="border-radius: 1rem; background: var(--primary); border:
none;">
          Save Purok
        </button>
      </div>
    </div>
  </div>
</div>
{% endblock %}

{% block extra_js %}
<script src="https://unpkg.com/maplibre-gl/dist/maplibre-gl.js"></script>
<script>
document.addEventListener('DOMContentLoaded', function () {
  // --- Map Configuration ---
  const BRGY_CENTER = [125.7222, 9.4898]; // [lng, lat]
  const defaultLat = parseFloat("{ household.latitude|default:'' }");
  const defaultLng = parseFloat("{ household.longitude|default:'' }");
  const hasCoords = !isNaN(defaultLat) && !isNaN(defaultLng);

  const mapCenter = hasCoords ? [defaultLng, defaultLat] : BRGY_CENTER;
  const mapZoom = hasCoords ? 17 : 15;

  const TILE_STREETS = 'https://tile.openstreetmap.org/{z}/{x}/{y}.png';
  const TILE_SATELLITE = 'https://server.arcgisonline.com/ArcGIS/rest/services/World_Ima
gery/MapServer/tile/{z}/{y}/{x}';

  function makeStyle(tileUrl) {
    return {
      version: 8,
```

IoT-Enabled Barangay Resident and Attendance Management System

```
sources: {
  'raster-tiles': {
    type: 'raster',
    tiles: [tileUrl],
    tileSize: 256,
    attribution: '&copy; OpenStreetMap contributors'
  }
},
layers: [{
  id: 'raster-layer',
  type: 'raster',
  source: 'raster-tiles',
  minzoom: 0,
  maxzoom: 19
}]
};
}

const streetsStyle = makeStyle(TILE_STREETS);

const map = new maplibregl.Map({
  container: 'household-map',
  style: streetsStyle,
  center: mapCenter,
  zoom: mapZoom,
  maxZoom: 19,
  attributionControl: false
});

// Add navigation controls (Zoom)
map.addControl(new maplibregl.NavigationControl({ showCompass: false }), 'top-right');

let marker = null;
const latInput = document.getElementById('id_latitude');
const lngInput = document.getElementById('id_longitude');
const clearRow = document.getElementById('clear-coords-row');
const btnClearCoords = document.getElementById('btn-clear-coords');

function updateInputs(lat, lng) {
  latInput.value = lat.toFixed(6);
  lngInput.value = lng.toFixed(6);
  clearRow.style.display = 'block';
}

function removeMarker() {
  if (marker) {
    marker.remove();
    marker = null;
  }
  latInput.value = '';
  lngInput.value = '';
  clearRow.style.display = 'none';
}

function createMarker(lngLat) {
  if (marker) {
    marker.setLngLat(lngLat);
  } else {
    const el = document.createElement('div');
    el.className = 'pin-marker';
  }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
el.innerHTML = '<i class="fas fa-thumbtack"></i>';

marker = new maplibregl.Marker({
  element: el,
  draggable: true,
  anchor: 'center'
})
.setLngLat(lngLat)
.addTo(map);

marker.on('dragend', function() {
  const newLngLat = marker.getLngLat();
  updateInputs(newLngLat.lat, newLngLat.lng);
});
}
updateInputs(lngLat.lat, lngLat.lng);
}

// Load existing marker if coords exist
map.on('load', function() {
  if (hasCoords) {
    createMarker({ lat: defaultLat, lng: defaultLng });
  }
});

// Click on map to place/move pin
map.on('click', function(e) {
  createMarker(e.lngLat);
});

// Clear coordinates handler
if (btnClearCoords) {
  btnClearCoords.addEventListener('click', function() {
    removeMarker();
  });
}

// Purok Selection Change: pan/zoom map to Purok coordinates
const purokSelect = document.getElementById('id_purok');
if (purokSelect) {
  purokSelect.addEventListener('change', function() {
    const selectedOpt = purokSelect.options[purokSelect.selectedIndex];
    if (selectedOpt) {
      const pLat = parseFloat(selectedOpt.dataset.lat);
      const pLng = parseFloat(selectedOpt.dataset.lng);

      // Only fly to purok location if we don't already have coordinates pinned
      if (!isNaN(pLat) && !isNaN(pLng) && !latInput.value) {
        map.flyTo({
          center: [pLng, pLat],
          zoom: 17,
          duration: 1200
        });
      }
    }
  });
}

// --- Purok Add Modal Save Hook ---
const btnSavePurok = document.getElementById('btn-save-purok');
```

IoT-Enabled Barangay Resident and Attendance Management System

```
const newPurokInput = document.getElementById('new_purok_name');

if (btnSavePurok) {
  btnSavePurok.addEventListener('click', function() {
    const name = newPurokInput.value.trim();
    if (!name) {
      alert('Please enter a purok name');
      return;
    }

    btnSavePurok.disabled = true;
    btnSavePurok.innerHTML = '<i class="fas fa-spinner fa-spin mr-1"></i>
Saving...';

    fetch('% url "residents:purok_add_api" %}', {
      method: 'POST',
      headers: {
        'Content-Type': 'application/json',
        'X-CSRFToken':
document.querySelector('[name=csrfmiddlewaretoken]').value
      },
      body: JSON.stringify({ name: name })
    })
    .then(res => res.json())
    .then(data => {
      if (data.status === 'success') {
        const option = new Option(data.name, data.id);
        purokSelect.add(option);
        purokSelect.value = data.id;
        $('#addPurokModal').modal('hide');
        newPurokInput.value = '';
      } else {
        alert(data.message || 'Failed to save purok');
      }
    })
    .catch(err => {
      console.error(err);
      alert('An error occurred while saving');
    })
    .finally(() => {
      btnSavePurok.disabled = false;
      btnSavePurok.innerHTML = 'Save Purok';
    });
  });
}
});
</script>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/residents/household_list.html

```
{% extends 'base.html' %}

{% block title %}Households - Barangay Management System{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Households</span>
{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Household Records</h1>
    <span>Manage and view all registered households</span>
  </div>
  <div class="header-actions">
    <a href="{% url 'residents:household_add' %}" class="btn btn-primary">
      <i class="fas fa-plus"></i> Add Household
    </a>
  </div>
</div>

<div class="card">
  <div class="table-responsive">
    <table>
      <thead>
        <tr>
          <th>Household No.</th>
          <th>Address</th>
          <th>Purok</th>
          <th>Members</th>
          <th>Head of Family</th>
          <th>Actions</th>
        </tr>
      </thead>
      <tbody>
        {% for household in households %}
        <tr>
          <td><strong>#{{ household.household_no }}</strong></td>
          <td>{{ household.address }}</td>
          <td>{{ household.purok }}</td>
          <td>
            <span class="member-count">
              <i class="fas fa-users"></i> {{ household.member_count }}
            </span>
          </td>
          <td>
            {% if household.head %}
            {{ household.head.full_name }}
            {% else %}
            <span class="text-muted">Not Set</span>
            {% endif %}
          </td>
          <td>
            <div class="btn-group">
              <a href="{% url 'residents:household_view' household.pk %}" class="btn btn-
outline btn-icon"
                title="View">

```


IoT-Enabled Barangay Resident and Attendance Management System

```

        <i class="fas fa-eye"></i>
    </a>
    {% if request.user.is_superuser or request.user.profile.role == 'captain' or
    request.user.profile.role == 'secretary' or request.user.profile.role == 'treasurer' or
    request.user.profile.role == 'admin' %}
        <a href="{% url 'residents:household_edit' household.pk %}" class="btn btn-
outline btn-icon"
            title="Edit">
            <i class="fas fa-edit"></i>
        </a>
    {% endif %}
</div>
</td>
</tr>
{% empty %}
<tr>
    <td colspan="6" class="text-center">No households found.</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>
{% if households.has_other_pages %}
<div class="pagination">
    <!-- Pagination logic -->
</div>
{% endif %}
</div>

<style>
    .member-count {
        background: rgba(var(--primary-rgb), 0.1);
        padding: 4px 8px;
        border-radius: 4px;
        font-weight: 600;
        font-size: 13px;
    }

    .member-count i {
        margin-right: 4px;
        color: var(--primary);
    }
</style>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/residents/household_view.html

```
{% extends 'base.html' %}

{% block title %}Household #{{ household.household_no }} - Details{% endblock %}

{% block extra_head %}
{% if household.latitude and household.longitude %}
<link rel="stylesheet" href="https://unpkg.com/maplibre-gl/dist/maplibre-gl.css" />
<style>
/* Custom mini marker */
.pin-marker-view {
width: 26px;
height: 26px;
background: linear-gradient(135deg, var(--primary), #3da667);
border: 2px solid #fff;
border-radius: 50%;
box-shadow: 0 2px 8px rgba(44,110,73,0.4);
display: flex;
align-items: center;
justify-content: center;
color: #fff;
}
.pin-marker-view i {
font-size: 10px;
}
</style>
{% endif %}
{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'residents:household_list' %}">Households</a> / #{{
household.household_no }}</span>
{% endblock %}

{% block content %}
{% url 'residents:household_list' as household_list_url %}
{% include 'components/back_button.html' with back_url=household_list_url
back_label='Back' %}
<div class="page-header">
<div class="page-title">
<h1>Household Details</h1>
<span>Information for Household #{{ household.household_no }}</span>
</div>
{% if request.user.is_superuser or request.user.profile.role == 'captain' or
request.user.profile.role == 'secretary' or request.user.profile.role == 'treasurer' or
request.user.profile.role == 'admin' %}
<div class="header-actions" style="display: flex; gap: 10px;">
<button type="button" class="btn btn-primary" style="background: var(--primary);
border: none; border-radius: 0.75rem;" data-toggle="modal" data-target="#addMemberModal">
<i class="fas fa-user-plus mr-1"></i> Add Member
</button>
<a href="{% url 'residents:household_edit' household.pk %}" class="btn btn-outline-
primary" style="border-radius: 0.75rem; border-color: var(--primary); color:
var(--primary);">
<i class="fas fa-edit mr-1"></i> Edit Household
</a>
</div>
{% endif %}
</div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</div>

<div class="profile-grid">
  <div class="profile-sidebar">
    <div class="card profile-card">
      <div class="profile-photo">
        <div class="photo-placeholder">
          <i class="fas fa-home"></i>
        </div>
      </div>
    </div>
    <div class="profile-info">
      <h2>Household #{ household.household_no }</h2>
      <span class="resident-id">Purok: {{ household.purok }}</span>
      <div class="status-tags">
        <span class="badge badge-primary">{{ members.count }} Members</span>
      </div>
    </div>
  </div>
</div>

<!-- Family Structure Card -->
<div class="card mt-4">
  <div class="card-header">
    <h3>Family Structure</h3>
  </div>
  <div class="card-body">
    <div style="display: flex; flex-direction: column; gap: 10px;">
      <div>
        <span class="text-muted" style="font-size: 12px; text-transform: uppercase; letter-spacing: 0.5px;">Head of Family</span>
        <div style="font-weight: 600; font-size: 15px; margin-top: 2px;">
          {% if household.head %}
            <i class="fas fa-user-tie" style="color: var(--primary); margin-right: 5px;"></i> {{ household.head.full_name }}
          {% else %}
            <span class="text-muted"><i class="fas fa-exclamation-circle" style="margin-right: 5px;"></i> Not Assigned</span>
          {% endif %}
        </div>
      </div>
      <div>
        <span class="text-muted" style="font-size: 12px; text-transform: uppercase; letter-spacing: 0.5px;">Total Members</span>
        <div style="font-weight: 600; font-size: 15px; margin-top: 2px;">
          <i class="fas fa-users" style="color: var(--primary); margin-right: 5px;"></i> {{ members.count }}
        </div>
      </div>
    </div>
  </div>
</div>

<!-- Pinned Location Card -->
<div class="card mt-4">
  <div class="card-header">
    <h3><i class="fas fa-map-marker-alt text-primary" style="margin-right: 5px;"></i> Pinned Location</h3>
  </div>
  <div class="card-body">
    <div style="display: flex; flex-direction: column; gap: 12px;">
      {% if household.latitude and household.longitude %}
```

IoT-Enabled Barangay Resident and Attendance Management System

```

<div>
  <span class="text-muted" style="font-size: 12px; text-transform: uppercase;
letter-spacing: 0.5px;">Coordinates</span>
  <div style="font-weight: 600; font-size: 14px; margin-top: 2px; font-family:
monospace;">
    {{ household.latitude|floatformat:6 }}, {{ household.longitude|floatformat:6
  }}
  </div>
</div>
<div id="mini-household-map" style="height: 180px; width: 100%; border-radius:
8px; border: 1px solid var(--border-light); background: #f8fafc;"></div>
{% else %}
<div class="text-center py-3">
  <i class="fas fa-map-marked-alt text-muted fa-2x mb-2"></i>
  <p class="text-muted small mb-0">No location pinned for this household.</p>
  {% if request.user.is_superuser or request.user.profile.role == 'captain' or
request.user.profile.role == 'secretary' or request.user.profile.role == 'treasurer' or
request.user.profile.role == 'admin' %}
    <a href="{% url 'residents:household_edit' household.pk %}" class="btn btn-
outline-primary btn-sm mt-2" style="border-radius: 0.5rem; border-color: var(--primary);
color: var(--primary);">
      <i class="fas fa-map-pin mr-1"></i> Pin on Map
    </a>
  {% endif %}
</div>
{% endif %}
</div>
</div>
</div>
</div>

<div class="profile-main">
  <div class="card">
    <div class="card-header">
      <h3>Family Members</h3>
    </div>
    <div class="table-responsive">
      <table>
        <thead>
          <tr>
            <th>Name</th>
            <th>Age</th>
            <th>Role</th>
            <th>Voter</th>
            <th>Actions</th>
          </tr>
        </thead>
        <tbody>
          {% for member in members %}
            <tr>
              <td>
                <strong>{{ member.full_name }}</strong>
              </td>
              <td>{{ member.age }}</td>
              <td>
                {% if member.is_household_head %}
                  <span class="status-badge badge-active">Head of Family</span>
                {% else %}
                  Member
                {% endif %}
              </td>
            </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
  </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

        </td>
        <td>
            {% if member.is_registered_voter %}
            <i class="fas fa-check text-success"></i>
            {% else %}
            <i class="fas fa-times text-muted"></i>
            {% endif %}
        </td>
        <td>
            <a href="{% url 'residents:view' member.pk %}" class="btn btn-outline btn-
icon">
                <i class="fas fa-eye"></i>
            </a>
        </td>
    </tr>
</tbody>
</table>
</div>
</div>

<!-- Family Tree Diagram -->
<div class="card mt-4 ft-card">
    <div class="card-header ft-header" style="display: flex; justify-content: space-
between; align-items: center; flex-wrap: wrap; gap: 10px;">
        <div>
            <h3><i class="fas fa-sitemap ft-icon"></i> Family Tree</h3>
            <span class="ft-subtitle">Hierarchical Overview</span>
        </div>
        {% if request.user.is_superuser or request.user.profile.role == 'captain' or
request.user.profile.role == 'secretary' or request.user.profile.role == 'treasurer' or
request.user.profile.role == 'admin' %}
        <div class="ft-actions">
            <button id="editTreeBtn" class="btn btn-sm" style="background: var(--primary);
color: white; border-radius: 0.5rem; border: none;">
                <i class="fas fa-project-diagram mr-1"></i> Edit Structure
            </button>
            <button id="saveTreeBtn" class="btn btn-sm btn-success" style="border-radius:
0.5rem; display: none;">
                <i class="fas fa-save mr-1"></i> Save Layout
            </button>
            <button id="cancelTreeBtn" class="btn btn-sm btn-light" style="border-radius:
0.5rem; display: none;">
                Cancel
            </button>
        </div>
        {% endif %}
    </div>
    <div class="card-body ft-body">
        {% if members %}
        <div class="family-tree-container" id="tree-container">
            <!-- Rendered by JS -->
        </div>
        {% else %}
        <div class="text-center py-4 text-muted">
            <i class="fas fa-tree fa-3x mb-3" style="opacity: 0.2;"></i>
            <p>No members in this household yet.</p>
        </div>
        {% endif %}
    </div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```

    </div>
  </div>
</div>

<!-- Add Member Modal -->
<div class="modal fade" id="addMemberModal" tabindex="-1" role="dialog" aria-
hidden="true">
  <div class="modal-dialog modal-dialog-centered" role="document">
    <div class="modal-content border-0 shadow-lg" style="border-radius: 1.5rem;">
      <div class="modal-header border-0 pt-4 px-4">
        <h5 class="modal-title font-weight-bold">Add Member to Household</h5>
        <button type="button" class="close" data-dismiss="modal" aria-label="Close">
          <span aria-hidden="true">&times;</span>
        </button>
      </div>
      <div>
        <form method="POST" action="{% url 'residents:household_add_member' household.pk
%}">
          {% csrf_token %}
          <div class="modal-body px-4 pb-4">
            <div class="form-group">
              <label class="form-label font-weight-bold">Select Resident</label>
              <select name="resident_id" class="form-control custom-select" required
style="border-radius: 0.75rem; height: 45px;">
                <option value="">-- Choose Resident --</option>
                {% for res in available_residents %}
                <option value="{{ res.pk }}">{{ res.full_name }} ({{ res.age }}
yrs)</option>
                {% endfor %}
              </select>
              <small class="form-text text-muted mt-2">Only unassigned residents are listed
here.</small>
            </div>
            <div class="form-group mt-3">
              <label class="form-label font-weight-bold">Relationship to Head</label>
              <select name="relationship" class="form-control custom-select" required
style="border-radius: 0.75rem; height: 45px;">
                <option value="">-- Select Relationship --</option>
                <option value="head">Head</option>
                <option value="spouse">Spouse</option>
                <option value="child">Child</option>
                <option value="parent">Parent</option>
                <option value="sibling">Sibling</option>
                <option value="grandchild">Grandchild</option>
                <option value="grandparent">Grandparent</option>
                <option value="other">Other Relative</option>
              </select>
            </div>
          </div>
          <div class="modal-footer border-0 px-4 pb-4">
            <button type="button" class="btn btn-light" data-dismiss="modal" style="border-
radius: 0.75rem; font-weight: 600; padding: 10px 20px;">Cancel</button>
            <button type="submit" class="btn btn-primary shadow-sm" style="background:
var(--primary); border: none; border-radius: 0.75rem; font-weight: 600; padding: 10px
20px;">Add Member</button>
          </div>
        </form>
      </div>
    </div>
  </div>
</div>
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<style>
/* Styles matching core layout */
.profile-grid {
  display: grid;
  grid-template-columns: 350px 1fr;
  gap: 25px;
}

@media (max-width: 992px) {
  .profile-grid {
    grid-template-columns: 1fr;
  }
}

.profile-card {
  text-align: center;
  padding: 30px 20px;
}

.profile-photo {
  width: 150px;
  height: 150px;
  margin: 0 auto 20px;
  border-radius: 50%;
  overflow: hidden;
  background: var(--border);
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 60px;
  color: var(--text-muted);
}

.status-badge {
  padding: 4px 10px;
  border-radius: 4px;
  font-size: 11px;
  font-weight: 600;
}

.badge-primary {
  background: rgba(var(--primary-rgb), 0.1);
  color: var(--primary);
}

/* Family Tree Distinctive Aesthetics */
.ft-card {
  border: none;
  background: linear-gradient(145deg, #ffffff, #f3f8f5);
  box-shadow: 0 10px 30px rgba(44, 110, 73, 0.08);
  border-radius: 20px;
  overflow: hidden;
}
.ft-header {
  background: transparent;
  border-bottom: 1px solid rgba(44, 110, 73, 0.1);
  padding: 25px 30px 15px;
}
.ft-header h3 {
  margin: 0;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
font-weight: 800;
color: #1a3c28;
letter-spacing: -0.5px;
display: flex;
align-items: center;
gap: 10px;
}
.ft-icon {
color: var(--primary);
background: rgba(var(--primary-rgb), 0.1);
padding: 8px;
border-radius: 8px;
font-size: 18px;
}
.ft-subtitle {
font-size: 12px;
text-transform: uppercase;
letter-spacing: 1px;
color: #6c8a7b;
font-weight: 600;
display: block;
margin-top: 5px;
}

/* Tree CSS Hierarchy */
.family-tree-container {
width: 100%;
overflow-x: auto;
padding: 30px 10px 50px;
display: flex;
justify-content: center;
}
.tree ul {
padding-top: 20px; position: relative;
transition: all 0.5s;
-webkit-transition: all 0.5s;
-moz-transition: all 0.5s;
display: flex;
justify-content: center;
padding-left: 0;
}
.tree li {
float: left; text-align: center;
list-style-type: none;
position: relative;
padding: 20px 10px 0 10px;
transition: all 0.5s;
-webkit-transition: all 0.5s;
-moz-transition: all 0.5s;
}
.tree li::before, .tree li::after {
content: '';
position: absolute; top: 0; right: 50%;
border-top: 2px solid #a6d1b7;
width: 50%; height: 20px;
}
.tree li::after {
right: auto; left: 50%;
border-left: 2px solid #a6d1b7;
}
```


IoT-Enabled Barangay Resident and Attendance Management System

```
.tree li:only-child::after, .tree li:only-child::before {
    display: none;
}
.tree li:only-child { padding-top: 0; }
.tree li:first-child::before, .tree li:last-child::after {
    border: 0 none;
}
.tree li:last-child::before {
    border-right: 2px solid #a6d1b7;
    border-radius: 0 5px 0 0;
    -webkit-border-radius: 0 5px 0 0;
    -moz-border-radius: 0 5px 0 0;
}
.tree li:first-child::after {
    border-radius: 5px 0 0 0;
    -webkit-border-radius: 5px 0 0 0;
    -moz-border-radius: 5px 0 0 0;
}
.tree ul ul::before {
    content: '';
    position: absolute; top: 0; left: 50%;
    border-left: 2px solid #a6d1b7;
    width: 0; height: 20px;
}
.tree-node {
    border: 1px solid rgba(44, 110, 73, 0.15);
    padding: 15px 20px;
    text-decoration: none;
    color: #1a3c28;
    font-family: inherit;
    font-size: 13px;
    display: inline-block;
    border-radius: 12px;
    background: white;
    transition: all 0.3s ease;
    box-shadow: 0 4px 15px rgba(0,0,0,0.03);
    position: relative;
    min-width: 130px;
}
.tree-node:hover {
    background: var(--primary);
    color: white;
    border: 1px solid var(--primary);
    transform: translateY(-5px);
    box-shadow: 0 10px 25px rgba(44, 110, 73, 0.3);
}
.tree-node:hover .node-name, .tree-node:hover .node-role {
    color: white;
}
.node-info {
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 4px;
}
.node-name {
    font-weight: 700;
    font-size: 14px;
    letter-spacing: -0.2px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.node-role {
    font-size: 11px;
    text-transform: uppercase;
    letter-spacing: 0.5px;
    color: #6c8a7b;
    font-weight: 600;
}

.head-node {
    border: 2px solid var(--primary);
    background: rgba(var(--primary-rgb), 0.05);
}
.head-node .node-role { color: var(--primary); }

.spouse-node {
    border-color: #f59e0b;
}
.child-node {
    border-color: #3b82f6;
}
.other-node {
    border-color: #8b5cf6;
}

/* Drag and Drop Interactivity Styles */
.draggable-node {
    cursor: grab;
    user-select: none;
}
.draggable-node:active {
    cursor: grabbing;
}
.drag-over {
    border: 2px dashed #10b981 !important;
    background: rgba(16, 185, 129, 0.1) !important;
    transform: scale(1.05);
    box-shadow: 0 10px 25px rgba(16, 185, 129, 0.2);
}
.edit-hint {
    display: none;
    font-size: 10px;
    color: #6b7280;
    margin-top: 8px;
    background: rgba(0,0,0,0.05);
    padding: 3px 8px;
    border-radius: 12px;
    font-weight: 600;
    text-transform: uppercase;
    letter-spacing: 0.5px;
}
.edit-mode-active .edit-hint {
    display: inline-block;
}
.edit-mode-active .tree-node {
    border: 1px dashed rgba(44, 110, 73, 0.4);
}
.edit-mode-active .head-node {
    border: 2px dashed var(--primary);
}
.edit-mode-active .tree-node:hover {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        border-style: solid;
    }
    .dragging {
        opacity: 0.5;
        transform: scale(0.95);
    }
</style>
{% endblock %}

{% block extra_js %}
<script>
document.addEventListener('DOMContentLoaded', function () {
    {% if members %}
    const membersData = [
        {% for member in members %}
        {
            id: {{ member.pk }},
            name: "{{{ member.full_name|escapejs }}}",
            role: "{{{ member.get_household_relationship_display|default:'Member'|escapejs }}}",
            relationship: "{{{ member.household_relationship|default:''|escapejs }}}",
            isHead: {{ member.is_household_head|yesno:"true,false" }},
            parentId: {% if member.parent_member %}{{ member.parent_member.pk }}{% else
%}null{% endif %}
        },
        {% endfor %}
    ];

    let isEditMode = false;
    let currentTreeData = JSON.parse(JSON.stringify(membersData));

    function buildTreeData(data) {
        let head = data.find(m => m.isHead);
        let rootNodes = [];

        let map = {};
        data.forEach(m => {
            m.children = [];
            map[m.id] = m;
        });

        data.forEach(m => {
            if (!m.isHead) {
                let pId = m.parentId;
                // If no parent and there is a head, link to head by default for display
                if (!pId && head) pId = head.id;

                if (pId && map[pId]) {
                    map[pId].children.push(m);
                } else {
                    rootNodes.push(m);
                }
            }
        });

        if (head) {
            return [head, ...rootNodes];
        }
        return rootNodes;
    }
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
function getNodeClass(m) {
  if (m.isHead) return 'head-node';
  if (m.relationship === 'spouse') return 'spouse-node';
  if (m.relationship === 'child') return 'child-node';
  return 'other-node';
}

function renderTree(data) {
  const container = document.getElementById('tree-container');
  if (!container) return;
  container.innerHTML = '';

  const treeData = buildTreeData(data);
  if (treeData.length === 0) return;

  const ul = document.createElement('ul');
  treeData.forEach(rootNode => {
    ul.appendChild(createNodeElement(rootNode));
  });

  const div = document.createElement('div');
  div.className = 'tree';
  div.appendChild(ul);
  container.appendChild(div);
}

function createNodeElement(node) {
  const li = document.createElement('li');

  const nodeDiv = document.createElement('div');
  nodeDiv.className = `tree-node ${getNodeClass(node)}`;
  nodeDiv.dataset.id = node.id;

  if (isEditMode) {
    nodeDiv.draggable = true;
    nodeDiv.classList.add('draggable-node');

    nodeDiv.addEventListener('dragstart', (e) => {
      e.dataTransfer.setData('text/plain', node.id);
      setTimeout(() => nodeDiv.classList.add('dragging'), 0);
    });

    nodeDiv.addEventListener('dragend', () => {
      nodeDiv.classList.remove('dragging');
      document.querySelectorAll('.drag-over').forEach(el =>
el.classList.remove('drag-over'));
    });

    nodeDiv.addEventListener('dragover', (e) => {
      e.preventDefault(); // allow drop
      if (!nodeDiv.classList.contains('dragging')) {
        nodeDiv.classList.add('drag-over');
      }
    });

    nodeDiv.addEventListener('dragleave', () => {
      nodeDiv.classList.remove('drag-over');
    });

    nodeDiv.addEventListener('drop', (e) => {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
e.preventDefault();
nodeDiv.classList.remove('drag-over');
const draggedId = parseInt(e.dataTransfer.getData('text/plain'));
const targetId = node.id;

if (draggedId && draggedId !== targetId) {
  if (!isDescendant(targetId, draggedId)) {
    updateParent(draggedId, targetId);
  } else {
    alert("Invalid move! You cannot place a parent inside its own
descendant.");
  }
}
});
}

const info = document.createElement('div');
info.className = 'node-info';
info.innerHTML = `${node.name}</span><span class="node-
role">${node.role}</span>`;
if (isEditMode) {
  info.innerHTML += `

Page 725


```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

renderTree(currentTreeData);

const editBtn = document.getElementById('editTreeBtn');
const saveBtn = document.getElementById('saveTreeBtn');
const cancelBtn = document.getElementById('cancelTreeBtn');

if (editBtn) {
  editBtn.addEventListener('click', () => {
    isEditMode = true;
    document.querySelector('.ft-card').classList.add('edit-mode-active');
    editBtn.style.display = 'none';
    saveBtn.style.display = 'inline-block';
    cancelBtn.style.display = 'inline-block';
    renderTree(currentTreeData);
  });

  cancelBtn.addEventListener('click', () => {
    isEditMode = false;
    document.querySelector('.ft-card').classList.remove('edit-mode-active');
    currentTreeData = JSON.parse(JSON.stringify(membersData));
    editBtn.style.display = 'inline-block';
    saveBtn.style.display = 'none';
    cancelBtn.style.display = 'none';
    renderTree(currentTreeData);
  });

  saveBtn.addEventListener('click', () => {
    const connections = currentTreeData.map(m => ({
      child_id: m.id,
      parent_id: m.parentId
    }));

    saveBtn.disabled = true;
    saveBtn.innerHTML = '<i class="fas fa-spinner fa-spin mr-1"></i> Saving...';

    fetch(`${url} 'residents:household_update_tree' household.pk %}`, {
      method: 'POST',
      headers: {
        'Content-Type': 'application/json',
        'X-CSRFToken': '{ csrf_token }'
      },
      body: JSON.stringify({ connections })
    })
    .then(res => res.json())
    .then(data => {
      if (data.status === 'success') {
        window.location.reload();
      } else {
        alert('Error: ' + data.message);
        saveBtn.disabled = false;
        saveBtn.innerHTML = '<i class="fas fa-save mr-1"></i> Save Layout';
      }
    })
    .catch(err => {
      console.error(err);
      alert('An error occurred.');
```

IoT-Enabled Barangay Resident and Attendance Management System

```
    });
  });
}
{% endif %}
});
</script>
{% if household.latitude and household.longitude %}
<script src="https://unpkg.com/maplibre-gl/dist/maplibre-gl.js"></script>
<script>
document.addEventListener('DOMContentLoaded', function () {
  const lat = parseFloat("{{ household.latitude }}");
  const lng = parseFloat("{{ household.longitude }}");

  const TILE_STREETS = 'https://tile.openstreetmap.org/{z}/{x}/{y}.png';
  const streetsStyle = {
    version: 8,
    sources: {
      'raster-tiles': {
        type: 'raster',
        tiles: [TILE_STREETS],
        tileSize: 256,
        attribution: '&copy; OpenStreetMap'
      }
    },
    layers: [{
      id: 'raster-layer',
      type: 'raster',
      source: 'raster-tiles',
      minzoom: 0,
      maxzoom: 19
    }]
  };

  const map = new maplibregl.Map({
    container: 'mini-household-map',
    style: streetsStyle,
    center: [lng, lat],
    zoom: 16,
    maxZoom: 19,
    attributionControl: false
  });

  map.on('load', function() {
    const el = document.createElement('div');
    el.className = 'pin-marker-view';
    el.innerHTML = '<i class="fas fa-home"></i>';

    new maplibregl.Marker({
      element: el,
      anchor: 'center'
    })
      .setLngLat([lng, lat])
      .addTo(map);
  });
});
</script>
{% endif %}
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/residents/list.html

```
{% extends 'base.html' %}

{% block title %}Residents - Barangay Management System{% endblock %}

{% block breadcrumb %}
<span>Brgy. Sico Sico Management System / Residents</span>
{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Resident Records</h1> <span>Manage and view all registered residents</span>
  </div>
  <div class="header-actions"> <a href="{% url 'residents:add' %}" class="btn btn-primary"> <i class="fas fa-plus"></i>
    Add Resident
  </a> </div>
</div>

<div class="card">
  <div class="card-header filter-card-header">
    <form method="GET" class="filter-form">
      <div class="filter-inputs-group">
        <div class="search-input-wrapper"> <i class="fas fa-search search-icon"></i>
<input type="text" name="q"
      value="{{ query }}" placeholder="Search by name..." class="form-control
search-input"> {% if query %} <a
      href="{% url 'residents:list' %}" class="search-clear" title="Clear search">
        <i class="fas fa-times-circle"></i> </a>
        {% endif %}
      </div>

      <div class="filter-select-wrapper"> <i class="fas fa-filter filter-icon"></i>
<select name="purok"
      class="form-control filter-select">
        <option value="">All Puroks</option>
        {% for p in puroks %}
        <option value="{{ p.id }}" {% if p.id|stringformat:"i" == selected_purok
%}>selected{% endif %}>{{ p.name }}
        </option>
        {% endfor %}
      </select>
    </div>
  </div>

  <button type="submit" class="btn btn-primary search-btn"> <i class="fas fa-
search"></i> <span>Search</span>
</button>
</form>
</div>
<div class="table-responsive">
  <table>
    <thead>
      <tr>
        <th>Photo</th>
        <th>Name</th>
        <th>Age</th>
      </tr>
    </thead>
  </table>
</div>
```


IoT-Enabled Barangay Resident and Attendance Management System

```

        <th>Gender</th>
        <th>Purok</th>
        <th>Voter</th>
        <th>Actions</th>
    </tr>
</thead>
<tbody>
    {% for resident in residents %}
    <tr>
        <td>
            <div class="user-avatar-sm">
                {% if resident.photo %}
                
                {% else %}
                
                {% endif %}
            </div>
        </td>
        <td>
            <div class="resident-name"><strong>{{ resident.last_name }}, {{
resident.first_name }}{% if resident.middle_name %} {{ resident.middle_name }}{% endif
%}{% if resident.suffix %} {{ resident.suffix }}{% endif %}</strong></div>
        </td>
        <td>{{ resident.age }}</td>
        <td>{{ resident.get_gender_display }}</td>
        <td>{{ resident.purok }}</td>
        <td>
            {% if resident.is_registered_voter %}
            <span class="status-badge badge-active">Yes</span>
            {% else %}
            <span class="status-badge badge-inactive">No</span> {% endif %}
        </td>
        <td>
            <div class="btn-group"> <a href="{% url 'residents:view' resident.pk %}"
class="btn-icon btn-view"
                title="View"><i class="fas fa-eye"></i></a> <a href="{% url
'residents:edit' resident.pk %}"
                class="btn-icon btn-edit" title="Edit"><i class="fas fa-edit"></i></a>
                <a href="javascript:void(0)" class="btn-icon btn-delete" title="Delete"
                onclick="confirmDelete('{{ resident.pk }}', '{{ resident.last_name }}', {{
resident.first_name }})">
                    <i class="fas fa-trash-alt"></i>
                </a>
            </div>
        </td>
    </tr>
    {% empty %}
    <tr>
        <td colspan="7" class="text-center">No residents found.</td>
    </tr>
    {% endfor %}
</tbody>
</table>
</div>
{% if residents.has_other_pages %}
<div class="pagination"> <!-- Pagination logic here --> </div>
{% endif %}
</div>

```

IoT-Enabled Barangay Resident and Attendance Management System

```
<!-- Delete Confirmation Modal -->
<div id="deleteModal" class="modal">
  <div class="modal-content card" style="max-width: 450px;">
    <div class="modal-header">
      <h3 style="color: var(--danger);"><i class="fas fa-exclamation-triangle"></i>
Permanent Delete</h3>
      <button class="close-modal" onclick="closeModal()">&times;</button>
    </div>
    <div class="modal-body" style="padding: 20px 0; text-align: center;">
      <p>Are you sure you want to <strong>PERMANENTLY DELETE</strong> the record for
<strong
      id="deleteResidentName"></strong>?</p>
      <p style="font-size: 13px; color: var(--text-muted); margin-top: 10px;">This action
will also delete their user
      account and cannot be undone.</p>
    </div>
    <div class="modal-footer" style="display: flex; gap: 12px; justify-content: center;
padding-top: 10px;">
      <form id="deleteForm" method="POST">
        {% csrf_token %}
        <button type="submit" class="btn btn-danger">
          <i class="fas fa-trash-alt"></i> Delete Permanently
        </button>
      </form>
      <button class="btn btn-secondary" onclick="closeModal()">Cancel</button>
    </div>
  </div>
</div>

<style>
.modal {
  display: none;
  position: fixed;
  z-index: 1000;
  left: 0;
  top: 0;
  width: 100%;
  height: 100%;
  background-color: rgba(0, 0, 0, 0.5);
  backdrop-filter: blur(4px);
  align-items: center;
  justify-content: center;
}

.modal-content {
  background: var(--bg-card);
  padding: 30px;
  border-radius: var(--radius-lg);
  box-shadow: var(--shadow-md);
  animation: modalSlide 0.3s ease;
}

.modal-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-bottom: 1px solid var(--border);
  padding-bottom: 15px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
.close-modal {
  background: none;
  border: none;
  font-size: 24px;
  color: var(--text-muted);
  cursor: pointer;
}

@keyframes modalSlide {
  from {
    transform: translateY(-20px);
    opacity: 0;
  }

  to {
    transform: translateY(0);
    opacity: 1;
  }
}
</style>

<script>
function confirmDelete(pk, name) {
  const modal = document.getElementById('deleteModal');
  const form = document.getElementById('deleteForm');
  const nameSpan = document.getElementById('deleteResidentName');

  nameSpan.textContent = name;
  form.action = `/residents/${pk}/delete/`;
  modal.style.display = 'flex';
}

function closeModal() {
  document.getElementById('deleteModal').style.display = 'none';
}

window.onclick = function (event) {
  const modal = document.getElementById('deleteModal');
  if (event.target == modal) {
    closeModal();
  }
}
</script>

<style>
.user-avatar-sm {
  width: 32px;
  height: 32px;
  border-radius: 50%;
  background: var(--border);
  display: flex;
  align-items: center;
  justify-content: center;
  overflow: hidden;
}

.user-avatar-sm img {
  width: 100%;
  height: 100%;
  object-fit: cover;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

.resident-name span {
  display: block;
  font-size: 11px;
  color: var(--text-muted);
}

.btn-icon {
  padding: 5px 10px;
  font-size: 12px;
}
</style>
{% endblock %}
```

File: templates/residents/registration_detail.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Review Registration: {{ registration.reference_number }} ? Barangay
Management System{% endblock %}

{% block extra_head %}
<style>
    .review-grid {
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 24px;
    }

    .review-section {
        background: var(--bg-card);
        border: 1px solid var(--border);
        border-radius: 16px;
        padding: 24px;
        margin-bottom: 24px;
    }

    .review-section-title {
        font-size: 14px;
        font-weight: 700;
        color: var(--primary);
        text-transform: uppercase;
        letter-spacing: 0.5px;
        margin-bottom: 20px;
        display: flex;
        align-items: center;
        gap: 8px;
    }

    .review-item {
        margin-bottom: 16px;
    }

    .review-label {
        font-size: 12px;
        color: var(--text-muted);
        display: block;
        margin-bottom: 4px;
    }

    .review-value {
        font-size: 15px;
        font-weight: 600;
        color: var(--text-main);
    }

    .doc-preview {
        width: 100%;
        border-radius: 12px;
        border: 1px solid var(--border);
        margin-top: 8px;
        transition: transform 0.3s ease;
    }
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        cursor: pointer;
    }

    .doc-preview:hover {
        transform: scale(1.02);
    }

    .status-banner {
        padding: 16px 24px;
        border-radius: 16px;
        margin-bottom: 24px;
        display: flex;
        align-items: center;
        justify-content: space-between;
    }

    .status-banner.pending {
        background: rgba(255, 193, 7, 0.1);
        border: 1px solid #FFC107;
        color: #856404;
    }

    .status-banner.approved {
        background: rgba(40, 167, 69, 0.1);
        border: 1px solid #28A745;
        color: #155724;
    }

    .status-banner.rejected {
        background: rgba(220, 53, 69, 0.1);
        border: 1px solid #DC3545;
        color: #721c24;
    }

    .action-buttons {
        position: sticky;
        top: calc(var(--header-height) + 24px);
        display: flex;
        flex-direction: column;
        gap: 12px;
    }
</style>
{% endblock %}

{% block content %}
{% url 'residents:registration_list' as registration_list_url %}
{% include 'components/back_button.html' with back_url=registration_list_url
back_label='Back' %}

<div class="page-header mt-3">
    <div class="page-title">
        <h1>Registration Review</h1>
        <span>Reference: <strong>{{ registration.reference_number }}</strong></span>
    </div>
</div>

<div class="status-banner {{ registration.status }}">
    <div class="d-flex align-items-center gap-3">
        <i
            class="fas {% if registration.status == 'pending' %}fa-clock{% elif
```

IoT-Enabled Barangay Resident and Attendance Management System

```
registration.status == 'approved' %}fa-check-circle{% else %}fa-times-circle{% endif %}
fa-2x"></i>
    <div>
        <h4 class="mb-0">Status: {{ registration.get_status_display }}</h4>
        <small>Submitted on {{ registration.created_at|date:"F d, Y P" }}</small>
    </div>
</div>
</div>

<div class="review-grid">
    <div class="review-left">
        <!-- Step 1: Personal Info -->
        <div class="review-section">
            <h3 class="review-section-title"><i class="fas fa-user"></i> Personal
Information</h3>
            <div class="row">
                <div class="col-md-6 review-item">
                    <span class="review-label">Full Name</span>
                    <span class="review-value">
                        {{ registration.first_name }} {{ registration.middle_name }} {{
registration.last_name }}{% if registration.suffix %} {{ registration.suffix }}{% endif %}
                    </span>
                </div>
                <div class="col-md-6 review-item">
                    <span class="review-label">Birthdate</span>
                    <span class="review-value">{{ registration.birthdate|date:"M d, Y"
}}</span>
                </div>
                <div class="col-md-6 review-item">
                    <span class="review-label">Gender</span>
                    <span class="review-value">{{ registration.get_gender_display
}}</span>
                </div>
                <div class="col-md-6 review-item">
                    <span class="review-label">Civil Status</span>
                    <span class="review-value text-capitalize">{{
registration.civil_status }}</span>
                </div>
                <div class="col-md-6 review-item">
                    <span class="review-label">Nationality</span>
                    <span class="review-value">{{ registration.nationality }}</span>
                </div>
                <div class="col-md-6 review-item">
                    <span class="review-label">Occupation</span>
                    <span class="review-value">{{ registration.occupation|default:"N/A"
}}</span>
                </div>
            </div>
        </div>
        <!-- Step 2: Contact & Address -->
        <div class="review-section">
            <h3 class="review-section-title"><i class="fas fa-map-marker-alt"></i> Contact
& Address</h3>
            <div class="row">
                <div class="col-md-6 review-item">
                    <span class="review-label">Mobile Number</span>
                    <span class="review-value">{{ registration.mobile_number }}</span>
                </div>
                <div class="col-md-6 review-item">
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        <span class="review-label">Email</span>
        <span class="review-value">{{ registration.email|default:"N/A"
    }}</span>
</div>
<div class="col-12 review-item">
    <span class="review-label">Complete Address</span>
    <span class="review-value">
        {% if registration.house_number %}{{ registration.house_number }}
    {% endif %}
        {% if registration.street %}{{ registration.street }}, {% endif %}
        {% if registration.purok %}{{ registration.purok }}, {% endif %}
        {% if registration.barangay %}{{ registration.barangay }}, {%
    endif %}
        {% if registration.municipality %}{{ registration.municipality
    }}{% if registration.city %}, {% endif %}{% endif %}
        {% if registration.city %}{{ registration.city }}{% endif %}
        {{ registration.zip_code }}
    </span>
</div>
<div class="col-md-6 review-item">
    <span class="review-label">Purok</span>
    <span class="review-value">{{ registration.purok }}</span>
</div>
<div class="col-md-6 review-item">
    <span class="review-label">Years of Residency</span>
    <span class="review-value">{{ registration.years_of_residency }}
Year(s)</span>
</div>
</div>
</div>

<!-- Step 3: Household & Classification -->
<div class="review-section">
    <h3 class="review-section-title"><i class="fas fa-home"></i> Household &
Classification</h3>
    <div class="row">
        <div class="col-md-6 review-item">
            <span class="review-label">Household Number</span>
            <span class="review-value">{{
registration.household_number|default:"New Household" }}</span>
        </div>
        <div class="col-md-6 review-item">
            <span class="review-label">PWD Status</span>
            <span class="review-value">{% if registration.is_pwd %}Yes{% else
%}No{% endif %}</span>
        </div>
        <div class="col-md-6 review-item">
            <span class="review-label">Senior Citizen</span>
            <span class="review-value">{% if registration.is_senior_citizen
%}Yes{% else %}No{% endif %}</span>
        </div>
        <div class="col-md-6 review-item">
            <span class="review-label">4Ps Member</span>
            <span class="review-value">{% if registration.is_4ps_member %}Yes{%
else %}No{% endif %}</span>
        </div>
    </div>

    {% if registration.guardian_name %}
    <hr class="my-4">

```


IoT-Enabled Barangay Resident and Attendance Management System

```
<h4 class="mb-3" style="font-size: 14px; color: var(--text-muted);">Guardian
Information (Minor)</h4>
<div class="row">
  <div class="col-md-6 review-item">
    <span class="review-label">Guardian Name</span>
    <span class="review-value">{{ registration.guardian_name }}</span>
  </div>
  <div class="col-md-6 review-item">
    <span class="review-label">Relationship</span>
    <span class="review-value">{{ registration.guardian_relationship
}}</span>
  </div>
  <div class="col-md-6 review-item">
    <span class="review-label">Guardian Contact</span>
    <span class="review-value">{{ registration.guardian_contact }}</span>
  </div>
</div>
{% endif %}
</div>
</div>

<div class="review-right">
  <!-- Documents -->
  <div class="review-section">
    <h3 class="review-section-title"><i class="fas fa-file-alt"></i> Uploaded
Documents</h3>

    <div class="review-item">
      <span class="review-label">Resident Photo</span>
      {% if registration.photo %}
      
      {% else %}
      <span class="text-muted">No photo uploaded</span>
      {% endif %}
    </div>

    <div class="review-item">
      <span class="review-label">Valid ID</span>
      {% if registration.id_card %}
      
      {% else %}
      <span class="text-muted">No ID uploaded</span>
      {% endif %}
    </div>

    {% if registration.birth_certificate %}
    <div class="review-item">
      <span class="review-label">Birth Certificate (Minor)</span>
      
    </div>
    {% endif %}

    {% if registration.proof_of_residency %}
    <div class="review-item">
      <span class="review-label">Proof of Residency</span>
      
    </div>
    {% endif %}
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
</div>
{% endif %}
</div>

{% if registration.status == 'pending' %}
<div class="action-buttons">
  <form action="{% url 'residents:approve_registration' registration.pk %}"
method="post">
    {% csrf_token %}
    <button type="submit" class="btn btn-success btn-lg w-100"
      onclick="return confirm('Are you sure you want to approve this
registration? This will create a new Resident and User account.')">
      <i class="fas fa-check-circle mr-2"></i> Approve Registration
    </button>
  </form>

  <form action="{% url 'residents:reject_registration' registration.pk %}"
method="post">
    {% csrf_token %}
    <button type="submit" class="btn btn-danger btn-lg w-100"
      onclick="return confirm('Are you sure you want to reject this
registration?')">
      <i class="fas fa-times-circle mr-2"></i> Reject Registration
    </button>
  </form>
</div>
{% endif %}
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/residents/registration_list.html

```
{% extends 'base.html' %}
{% load static %}

{% block title %}Pending Registrations ? Barangay Management System{% endblock %}

{% block content %}
<div class="page-header">
  <div class="page-title">
    <h1>Resident Registrations</h1>
    <span>Review and verify new resident sign-ups</span>
  </div>
  <div class="header-actions d-flex align-items-center gap-3">
    {% if registrations %}
      <form action="{% url 'residents:registration_bulk_delete' %}" method="post"
onsubmit="return confirm('Are you sure you want to delete ALL {{ status_filter }}
registrations? This cannot be undone.')">
        {% csrf_token %}
        <input type="hidden" name="status" value="{{ status_filter|default:'all' }}">
        <button type="submit" class="btn btn-outline-danger shadow-sm d-flex align-
items-center gap-2" style="border-radius: 12px; padding: 10px 15px; font-weight: 600;">
          <i class="fas fa-trash-sweep"></i>
          <span>Clear {{ status_filter|default:'All'|title }}</span>
        </button>
      </form>
    {% endif %}

    <div class="dropdown">
      <button class="btn dropdown-toggle shadow-sm d-flex align-items-center gap-2
      {% if status_filter == 'pending' %}btn-warning text-white
      {% elif status_filter == 'approved' %}btn-success text-white
      {% elif status_filter == 'rejected' %}btn-danger text-white
      {% else %}btn-primary text-white{% endif %}"
      type="button" id="statusDropdown" data-toggle="dropdown"
      aria-haspopup="true" aria-expanded="false"
      style="border-radius: 12px; padding: 10px 20px; font-weight: 700; border:
none; transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);">

        <i class="fas fa-filter mr-1"></i>
        <span>
          {% if status_filter == 'pending' %}
            <i class="fas fa-clock mr-1"></i> Pending
          {% elif status_filter == 'approved' %}
            <i class="fas fa-check-circle mr-1"></i> Approved
          {% elif status_filter == 'rejected' %}
            <i class="fas fa-times-circle mr-1"></i> Rejected
          {% else %}
            <i class="fas fa-list mr-1"></i> All Registrations
          {% endif %}
        </span>
      </button>
      <div class="dropdown-menu dropdown-menu-right shadow-lg border-0" aria-
labelledby="statusDropdown" style="border-radius: 16px; padding: 10px; margin-top: 10px;
min-width: 200px;">
        <a class="dropdown-item d-flex align-items-center {% if not status_filter
}%active bg-primary text-white{% endif %}" href="?status=" style="border-radius: 10px;
padding: 12px 16px; font-weight: 600; margin-bottom: 4px;">
          <i class="fas fa-list mr-3 {% if not status_filter %}text-white{% else
```

IoT-Enabled Barangay Resident and Attendance Management System

```

    %}text-muted{% endif %}"></i> All Registrations
    </a>
    <div class="dropdown-divider my-2"></div>
    <a class="dropdown-item d-flex align-items-center {% if status_filter ==
'pending' %}active bg-warning text-white{% endif %}" href="?status=pending" style="border-
radius: 10px; padding: 12px 16px; font-weight: 600; margin-bottom: 4px;">
        <i class="fas fa-clock mr-3 {% if status_filter == 'pending' %}text-
white{% else %}text-warning{% endif %}"></i> Pending
    </a>
    <a class="dropdown-item d-flex align-items-center {% if status_filter ==
'approved' %}active bg-success text-white{% endif %}" href="?status=approved"
style="border-radius: 10px; padding: 12px 16px; font-weight: 600; margin-bottom: 4px;">
        <i class="fas fa-check-circle mr-3 {% if status_filter == 'approved'
%}text-white{% else %}text-success{% endif %}"></i> Approved
    </a>
    <a class="dropdown-item d-flex align-items-center {% if status_filter ==
'rejected' %}active bg-danger text-white{% endif %}" href="?status=rejected"
style="border-radius: 10px; padding: 12px 16px; font-weight: 600;">
        <i class="fas fa-times-circle mr-3 {% if status_filter == 'rejected'
%}text-white{% else %}text-danger{% endif %}"></i> Rejected
    </a>
</div>
</div>
</div>
</div>

<div class="card">
    <div class="table-responsive">
        <table class="table">
            <thead>
                <tr>
                    <th>Ref Number</th>
                    <th>Name</th>
                    <th>Date Submitted</th>
                    <th>Purok</th>
                    <th>Status</th>
                    <th class="text-right">Actions</th>
                </tr>
            </thead>
            <tbody>
                {% for reg in registrations %}
                <tr>
                    <td><span class="font-weight-bold">{{ reg.reference_number
}}</span></td>
                    <td>
                        <div class="d-flex align-items-center">
                            {% if reg.photo %}
                            
                            {% else %}
                            
                            {% endif %}
                            {{ reg.first_name }} {{ reg.last_name }}
                        </div>
                    </td>
                    <td>{{ reg.created_at|date:"M d, Y" }} <small class="text-muted">{{

```

IoT-Enabled Barangay Resident and Attendance Management System

```
reg.created_at|time:"H:i" }}</small></td>
    <td>{{ reg.purok }}</td>
    <td>
        {% if reg.status == 'pending' %}
        <span class="badge badge-warning">Pending</span>
        {% elif reg.status == 'approved' %}
        <span class="badge badge-success">Approved</span>
        {% else %}
        <span class="badge badge-danger">Rejected</span>
        {% endif %}
    </td>
    <td class="text-right">
        <div class="btn-group">
            <a href="{% url 'residents:registration_detail' reg.pk %}"
                class="btn btn-sm btn-outline-primary" title="Review">
                <i class="fas fa-search"></i>
            </a>
            <form action="{% url 'residents:registration_delete' reg.pk
                %}" method="post" style="display:inline;" onsubmit="return confirm('Are you sure you want
                to delete this registration?')">
                {% csrf_token %}
                <button type="submit" class="btn btn-sm btn-outline-
                danger" title="Delete">
                    <i class="fas fa-trash-alt"></i>
                </button>
            </form>
        </div>
    </td>
</tr>
{% empty %}
<tr>
    <td colspan="6" class="text-center py-5">
        <i class="fas fa-user-clock fa-3x text-muted mb-3 d-block"></i>
        <p class="text-muted">No resident registrations found.</p>
    </td>
</tr>
{% endfor %}
</tbody>
</table>
</div>

{% if registrations.has_other_pages %}
<div class="card-footer">
    <nav aria-label="Page navigation">
        <ul class="pagination mb-0">
            {% if registrations.has_previous %}
            <li class="page-item"><a class="page-link"
                href="?page={{ registrations.previous_page_number }}&status={{
status_filter }}">Previous</a>
            </li>
            {% endif %}

            <li class="page-item active"><span class="page-link">{{
registrations.number }} of {{ registrations.paginator.num_pages }}</span></li>

            {% if registrations.has_next %}
            <li class="page-item"><a class="page-link"
                href="?page={{ registrations.next_page_number }}&status={{
status_filter }}">Next</a></li>
            {% endif %}
        </ul>
    </nav>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
        </ul>
    </nav>
</div>
{% endif %}
</div>
{% endblock %}
```

IoT-Enabled Barangay Resident and Attendance Management System

File: templates/residents/view.html

```
{% extends 'base.html' %}

{% block title %}{{ resident.full_name }} - Resident Details{% endblock %}

{% block breadcrumb %}
<span><a href="{% url 'residents:list' %}">Residents</a> / {{ resident.full_name }}</span>
{% endblock %}

{% block content %}
{% url 'residents:list' as residents_list_url %}
{% include 'components/back_button.html' with back_url=residents_list_url
back_label='Back' %}
<div class="page-header">
    <div class="page-title">
        <h1>Resident Details</h1>
        <span>Profile information for {{ resident.full_name }}</span>
    </div>
</div>

<div class="profile-grid">
    <div class="profile-sidebar">
        <div class="card profile-card">
            <div class="profile-photo">
                {% if resident.photo %}
                
                {% else %}
                
                {% endif %}
            </div>
            <div class="profile-info">
                <h2>{{ resident.full_name }}</h2>
                <span class="resident-id">ID: #{{ resident.pk|stringformat:"05d" }}</span>
                <div class="status-tags">
                    {% if resident.is_registered_voter %}
                    <span class="badge badge-success">Voter</span>
                    {% endif %}
                    {% if resident.is_pwd %}
                    <span class="badge badge-warning">PWD</span>
                    {% endif %}
                    {% if resident.is_4ps_member %}
                    <span class="badge badge-info">4Ps</span>
                    {% endif %}
                </div>
            </div>
        </div>
    </div>

    <div class="card mt-4">
        <div class="card-header">
            <h3>Contact Information</h3>
        </div>
        <div class="card-body">
            <div class="info-item">
                <i class="fas fa-phone"></i>
                <div>
                    <label>Phone</label>
                    <span>{{ resident.contact_number|default:"Not provided" }}</span>
                </div>
            </div>
        </div>
    </div>
</div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```

        </div>
    </div>
    <div class="info-item">
        <i class="fas fa-envelope"></i>
        <div>
            <label>Email</label>
            <span>{{ resident.email|default:"Not provided" }}</span>
        </div>
    </div>
</div>
</div>
</div>
</div>
</div>
<div class="profile-main">

    <div class="card mt-4">
        <div class="card-header">
            <h3>Personal Information</h3>
        </div>
        <div class="card-body">
            <div class="detail-grid">
                {% if resident.user_profile %}
                <div class="detail-item">
                    <label>System Username</label>
                    <span class="text-primary" style="font-family: monospace; font-size:
1.1em;">{{ resident.user_profile.user.username }}</span>
                </div>
                <div class="detail-item">
                    <label>System Password</label>
                    <div style="display: flex; gap: 10px; align-items: center; margin-top: 2px;">
                        <div style="position: relative; display: flex; align-items: center;">
                            <span class="text-muted" id="passwordMask" style="font-family: monospace;
font-size: 1.1em; letter-spacing: 2px;">????????</span>
                            <i class="fas fa-eye" style="margin-left: 8px; font-size: 12px; color:
var(--text-muted); cursor: help;" title="Password is encrypted for security"></i>
                        </div>
                    </div>

                    {% if request.user == resident.user_profile.user %}
                    <button type="button" class="btn btn-outline-primary" data-toggle="modal"
data-target="#changePasswordModal" style="padding: 2px 8px; font-size: 12px; border-
radius: 4px; border: 1px solid var(--primary); background: transparent; cursor: pointer;">
                        Change
                    </button>
                    {% elif request.user.is_superuser or request.user.profile.role != 'resident'
%}

                    <form method="POST" action="{% url 'residents:reset_password' resident.pk
%}" style="display: inline;">
                        {% csrf_token %}
                        <button type="submit" class="btn btn-outline-warning" onclick="return
confirm('Are you sure you want to reset this resident\'s password to a temporary one?')">
                            style="padding: 2px 8px; font-size: 12px; border-radius: 4px; border: 1px solid #ffc107;
background: transparent; cursor: pointer; color: #856404; font-weight: 600;">
                                Reset
                            </button>
                        </form>
                    {% endif %}
                </div>
            </div>
        </div>
    </div>
    <div class="detail-item">

```


IoT-Enabled Barangay Resident and Attendance Management System

```
<label>System Role</label>
<span class="badge badge-info">{{ resident.user_profile.user.get_role_display
}}</span>
</div>
{% endif %}
<div class="detail-item">
  <label>Full Name</label>
  <span>{{ resident.last_name }}, {{ resident.first_name }}{% if
resident.middle_name %} {{ resident.middle_name }}{% endif %}{% if resident.suffix %} {{
resident.suffix }}{% endif %}</span>
</div>
<div class="detail-item">
  <label>Gender</label>
  <span>{{ resident.get_gender_display }}</span>
</div>
<div class="detail-item">
  <label>Birthdate</label>
  <span>{{ resident.birthdate }} (Age: {{ resident.age }})</span>
</div>
<div class="detail-item">
  <label>Civil Status</label>
  <span>{{ resident.get_civil_status_display }}</span>
</div>
<div class="detail-item">
  <label>Nationality</label>
  <span>{{ resident.nationality }}</span>
</div>
<div class="detail-item">
  <label>Religion</label>
  <span>{{ resident.religion|default:"---" }}</span>
</div>
<div class="detail-item">
  <label>Occupation</label>
  <span>{{ resident.display_occupation }}</span>
</div>
</div>
</div>
</div>
<div class="card mt-4">
  <div class="card-header">
    <h3>Address & Household</h3>
  </div>
  <div class="card-body">
    <div class="detail-grid">
      <div class="detail-item col-span-2">
        <label>Address</label>
        <span>{{ resident.address }}</span>
      </div>
      <div class="detail-item">
        <label>Purok / Zone</label>
        <span>{{ resident.purok }}</span>
      </div>
      <div class="detail-item">
        <label>Household</label>
        <span>
          {% if resident.household %}
          <a href="{% url 'residents:household_view' resident.household.pk %}">
            Household #{{ resident.household.household_no }}
          </a>
          {% else %}
            No Household
          {% endif %}
        </span>
      </div>
    </div>
  </div>
</div>
```

IoT-Enabled Barangay Resident and Attendance Management System

```
{% if resident.is_household_head %}(Head){% endif %}
{% else %}
    Not assigned to a household
{% endif %}
</span>
</div>
</div>
</div>
</div>

{% if certificates %}
<div class="card mt-4">
    <div class="card-header">
        <h3>Recent Certifications</h3>
    </div>
    <div class="table-responsive">
        <table>
            <thead>
                <tr>
                    <th>Type</th>
                    <th>Date Issued</th>
                    <th>Status</th>
                </tr>
            </thead>
            <tbody>
                {% for cert in certificates %}
                <tr>
                    <td>{{ cert.get_cert_type_display }}</td>
                    <td>{{ cert.date_issued|date:"M d, Y" }}</td>
                    <td><span class="status-badge badge-{{ cert.status }}">{{
cert.get_status_display }}</span></td>
                </tr>
                {% endfor %}
            </tbody>
        </table>
    </div>
</div>
{% endif %}

<div class="card mt-4">
    <div class="card-header">
        <h3>Actions</h3>
    </div>
    <div class="card-body">
        <div class="action-row">
            <a href="{% url 'residents:edit' resident.pk %}" class="btn btn-primary">
                <i class="fas fa-edit"></i> Edit Profile
            </a>
        </div>
    </div>
</div>
</div>
</div>

<!-- Change Password Modal -->
<div class="modal fade" id="changePasswordModal" tabindex="-1" role="dialog" aria-
hidden="true">
    <div class="modal-dialog modal-dialog-centered" role="document">
        <div class="modal-content border-0 shadow-lg" style="border-radius: 1.5rem;">
```

IoT-Enabled Barangay Resident and Attendance Management System

```
<div class="modal-header border-0 pt-4 px-4">  
  <h5 class="modal-title font-weight-bold">Change System Password</h5>  
  <button type="button" class="close" data-dismiss="modal" aria-label="Close">  
    <span aria-hidden="true">&times;</span>  
  </button>  
</div>  
<form method="POST" action="{% url 'residents:change_password' resident.pk %}">  
  {% csrf_token %}  
  <div class="modal-body px-4 pb-4">  
    <div class="form-group mb-4">  
      <label class="text-muted small font-weight-bold text-uppercase">New  
Password</label>  
      <div style="position: relative;">  
        <input type="password" name="new_password" id="newPasswordInput"  
class="form-control py-3" placeholder="Enter new password" style="border-radius: 0.75rem;  
padding-right: 45px;" required minlength="8">  
        <button type="button" onclick="togglePasswordVisibility()" style="position:  
absolute; right: 15px; top: 50%; transform: translateY(-50%); background: none; border:  
none; color: var(--text-muted); cursor: pointer;">  
          <i class="fas fa-eye" id="toggleIcon"></i>  
        </button>  
      </div>  
      <small class="form-text text-muted mt-2">Password must be at least 8  
characters long.</small>  
    </div>  
    <button type="submit" class="btn btn-primary btn-block py-3 font-weight-bold"  
style="border-radius: 1rem;">  
      Update Password  
    </button>  
  </div>  
</form>  
</div>  
</div>  
</div>  
  
<script>  
function togglePasswordVisibility() {  
  const input = document.getElementById('newPasswordInput');  
  const icon = document.getElementById('toggleIcon');  
  if (input.type === 'password') {  
    input.type = 'text';  
    icon.classList.remove('fa-eye');  
    icon.classList.add('fa-eye-slash');  
  } else {  
    input.type = 'password';  
    icon.classList.remove('fa-eye-slash');  
    icon.classList.add('fa-eye');  
  }  
}  
</script>  
  
<style>  
.profile-grid {  
  display: grid;  
  grid-template-columns: 340px 1fr;  
  gap: 20px;  
  align-items: start;  
}  
  
.profile-card {
```

IoT-Enabled Barangay Resident and Attendance Management System

```
text-align: center;
padding: 30px 20px;
}

.profile-photo {
width: 150px;
height: 150px;
margin: 0 auto 20px;
border-radius: 50%;
overflow: hidden;
border: 5px solid var(--bg-card);
box-shadow: 0 0 20px rgba(0, 0, 0, 0.2);
}

.profile-photo img {
width: 100%;
height: 100%;
object-fit: cover;
}

.photo-placeholder {
width: 100%;
height: 100%;
background: var(--border);
display: flex;
align-items: center;
justify-content: center;
font-size: 60px;
color: var(--text-muted);
}

.profile-info h2 {
font-size: 20px;
margin-bottom: 5px;
}

.resident-id {
display: block;
color: var(--text-muted);
font-size: 13px;
margin-bottom: 15px;
}

.status-tags {
display: flex;
gap: 8px;
justify-content: center;
}

.badge {
padding: 4px 10px;
border-radius: 4px;
font-size: 11px;
font-weight: 600;
text-transform: uppercase;
}

.badge-success {
background: rgba(16, 185, 129, 0.1);
color: #10b981;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

.badge-warning {
  background: rgba(245, 158, 11, 0.1);
  color: #f59e0b;
}

.badge-info {
  background: rgba(59, 130, 246, 0.1);
  color: #3b82f6;
}

.info-item {
  display: flex;
  align-items: flex-start;
  gap: 15px;
  margin-bottom: 20px;
  width: 100%;
  overflow-wrap: break-word;
}

.info-item i {
  font-size: 20px;
  color: var(--primary);
  width: 32px;
  height: 32px;
  background: rgba(79, 70, 229, 0.1);
  border-radius: 8px;
  display: flex;
  align-items: center;
  justify-content: center;
  flex-shrink: 0;
}

.info-item label {
  display: block;
  font-size: 11px;
  text-transform: uppercase;
  letter-spacing: 0.5px;
  color: var(--text-muted);
  margin-bottom: 2px;
}

.info-item span {
  font-size: 15px;
  font-weight: 600;
  color: var(--text-main);
}

.detail-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 35px 25px;
}

.detail-item label {
  display: block;
  font-size: 12px;
  color: var(--text-muted);
  margin-bottom: 5px;
}
```

IoT-Enabled Barangay Resident and Attendance Management System

```
}

.detail-item span {
  font-size: 15px;
  font-weight: 600;
  color: var(--text-main);
  display: block;
  margin-top: 2px;
}

.col-span-2 {
  grid-column: span 2;
}

.mt-4 {
  margin-top: 30px;
}

.action-row {
  display: flex;
  justify-content: flex-end;
  gap: 12px;
  flex-wrap: wrap;
}

.btn-outline-danger {
  color: #ef4444;
  border-color: #ef4444;
  background: transparent;
  transition: background 0.15s ease, color 0.15s ease, border-color 0.15s ease;
}

.btn-outline-danger:hover,
.btn-outline-danger:focus {
  background: #ef4444 !important;
  color: white !important;
  border-color: #ef4444 !important;
}

@media (max-width: 992px) {
  .profile-grid {
    grid-template-columns: 1fr;
  }
}
</style>
{% endblock %}
```

File: test_fpdf.py

```
from fpdf import FPDF
pdf = FPDF()
print("fpdf version:", getattr(pdf, '__version__', 'unknown'))
if hasattr(pdf, 'insert_toc_placeholder'):
    print("insert_toc_placeholder is available!")
else:
    print("insert_toc_placeholder is NOT available.")
```

IoT-Enabled Barangay Resident and Attendance Management System

File: urls.py

```
from django.contrib import admin
from django.urls import path, include
from django.conf import settings
from django.conf.urls.static import static
from core import views as core_views

urlpatterns = [
    path("admin/", admin.site.urls),
    path("", include("residents.urls")),
    path("census/", include("census.urls")),
    path("officials/", include("officials.urls")),
    path("attendance/", include("attendance.urls")),
    path("certifications/", include("certifications.urls")),
    path("reports/", include("reports.urls")),
    path("philsys/", include("philsys.urls")),
    path("documents/", include("documents.urls")),
    path("login/", core_views.login_view, name="login"),
    path("logout/", core_views.logout_view, name="logout"),
    path("signup/", core_views.signup_view, name="signup"),
    path("resident-signup/", core_views.resident_signup_view, name="resident_signup"),
    path("registration-success/<str:ref>/", core_views.registration_success_view,
name="registration_success"),
    path(
        "dashboard/",
        lambda request: __import__("residents.views", fromlist=["dashboard"]).dashboard(
            request
        ),
        name="dashboard",
    ),
]

if settings.DEBUG:
    urlpatterns += static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT)
```


IoT-Enabled Barangay Resident and Attendance Management System

File: wsgi.py

```
"""
WSGI config for barangay_system project.

It exposes the WSGI callable as a module-level variable named ``application``.

For more information on this file, see
https://docs.djangoproject.com/en/6.0/howto/deployment/wsgi/
"""

import os

from django.core.wsgi import get_wsgi_application

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'barangay_system.settings')

application = get_wsgi_application()
```